The Palgrave Handbook of Supply Chain and Disruptive Technologies

Edited by
Nachiappan Subramanian
Yasanur Kayikci · Atanu Chaudhuri
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1

Mapping the Supply Chain with Disruptive Technologies: Parts and Overview

Nachiappan Subramanian, Yasanur Kayikci, Atanu Chaudhuri, and Michael Bourlakis

Digitalisation refers to the strategic use of information to enhance both inter- and intra-organizational decision-making and processes (Holmstrom et al., 2019). In today's rapidly evolving landscape, emerging and disruptive technologies are reshaping supply chains, compelling stakeholders to explore innovative pathways for generating and capturing value. These advancements are not only driving radical growth but also fostering operational excellence.

A critical discussion revolves around the need to develop robust digital capabilities to fully harness the potential of transformative technologies such as blockchain, big data analytics, cloud computing, artificial intelligence, drones, autonomous vehicles, 5G, and beyond. These technologies are revolutionizing key supply chain functions, including material management,

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Table 1 Classification of technology and description (Adapted from Pagano & Liotine, 2020)

Emerging technologies (These are disruptive in nature and significantly alter the logistics landscape. The current adaption level is 10%)	
3D printing	On demand manufacturing and quick production of more customized products
Drones	Drones are an advanced version of unmanned aerial vehicles (UAVs) that are important transportation systems to fulfil logistics needs during emergency situations where conventional logistical methods are not effective
Autonomous vehicles & robotics	Use of intelligence in smart IoT systems in business operations, production, and logistics enabling track and trace of real-time data and autonomous decision-making
Exponential technologies (These technologies build intelligence are in the early stage of adoption)	
Blockchain	Blockchain is a distributed ledger technology that can improve the speed of transaction execution and validation of information with multiple parties
Internet of Things	It is the connection of devices and equipment to the internet
Virtual reality	Virtual scenario to enhance real-time experience
AI and Machine learning	Use of statistical technique used to predict the outcome based on a large data set and the evolution of large language models revolutionized generative AIs

inventory control, transportation, distribution, workflow optimization, plant maintenance, quality assurance, and environmental and health safety. By enhancing value creation and improving operational efficiency, digitalization is fundamentally redefining supply chain dynamics.

The pervasive role of these technologies in reshaping the supply chain, categorized by their adoption levels, is illustrated in Table 1 (Pagano & Liotine, 2020).

Smart supply chain functions encompass several critical capabilities, including sensing (real-time data collection through advanced technologies), integration (seamless data connectivity), intelligent analysis (leveraging algorithms and simulators for insights), optimization decisions (using predictive analytics to determine effective solutions), timely feedback (delivering real-time, actionable information), and automated error correction (self-regulating mechanisms to address deviations). These functions collectively

enable supply chains to adapt dynamically to evolving scenarios and disruptions.

However, as Bjorkdahl et al. (2020) highlight, several open questions remain from the perspective of focal firms, requiring further exploration:

- What are the key barriers, enablers, and capabilities necessary to support digitalization efforts?
- At what stage do digitalization initiatives create the most value, and how urgent are these efforts?
- How can organizations identify and implement a suitable operating model that integrates cultural, organizational, and managerial processes to drive digital transformation effectively?

Addressing these questions is crucial for organizations striving to unlock the full potential of digitalization and build resilient, future-ready supply chains.

In addition, further investigation from the focal firm's supply chain perspective in terms of product, manufacturing processes and logistics are needed to understand the following (Holmstrom et al., 2019): (i) digitalisation's role in shifting economies of scale from maximizing manufacturing asset reuse to design knowledge reuse. (ii) challenges in accommodating dynamic changes in product design. (iii) role of economies of scale and scope in digitalisation of manufacturing when product diversity increases. (iv) digital encapsulation impact on inventory management and lot sizing, and (v) higher autonomy and value creation or service delivery of supply chain actors.

From the supply chain point of view, digitalisation in terms of interactivity and redistribution of activities to other regional and global organizations will have varied impacts on design, manufacturing and delivery and use. Few open questions are as follows: (i) new emerging design practices, (ii) to what extent inclusion of user experience opens up a new avenue for incremental customization, (iii) intervention of technology between build-to-model and conventional manufacturing, (iv) new practices involving interactive design for product use, (v) assessing the capabilities through the interaction between individuals and structure. From the redistribution angle few open questions are (i) the use of digitalisation to improve the performance of community designers, (ii) risks associated with the transfer of digital design and manufacturing, (iii) influence of power, disintermediation, entry of new players in redistribution and restructuring of regional manufacturing through digitalisation and (iv) digitalisation support to business models for the sharing economy.

Building on these developments, Mithas et al. (2022) mapped emerging technologies to key operations management research challenges, urging scholars to explore critical questions—particularly, how these advancements contribute to sustainability and resilience. Similarly, Choi et al. (2022) examined the advantages and limitations of disruptive technologies in the digital era and beyond, with a specific focus on sustainable social welfare. Their research highlights key concerns, including employee well-being, data privacy, and the role of policymakers in shaping responsible digital transformation.

This handbook seeks to address some of the pressing open questions surrounding supply chain digitalization, as outlined in the Part-technology mapping presented in Table 2. Overall, it delves into six core Parts, each corresponding to different stages of the supply chain, offering a comprehensive exploration of digital transformation in this evolving landscape.

- Part I: Supply Chain digitalization—Key barriers, enablers and capabilities
- Part II: Technological effort vs value in digital supply chains
- Part III: Transformational model and process in digital supply chains
- Part IV: Digital design capabilities for various technologies in digital supply chains
- Part V: Digital manufacturing capabilities for emerging technologies in digital supply chains
- Part VI: Digital delivery and use capabilities for emerging technologies in digital supply chains

The technology coverage includes Blockchain, Big data analytics, cloud/edge/fog computing, artificial intelligence, machine/deep learning, 3D/4D printing, drones, augmented and virtual reality, cyber physical system, cybersecurity, digital twins, autonomous vehicles, integrated systems, 5G/6G, IoT/IIoT, RFID/QR code, Bluetooth, and cognitive computing.

Part I: Supply Chain Digitalization—Key Barriers, Enablers, and Capabilities

This Part explores the critical barriers, enablers, and capabilities required to support digitalization efforts in supply chains. Technologies such as blockchain, big data analytics, cloud computing, artificial intelligence, drones, autonomous vehicles, and 5G are considered within this Part. However, particular emphasis is placed on prominent technologies like 3D printing, the Internet of Things (IoT), and big data, which are examined in depth across four dedicated chapters.

Table 2 Part- Technology mapping

Part vs Technology	Blockchain	Big Data & Analytics	Cloud /Edge/ Fog Computing	Artificial Intelligence	Machine Learning/ Deep Learning	3D Printing/ Additive Manufacturing	Drones	Augmented and Virtual Reality	Advanced Robotics and AGVs
(1) Key enablers and capabilities	X	X		X			X		
(2) Value of digitalisation efforts			X						X
(3) Transformational model and process					X				
(4) Digital design capabilities	X					X	X	X	X
(5) Digital manufacturing capabilities					X	X		X	
(6) Digital delivery and use capabilities		X		X			X		

Cyber Physical System	Cybersecurity	Simulation/ Digital Twin	Autonomous Vehicles	Integrated systems (M2M)	5G/ 6G	IoT/IioT/ Sensors	RFID/ Barcode	NFC/ QR-Code	Bluettooth/ ZigBee/WiFi	Cognitive Computing	Others
X						X				X	
				X				X			
	X				X				X		
		X					X	X			
	X		X								

The chapters in this Part provide a comprehensive literature review and framework development, offering insights into the impact of digitalization across end-to-end supply chains. As editors, we have ensured that the coverage spans various stages of supply chains, rather than focusing solely on specific companies or manufacturers. Below is a summary of the four chapters:

Chapter 2: 3D Printing in Supply Chains—Benefits, Challenges, and Future Directions

This chapter systematically reviews 139 studies to examine the role of 3D printing in manufacturing supply chains (SCs). The analysis focuses on:

1. Trends in 3D printing research within manufacturing SCs.
2. Benefits and challenges of implementing 3D printing in manufacturing SCs.
3. Future research directions in the field.

The findings highlight key advantages, such as greater supply chain flexibility, shorter lead times through localized manufacturing, opportunities for mass customization, and a reduced carbon footprint. However, the study also identifies notable challenges, including the need for new skill sets, the high cost of printers, and structural adaptations required in supply chains. The chapter presents a structured framework addressing these complexities.

Chapter 3: The Role of IoT and Big Data in Supply Chain Processes

This chapter explores the transformative impact of Internet of Things (IoT) and Big Data (BD) on supply chain operations, particularly in inventory optimization and food traceability. Using real-world examples, the authors highlight the value of these technologies while acknowledging significant technological challenges related to data quality, security, and governance.

The study identifies key enablers, benefits, and challenges of adopting IoT and BD in supply chains, emphasizing their potential to enhance operational efficiency and competitiveness.

Chapter 4: Disruptive Technologies and Sustainable Development Goals in Supply Chains

This chapter investigates how disruptive technologies contribute to achieving Sustainable Development Goals (SDGs) in supply chains. The authors employ a causal-loop diagram to analyze the causal relationships between sustainability impacts and technological advancements.

Key insights include:

- The lack of climate change action due to strict enforcement of environmental policies.
- Challenges in renewable energy access, electricity distribution, and smart city development, which directly affect carbon emission reductions.
- The crucial role of governments in creating a sustainable ecosystem that fosters the adoption of green technologies.

Chapter 5: Digital Supply Chain Activities and Sustainable Performance—The Mediating Role of Supply Chain Capabilities

Focusing on Iranian manufacturing companies, this chapter examines the mediating role of supply chain capabilities (SCCs) in the relationship between digital supply chain (DSC) activities and sustainable performance.

Based on empirical evidence from senior managers and high-level employees, the study investigates:

1. The impact of digital supply chain activities on sustainable performance.
2. How supply chain capabilities mediate this relationship.

The chapter presents a conceptual framework linking DSC activities, SCCs, and sustainable performance. From a managerial perspective, the findings suggest that companies should allocate more resources to enhance the benefits of digital technologies, particularly in communication and information management. Additionally, the authors confirm that digital technologies play a pivotal role in developing SCCs, enabling firms to achieve greater responsiveness, integration, coordination, and information exchange.

Part II: Technological Effort vs. Value in Supply Chains

This Part examines the value generated by digitalization efforts across various stages of the supply chain. It explores the impact of technologies such as blockchain, big data analytics, cloud computing, artificial intelligence, drones, autonomous vehicles, and 5G. To ensure a practical perspective, the editors encouraged authors to illustrate technology applications through

pilot studies. Each chapter focuses on a specific technology, detailing current challenges, real-world applications, and future research opportunities.

Chapter 6: Procurement Digitalisation on International Procurement and Supply Chain Management Strategies

This chapter investigates the impact of procurement digitalization (PD) efforts on international procurement and supply chain management (IPSCM) strategies within the fast fashion sector. Through a systematic literature review, the study addresses the critical question:

- *How do digitalization efforts contribute to value creation in international procurement and supply chain management strategies?*

Findings highlight that real-time data utilization, sustainability, supply chain transparency, digital transformation, and data quality are key value drivers. However, resistance to change, high implementation costs, cybersecurity risks, and legacy system incompatibilities pose significant challenges. The study underscores the importance of collaboration, change management, and continuous improvement in fostering successful digitalization efforts.

Additionally, it calls for quantitative assessments of digital transformation efforts within leading fast fashion firms for benchmarking, along with a deeper exploration of innovative sustainability technologies and circular business models to unlock transformative possibilities.

Chapter 7: Customer Switching Behavior and the Adoption of Disruptive Technologies

This chapter explores how customer dissatisfaction and comparative economic value influence switching behavior toward disruptive technologies in B2B markets. The study investigates:

1. *How do user dissatisfaction and comparative economic value shape customer preferences and switching behavior?*
2. *What role does social influence play in moderating this relationship?*
3. *How does perceived preference mediate the link between dissatisfaction, economic value, and switching behavior?*

Empirical findings reveal that user-perceived preference acts as a key mediator between dissatisfaction, economic value, and switching decisions. The study

provides critical insights into both the advantages and challenges of disruptive technologies, shedding light on their impact on existing and potential customers.

Chapter 8: Chief Privacy Officer (CPO) Appointments in Digital Supply Chains—Antecedents and Financial Implications

This chapter examines the drivers and financial consequences of appointing a Chief Privacy Officer (CPO) in response to rising concerns over data privacy risks in new technology implementations. By analyzing 50 CPO appointments in U.S. firms (2010–2019), the study addresses the following questions:

1. *What factors influence a firm's decision to appoint a CPO?*
2. *What are the immediate financial benefits of CPO appointments?*

Findings indicate that firms with high inherent data risks and media exposure are more likely to appoint a CPO, whereas those that have already experienced prior data breaches are less inclined to do so. Interestingly, CPO appointments do not generate an immediate positive stock market reaction.

The study identifies key future research directions, including:

- Firm value—Under what conditions can a CPO appointment positively impact long-term firm value?
- Financial performance—How can CPOs drive financial and operational benefits?
- Data breaches—What strategies can firms employ to mitigate the negative effects of privacy breaches?
- Top management integration—How do CPOs interact with other senior executives, and what criteria are used for their selection?

Chapter 9: Social Implications of Autonomous Vehicle (AV) Adoption in Supply Chains

This chapter assesses the social sustainability impact of Unmanned Ground Vehicles (UGVs) and Unmanned Aerial Vehicles (UAVs) in healthcare, humanitarian, and commercial supply chains. Through a systematic literature review, the study identifies:

- 13 positive and 7 negative social impacts of AV adoption.

- The relationship between these impacts and six key social sustainability indicators:
  - Health & Safety
  - Customer Satisfaction
  - Employment
  - Transparency
  - Diversity & Equality
  - Welfare

The study emphasizes the need for updated regulations to accommodate broader AV integration in supply chains and recommends that policy-makers review and refine existing laws to align with emerging technological advancements.

Chapter 10: The Impact of Digitalization on Firm-Level Innovation Performance During the Pandemic

Utilizing cross-country survey data from 45 economies across Africa, East Asia Pacific, Europe & Central Asia, Latin America & the Caribbean, and the Middle East & North Africa, this study examines the causal link between government support for digital technology adoption and firm-level innovation performance.

The study addresses:

- *How have firms leveraged digitalization to transform goods, services, and business models during the pandemic?*
- *To what extent have government initiatives influenced digital technology adoption?*

By leveraging World Bank Enterprise Survey data, the chapter provides empirical insights into how digitalization efforts have driven business model innovation, technological adoption, and competitive advantage across various global economies.

Part III: Transformational Models and Processes in Supply Chains

This Part explores the development of transformational models and processes across various stages of the supply chain, driven by cutting-edge technologies such as blockchain, big data analytics, cloud computing, artificial intelligence, drones, autonomous vehicles, and 5G. The chapters document process-level

innovations through case studies and empirical evidence, offering insights into how technology is reshaping supply chain dynamics.

Chapter 11: Deep Learning Applications in Inventory Management

This chapter reviews twelve key deep learning applications that are revolutionizing inventory management, including:

- Automating inventory inspection
- Reducing inventory costs and improving decision-making
- Minimizing the bullwhip effect across multi-tier supply chains
- Mitigating risks associated with ineffective inventory management
- Extracting and optimizing inventory features for better forecasting
- Developing dynamic inventory policies to adapt to changing environments
- Providing real-time inventory information for autonomous vehicles, robots, and drones

The chapter concludes with an insightful future research agenda, emphasizing methodological advancements and strategies to transform inventory management using deep learning techniques.

Chapter 12: Smart Farming, Smart Distribution, and Smart Retailing in Agri-Food Supply Chains

This chapter examines the digital transformation of agri-food supply chains, highlighting the critical role of emerging technologies in enhancing:

- Food quality and safety
- Regulatory compliance and labeling
- Globalization in agri-food trade
- Climate change adaptation and resource management

A key finding is the limited application of digital technologies in the upstream agricultural supply chain. The chapter underscores the need for integrating and ensuring interoperability among different technologies to address climate-related risks and enhance supply chain resilience.

Chapter 13: Securing the Supply Chain with Blockchain and IoT

This chapter explores the complexities of building a securely auditable supply chain using IoT sensors and a decentralized blockchain distribution network. It highlights the challenges and opportunities in creating a transparent, tamper-proof supply chain, with a focus on:

- Cybersecurity risks and trust mechanisms
- Reliability, accuracy, and data integrity in decentralized networks
- Interdisciplinary research to improve security frameworks
- AI-driven governance models to prevent monopolization in highly regulated sectors (e.g., pharmaceuticals and food industries)

The chapter emphasizes the crucial role of AI in enhancing supply chain security and ensuring fair stakeholder participation in regulated industries.

Chapter 14: Industry 5.0 and the Future of Supply Chains

This chapter introduces the concept of Industry 5.0, which extends beyond profit-driven supply chains to emphasize human-centric, sustainable, and resilient supply chain models. Through a comprehensive literature review, the study aims to:

- Define the key constructs of Supply Chain 5.0
- Develop a conceptual framework for the transition to human-centric supply chains
- Analyze the influence of industry and regional dynamics on emerging Supply Chain 5.0 paradigms

By leveraging disruptive technologies, the chapter envisions a future-ready supply chain that balances technological efficiency with human well-being, environmental responsibility, and long-term sustainability.

Part IV: Digital Design Capabilities for Emerging Technologies in Supply Chains

This Part delves into the critical role of digital design capabilities in enabling the successful adoption of blockchain, big data analytics, cloud computing, artificial intelligence, drones, autonomous vehicles, 5G, and other transformative technologies in supply chains. The chapters explore key aspects such as knowledge reuse, dynamic capabilities in new product development, emerging design practices, user experience, incremental customization, and digitalization strategies to enhance the performance of community designers.

Chapter 15: Blockchain Capabilities for Food Supply Chain Traceability

This chapter investigates the application of blockchain technology in ensuring end-to-end traceability within food supply chains. Through a systematic literature review, the study addresses two fundamental questions:

1. What are the key traceability requirements in food supply chains?
2. How can blockchain capabilities facilitate farm-to-fork visibility, smart contracts, and integration?

The chapter identifies nine key themes linked to the triple bottom line performance, with food logistics, provenance, and quality emerging as top priorities. It concludes with promising research directions, including food system audits, due diligence, carbon accounting, and decarbonization strategies.

Chapter 16: Trust Design Framework in Blockchain-Enabled Supply Chains

This chapter explores the role of trust in supply chain operations, specifically investigating:

- What are the trust-related technical design and business affordances of blockchain technology in supply chains?

The study proposes a trust design framework that optimizes blockchain architectures and ecosystems, ensuring seamless interaction among multiple heterogeneous stakeholders across end-to-end supply networks. The chapter highlights the importance of behavioral dynamics among stakeholders to maximize blockchain's potential in enhancing transparency and security in supply chain transactions.

Chapter 17: Blockchain & IoT Integration for Vaccine Supply Chain Trust & Traceability

Building on the previous chapter, this study examines the combined role of blockchain and IoT in ensuring trust and traceability in vaccine distribution. The chapter presents a proof of concept, utilizing a blockchain technology platform enabled by smart contract language system (DAML) to monitor and maintain temperature conditions across the vaccine supply chain. Key contributions include:

- Ensuring regulatory compliance and secure bookkeeping
- Enhancing stakeholder collaboration through well-defined smart contracts
- Future research opportunities for technology-driven authenticity and trust-building in healthcare supply chains

Chapter 18: Trust Dimensions in Blockchain and Process Industries

This chapter empirically validates three dimensions of trust—contractual, performance, and goodwill—within process industries, using flour mills as a contextual case study. The findings highlight:

- Blockchain effectively supports trust through contractual and performance dimensions
- Blockchain has limited impact on goodwill trust
- Data design capabilities are essential for achieving operational efficiency, cost reduction, visibility, and enhanced collaboration

Additionally, the study links blockchain adoption to several United Nations Sustainable Development Goals (SDGs), including:

- SDG 3 (Good Health & Well-being)
- SDG 9 (Industry, Innovation & Infrastructure)
- SDG 12 (Sustainable Consumption & Production)

Chapter 19: Social Media Engagement in Container Logistics

This chapter examines how logistics firms leveraged social media to enhance container supply chain engagement during the pandemic. Using a strategic operations and affordance theoretical perspective, the study analyzes communication strategies employed by Fortune 500 and Alphaliner 100 companies. The shared information is categorized into three engagement themes:

- Combat—Crisis response and operational resilience
- Resolute—Strategic adaptation and business continuity
- Acknowledgements—Stakeholder engagement and corporate responsibility

The findings provide insights into how digital platforms can enhance B2B engagement, supply chain transparency, and crisis communication strategies in global logistics networks.

Part V: Digital Manufacturing Capabilities for Emerging Technologies in Supply Chains

This Part explores the awareness, development, and implementation of digital manufacturing capabilities to leverage blockchain, big data analytics, cloud computing, artificial intelligence, drones, autonomous vehicles, 5G, and

other cutting-edge technologies. Each chapter focuses on a critical aspect of digital manufacturing, including:

- Product diversity in digital manufacturing
- Impact of digitalization on inventory management and lot sizing
- The role of digitalization in build-to-model manufacturing
- Risks associated with digital design and manufacturing transfer
- The influence of new players in the redistribution and restructuring of regional manufacturing

Chapter 20: Digitalization in Bioproduction—AI & Digital Twins for Process Optimization

This chapter examines the digitalization efforts in bioproduction within the food industry, focusing on compliance with food safety regulations. Specifically, the study investigates the role of digital twins and artificial intelligence (AI) in virtually conducting experiments to optimize temperature control in production processes.

Using real-time company data, the study simulates production environments with digital twins and applies artificial neural networks (ANNs) to predict key process parameters such as pH levels, temperature, and carbon sources. The findings position digital twins as a powerful planning tool for future process industries, enhancing efficiency, compliance, and sustainability.

Chapter 21: Predictive Maintenance—Digital Twins in CNC Machine Failure Detection

Building on the previous chapter, this study delves into the application of digital twins for fault detection in computer numerical control (CNC) machines, with a specific focus on stepper motors. The chapter emphasizes predictive maintenance planning through the integration of machine learning algorithms and real-time data analytics.

Employing techniques such as:

- Artificial Neural Networks (ANNs)
- Support Vector Machines (SVMs)
- Statistical Quality Control (SQC)

The study characterizes fault patterns and recommends implementing risk control frameworks for future predictive maintenance strategies. The chapter

also presents a detailed case study demonstrating the effectiveness of digital twin models in industrial maintenance optimization.

Chapter 22: Implementing Smart Production Through Lean Digitalization

This chapter investigates how companies can transition to smart manufacturing by acquiring real-time data through lean methodologies, as outlined by Womack and Jones (1996). The approach aligns company goals with an assessment logic framework, enabling firms to develop optimal data acquisition strategies for enhanced manufacturing intelligence.

Key insights include:

- Mapping stakeholder requirements to data transformation strategies
- Digitalization-driven process optimization
- A proposed application of the approach to SMEs, ensuring broader industrial adoption

This chapter underscores real-time data acquisition as a cornerstone of next-generation manufacturing, driving efficiency, agility, and competitiveness.

Chapter 23: Additive Manufacturing & Industry 4.0—A Path to Digital Transformation

This chapter delves into the role of additive manufacturing (AM) and Industry 4.0 technologies in enhancing productivity and efficiency in contemporary manufacturing. Through an extensive literature review, the study identifies the contributions of:

- Big data analytics
- Robotics & AI
- Cybersecurity & IoT
- Simulation techniques & blockchain
- Cloud computing & system integration

The findings highlight key challenges, including high costs, security vulnerabilities, and skill shortages, while also presenting exciting research questions for future exploration. The study emphasizes the need for interdisciplinary collaboration to fully harness the transformative potential of Industry 4.0-driven additive manufacturing.

This collection of studies underscores the pivotal role of digital manufacturing capabilities in revolutionizing modern supply chains. From

digital twins optimizing bioproduction to predictive maintenance in CNC machining and Industry 4.0-driven additive manufacturing, these chapters offer rich insights into the future of intelligent, resilient, and data-driven production ecosystems.

Part VI: Digital Delivery and Use Capabilities for Emerging Technologies in Supply Chains

This Part delves into the logistical challenges faced by modern supply chains and examines how emerging technologies are shaping digital delivery and use capabilities. Each chapter explores the application of digital technologies across key industries such as food, retail, and automotive, covering critical aspects such as:

- Autonomy and value creation in logistics services
- Business models for the sharing and on-demand economy (e.g., Mobility-as-a-Service, on-demand freight services)
- The role of the Physical Internet in supply chains
- Crowd logistics and synchromodality servitization
- Business process management and system integration (horizontal and vertical)
- Sustainability in logistics (e.g., zero-emission logistics, circular economy, smart cities)

Chapter 24: The Role of UAV Automation in Inbound Logistics & Warehousing

This chapter explores the transformative role of unmanned aerial vehicles (UAVs) in automating inbound logistics operations, particularly in warehouse management. The study examines UAV applications across ten key warehouse activities, including:

- Inventory accuracy assurance
- Buffer stock maintenance & cycle counting
- Data collection, processing & inventory turnover
- Item search & real-time surveillance
- Space optimization & stock management

Beyond operational efficiency, the chapter critically evaluates the social and ethical implications of UAV deployment, highlighting concerns such

as privacy risks, job displacement, and psychological effects on warehouse workers.

Chapter 25: The Physical Internet & Sustainable Urban Logistics

With the exponential growth of e-commerce, urban logistics face increasing pressure from traffic congestion and sustainability concerns. This chapter provides an in-depth review of the Physical Internet and its role in optimizing urban logistics and last-mile delivery operations.

Key research questions addressed include:

- How can the Physical Internet improve efficiency and effectiveness in urban logistics?
- What are the barriers to implementing the Physical Internet in last-mile delivery?
- How do the Physical Internet and Digital Internet influence last-mile logistics?

The chapter highlights potential solutions, challenges, and future research directions for integrating Physical Internet frameworks into sustainable and intelligent urban logistics ecosystems.

Chapter 26: Blockchain for Hydrogen Supply Chain Optimization

This chapter explores the capabilities of blockchain technology in optimizing the hydrogen supply chain, providing insights into secured data access and automation of key operations. The study underscores the importance of:

- Stakeholder collaboration for a seamless hydrogen supply chain
- Energy-efficient blockchain adoption
- Standardized regulatory frameworks for blockchain implementation

The chapter introduces tokenized hydrogen assets as a means to ensure transparency and traceability and suggests the amalgamation of artificial intelligence within blockchain ecosystems. Additionally, the authors highlight the critical need for training and education to address skill shortages in the hydrogen sector.

For future research, the study calls for exploring blockchain integration with emerging technologies to further enhance supply chain sustainability, transparency, and efficiency.

Chapter 27: Late Last-Mile Delivery in a Physical Internet Enabled Context

Addressing challenges such as traffic congestion, limited parking availability, and inefficiencies in delivery handovers, this chapter introduces a Parcel Reshuffling Algorithm designed to enhance last-mile delivery efficiency. The algorithm dynamically updates schedules using real-time data, redistributing parcel loads among delivery vehicles.

The chapter presents a case study from Madrid, Spain, detailing the algorithm's four-step approach:

1. Identifying delivery vehicles in transit
2. Reshuffling parcels to vehicles experiencing delays
3. Identifying an optimal exchange point between vehicles
4. Rerouting both vehicles to complete deliveries efficiently

The study demonstrates significant benefits, including:

- Reducing second delivery attempts
- Lowering additional delivery costs by 20%
- Decreasing environmental impact through reduced emissions
- Enhancing ecological and social sustainability in urban logistics

Chapter 28: Disruptive Technologies & Strategies in Last-Mile Delivery

This chapter offers a comprehensive review of emerging disruptive technologies and strategies reshaping last-mile delivery operations. The study addresses two fundamental research questions:

1. What disruptive technologies and innovative solutions exist for last-mile delivery?
2. What are the key adoption challenges associated with these technologies?

The chapter identifies a range of transformative technologies, including:

- Artificial Intelligence (AI) & Big Data

- Physical Internet & Blockchain
- Digital Twins & IoT
- 3D Printing, AR/VR, and Autonomous Vehicles
- Microhubs & Mobile Parcel Solutions

However, the chapter also outlines barriers to adoption, such as:

- Legal and regulatory constraints
- Infrastructure limitations
- Data security and management challenges
- Resistance from top management

The findings emphasize the need for cross-sector collaboration, technological standardization, and supportive policy frameworks to fully unlock the potential of disruptive innovations in last-mile logistics.

This Part underscores the critical role of digital delivery and use capabilities in modern supply chains, highlighting UAVs, blockchain, the Physical Internet, and AI-driven algorithms as game-changers in logistics and last-mile delivery. From enhanced warehouse automation to revolutionizing urban logistics through parcel reshuffling and decentralized freight management, these chapters provide valuable insights into the future of sustainable, efficient, and technology-driven supply chains.

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Part I

Supply Chain Digitalization—Key Barriers, Enablers, and Capabilities

2

3D Printing in Supply Chains—Benefits, Challenges, and Future Directions

Kamrul Ahsan and Shams Rahman

1 Introduction

Three-dimensional (3D) printing is a digital technology based on additive manufacturing in which products are manufactured layer by layer by adding materials using a digital 3D computer-aided design (CAD) file (Waller & Fawcett, 2014).

Unlike traditional manufacturing, 3D printing is a tool-less manufacturing system (Attaran, 2017b). It uses the digital design of products from open or closed source (Laplume et al., 2016), special raw materials, and dedicated printers for manufacturing the product at a convenient location and suitable time.

The global market for 3D-printing-based businesses is estimated to hit US$23.79 billion by 2025 (GlobalMarketInsights, 2018). It has been suggested that the adoption of 3D printing manufacturing technology may have the potential to change the way we manage traditional manufacturing supply chains (SCs) in terms of sourcing, manufacturing, warehousing, transportation and packaging, and other functional areas of SCs (Beltagui et al., 2023; Despeisse et al., 2017; Durach et al., 2017; Kunovjanek et al., 2020).

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Although a substantial volume of research has been conducted on the technological aspects of 3D printing (Bak, 2003; d'Aveni, 2015; Mellor et al., 2014; Rengier et al., 2010; Schniederjans, 2017; Vinodh et al., 2009; Wu et al., 2015; Zeltmann et al., 2016), research on 3D printing's impact on the benefits and challenges of SCs is limited. In recent times, there has been increased research interest in 3D printing in SCs (Davies et al., 2022; Durach et al., 2017; Kunovjanek et al., 2020; Meyer et al., 2020; Ryan et al., 2017; Savolainen & Collan, 2020; Yeh & Chen, 2018;). A thorough understanding of the consequences of implementing 3D printing will assist firms to develop appropriate strategies to enhance their competitiveness in the future business environment. Hence, the objectives of this chapter are to

- investigate the trends of 3D printing research in manufacturing SCs,
- assess the benefits and challenges of implementing 3D printing in manufacturing SCs, and
- suggest future research directions.

The remainder of the chapter is organised as follows. In Sect. 2, an overview of 3D printing is provided. Section 3 explained the systematic literature review of academic journal articles pertaining to 3D printing, additive manufacturing, and supply chain management. We identified and discussed 3D printing supply chain benefits and challenges in Sect. 4 and Sect. 5 respectively. Section 6 concluded the chapter with a summary and future research directions.

2 Overview of 3D Printing

3D printing or additive manufacturing consists of processes that include planning and acquisition of designs or images, post-processing of images by CAD, sourcing raw materials, manufacturing of products and, finally, delivering the product to the customer (Berman, 2012).

3D printing technology has the flexibility to design complex and tailored products and parts (Strong et al., 2018). For 3D printers, the CAD file may be generated from a design process or through a 3D scan of an existing object or body (Zijm et al., 2019). Generally, in the manufacturing process, 3D printing uses raw materials such as plastics, alloys and resins (Waller & Fawcett, 2014). However, with the advancement of technology, an increasing variety of raw materials has become available, such as metallic superalloys,

ceramic powder, metal or even glass, and polymers (Strong et al., 2018; Zijm et al., 2019). These raw materials along with digital files are the inputs to the printer for manufacturing products.

The manufacturing process is the core of 3D printing SCs. At this stage, 3D printing machines read data from computer drawings and layer the raw materials in a series of cross-sections to form the desired product (Zeltmann et al., 2016). Most consumer 3D printers use fused filament fabrication, in which the printer head lays down consecutive layers of materials at high temperatures (Halassi et al., 2019). At the process level, 3D printing technology is used as a standalone configuration or a combination of configurations (Braziotis et al., 2019). Standalone configurations use functional-oriented 3D printing production technology operating alone; in such cases, a business, or an individual uses 3D printing as the core manufacturing process to produce parts or products on demand (Tziantopoulos et al., 2019). An overview of a typical 3D printing SC processes is shown in Fig. 1. In combinational configurations, 3D printing works side by side with other traditional tool-based production technologies to achieve objectives that are technically or economically more viable or to maximise the customer order benefits (Braziotis et al., 2019).

Manufacturing processes rely on the intersection of two SCs: the SC of a machine and materials vendor, and the SC of 3D printer manufacturers, material providers, model designers, and consumers intending to purchase 3D print products (Mellor et al., 2014; Rong et al., 2018). The deployment of 3D printing may take place under both centralised and distributed SC configurations (Braziotis et al., 2019). Under a centralised configuration, 3D printing facilities are located in a centralised location to serve aggregate

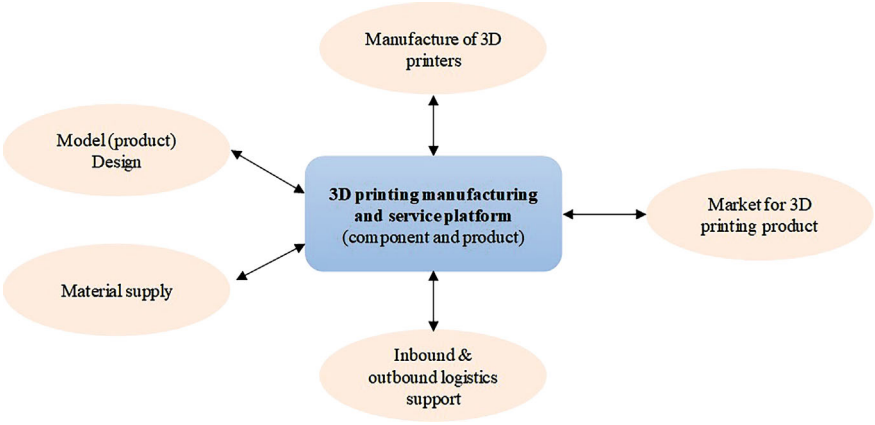

Fig. 1 3D printing manufacturing supply chain

demand from multiple demand locations (Khajavi et al., 2014). In this case, customers' design orders are pooled via e-commerce websites, and then a 3D printing facility is used to produce the product; finally, orders are packaged and shipped to the customer through logistics services (Ryan et al., 2017). In a distributed SC configuration, 3D printing facilities are deployed close to service locations or onsite (Braziotis et al., 2019).

Adoption of 3D printing is increasing with the expansion of knowledge about this technology and its capabilities (Schniederjans, 2017). Business and manufacturing organisations are now making more 3D printed products for consumers. Consumers are also becoming more aware of 3D printing uses as well as its potential applications for their everyday lives. While these benefits and uses have prompted, 3D printing in manufacturing SCs has yet to reach high levels of adoption. The biggest areas of concern that should be investigated are how to increase industrial applications and how 3D printing will influence SCs of the future (Mohr & Khan, 2015). To assess the worthiness of 3D printing adoption in manufacturing SCs, along with the benefits, the challenges of implementing 3D printing in manufacturing SCs must be considered (Beltagui et al., 2023; Schniederjans, 2017). Hence, in this research we have investigated the benefits and challenges of 3D printing in SCs.

3 Identification of 3D Printing SC Benefits and Challenges

A systematic review of 3D printing research in relation to SC management is conducted. There are several reasons for using such a research tool.

- First, a literature review is a systematic, explicit and reproducible research tool for identifying, evaluating, interpreting or synthesising and refining the existing body of scattered knowledge in recorded documents (Fink, 1998).
- Second, a literature review can summarise patterns, themes and issues of existing research (Seuring & Muller, 2008).
- Third, it helps to identify the knowledge gaps to be filled in order to develop the existing body of knowledge (Tranfield et al., 2003).
- Last, through refining and summarising the existing body of scattered knowledge, a literature review can contribute to theory development (Meredith, 1993).

Generally, a systematic literature review is conducted through content analysis of articles. Content analysis is a class of methods within empirical social science (Seuring & Gold, 2012), in which quantitative and qualitative aspects are mixed to assess descriptive as well as content criteria. It is conducted based on the development of a coding framework and on the categorisation of the information according to that framework (Schilling, 2006). Thereafter, analysis is performed to convert some raw data into a meaningful format (Denyer & Tranfield, 2009; Dixon-Woods et al., 2005).

We followed the five-step process recommended by Denyer and Tranfield (2009) for a systematic literature review. These steps entail:

(i) research question definition,
(ii) literature review protocol,
(iii) selection and evaluation of studies,
(iv) data extraction (synthesis) and
(v) analysis.

In the following subsections, we discuss the steps.

3.1 Literature Review Protocol

In conducting the literature review, we considered peer-reviewed journal articles published in English up until 2020 and listed in major database Scopus. The unit of analysis was a single article related to 3D printing or additive manufacturing (AM) related to SCs and logistics.

3.2 Selection and Evaluation of Studies

A 'structured keyword search' was carried out with the key words (mentioned in the Fig. 2) being present in the title, and/or keywords, and/or abstract. Following Glock (2017), we used specific inclusion and exclusion criteria to refine our search and to obtain articles that are consistent with the research theme or within the scope of this research. The inclusion criteria are of the search is explicitly related to 3D printing and SC activities, such as supply/supplier/sourcing, manufacturing, operations, warehousing, and distribution. Hence, as a preliminary inclusion criterion, we restricted our search to articles that include "3D printing" or "additive manufacturing (AM)" and any of the key words "supply chain", or "logistics", or "sustainability", or 'business model' within the abstract, or title of the article or keyword.

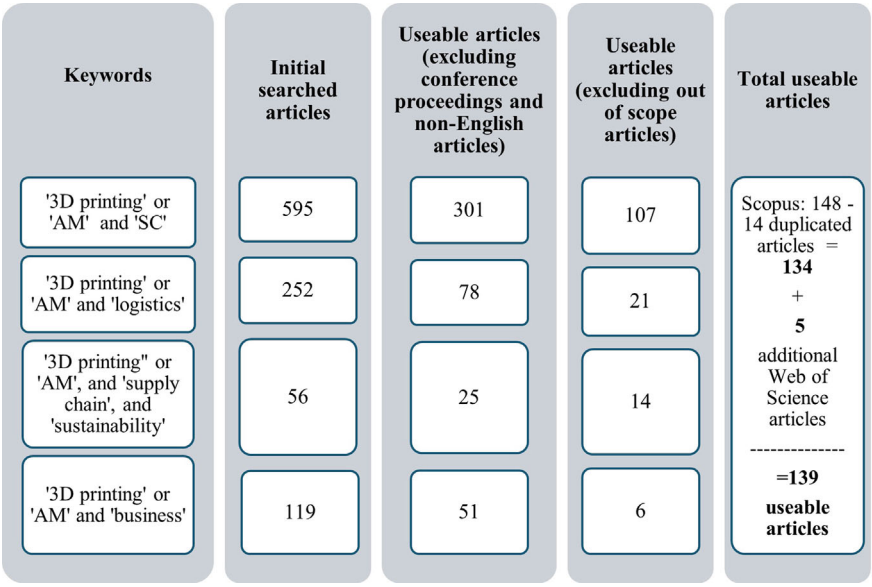

Fig. 2 Summary of article selection process (*Source* Authors)

As exclusion criteria, we considered the article is not in English, the article is not an academic article (i.e., theses, conference articles, magazine articles and forewords were not included) or the article is only loosely related. A loosely related article is one that examines 3D printing or additive manufacturing but does not focus on SC and logistics, for example, a 3D printing article that focuses on purely on materials science, medical science/application, or one in which 3D printing is mentioned only in the reference list or only in the literature review but is not the focus of the article.

From Scopus search, we obtained total 148 articles. Then after removing the 14 duplicated articles, we obtained 134 articles which was saved in Scopus. Lastly, to minimise missing work relevant to the review, Scopus articles were cross-examined through Web of Science (WoS). Five additional articles were obtained following the WoS search, resulting in 139 useable articles being used for our systematic review. For each of the 139 selected articles, the bibliographic information, abstract, keywords and references were converted to comma-separated values file format for use in Excel. The procedure was designed with the consensus of the authors, and all information is auditable through review and track.

3.3 Data Extraction

Identified articles were further reviewed through content analysis of various quantitative and qualitative categorisations. We then extracted relevant content data for coding, extraction, synthesis and analysis following the steps outlined by Denyer and Tranfield (2009) such as:

(i) development of content coding framework for analysis,
(ii) extraction of information from 139 selected articles according to the classification scheme, and
(iii) synthesising/clustering/classification and interpretation of key results.

4 Results and Analysis

4.1 What is the Trend of 3D Printing Research in Supply Chain?

3D printing research in the context of SCs grew exponentially over the same period (see Fig. 3). Between 2009 and 2020, a total of 139 articles were identified that addressed issues of research on 3D printing in SCs. A descriptive analysis of the published articles revealed that since 2007, on average, approximately 14 articles were published per year.

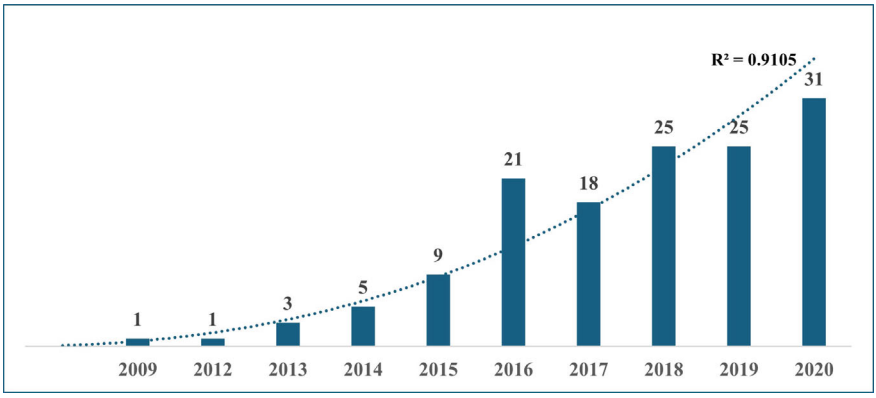

Fig. 3 Publication of 3D printing research with a supply chain focus over time (*Source* Authors. *Note* The number against each theme means the proportion of frequency of a theme to the total number frequency of the theme, and the number within parentheses means the total number of articles covering a theme)

Only 20 articles (14%) were published between 2007 and 2015 (about three articles were published per year). During this period (2007–2012), published articles focused on topics such as agility (Vinodh et al., 2009), highlighted the benefits of mass customisation and predicted the future of 3D printing technology (Berman, 2012). The focus of the research was mainly on 3D printing conceptual research; examples fell into categories such as emphasising the impact on SCs (Khajavi et al., 2014; Liu et al., 2013), and the societal (Huang et al., 2013) and economic (Petrick & Simpson, 2013; Waller & Fawcett, 2014) impacts of 3D printing.

Research on 3D printing implications for SCs started to increase from 2016. During 2016–2020, 124 articles (86%) were published, which means an average of 25 articles were published per year. The focus of current research on 3D printing in SCs is still on how 3D printing technology has emerged (Halassi et al., 2019; Ryan et al., 2017) and adaptation challenges, issues or determinants (Kapetaniou et al., 2018; Oettmeier & Hofmann, 2017; Schniederjans, 2017; Thomas, 2016; Wu & Chen, 2018). However, the data suggest that the research is shifting towards other contemporary issues such as the impact on SCs (Chan et al., 2018; Jia et al., 2016; Li et al., 2017a), global trade and value chains (Laplume et al., 2016; Long et al., 2017), reshoring or near-shoring manufacturing (Ben-Ner & Siemsen, 2017; Moradlou & Tate, 2018; Rayna & Striukova, 2016), sustainable SCs (Despeisse et al., 2017; Peng et al., 2018), and changing to new business models (Jia et al., 2016; Rong et al., 2018).

4.2 Analysis of Benefits and Challenges of 3D Printing in Supply Chains

4.2.1 Relative Frequency of Benefits and Challenges

After synthesising the qualitative data and identifying themes for the benefits and challenges, we identified the frequency of articles that addressed a particular theme. A summary of themes identified as benefits is shown in Fig. 4. The identified themes, explanation of these themes and related selective references are given in Table 1. Themes related to 'benefit' were identified 284 times, out of which the 'flexibility' theme was identified 40 times. Hence, the proportion of times 'flexibility' was identified with respect to total frequency was 40/284, which is 0.14. The mean of all weights related to 'benefit' was 0.0714, where the standard deviation was 0.037.

Following the same procedure, we identified the frequency of articles covering the challenges and the proportion of coverage frequency of a theme

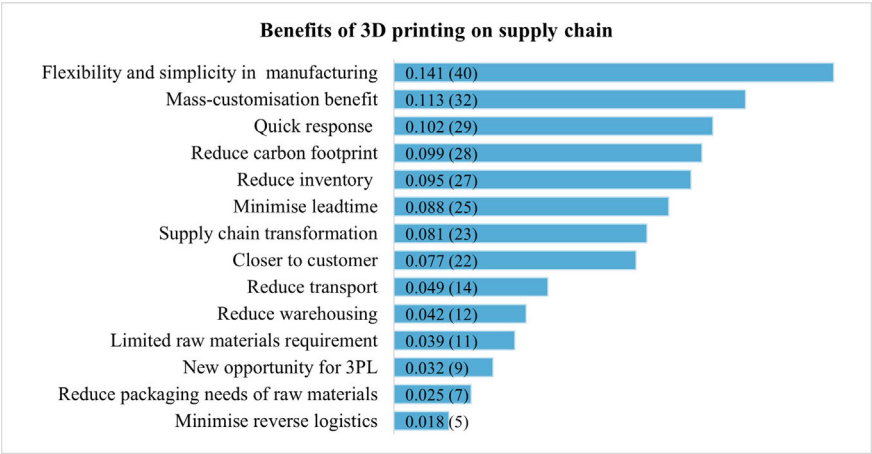

Fig. 4 Relative frequency of the benefit themes of 3DP on supply chain (*Source* Authors)

to the total number frequency of the theme. A summary of themes identified as challenges is shown in Fig. 5. The identified challenge themes, explanation of these themes and related selective references are given in Table 2. Themes related to 'challenges' were identified 183 times, out of which the 'required new skills' theme was identified 27 times. Hence, the proportion of times 'high cost of printers and raw materials' was identified with respect to total frequency was 27/183, which is 0.148. The mean of all the weights related to 'challenges' was 0.0833, where the standard deviation was 0.0369.

4.2.2 Matrix Analysis of 3D Printing Benefits

Through literature review on supply chain performance measures we grouped the benefits of 3D printing in accordance with triple bottom line measures such as economic, social, and environmental (Acquaye et al., 2017; Qorri et al., 2018).

Following Ahsan and Rahman (2017) and Rahman et al., (2019) we classified the benefits in the categories of 'most frequently mentioned', frequently mentioned' and 'least frequently mentioned'. The weights of the benefit themes that fell beyond 'mean + 1 standard deviation' were considered to belong in the most frequently mentioned cluster. The weights of the themes that fell below the 'mean' were considered to belong in the least frequently mentioned cluster. The weights of the themes that fell between the ranges 'greater than mean but less than equal to mean + 1 standard deviation' were considered to be part of the frequently mentioned cluster.

Table 1 Identified benefit themes, explanation of themes and selective references

No	Cluster of benefits themes	Explanation	Selective references
1	Flexibility and simplicity in manufacturing	Flexibility in 3D printing manufacturing means flexibility in design, time to production, the location of the manufacturing facility. Regarding simplicity, 3DP manufacturing requires few manufacturing steps with no tooling or moulding	Cooper (2016); Rogers et al. (2016); Ben-Ner and Siemsen (2017); Oettmeier and Hofmann (2017); Ryan et al. (2017); Srinivasan et al. (2018); Delic et al. (2019); Heinen and Hoberg (2019); Li et al. (2019); Cox and Koepsell (2020); Delic and Eyers (2020); Franco et al. (2020); Holzmann et al. (2020); Meyer et al. (2020)
2	Quick response	Quick response to the customer means 3DP can respond to customer demand very quickly. As 3DP manufacturing requires no tooling and no sub-assembly of parts, it is possible to change the design quickly	Liu et al. (2013); d'Aveni (2015); Laplume et al. (2016); Durach et al. (2017); Ghadge et al. (2018); Rong et al. (2018); Chandima Ratnayake (2019); Khorram Niaki et al. (2019); Zanoni et al. (2019); Cox and Koepsell (2020); Meyer et al. (2020); Yılmaz (2020)

(continued)

Table 1 (continued)

No	Cluster of benefits themes	Explanation	Selective references
3	Mass customisation benefit	For a single or small quantity of customer order products, benefits of mass customisation such as less cost of product and variety can be achieved through 3D printing	Mellor et al. (2014); d'Aveni (2015); Kietzmann et al. (2015); Weller et al. (2015); Jia et al. (2016); Rayna and Striukova (2016); Durach et al. (2017); Dwivedi et al. (2017); Shukla et al. (2018); Yeh and Chen (2018); Baumers and Holweg (2019); Martinelli and Christopher (2019); Tziantopoulos et al. (2019); Arbabian and Wagner (2020); Boer et al. (2020); Chaudhuri et al. (2020); Franco et al. (2020)
4	Closer to customer	3D printing facilities can be processed at the customer's home or in nearby facilities, which brings manufacturing close to home or near home	Berman (2012); Petrick and Simpson (2013); Weller et al. (2015); Ford and Despeisse (2016); Zeltmann et al. (2016); Durach et al. (2017); Durão et al. (2017); Khorram and Nonino (2017); Garmulewicz et al. (2018); Peng et al. (2018); Halassi et al. (2019); Arora et al. (2020); Florén et al. (2020)

(continued)

Table 1 (continued)

No	Cluster of benefits themes	Explanation	Selective references
5	Minimise lead time	3D printing requires no tooling; hence, no set-up time is required. If desired raw materials are available, with the digital design a small lot size of products can be made within a short period	Huang et al. (2013); Mellor et al. (2014); d'Aveni (2015); Kietzmann et al. (2015); Tatham et al. (2015); Sasson and Johnson (2016); Durach et al. (2017); Li et al. (2017a); Busachi et al. (2018); Muir and Haddud (2018); Wu and Chen (2018); Zhou et al. (2018); Arora et al. (2020); Berman (2020); Rodríguez-Espíndola et al. (2020)
6	Reduce inventory	Goods are produced only when a customer needs them; hence, less work-in-process inventory or finished goods inventory is needed	Berman (2012); Huang et al. (2013); Waller and Fawcett (2014); Achillas et al. (2015); d'Aveni (2015); Bogers et al. (2016); Emelogu et al. (2016); Holmström and Gutowski (2017a); Kellens et al. (2017); Muir and Haddud (2018); Chandima Ratnayake (2019); Boer et al. (2020); Haleem et al. (2020)
7	Reduce warehousing	Less or no finished goods stock will reduce the need for warehousing and the number of warehouses in the SC	Waller and Fawcett (2014); Oettmeier and Hofmann (2016); Ghadge et al. (2018); Chandima Ratnayake (2019); Heinen and Hoberg (2019); Baiano (2020); Franco et al. (2020)

(continued)

Table 1 (continued)

No	Cluster of benefits themes	Explanation	Selective references
8	Reduce transport effort	3D printing uses digital product design, requires a limited amount of raw materials from suppliers, and the final product can be shifted to the customer with minimum or no transportation effort	Gebler et al. (2014); Doicin and Ulmeanu (2015); Wagner and Walton (2016); Ben-Ner and Siemsen (2017); Chen (2017); Holmström and Gutowski (2017a); Heinen and Hoberg (2019); Akbari and Ha (2020); Chowdhury et al. (2020); Hecker (2020); Martens et al. (2020)
9	Reduce packaging needs for raw materials	As 3D printing upstream of the SC requires only the raw materials as resin, no other semi-finished raw materials are necessary. Hence, 3D printed products have less packaging needs for raw materials	Waller and Fawcett (2014); Mckinnon (2016); Peng et al. (2018); Chandima Ratnayake (2019); Baiano (2020); Wang et al. (2020)
10	Minimise reverse logistics	Minimised reverse logistics means fewer product returns, minimum flows of unwanted, unsuitable, or damaged products	Ford and Despeisse (2016); Despeisse et al. (2017); Khorram and Nonino (2017); Boon and Van Wee (2018); Strong et al. (2020)

(continued)

Table 1 (continued)

No	Cluster of benefits themes	Explanation	Selective references
11	Reduce carbon footprint	3D printing creates less waste in manufacturing, requires less inventory (no work-in-process or finished goods inventory), minimum transportation and warehousing, thus reducing the carbon footprint	Petrick and Simpson (2013); Achillas et al. (2015); Kietzmann et al. (2015); Jia et al. (2016); Tang et al. (2016); Cerdas et al. (2017); Li et al. (2017a); Sun and Zhao (2017); Ghadge et al. (2018); Unruh (2018); Turner et al. (2019); Baiano (2020); Florén et al. (2020); Isasi-Sanchez et al. (2020); Pilz et al. (2020)
12	Supply chain transformation	New models of supply chains will emerge because manufacturing will be close to customers, which will require digital product designs, few raw materials (resin) and no sub-assembly or assembly of final products, and products can be made when they are required	Petrick and Simpson (2013); Eyers and Potter (2015); Kietzmann et al. (2015); Scott and Harrison (2015); Chen (2016); Knofius et al. (2016); Mckinnon (2016); Sasson and Johnson (2016); Ben-Ner and Siemsen (2017); Holmström and Gutowski (2017); Do Chung et al. (2018); Rejeski et al. (2018); Ashour Pour et al. (2019); Holzmann et al. (2019); Boer et al. (2020)

(continued)

Table 1 (continued)

No	Cluster of benefits themes	Explanation	Selective references
13	New opportunity for 3PL	A new opportunity for 3PL services will emerge to facilitate 3D printing using digital product design files, small local deliveries, bringing the raw materials from local or international sources or supporting them by providing 3D printers	Jin and Ji (2013); Khajavi et al. (2014); Waller and Fawcett (2014); Kietzmann et al. (2015); Mckinnon (2016); Attaran (2017b); Durach et al. (2017); Chan et al. (2018); Kapetaniou et al. (2018)
14	Limited raw materials requirement	3DP manufacturing requires less types of raw materials such as resin	Waller and Fawcett (2014); Achillas et al. (2015); Ford and Despeisse (2016); Ben-Ner and Siemsen (2017); Khorram and Nonino (2017); Knofius et al. (2019); Isasi-Sanchez et al. (2020); Kunovjanek and Reiner (2020)

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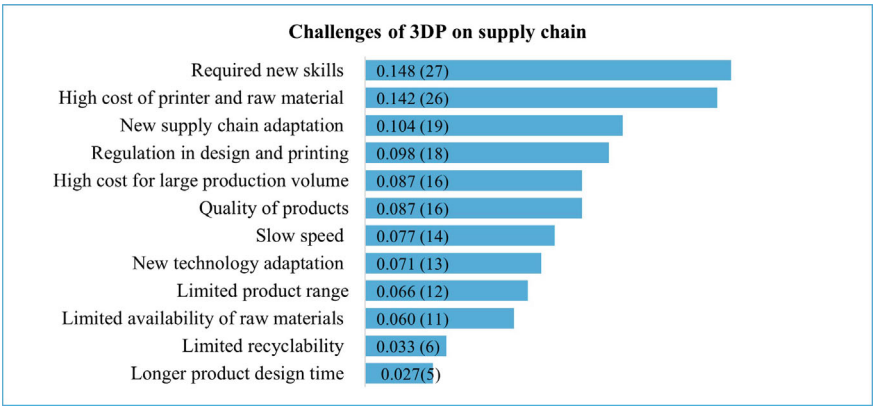

Fig. 5 Relative frequency of the challenge themes of 3DP on supply chain (*Source* Authors. *Note* The number against each theme means the proportion of coverage of frequency of a theme to the total number frequency of the theme, and the number within the parentheses means the total number of articles covering a theme)

Table 2 Identified challenge themes, explanation of themes and selective references

No	Themes of challenge	Explanation	References
1	Required new skills	3D printing develops a knowledge economy, and because of the adaptation of new technology in manufacturing, new skills are required for product design, materials development and sourcing, and to work in new SC orientation	Mellor et al. (2014); Axtman and Wilck (2015); Oettmeier and Hofmann (2016); Dwivedi et al. (2017); Martinsuo and Luomaranta (2018); Afshari et al. (2019); Emelogu et al. (2019); Friesike et al. (2019); Öberg and Shams (2019); Afshari et al. (2020); Boer et al. (2020)
2	Limited product range	Because of the limited types of raw materials available and the additive nature of manufacturing, 3D printing is not ready to produce a wide range of products and large size products	(Waller and Fawcett (2014); Jia et al. (2016); Mckinnon (2016); Attaran (2017b); Chekurov et al. (2018); Hannibal and Knight (2018); Martinsuo and Luomaranta (2018); Rong et al. (2018); Shukla et al. (2018)
3	Quality of products	3D manufacturing produces products layer by layer. Hence, it involves issues of surface finish, use of different types of colours and product strength	Berman (2012); Petrick and Simpson (2013); Weller et al. (2015); Chiu and Lin (2016); Long et al. (2017); Boon and Van Wee (2018); Wu and Chen (2018); Chowdhury et al. (2019); Kleer and Piller (2019); Corsini et al. (2020)

(continued)

Table 2 (continued)

No	Themes of challenge	Explanation	References
4	Limited use of raw materials	3D printing technology is still limited to using certain raw materials, such as resin and plastics	Mellor et al. (2014); Rylands et al. (2016); Despeisse et al. (2017); Long et al. (2017); Boon and Van Wee (2018); Martinsuo and Luomaranta (2018); Strong et al. (2018); Tziantopoulos et al. (2019); Savolainen and Collan (2020)
5	High cost for large production volume	3D printing takes time to produce a large volume of products; hence, for mass production, 3D printing facilities are still not cost effective	Baumers et al. (2013); Achillas et al. (2015); Baumers et al. (2016); Oettmeier and Hofmann (2016); Sasson and Johnson (2016); Khorram and Nonino (2017); Chan et al. (2018); Yeh and Chen (2018); Boer et al. (2020)
6	New technology adaptation	Use of 3D printing is spreading; however, there are issues in adopting the technology. Challenges of adopting 3D printing manufacturing are technological, organisational, environmental, and financial	Vinodh et al. (2009); Petrick and Simpson (2013); Mellor et al. (2014); Ben-Ner and Siemsen (2017); Holzmann et al. (2017); Oettmeier and Hofmann (2017); Garmulewicz et al. (2018); Yeh and Chen (2018); Luomaranta and Martinsuo (2019); Kunovjanek et al. (2020); Martens et al. (2020)

(continued)

Table 2 (continued)

No	Themes of challenge	Explanation	References
7	Longer product design time compared with manufacturing time	Product design in 3D printing is cloud based and separated from product manufacturing. Consumers have a wide range of design freedom. They can be involved in a custom design process, or be able to purchase designs from an online market and then build products at home. This freedom introduces challenges of finding the right designer and design, which involves more time compared with manufacturing time	Berman (2012); Baumers et al. (2016); Dwivedi et al. (2017); Beltrametti and Gasparre (2018)
8	Slow speed compared with mass production	3D printing is suitable for a small quantity of products. Because of the slow speed of manufacturing, when it needs mass production, 3D printing takes more time than mass production	Mellor et al. (2014); Weller et al. (2015); Ford and Despeisse (2016); Thomas (2016); Shukla et al. (2018); Tziantopoulos et al. (2019); Longhitano et al. (2020)
9	High cost of printer and raw material	3D printer costs are still very high, and the cost of raw materials is also still high compared with traditional manufacturing	Berman (2012); Liu et al. (2013); Mellor et al. (2014); Scott and Harrison (2015); Thomas (2016); Li et al. (2017a); Khajavi et al. (2018); Rong et al. (2018); Kemsaram and Maley (2019); Kunovjanek et al. (2020)

(continued)

Table 2 (continued)

No	Themes of challenge	Explanation	References
10	Limited recyclability of end-of-life 3D printed products	Most of the raw materials used in 3D manufacturing cannot be reused because of the change of materials' properties	Waller and Fawcett (2014); Ford and Despeisse (2016); Meisel et al. (2016); Garmulewicz et al. (2018); Unruh (2018); Westerweel et al. (2018); Zanoni et al. (2019)
11	New supply chain adaptation issues	3D printing is bringing manufacturing close to the customer and the new manufacturing technology is changing the existing SC structure of raw materials supply, product design, manufacturing, and distribution	Petrick and Simpson (2013); Waller and Fawcett (2014); d'Aveni (2015); Rogers et al. (2016); Thomas (2016); Ryan et al. (2017); Moradlou and Tate (2018); Corsini et al. (2020); Martens et al. (2020)
12	Regulation in design and printing	In 3D printing, consumers have the freedom to purchase designs from an online market and then build products at home. This freedom could result in easy access to make counterfeit and illegal products, and there are chances of violating the intellectual property rights of product design. Hence, regulations in design and printing are important to control personalisation and printing	Kietzmann et al. (2015); Chen (2016); Cerdas et al. (2017); Gibson (2017); Garmulewicz et al. (2018); Bonnín Roca et al. (2019); Halassi et al. (2019); Zanoni et al. (2019); Baiano (2020); Longhitano et al. (2020); Savolainen and Collan (2020)

Source Authors

We then developed a 3 X 3 matrix using the frequency of citation of benefit and triple bottom line measure categorisation. Based on the categorisation, we grouped the benefits of 3D printing in supply chain into nine clusters; these are shown in Fig. 6.

Figure 6 shows that six, four, and four benefit themes fell under economic, social, and environmental categories respectively. The decision of allocating benefits to a particular category of the triple bottom line measure was ensured using literature review (see Table 3).

Under the economic category the most frequently mentioned benefits were flexibility and simplicity in manufacturing; and the least frequently mentioned benefits was 'reduce warehousing'. Most of the benefits of 3D printing under this category were related to its novelty in manufacturing technology and the advantages of the production or operations process, which can bring economic benefits.

The benefit themes considered under the social category are based on the value creation for the customers and end consumers. Under this category the most frequently mentioned benefit was 'mass customisation' and the least frequently mentioned was 'new opportunity for 3PL'. Whereas the frequently

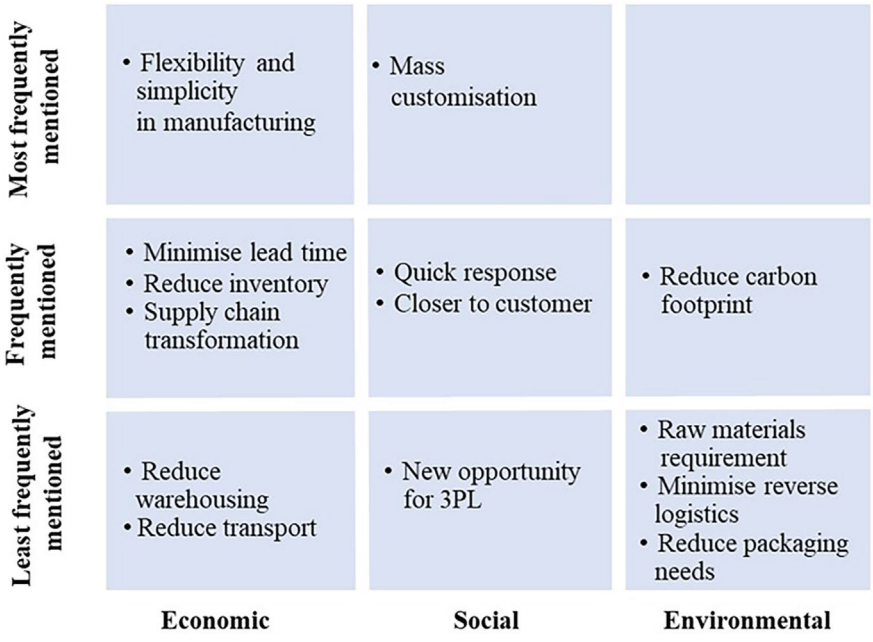

Fig. 6 3D printing benefits matrix based on the citation of benefit themes and triple bottom line measures (*Source* Authors)

Table 3 References related to benefit and challenge themes under the triple bottom line measures and organisational decision-making hierarchy category

Benefits			Challenges		
Category	Theme	Reference	Category	Theme	Reference
Economic	Flexibility and simplicity in manufacturing	Son and Park (1990); Weller et al. (2015)	**Operational**	Quality of products	Despeisse et al. (2017)
	Minimise lead time	Ouyang et al. (1999); Qorri et al. (2018)		Slow speed	Attaran (2017b)
	Reduce inventory	Callioni et al. (2005); Weller et al. (2015)		Limited recyclability	Despeisse et al. (2017)
	Reduce transport effort	Peng et al. (2018)	**Tactical**	High cost of printer and raw material	Malhotra et al. (1994)
	Supply chain transformation	Handfield and Nichols (2002)		High cost of large production volume	Ahumada et al. (2012)
	Reduce warehousing	Matthews and Hendrickson (2002)		Limited availability of raw materials	Malhotra et al. (1994)
Social	Mass customisation	Zhang et al. (2015)		Limited product range	Malhotra et al. (1994)
	Quick response	Suri (1998)		Longer product design time	Malhotra et al. (1994)
	Closer to customer	Reeves (2009)	**Strategic**	Required new skills	Jiang et al. (2017); Sodhi and Tang (2017)

(continued)

Table 3 (continued)

Benefits			Challenges		
Category	Theme	Reference	Category	Theme	Reference
	New opportunity for 3PL	Zacharia et al. (2011); Marchet et al. (2017)		New supply chain adaptation	Büyüközkan and Göçer (2018)
Environmental	Reduce carbon footprint	Reeves (2009)		Regulation in design and printing	Heizer et al. (2016)
	Limited raw materials requirements	Peng et al. (2018)		New technology adaptation	Dalmarco and Barros (2018)
	Minimise reverse logistics	Shaharudin et al. (2015); Zailani et al. (2017)			
	Reduce packaging needs for raw materials	Verghese and Lewis (2007)			

Source Authors

mentioned benefit themes were quick response to customer need and having the manufacturing closer to customers.

The benefits under the environmental category were reduction of packaging needs; minimised need for reverse logistics activities; reduction in carbon footprint; supply chain transformation, and limited requirement of raw materials for production.

4.2.3 Matrix Analysis of 3D Printing Challenges

As before, employing the guideline of Ahsan and Rahman (2017) and Rahman et al. (2019) we clustered challenges in the categories of 'most frequently mentioned', 'frequently mentioned', and 'least frequently mentioned'. Following Jurkovich (Jurkovich, 1974) and Jacobs and Chase (Jacobs et al., 2014) we considered the organisational decision-making hierarchy structure as strategic, tactical, and operational. Using the frequency of citation of challenges and organisational decision-making hierarchy structure, we then developed a 3 χ 3 matrix and grouped the 3D printing implementation challenges into nine categories, which is shown in Fig. 7.

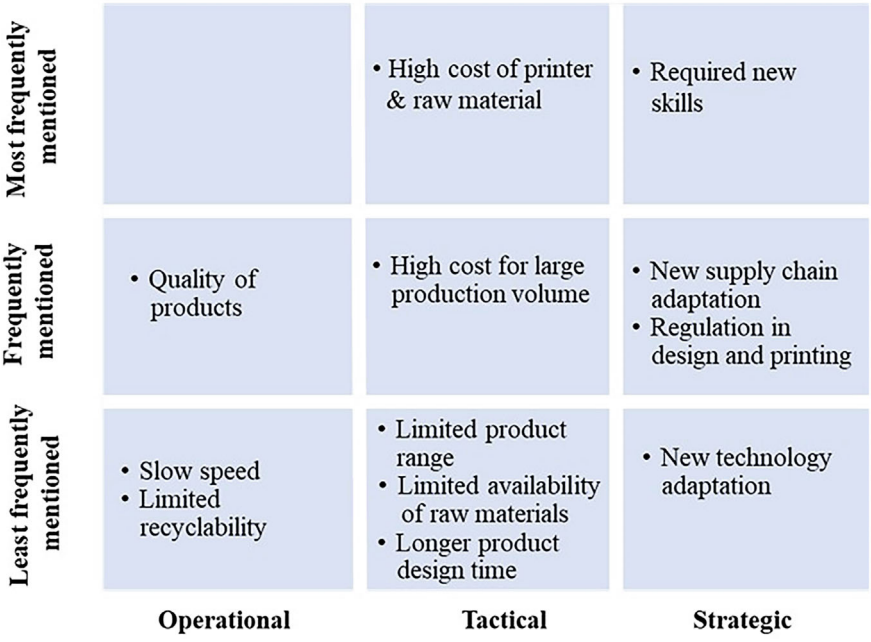

Fig. 7 3D printing implementation challenge matrix based on the citation of challenges and organisational decision-making hierarchy structure (*Source* Authors)

The analysis of the challenges revealed that new skills required, and the high cost of printer and raw materials were the two most frequently cited challenges of 3D printing manufacturing SCs. The challenges such as longer product design time compared to manufacturing time; limited product range and recyclability potential; and limited use of raw materials were the least frequently mentioned challenges. The analysis also revealed that three, five, and four challenges were categorised under the operational, tactical, and strategic decision-making realm respectively. For example, building capacity through recruitment of new skilled personnel or adaptation of new technology are the strategic level decision (De Jong & De Bruijn, 2013; Jiang et al., 2017), whereas tackling the quality issues of surface finish, use of different types of colour is an operational level decision (Despeisse et al., 2017). Some of the other challenges of 3D printing related to costs associated with materials, and limitations of raw materials were belonged to tactical level decision to enhance business prosperity (Vinodh et al., 2009). The decision of allocating challenges to a particular organisational decision-making hierarchy was ensured using literature review (see Table 3).

5 Discussion

We have identified the themes of benefits and challenges for the implementation of 3D printing manufacturing in SCs. Furthermore, we have explored the benefits and implementation challenges using matrix analysis. Using triple bottom line measures and organisational decision-making hierarchy structure against benefit and challenge citation respectively, this analysis has produced nine clusters for benefits and challenges. In the following, we discuss the results of these analyses and their implications.

5.1 3D Printing Benefits and Triple Bottom Line Measures

5.1.1 Economic Benefit Themes

The economic benefits are mainly related to the efficiency or lean dimensions of supply chains which include reduction in inventory, distribution, warehousing, and lead time (Li et al., 2017a; Weller et al., 2015; Zhu & Sarkis, 2004). The most frequently cited benefits of 3D printing in SCs under the economic measure have been identified as flexibility and simplicity in manufacturing. Using 3D printing, one can produce any feasible product

as long as it can be designed. 3D printing eliminates modular production and assembly lines from the manufacturing system (Durach et al., 2017; Weller et al., 2015). Therefore, setting up or change-over time in the production process will be reduced considerably, which will enhance flexibility and simplicity in manufacturing. The process of manufacturing is less complex and requires fewer production steps than the traditional manufacturing system (Achillas et al., 2015; Ben-Ner & Siemsen, 2017; Durach et al., 2017; Weller et al., 2015). In a 3D printing manufacturing environment, products can be produced in random order, when production planning is performed in real time. Further, 3D printing technology would replace the physical flow of raw materials (Achillas et al., 2015; Waller & Fawcett, 2014) because of the digital design files (Wu et al., 2015).

3D printing reduces manufacturing lead time by using a digital design of the product and reducing SC dependency by purchasing raw materials only from the suppliers and bringing manufacturing closer to the customer. Transportation needs are shifted from semi-finished products to raw materials, manufacturing operations can be handled in a single facility close to customers with a significant savings in inventory holding costs (Achillas et al., 2015). The benefits of simplicity in manufacturing, being close to the customer and reduced lead time may add the benefits of a quick response feature in 3D printing SCs (Attaran, 2017a; Huang et al., 2013; Sodhi & Tang, 2017), close to the customer will lessen the complexity of SCs, which in turn has the potential to reduce emissions along the SC (Baumers et al., 2016).

3D printing manufacturing may also introduce simpler and new SC structures. The centre point of SCs will be the 3D printing manufacturer (an individual or an organisation). SCs are becoming more circular, flexible and responsive, and less complex. In circular SCs, the focal company is the 3D printing manufacturer, which is surrounded by the 3D printer supplier, product designer (from cloud sourcing), raw materials supplier, customers (consumer, retailer, spare parts supplier) and 3PL service providers. Instead of the traditional complex SC, 3D printing will link up directly with the product designer, printer supplier, retailer or end customer, raw material supplier and inbound and outbound distribution.

5.1.2 Social Benefit Themes

The benefit themes considered under the social category are based on the value creation for the customers, hence these benefits are mainly related to the effectiveness or agility dimensions of supply chain (Jiang et al., 2017; Mefford, 2011). The most cited benefit of 3D printing in SCs under social was enabling mass customisation with low cost**.** Literature suggested that 3D printing is revolutionary in regard to providing the benefits of mass customisation even for a single unit of an item (Berman, 2012; Boon & Van Wee, 2018; Jia et al., 2016; Waller & Fawcett, 2014). In a 3D printing production environment, customisation can start early in the design process and materials selection stage without incurring additional costs in manufacturing. It is possible to obtain the benefits of large varieties (of stock keeping units) of make-to-order products in small batches, where manufacturers can respond to customer-specific requirements quickly as long as the design of the product is ready. Because 3D printing products can be manufactured close to the customer (Huang et al., 2013; Sodhi & Tang, 2017) with no time restriction, and when the customer needs them with minimum lead time. Customisation is postponed until the most accurate demand information is available. Thus, 3D printing could bring mass customisation benefits for a small quantity of products in a highly responsive manner. Furthermore, inbound logistics will be limited and mostly related to raw materials suppliers, which will be a specific type of raw materials for the 3D printer. Consumers can have their own printer to manufacture their products so the manufacturing will be close to the customer, and there will be fewer outbound logistics requirements. Having the manufacturing facility closer to the customer along with the benefits of mass customisation, and local 3PL service provider support, 3D printing manufacturing SCs will become more responsive to customer needs.

One of the important benefits of 3D printing manufacturing technology is bringing the manufacturing closer to the customer. The 3D printing production process is simple because it has only a few production steps and a less complex SC (Weller et al., 2015). Because of the simplicity in the production and SC process, 3D printing facilities can operate as standalone entities in strategic locations globally or close to customers (Ford & Despeisse, 2016; Huang et al., 2013). Location or relocation of the manufacturing facility closer to the customer may reduce the manufacturing lead time (Ben-Ner & Siemsen, 2017), and that may assist the SC to respond quickly to customer demand (Laplume et al., 2016; Rogers et al., 2016).

The least cited benefits of 3D printing in SCs under the social benefit category was related to the emergence of a new third-party logistics (3PL) role. New orientation of 3D printing manufacturing will lead to fewer raw material requirements (Ben-Daya et al., 2017) and reduced transport and warehousing efforts (Mckinnon, 2016). There will be opportunities for local 3PL firms to offer new remanufacturing services (Chan et al., 2018; Kietzmann et al., 2015), hence creating job opportunity locally. The 3PL role will be to focus more on managing product design and development, raw materials procurement and delivery services, and partner and contract management (Khajavi et al., 2014). Hence, the role of 3PL firms will be changed and new business relationships will emerge.

5.1.3 Environmental Benefit Themes

The environmental related benefits are among the least mentioned benefits which include minimum 'raw material requirement' and 'reverse logistics' activities, and reduced 'packaging needs'(Rejeski et al., 2018). Manufacturing closer to customers also eliminates the distances between the production and point of consumption, reduces the 'transportation efforts' related to moving finished goods and reduces outbound logistics requirements (Mckinnon, 2016; Rehnberg & Ponte, 2018). Because consumers can receive their product without outbound logistics, there will be fewer or no product returns or reverse flow of materials from consumer to retailers or manufacturers (Boon & Van Wee, 2018). If 3D printing facilities are located closer to customers, customers will be able to make the 3D printing products when they require them, which can minimise the manufacturing lead time and can reduce the level of finished goods inventory. Proximity to the market also enables production of small batch quantities, which can also reduce the finished goods inventory.

Because of the limited raw materials requirements, the need for packaging of raw materials will be limited or eliminated (Mckinnon, 2016; Peng et al., 2018). Reducing the efforts involved in the transport logistics of finished goods, requiring fewer warehouses to store raw materials or semi-finished or finished goods, minimising the requirements of packaging for transport or storage and avoiding product returns or reverse flow of materials can eliminate waste, and thus assist in reducing the carbon footprint of SCs (Mckinnon, 2016; Mohr & Khan, 2015).

5.2 3D Printing Implementation Challenges and Organisational Decision-Making Hierarchy Structure

5.2.1 Operational Level Challenges

Operational level decisions are routine issues which are generally related to the improvement of efficiency and productivity of production systems (Heizer et al., 2016). In this study, challenges such as slow speed in case of mass production, and limited recyclability of end-of-life 3D printed products, and quality of products are identified as the operational level challenges.

In the implementation of 3D printing product quality is considered as one of the key operational level challenges due to new technological-related issues. For example, materials used for 3D printing such as liquid polymers, powder comprised of resin plaster lack material strength. Plastic beads or large-sized particles create a rough and ribbed surface, which produces an uneven and unfinished texture to the end product (Huang et al., 2013). Since 3D printing products are manufactured layer by layer basis, the product strength is not uniform. Furthermore, often the parts from different machines exhibit varying properties. These quality related issues restrain the application of 3D printing technology in its current level of sophistication (Wu & Chen, 2018).

3D printing is technologically constrained to produce a large quantity of products quickly (Shukla et al., 2018). Compared to the traditional mass production system, 3D printing manufacturing is relatively slow to produce a large volume of products (Berman, 2012; Rengier et al., 2010)**,** and an employment of large numbers of 3D printing machines to achieve the high-volume production **will be expensive affair** (Kapetaniou et al., 2018). It is therefore obvious that at this stage it is unlikely to use 3D printing for mass production (Kapetaniou et al., 2018).

Since 3D printed products use a particular type of resin (mixed materials), they have limited recyclability opportunity at their end-of-life cycle (Ford & Despeisse, 2016; Garmulewicz et al., 2018). Therefore, although 3D printing has the potential to make SC systems more environmentally friendly, product composition such as the use of some raw materials limits the recyclability (Ford & Despeisse, 2016; Weller et al., 2015). Further research and innovation are essential to improve the production process to enhance the recyclability of 3D printed products.

5.2.2 Tactical Level Challenges

Tactical level decisions are taken by mid-level management and are aimed at identifying medium-term policies related to production planning and control, material planning, purchasing, distribution, and storage in ensuring efficiency and cost-effectiveness to add value to the product and service among others (Malhotra et al., 1994). The key tactical level challenge of 3D printing in SCs is identified as the high cost of printers. High cost of printer is a significant barrier for the success of 3D printing adaptation within SCs. Previous studies found that cost issues are related to machine costs, labour costs and material costs (Garmulewicz et al., 2018; Rong et al., 2018). Use of 3D printing is still restricted by the high cost of printers and raw materials. Industrial category 3D printers are still an expensive investment, which is a setback for producing a small quantity of products because of the high initial investment required (Gebler et al., 2014; Long et al., 2017). Moreover, because of the additive nature of 3D production, production of a large quantity of products is time-consuming and expensive compared with traditional production methods (Garmulewicz et al., 2018; Ryan et al., 2017). Further, product design can take significantly longer than manufacturing time (Beltrametti & Gasparre, 2018; Dwivedi et al., 2017), increasing the risk of a product launch.

3D printing has relatively high unit production costs, particularly for large volume production runs (Achillas et al., 2015). Due to the additive nature of the 3D printing, production of a large quantity of products is time-consuming and expensive compared to the traditional mass production methods (Berman, 2012; Long et al., 2017).

The lack of various options for raw materials limits the implementation of 3D printing manufacturing. However, some industries are ahead of others, for example, manufacturers of rubber products, non-metallic mineral products, machinery, jewellery, dental and medical devices, low-turnover repair or replacement parts and sports items (Ahsan & Rahman, 2019; Berman, 2012; Laplume et al., 2016;). In contrast, some industries are yet to use 3D printing because of the limited types of raw materials available.

The materials used for 3D printing are liquid polymers, powder comprised of resin or plaster, or metal. Due to limitation of varieties of raw materials use, 3D printing technology is limited to produce a wide range of products. 3D printed products are yet to start in industries are such as manufacturing of solid wood, cork, leather, natural textiles, petroleum products, tobacco

product, reproduction of recorded media, fabricated metal products, manufacturing of computers, electrical and transport equipment. Moreover, 3D printing takes longer product design time compared with manufacturing time (Long et al., 2017), which delays the product lunch.

5.2.3 Strategic Level Challenges

Strategy is an organisational action plan to achieve its mission and exploit opportunities and strengths, overcome weaknesses and neutralise threats (Heizer et al., 2016). Strategic issues in manufacturing are related to planning for human resource, manufacturing strategy, process technology, process planning, and dealing uncertainties related to external environment (Malhotra et al., 1994).

The planning and development of new skills required for 3D prinitng manufacturing have been identified as the most frequently cited challenge that require strategic decision. With the introduction of 3D printing manufacturing technology, new skills related to the new technology management need to be developed (Rong et al., 2018). A challenge will be to also find the right designer and design (Durach et al., 2017; Ford & Despeisse, 2016; Schniederjans & Yalcin, 2018; Thomas, 2016). Moreover, there are complexities in CAD software (Shukla et al., 2018). A newly skilled workforce will be required that can design (Shukla et al., 2018), model and manufacture different products (Ford & Despeisse, 2016; Garmulewicz et al., 2018). New skills are also important for SC managers because of the likely changes in the exiting SC structure. 3D printing is part of Industry 4.0; hence, skills for adapting new types of technology are required. New hard and soft skills from the workforce will be required. The hard skills are related to CAD product design and using 3D printing technology to manufacture different products. The soft skills are related to managing the SC. As the SC system changes, relationships between SC partners change and new forms of relationships, such as collaboration, emerge. To handle this new SC relationship, new soft skills are needed. Significant training on technical aspects of 3D printing will be required because most of the non-expert owners of 3D printing will start printing at home (Schniederjans & Yalcin, 2018; Weller et al., 2015). In the new 3D printing market, where the competitive landscape is rapidly shifting, dynamic capability needs to be developed by developing skills in 3D printing design and development/innovation of raw materials that create value for firms within dynamic markets by manipulating resources into new value-creating strategies. These issues are related to organisational readiness and strategy to adopt the new technology and manage related SC issues.

3D printing manufacturing will require to strategically integrate an entirely new supply chain model (Sodhi & Tang, 2017). Therefore, the new SC structure needs to define the roles and responsibilities of design and manufacturing, and new local collaboration (Petrick & Simpson, 2013). 3D printing requires a digital design, a new raw materials chain and distributed manufacturing, which may disrupt the traditional SC. Raw material sources will be highly concentrated in a few specialised suppliers (Despeisse et al., 2017), and there is a lack of material variety to choose from. The deployment of 3DP production technology will change the existing supply chain structure of raw materials sourcing and supply. The raw materials will only be consumed at downstream supply chain, mostly purchased items will be raw materials and CAD. The digital raw materials chain (source digital design) will create disruption in traditional part supply (Berman, 2012; Kietzmann et al., 2015; Waller & Fawcett, 2014). Raw material sources will be highly concentrated on a few specialised suppliers (Despeisse et al., 2017), and at this stage there are lack of material variety to choose. Firms need to redesign and establish relationship with the new supply chain partners in new digital business environment.

Consumers or 3D printing users have the freedom to find designs for the products they want to make. They can purchase designs from online markets and then build products at home. These designs can be easily copied and resold (Long et al., 2017). This freedom introduces challenges of making counterfeit products and illegal products, along with issues of intellectual property rights of design (Chan et al., 2018; Dwivedi et al., 2017; Zeltmann et al., 2016). Hence, new challenges will emerge involving ethical issues of using the new technology and regulating piracy or intellectual copyright issues through government policy (Schniederjans & Yalcin, 2018).

It is being suggested that the adoption of 3D printing technology shifting the current manufacturing paradigm from mass production to mass customisation and personalisation and bringing manufacturing close to customer. Since 3D printing is an emerging technology, there are issues in adapting the new technology by firms, and required new skills to use the technology and related supply chain. The adaptation of new manufacturing technology needs firms' capabilities of digital infrastructure to use of CAD software, as well as digital imaging technology to produce goods (Yeh & Chen, 2018). Overall, the 3D printing technology adaptation may inflict striking impacts on supply chain members such as machine vendors, material manufacturers, software providers, logistics operators, and research centres.

6 Contribution, Limitations, and Future Research Directions

This study examined the current research trends on 3D printing in SCs, explored how 3D printing is affecting the manufacturing SC and suggested future areas of growth and research direction. The study contributes to academic literature and managerial implications in the following ways.

Through the literature review, this study has identified the 3D printing research trends, gaps, and developed frameworks to conceptualise SC benefits and challenges of 3D printing manufacturing which may contribute both SC management and advanced manufacturing literature. The findings reveal a growing trend in 3D printing research within SCs. The synthesis of the literature indicates that the research on 3D printing in SCs is still at a very early stage. In the early stages of adopting 3D printing, challenges are mostly related to technology and support services. To understand the benefits and challenges of 3D printing manufacturing in SCs, additional research is potentially critical. We identified that survey-based empirical studies on 3D printing in SCs are limited at this stage; however, this category of research has the potential to grow when the popularity of 3D printing increases in different industrial sectors. As 3D printing manufacturing is increasing its application, many industries are adopting the new technology and hence there is scope for empirical research with large sample sizes to capture the key challenges and critical success factors of 3D printing manufacturing SC performance.

This study contributes to academic literature by identifying 14 themes pertaining to the advantages and 12 themes related to the challenges of 3D printing in SCs within a triple bottom line matrix and organisational decision-making hierarchy structure respectively. Additionally, this research highlights potential gaps in 3D printing within SCs that researchers can evaluate for future study. In the following, we outline potential avenues for future research relating to the benefits and challenges of 3D printing within supply chains. These areas encompass:

- With the advancement of 3D printing manufacturing methods, the community considers higher environmental sustainability as a key advantage. It is expected that 3D printing manufacturing system can reduce environmental impact and energy consumption significantly compared to traditional manufacturing practices. Our study demonstrates that to-date the environmental sustainability is an under-researched area in the body of research on 3D printing manufacturing SCs. Further research can be

extended to analyse how the reduction of packaging, transport and warehousing, reverse logistics, and limited raw material requirements can be harnessed to reduce carbon footprints to achieve sustainability.

- Since 3D printing is a disruptive technology and new soft and hard skills and competencies will be required to realise the needs of the new technology. Recently there have been few attempts to assess the skills need, however these studies are exploratory in nature (Benešová & Tupa, 2017; Grzybowska & Łupicka, 2017). Further research is required to investigate the required competencies to handle the adaptation of new technology and organisational strategies of hiring, developing, and retaining talents in this area.
- 3D printing has gone through vast technological development in the past 20 years, and the cost of both personal and industrial 3D printers has reduced substantially (Li et al., 2017b). In spite of this development, our study found that the high cost of printers and raw material is still limiting the use of 3D printing in supply chains. Further research in the technology domain of 3D printing is necessary to investigate how the cost of printers and raw materials can be further reduced to reduce total SC costs.
- Although the quality of 3D printing products has improved over the years (Chua et al., 2017), quality relating to surface finish, standard, strength, colour is still remained one of the critical challenges of 3D printing manufacturing. Further research is required to investigate whether the quality challenges are different for different type of raw materials or different type of products.
- Previous research identified mostly the economic benefits of 3D printing manufacturing. 3D printing manufacturing may create new social benefits and opportunities. There is a lack of research in the social benefit domain of 3D printing manufacturing. Creating new opportunities for local 3PLs in terms of new job creation locally and providing higher level of delivery services through designing innovative last mile service delivery systems under 3D printing manufacturing, will add social value to communities and end customers. Further research can be conducted in these areas.

This study contributes to managerial implications by identifying benefits and challenges of 3D printing in SCs. The findings provide valuable insights for 3D printing manufacturers to restructure their SCs in alignment with the new manufacturing set-up. The identification of benefits related to 3D printing could incentivise new businesses to adopt this technology, while also aiding policy makers in considering its implementation within a new manufacturing strategy at a national level. Additionally, the identification of

challenges and their respective clusters could assist managers in addressing these barriers at an appropriate decision-making level. Strategic decisions play a crucial role in addressing the challenges of planning and developing new skills requirements within the rapidly evolving technological landscape. Additionally, tactical decisions are necessary to identify ways of reducing production costs, while operational level decisions are crucial for enhancing the speed and quality of end products.

This study has some limitations. It was limited to journal articles published in English up to 2020, which means that we may have missed noteworthy papers published in other languages or after 2020. Conducting a systematic review is a time-consuming process. By the time the review is completed in 2020 and the article is written and published, there may be new research that is not included in this rapidly evolving field of 3D printing. In addition, despite our best efforts to capture all relevant studies, our search process may have been slightly affected by the inconsistent use of keywords in published articles to capture all relevant articles using only the search and exclusion terms (Ahsan & Rahman, 2022). This may also be due to differences in terminology and database coverage.

The full extent of the impact of 3D printing on SCs is not yet understood. Synthesising the identified the 3D printing research on SC this study assesses the benefits and challenges of implementing 3D printing in manufacturing SCs. We believe this study will provide substantial value to readers in the additive manufacturing or 3D printing field by comprehensively examining the benefits and challenges of 3D printing in SCs. This will enable businesses to redesign their SC structure in response to the transformative effects of 3D printing.

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3

The Role of IoT and Big Data in Supply Chain Processes

Walid Khalid Al Saad and Moayad Al Talib

1 Introduction

The twenty-first century has experienced a significant transformation in the way supply chain management (SCM) functions, primarily due to the convergence of two remarkable technological forces: the Internet of Things (IoT) and Big Data (BD). The cooperation between IoT and BD has ushered in a new era differentiated by increased efficiency, transparency, and agility in supply chain operations. This chapter explores the distinct characteristics and impact of IoT and BD on SCM.

We start by delving into the realm of IoT, a concept initially introduced in 1999, which has since garnered considerable attention across diverse industry sectors. At its core, IoT represents a network of interconnected objects enriched with sensors and global connectivity, enabling the real-time collection, analysis, and utilization of data for more convenient and efficient decision-making. We will examine the definition of IoT, its characteristics, and its transformative role in enhancing supply chain processes. Subsequently, we study BD, referring to extensive and intricate datasets that challenge conventional data processing methods. These datasets originate from various sources, spanning structured, semi-structured, and unstructured data. We will also explore the three fundamental "V's"of BD: Volume, Velocity, and Variety, which serve as its foundational pillars.

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Throughout this chapter, we will not only uncover the enablers, benefits, and challenges associated with adopting IoT and BD in supply chain operations but also provide real-world examples of organizations that have embraced these technologies to enhance the efficiency of their supply chains. Ultimately, this chapter aims to provide insight for supply chain professionals as they navigate the ever-evolving landscape of IoT and BD in pursuit of improved efficiency and competitiveness.

2 Comprehensive Review on IoT and BD Technologies

2.1 Internet of Things

Introduced for the first time in 1999, IoT has garnered significant interest due to its capacity to provide convenience, enhance efficiency, and yield competitive benefits for various business sectors (Aryal et al., 2018).

IoT can be defined as a network of interconnected objects, a capability, or a technology that enables interaction with the environment:

- A network of "readable, recognizable, locatable, and manageable" objects over the Internet.
- A platform where physical objects are equipped with sensors and interconnected in a worldwide unifying network, so that real-time information is collected, stored, transmitted, accessed, and analysed through the internet to support decision-making (Li et al., 2018).
- A network comprised of various nodes belonging to the technological, physical, and broad socioeconomic environments (Krotov, 2017).

2.2 Characteristics of IoT

Considering all perspectives of modern-day IoT systems, a few generic and vital characteristics are (Rose et al., 2015):

- *Sensor Data Gathering,Storage, and Processing:* A multitude of distributed sensors, often referred to as smart devices, collect data about the physical environment or objects and send this data to the cloud for storage and analysis. This process aims to enhance business operations.

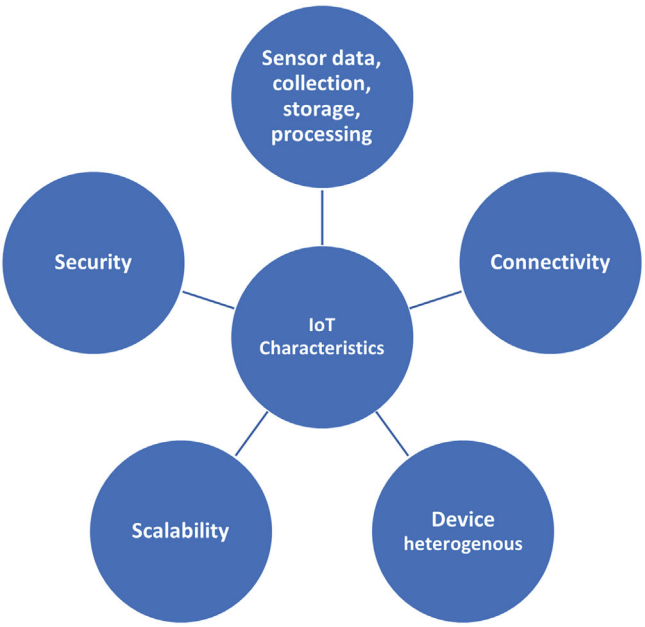

Fig. 1 IoT characteristics (*Source* Authors)

- *Connectivity:* IoT enables the connection of physical and virtual objects through a global communication infrastructure, which relies on both wired and wireless technologies.
- *Device Heterogeneity:* The ability of devices, which may have different hardware and network specifications, to work together seamlessly at both the hardware and software levels, supports intelligent interactions.
- *Scalability:* The growing number of IoT devices leads to a transition from human-to-human interactions to interactions between devices.
- *Security:* Security measures need to be in place at both the network and device levels to ensure the protection of data.

An overview of the IoT characteristics is illustrated in Fig. 1.

2.3 Big Data

The term BD has gained prominence in recent years, indicating a significant shift in how we gather, analyse, and utilize information. Fundamentally, BD refers to extremely extensive and complex datasets that cannot be effectively managed or analysed using conventional data processing technologies (Hulsen et al., 2019).

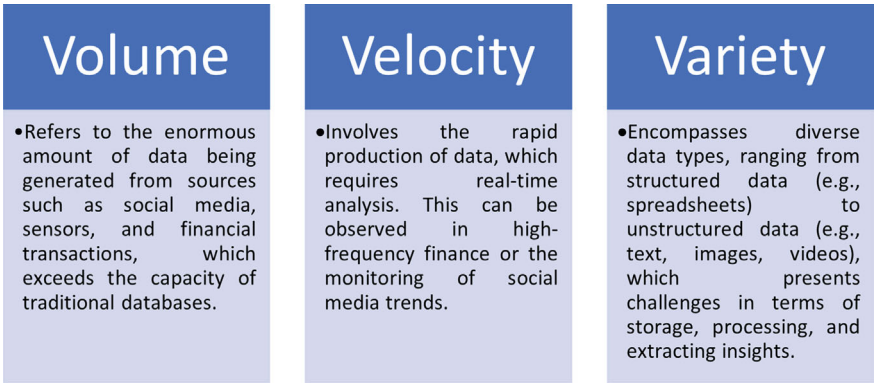

Fig. 2 The three V's (*Source* Authors)

Instances of BD originate from various sources, including social media interactions, financial transactions, sensor-generated data, and medical records. This data can take on various forms, falling into the categories of structured, semi-structured, or unstructured.

- Structured data refers to information that has been systematically organized such as data stored in a well-structured database table.
- Semi-structured data, typified by formats such as JSON, possesses a more fluid and adaptable structure.
- Unstructured data, such as text, images, and videos, lacks any predefined organizational structure.

BD is characterized by the three V's: Volume, Velocity, and Variety (Ghasemaghaei & Calic, 2019) as seen in Fig. 2.

BD represents a qualitative shift in how we handle information, offering significant promise for improved decision-making and creativity. It does, however, raise questions regarding security, privacy, and ethical use. This chapter will delve into the complexities of BD, including its origins, applications, and future implications.

2.4 Sources of Big Data

BD can be used in a variety of fields, including supply chain management, to enhance decision-making and maximize performance. Supply chain management is the process of planning, organizing, and carrying out the flow of information, materials, and services from suppliers to customers. Supply

chain managers can enhance their competitiveness, efficiency, and effectiveness by utilizing BD to gain valuable insights into various aspects of the supply chain.

Some of the sources of BD in supply chain management include:

- Data from social media platforms and online sources can provide a more comprehensive view of market trends, customer sentiment, and competitive intelligence. This data can help businesses understand customer needs, preferences, satisfaction levels, and sentiments. It can also help them identify trends, opportunities, and threats, as well as monitor brand reputation and competitive intelligence.
- Enterprise Resource Planning (ERP) systems are essential for managing a corporation's supply chain. They generate a vast amount of inventory, sales, procurement, and production information, providing a comprehensive understanding of the entire supply chain process. This internal dataset is gathered and maintained within operational platforms such as ERP and Supply Chain Management (SCM). It includes details about current suppliers, order histories, quality, expenses, delivery efficiency, stock levels, manufacturing timetables, sales projections, and client orders.
- IoT devices are seamlessly integrated into various aspects of the supply chain, including tracking goods in transit, managing warehouse operations, and monitoring vehicle fleets. They generate a valuable amount of data encompassing factors such as temperature, location, humidity, and a range of other environmental and logistical variables. This external data is primarily derived from sensors, monitors, and devices embedded within products, vehicles, machinery, and infrastructure. It delivers real-time insights into the status, location, condition, and performance of physical assets and operational processes.
- Customers and suppliers' data: When it comes to customers, data collected from their interactions provides invaluable insights, offering a window into their purchase histories, preferences, and feedback. On the supplier side, data covers various aspects, including delivery performance, quality metrics, and contractual obligations.
- Transaction and financial data within the supply chain are indispensable. It helps assess the cost-effectiveness and profitability of various operations. This data encompasses details such as invoices, payments, and budgetary information, providing a comprehensive financial perspective on supply chain activities.
- RFID tags are integral components in supply chain operations. They are small electronic devices that utilize radio waves to store and transmit data.

By attaching RFID tags to products or assets, the supply chain can efficiently identify and track them. The real-time data emitted by RFID tags encompasses crucial information related to product identification, movement, and inventory levels. This high-volume and high-speed data proves instrumental in enhancing supply chain visibility, traceability, inventory management, and compliance. It is an indispensable tool for efficient and effective supply chain operations.

3 Enablers, Benefits, and Challenges of Adopting IoT and BD in Supply Chain

3.1 IoT Enablers

Incorporating IoT technology into our daily lives brings forth a range of new opportunities and challenges for consumers, businesses, and governments. IoT products and digital services have the potential to enhance productivity, reduce time-to-market, and create numerous new enterprises and job opportunities. This results in an improvement in our overall standard of living, but it also comes at the expense of increased energy consumption, which directly impacts the environment. IoT technology has the potential to revolutionize our lives. However, its implementation should be approached with careful consideration of its impact on our quality of life and sustainability (Gubbi et al., 2013; Gupta & Gupta, 2016; Lin et al., 2017). Figure 3 illustrates IoT enabling technologies. The definitions are summarized in Table 1.

3.1.1 Sensing Technologies

Effective sensor and actuator identification, as well as the detection of physical phenomena, are fundamental characteristics of IoT systems at the system level. In IoT, sensing involves gathering data from interconnected smart devices using sensors and actuators. Sensors, which are essentially electronic devices, are responsible for generating electrical, optical, and digital data derived from the physical environment. This data is then electronically transformed into valuable information for intelligent devices or human interpretation. Actuators, on the other hand, complement sensors by converting electrical signals into non-electrical energy.

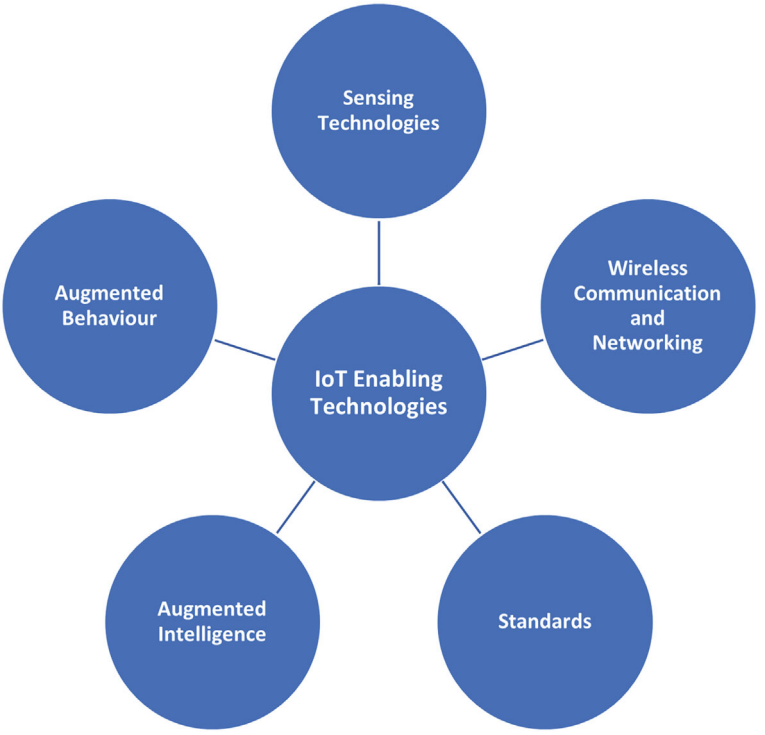

Fig. 3 IoT enabling technologies (*Source* Authors)

Table 1 IoT enabling technologies definitions

Technology	Definition
Sensing Technologies	Devices that generate an electronic signal from a physical condition or event
Wireless Communication and Networking	A mechanism for communicating an electronic signal
Standards	Commonly accepted prohibitions or prescriptions for action
Augmented intelligence	Commonly accepted prohibitions or prescriptions for action
Augmented behaviour	Technologies and techniques that improve compliance with prescribed action

3.1.2 Wireless Communication and Networking

Wireless communication and networking are pivotal components of the IoT ecosystem, with a focus on technologies such as RFID-based sensors and Wireless Sensor Networks (WSNs). RFID, or Radio Frequency Identification, is a wireless communication technology that utilizes electromagnetic fields to automatically identify tags attached to physical objects. In IoT, data transmission typically involves a variety of wireless communication and network technologies. Wireless communication in IoT emphasizes how diverse devices can communicate with each other in a sustainable manner, ensuring mutual understanding. In contrast, wireless networking is focused on efficiently connecting devices to facilitate the transmission of sensed data (Ruiz-Garcia et al., 2009).

3.1.3 Aggregation and Standardization

Data aggregation within the IoT context brings the issue of standardization to the forefront. While relational databases and SQL have traditionally been used for storing and querying structured data, there is currently no standard for handling unstructured data.

With the IoT promising the scalability of billions of devices, it is essential to have common standards in place to enable communication and data aggregation across diverse data types. Existing Internet standards were developed without considering the IoT vision, leading to the use of proprietary protocols for IoT systems. This can hinder communication between IoT devices. Standardization is essential in the IoT domain as it ensures interoperability, scalability, consistent data semantics, security, and privacy. While there is no universal standardization body for IoT technology, there are several standardization organizations that are active at the international, regional, and national levels (Pal et al., 2018; Sen, 2018).

Regulatory standards are also crucial in the evolution of IoT, as they address data ownership, use, and sale. The US Federal Trade Commission has outlined recommendations, known as Fair Information Practice Principles (FIPPs), to guide IoT applications (Holdowsky et al., 2015).

3.1.4 Augmented Intelligence

The analysis of collected data requires the application and advancement of various augmented cognitive technologies. Augmented intelligence automates

systems to perform descriptive (representing data to identify insights), predictive (anticipating future outcomes), and prescriptive (related to optimization) analyses. Predictive analysis uses machine learning techniques to identify future trends from historical data without the need for explicit programming instructions. However, these technologies need further refinement as predicting future trends in practical applications can be challenging, even with a strong correlation between entities (Pal et al., 2018).

3.1.5 Augmented Behaviour

Augmented behaviour encompasses actions that consider all phases of the information value loop, from sensing to data analysis. This phase supports the execution of recommended actions using advanced technologies such as Machine-to-Machine (M2M) and Machine-to-Human (M2H). IoT transitions from data science to behavioural sciences during this phase, tracking changes in human behaviour and organizational processes. Advancements in machine-to-machine (M2M) and machine-to-human (M2H) technologies are the main driving forces that enable machines to comprehend their surroundings and make logical decisions (Holdowsky et al., 2015).

3.2 BD Analytics and the Data Information Knowledge Wisdom (DIKW) Pyramid

Technologies such as IoT, RFID, and sensors enable the collection of BD in the supply chain. However, they also require advanced analytical methods to extract meaningful information from this data. Some of the analytical methods used to process and analyse BD in the supply chain are (Soltanpoor & Sellis, 2016).

- Descriptive analytics, a pivotal method in supply chain analysis, involves summarizing and visually representing historical or present data. By employing a range of techniques, including statistics, charts, graphs, dashboards, and detailed reports, this approach reveals key performance indicators (KPIs), highlights prevailing trends, and patterns, and identifies potential outliers. It essentially provides a comprehensive view of the supply chain's historical performance, allowing stakeholders to gain valuable insights into past and current operations.
- Predictive analytics, an indispensable method in the field of supply chain management, utilizes historical data to anticipate future developments and

events. By utilizing techniques such as regression, classification, clustering, simulation, and the capabilities of machine learning, this approach builds models that provide predictive insights based on historical lessons. These predictive models are skilled at forecasting emerging supply chain trends, future demand patterns, and potential issues. Consequently, the careful implementation of machine learning and statistical techniques serves as the foundation of this approach, allowing organizations to proactively manage the intricacies of their supply chain operations.

- Prescriptive analytics goes beyond prediction by utilizing past or current data to recommend optimal actions or decisions in the supply chain. It employs techniques such as optimisation, decision analysis, game theory, and artificial intelligence to generate solutions that maximize a specific objective or criterion. This method helps identify the optimal course of action in response to various scenarios, enabling businesses to make informed decisions that optimize their supply chain operations.

3.2.1 The DIKW Pyramid

BD can be either challenging or straightforward to work with, depending on how we handle, analyse, and utilize it. It can also help us learn new things and make better decisions. But not all data is important or relevant. To understand how we can transform BD into valuable knowledge and wisdom, we can refer to the DIKW Pyramid as a reference (see Fig. 4).

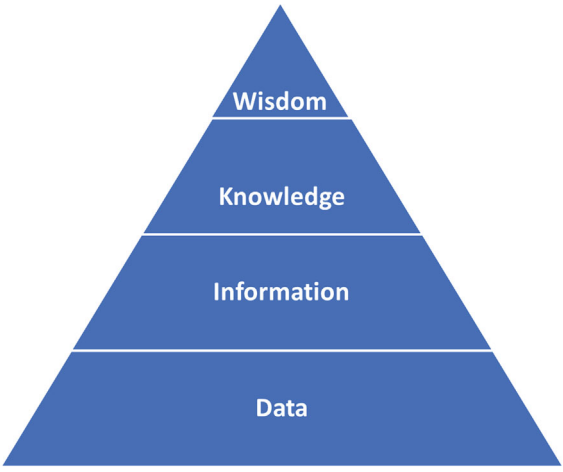

Fig. 4 DIKW Pyramid (*Source* Authors)

The DIKW Pyramid is a model that illustrates the relationship between various levels of information processing, ranging from basic data to advanced wisdom. The concept is also known by other names, such as the DIKW hierarchy, wisdom hierarchy, knowledge hierarchy, information hierarchy, information pyramid, and data pyramid1. The DIKW Pyramid has four main components: data, information, knowledge, and wisdom. Each part has its own features, roles, and problems (Van Meter, 2020).

- *Data:* At the base of the pyramid, we find data. This represents raw facts, symbols, numbers, characters, images, sounds, and other pieces of information that are collected or observed. Data, in its raw form, lacks structure, organization, and context. It is like the fundamental building blocks, offering no inherent meaning or value.
- *Information:* Moving up the pyramid, we reach the level of information. This is data that has been processed, organized, structured, or presented within a specific context to make it useful. Information is all about answering the "who," "what," "when," "where," and "how" questions. It serves as the cornerstone of knowledge.
- *Knowledge:* The third tier, knowledge, is where information is taken to a deeper level. Information is analysed, synthesized, understood, or applied to a specific domain or purpose. Knowledge goes beyond the 'what' and 'how' questions and delves into the 'why,' 'how much,' 'how often,' 'how likely,' and 'what if' inquiries. It is the foundation upon which wisdom is built.
- *Wisdom:* The pinnacle of the pyramid is wisdom. This represents knowledge that has been evaluated, refined, generalized, or integrated with other forms of knowledge and experience. Wisdom deals with questions of "what is best," "what is right," "what is ethical," and "what is valuable." The goal is to process information.

This hierarchy takes us on a journey from basic data to profound wisdom, illustrating how information evolves and gains depth. Ultimately, it contributes to wise decision-making and valuable insights.

4 Best Practices of IoT and BD in Supply Chain Digitalisation

4.1 IoT Roles in Supply Chain Management Processes

The roles of IoT technologies in industrial applications hold the potential to significantly enhance production efficiency and facilitate improved communication and networking between operators and machines. This, in turn, can lead to companies becoming more competitive in the market, with superior quality control and minimized losses. A crucial aspect of this transformation involves the development, design, and integration of various sensors specifically designed for industrial applications, creating comprehensive and effective management systems (Aryal et al., 2018; Li et al., 2018).

In the realm of supply chain management, IoT offers various capabilities that can result in cost savings, improved inventory accuracy, and enhanced product tracking. One valuable framework for understanding supply chains is the process-centric perspective, with the SCOR model being a widely recognized framework. The SCOR model categorizes supply chain processes into Plan, Source, Make, Deliver, Return, and Enable (APICS, 2015).

As the planning process is interconnected with all other SCOR processes, it is not analysed separately. In the context of major supply chain processes (source, make, deliver, and return), IoT plays distinct roles and has notable impacts, as summarized in the following figure (Ben-Daya et al., 2019). Figure 5 outlines the specific contributions of IoT to each of these processes and highlights the resulting impacts.

4.2 BD Barriers and Capabilities in Supply Chain

Supply chain managers can greatly benefit from BD, as it offers improved demand forecasting, increased operational efficiency, a platform for innovation, and enhanced customer satisfaction. But it also brings with it a multitude of barriers and challenges, including those related to data governance, data security, data integration, data quality, and data analysis. This section will examine the main barriers and capabilities that BD presents in the supply chain.

BD has the power to completely change how supply chains are managed. But despite its potential, several barriers may hinder its widespread use. These barriers fall into four primary groups (see Fig. 6):

BD is like a superpower for supply chains. It grants individuals the ability to see through uncertainty and make informed decisions that can result in

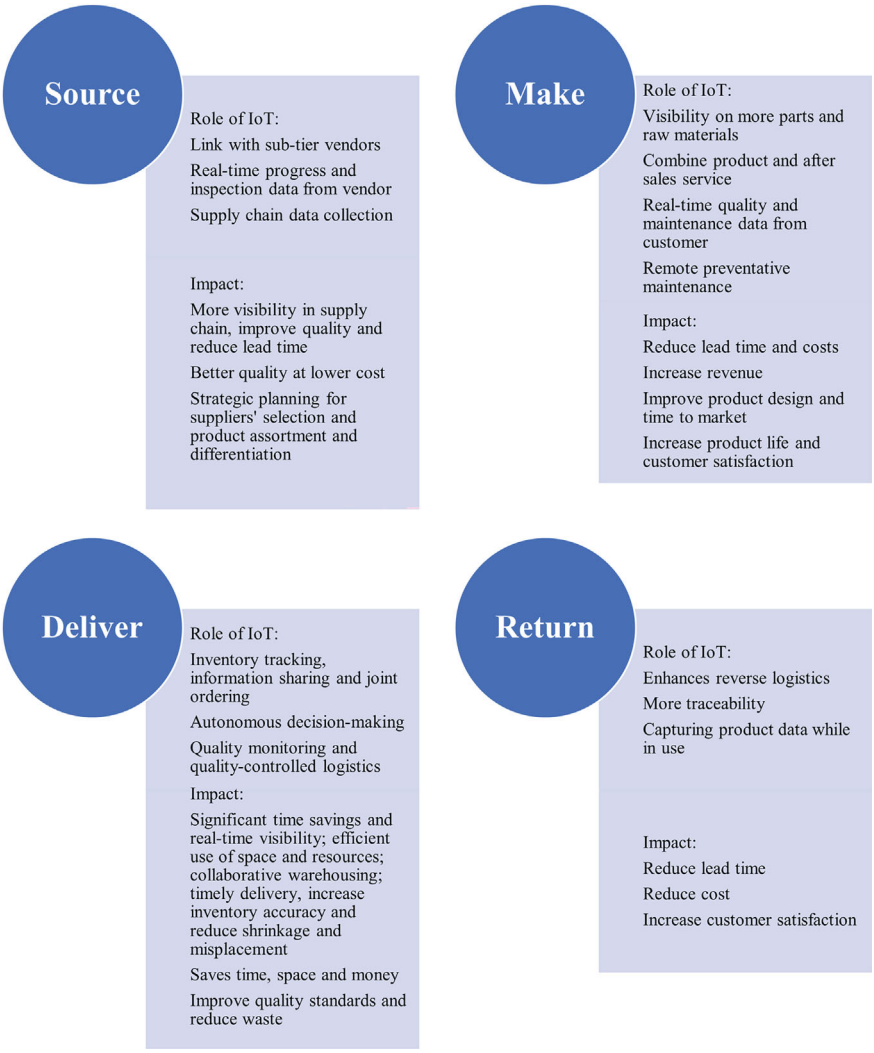

Fig. 5 IoT roles in supply chain management processes (*Source* Authors)

significant improvements in efficiency, agility, and profitability (Raman et al., 2018). Figure 7 clarifies the capabilities of BD.

To conclude, integrating BD into the supply chain offers significant opportunities for enhancing efficiency and decision-making. Addressing the barriers associated with and capitalizing on the enablers, on the other hand, are critical steps towards realizing the full potential of BD in supply chain management. Organizations that effectively navigate these issues will gain a competitive advantage in the evolving, data-driven supply chain world.

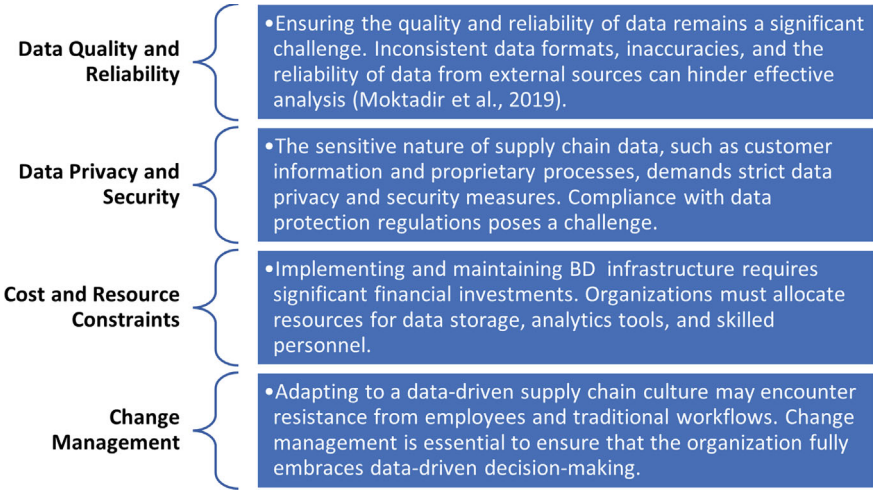

Fig. 6 BD barriers (*Source* Authors)

Fig. 7 BD capabilities (*Source* Authors)

4.3 Examples of Application Areas of IoT

The application areas of IoT are diverse and are built upon existing technological solutions. Beyond industrial applications, some of the most significant and rapidly evolving areas of application are related to the concept of smart cities, transportation, healthcare, and agriculture (Nižetić et al., 2020).

The development and success of specific IoT application areas depend on several key factors, including:

- Advancements in electronic components available in the market.
- Availability of software solutions and user-friendly interfaces.
- Solutions related to sensor technologies and data acquisition.
- Quality of network connectivity and infrastructure.
- Sufficient energy supply for the production and operation of IoT devices.

Following is a closer look at some key IoT application areas:

4.3.1 IoT in Agriculture

Efficient agricultural production is crucial for meeting the increasing global demand for food resources. Challenges such as population growth and climate change have the potential to decrease crop yields. Additionally, food waste is a global concern. IoT technologies can play a vital role in ensuring an adequate food supply and enhancing the efficiency of agricultural processes. IoT can provide valuable data for crop monitoring, disease detection, soil and nutrient management, water conservation through smart irrigation, and precise seeding, among other benefits (Farooq et al., 2020).

4.3.2 IoT in Healthcare

IoT has a significant role to play in healthcare through the concept of e-health. IoT can enhance the quality of healthcare services by collecting and analysing patient health data, ultimately improving patient safety and care. Smart medical devices can monitor vital signs and functions, enabling remote health monitoring, early symptom prediction, disease prevention, and assistance for the elderly. IoT technologies can support rehabilitation, especially in home-based rehabilitation scenarios.

One challenging field of implementing IoT technologies has been identified in the healthcare system, specifically through the concept of e-health (Papa et al., 2020). An increase in the service quality of healthcare systems could be achieved through IoT support, primarily through the collection of patient health data. This improvement could result in enhanced patient safety and care, ultimately leading to an increase in patient life expectancy. There is enormous potential in smart medical devices for various purposes. These devices can be used to monitor vital human functions such as heart rate, skin temperature, and movement (Muthu et al., 2020).

4.3.3 IoT in Transportation

The transportation industry is undergoing significant changes, driven by the increasing adoption of electric vehicles and the environmental push to phase out diesel-powered vehicles. Alternative transportation technologies, such as hydrogen-based vehicles, are also being developed. IoT is pivotal in the concept of the "Internet of Vehicles," with smart cars being a prominent application. IoT enhances the driving experience, safety, and comfort by collecting data related to vehicle parameters such as tire pressure, fuelling, early detection of potential failures, and maintenance indicators. It adds value for customers and enhances competition among vehicle manufacturers (Brincat et al., 2019).

In summary, IoT technology is transforming various sectors by offering solutions to complex challenges and improving efficiency and effectiveness across different industries.

4.4 Examples of Organizations Adopting and Promoting IoT and BD

IoT and BD have been effectively embraced and promoted by several businesses in their supply chain operations, yielding notable results. Here are some examples from the real world:

- *Amazon Go:* In 2018, Amazon launched its Amazon Go establishment in Seattle, Washington. This store operates with a nearly automated system, allowing customers to make purchases without the need for cashiers or traditional payment terminals. This innovative shopping experience is made possible by the continuous evolution of IoT. To shop at the Amazon Go store, customers use the Amazon Go app, which effectively creates a virtual cart containing their selected items. This technology, driven by IoT vision and deep learning, has the capability to effortlessly recognize the products customers choose to buy or return. As a result, once customers exit the store, their Amazon account is automatically charged, and they can promptly access their receipt, which includes payment details, through the app (Aznag & Tahanout, 2022).
- *Walmart:* Walmart is a leader in utilizing BD for supply chain optimization. The company collects and analyses data from various sources, such as sales, weather, social media, etc., to predict customer demand and manage inventory and transportation. By using BD, Walmart can effectively maintain optimal product inventory levels, ensuring that shelves are consistently

stocked with the appropriate items at the precise time. This not only enables cost savings on storage and delivery expenses but also enhances overall operational efficiency. Walmart also uses BD to monitor its supply chain operations. This involves the use of sensors, RFID tags, GPS devices, and other technologies to track the location and status of its products, vehicles, and assets. This helps Walmart enhance its operational efficiency, reduce waste, and mitigate risks (Sanders, 2014).

- *Amazon:* Amazon uses BD to improve its supply chain performance. The company personalizes the shopping experience for its customers by using BD to recommend products based on their past purchases. This increases sales and customer satisfaction. Amazon also uses BD to enhance its shipping efficiency. The company forecasts customer demand, optimizes inventory, and predicts shipping routes. This reduces shipping costs and delivery times (Fernando et al., 2018).
- *The United Parcel Service (UPS)* leverages BD to improve its delivery operations. The company analyses data on traffic, weather, and customer locations to find the best routes for delivering packages. This reduces fuel costs and emissions. UPS also uses BD to predict demand for its services and allocate the appropriate resources to fulfil customer requirements. This increases efficiency and customer satisfaction. UPS also uses BD to identify and prevent fraud, as well as predict and prevent vehicle breakdowns (Marr, 2018).
- *Procter & Gamble:* They leverage BD to enhance its supply chain performance. The company collects and analyses data from various sources, such as supplier performance, inventory levels, transportation costs, point-of-sale data, and social media. This helps the company identify areas for improvement and adjust its production and distribution strategies. Procter & Gamble also uses BD to forecast customer demand and ensure that it has the right number of products on hand (Sanders, 2014).

These are just a few examples of how IoT and BD can assist in supply chain management. These technologies can also be used for various purposes, including supplier relationship management, customer relationship management, and risk management. They can give organizations competitive advantages by harnessing their potential and overcoming their challenges.

5 Conclusion, Limitations, and Future Research

In this chapter, we have explored the game-changing impact of the IoT and BD analytics on SCM. Our journey has encompassed insights into the evolving trends, applications, and challenges within IoT and BD in SCM, supported by real-world examples that demonstrate their practical value. Companies have used these technologies to optimize inventory, enhance food traceability, provide valuable insights, and promote intelligent supply chain collaboration.

The relationship between IoT and BD is mutually beneficial. IoT serves as a significant source of BD, with IoT devices generating vast amounts of data on various aspects. This data enables better decision-making and process optimization. Simultaneously, BD analytics can enhance the performance of IoT devices by identifying patterns and anomalies within the data. This allows for the prediction of future events and the prevention of issues, improving the overall efficiency of IoT devices.

We have underscored the diverse capabilities of IoT and BD, ranging from demand forecasting to risk management. However, it is crucial to acknowledge the technical, organizational, and ethical challenges they bring, including data quality, security, and governance issues.

In conclusion, the IoT and BD serve as disruptive forces that provide a competitive advantage to organizations that can effectively navigate their complexities. We encourage future research to focus on advanced data analysis methods for IoT generated data in SCM. We also advise practitioners to strategically implement IoT and BD solutions that are aligned with their business objectives and customer needs.

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4

Disruptive Technologies and Sustainable Development Goals in Supply Chains

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1 Introduction

The Sustainable Development Goals (SDGs) suggest a holistic approach to developing sustainable practices and solutions by addressing society's fundamental problems (Dantas et al., 2021). Accelerated industrialization has led to challenges with access to high-quality goods and services in health care, education, agriculture, finance, etc. (Truby, 2020; Vinuesa et al., 2020). In addition, raising awareness of the environmental and social impacts of supply chain operations in all areas has made the necessity of effective solutions inevitable to challenge dynamic environments (Meneghetti & Monti, 2015). Thus, various suggestions supported by disruptive technologies such as blockchain, Internet of Things (IoT), robotics, etc., have been developed to deal with the constantly increasing difficulties that affect supply chains in each area (De Sousa Jabbour et al., 2018).

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Disruptive technologies are causing fundamental changes in existing markets or industries, disrupting traditional business models and creating new opportunities (Di Vaio et al., 2020). Such technologies change existing regulations, habits, and business ways and cause industries to operate differently. Disruptive technologies can enhance each United Nations SDGs, such as reducing poverty and hunger, improving health, creating new job opportunities, reducing climate change and reducing disruptions in supply chains (Mhlanga, 2021; Mondejar et al., 2021). Besides, disruptive technologies ensure SDGs, particularly on poverty reduction (SDG9), improving the accuracy and reliability of infrastructure, such as transport, that enable economic growth and development in developing economies (SDG9, SDG10). Thus, adopting disruptive technologies is needed to achieve green digitalization to SDGs from a disruptive technologies perspective (Kayikci et al., 2022). For instance, in the health sector, artificial intelligence (AI) technologies are assisting in saving lives and diagnosing diseases rapidly and accurately (SDG 3) (Di Vaio et al., 2020). Green digitalization means using digital technologies to promote environmentally friendly and sustainable practices (Mendes & Viola, 2023). This concept includes the combination of environmental sustainability and digital transformation. Green digitalization aims to increase environmental sustainability and reduce carbon footprint for businesses and society. This approach can solve economic growth and environmental and social problems of technology (Lee et al., 2023).

Food and agricultural practices must be more data-driven and efficient to increase productivity, decrease energy usage and improve supply chain operations (Kamble et al., 2020). The United Nations Food and Agriculture Organisation aims to support increased innovative technologies in the food and agricultural systems (SDG 2). While disruptive technologies significantly cause the increasing usage of energy and gas emissions, it is significant to investigate sustainable and greener energy and more energy-efficient solutions, especially in supply chains (Mondejar et al., 2021). Thus, affordable and clean energy, one of the SDGs, can be achieved by digital technologies adapted from sustainable perspectives (SDG 7). Besides, disruptive technologies encourage community social and economic development by creating new job opportunities that can contribute to decent work and economic growth (SDG 8) (Sharma et al., 2021). Another remarkable contribution of disruptive technologies is increasing collaboration between stakeholders such as suppliers, producers, distributors, and customers because of the increasing interactions between them in the supply chains and enhancing synchronized decisions and actions in whole processes (Mhlanga, 2021). Therefore, this study aims to investigate the role of disruptive technologies in SDGs in

supply chains. Firstly, the relationships between disruptive technologies and SDGs have been given deep insight by providing systematic approaches to achieve this purpose. Then, how different disruptive technologies can achieve SDGs, especially in supply chains, to deal with the environmental, social, and economic impacts will be discussed.

2 The Importance of Integration of SDGs and Disruptive Technologies

Sustainability is defined as maintaining the ability of people to be permanent, as well as ensuring the continuity of diversity and productivity (Dinis et al., 2022). According to the Brundtland Report (1987), sustainable development is defined as "development that meets the needs of the present without compromising the ability of future generations to meet their needs" (UN, 2019). By the most popular definition, sustainable development is a way of managing our planet so that the well-being of modern humans does not mean limiting the extent to which future generations can meet their needs (Sanchez-Planelles et al., 2022).

In this context, the SDGs are a call to action consisting of 17 universal goals applicable to countries at different levels of development (Voola et al., 2022). SDGs reveal the relationships and common aspects between complex problems (Zakari et al., 2022). The goals cover issues to be considered at all scales and levels and aim to improve the living conditions of present and future generations worldwide through practices from the local to the global (Bohnsack et al., 2022). SDGs are briefly summarized in Table 1.

The common goal of the 17 different purposes of the United Nations for sustainable development is to provide environmental, social and economic development (Argyroudis et al., 2022). Moreover, one of the most important benefits of implementing SDGs is the reduction of impacts on environmental, social, and economic sustainability (Dantas et al., 2021). Disruptive technologies are needed to reduce sustainability effects and implement the SDGs. Disruptive technologies are vital in realizing SDGs (El-Haddadeh et al., 2021). Moreover, these technologies contribute to sustainability by optimizing resource use and cooperation between operations (Secundo et al., 2022). Big data, IoT, network technologies, AI, and cloud systems are some of the leading technologies in this field (Pigola et al., 2021). In addition, Blockchain technology increases transparency in supply chain management, food safety and sourcing, making the process more accurate (Chandan et al.,

Table 1 17 SDGs and explanations

SDG(s)	Explanations
No Poverty (SDG 1)	It means eliminating all forms of poverty everywhere and for all, ensuring equality in access to essential resources and financial services, and developing development strategies that are both politically and financially inclusive Krannich and Reiser (2021)
Zero Hunger (SDG 2)	It means ending hunger and malnutrition, ensuring everyone has sufficient food throughout the year, and implementing resilient agriculture practices Voola et al. (2022)
Good Health and Well-Being (SDG 3)	It means ensuring a healthy life and promoting well-being for all ages Zakari et al. (2022)
Quality Education (SDG 4)	It means providing quality education for all and promoting lifelong learning opportunities Voola et al. (2022)
Gender Equality (SDG 5)	It means achieving gender equality and empowering all women and girls Zakari et al. (2022)
Clean Water and Sanitation (SDG 6)	It means ensuring everyone can access safe and healthy drinking water, protecting aquatic ecosystems, managing resources well, and promoting water efficiency Krannich and Reiser (2021)
Affordable and Clean Energy (SDG 7)	It means increasing investment in clean energy sources such as solar, wind and thermal, developing infrastructure and technology, maintaining cost standards, and promoting energy efficiency so everyone can access renewable energy Zakari et al. (2022)
Decent Work and Economic Growth (SDG 8)	It promotes inclusive, sustainable economic growth, productive employment and decent work for all Krannich and Reiser (2021)

(continued)

Table 1 (continued)

SDG(s)	Explanations
Industry, Innovation and Infrastructure (SDG 9)	It means building resilient infrastructures, supporting inclusive and sustainable industrialization and strengthening innovation UN (2019)
Reduced Inequalities (SDG 10)	It refers to the reduction of the elements of inequality within and between countries Voola et al. (2022)
Sustainable Cities and Communities (SDG 11)	It means making cities and settlements inclusive, safe, resilient and sustainable UN (2019)
Responsible Consumption and Production (SDG 12)	It means providing sustainable consumption and production constraints Krannich and Reiser (2021)
Climate Action (SDG 13)	It means urgently combating climate change and its effects Krannich and Reiser (2021)
Life Below Water (SDG 14)	It means protecting the marine and coastal ecosystems against pollution, protecting the living life in the waters, and using all resources sustainably UN (2019)
Life on Land (SDG 15)	It means protecting terrestrial ecosystems (forests, mountains, wet and dry areas) and arable land, stopping deforestation, and reducing the loss of natural habitats and biodiversity Voola et al. (2022)
Peace, Justice and Strong Institutions (SDG 16)	It means ensuring everyone's access to justice and establishing effective, accountable, inclusive institutions at all levels Krannich and Reiser (2021)
Partnerships for the Goals (SDG 17)	It means strengthening the means of implementation and bringing vitality to global cooperation for sustainable development Voola et al. (2022)

Source Authors

2023). Therefore, it is one of the critical technologies in achieving the objectives of the SDGs.

3 Proposing Conceptual Framework for Matching Disruptive Technologies with SDGs in Supply Chains in Terms of Impacts on Sustainability

A conceptual framework is presented in this section to match digital technologies with considering their impacts on sustainability. SDGs include interrelated components, and their impacts must be handled holistically and systematically (Merino-Saum et al., 2018). Thus, a conceptual framework is developed using a causal loop diagram by considering sustainability impacts' causality relations.

Disruptive technologies are classified into nine different categories. AI, big data and cloud technologies, blockchain, radio frequency identification (RFID) and remote sensors, Augmented Reality/Virtual Reality (AR/VR), robotics, IoT, drones and autonomous vehicles, additive manufacturing & 3D printing. According to this table, AI can apply to achieving SDGs 2, 3, 5, 8 and 16 (Behl et al., 2021; Pigola et al., 2021). AI simplifies work by systematizing repetitive tasks, increasing efficiency and productivity across different industries such as food, healthcare and manufacturing. Automating administrative tasks can track medical records; AI simplifies work by systematizing repetitive tasks, increasing efficiency and productivity across different industries such as food, healthcare and manufacturing. By automating administrative tasks, it can track the records related to critical products and processes. Big data & cloud technologies can be implemented to succeed in SDGs 1, 3, 5, 9, 10, 11, 15, and 16 (Dantas et al., 2021). Big data technologies can be used to develop predictive models to anticipate the impacts of potential risks by collecting data on social components. Data-driven actions are critical to determine improvement areas in society. Accessing adequate data on and tracking hidden patterns in disparities is required for crucial business decisions. Blockchain is a vital technology for the accomplishment of SDGs 6, 7, 9, 10, 11, 12, 13, 14, and 15 (Parmentola et al., 2022; de Viliiers et al., 2021). Moreover, blockchain technology enables various green digitization applications by providing a reliable, transparent, decentralized database. Factors such as reduced data manipulation, increased collaboration opportunities, and more transparent management of records increase the potential of blockchain technology to contribute to sustainability goals. SDGs 2,11, and 14 can be achieved with RFID and remote sensors (Benyam et al., 2021; Zhou et al., 2022). RFID and remote sensors can improve resource efficiency by tracking and identifying operations and processes to increase quality and traceability. It also can enhance human

health and well-being by reducing waste pollution and improving data accuracy and availability in the entire supply chain. AR/VR can be implemented for SDGs 4 and 11(Dantas et al., 2021; Hořejší et al., 2020). It helps develop employees' technical skills by providing training and enhancing the learning experience. Besides, Robotics is required to provide SDGs 2, 8, 9, 13, and 14 (Pan et al., 2018; Secundo et al., 2022). It ensures increased productivity, consistency and quality in the operations by handling responsibilities that are challenging for humans. SDGs 6, 7, 9 and 13 can be achieved with the IoT (de Viliiers et al., 2021; Secundo et al., 2022). The IoT can enable real-time data and resource visibility. It provides data-driven insights for making rapid decision-making decisions—improved operational efficiency. In addition, drones and autonomous vehicles can be used for SDGs 3, 14 and 15 (Benarbia & Kyamakya, 2021). Autonomous drones provide a significant increase in efficiency and productivity in various industries. Thanks to their advanced sensors and imaging capabilities, these drones can collect data quickly and accurately and monitor large areas more effectively. Valuable data through autonomous drones can provide remote monitoring in hazardous places in the urban. Finally, Additive manufacturing & 3D Printing is crucial for achieving SDG 9 and SDG 17 (Malik et al., 2022; Secundo et al., 2022). Additive manufacturing technologies are critical for material and resource efficiency and production flexibility. Sustainability impacts by matching disruptive technologies & SDGs are presented in Table 2.

This framework shows that digital technologies match SDGs and their impacts on sustainability. SDGs consist of related parameters, and their impacts need to be handled systematically. SDGs and system thinking are closely related concepts that offer a holistic and interconnected approach to addressing global challenges and advancing sustainability. System thinking helps to comprehend, analyse, and manage relationships and dependencies among numerous aspects of sustainable development, while the SDGs provide a set of globally agreed-upon goals and targets. System thinking recognizes that the SDGs are interrelated and interdependent. Changes or actions in one area can have ripple effects across multiple goals and targets. It emphasizes the need to consider the entire system rather than focusing solely on individual goals. This approach helps identify leverage points where interventions can have the most critical impact across multiple SDGs. System thinking recognizes that the SDGs are profoundly interconnected and achieving one goal often depends on the success of others. For example, addressing clean energy (SDG 7) contributes to climate action (SDG 13). Besides, industry, innovation and infrastructure (SDG 9) can positively impact the health (SDG 3) and quality education (SDG 4), decent

Table 2 Proposed sustainability impacts by matching disruptive technologies & SDGs

Disruptive Technologies	SDGs	Impact	References
Artificial Intelligence	SDG 2	Undernourishment	Pigola et al. (2021); Behl et al. (2021)
	SDG 3	Lack of medicine availability	
	SDG 5	Discrimination	
	SDG 8	Lack of productivity	
	SDG 16	Lack of intelligent and low-carbon cities	
Big Data & Cloud Technologies	SDG 1	Lack of regular observations for income levels	Dantas et al. (2021)
	SDG 3	Lack of health expenditure	
	SDG 5	Lack of tracking gender equality	
	SDG 9	Lack of R&D researchers	
	SDG 10	Increase in migration level	
	SDG 11	Lack of urban air pollution	
	SDG 15	Increase in wildlife poaching	
	SDG 16	The threat to public safety	
Blockchain	SDG 6	Inefficient water usage	Parmentola et al. (2022); de Viliiers et al. (2021)
	SDG 7	Lack of electricity access	
	SDG 9	Intensity of CO_2 emissions	
	SDG 10	Income growth inequality	
	SDG 11	Higher risks of solid waste	
	SDG 12	Huge global food loss	
	SDG 13	Disaster risks	
	SDG 14	Insufficient research for marine technologies	
	SDG 15	Lack of biodiversity planning	

(continued)

Table 2 (continued)

Disruptive Technologies	SDGs	Impact	References
RFID & Remote Sensors	SDG 2	Food Insecurity	Zhou et al. (2021); Benyam et al. (2021)
	SDG 11	Lack of prediction of transportation access	
	SDG 14	Lack of prediction of marine pollution	
Augmented Reality/ Virtual Reality	SDG 4	Lack of accessibility to further education	Dantas et al. (2021); Hořejší et al. (2020)
	SDG 11	Lack of urban planning	
Robotics	SDG 2	Lack of sustainable production	Secundo et al. (2022); Pan et al. (2018)
	SDG 8	Higher unemployment rate	
	SDG 9	Lack of observation of rural road access	
	SDG 13	Lack of integration climate change policies	
	SDG 14	Lack of marine ecosystem observation	
Internet of Things	SDG 6	Insufficient control of water resources	de Viliiers et al. (2021); Secundo et al. (2022)
	SDG 7	Lack of access to renewable energy	
	SDG 9	Less manufacturing value	
	SDG 13	Lack of climate change actions	
Drones & Autonomous Vehicles	SDG 3	Health and safety issues	Benarbia and Kyamakya (2021)
	SDG 14	High ocean acidification	
	SDG 15	More degraded land	
Additive manufacturing & 3D Printing	SDG 9	Lack of R&D spending	Secundo et al. (2022); Malik et al. (2022)
	SDG 17	Lack of sustainable technologies	

Source Authors

work and economic growth (SDG 8) and partnerships for the goals (SDG 17) and system thinking, particularly when combined with causal loop diagrams, provide a powerful framework for understanding, analysing, and addressing complex global sustainability challenges.

Thus, the conceptual framework is developed using a causal loop diagram by focusing on the causality relations of sustainability impacts. The causal loop diagram is presented in Fig. 1. Detailed explanations of the causal loop diagram are also discussed in the discussion section.

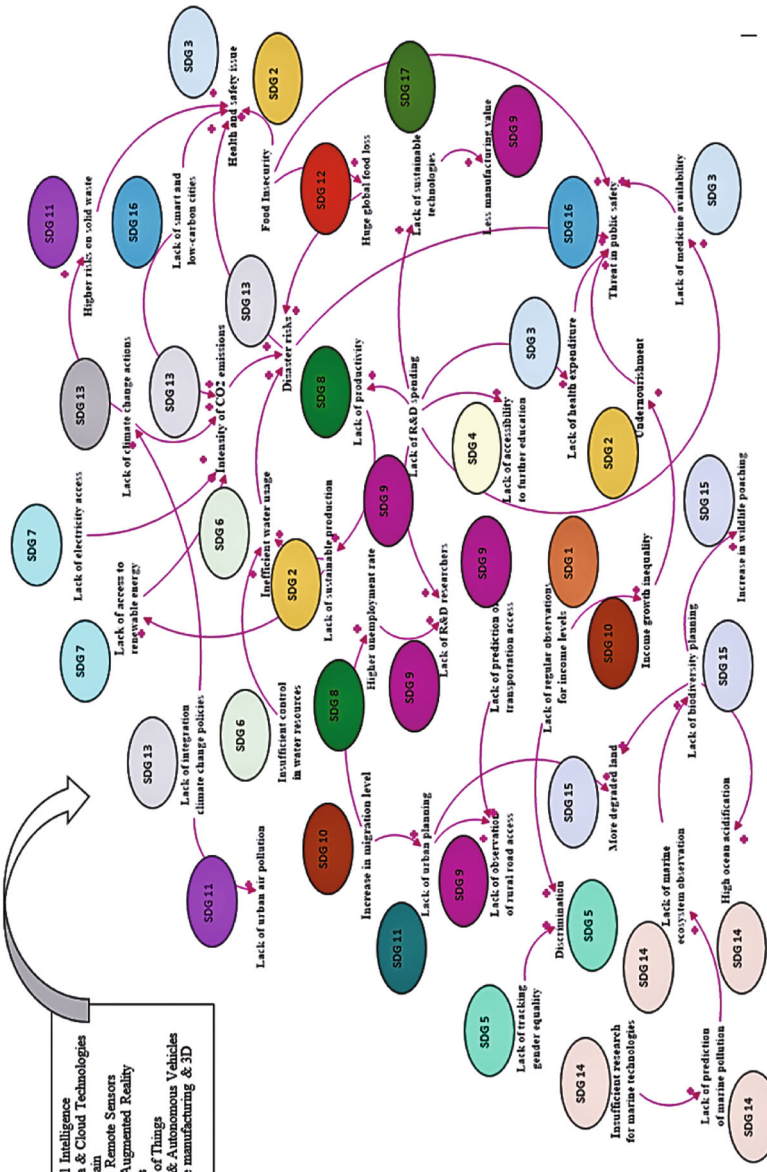

Fig. 1 Casual loop diagram (*Source* Authors)

4 Discussion and Conclusions

A causal loop diagram that shows proposed sustainability impacts by matching disruptive technologies & SDGs is presented in Fig. 1. As indicated in Fig. 1, the lack of climate change actions is directly influenced by the lack of climate change policies. Measures on climate change are critical to decreasing the higher risk of solid waste and the intensity of carbon emissions (Sharma et al., 2021). Besides, the power of carbon emissions is affected by different variables, such as lack of access to renewable energy, lack of electricity access, and lack of intelligent and low-carbon cities (Sun et al., 2022). Affordable and clean energy, one of the SDGs, can be accomplished through digital technologies adapted from sustainable perspectives (SDG 7).

In addition, health and safety issues have been caused by higher risks of solid waste, food insecurity (SDG 2), lack of intelligent and low-carbon cities (SDG 16) and disaster risks (SDG 13). Huge global food loss (SDG 12) also increases disaster risks (Ali et al., 2019). Moreover, lack of R&D spending (SDG 9) is a critical parameter that has a significant impact on health expenditure (SDG 3), productivity (SDG 8), sustainable technologies (SDG 17) and lack of researchers (Kocak & Alnour, 2022). Furthermore, inadequacy in urban planning (SDG 11) and higher unemployment rate (SDG 8) are also relationships between lack of R&D spending (SDG 9). Government incentives are significant for increasing health expenditure, productivity, and sustainable technologies (Guo et al., 2018; Song et al., 2022). In addition, training and education are needed to deal with the higher unemployment rate (Fenitra et al., 2022; Liu et al., 2021).

Insufficient research for maritime technologies is related to a lack of prediction of maritime pollution (SDG 14), and it causes a lack of marine ecosystem observation (SDG 14) (Del Giudice et al., 2022). Moreover, the lack of marine ecosystem observation (SDG 14) is caused by a lack of biodiversity planning and high ocean acidification (SDG 14) (Peña et al., 2022). In addition, inadequacy in urban planning and lack of biodiversity planning increase degraded land (SDG 15). These objectives can be achieved by using blockchain technology in supply chain processes as this technology enables waste seas or harmful products to be traced and traced throughout the supply chain.

Public safety threats are caused by disaster risk, food insecurity, lack of health expenditure, lack of medicine availability, and undernourishment (Khan et al., 2022). Effective policies for urban planning are required to protect public safety and health and to minimize environmental impacts on supply chains (Tseng et al., 2022; Yu et al., 2022). In addition, disruptive technologies warrant SDGs by improving the accuracy and reliability of infrastructure. It provides effective urban planning (Mondejar et al., 2021).

Thus, adopting disruptive technologies is needed to accomplish green digitalization to SDGs from a disruptive technologies perspective (Kayikci et al., 2022; Voola et al., 2021).

Moreover, a positive relationship exists between the lack of tracking gender equality and discrimination (SDG 5) (Khamzina et al., 2022). In addition, the lack of regular observation of income level (SDG 1) is also caused by increasing discrimination and income growth inequality. Thus, when the lack of regular observations for income level rises, income growth inequality increases (Langnel et al., 2021; Rosés & Wolf, 2021). This leads to an increase in undernourishment. Finally, it causes an increasing threat to public safety (Khan et al., 2022; Sam et al., 2021). An improved tracking system supported by digital technologies is crucial to eliminate undesirable conditions and reach SDGs (Dwivedi et al., 2022; Khan et al., 2021).

In conclusion, this study aims to analyse disruptive technologies' role in achieving SDGs in supply chains. To achieve this purpose, a system thinking approach is proposed in this study. Firstly, the mentioned sustainability impacts by matching disruptive technologies & SDGs are interrelated, and the relationships between disruptive technologies and SDGs have been investigated holistically. Then, how different disruptive technologies can achieve SDGs, especially in supply chains, to deal with the environmental, social and economic impacts are presented. Then, a conceptual framework is developed to consider their effects on sustainability. Therefore, this study is crucial to analyse relationships between disruptive technologies and SDGs and their sustainability impacts by presenting the bigger picture. SDGs and their sustainability impacts are interrelated, and significant relationships exist. The disruptive technologies mentioned above should be handled to analyse sustainability impacts systematically and holistically.

As a future work, the framework can be quantified by a system dynamics modelling approach. The proposed framework can be improved by adding numerous indicators related to SDGs. The proposed framework can be applied in the different sectors specifically.

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5

Digital Supply Chain Activities and Sustainable Performance—The Mediating Role of Supply Chain Capabilities

Changiz Valmohammadi and Mahdieh Ahmad Amouei

1 Introduction

Today, with the digital revolution in the industry 4.0 age (I4.0), new technologies based on information and communication technologies (ICT) have received much attention in the research and practice communities related to supply chain management (SCM). ICT can support it by integrating processes and information across the supply chain (SC) (Barbieri et al., 2021). Also, ICT development is causing digital disruption that can affect supply chains. Thus, traditional supply chains need new manufacturing strategies to improve themselves (Queiroz et al., 2021). The whole structure, managerial components, processes, and SC flows are changing because of the customized and emergent markets that require quick responses (Garay-Rondero et al., 2019). The digitalization of SC processes can be an answer to address SC concerns (Farahani et al., 2017). SCM with the I4.0 approach includes a new stage of development so that using digital technologies, SC activities can be largely automated (Hofmann et al., 2019). Digital Supply Chain (DSC) can support different parts of the SC such as new production models,

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customer relationships and experiences, transportation modes, etc., based on real-time information exchange (Queiroz et al., 2021). The DSC can reduce any bullwhip effects by increasing the visibility of material flows along the SC (Zekhnini et al., 2022). According to Hahn (2020), digital technologies such as autonomous guided vehicles (AGVs), 3D printing, and robots are expected to complement I4.0 SC systems.

On the other hand, in the present era, due to increasing public awareness of the importance of sustainability, businesses that include environmental and social goals in their strategies, are given more attention and appreciation. For this reason, companies try to change their performance criteria from economic purposes to sustainability goals (To et al., 2021).

According to Chiappetta Jabbour et al. (2020), one of the components of supply chains in this era is sustainability. So that without considering the dimensions of sustainability, the discussion in the field of supply chains remains incomplete. The increased global pressures on environmental, social, and human rights issues have led companies to increase their responsibility in managing their supply chain (Ebinger & Omondi, 2020). To achieve sustainable performance and maintain their competitive advantage, they must consider three dimensions of sustainability (economic, social, and environmental). So the challenge for companies is not only to meet the needs of their stakeholders by achieving positive economic performance but also to find the right balance among the three dimensions of sustainability (Cherrafi et al., 2017).

Di Vaio and Varriale (2020) mention that knowing and benefiting from the advantages of using digital technologies in the supply chain and paying attention to their sustainability effects can help companies to achieve a sustainable supply chain. Digital technologies such as artificial intelligence (AI) can create innovative business models that increase sustainability in all three dimensions (Sharma et al., 2020). For most companies, focusing on sustainability can have consequences beyond SCM sustainability (Carter et al., 2019). For this reason, understanding how production practices affect sustainable performance is critical (Abdul-Rashid et al., 2017). Thus, it can be said that DSCs have gained commercial importance for developing a competitive and sustainable SC (Edwin Cheng et al., 2022).

Another point that has been considered in this study is the role of using supply chain capabilities (SCCs) in DSC management to achieve sustainable performance. By using SCCs, companies can improve their SC performance and outperform their competitors (Agyabeng-Mensah et al., 2020). These capabilities can increase the market share and sales volume of the companies by accelerating the delivery of products to consumers efficiently, as well as

permit SC partners to respond effectively to customer enquiries (Rajaguru & Matanda, 2019).

Given the high potential for SC development, SCM researchers must examine the effects of DSCs on the main theoretical concepts in SCM (Stank et al., 2019). Accordingly, in this study, the mediating role of SCCs in DSC activities and sustainable performance is investigated based on the research of (Ahmad Amouei et al. (2023a, 2023b). Then, a conceptual framework is presented.

The rest of this chapter is as follows: Sect. 2 discusses the theoretical background and development of hypotheses based on a literature review. In Sect. 3, the research methodology as well as data collection and analysis are presented. Followed by Sect. 4 which presents discussion of the findings. Finally, Sect. 5 wraps up the chapter with theoretical and practical contributions of the study and research limitations as well as recommendations for future research.

2 Theoretical Background and Research Hypotheses

2.1 Digital Supply Chain Activities

The digital supply chain means managing a set of related activities in SC processes using new technologies (Nasiri et al., 2020). DSC can increase the quantity and quality of information, communication, and collaboration by using digital platforms, thus increasing reliability, efficiency, and agility (Büyüközkan & Göçer, 2018). For example, using a blockchain in a manufacturing company makes it possible to improve the tracking speed of problems related to a part, product, or supplier. Using blockchain in the SC also helps to increase product safety, reduce maintenance costs, and improve service levels (Cole et al., 2019).

In this chapter, based on the research of Ahmad Amouei et al. (2023a), four dimensions are considered for the main activities of DSC, which are as follows: Digital supplier; is the integration of the company with stakeholders in a supply network, so that competition gives way to cooperation, and increases transparency and security in transactions and relationships (Korpela et al., 2017). These transactions and relationships can be created using digital technology and based on various data such as on-time delivery, supplier performance, improved sourcing decisions, added value, etc. (Farahani et al., 2017). Digital manufacturing means using digital technologies such as cyber

physics systems (CPS) and IoT to manage, monitor, control product lifecycle, and decentralized production (Queiroz et al., 2021). Digital logistics refers to the more efficient use of resources by using digital technologies to make transportation activities smarter such as AGVs (Queiroz et al., 2021). This increases vertical and horizontal integration between SC partners (Kayikci, 2018). Digital customers mean creating intelligent communication between companies and their customers aimed at increasing customer information and enhancing the accuracy of this information (Li et al., 2016).

Support activities are activities that are done to support main activities. According to Ahmad Amouei et al. (2023a), these activities in the DSC include three dimensions: Digital performance which means using digital technologies to support the organization's strategies and mission toward achieving a competitive advantage (Gawankar et al., 2020). Digital technology includes those enabler technologies that are related to DSC, such as IoT, blockchain, AI, big data analytics, CPS, and cloud computing (CC) (Attaran, 2020; Büyüközkan & Göçer, 2018). The company's human resources include knowledge, experience, problem-solving skills, business intelligence (BI), leadership qualities, etc. (Jeble et al., 2018). A digital human resource is a holistic approach. This approach includes all the human resources management functions such as, selection, recruitment, training and development, performance appraisal, etc. (Salim & Sulphey, 2021).

2.2 Sustainable Performance

Sustainability refers to the triple-bottom-line concept defined by Elkington (1994). This term means balancing economic, environmental, and social aspects from the company's perspective (Abdul-Rashid et al., 2017). In this chapter, according to Ahmad Amouei et al. (2023a), three dimensions have been considered for sustainable performance, including Economic performance, which means measuring performance based on economic growth while increasing life's quality and maintaining the environment (Abdul-Rashid et al., 2017). Companies with economic sustainability should strive to promote economic efficiency and economic development (Pathak et al., 2019). Environmental performance is the optimal use of efficient sustainable energy sources and ensuring the renewability of resources used during production with minimal CO_2 emissions (Abdul-Rashid et al., 2017). Environmental sustainability must preserve environmental integrity for present and future generations by reducing pollution, promoting resource conservation, and preventing climate change (Junge & Straube, 2020; Pathak et al., 2019). Social performance is defined as the success of companies in

maintaining and improving life's quality, taking into account environmental aspects (Abdul-Rashid et al., 2017).

2.3 Supply Chain Capabilities

Supply chain capabilities mean the ability of a company to facilitate SC activities by identifying, using, and absorbing internal and external resources/information (Asamoah et al., 2021). Companies can use SCCs to gain a competitive advantage (Yu et al., 2018). In this chapter, according to Yeniyurt et al. (2019), we have considered four dimensions for SCCs which are: Information exchange is the company's ability to share knowledge/information with its SC partners (Mehdikhani & Valmohammadi, 2019; Yeniyurt et al., 2019). Knowledge sharing means learning inside and outside the company's work area (Valmohammadi & Ahmadi, 2015). Coordination means firm-internal and inter-organizational coordination (Hautala-Kankaanpää, 2022). It indicates the extent to which a company can effectively coordinate various SC activities and transactions with its partners (Asamoah et al., 2021). This coordination is achieved by creating, sharing, and applying knowledge area (Valmohammadi & Ahmadi, 2015). Integration of activities refers to establishing strategic relationships with SC partners to align the company's activities and technologies with them (Asamoah et al., 2021). SC responsiveness is defined as the measure of the dynamic ability of SC partners to respond to environmental change, react to changes in demand and supply, and compete effectively (Yeniyurt et al., 2019; Yu et al., 2018).

2.4 Research Hypotheses and Conceptual Model

Many studies have been conducted on using digital technologies in SC activities (main and support) and their impact on sustainable performance. Khan et al. (2020), for instance, found a significant relationship between the integration process, economic performance, and supplier integration, using CC technology. The studies of Gillani et al. (2020) revealed that digital manufacturing technologies have a positive effect on operational performance. Kayikci (2018) studied the digital logistics process, its benefits, and its impact on sustainability. Using a logistical perspective, the author proposed a new approach to understanding sustainability concepts. According to Junge and Straube (2020), the introduction of digital transformation technologies (DTT) in logistics has a moderate positive impact on social and environmental sustainability. Mihardjo et al. (2019) reported that the digital

customer experience could have a positive impact on sustainable performance through brand and SC image mediation. Eslami et al. (2021) revealed that I4.0 digital technologies enhance financial performance by affecting SC agility. These findings could help understand the role of digital technologies in achieving the SC competitive advantage. According to Sharma et al. (2020), using machine learning in the agricultural SC could have a positive effect on sustainable performance. Gawankar et al. (2020), and Edwin Cheng et al. (2021), reported the positive effect of using big data analysis on chain performance and subsequently, its impact on firm performance. Salim and Sulphey (2021) concluded that there is a positive and significant relationship between SCM performance and the digitization of human resource information. Saryatmo and Sukhotu (2021)reported that the DSC could have a positive impact on the dimensions of operational performance (quality, efficiency, and cost reduction). By summarizing the studies conducted and based on the research of Ahmad Amouei et al. (2023a), the following hypotheses are put forth.

H1: DSC Main Activities have a positive impact on Sustainable Performance.

H2: DSC Support Activities have a positive impact on Sustainable Performance.

Farahani et al. (2017) defined six dimensions for DSC management. They considered digital performance measurement and digital technology as support activities. They also identified main activities, including digital suppliers, digital manufacturing, digital logistics, and digital customers. In another study, Smidt and Jokonya (2022), presented a schematic representation of the vertical integration of the digital agriculture value chain and available DT services, showing that digital technologies such as AI, IoT, and blockchain can influence supplier and customer practices. López et al. (2022), showed the positive impact of using digital technologies such as RFID on manufacturing and logistics in the study of digital value chain restructuring and labor process transformations in the fast-fashion sector. Considering the relationships obtained in the studies conducted and based on the research of Ahmad Amouei et al. (2023a), the third hypothesis is proposed as follows:

H3: *DSC Support Activities have a positive impact on DSC Main Activities.*

Although few studies have been done on the impact of DSC activities on SCCs, we can refer to the results obtained by Yu et al. (2018). They reported

that data-driven SC had a significant positive effect on SCCs. Also, Abdul-Rashid et al. (2017) revealed that the sustainable manufacturing process had the most significant impact on sustainable performance. Therefore, the following hypotheses are suggested:

H4: *DSC Main Activities have a positive impact on SC Capabilities.*

H5: *DSC Support Activities have a positive impact on SC Capabilities.*

Several studies have been conducted to investigate the relationship between SCCs and sustainable performance. These include Backstrand and Powell (2021) and Oh et al. (2019). These studies indicated a significant relationship between SCCs and firm performance. According to Cheung et al. (2018), the SCC as an enabler can increase SC performance through information technology. Yeniyurt et al. (2019) defined IT resources, innovation, and SCCs as drivers of business performance. To et al. (2021) stated that sustainable SCM and dynamic SCCs have positively affected the firm's sustainable development. They considered sustainable development, including economic efficiency, social efficiency, and environmental performance. Therefore, the following hypothesis is suggested:

H6: *SC Capabilities have a positive impact on Sustainable Performance.*

In many SC studies, SCCs have been considered a mediator (Asamoah et al., 2021; Hong et al., 2019; Nandi et al., 2020; Rajaguru & Matanda, 2019; Tigga et al., 2021). Queiroz et al. (2021), for instance, argued that the digitization of supplier, manufacturing, transportation, customer, human resource, and digital technology enablers could have a positive impact on SCCs and thus improve SC performance. Also, Hautala-Kankaanpää (2022), in his study investigated the impact of digital platforms and SCCs on operational performance. In this study, the author showed that SCCs mediate the relation between digital platforms and operational performance. Based on these findings, it can be guessed SC Capabilities can mediate the relationship between DSC activities and sustainable performance. Therefore, the following hypotheses are proposed:

H7: *SC Capabilities mediate the relationship between DSC Main Activities and Sustainable Performance.*

H8: *SC Capabilities mediate the relationship between DSC Support Activities and Sustainable Performance.*

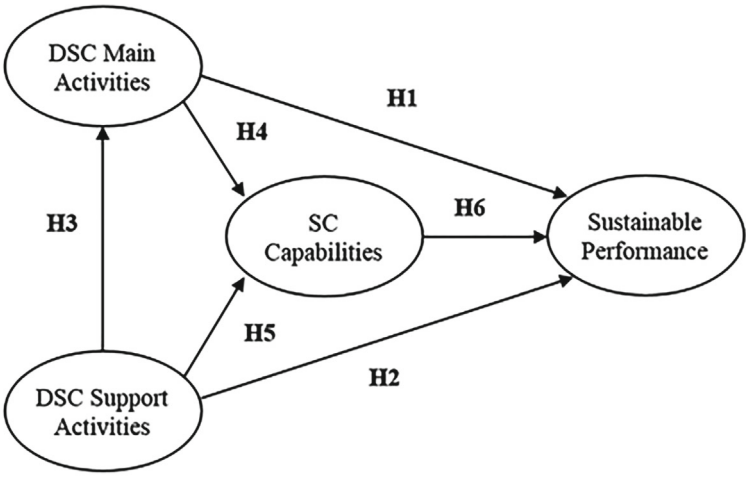

Fig. 1 Proposed conceptual model of research (*Source* Authors)

Accordingly, based on the mentioned hypotheses our proposed conceptual model is presented in Fig. 1.

3 Research Methodology

To achieve this chapter's goal, quantitative methods were utilized. Thus, hypotheses were tested by collecting and analyzing survey data. All data were analyzed using IBM SPSS 26 and Smart PLS2.

3.1 Data Collection

Iran is one of the largest economies in the Middle East. The country has some of the largest oil and natural gas reserves in the world. Therefore, Iran's economy is dominated by the oil and gas sector and the noticeable governmental presence in the manufacturing sector. As a result, economic activity and government revenues to a large extent depend on oil revenues. On the other hand, due to the geographical and strategic location of Iran, this country can be considered an entry point to other upcoming regional markets in the Middle East and Central Asia. Nevertheless, the policies of this country and the trade sanctions imposed by the USA and other countries have caused problems in Iran's economy.

Due to the growing restrictions on the one hand and the eagerness of the Iranian business community to cooperate with foreign businesses and investments on the other hand, Iranian authorities have defined new strategies to reform the market in recent years. The purpose of these strategies is to reduce Iran's economic dependence on oil revenues and increase manufacturing output in the private sector. To achieve this goal, it is necessary to use high-tech and knowledge-based solutions to strengthen the economic and manufacturing sectors (https://iran.um.dk/en/the-trade-council/iran-as-a-market). As Valmohammadi (2013) argues due to the important role of SCM practices in enhancing the competitive advantage of Iranian manufacturing companies, most manufacturing industries such as petrochemical and automotive companies consider SCM from a strategic point of view. Therefore, in this study, Iranian manufacturing companies were selected as the statistical population. The list of these companies was obtained from the website of the Tehran Stock Exchange (TSE.ir). Thus, the respondents were manufacturing companies in various fields of activity to ensure generalizability. In this research, with the experts' recommendation, top managers with at least two years of experience in their position were selected as the major respondents. Thus, ensuring that they have enough experience in making decisions in their organizations. It can be said that these managers are familiar with different aspects of the supply chain and are actively involved in SC processes.

350 questionnaires were sent to them through the companies' emails and websites, along with explanations about the research objectives and the confidentiality of the answers. Returned questionnaires were reviewed for missing data, and incomplete questionnaires were removed. Finally, 126 correctly completed questionnaires were received, yielding a response rate of 36%. As Haddadi Harandi et al. (2019) by literature argue to minimize self-report bias in the data, the respondents were informed that it is not necessary to write their names and based on the recommendation of Baruch and Holtom (2008), the non-response bias test (wave analysis) was performed. Thus, the responses of early and late waves of returned surveys were compared, based on the assumption that the opinions of late respondents are representative of the opinions of non-respondents. The results of the student's t-test indicated that there are no statistically significant differences between early-wave and late-wave groups; therefore, non-response bias was not a problem (Prajogo & McDermott, 2011). Also, to minimize the likelihood of common method variance in the design of the questionnaire, we followed the procedure used by Haddadi Harandi et al. (2019).

In the first step, multiple-item scales were used to measure the constructs and scattered questions about the predictor and criterion variables throughout the instrument. Second, Harman's single-factor test was conducted and found that no single factor or a general factor accounted for the majority of the variance in the measures. Ultimately, the effects of a single unmeasured latent method factor added to our measurement model were analyzed by loading all items originating from the same informant onto both the method variable and its respective latent variable. The comparison of standardized parameter estimates with and without the method variable showed that the significance of the substantive relationship was not affected. Therefore, based on the obtained results, common method bias is not a concern.

The designed questionnaire for this research consisted of four parts. The measurements were mainly adapted from the research literature, and the required changes were made through interviews with nine SC experts. For the items to measure the main and support activities of DSC, Ahmad Amouei et al. (2023a), was used. To measure SC capabilities, Asamoah et al. (2021) and Yu et al. (2018) were adapted. It was also adapted from Abdul-Rashid et al. (2017), to measure sustainability performance. The items were then modified accordingly to suit the research field. All items were measured using a 5-point Likert scale (from "1 = strongly disagree" to "5 = strongly agree"). Then, using the opinion of experts, face validity was performed. Finally, after analyzing the preliminary studies, a scale of 63 items was obtained.

According to Bekaroglu and Danayiyen (2020), because the questionnaire was modified according to the present research, a preliminary test was performed to confirm its preliminary reliability and validity by relying on the new scale. According to Ghasemi and Valmohammadi (2018), the minimum number of samples to perform explanatory factor analysis (EFA) must be five times the variable number. Thus, in the pilot test, according to the 14 variables considered, 70 samples were selected. By examining Cronbach's alpha, 5 items with values less than 0.7 were removed (per 5, per 6, sup 5, sup 6, cus 5). Also, in the preliminary test, EFA was performed. As a result, 2 items (hum 5 = 0.4, man 5 = 0.1) with a factor-loading of less than 0.5 were removed (Gawankar et al., 2020). According to the results of the Kaiser-Meyer-Olkin test and acceptable values (KMO > 0.6 and Sig < 0.05) (Ghasemi & Valmohammadi, 2018), the adequacy of the test samples was confirmed. Finally, a questionnaire consisting of 56 items with four main components was obtained. (The final questionnaire is provided in Appendix A.) This questionnaire was used for the main sample, which included 126 participants. The information of the participants is presented in Table 1.

Table 1 Respondent and organization profile

Percentage (%)	Frequency	Category	Demographic variables
34	43	Graduate	Educational level
47	59	Postgraduate	
19	24	Doctorate	
14	18	CEO/general manager	Designation
25	31	Executive managers	
20	25	Middle manager	
33	42	Supervisors	
8	10	Other	
46	58	2–5	Years in current position
36.5	46	6–10	
17.5	22	> 10	
35	44	Auto and parts	Industry type
21	26	Manufacturing	
13.5	17	Machinery and industrial	
24.5	31	products	
6	8	Food and beverage	
		Packaging	
		Others	
21.5	27	$\leq$ 500	**Number of Employees**
38	48	501–1000	
32.5	41	1001–3000	
8	10	> 3000	
100	126	**Total**	

Source Authors

According to the information presented in Table 1, the highest percentage of respondents had a post-graduate degree (47%) and the lowest percentage had a doctorate (19%). Supervisors had the highest response rate (33%). Also, 46% of respondents had between 2 to 5 years of experience in their current position. Manufacturing companies in automobiles and auto parts had the highest percentage among other companies (35%). While companies producing pharmaceuticals and cosmetics, electronics, construction equipment, etc., had the lowest participation rate with 6%. 38% of respondents worked in manufacturing companies with 500 to 1000 employees, and only 8% worked in companies with more than 3000 employees.

3.2 Data Analysis

EFA is one of the advanced multivariate statistical techniques to achieve the scientific goals of research and assess the validity of the questionnaire. EFA is used to retain items that meet objectives and remove ambiguous

variables (Francis et al., 2021). In this research, EFA and Varimax rotation was performed for the first-order and second-order to extract the main dimensions in each construct. By forming the Varimax matrix, it could be stated that 56 questions used in the research questionnaire are categorized into 14 observed variables (see Appendix B, Table 10. According to the obtained results, it can be said that the observed variables are categorized by four factors (latent variable). Three observed variables (digital performance, digital technology, and digital human resources) determine the DSC support activities as an independent variable. Four variables (digital suppliers, digital manufacturing, digital logistics, and digital customers) determine the DSC main activities as a dependent variable of the research, as well as three variables (economic performance, environmental performance, and social performance) determine sustainable performance as a dependent variable. Four variables (information exchange, coordination, integration of activities, and SC responsiveness) determine SC capabilities as a mediating variable. All items used in the final questionnaire had Cronbach's alpha value in the range of 0.879–0.945, and the reliability of the questionnaire was highly acceptable (Table 2).

Table 2 Cronbach's α coefficients for each component

Construct	Sub-construct	Cronbach's α	Mean	Variance	N of Items
DSC support activities	Digital performance	0.930	17.651	6.71	4
	Digital technology	0.931	17.656	3.99	4
	Digital human resources	0.936	17.627	5.04	4
DSC main activities	Digital suppliers	0.915	17.302	4.80	4
	Digital manufacturing	0.945	17.659	5.33	4
	Digital logistics	0.929	17.611	5.30	4
	Digital customers	0.900	17.278	4.71	4
SC capabilities	Information exchange	0.924	17.476	5.18	4
	Coordination	0.879	17.452	4.14	4
	Activity integration	0.925	17.627	4.84	4
	SC Responsiveness	0.924	17.744	4.27	4
Sustainable performance	Economic performance	0.926	17.645	4.95	4
	Environmental performance	0.924	17.841	4.04	4
	Social performance	0.911	17.800	4.42	4
				Total	56

Source Authors

Table 3 Overview of EFA results

Bartlett's test			KMO index	EFA	
Sig	df	Chi-square			
0.000	66	1405.072	0.884	First- order	DSC support activities
0.000	3	120.330	0.672	Second-order	
0.000	120	1839.816	0.868	First-order	DSC main activities
0.000	6	147.867	0.760	Second-order	
0.000	120	1851.547	0.881	First-order	SC capabilities
0.000	6	193.176	0.809	Second-order	
0.000	66	1358.731	0.881	First-order	Sustainable performance
0.000	3	135.929	0.698	Second-order	

Source Authors

According to the values obtained from KMO and Bartlett tests, the validity of the questionnaire and the appropriate number of samples were confirmed (Ghasemi & Valmohammadi, 2018). These results are presented in Table 3.

According to Agyabeng-Mensah et al. (2020), the most appropriate and acceptable technique for testing various hypotheses in the social sciences is structural equation modeling (SEM), and partial least squares (PLS) has been widely used in SC research (Delic & Eyers, 2020; Gawankar et al., 2020). Thus, in this research, we used Smart PLS2 software for the PLS-SEM approach. One of the advantages of the PLS is the ability to simultaneously examine a set of related dependency relationships between latent constructs and measured variables. Also, the PLS method has fewer requirements on distribution and sample size compared to the analysis of covariance, does not require input data with normal-distributed, and provides reliable and consistent results (Khin & Ho, 2019).

On the other hand, according to Sarstedt et al. (2019), because each construct is measured reflectively, all relationships in the proposed conceptual model are of the reflective-reflective type. To analyze the data, we first examined the reliability and validity of the constructs with confirmatory factor analysis (CFA). Thus, the results were analyzed based on acceptable values for Cronbach's alpha (> 0.7) (Khin & Ho, 2019), mean–variance (AVE > 0.5) (Asamoah et al., 2021), and composite reliability (0.7 < CR < 0.9) (Dadashnejad & Valmohammadi, 2018). Also, the coefficient of determination (R2) was calculated. According to Asamoah et al. (2021), the explanatory power of the model concerning the values of 0.25, 0.50, and 0.75 for R2 are considered substantial, moderate, and weak, respectively. Thus, according to R2 obtained for DSC Main Activities (0.514), SC Capabilities (0.693), and Sustainable

Performance (0.703), explanatory power of the proposed model is moderate. These results are presented in Table 4.

The results of the Fornell and Larcker criteria displayed that AVE's square root of any construct was greater than the common variance of this and other constructs of the model (Dadashnejad & Valmohammadi, 2018). Thus, the divergent validity of the model is acceptable (Table 5).

The amount of factor-loading and cross-loading were calculated to check the reliability of the items. According to Valmohammadi et al. (2019), factor-loading values are greater than 0.7 and acceptable (Table 6).

According to Aslam et al. (2020), to ensure that hypothetical direct paths are supported, we examined path coefficients (β) and t-values for each of them. We used a bootstrapping technique with 5000 bootstrap samples with a 95% confidence interval. The results are presented in Fig. 2 and Table 7.

As recommended by Rajaguru and Matanda (2019) and Asamoah et al. (2021), the mediating effect of SCCs on the relationship between DSC activities on sustainable performance, by fitting the direct and indirect models, in the SEM was examined. To test whether the mediating effect of SC capabilities is significant, we calculated the z-value using the Sobel test. The results of the mediating effect are presented in Table 8. Given that the values obtained

Table 4 Construct reliability and validity

Cronbach's alpha	R^2	Composite Reliability	AVE	
0.797	0.514	0.868	0.622	DSC main activities
0.791	-	0.878	0.706	DSC support activities
0.840	0.693	0.893	0.677	SC capabilities
0.827	0.703	0.897	0.744	Sustainable performance

Source Authors

Table 5 Fornell-Larcker criterion

Sustainable performance	SC capabilities	DSC support activities	DSC main activities	
			0.789	DSC main activities
		0.840	0.717	DSC support activities
	0.823	0.801	0.733	SC capabilities
0.863	0.769	0.724	0.784	Sustainable performance

Source Authors

Table 6 Factor loadings and cross loadings

Sustainable performance	SC capabilities	DSC support activities	DSC main activities	
0.6065	0.6126	0.6335	**0.7602**	Suppliers
0.5973	0.5354	0.5402	**0.7903**	Manufacturing
0.6730	0.6040	0.5903	**0.8233**	Logistics
0.5904	0.5528	0.4845	**0.7788**	Customers
0.6718	0.7568	**0.8980**	0.6712	Performance
0.5651	0.5281	**0.7710**	0.5594	Technology
0.5833	0.7140	**0.8470**	0.5703	Human resource
0.6559	**0.7891**	0.6163	0.5252	Information
0.6525	**0.8400**	0.6397	0.6042	Coordination
0.5649	**0.7981**	0.6401	0.6208	Integration
0.6553	**0.8616**	0.7335	0.6593	Responsiveness
0.8330	0.7094	0.7064	0.6387	Economic
0.8834	0.6246	0.5879	0.7078	Environmental
0.8699	0.6532	0.5759	0.6818	Social

Source Authors

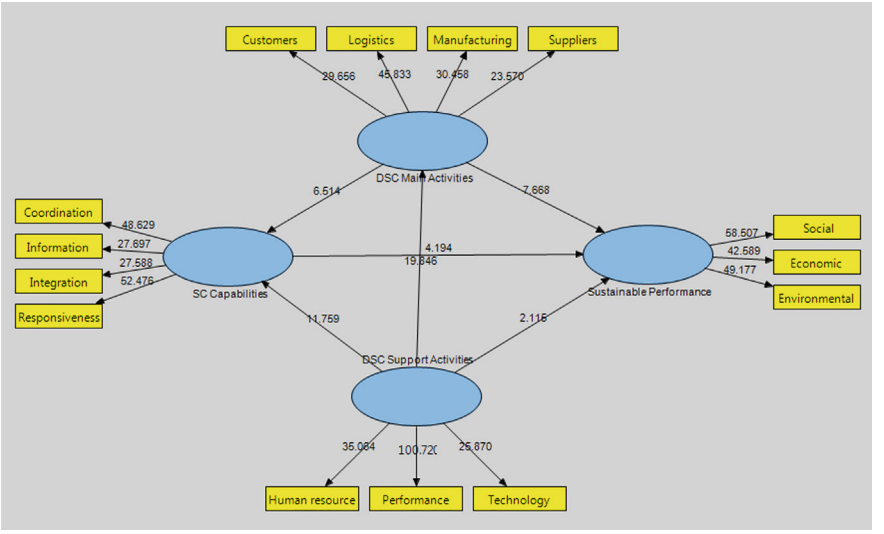

Fig. 2 Bootstrapping and t-statistic (*Source* Authors)

were greater than the accepted values (z-value > 1.96) (Nazari et al., 2021), it can be said that SCCs mediate partially the relationship between DSC main activities and sustainable performance, and similarly, SCCs mediate partially the relationship between DSC support activities and sustainable performance. According to Mehdikhani and Valmohammadi (2019), partial mediation

Table 7 The results of the research hypotheses

Decision	T-statistics	Standard Error	Path coefficient (β)	Path	Hypothesis
Supported	6.514	0.051	0.328	DSC Main Activities➔ SC Capabilities	**H4**
Supported	7.668	0.058	0.436	DSC Main Activities➔ Sustainable Performance	**H1**
Supported	19.646	0.037	0.717	DSC Support Activities➔ DSC Main Activities	**H3**
Supported	11.759	0.048	0.566	DSC Support Activities➔ SC Capabilities	**H5**
Supported	2.116	0.067	0.145	DSC Support Activities➔ Sustainable Performance	**H2**
Supported	4.194	0.081	0.334	SC Capabilities➔ Sustainable Performance	**H6**

Source Authors

means that in addition to a significant relationship between the mediating variable and the dependent variable, there is also, some direct relationship between the independent and dependent variables.

4 Discussion of Findings

The results of this research are based on examining the direct relationship (among DSC activities, SC capabilities, and sustainable performance) and the mediating relationship of SC capabilities (between DSC activities and sustainable performance). Based on the hypotheses H1 ($\beta = 0.436$, $t = 7.668$) and H2 ($\beta = 0.145$, $t = 2.116$), the main and support activities have a positive effect on sustainable performance. This is consistent with previous studies that had reported positive and significant effects of using digital technologies in SC activities on sustainable performance (Gillani et al., 2020; Eslami et al., 2021; Kayikci, 2018; Gawankar et al., 2020; Edwin

Table 8 The result of the mediation test

Hypothesis	Path	Sobel test							Result
		a	b	Sa	Sb	*Z*-Value	std. Error	p-value	
H7	DSC Main Activities➔ SC Capabilities➔ Sustainable Performance	0.73	0.42	0.03	0.06	6.73	0.05	0.00	**Partial mediation**
H8	DSC Support Activities➔ SC Capabilities➔ Sustainable Performance	0.80	0.53	0.02	0.07	7.44	0.06	0.00	**Partial mediation**

Source Authors

Cheng et al., 2021; Saryatmo & Sukhotu, 2021). For instance, Li et al. (2020) stated that DSC platforms as mediators can increase the impact of digital technologies on environmental and economic performance. Shee et al. (2018) reported that using CC for customer integration, supplier integration, and internal integration could have a positive impact on the company's sustainable performance. In this study, they considered top management support as a moderating variable. Based on the results of H3 ($\beta = 0.717$, t $= 19.646$), it was confirmed that DSC support activities have a positive and significant effect on the DSC main activities, as Farahani et al. (2017), and Agrawal and Narain (2018), stated in their studies. The results obtained for the first three hypotheses of this research supports the results obtained from the study of Ahmad Amouei et al. (2023a, 2023b). The results obtained for H4 ($\beta = 0.328$, $t = 6.514$) and H5 ($\beta = 0.566$, $t = 11.759$) prove that DSC activities have a positive and significant effect on SCCs. The results obtained in this research support the findings of Abdul-Rashid et al. (2017), and Yu et al. (2018). According to H6 ($\beta = 0.334$, $t = 4.194$), SCCs have a positive and significant effect on sustainable performance. This result is in line with previous studies (Backstrand & Powell, 2021; Cheung et al., 2018; Khin & Ho, 2019; To et al., 2021; Yeniyurt et al., 2019). For instance, Yu et al. (2018) found that SC coordination capability and responsiveness capability were positively and significantly related to financial performance.

The Sobel test was used to investigate the mediating role of SC capabilities. Based on the results of this test, H7 (Z-value $= 6.73$, $p = 0.00$) and H8 (Z-value $= 7.44$, $p = 0.00$) are confirmed. This means that SCCs mediate partially the relationship between DSC activities and sustainable performance. The mediating role of SCCs has been confirmed in many previous studies (Hautala-Kankaanpää, 2022; Hong et al., 2019; Nandi et al., 2020; Queiroz et al., 2021; Rajaguru & Matanda, 2019; Tigga et al., 2021). For instance, Agyabeng-Mensah et al. (2020) examined the potential impact of SC integration on the development of firm capabilities and performance by considering SCCs as a mediating variable. They found that companies that share knowledge, skills, information, and resources with SC partners develop SCCs and improve performance. Asamoah et al. (2021) acknowledged the mediating role of SCCs in the relationship between IOS usage and SC performance.

5 Conclusion

The purpose of this chapter was to propose a conceptual model to investigate the mediating role of SCCs in the relationship between DSC activities on the sustainable performance of manufacturing companies. Analysis of the responses obtained from a survey of top managers of manufacturing companies demonstrated that using digital technologies in SCCs enhances sustainable performance not only directly but also through SCCs. This means that the different dimensions of SCCs represent the firm's ability to manage and interact with supply chains and, therefore, can be considered essential drivers for the firm's operations (Hautala-Kankaanpää, 2022).

Based on the results obtained in this chapter, it can be said that using digital technologies in the main and support activities of the supply chain increases the ability of a company to share its knowledge and information with other supply chain partners. As a result, information exchange improves qualitatively and quantitatively. Using digital technologies enables better and faster coordination of various supply chain activities with partners. Similarly, the alignment and integration of the company's activities with its partners are facilitated by using digital technologies in the supply chain.

In today's digital era, despite the speed of environmental changes and intensely competitive markets, the company's flexibility and the response speed of the supply chain partners can be important factors in maintaining/improving the company's position.

On the other hand, according to the results obtained in this chapter, it can be said that using SCCs helps companies to improve their sustainable performance. Different dimensions of SCCs can increase the company's sustainable performance by reducing costs, optimal use of sustainable energy sources, reducing pollution, and improving and maintaining the quality of life.

5.1 Contribution to Theory

This research contributes to the DSC literature, by proposing a conceptual framework regarding the DSC activities, SCCs as a mediator variable, and sustainable performance which can be helpful for both practitioners and academicians. Although prior research by Ahmad Amouei et al. (2023a, 2023b) has established that using digital technologies in SC activities affects sustainable performance, the current study highlights the mediating role of SCCs in these relationships. Therefore, the results of the study expand the

frontiers of supply chain management in the context of using digital technologies and SCCs and confirm that digital technologies combined with SCCs can increase sustainable performance.

The findings of this research provide insights into the relationship between using digital technologies in SC activities and sustainable performance in Iranian manufacturing companies as a developing country.

Valmohammadi (2013) argues that the most important reason for the low utilization of IT in Iranian organizations, unlike companies in developed countries, is the poor ICT infrastructure, the lack of skilled people, and the scarcity of financial resources. Therefore, it is required to increase training programs among various levels of employees and managers in Iranian organizations to become more familiar with the concepts, practices, and functioning of digital technologies and the benefits and opportunities that can be created by these technologies in supply chains.

This study extends the supply chain-related literature and points to the need for more empirical work focusing on the role of different dimensions of SCCs. Many top managers and high-level experts of Iranian manufacturing companies have little familiarity with the concepts and applications of digital technologies in their company's SC activities. This issue indicates the need for supply chain scholars to emphasize more on non-technical and operational factors, as well as search for more appropriate theoretical frameworks to explain how to improve the sustainable practices of companies using digital technologies and considering supply chain capabilities. Therefore, according to Hautala-Kankaanpää (2022), academics could widen the analysis of digitalization to cover organizational aspects to increase the understanding of this complex phenomenon.

5.2 Implications for Practice

The results obtained in this chapter bring out a few key implications for managers. Given the dimensions of sustainable performance, the conceptual framework developed in this research can create a new approach to strategic prioritization.

After the deregulation and privatization of most industries by the government, Iranian manufacturing companies showed a great desire for SCM methods. This tendency is strongly related to the rising pressure caused by the intense completion in the country's market. Therefore, most manufacturing industries consider SCM as a strategy to increase their competitive advantage, which can increase market share both at national and international levels (Valmohammadi, 2013). This approach can be used as a guide

by manufacturers. The findings can also be used by policymakers to identify and prioritize using digital technologies in different parts of the manufacturing companies' SC, to support regulations, policies, and infrastructure, as well as technical and financial assistance.

According to Nandi et al. (2020), a major concern for supply chain practitioners is to identify and find problems in their organizations and environments that they can solve using technology. Therefore, our study offers several implications for practice. This study presents the current scenario of digital technologies applications in SC activities to managers, considering SCCs as a mediator, based on the answers obtained from 126 top managers in different Iranian manufacturing companies. Supply chain managers could use the findings presented in this chapter in their respective industry context.

The finding that using digital technologies in SC activities positively affects sustainable performance, suggests that managers need to allocate more resources to achieve the benefits of using digital technologies in communication and information functions. In addition, using digital technologies directly leads to the development of important SCCs for companies and enables them to achieve more responsiveness, integration, coordination, and information exchange. This research also highlights the critical role of SCCs in the results of using digital technologies in SC for managers and other professionals. While using digital technology can directly enhance the sustainable performance of companies, SCCs have been identified as another way in which digital technologies can enhance sustainable performance.

Better SC coordination depends on the effectiveness of information exchange between members and the supplier's ability to meet the needs of flexible supplies. Therefore, managers need to simultaneously focus on establishing effective information exchange and ensuring suppliers' ability to offer flexible resources (Tigga et al., 2021). By using digital technologies in the supply chain, managers can ensure that information exchange through these technologies is tightly linked to SC coordination requirements. Thus, companies can use higher levels of SCCs resulting from using digital technologies to modify the sustainable performance of their SC.

Today, Iran can be on the verge of a significant digital transformation. Because the implementation of global technology in Iran and the improvement of the domestic technological situation of this country have created the potential and significantly increased the share of the GDP of information technology (Aghayari et al., 2022). Based on the impact of using

digital technologies and SCCs on increasing sustainable performance, policy-makers and manufacturing managers might change their focus from reducing costs to activating new processes and business models to achieve competitive advantage. They can also simultaneously consider the management, implementation, and application of digital technologies and SCCs in SC. Because according to the results of the present research, this can provide many benefits in terms of achieving sustainable performance.

Building a digital culture can support different digital resources. Digital culture can help companies benefit from the investments associated with digitalization (Hautala-Kankaanpää, 2022).

5.3 Limitations and Future Lines of Research

Like any study, this research has a few limitations that can provide opportunities for further research. One of them relates to the scope of this research which is limited to the context of Iran. Thus, caution should be exercised in generalizing the findings of this research. Although this research examines the views of managers of Iranian manufacturing companies, it can be helpful for researchers who study the DSC, and manufacturing companies' managers and practitioners, in other developing countries and developed countries as well.

Also, future research is recommended to investigate the impact of using digital technologies in service companies.

Appendix A: Questionnaire-Measurement Scales of Constructs

DSC Support Activities (adapted from Ahmad Amouei et al. [2023a]).

Performance

per1: By using digital technology, the company's performance is improved in various ways.
per2: Using digital technology in re-engineering processes and digitizing them can reduce parallel activities.
per3: Due to competitive pressures, lack of investment in digital technology hinders the company's profitability and competitiveness.
per4: Using digital technology enables organizational links and integration within the organization and with other organizations and companies.

Technology

tech1: Intelligent control (cameras, sensors, etc.) is one of the automation capabilities in digital technology.
tech2: Agility, flexibility, and increased supply chain resilience are the hallmarks of digital technologies.
tech3: Protocols, security prerequisites, and proper standardization are part of the digital technology infrastructure.
tech4: Having a good internet platform requires using digital technology (such as IoT) in the supply chain.

Human Resources

hum1: A change-receptive organizational culture improves implementing and utilizing digital technology.
hum2: A proper understanding of human resource management from the concept of industry 4.0 can accelerate the company's transition to digitalization (digital transformation).
hum3: The company can use in-service training courses to update the knowledge of employees, engineers, and managers in the field of digital technology.
hum4: To take advantage of digital technology, using specialized and skilled personnel in digital technologies should be one of the recruitment programs of the company.

DSC Main Activities (adapted from Ahmad Amouei et al. [2023a])
Suppliers

sup1: Creating a supplier-related database with accurate and up-to-date information can speed up and facilitate the evaluation and selection process.
sup2: Digital technology can help manufacturing companies identify and select alternative supplier options in an emergency.
sup3: The type of raw materials (general-specific) is influential in policy-making to segment suppliers.
sup4: Using digital technology, it is possible to measure and verify the fulfilment of supplier obligations (by the company) more quickly and accurately.

Manufacturing

man1: Using digital technology, the required information about each product will be available in real-time and throughout the supply chain.
man2: By using digital technology while reducing the time of the custom product production process, the quality of the custom product can be increased.
man3: Using digital technology can help to identify and standardize the various stages of the manufacturing process in the company's supply chain.
man4: With using digital technology, flexibility in designing and producing new and mass-produced products increases.

Logistics

log1: Using digital technology contributes to the development of logistics (increasing accuracy in transportation planning and accelerating coordination between stakeholders).
log2: Using digital technology, it is possible to track the items and components used in each product accurately.
log3: The development and modification of business models such as e-commerce and e-business are one of the effects of using digital technology in the supply chain.
log4: Using digital technology, logistics and SC management can be improved by automating repetitive tasks.

Customers

cus1: Using digital technology, it is possible to follow up faster on customer requests, suggestions, and complaints.
cus2: Using digital technology can speed up and facilitate the customer ordering process.
cus3: Digital technology can help measure customer satisfaction in real-time with high speed and accuracy.
cus4: Sharing information using digital technology such as social media is effectively increases customer awareness of product quality.

Supply Chain Capabilities (adapted from Asamoah et al. [2021; Yu et al. 2018])

Information Exchange

info1: Using digital technology, a company can exchange more information with its partners.
info2: Using digital technology helps information flow more freely between the company and its partners.
info3: The company benefits from exchanging information with SC partners.
info4: With using digital technology, it is possible to exchange information with supply chain partners accurately and on time.

Coordination

coor1: By using digital technology, the company can become more efficient in coordinating activities with its partners.
coor2: Using digital technology, the company can conduct transaction follow-up activities with its partners more efficiently.
coor3: With digital technology, a company can spend less time coordinating transactions with its partners.
coor4: Using digital technology can help reduce corporate coordination costs.

Activity Integration

integ1: Using digital technology, the company can develop strategic plans in collaboration with its partners.
integ2: Using digital technology, the company can actively collaborate with its partners in forecasting and planning.
integ3: Using digital technology, the company can project and plan future demand collaboratively with its partners.
integ4: Using digital technology, the company can forecast and plan collaboratively with its partners.

Supply Chain Responsiveness

resp1: Using digital technology, a company's supply chain can respond more quickly and effectively to changing customer and supplier needs.
resp2: Using digital technology, the company's supply chain can develop and market new products more rapidly and effectively.
resp3: Using digital supply chain technology, a company can compete effectively in most markets.

resp4: Companies and supply chain partners using digital technology can use information from different partners in multiple ways depending upon various requirements.

Sustainable Performance (adapted from Abdul-Rashid et al. [2017])
Economic

eco1: Using digital technology improves the company's position in the marketplace.
eco2: Using digital technology increases profitability.
eco3: Using digital technology decrease costs (purchase of materials, manufacturing, waste treatment).
eco4: Using digital technology improved product quality.

Environmental

env1: Using digital technology reduces wastewater and solid waste.
env2: Using digital technology reduces energy consumption and material usage.
env3: Using digital technology reduces CO2 emissions and the production of toxic/ harmful/ hazardous/ flammable substances.
env4: Using digital technology leads to improved compliance with environmental standards.

Social

soc1: Using digital technology helps to improve work safety.
soc2: Using digital technology helps to improve the work environment.
soc3: Using digital technology helps to enhance the quality of living the surrounding community.
soc4: Using digital technology helps to strengthen relationships with the community and stakeholders.

Appendix B: First- and Second-Order Principal Constructs Analysis and Total Variance Explained

See Tables 9 and 10

Table 9 First- and second-order EFA for constructs

Variables	Principal components (first-order)				Component (second-order)	
	Manufacturing	Suppliers	Logistics	Customers	Variables	DSC main activities
sup1	.123	.858	.162	.178	Suppliers	.740
sup2	.164	.872	.119	.157		
sup3	.250	.821	.146	.147		
sup4	.250	.813	.151	.186		
man1	.868	.239	.186	.148	Manufacturing	.796
man2	.841	.223	.193	*.226*		
man3	.874	.178	.247	.130		
man4	.834	.197	.311	.162		
log1	.190	.151	.822	.311	Logistics	.824
log2	.202	.120	.820	.308		
log3	.331	.212	.795	.220		
log4	.248	.173	.849	.169		
cus1	.206	.100	.291	.783	Customers	.790
cus2	.196	.088	.256	.839		
cus3	.203	.238	.190	.813		
cus4	.038	.314	.193	.804		
Variables	*Integration*	*Information*	*Responsiveness*	*Coordination*	*Variables*	*SC capabilities*
info1	.225	.845	.146	.114	Information	.780
info2	.245	.878	.079	.157		
info3	.102	.805	.262	.305		
info4	.128	.789	.355	.299		

(continued)

Table 9 (continued)

Variables	Principal components (first-order)				Component (second-order)	
	Manufacturing	Suppliers	Logistics	Customers	Variables	DSC main activities
coor1	.148	.370	.247	.751	Coordination	.839
coor2	.150	.242	.198	.795		
coor3	.256	.113	.210	.801		
coor4	.342	.167	.214	.715		
integ1	.818	.180	.224	.265	Integration	.812
integ2	.858	.187	.220	.203		
integ3	.836	.186	.176	.242		
integ4	.777	.172	.345	.141		
resp1	.411	.194	.710	.334	Responsiveness	.860
resp2	.268	.170	.770	.391		
resp3	.237	.184	.870	.158		
resp4	.232	.307	.775	.191		
Variables	*Human*	*Technology*	*Performance*		*Variables*	*DSC support activities*
per1	.313	.213	.826		Performance	.891
per2	.296	.270	.834			
per3	.367	.248	.786			
per4	.273	.314	.811			
tech1	.148	.853	.318		Technology	.790
tech2	.139	.838	.278			
tech3	.198	.883	.169			

Variables	Principal components (first-order)				Component (second-order)	
	Manufacturing	Suppliers	Logistics	Customers	Variables	DSC main activities
tech4	.237	.868	.171			
hum1	.877	.228	.212		Human	.845
hum2	.847	.191	.274			
hum3	.867	.170	.319			
hum4	.790	.167	.381			
Variables	*Environmental*	*Economic*	*Social*		*Variables*	*Sustainable performance*
eco1	.136	.891	.171		Economic	.812
eco2	.140	.876	.279			
eco3	.283	.868	.200			
eco4	.474	.725	.179			
env1	.825	.361	.230		Environmental	.890
env2	.812	.245	.355			
env3	.809	.195	.271			
env4	.774	.138	.480			
soc1	.432	.262	.697		Social	.873
soc2	.280	.230	.829			
soc3	.264	.163	.854			
soc4	.258	.235	.819			

Extraction Method: Principal Component Analysis
Rotation Method: Varimax with Kaiser Normalization

Table 10 Total variance explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
Human	7.103	59.188	59.188	7.103	59.188	59.188	3.391	28.259	28.259
Technology	1.839	15.324	74.513	1.839	15.324	74.513	3.386	28.213	56.472
Performance	1.093	9.105	83.618	1.093	9.105	83.618	3.257	27.146	83.618
Manufacturing	8.100	50.624	50.624	8.100	50.624	50.624	3.460	21.623	21.623
Suppliers	1.988	12.427	63.052	1.988	12.427	63.052	3.294	20.588	42.211
Logistics	1.813	11.330	74.382	1.813	11.330	74.382	3.239	20.247	62.458
Customers	1.213	7.580	81.962	1.213	7.580	81.962	3.121	19.504	81.962
Integration	8.652	54.078	54.078	8.652	54.078	54.078	3.424	21.402	21.402
Information	1.792	11.202	65.280	1.792	11.202	65.280	3.318	20.739	42.141
Responsiveness	1.323	8.268	73.548	1.323	8.268	73.548	3.116	19.473	61.613
Coordination	1.175	7.346	80.894	1.175	7.346	80.894	3.085	19.280	80.894
Environmental	7.173	59.772	59.772	7.173	59.772	59.772	3.337	27.806	27.806
Economic	1.658	13.814	73.587	1.658	13.814	73.587	3.290	27.415	55.221
Social	1.033	8.605	82.192	1.033	8.605	82.192	3.236	26.970	82.192

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Part II

Technological Effort vs. Value in Supply Chains

6

Procurement Digitalisation on International Procurement and Supply Chain Management Strategies

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1 Introduction

In the era of digitisation, the landscape of competition is swiftly changing, as new business models emerge, industries evolve dramatically, and traditional boundaries of corporations continue to blur (Fitzgerald et al., 2022). The procurement and supply chain management (PSCM) strategies in the international fast fashion sector are amidst this transition, steering towards a future characterised by Procurement 4.0 or Industry 4.0 technologies (Ivanov & Dolgui, 2020). Procurement has undergone a significant transformation, shifting towards a more strategic role involving activities such as

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strategic sourcing, supplier relationship management, and supply networks (Tassabehji & Moorhouse, 2008). This change has been facilitated by the adoption of advanced digital technologies, including artificial intelligence (AI) and blockchain, which have replaced traditional e-procurement solutions from the 90s. This trend, known as procurement digitalisation, has the potential to revolutionise the procurement organisation and its processes (Srai & Lorentz, 2019). This research explores this transformative landscape, focussing on the impact of procurement digitalisation (PD) on international procurement and supply chain management (IPSCM) strategies in the fast fashion sector.

The fast fashion industry's unprecedented pace necessitates agile supply chains and quick response strategies (Christopher et al., 2004). Although digital technologies promise transformative possibilities for PSCM, their effective integration remains a significant challenge (Sweeney et al., 2020). Firms face issues such as technological infrastructure, digital literacy, security concerns, and regulatory barriers (Mikalef & Pateli, 2017). Addressing this problem, the research will ascertain the impact of technologies like blockchain, AI, big data, augmented reality (AR), three-dimensional (3D) printing, robotic process automation (RPA), and Internet of Things (IoT) on PSCM strategies in the international fast fashion sector. A recent example of the problem is during the COVID-19 pandemic, fast fashion retailer Primark faced procurement difficulties, as detailed in its 2022 annual report. Supplier production disruptions led to delayed procurement of raw materials and goods. Transitioning to a more digital procurement process was slow due to issues like technological infrastructure gaps and data security. However, Primark recognised the value of a digital strategy incorporating AI, Blockchain, and IoT to address these procurement challenges (Primark, 2022).

Digitalisation underpins what is often referred to as Industry 4.0, leading to the forefront of the industrial revolution. It bears transformative potential for business processes, particularly for PSCM, promising to enhance efficiency, reduce costs, and improve transparency. The fast fashion industry, characterised by rapid product cycles and a consumer-centric approach, presents a sector where the challenges and benefits of digitalisation are distinctly apparent (Christopher et al., 2004).

The term "fast fashion" refers to a type of business model that prioritises speed and cost efficiency in the manufacturing and distribution of clothing. Its distinguishing features include regular changes in clothing styles, inexpensive prices, and an emphasis on the latest fashion trends over durability or high-quality materials. Some examples of international fast fashion brands are

Zara, ASOS, H&M and Primark. The fast fashion sector thrives on its ability to produce and distribute products quickly to respond to changing consumer trends (Caro & Martínez-de-Albéniz, 2015). This sector's competitiveness largely depends on the supply chain's efficiency and responsiveness, necessitating strategies that can deal with high levels of uncertainty and complexity (Christopher et al., 2004). The competitive edge in this dynamic environment often hinges on the effective use of digital technologies in PSCM (MacCarthy et al., 2016).

2 Methodology

Due to the nascent nature of the research topic, this study followed the Systematic Review (SLR) guidelines proposed by Transfield et al. (2003), which consist of five steps: (1) Planning, (2) Searching, (3) Screening, (4) Extraction and Synthesis, and (5) Reporting. The primary advantage of SLR is that it affords a transparent and repeatable process to select relevant literature to investigate the research phenomenon (Habib et al., 2015).

2.1 Step 1: Planning

At the planning stage following research questions were proposed:

1. What are the key factors that influence the impact of digital technologies on Procurement and Supply Chain Management (PSCM) strategies in the international fast fashion sector?
2. What challenges are encountered when implementing digital technologies in PSCM strategies in the international fast fashion sector?
3. What are the good practices for implementing digital technologies in PSCM strategies in the international fast fashion sector?
4. What role do key stakeholders play in implementing digital technologies in PSCM strategies in the international fast fashion sector?
5. What actions can be recommended to successfully manage and control the implementation of digital technologies in PSCM strategies in the international fast fashion sector?

Table 1 Key terms and search terms (*Source* Authors)

Key terms	Definition
Procurement	"Procurement is defined as the buying of goods and services that enable an organisation to operate its supply chains, in a profitable and ethical manner" (CIPS, 2023, p. 1)
Supply chain	A supply chain refers to the interconnected network of organisations, linked through various upstream and downstream operations, that contribute to an array of processes and activities. These collective efforts aim to create value manifested in the form of goods and services, eventually reaching the final customer (Christopher, 2016)
Digital technologies	Digital technologies involve a variety of tech-based tools and resources, including AI, blockchain, big data, and IoT. These technologies play a crucial role in creating, storing, managing, and distribution of digital data (Treiblmaier, 2018)
Fast fashion	Fast fashion is a business model in the fashion industry where the production process is expedited in order to get new trends to the market as quickly as possible, often resulting in affordable clothing, but also raising concerns about the environmental and social impact of the process (Joy et al., 2012)
	Search string
Procurement	(Procurement OR buying OR sourcing OR purchas*) AND
Supply chain	("supply chain" OR "demand chain" OR "supplier network" OR suppl*) AND
Digitalisation	(Digital* OR tech* OR futur* OR disrupt* OR innovat*) AND
Industry	("Fast fashion" OR "fast cloth" OR apparel OR garment OR cloth*) AND
Technologies	("Blockchain" OR IoT OR automation OR "artificial intelligence" OR data*)

2.2 Step 2: Searching

The searching stage, the following search terms (see Table 1) were used to develop the search strings.

2.3 Step 3: Screening

At the screening stage, first, inclusion and exclusion criteria were used to short list articles based on titles and abstract based screening (see Table 2).

Next, the following quality criteria (Table 3) was used to short list articles based on full text-based screening (Habib et al., 2015).

Table 2 Inclusion/exclusion criteria (*Source* Authors)

#	Characteristic	Inclusion criteria	Rationale
1	Language	English	Academic standard and global accessibility
2	Publication date	Articles published between 2010 and 2023	To keep the paper more relevant to the topic
3	Article type	Academic journals, industry reports, government policy documents	Credibility and Authority, Depth of Analysis, Current and Relevant Information, Variety of Perspectives
4	Subject area	Articles focussing on digital technologies, procurement, strategy, and supply chain management in the international fast fashion sector	Articles that are within the research scope
5	Unit of reference	Fast fashion, Apparel	Reference industry chosen for research
	Characteristic	Exclusion Criteria	Rationale
1	Language	Non-English	Not academic standard
2	Publication date	Articles published before 2010	Outdated papers
3	Article type	News articles, editorials, meetings, books, working papers	Not relevant information and no depth of analysis
4	Subject area	Articles not relevant to digital technologies, procurement and supply chain management in the international fast fashion sector	Articles that are not within the research scope
5	Unit of reference	Other sectors	Not relevant to this research

2.4 Step 4: Extraction and Synthesis

After short listing the 60 academic sources, the following data extraction fields were used to extract the data related to key themes: key factors influencing the impact of digital technologies, challenges in implementation, good practices, the role of stakeholders, and recommendations to effectively manage digitalisation. After extracting this data, the emerging themes were identified and synthesised based on independent coding conducted by the authors to ensure transparency, consistency, and objectivity in the process.

Table 3 Quality criteria for full text-based screening. Source Authors

#	Criteria	Level			
		0	1	2	3
1	Contribution to the theory of procurement digitalisation and procurement fast fashion	Sufficient details were not provided in the publication to evaluate this criterion adequately	There is a lack of theoretical foundation	The publication has a limited theoretical foundation	The publication demonstrates a solid theoretical foundation and contributes to the theory of practice
2	Methodology	Sufficient details were not provided in the publication to evaluate this criterion adequately	The execution of the study's idea is inadequate, and the utilisation of inappropriate quantitative methods is observed	The research design is reasonable and well-justified, but the execution of the study idea is incomplete	The research design thoroughly evaluates the research argument
3	Data analysis	Sufficient details were not provided in the publication to evaluate this criterion adequately	The data sample used in the study is inadequate, leading to inconclusive research results	The study utilises an appropriate data sample, resulting in adequate findings. However, the explanation provided for these results is insufficient	The study incorporates an adequate data sample, and the research results are well-explained, effectively supporting the theoretical arguments

(continued)

Table 3 (continued)

#	Criteria	Level			
		0	1	2	3
4	Implications for practice	Sufficient details were not provided in the publication to evaluate this criterion adequately	The study does not effectively inform or provide practical implications	The study offers limited implications for practical application	The study presents critical implications that are valuable for practical application
5	Level of citations	Sufficient details were not provided in the publication to evaluate this criterion adequately	The publication lacks citations	The publication includes at least some level of citation	The publication includes a comprehensive list of citations

3 Summary of SLR Process

Figure 1 demonstrates the steps followed in the SLR to identify the fine section of 60 sources identified for this research.

4 Findings and Discussion

4.1 Step 5: Reporting

This section is based on Step 5 of the SLR process i.e., reporting the key themes emerging from the research including: (1) key factors influencing the impact of digital technologies; (2) challenges in implementation; (3) good practices; (4) role of stakeholders; and (5) recommendations to effectively manage digitalisation as follows:

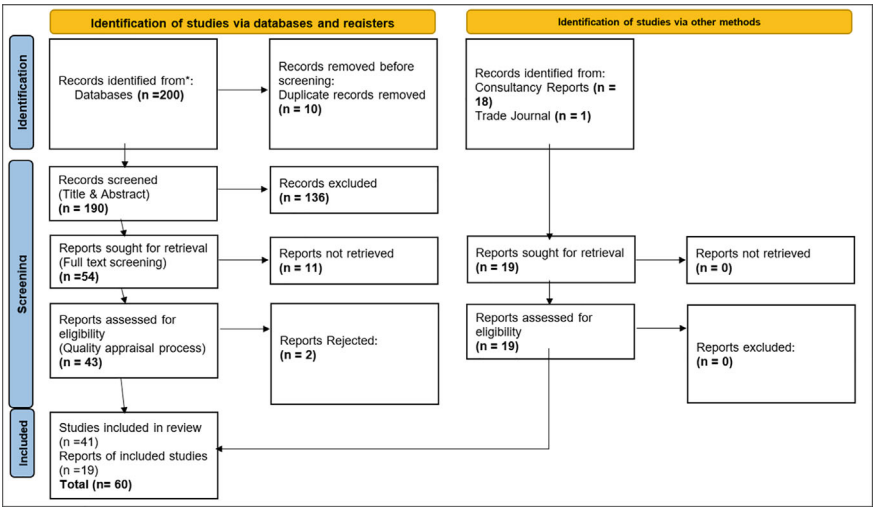

Fig. 1 Generation of articles for SLR (*Source* Author)

4.2 Key Factors Influencing the Impact of Digital Technologies

4.2.1 Real-Time Data Utilisation

PwC (2016) emphasises that fashion companies can harness data analytics and agile IT platforms to implement dynamic pricing, enhance transparency, and improve forecasting accuracy. Real-time data utilisation, particularly leveraging sell-out data from gatekeepers' websites, aids in reordering management, preventing stockouts, and fostering agility. Agrawal et al. (2021) proposed a blockchain-based traceability framework for the textile and clothing industry, aimed at addressing information asymmetry and low visibility in supply chains. By utilising smart contracts and mass balancing validation mechanisms, secure data sharing among stakeholders is facilitated (Agrawal et al., 2021). This digital framework ensures data accuracy, traceability, and technology-based trust. Essential traceability information, determined through consensus among supply chain partners, is categorised into private shareable, public shareable, secured, and linking information sets. This enhances transparency and regulatory compliance (Agrawal et al., 2021).

Digital technologies, involving big data analytics and cloud computing empower organisations to oversee complex procurement processes efficiently. The utilisation of big data analytics aids in identifying patterns and

optimising procurement processes, while cloud-based systems provide real-time access, enhancing collaboration. Increased openness and traceability in procurement transactions through digital technologies strengthen buyer–supplier relationships, elevate confidence levels, and yield strategic benefits. Successful implementation of digital technologies in procurement leads to competitive advantage, supporting organisational goals and generating sustainable competitive advantages (CADs) that are challenging to replicate (Alabdali & Salam, 2022). These technologies enable agility, data accuracy, collaborative relationships, and strategic decision-making, contributing to competitive advantage and sustainable CADs in the fast-paced fashion industry.

4.2.2 Sustainability and Environmental Impact

The integration of digital technologies has significantly transformed PSCM strategies within the international fast fashion sector, with a distinct focus on sustainability and environmental impact. Blockchain-enabled apparel retail platforms, leverage peer-to-peer transactions, eliminating intermediaries and fostering efficiency. This not only enhances transparency but also promotes sustainability, as the decentralised ledger stores records, enabling direct interactions between buyers and sellers. This approach reduces costs and elevates profits, aligning with sustainable practices. Digital ecosystems, play a pivotal role in ensuring ethical and sustainable production. By digitising the supply chain and fostering transparency, companies gain insights into sustainable material sourcing and ethical production methods. Such practices cater to the rising consumer demand for ethically produced fashion and position companies to meet evolving regulations (Hedrich et al., 2021). Moreover, a sustainable fashion supply chain, enhances brand reputation and appeals to a broader range of ethically conscious consumers. For firms, sustainability signifies responsibility, providing a competitive edge, while for consumers, it reflects values of equality and environmental consciousness (Shen, 2014). Ultimately, sustainability becomes an integral aspect of brand identity, resonating with consumers who prioritise ethical practices and contributing to the sector's evolution towards a more environmentally conscious future.

4.2.3 Supply Chain Traceability and Transparency

Growing customer demands for information about raw material processes and the upcoming need to digitise value chains to comply with future regulations. To meet these demands, companies are adopting Product Lifecycle Management (PLM) systems to manage suppliers and take control of manufacturing and delivery timelines. This ensures a clear view of production status, which is crucial. Creating a digital ecosystem through interconnected systems promotes full transparency deeper into the supply chain, helping to identify risks and opportunities (Hedrich et al., 2021). Traceability enhances transparency by establishing an information trail for tracking products and materials at each production stage. This data can be securely shared using blockchain technology, reducing the risk of data manipulation and loss. Real-time data acquisition and monitoring, along with product quality control, aid in identifying issues related to authenticity, ethical production, and sustainability, contributing to a more transparent and traceable supply chain (Agrawal et al., 2021).

Jahani et al. (2021) emphasises data availability, quality, collaboration, and communication among supply chain stakeholders. Emerging technologies such as IoT, cloud computing, AI, blockchain, and smart contracts further enhance transparency and efficiency across the supply chain. IoT sensors collect real-time data, cloud computing streamlines data storage and sharing, AI improves coordination, and blockchain technology ensures data security and trustworthiness (Jahani et al., 2021). These technologies facilitate informed decision-making, supplier collaboration, and ethical production practices, resulting in a more accountable, efficient, and trustworthy supply chain.

4.2.4 Digital Transformation

Digital technologies are reshaping PSCM strategies in the global fast fashion sector, driven by a wave of digital transformation like shown in Fig. 2. The transformative potential of Industry 4.0 technologies, like IoT and AI, which enable organisations to become more sustainable, economical, and competitive. Strategic procurement 4.0, aligning digital technologies with sustainability targets, strengthens competitiveness in the circular economy. Ahmad et al. (2020) highlights that Industry 4.0 revolutionises the Textiles & Apparel (T&A) industry by integrating modern manufacturing with digital infrastructure, enhancing decision-making and supply chain operations through

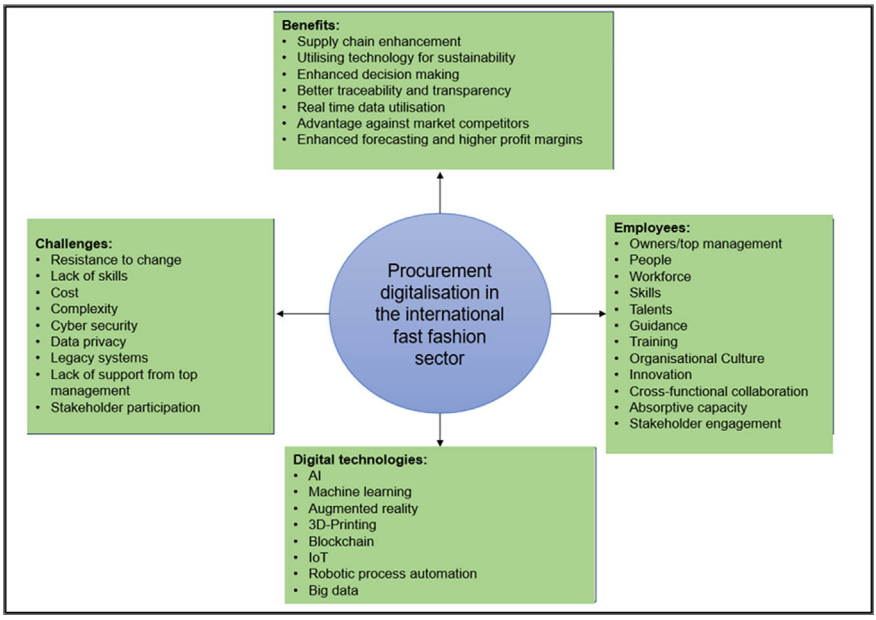

Fig. 2 Digitalisation of procurement (*Source* Authors)

advanced analytics. Utilising big data empowers T&A companies to optimise supply chain operations in real-time (Ahmad et al., 2020). Business Intelligence Systems (BIS) analyse data, pinpointing areas for sustainability improvements, such as waste reduction and energy efficiency. Digitisation has the potential to enhance sourcing processes, reducing lead times, fostering innovation, and catering to customer needs (Berg et al., 2017). Digital technologies are changing how fast fashion companies manage their supply chains. These technologies, like IoT and AI, help organisations be more sustainable and competitive. They make decisions smarter and supply chains more efficient. Industry 4.0 connects modern manufacturing with digital tools, revolutionising the T&A industry (Ahmad et al., 2020).

4.2.5 Data Quality and Integration

Digital technologies are revolutionising PSCM strategies in the global fast fashion sector by enhancing data quality and integration. PwC (2016) seamlessly incorporating data across the entire supply chain in real-time, often without manual intervention, results in quicker deliveries and enhanced management of freight and inventory. This boosts efficiency and revenue. The actions that merge data from different internal sources, resulting in enhanced

Table 4 Categorisation of influencing factors (*Source* Authors)

Nature of impact of factors	Location of factors	
	Internal	External
Strategic (long term)	Digital transformation	Supply chain traceability and transparency (Agrawal et al., 2021) Sustainability and environmental impact
Operational (short term)	Real-time data utilisation (Shah et al., 2023) Data quality and integration (Lorentz et al., 2021)	

decision-making. These efforts involve merging diverse data types from different sources. Advanced applications utilise algorithms to enhance data quality and facilitate real-time monitoring, thereby enabling forward-looking spend planning. These applications achieve this by combining spending data with contracts and orders (Lorentz et al., 2021). Purchasing departments generate substantial data, and technological advancements enable powerful algorithms, like AI, to analyse this data. AI helps purchasers find opportunities and threats in data. By comparing internal and external data, purchasers can uncover inconsistencies in supplier practices related to innovation, ethics, sustainability, pricing, and trends (Allal-Chérief et al., 2021). In short, digital technologies are changing how the fast fashion sector manages its supply chain. They improve data quality by combining information from different places and using smart algorithms. This helps companies make better decisions, find opportunities, and avoid problems. Table 4 provides the categorisation of influencing factors.

4.3 Key Challenges in Implementing Digital Technologies

4.3.1 Resistance to Change and Lack of Skills

Implementing digital technologies in PSCM strategies within the global fast fashion sector comes with notable challenges, particularly concerning resistance to change and the lack of necessary skills. The successful adoption requires more than just acquiring technologies; it entails finding skilled individuals and fostering a cultural shift towards embracing digital efforts, effectively transforming the entire organisation. The need for a cultural

and process transformation when integrating digital technologies can be met with resistance from employees accustomed to traditional methods. Moreover, deficiencies in technical skills like data analytics and soft skills like change management hinder effective implementation (Biltoft-Knudsen et al., 2022). The challenge of creating lasting and user-friendly solutions for various processes highlights the need for new procurement models and agile approaches to adapt to digitisation's changes. The limited success in substantially improving process efficiency through digitisation and completing planned digital projects hints at the potential influence of resistance to change and skill gaps on these challenges (Biltoft-Knudsen et al., 2022). A shift in culture, along with fostering the right expertise, is crucial to navigate these challenges and unlock the full potential of digital transformation.

4.3.2 Cost and Complexity

Companies in the international fast fashion sector encounter challenges when incorporating digital technologies into their PSCM strategies, particularly related to cost and complexity. Implementing these technologies can directly impact merchandise costs at the individual item level, necessitating substantial investments in employee training, IT infrastructure upgrades, and technology integration (Ahmad et al., 2020). The intricacy of digital technologies also presents hurdles, notably in data management and analysis. The significant volume of data generated can overwhelm companies, especially those lacking expertise and resources (Ahmad et al., 2020). This becomes more pronounced for small and medium-sized enterprises (SMEs), which may lack the necessary means to handle costs and complexities (Ahmad et al., 2020). The expense associated with technology implementation includes investments in hardware, software, and employee training (Biltoft-Knudsen et al., 2022). Hiring specialised staff to manage new technologies further adds to costs. The complexity arises from choosing suitable technologies amidst a plethora of options and integrating them with existing systems. The implementation process itself requires meticulous planning, testing, and coordination across various departments (Biltoft-Knudsen et al., 2022).

4.3.3 Cybersecurity and Data Privacy

Companies in the global fast fashion sector confront challenges concerning cybersecurity and data privacy when integrating digital technologies into their PSCM strategies. As sourcing processes become digitised, substantial

volumes of data, including sensitive information like supplier contracts and customer data, are generated, and processed. The security and integrity of this data are crucial to fend off cyber threats and data breaches, which can lead to severe financial and reputational consequences (Berg et al., 2017). The National Association of State Procurement (NASPO) emphasises security concerns in adopting digital technologies like the IoT. Cloud computing raises concerns regarding the security and confidentiality of data for both companies and individuals (Motaung & Sifolo, 2023). Cybersecurity acts as a obstacle, impacting the implementation of digital procurement. The exposure to cybercrime raises the risk of external parties accessing a company's systems and data. Thus, businesses must adopt adaptable strategies to manage security risks associated with not adopting digital procurement (Motaung & Sifolo, 2023). Storing and processing large amounts of data can be risky if not properly secured. Hackers and data breaches can harm a company's finances and reputation. Even the use of new technologies like the IoT can raise security concerns. Businesses must take strong measures to protect their data from cyber threats.

4.3.4 *Legacy Systems and Interoperability*

Companies in the international fast fashion sector encounter challenges related to legacy systems and interoperability when incorporating digital technologies into their PSCM strategies. Many businesses have invested heavily in outdated legacy systems that aren't compatible with newer digital technologies. This mismatch makes it tough to seamlessly integrate data throughout the supply chain, leading to a lack of real-time visibility and optimisation. This can harm the supply chain's efficiency and effectiveness. Moreover, different partners in the supply chain might use systems that can't communicate with each other, creating data silos and hindering information sharing. To overcome these hurdles, a comprehensive approach to digital transformation is advised. This means not just adopting new technologies but also rethinking the entire supply chain structure to ensure that all systems work together smoothly. Achieving this may require substantial investment in new technologies, along with policy and procedure adjustments. Close collaboration with supply chain partners is also vital to ensure interoperability and effective data sharing. To solve this, companies need to change how their systems work together. This could mean using new technologies and talking with partners to make sure everything works together smoothly. Table 5 shows the categorisation of key challenges.

Table 5 Categorisation of challenges (*Source* Authors)

Nature of impact of challenges	Location of challenges	
	Internal	External
Strategic (long term)	Resistance to change (Biltoft-Knudsen et al., 2022) Lack of digital skills (Motaung & Sifolo, 2023)	Uneven digital maturity across supply chain (Biltoft-Knudsen et al., 2022)
Operational (short term)	Legacy systems Integration complexity	Cybersecurity threats (Motaung & Sifolo, 2023) High costs (Ahmad et al., 2020)

4.4 Good Practices in Implementing Digital Technologies

4.4.1 Integration of IoT for Supply Chain Enhancement

In the international fast fashion sector, adopting IoT technology holds promise for enhancing supply chain management by exchanging information among retail outlets, distribution hubs, storage facilities, and production units. IoT involves connecting physical devices like appliances and vehicles with electronics, sensors, and software, allowing them to exchange data (Shah et al., 2023). IoT can track goods, inventory, and data in real-time, improving supply chain operations. However, effective information sharing relies not only on technology but also on company policies. Top management's role is pivotal here. They can establish policies and guidelines guaranteeing that relevant information reaches the right individuals in a timely manner. This adds clarity to information sharing, even in cases where human intervention is involved (Shah et al., 2023). IoT is a game-changer for fast fashion's supply chain management, enabling real-time data sharing. For successful implementation, it's crucial to have clear policies guided by top management that ensure the accurate information reaches the right stakeholders on time.

4.4.2 Utilising Blockchain for Sustainability

In the international fast fashion sector, companies can use blockchain technology for sustainable practices. Blockchain helps reduce waste and promote sustainability by enabling efficient management of excess apparel inventory. It also aids in capturing the value of resources throughout the supply chain, thus minimising various types of waste. This technology is not only about waste

reduction; it also assists in tracking environmental impacts. Participants in the supply chain can monitor emissions, such as carbon emissions and toxic pollutants, at each stage and ensure compliance with environmental regulations. This encourages responsible practices throughout the supply chain and reduces overall environmental emissions. Additionally, blockchain tracks the origins of raw materials, preventing overuse of natural resources like water and preventing issues like deforestation. It also helps businesses manage their waste effectively, facilitating recycling and proper disposal (Park & Li, 2021). Companies can use blockchain in fast fashion to be more eco-friendly. It helps manage excess clothes and prevents waste. Also, it tracks things like pollution and makes sure everyone follows environmental rules. By using blockchain, companies can take care of the environment and be more sustainable.

4.4.3 Machine Learning and AI for Enhanced Decision-Making

Leading fast fashion companies are leveraging machine learning and AI to rapidly analyse sales, social media, and customer data to identify the latest trends and adjust design and production accordingly (Papachristou et al., 2020). By forecasting demand more accurately, companies can optimise inventory levels and raw material procurement to match real-time needs (Papachristou et al., 2020). Supply chain transparency is improved through AI tracking of orders and inventory, reducing waste and costs. Smart automation of production via AI and robotics boosts efficiency and quality control (Papachristou et al., 2020). Real-time collaboration between teams is enhanced by shared AI insights on consumer preferences, streamlining decision-making. Overall, astute deployment of machine learning and AI enables fast fashion leaders to respond to shifting market demands with greater speed, flexibility, and precision (Papachristou et al., 2020). Key focus areas include demand forecasting, inventory and procurement optimization, supply chain transparency, smart manufacturing, and cross-functional team coordination. With the right data architecture and change management, AI and machine learning can drive improved resilience, sustainability, and innovation.

4.4.4 *Data-Driven Decision-Making*

Leading fast fashion companies are investing in advanced data analytics capabilities to support more agile and effective sourcing strategies (Seara et al.,

Table 6 Categorisation of good practices (*Source* Authors)

Nature of impact of good practices	Location of Good Practices	
	Internal	External
Strategic (long term)	AI and machine learning (Papachristou et al., 2020)	Blockchain for sustainability
Operational (short term)	IoT integration (Shah et al., 2023) Data-driven decisions (Berg et al., 2017) Digitisation for transparency (Hedrich et al., 2021)	

2022). By analysing components like cost, risk factors, and efficiency drivers, data modelling enables better supplier and country selection aligned to value priorities (Berg et al., 2017). Big data analysis of past performance and future scenarios allows optimisation of inventory, production volume, and logistics based on predicted demand (Seara et al., 2022). Supply chain transparency and compliance is enhanced through tracking and monitoring using digital tools like dashboards, interactive apps, and live data feeds (Berg et al., 2017). Rather than gut feel, fact-based decisions on sourcing location, partner selection, order quantities, and logistics are enabled by advanced analytics techniques including AI and machine learning algorithms applied to large internal and external datasets.

To fully leverage data-driven decision-making, fast fashion leaders need robust data architecture, upskilling, executive commitment, and change management across procurement, design, and production teams (Berg et al., 2017). The result is greater speed, flexibility, and efficiency in matching supply with the latest consumer demand signals. Categorisation of good practices is demonstrated in Table 6.

4.5 Role of Key Stakeholders in the Implementation of Digital Technologies

4.5.1 Top Management and Absorptive Capacity

In the international fast fashion sector, the involvement of key stakeholders like top management, suppliers, and technology providers is crucial when adopting new digital technologies. Top management, especially, holds a crucial responsibility in shaping these choices. They provide active backing and guidance, which is vital to initiate the adoption of emerging technologies,

including within supply chain operations. The decision to adopt a new technology ultimately rests with top management, and their commitment greatly influences how open the organisation is to embracing innovation (Nath et al., 2022). Another important concept is "absorptive capacity." This refers to a firm's capacity to understand, integrate, and employ new information adeptly. It resembles the company's proficiency in acquiring and applying new knowledge (Nath et al., 2022). This capacity is influenced by the knowledge a company already has and the effort it puts into learning new things. It helps companies predict how useful a new technology could be in a changing environment. For example, if a company knows a lot about a technology and makes a strong effort to learn more, it's more likely to succeed in using that technology (Nath et al., 2022). In fast fashion, top managers are key players in deciding to use new digital tools. They guide and support the process. Additionally, a company's "absorptive capacity," which is how well it can learn and use new information, is important. The more a company knows and tries to learn about a technology, the better it can use it (Nath et al., 2022).

4.5.2 Stakeholder Engagement and Sustainable Strategies

In the fast fashion sector, involving key stakeholders like management, suppliers, and technology providers is crucial when using new digital technologies for PSCM. Stakeholder engagement means working closely with these groups to understand their needs and ideas. This helps fashion companies develop strategies that are good for the environment and society (Ahmad et al., 2020). Stakeholder engagement is really important for sustainability in fashion. It means talking to different people involved, like managers, suppliers, and tech experts. This helps fashion companies figure out how to use new technologies in a way that's good for everyone. For example, suppliers might need to use eco-friendly materials, and tech experts might need to come up with ways to make things in a more sustainable way (Shen, 2014). Stakeholders also matter in how things are bought and used. Businesses involve people like managers, suppliers, and customers when making decisions about what to buy and how to do it in a way that's good for the planet. It's like a team effort to make sure things are done in a responsible way. This kind of teamwork helps businesses make better choices that use less energy and resources.

4.5.3 Cross-Functional Collaboration

In the fast fashion world, working together among important people like managers, suppliers, and tech experts is really important. They need to collaborate to make sure new digital technologies are used well in managing the supply chain. Using digital tools for supply chain means companies need to plan, organise, and track things like buying, storing, and delivering products using technology. This needs everyone to work together, like suppliers and tech providers, to make sure everything fits together smoothly (Deloitte, 2017). When different parts of the supply chain use digital tools that talk to each other, it's like putting a puzzle together. This helps companies serve customers better, be flexible, save money, and work efficiently. So, everyone needs to cooperate—managers to set plans and goals, suppliers to give their thoughts, and tech providers to make sure everything works smoothly. Also, when using digital technology, different parts of the company need to work together well. Managers need to plan how technology will be used, suppliers need to share their ideas, and tech experts need to make sure everything aligns with the company's tech approach. By working together, they can make sure digital technology is used well and brings benefits to the whole company (Deloitte, 2017).

4.5.4 Strategic Guidance and Regulatory Support

In the fast fashion industry, key people like managers, suppliers, and tech experts can provide valuable guidance when using new digital tools for supply chain management. Managers lead the way by planning how to use technology, and they also understand the regulatory landscape that needs to be navigated. They can guide the company in ensuring that digital tools are used correctly and in compliance with regulations. They can also help integrate technology smoothly with the company's existing practices (Deloitte, 2017). Suppliers also play a significant role. They possess deep knowledge of the regulatory requirements in their industry. They can help identify potential compliance challenges and offer guidance on how to address them. They can also contribute ideas on how to leverage technology in ways that align with regulatory standards (Deloitte, 2017). Tech experts are crucial for ensuring that the technology functions well and remains within regulatory boundaries. They select the right technology for the company and ensure it adheres to regulations. Moreover, they assist in utilising technology in a manner that is forward-looking for the company's growth (Deloitte, 2017). Policymakers and regulators also have an essential role to play. They can encourage the

Table 7 Categorisation of stakeholder role (*Source* Authors)

Nature of impact of stakeholder role	Location of stakeholder role	
	Internal	External
Strategic (long term)	Leadership backing (Nath et al., 2022) Absorptive capacity (Nath et al., 2022)	Policymaker incentives (Ahmad et al., 2020)
Operational (short term)	Tech integration expertise (Deloitte, 2017)	Supplier collaboration (Corboş et al., 2023)

utilisation of technology and sustainable practices in the sector by providing incentives and support. Through regulations and standards, they can create an environment that promotes responsible technology adoption and sustainable supply chain management (Ahmad et al., 2020). Categorisation of the stakeholders' role is provided in Table 7.

4.6 Recommended Actions and Strategies

4.6.1 Strategic Digital Transformation

To navigate the integration of digital technologies into PSCM in the international fast fashion sector, several strategic actions are recommended:

- *Thorough Analysis:* Begin by conducting a comprehensive assessment of the current PSCM processes. Identify areas where digital technologies can enhance efficiency, reduce costs, and strengthen relationships with suppliers (Deloitte, 2017).
- *Clear Vision and Strategy:* Develop a well-defined vision and strategy for the digital transformation of PSCM. This should encompass not only technological aspects but also consider regulatory requirements and compliance issues that may arise (Deloitte, 2017).
- *Prioritise and Agile Approach:* Prioritise digital projects based on their potential impact, cost-effectiveness, and speed of implementation. Embrace an Agile approach to execute projects in short, iterative cycles, ensuring adaptability and swift progress (Deloitte, 2017).
- *Collaboration with Stakeholders:* Engage with suppliers and technology providers to ensure that digital solutions align with industry standards and regulations. Verify that these solutions are scalable and sustainable over time, promoting long-term success (Deloitte, 2017).

- *Continuous Monitoring and Optimisation:* Monitor the implementation of digital technologies closely. Measure their effects on PSCM processes, supplier relationships, and overall business performance. Leverage this data to refine operations and make informed decisions (Deloitte, 2017).

4.6.2 Technology Integration and Training

To effectively manage the integration of digital technologies into PSCM within the international fast fashion sector, consider these recommended strategies:

- *Training and Education:* Provide comprehensive training and education to employees alongside the adoption of digital technologies. This ensures that the workforce is equipped with the skills and knowledge needed to embrace these advancements effectively (Seyedghorban et al., 2020).
- *Behavioural Change Initiatives:* Implement initiatives that encourage behavioural changes among employees. This can help them adapt to new processes and technologies more readily, fostering a smooth transition (Seyedghorban et al., 2020).
- *Phased Implementation:* Embrace a phased approach to implementing digital technologies. Begin with digitisation, progress to digitalisation, and ultimately integrate these technologies into the processes (Seyedghorban et al., 2020). This gradual approach allows for smoother adaptation.
- *Collaborative Culture:* Foster a culture of collaboration and continuous learning within the organisation. This encourages employees to work together and stay adaptable in the face of technological changes (Yang et al., 2023).

Through blending training, education, and behavioural change initiatives with a gradual integration approach, fast fashion firms can adeptly implement digital technologies in their PSCM strategies. Fostering a collaborative culture keeps employees adaptable to new tech and processes, allowing the sector to fully leverage digital progress (Yang et al., 2023).

4.6.3 Agile Approach and Collaboration

To effectively manage the implementation of digital technologies PSCM within the international fast fashion sector, consider these recommended actions and strategies:

- *Cross-Functional Teams:* Establish cross-functional teams with experts from various departments to lead the digital transformation (Deloitte, 2017). Collaborate with internal partners and suppliers to maintain a human touch and preserve collaborative relationships as technology advances (Biltoft-Knudsen et al., 2022).
- *Agile Approach:* Embrace an Agile approach for project execution, utilising iterative sprints to ensure prompt and cost-effective implementation (Deloitte, 2017). This method encourages flexibility and the willingness to adapt.
- *Collaboration with Partners:* Collaborate closely with suppliers and technology providers to recognise digital enhancement possibilities in PSCM strategies (Deloitte, 2017). Supplier input and technology expertise are essential for effective alignment.
- *Clear Vision and Strategy:* Develop a well-defined vision and strategy for digital transformation while addressing regulatory requirements (Deloitte, 2017). Compliance and industry standards alignment are crucial.
- *Project Prioritisation:* Prioritise digital projects based on impact, cost, and implementation speed (Deloitte, 2017). This ensures optimal resource allocation and tangible outcomes.
- *Continuous Monitoring and Optimisation:* Continuously monitor digital technology implementation's effects on processes, relationships, and overall performance to drive ongoing improvement (Deloitte, 2017).
- *Culture of Innovation:* Foster an innovative culture that embraces constant improvement within the organisation (Deloitte, 2017). This mindset encourages stakeholders to adapt to changes and drive the digital journey.

These combined actions empower organisations to efficiently adapt to new technologies, enhancing efficiency, collaboration, and overall competitiveness.

4.6.4 Data-Driven Decision-Making

To effectively manage and control the implementation of digital technologies for data-driven decision-making in PSCM within the international fast fashion sector, consider the following recommended actions and strategies:

- *Define Data Needs and Goals:* Start by understanding the organisation's specific data requirements and objectives. Identify the key data sources and types needed to support informed decision-making.

- *Select Appropriate Digital Solutions:* Invest in suitable digital solutions that can collect, analyse, and visualise data effectively. Ensure these solutions can seamlessly integrate with existing systems and processes for streamlined operation.
- *Develop Expertise:* Ensure the organisation possesses the necessary skills to manage and analyse data proficiently. Provide training to employees on how to effectively utilise digital tools for data-driven decision-making.
- *Establish Governance and Control:* Implement clear governance and control mechanisms to maintain the accuracy, reliability, and security of data. Align data-driven decisions with the organisation's overall strategy and goals.
- *Continuous Monitoring and Evaluation:* Regularly assess the effectiveness of digital solutions and data-driven processes. Make necessary adjustments to optimise performance and ensure desired outcomes.
- *Integrate Market and Frontline Insights:* Alongside data-driven strategies, incorporate market and frontline insights to enhance supply chain planning. Employ analytics-driven solutions to manage dynamic demand effectively and prioritise resources for critical goods (Scalco et al., 2023).

4.6.5 Strategic Investment and Roadmap

To ensure effective management and control of the adoption of digital technologies in PSCM within the international fast fashion industry, consider the following recommended actions and strategies:

- *Prioritise High-Value Opportunities:* Companies should carefully select digital applications that offer substantial business value and address existing performance gaps. Align these ideas with current challenges to identify areas of synergy and potential benefits. Develop comprehensive business cases for the most relevant and financially rewarding opportunities.
- *Launch Small-Scale Pilots:* Design and launch pilot projects to test digital initiatives on a smaller scale before full implementation. Begin by testing ideas in high-priority areas of the business to identify effective strategies and uncover emerging opportunities. This approach allows for iterative refinement and learning from successes and failures.
- *Segmented Rollout and Capability Development:* Following successful pilot projects, expand digitisation efforts to other segments of the PSCM process with the highest expected benefits. Consider factors like key suppliers or regions for segmentation. Evolve the required capabilities alongside the rollout to ensure a seamless integration.

- *Establish Clear Business Rules and Metrics:* Develop straightforward business rules that govern the management of the PSCM process. Define key performance indicators (KPIs) to measure outcomes, including metrics such as cost savings, inventory reduction, and lead time reduction. These metrics help assess the success of digitisation initiatives.
- *Enhance Collaboration and Partnerships:* Strengthen collaborative efforts with partners, suppliers, distributors, and technology providers. Recognise that a fully integrated PSCM process relies on collaboration. Identify key partners, such as technology providers or logistics partners, to support the implementation of digital technologies.

Companies in the fast fashion sector can foster cross-functional collaboration by engaging various departments to collectively develop a roadmap. This collaborative effort can identify prevailing and emerging digital trends, allowing strategic investments in technology adoption. By prioritising initiatives such as workforce training and acquiring cutting-edge technologies, companies can enhance their digitalisation efforts, gaining a competitive edge in the market. By following these strategies, companies in the fast fashion industry can efficiently navigate in the implementation of digital technologies in PSCM strategies. The identified recommendations are presented in Table 8.

Table 8 Categorisation of recommendation (*Source* Authors)

Nature of impact of recommendation	Location of recommendation	
	Internal	External
Strategic (long term)	Clear technology roadmap (Deloitte, 2017)	Stakeholder collaboration (Deloitte, 2017)
Operational (short term)	Training programs (Seyedghorban et al., 2020) Agile delivery approach (Deloitte, 2017) Performance measurement	Pilot testing

5 Conclusion

Following the SLR methodology based on 60 sources, this research investigated the impact of procurement digitalisation (PD) on international procurement and supply chain management (IPSCM) strategies in the international fast fashion sector by identifying key factors influencing this impact, challenges in implementation, good practices, the role of stakeholders, and recommendations to effectively manage digitalisation as shown in Fig. 2.

5.1 Research Limitations

While these findings represent valuable insights, some limitations are acknowledged. First, the sole reliance on secondary sources means inclusion of primary data through interviews and surveys could have enriched the analysis with direct experiences. Second, a broad perspective was taken rather than a focussed analysis of specific fashion firms' digitalisation journeys. Third, the limited research dedicated specifically to the fast fashion sector may have constrained the investigation's relevance and depth. Fourth, the study emphasised digital technologies, potentially overlooking the potential effects of other innovations like sustainability, transparency, circular economy, and equality practices influencing strategy evolution.

5.2 Future Research Opportunities

Several opportunities exist to build on this research and address its limitations. First, empirical studies examining implementation challenges and outcomes through extensive case studies would generate nuanced qualitative insights. Second, focussed quantitative assessments of leading fast fashion organisation's digital transformation results could provide measurable benchmarks. Third, thoroughly addressing the identified research gaps around implementation challenges and stakeholder roles presents a promising area for investigation. Fourth, comprehensively exploring innovative sustainability technologies and circular business models may illuminate new transformative possibilities.

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7

Customer Switching Behaviour and the Adoption of Disruptive Technologies

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1 Introduction

In B2B businesses, firms' switching behaviour reflects their desire to consider available alternatives in the market, resulting in a competitive activity by the supplier's offerings in the marketplace (Krafft et al., 2021). As new and competing technology emerges, B2B firms may like to switch from their previous preferential supplier to another one embedded with updated technology (Bhattacharya et al., 2022). Because of the development of disruptive technologies, the technological revolution has changed the way B2B business models operate (Haddad et al., 2021).

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B2B users at each level of their operation evaluate their outcomes, and they usually select the alternative technology if the newly adopted technology can offer higher levels of satisfaction than the previous one (Xu et al., 2021). The B2B customers' evaluation of the disruptive technology product's overall economic advantages relative to the customer's present product is known as comparative economic value (Lin et al., 2021). Moreover, prior researchers have mentioned that firms usually move to another supplier from where they get more economic facilities (Baloch & Rashid, 2022; Xu et al., 2023). Research also emphasize the significance of social influence on B2B customers' behaviours, as social influence also evidence to perform as moderation in the relationship between B2B customers' attitudes and behaviours (Yang et al., 2021).

Several prior research have been found on the preferences for disruptive technology. Most of the research has focused on disruptive technology from a different perspective like the adoption of advanced technology (Spanaki et al., 2022), a solution to disaster management (Munawar et al., 2022), support to games (de Rosa & Strode, 2022), digital supply chain (Mitra et al., 2023), cannibalization and survival strategy (Chandrasekaran et al., 2022), etc., Most of the research has focused on the positive aspect of the adoption of new technologies from the side of the customer, whereas, the inner causes of welcoming disruptive technology is not well studied. Moreover, the key question on disruptive technology from the B2B perspective is also underexplored (Xue & Li, 2022). An inner cause of user dissatisfaction drives switching behaviour (Zhang et al., 2022). Surprisingly, the relationship between user dissatisfaction and their perceived preferences and switching behaviour towards disruptive technology has not been established yet (Homayounfard & Zaefarian, 2022). Xu et al. (2021) mentioned that in the industry, technology disruption might happen when the outcome of previous technology doesn't meet the actual requirements. On the other hand, economic efficiency is another important parameter for industrial success (Shvindina, 2022). Alet Vilaginés (2022) mentioned that economic benefit leads business customers to switch from one supplier to another. Even though the issue of economic comparison is a well-established issue in the business world, the importance of comparative economic value in switching to disruptive technology has not been studied yet.

Thunberg and Arnell (2022) mentioned in their research that user dissatisfaction with any technology leads to another preference and it is very common in B2B and consumer-behaviour-related studies. On the other side, business customers usually prefer products or raw materials at a lower cost and that is why depending on multiple suppliers is a common fashion for the

producers (Shankar, 2022). Shankar (2022) mentioned that a lower opportunity cost than its competitive seller is an economic advantage. In B2B studies utilizing business, the relationship between customer dissatisfaction, comparative economic value, and perceived preferences for disruptive technology has received limited attention. Additionally, the relationship between perceived preferences with customer switching behaviour is well-defined in many pieces of literature (Campos Pimenta et al., 2022; Rawis et al., 2022). The argument shows that perceived preferences act as a mediator between the relationships between user dissatisfaction and B2B customer switching behaviour to disruptive technology and with comparative economic value and B2B customer switching behaviour to disruptive technology (Kamolsook et al., 2019). Business customers have easy access to information about different disruptive technology products and quickly develop a perceived preference for disruptive technology based on their present satisfaction/dissatisfaction level and the economic value they want to get from the current product and make the decision to switch (Naradipa et al., 2020).

Above all, several research gaps were found in comprehending the factors affecting the business customers' (B2B) switching behaviour towards disruptive technology. Business customers' technology switching behaviour towards disruptive technology has lacked attention of B2B marketing researchers (Kamolsook et al., 2019). A limited number of studies have also examined social influence (Fernandes & Panda, 2019) affecting customers' switching behaviour towards disruptive technology; in fact, to the best of authors' knowledge, social influence has never been used as a moderator to examine the influence of the relationship between user dissatisfaction, comparative economic value, and B2B customers' switching behaviour towards disruptive technology. The following research questions (RQs) are addressed by this study to fill in these considerable research gaps in the B2B customers' switching behaviour towards disruptive technology:

RQ1: What is the effect of user dissatisfaction and comparative economic value on customers' preferences and customers' switching behaviour towards disruptive technology?

RQ2: What is the moderating role of social influence between user dissatisfaction, comparative economic value, user perceived preferences, and customers' switching behaviour towards disruptive technology?

RQ3: What is the mediating role of perceived preferences between user dissatisfaction, comparative economic value, and customers' switching behaviour towards disruptive technology?

The study's objective in responding to these research questions is to investigate the customers' switching behaviour towards disruptive technology due to dissatisfaction and comparative economic value. The objectives of this study can be classified into two groups. **First**, we look at how customers switch from their current technology products to disruptive ones and the role of comparative economic value in forming B2B customer switching behaviour (Cao et al., 2022). **Secondly**, as the previous studies only focus on customers switching behaviour where users' new technology leads them to switch, but in this study, we will investigate customer's dissatisfaction and comparative economic value that can influence the customers switching behaviour towards disruptive technology. Third, we additionally considered social influence as a moderator and user perceived preference for disruptive technology as a mediator in the formation of B2B customers switching behaviour towards disruptive technology.

The present study considered the expectation disconfirmation theory (EDT) to develop a conceptual framework for explaining how expectations evolve in a business market (Rehman et al., 2021). Therefore, we integrate the expectation disconfirmation theory (EDT) constructs (expectations, perceived performance, disconfirmation of beliefs, and satisfaction) to build a new model where user dissatisfaction and comparative economic value influence user perceived preferences in B2B customer switching behaviour. Based on the core concept of this EDT theory, where disconfirmation is affected by expectations and perceived performance together and results in satisfaction (Chen et al., 2021), our study focused on creating a relationship between user dissatisfaction and comparative economic value which results in business customers' switching behaviour towards disruptive technology where social influence is a moderator and user perceived preferences for disruptive technology is a mediator.

Our study contributes three new findings to the literature on business customers' switching behaviour in the scope of disruptive technology. **First**, in contrast to prior research, we consider discontinuous technological advancement and add to the literature stating that customers switching behaviour towards disruptive technology involves expectations and performance of current technology products, user dissatisfaction and comparative economic value. **Secondly**, we contribute to the existing literature by highlighting social influence as a moderator between user dissatisfaction, comparative economic value and customers' switching behaviour towards disruptive technology products. Also, this study contributes to the existing literature by stating user perceived preferences for disruptive technology

as a mediator. **Thirdly**, by focusing on the understudied topic of disruptive technology products, we add to the literature on business customers' switching behaviour, for which our extended model yields meaningful results by applying EDT theory.

The remaining part of this study is organized as follows: Sect. 2 discusses the relevant literature. Section 3 presents the hypotheses development. Section 4 is research methodology. In Sect. 5, the analysis and results are discussed. Later in Sect. 6, a discussion of the findings was depicted. Finally, the conclusions, managerial implications, limitations, and future directions are discussed in Sect. 7.

2 Literature Review

2.1 Expectation Disconfirmation Theory

In understanding business customers switching behaviour, researchers have utilized several theories including the construal level theory (Wirtz et al., 2014), Push–Pull-Mooring Theory (Dogra et al., 2022), the Technology Acceptance Model (TAM) (Cai et al., 2022), expectation disconfirmation theory (EDT) (Mustafa et al., 2020). Among all, the expectation disconfirmation theory (EDT) is frequently used in information systems research to examine adoption behaviour and switching behaviour (Mustafa et al., 2020). In EDT theory, the term "expectations" refers to the qualities or traits that specific individuals anticipate from technology, information systems, products, or services. These expectations have a direct impact on both the disconfirmation of beliefs and perceptions of performance (Hasan, 2019). Although this theory was created to investigate physical objects, EDT is today widely used for a variety of technology-related activities (Rehman et al., 2021). Customer satisfaction literature in the retail and service industries has widely acknowledged and used the effects of expectation disconfirmation (Cai & Chi, 2021). According to EDT, customers' satisfaction depends on the magnitude and direction of disconfirmation (Venkatesh & Goyal, 2010). Whereas negative disconfirmation causes negative behaviour, bad attitudes, and user dissatisfaction (Zhang et al., 2021). The study would like to utilize EDT theory from the perspective of B2B practice. In B2B business, business customers access different amenities from different sources, however, any dissatisfaction related to technologies always leads to preferring another technology. On the other hand, cost efficiency and economic benefit also lead

to taking a better alternative. Thus, optimizing the EDT theory from this background will add some value to B2B and disruptive technology literature.

According to EDT theory, a comparison between perceived value and pre-expectations implicitly determines whether customers are satisfied with services (disconfirmation) (Wang & Fan, 2022). Thus, in this study, we have constructed five variables and established relationships among them. These variables are user dissatisfaction, comparative economic value, user perceived preference for disruptive technology, social influence, and customers' switching behaviour. The following section explains the constructs tested in this study separately.

2.2 Disruptive Technology

Disruptive technology refers to the selection and/or use of modern technology that fundamentally transforms how firms operate, often displacing existing processes, products, or business models, and creating entirely new ways of delivering value to customers (Singh & Hanafi, 2019). When technology disregards accepted paradigms or practices, it is seen as disruptive (Robertson et al., 2021). Artificial intelligence (AI) (Paschen et al., 2020), blockchain and distributed ledger technology (Morkunas et al., 2019), and virtual reality (Farshid et al., 2018) are a few examples of disruptive technologies that can disrupt established paradigms. Embracing disruptive technology in different business elements is a very common practice among industrial buyers (Teixeira et al., 2022).

2.3 User Dissatisfaction

In a B2B relationship, every transaction is outcome based (Korkeamäki et al., 2022). Business customers become dissatisfied when a product or service fails to match their expectations or has several flaws (Pascual-Nebreda et al., 2022). Dissatisfied industrial customers have the option to keep silent or switch suppliers without warning, which affects the supplier–buyer relationship (Ashok et al., 2002). When deciding whether to switch from a current technology product to a disruptive technology product, business customers contrast these products, as opposed to adopting new technology to satisfy demands that they had previously been unable to satisfy with prior products (Hsu, 2014; Kamolsook et al., 2019).

2.4 Comparative Economic Value (CEV)

Scholars have defined customer value in terms of the acquire (benefits) and provide (sacrifice) components (Kamolsook et al., 2019; Zeithaml, 1988). Comparative economic value is the customers' assessment of the disruptive technology items' total economic benefits relative to those of the customer's present product. It is composed of (1) comparative monetary cost, (2) comparative long-term benefit, and (3) comparative superiority (Kamolsook et al., 2019). The advantage received by a business from a good or service is measured in terms of economic value. A business may assess the economic value of its goods to estimate demand and set prices.

2.5 User Perceived Preference for Disruptive Technology

User preferences are complex, diverse, and heterogeneous (Li et al., 2021a, 2021b). Numerous factors affect business customer preferences, which include likes, dislikes, motivations, and dispositions (Kumar-Singh & Thirumoorthi, 2019). User preferences are shifting away from fixed technology and towards mobile technologies, which is a prominent trend in the world of ICT (Mugo et al., 2017). Prior research on user preferences has primarily focused on self-reported data rather than actual user behaviour observation (Tondello & Nacke, 2020).

2.6 Social Influence

Customers are typically impacted by the presence, behaviour, and expectations of others. Business customers are interconnected with each other, and they share their views with their views. One of the most important aspects of changing customer behaviour in a sustainable way is social factors (White et al., 2019). Social influence is the process by which social interactions cause a person's views, attitudes, ideas, and behaviours to change (Gani et al., 2023a). In a business group, business customers have an impact on decisions on their business operations (Charry & Tessitore, 2021). However, it is still unknown how strong this effect is in comparison to that of peers about disruptive technology.

2.7 Customers' Switching Behaviour Towards Disruptive Technology

Business customers' switching behaviour is a process in which customers temporarily or totally replace their relationship with their existing product service provider with one of their competitors' products (Hafandi et al., 2021). The possibility or certainty that a customer will migrate from their current product provider to a new product provider is known as switching intention (Satriadi et al., 2022). Product switching behaviour is very common in business customers' post-adoption behaviour (Cheng et al., 2019).

3 Hypotheses Development

3.1 User Dissatisfaction and Customers' Switching Behaviour

Due to dissatisfaction or other issues, business customers gradually shift from being loyal customers to a switcher (Satriadi et al., 2022). While business users think that they may get more profit from adopting a disruptive technology they switch from their existing one (Cui et al., 2021). Xu et al. (2021) mentioned that user dissatisfaction influences user switching behaviour positively. One of the constructs of EDT is dissatisfaction, which is the result of overall expectation and performance of existing products; however, this dissatisfaction is a significant factor for customer to switch their product (Kamolsook et al., 2019) Hence,

H1: User dissatisfaction is positively related to customers' switching behaviour towards disruptive technology.

3.2 Comparative Economic Value and Customers' Switching Behaviour

According to the earlier study, the perceived economic value would affect switching behaviour (Lin et al., 2022). Kamolsook et al. (2019) stated comparative economic value (CEV) is an expanded definition of perceived value that contrasts the disruptive technology product's anticipated benefits with the expected financial costs. Kamolsook et al. (2019) also stated that in the context of disruptive innovation, when the sacrifices are considerably

more evident than with continuous innovation, this comparison of two products and of the advantages and disadvantages probably plays a crucial role to switch the current product. In EDT, expectation and performance jointly develop disconfirmation which means the valuation of perceived performance if it is worthy or not compared to the previous product (Hasan et al., 2019); however, if customers can find more value offering from the existing product will determine to switch from their current digital product. We can posit that comparative economic value is positively related to customers' switching behaviour towards disruptive technology.

H2: Comparative Economic Value is positively related to customers' switching behaviour towards disruptive technology.

3.3 User Dissatisfaction and User Perceived Preference for Disruptive Technology

When perceived performance satisfies or exceeds expectations, satisfaction is attained; on the other hand, when perceived performance falls short of expectations, discontent results (Chen et al., 2022). Buyers judge the outcomes of the product, and while it doesn't match their expectations, they usually switch to other products or suppliers (Mohd Yusof et al., 2021). Denden et al. (2018) found a significant positive relationship between dissatisfaction and preferences for disruptive technology. Thus, we can posit:

H3: User dissatisfaction is positively related to user perceived preference for disruptive technology.

3.4 Comparative Economic Value and User Perceived Preference for Disruptive Technology

Even though business buyers expect quality goods for their operations, they expect proper value and economic price for their quantity purchase. Zhang et al. (2021) mentioned that in buying, B2B customers usually compare the current purchase experience with prior expectations (comparative economic value). Makarina et al. (2019) and Huang et al. (2019) stated that customers' perceived value (CEV) is the perception of customers' preferences based

on economic aspects. Makarina et al. (2019) and Huang et al. (2019) mentioned that economic benefit has a positive relationship with preferences for switching from one distributor to another. Hence,

H4: Comparative economic value is positively related to user perceived preference for disruptive technology.

3.5 User Perceived Preference for Disruptive Technology and Customer's Switching Behaviour towards Disruptive Technology

Business customers' attitudes or intentions regarding switching are positively correlated with their actual intention to switch (Msaed et al., 2017). Potra et al. (2018) found that due to business users' operational purpose, they compare the current product to other products, which affects their future purchase intention or switching intention. Satriadi et al. (2022) found that firms' attitude represents preferences that correlate with how firms rate a product, which affects both future purchase intentions and switching behaviour. Hence,

H5: User perceived preference for disruptive technology is positively related to customer's switching behaviour towards disruptive technology.

3.6 The Moderating Role of Social Influence

Within a business group, business buyers usually discuss their views with their peers and in B2B business, views from other business ties are highly influential (Joa & Magsamen-Conrad, 2021). Xu et al. (2021) stated that social influences positively moderate the relationship between user dissatisfaction and switching behaviour. Shah et al. (2018) found that social influence has a positive influence on user dissatisfaction and business buyers' switching behaviour. In B2B, the reference price is important for the business client (Tjokrosaputro & Cokki, 2020). Tjokrosaputro and Cokki (2020) found that comparative economic price (less price with more benefit) is always desired by the industrial buyers. Industrial buyers always depend on multiple suppliers for getting their raw materials at a convenient price. Kamolsook et al. (2019) mentioned that the influence of other business stakeholders positively moderates the relationship between price and switching behaviour.

Russo et al. (2017) mentioned that influence from other business associates moderates between business customers switching intention and switching behaviour. Misra et al. (2022) stated that social influence changes business buyers' behaviour and it influences preferences and intentions. Thus, using social influence as a mediator, the study would like to hypothesize that:

H6: Social influence moderates the relationship between user dissatisfaction and customers' switching behaviour toward disruptive technology.

H7: Social influence moderates the relationship between comparative economic value and customers' switching behaviour toward disruptive technology.

H8: Social influence moderates the relationship between user perceived preference of disruptive technology and customers' switching behaviour toward disruptive technology.

Here, Fig. 1. demonstrates the relationship between user dissatisfaction and customers' switching behaviour as well as the relationship between comparative economic value and customers' switching behaviour towards disruptive technology. It demonstrates the moderating role of social influence and the mediating role of user perceived preferences.

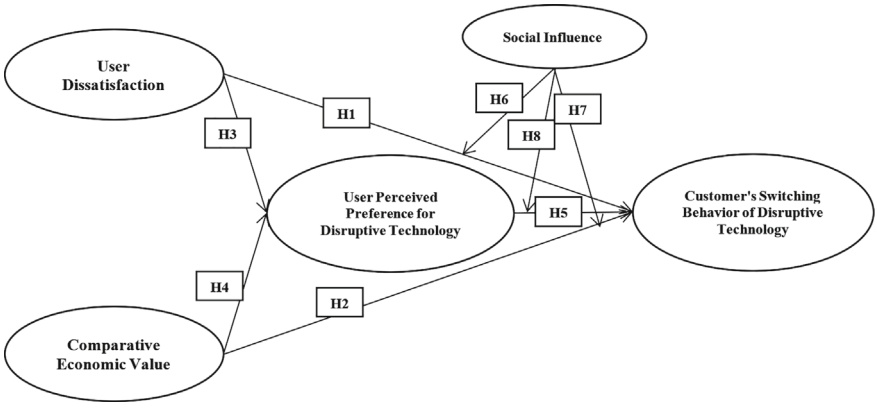

Fig. 1 Proposed model of the study (*Source* Authors)

4 Research Design and Method

4.1 Target Population

The study collected its primary data from 626 B2B supplier firms from Bangladesh. We have screened different industries from Yellow Pages. We have collected data from the manufacturers to understand their dissatisfaction and economic value-related factors that are directly related to switching behaviour.

4.2 Measures

To test the reliability and validity of the survey questionnaire, the measurement items for the research variables were adapted from previous studies. All the items were measured by using a 5-point Likert scale, where 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, and 5 = strongly agree. The measurement items are represented in Appendix A along with a list of their adapted sources. The items of user dissatisfaction adapted from Moon et al. (2017) study; items of comparative economic value adapted from Kamolsook et al. (2019) and Lin et al. (2022); items of user perceived preferences of disruptive technology adapted from Li et al., (2021a, 2021b); items of social influence were sourced form Venkatesh et al., (2003, 2012) and Kamolsook et al. (2019). Lastly, the items of customers' switching Behaviour of disruptive technology adapted from Davis (1989), Venkatesh et al. (2003, 2012) and Kamolsook et al. (2019) research. Before administering the survey instrument, we performed pretesting of the instrument, and feedback was taken from five academics of industrial marketing field, and five professionals from the firms under technology industry. The present study applied these sessions with the experts separately to ensure the content and face validity of the survey questionnaire. In this process, we also applied a snowball sampling method and the questionnaire was distributed among 50 additional respondents to further confirm the measurement's reliability and overall functionality of the measurement scale under each variable. For instance, Cronbach's alpha scores were checked for reliability. The results showed meritorious and supported the internal consistency of the collected data set. After pretesting, with a few minor modifications were made to the measurement scales and wording, and then we applied a face to face survey, also known as in person data collection method was conducted.

4.3 Questionnaire Design and Data Collection

For this study, the respondents were selected based on business customers' switching behaviour towards technology products as innovative products are being emerged at a rapid pace. To get actual respondents who used technology products and keep changing them if they are dissatisfied, in the questionnaire, we included a screening question that directly asked if they recently switched to any other technology product. Additionally, the response to this question was sent in a yes-or-no format. This question helped us to identify the actual users (B2B firms) of disruptive technology products. The questionnaire was distributed to the respondents via face-to-face survey by using the convenience sampling method. Three sections comprised with the survey questionnaire: (a) screening questions (b) questions on respondents' demographics; (c) scales of each variables in the conceptual framework. A cover letter detailing with the research objectives were distributed to the respondents, along with the instructions before they started each section of the survey instrument. In addition, we also ensure respondents' privacy, anonymity and confidentiality to encourage them to provide honest responses. Respondents were given adequate time to complete the survey question. Around 1500 questionnaires were distributed to users of disruptive technology products, and 626 surveys were returned. Thus, 626 surveys were usable for final data analysis.

4.4 Non-response Bias

To assess non-response bias, we compared early and late waves of returned surveys (Armstrong & Overton, 1977). We compared the returned questionnaires from the first wave and the last wave data (based on the dates the responses were received). We received 307 responses at the end of the first wave (15 April 2022) and 319 responses at the end of the second wave (16 July 2022). T-tests on the two groups were used as the foundation for the comparison study. The analyses did not reveal any statistically significant differences ($p > 0.5$). The absence of non-response bias in our data allowed us to draw this conclusion (Lambert & Harrington, 1990).

4.5 Common Method Variance

The common technique variance was put to the test in this investigation in accordance with Podsakoff et al. (2003). This study used a 5-point Likert scale format to measure the independent variables and dependent variables

(Rahman et al., 2021; Gani et al., 2022). During the data collection period, all the respondents' identities and answers were kept anonymous due to their privacy, and they were aware of it. Applying exploratory factor analysis, the study employed Harman's (1967) single-factor test to examined the common method variance in the collected data set. Less than the required criterion of 50.00%, the result showed an explained variance of 20.39%. In conclusion, the results confirmed that our collected data had no such issue of common method variance.

5 Data Analysis and Result

5.1 Demographic Profile

Respondents' demographic details are highlighted in Table 1. Most of the respondents were males (50.80%) and were aged between 26 and 35 (37.53%). The remaining participants ranged in age from 26 to 35 (37.53%) and from 18 to 25 (29.55%). Among the respondents, from the perspective of job position, supply chain managers hold 28.75%. The highest number of firms belong to 5–10 years which is 32.10% and firms are medium sized 46.16% from different industries.

5.2 Measurement Model

A two-step process combining a measurement model and a structural model were used, following the recommendations of Gerbing (1988), to confirm the accuracy and validity of the constructs before assessing the structural relationship of the proposed model (see Table 2). The results from the test (see Table 2) demonstrated that each construct of the research model showed an adequate level of Cronbach's alpha values greater than 0.70, above the threshold limit (0.70) proposed by Fornell and Larcker (1981). The measurement model's components were also found to be sufficiently dependable. The results from measurement model (Table 2) indicates that, the estimated construct loadings ranged from 0.701 to 0.879, the composite reliability (CR) ranged from 0.795 to 0.871, and the average variance extracted (AVE) varied from 0.565 to 0.692, all of which are greater than the threshold level. Additionally, the values of VIFs should be lower than 3.3 to avoid vertical or multiple co-linearity (Kock & Lynn, 2012). Because every VIF, in this case, is below 3.3, the discriminant validity of the data set was also satisfied. As a result, the convergent validity requirement's parameters were validated. The

Table 1 Participant description ($N = 626$)

Characteristics	Categories	Frequencies	Percentage
Gender	Male	318	50.80
	Female	308	49.20
Age	18–25	185	29.55
	26–35	235	37.53
	36–45	91	14.53
	46–55	68	10.86
	55–above	47	07.53
Job Position	Marketing Manager	160	25.59
	Procurement manager	152	24.28
	Supply chain manager	180	28.75
	IT manager	134	21.40
Firms age	Less than 5 years	165	26.35
	5 to 10 years	201	32.10
	11 to 15 years	188	30.03
	16 to above	72	11.50
Switching frequency	> 5 times a year	56	8.95
	3–5 times a year	87	13.90
	1–3 times a year	258	41.21
	< 1 times a year	225	35.94
Size (Employees)	Small (10–49 employees)	237	37.85
	Medium (50–249 employees)	289	46.16
	Large (250 employees and more)	100	15.97
Type of firm	Beverage producers	72	11.50
	Furniture manufacturers	67	10.70
	Leather products	89	14.21
	Steel mills	21	3.35
	Cement producers	70	11.18
	Foundry	88	14.05
	Agro processors	98	15.65
	Jute mills	43	6.86
	Rubber goods manufacturer	78	12.46

results from testing the measurement model (Table 3) also showed the inner VIF, which can be used to test for collinearity.

Additionally, the outcome supports discriminant validity. The study calculated AVE's square root (see Table 4) to confirm discriminant validity (Fornell & Larcker, 1981). The measurement model's results provided a sufficient proof of its reliability, convergent validity, and discriminant validity. The results from the measurement model analysis provide us with the assurance

Table 2 Convergent validity and internal reliability

Constructs	Items	Standard load-ings	Cronbach's Alpha	Composite reliability (CR)	Average Variance Extracted (AVE)	VIF
Social Influence (SI)	SI1	0.806	0.767	0.806	0.581	1.221
	SI2	0.732				1.268
	SI3	0.745				1.313
Comparative economic value (CEV)	CEV1	0.815	0.778	0.871	0.692	1.483
	CEV2	0.850				1.811
	CEV3	0.830				1.651
Customers switching behaviour towards disruptive technology (CSBDT)	CWBDT1	0.701	0.715	0.795	0.565	1.164
	CWBDT1	0.793				1.264
	CWBDT3	0.759				'1.260
User Dissatisfaction (UD)	UD1	0.739	0.754	0.860	0.673	1.278
	UD2	0.879				1.992
	UD3	0.836				1.861
User perceived preferences of disruptive technology (UPPDT)	UPPDT1	0.831	0.770	0.867	0.685	1.618
	UPPDT2	0.859				1.738
	UPPDT3	0.792				1.457

Table 3 Inner Variance inflator factor

	CSBDT	UPPDT
CEV	2.502	2.059
SI	1.950	
UD	2.608	2.059
UPPDT	2.171	

to confirm the constructs and proceed with testing the proposed structural model for hypothesized relationships.

Furthermore, the discriminant validity was cross-checked using the heterotrait-monotrait ratio of correlations (HTMT). According to Ali et al. (2018), researchers should employ the HTMT criterion of Henseler et al. (2015) rather than the established techniques for evaluating discriminant validity, such as cross-loadings and the Fornell-Larcker criterion (Fornell & Larcker, 1981). The HTMT ratio needs to be below 1.00. (Henseler et al.,

Table 4 Constructs' correlation coefficients and the square root of AVE (in bold on diagonal)

	CEV	CSBDT	SI	UD	UPPDT
CEV	*0.832*				
CSBDT	0.532	*0.752*			
SI	0.577	0.588	*0.762*		
UD	0.717	0.565	0.658	*0.820*	
UPPDT	0.680	0.569	0.594	0.640	*0.828*

Table 5 Heterotrait–Monotrait (HTMT) ratio

	CEV	CSBDT	SI	UD	UPPDT
CEV					
CSBDT	0.760				
SI	0.824	0.809			
UD	0.832	0.827	0.841		
UPPDT	0.877	0.827	0.846	0.843	

2015). Therefore, it is proven that all constructs satisfied the criteria for discriminant validity. Table 5 displays the HTMT findings which also support the recommendation of the previous scholars prescribed threshold.

5.3 Structural Model

The structural model identifies the relationships between latent variables and enables us to assess the degree of influence between them (calculated as path coefficients). Using Smart PLS 3.3, the structural model's explanatory power was assessed as well as the structural path and R^2 scores of endogenous variables. The bootstrapping approach was used to test the hypothesized association at a significance level of 0.05 ($p < 0.05$) (Buckland et al., 1997). Path coefficient (β) and corresponding p-values were used in this study to examine the proposed relationship between the independent and dependent variables. The PLS findings of the hypothesis testing are reported in Tables 6 and 7, and the structural equation model results are displayed in Fig. 2. With an R^2 adjusted value of 0.438, a significant percentage of the variation in customers' switching behaviour towards disruptive technologies was predicted by the conceptual model, as seen in Table 6. For user perceived preferences of disruptive technology, a good R^2 adjusted value was also reported (0.509). The predictive validity of the current study model is therefore confirmed. Table 6 displays the testing outcomes for the structural model. Additionally, the Stone-Geisser Q^2 has been established to evaluate the predictive relevance

of the model. According to Hair et al. (2019), the redundancy value in Q^2 must be higher than zero.

The results show that the relationships between user dissatisfaction (UD) and B2B customer switching behaviour towards disruptive technology (CSBDT) ($\beta = 0.147$, $p < 0.05$), comparative economic value (CEV) and B2B customer switching behaviour towards disruptive technology (CSBDT) ($\beta = 0.115$, $p < 0.05$), user dissatisfaction (UD) and user perceived preferences for disruptive technology (UPPDT) ($\beta = 0.314$, $p < 0.05$), comparative economic value (CEV) and user perceived preferences for disruptive technology (UPPDT) ($\beta = 0.455$, $p < 0.05$) and user perceived preferences for disruptive technology (UPPDT) and B2B customers switching behaviour towards disruptive technology (CSBDT) ($\beta = 0.207$, $p < 0.05$) were significant. Thus, H1, H2, H3, H4 and H5 were supported.

Additionally, this research also examined the moderating effect of social Influence (SI) between UPPDT, CEV and UD, as shown in Table 8. Here, social influence was found with a considerable influence on the relationship between UD and CSBDT, as well as between CEV and CSBDT and between UPPDT and CSBDT. The value here discovered of the moderating effect of SI between UD and CSBDT ($\beta = 0.121$, $p < 0.010$) (H8), again moderating effect of SI between CEV and CSBDT ($\beta = 0.222$, $p < 0.00$) (H7) and the moderating effect of SI between UPPDT and CSBDT ($\beta = 0.113$, $p < 0.005$) (H6). Here, H6, H7 and H8 supported as p-value meets the suggested level.

Likewise, this study further examined the mediating effect of UPPDT between UD and CSBDT as well as CEV and CSBDT as shown in Table 9.

Table 6 Result of Adjusted R^2 and Q^2

Endogenous latent construct	R^2	R^2 adjusted	Q^2
CSBDT	0.442	0.438	0.244
UPPDT	0.511	0.509	0.346

Table 7 Result of hypotheses testing

Relation		β	P Values	Decisions	95% BC-CI	
					Lower	**Upper**
H1	UD - > CSBDT	0.147	0.001	Supported	0.051	0.220
H2	CEV - > CSBDT	0.115	0.024	Supported	0.014	0.207
H3	UD- > UPPDT	0.314	0.000	Supported	0.221	0.392
H4	CEV - > UPPDT	0.455	0.000	Supported	0.367	0.536
H5	UPPDT- > CSBDT	0.207	0.000	Supported	0.120	0.293

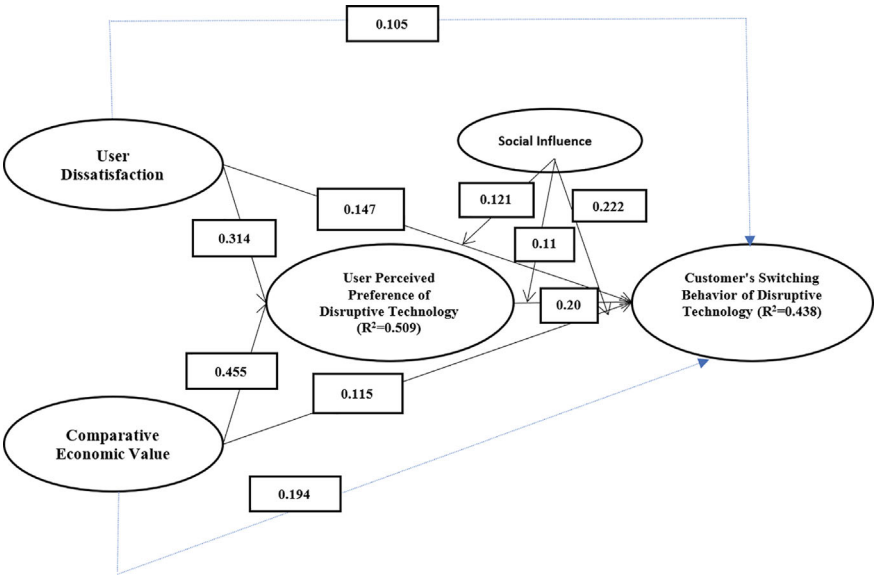

Fig. 2 Direct, indirect and moderation relationship (*Source* Authors)

Table 8 Test of moderation

Relation		β	P Values	Decisions	95% BC-CI	
					Lower	**Upper**
H6	UPPDT*SI- > CSBDT	0.113	0.005	Supported	0.103	0.345
H7	CEV* I - > CSBDT	0.222	0.000	Supported	0.114	0.279
H8	UD*SI - > CSBDT	0.121	0.010	Supported	0.101	0.296

Because the relationship between the dependent and independent variables is significant in this case, as recommended by Preacher and Hayes, (2008), user perceived preference for disruptive technology (UPPDT) was found to have a significant and partial mediating effect on the connection between UD and CSBDT as well as CEV and CSBDT. The values here was found ($\beta = 0.105$, $p < 0.00$) and ($\beta = 0.194$, $p < 0.000$) respectively, showed in Fig. 2. Thus, the study accepted hypotheses H9 and H10.

Table 9 Indirect effect (test of mediation)

Relation		β	P Values	Decisions	95% BC-CI	
					Lower	**Upper**
H9	UD - > CSBDT	0.105	0.000	Supported	0.094	0.192
H10	CEV- > CSBDT	0.194	0.000	Supported	0.135	0.234

6 Discussion

The study's results demonstrate that user dissatisfaction has a positive relationship with B2B customers' switching behaviour, which agrees with the earlier arguments proposed by Wu and Cheng (2018). This study also revealed that comparative economic value has a positive relationship with customers' switching behaviour towards disruptive technology (Satriadi et al., 2022). Between these two predictors, the results from the study shows that industrial users' dissatisfaction is more dominant than comparative economic value. Here, the result demonstrates that industrial buyers are ready to sacrifice price, but ultimate dissatisfaction comes from the lack of operational inefficiency. Our result presents interesting findings through the study that UPPDT is significantly affected by CEV compared to UD. It shows that price comparison comes initially regarding the alternatives (Ghasri & Vij, 2021).

This study also assessed the moderating role of social influence on the relationship between user dissatisfaction, comparative economic value, user perceived preferences of disruptive technology, and customers' switching behaviour towards disruptive technology. Customers switch towards disruptive technology because they get dissatisfied with their current product and societal factors such as the influence of other business organizations also encourage them to switch the product (Shah et al., 2018).

According to Kamolsook et al. (2019) and Gani et al. (2023b), social influence constructs influence both how well customers perceive a product (like the comparative economic value) and how likely they are to make a purchase (such as the disruptive technology product switching intent). This study also posits that perceived user preferences and customers' switching behaviour are influenced by other market externalities, such as social influence (Ghasri & Vij, 2021).

This study also demonstrates the mediating role of user perceived preferences for disruptive technology. User perceived preferences for existing products impact customers' level of satisfaction and result in switching to their current product (Naradipa et al., 2020). According to Garga et al. (2019), if perceived preferences for the current product fail to perform or do not meet the anticipated threshold, users get dissatisfied with the existing product and decide to switch products. Also, user perceived preferences influence the customer to evaluate current and potential products' performance and economic value; if the current product fails to fulfil the level of economic value users should get, they will search for alternatives and switch the product (Garga et al., 2019). Higher perceived preference raises the perceived complexity for customers to compare the economic value of various

products and select a product that matches their preferences, increasing the number of product alternatives to consider and switching from the current product (Whitley et al., 2018).

6.1 Theoretical Contribution

This research applied the philosophy of expectation disconfirmation theory (EDT) to examine why B2B users switch to disruptive technology. The core constructs of expectation disconfirmation theory are—expectation, performance, disconfirmation, and satisfaction. This theory links disruptive technology with user dissatisfaction and comparative economic value, which causes the B2B customers to switch. The model articulates a theoretical justification for why users' intrinsic intent to switch to a disruptive technology is affected by both their dissatisfaction with the current technology product and their high expectations for the disruptive technology as economic value (Fan et al., 2014). According to EDT, expectations describe the features or characteristics that particular people expect from technology and performance is an individual's post-usage perception of how the technology managed the features that were expected throughout the use time (Lankton & McKnight, 2012). Disconfirmation, on the other hand, is a post-usage comparison that is purely subjective and can lead one to believe that performance was better, the same as, or worse than anticipated.

In this study, EDT is the base to measure the level of B2B customer's satisfaction. If the B2B technology user is dissatisfied with the product's expected features and anticipated performance, firms will switch to disruptive technology (Kamolsook et al., 2019). It indicates that B2B user dissatisfaction has a significant influence on customers' switching behaviour towards disruptive technology (Lin et al., 2021). According to the customer's perceptions of what they are gaining and giving up, perceived value refers to their overall appraisal of the efficiency of a product (Zeithaml, 1988). We argue that the constructs of expectation disconfirmation theory demonstrate expectations from existing products and evaluate their performance and determine if the product is better or worse or the same compared to the previous one and is offering more economic value, then finally will make the decision to switch the current technology product.

The expectation disconfirmation theory's construct-disconfirmation, which is the belief of an individual that the previous product's performance was better or same or worse, establishes the B2B user's perceived preference for technology products. If the perceived preference for a certain product does not meet up the expectation, and performance, and leaves room for disbelief,

then it will result in user dissatisfaction (Naradipa et al., 2020); also, users will compare the economic value between their current product and other products (Garga et al., 2019). It reveals that user perceived preference is a mediator between user dissatisfaction and B2B customers' switching behaviour as well as comparative economic value and customers' switching behaviour towards disruptive technology.

This study finds that social influence is a moderator in customers' switching behaviour to acquire disruptive technology. If services do not meet their needs in the manner that their peers have stated, customers are prone to negative confirmation if they lack the informed expectations that come from independent decision processing (Oliver et al., 1997). It implies that users get dissatisfied and feel deprived of economic value if their expectations and the performance of existing products do not match as stated by reference groups. It also indicates that the relationship between B2B user perceived preferences, which is generated by the disconfirmation of products, and the customers' switching behaviour is correlated and can be affected by social influence. Thus, social influence has a vital role in strengthening the relationship between user dissatisfaction and switching behaviour as well as between comparative economic value and switching behaviour towards disruptive technology.

6.2 Managerial Implication

Managers of B2B firms should focus on enhancing disruptive technology and innovations to provide a continuation of accessible solutions for everyone that will improve the lives of users. They should enhance the economic value offering and attributes that can satisfy B2B customers so that B2B customer switching will decrease. To achieve customer satisfaction and decrease customer switching, they should provide continuous disruptive innovation in technology to meet the varied needs of users (Xu et al., 2021). To prevent B2B customers from switching from existing technology products, managers should intentionally develop and impose switching barriers, including various complexities, to discourage customers from switching or migrating to other service providers (Amani, 2022).

7 Conclusion

This study explored the factors influencing business to business (B2B) customers to switch from their current technology product to any other updated technology. By embracing current research trends in the technology sector and the application of disruptive technology products, this study contributes to the literature. Modern firms in developing countries are increasingly accepting disruptive technology products (such as 5G and improved connectivity, cyber security advances, 3D printing, automation and robotics, artificial intelligence and machine learning, virtual and augmented reality), which has given the technology industry a great opportunity. If this kind of technology is deployed, both B2B customers would greatly benefit. Innovative technology transforms an entire industry in a good way. Therefore, if technology is adopted and exploited appropriately, like the internet, it may provide enormous inventive benefits to both the industry's customers and the producers.

The present study also embrace with some shortcomings and provide directions for further research. The first limitation on the generalizability of this study's findings is that it only examines one specific type of customer within a single text (e.g., disruptive technology). This limitation needs further research. Another significant shortcoming of this study is that it is a cross-sectional study, and the exposure and outcome are measured simultaneously. Although cross-sectional data can show the temporary relationship between user dissatisfaction and customers' switching behaviour towards disruptive technology and the relationship between comparative economic value and B2B customers' switching behaviour towards disruptive technology, it is best to pass the verification of longitudinal data to determine whether the relationship between user dissatisfaction and customers' switching behaviour will change over time (Lin et al., 2021).

Furthermore, in this study we have found that social influence has a moderating effect and user perceived preferences has a mediating effect on customers' switching behaviour who expects updated technology that can satisfy them and offer greater economic value compared to other technology products; however, the impact of other factors such as brand image, quality of product, perceived enjoyment on customers' switching behaviour is still unexplored. This study will assist the researchers to investigate the unexplored area of customers' switching behaviour towards disruptive technology to understand why customers make decisions to switch their existing technology

products. Moreover, managers of B2B firms can also apply the research findings to better understand the advantages and disadvantages of disruptive technology and its impact on existing and potential customers' switching behaviour.

Appendix A: Operationalization of Constructs

Constructs	Measures	Sources
1. User Dissatisfaction	1. I am very dissatisfied with the current technology used in our firm 2. I feel uncomfortable with my recent usage of technology used in our firm 3. I regret purchasing/ using current technology	Moon et al. (2017)
2. Comparative Economic Value	Compared with my current technology, new technology will… 1. Provide more diversified functionality 2. Provide higher quality 3. Provide more functions	Lin et al. (2022), Kamolsook et al. (2019)
3. User Perceived Preferences of Disruptive Technology	1. I feel that the current technology should be changed 2. The current technology doesn't suit me well 3. My perception towards the current technology needs to be installed	Li et al. (2021a, 2021b)

(continued)

(continued)

Constructs	Measures	Sources
4. Social Influence	1. People from the business peer group who are important to me think that my firm should use new technology 2. People from the business group who influence my behaviour think that my firm should use new technology 3. My neighbours think that in my firm, I should use new technology	Venkatesh et al. (2003), Venkatesh et al. (2012), and Kamolsook et al. (2019)
5. Customers' Switching Behaviour of disruptive technology	1. I intend to use new technology soon 2. I predict that I would use technology in the near future 3. I plan to use technology in the near future	Davis (1989), Venkatesh et al. (2003, 2012), and Kamolsook et al. (2019)

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8

Chief Privacy Officer (CPO) Appointments in Digital Supply Chains—Antecedents and Financial Implications

Christoph G. Schmidt

1 Introduction

Data privacy has received increasing attention from managers, customers, and regulatory entities in recent years. Data privacy can be understood as a measure of control that people have over who can access their personal information and how this information is used (Pandit et al., 2019). In the digital society and shared economy era, multinational firms collect and store personal information, increasing data privacy concerns.

To ensure the privacy of personal information, customers and policy-makers are putting the pressure back on firms (Redman & Waitman, 2020). According to a PwC study, only 25% of consumers trust their sensitive data to be handled responsibly (PwC, 2017). Customers are increasingly aware of a firm's data privacy policies and change their buying behavior accordingly (Gwebu et al., 2018). The number of policies regarding data privacy and security is rising. The European GDPR or the California Consumer Privacy Act (CCPA), for example, specifically regulate an organization's processing of personal data (Pandit et al., 2019). Firms are held responsible for their data's confidentiality, availability, and integrity. Failing to uphold data privacy responsibilities can result in a breach, in the form of misuse or theft of data, with potentially detrimental consequences for a firm's reputation and

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financial performance (Gwebu et al., 2018; Modi et al., 2015). Consequently, managing data privacy has become a significant challenge for firms implementing digitalization initiatives.

To handle the responsibilities associated with managing data privacy, firms increasingly appoint a dedicated management position, a Chief Privacy Officer (CPO). A CPO is a senior-level executive responsible for developing and implementing data privacy policies and ensuring regulatory compliance along the supply chain (Fusaro, 2000). For example, Facebook appointed a CPO after another costly data privacy lawsuit in 2019 (Rodriguez, 2019). In 2018, the ride-sharing app Uber appointed a Chief Privacy Officer about eight months after a data breach that affected about 57 million users (Stewart, 2018).

As data privacy in a digitalization context is not a firm-internal issue, the CPO should take on a supply chain perspective for two reasons. First, most personal information, including credit card information, home address, and social security numbers, is shared at retail (Martin & Palmatier, 2020). When customers purchase goods and services on the Internet, personal data might be shared with suppliers to facilitate the efficient delivery of the product or service (Massimino et al., 2018). Second, CPOs are often actively involved in decisions on data processing. New technologies, such as Artificial Intelligence (AI), often utilize large sets of customer data to generate insights for operations and supply chain management (Araz et al., 2020; Choi et al., 2021). Ensuring data privacy in AI-based technologies is another area of debate. In addition, especially in high-risk sectors, including transportation and healthcare, data privacy must be guaranteed along the entire supply chain.

Given the importance of data privacy and its potential consequences, we investigate the firm appointments of Chief Privacy Officers (CPO). In this study, we ask: (1) *When does a firm appoint a CPO?* and (2) *What are the immediate financial benefits of appointing a CPO?* Specifically, we propose a set of factors influencing a firm's decision to appoint a CPO.

To answer our research questions, we collect data on 50 CPO appointments by U.S. firms between 2010 and 2019. We build a control group of non-CPO-announcing firms and examine the influence of factors related to data risks, media attention, and prior data breaches on the likelihood of a CPO appointment. As the overall effects of top management appointments are hard to measure, we start by conducting a short-term event study to assess the financial benefits associated with appointing a CPO (Arora et al., 2019; Roh et al., 2016).

Our findings contribute to a variety of literature streams as well as provide insights for practitioners. First, we are among the first to explore the appointment of a CPO as a formal measure to signal data privacy efforts to stakeholders. We quantify the financial consequences of CPO appointments and surprisingly find no immediate positive financial benefits. We believe this noteworthy finding requires further investigation and encourage future research. In line with prior studies on top management team changes (Arora et al., 2019; Hendricks et al., 2015), we argue that there are indirect and often long-term effects of a CPO appointment, for example, after the materialization of an adverse event. These findings add to the ongoing discussion on CPOs and data privacy directors (e.g., Elahi et al., 2015; Fehr & French, 2022). Our results are also closely linked to general management issues regarding top management teams and board compositions (e.g., Arora et al., 2019; Roh et al., 2016).

Second, we examine potential drivers for a firm's decision to appoint a CPO. Specifically, we find that more data-intensive firms are more likely to appoint a CPO, and media attention can be a significant driver by highlighting potential data privacy-related risks. The study thus extends the literature on data privacy practices (e.g., Martin & Palmatier, 2020) and adds to the body of literature examining the influence of media scrutiny on firm decisions (e.g., Beattie et al., 2021; Eftekhar et al., 2017). Surprisingly, we find that prior data privacy-related incidents do not affect the likelihood of a CPO appointment, contradicting most anecdotal evidence (Rodriguez, 2019; Stewart, 2018). We encourage future research to explore this challenging finding in more detail.

Third, our study adds to the emergent stream of new technologies in operations and supply chain management, especially regarding data privacy (e.g., Massimino et al., 2018). We also contribute to the more general discussion on corporate digitalization initiatives (Gökalp & Martinez, 2022; Jeong et al., 2022; Verhoef et al., 2021). This chapter highlights the data privacy risks associated with digitalization efforts and the need for adequate firm and supply chain capabilities to realize the potential benefits of new technologies. New technologies to improve existing and enable new products and processes can create substantial value for supply chains but also generate a new set of risks, often related to data management and governance. We highlight their importance for practitioners and suggest several areas for future research.

2 Related Literature

We present prior research on digitalization and data privacy to contextualize the study. We also define data privacy and elaborate on the CPO role. We then discuss the limited set of studies that examine antecedents and financial benefits of top management team appointments.

2.1 Digital Technologies and Data Privacy

Digitalization describes how information or digital technologies can alter existing business processes (Gökalp & Martinez, 2022; Verhoef et al., 2021). General management literature covers topics related to digital technologies and marketing and customer relationships (e.g., Chen et al., 2021a), administration and human resources (e.g., Kellogg, 2022), innovation (Jeong et al., 2022), as well as new business models (Chen et al., 2021b; Friedrich et al., 2022). Given their interconnected nature, digital technologies are frequently applied to improve firm operations and supply chain management. Studies explore operational efficiency and sustainability implications (e.g., Gebhardt et al., 2021; Nunez-Merino et al., 2020). In addition, multiple studies examine the general effects of digital technologies on a firm's financial and operational performance (Li et al., 2020).

Other streams of literature cover the risks and opportunities associated with specific technologies, including blockchain (Klöckner et al., 2022; Pun et al., 2021), social media platforms (Schmidt et al., 2020), Internet of Things (IoT) (Christidis & Devetsikiotis, 2016), big data and artificial intelligence (AI) (Araz et al., 2020; Kache & Seuring, 2017), and drones (Shen et al., 2021). All these tools and technologies create, store, and share data within and across organizational boundaries (Araz et al., 2020). In addition to firm data, using platforms in shared economy systems, people increasingly share personal information that is subsequently collected and stored by firms. As a result, the theft of private customer data has become a significant operational risk for firms across all industries.

Data privacy poses a substantial operational risk for firms (Kumar & Mallipeddi, 2022). Data privacy can be understood as a measure of control that people have over who can access their personal information and how this information is used (Pandit et al., 2019). Firms are challenged with managing large amounts of customer data, including credit card information, home address, and social security numbers (Redman & Waitman, 2020). Consequences for firms can be dramatic. The main threat is the occurrence of a

data breach. Data breaches can damage a firm's reputation and deter potential customers (Janakiraman et al., 2018). On the one hand, such an incident can damage a firm's reputation and deter potential customers (Massimino et al., 2018). On the other hand, the potential adverse financial effects can be substantial, for example, on the stock market (Modi et al., 2015).

Kumar and Mallipeddi (2022) conceptually introduce data privacy as a risk for firms and argue for the importance of cybersecurity to combat the associated risks. Massimino et al. (2018) criticize the privacy and confidentiality of sensitive data, including customer information, which has received less attention, especially compared to traditional performance categories in operations and supply chain management. They theorize on the impact of data privacy on decision-making regarding global operations and supply chain strategy, the inter- and intra-organizational management of technology, and public policy. While the topic is gaining traction, there is still a need to understand the implications of data privacy accompanying increasing levels of digitalization across all industries and business functions.

2.2 The Chief Privacy Officer (CPO)

Given the potentially devastating consequences for a firm's reputation and financial performance, as well as the general interest in and public awareness of the topic, firms are starting to increase their data privacy efforts (Gwebu et al., 2018; Massimino et al., 2018). Firms implement practices and management procedures to improve data privacy, update technologies and data infrastructures, and hire dedicated personnel (Daneshvar Kakhki & Gargeya, 2019; Redman & Waitman, 2020). According to the Cisco Data Privacy Benchmark Study 2020, businesses see an average return of 2.7 times on their original investment when implementing data privacy practices (Cisco, 2020). Another way of signaling to stakeholders, including investors but also customers and the public, that a firm is aware of the topic and is willing and able to manage data privacy sufficiently is to hire dedicated top management executives (Arora et al., 2019; Hardcopf et al., 2019). Several firms, including Facebook and Uber, have hired a Chief Privacy Officer (CPO) to demonstrate the importance of data privacy.

The CPO is a senior-level executive responsible for developing and implementing data privacy policies and ensuring regulatory compliance along the supply chain (Fusaro, 2000). The position is not new. In fact, it's more than two decades old, but it has become more prevalent in recent years (Fehr & French, 2022). The primary responsibilities of a CPO include the development and implementation of internal data policies as well as the

communication and coordination of privacy policies to business partners along the supply chain (e.g., Martin & Palmatier, 2020). Privacy policies include processes and organizational structures for dealing with data collected from customers, clients, and employees (Wigmore, 2014). In addition, CPOs are involved in planning IT infrastructure and landscape (Elahi et al., 2015). The CPO also deals with compliance with the regulatory environment. The role is often intertwined with other data and technology-related top management positions, such as the Chief Information Officer (CIO), the Chief Security Officer (CSO), the Chief Data Officer (CDO), and the Chief Compliance Officer (CCO), and the responsibilities of a CPO can in practice also be part of another role. Despite increasing popularity, partially fostered by changing regulatory environments, research on the appointment decision and impact of CPOs is sparse.

2.3 Appointments of Top Management Positions

Our study also draws from the general management literature on top management teams. For example, multiple literature streams exist on candidate selection or board composition and interaction (El Nayal et al., 2019; Krause et al., 2017; Zhu & Westphal, 2014). Studies examine, for example, top management communication (Johnson & Lederer, 2014) or incentives (Gore et al., 2011). Literature also covers a wide range of top management positions, including Chief Marketing Officers (Nath & Mahajan, 2011), Chief Information Officers (Chen et al., 2021a), Chief Sustainability Officers (Arora et al., 2019), and Chief Supply Chain Officers (Roh et al., 2016; Wagner & Kemmerling, 2014). A vast body of literature also explores antecedents of appointment and the financial consequences of new top management position appointments.

Responding to the discourse on the financial and marketing performance implications related to Chief Marketing Officers, Germann et al. (2015) find that Chief Marketing Officer appointments are associated with about a 15% higher Tobin's q performance compared to firms without a Chief Marketing Officer. For example, Roh et al. (2016) study the antecedents and consequences of recent Chief Supply Chain Officer appointments. They find that this is often a direct result of supply chain integration issues or diversification pressure. Chief Supply Chain Officer appointments positively affect the financial performance of firms with high internationalization and diversification characteristics. Arora et al. (2019) investigate the effects of Chief Sustainability Officer appointments. Using an event study methodology, they find that appointing a Chief Sustainability Officer does not affect a firm's

stock market performance but drives long-term operational performance. Further, they identify a positive stock market reaction when the sustainability responsibilities associated with the position are narrow and the firm previously experienced sustainability accidents. While there is a body of research on top management teams and the antecedents and implications of their appointments, the CPO position has thus far been largely neglected. Given that the responsibilities of a CPO primarily include managing data privacy, more exploration is needed.

3 Hypotheses Development

To gain additional insights into the impact of the CPO role, we develop our set of hypotheses in two steps. First, we explore the antecedents to the appointment of a CPO related to data risks, media attention, and prior data breaches. Second, we study the appointment's impact on the firm's financial performance, understood as the resulting stock market reaction. Figure 1 depicts our conceptual framework.

3.1 Antecedents of CPO Appointments

Data privacy poses a significant operational risk for firms (Massimino et al., 2018). Sensitive consumer data can be stolen and then used to cause harm to the firm or the individual. Examples include credit card fraud or identity theft (Lai et al., 2012). While firms across all industries around the world are

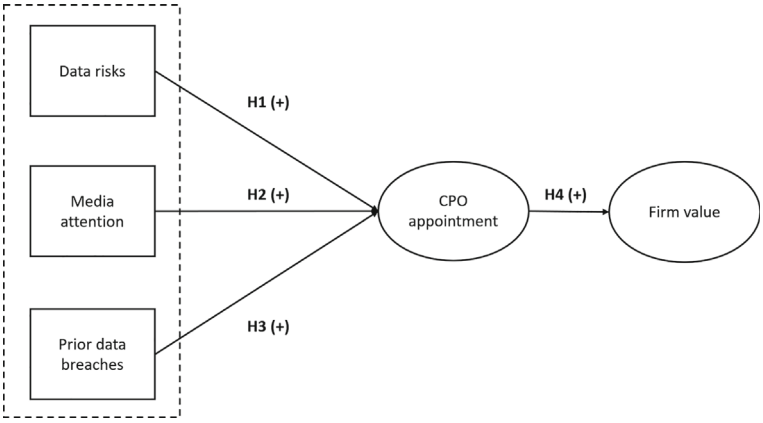

Fig. 1 Conceptual framework (*Source* Author)

affected by such risks, some firms might be more exposed to data risks than others. Especially with people increasingly shopping online, retailers collect the most extensive and broad set of personal information, such as credit card information or home address (Martin & Palmatier, 2020). In contrast, firms operating mainly in B2B industries or supplying raw materials are less engaged with customer data.

Even within industries, individual firms have differences in data risk exposure, depending on their business model. Firms relying more on a data-intensive business model are inherently more exposed to privacy risks (Araz et al., 2020; Gwebu et al., 2018). In addition, business processes can also rely on the flow of data differently. Firms that continuously exchange data with business partners in and beyond the supply chain are also more exposed to data privacy and protection risks than firms that share fewer data (Feng & Shanthikumar, 2018; Kwak et al., 2018; Modi et al., 2015). Every new actor in the data flow network creates new vulnerabilities and challenges for safeguarding data privacy. Creating a joint approach with supply chain partners is costly and requires substantial coordination efforts (Schmidt & Wagner, 2019; Um & Kim, 2019). Thus, we suspect those firms to be more proactive when implementing measures, often accompanied by a CPO appointment, to deal with potential risks.

Hypothesis 1: Firms that exhibit more data risks are more likely to appoint a CPO.

Literature shows that media exposure influences a firm's decision-making. For example, Eftekhar et al. (2017) find that media exposure greatly affects operations and supply chain decisions. They also find media attention to influence the initiation and organization of humanitarian initiatives. In the context of automotive recalls, Beattie et al. (2021) find that news coverage affects how firms conduct their recall campaign. Similarly, we argue that the increasing public discourse on data privacy and security and the rising awareness of customers will also affect how firms decide regarding their top management team composition.

In addition to affecting firm decisions in general, media attention can accentuate the negative consequences of various other firm events (Graf-Vlachy et al., 2020). Schmidt et al. (2020) find that in addition to traditional media outlets, public discourse on social media platforms accentuates the negative consequences of supply chain disruptions. Kölbel et al. (2017) find that media attention affects how corporate social irresponsibility is perceived and valued by firm stakeholders. As public pressure regarding data privacy rises (Massimino et al., 2018), we expect a similar effect. Firms that receive

more media attention might be inclined to act more carefully and take potential risks more seriously, as they do not want to experience an adverse data privacy-related event with high media attention and argue why the topic was not taken seriously enough. Similarly to developments regarding sustainability efforts and corporate social responsibility debate (e.g., Arora et al., 2019), firms are inclined to take a proactive approach and signal awareness to respond to stakeholder pressure (Schmidt et al., 2017).

Hypothesis 2: Firms with higher media attention are more likely to appoint a CPO.

In practice, firms are often forced to react after a data breach has already happened and has thus made data privacy a pressing issue for the firm (D'Arcy et al., 2020). Having experienced an adverse data privacy-related event can raise awareness of the problem within a firm and potentially remove internal barriers to new practice adoption (Gwebu et al., 2018). A high cost is also associated with such adverse events, potentially persuading management teams to make a case for improved data privacy.

Especially for firms that have (partially) neglected the topic before, the appointment of a CPO after a data breach can be a strong signal to the firm's share- and stakeholders that data privacy will be a more serious issue within the firm moving forward. For example, Facebook was early in appointing a CPO after a series of privacy scandals and remaining concerns by platform users. Facebook then appointed a CPO after another costly privacy lawsuit (Rodriguez, 2019). The responsibilities of the new CPO often include clarifying the previous incident, understanding the issues, and developing new structures and processes to ensure data privacy in the future. Overall, we propose that creating a dedicated position to deal with privacy risks is more likely when the firm has already experienced the potential negative implications.

Hypothesis 3: Firms with prior data breaches are more likely to appoint a CPO.

3.2 Financial Consequences of CPO Appointments

This study also explores the financial consequences of appointing a CPO. Prior research on top management team appointments shows mixed results regarding the direct and indirect effects of the appointment on a firm's financial performance (e.g., Arora et al., 2019; Germann et al., 2015). Roh et al.

(2016), for example, find that the appointment of a chief supply chain officer has potentially positive implications for a firm's Return on Assets (ROA) in the years after the announcement. They find this effect to be moderated by internationalization and diversification. Germann et al. (2015) also find a positive effect of Chief Marketing Officer appointments on firm performance. Specifically, they identify that firms that employ a Chief Marketing Officer experience an increase of about 15% (Tobin's q) compared to firms without a Chief Marketing Officer. In contrast, Arora et al. (2019) find that appointing top management sustainability executives does not directly increase firm performance, as measured by stock market reaction.

However, we argue that the appointment of a CPO will evoke a positive reaction in the stock market. Stock market reactions are based on anticipating future cash flows (Klöckner et al., 2022; Modi et al., 2015). On the one hand, even though a new top management position is always associated with initial costs, including organizational restructuring, the focus on data privacy issues could save costs in the long term. Research suggests that negative data privacy-related events can be very costly, as firms have to restore operations and processes, fix privacy issues, and potentially pay fines and reparations to affected customers and regulatory authorities (Martin et al., 2017; Modi et al., 2015). Reputational damage can also be very costly for the firm regarding customer behavior and future sales (Gwebu et al., 2018). On the other hand, taking data privacy can potentially improve future revenues. Managing data privacy well can prevent the associated risks from materializing. Early adoption can allow a firm to shape regulatory directives in the future and gain a competitive edge regarding aware customers. Overall, we believe CPO can positively affect a firm's bottom line.

Hypothesis 4: CPO announcements are associated with a positive stock market reaction.

4 Method and Results

To empirically examine the antecedents of CPO appointments (Hypothesis 1–3), we conduct a regression analysis. To model the appointment decision, our U.S. sample includes firms with and without a CPO position. We then conducted an event study to examine the effects of the CPO appointment on firm value (Hypothesis 4).

4.1 Data Collection

To study the antecedents of and the effects of CPO announcements on firm value, we collect CPO appointments that were publicly announced between 2010 and 2019. Specifically, we leverage data from the leading news agencies *PR Newswire* and *Business Wire* to identify the announcements (Klöckner et al., 2022). We only consider U.S. firms publicly listed on either the NYSE or NASDAQ (Schmidt et al., 2020). To facilitate the event study, we removed firms without sufficient firm financial information available (Hendricks et al., 2015). Our final sample includes 50 CPO appointments from 50 unique firms. Table 1 provides the distribution of our sample across years and industries.

To examine the antecedents of firms appointing a CPO, we need to extend our sample to include firms that did not announce a CPO appointment. As commonly used, we utilize a propensity score matching (PSM) approach to build the combined sample (Hendricks et al., 2015). The PSM approach allows us to identify the firms that are structurally closest to our sample firms in terms of firm size, firm growth, profitability, and industry. For more details on the PSM approach, please see Klöckner et al. (2022). In addition, the approach identifies the closest non-announcing match, resulting in a combined sample of 100 observations.

News data was retrieved from Google News using the *SerpAPI*[1] interface in Python. We use the 'edgar' R package to access annual firm reports (Lonare et al., 2020), specifically the section on risk factors in *ITEM 1a* of a firm's annual 10 K filing with the SEC (Berger et al., 2020). We also retrieve daily

Table 1 Sample summary statistics

Year	Frequency	Percentage	Industry	Frequency	Percentage
2019	11	22%	Manufacturing	9	18%
2018	7	14%	Communication	6	12%
2017	8	16%	Trade	3	6%
2016	3	6%	Finance & real estate	16	32%
2015	6	12%	Services	16	32%
2014	6	12%	*Total*	*50*	100%
2013	5	10%			
2012	1	2%			
2011	3	6%			
Total	*50*	*100%*			

[1] https://www.serpapi.com/.

stock prices and additional firm financial controls from the Thomson Reuters Eikon database.

4.2 Variables and Measures

To study the antecedents of and the effects of CPO announcements on a firm's financial performance, we examine a set of variables. Table 2 provides an overview of the variables and measures used in this study. To test how the firm's data risks affect CPO appointments, we count the number of data risk-related keywords in the firm's risk description. Specifically, we use *ITEM 1a* from the 10 K filings, accessed via the EDGAR SEC filings database.

Media attention is measured as the number of Google News hits mentioning the firm name and shorter variants. We include news from the past 180 days. To study the effects of prior data breaches on the appointment

Table 2 Measurement details for hypothesized and control variables

Variable (label)	Description	Data source
Data risks (DATARISK)	The number of data-related keywords occurring in the firm's annual report risk section (Keywords: data, data security, data regulation, information system, information technology, technology)	EDGAR SEC filings
Media attention (MEDIA)	Number of Google news hits mentioning the firm name within half a year (180 days) prior to the CPO announcement	Google News
Prior data breach (BREACH)	Dummy variable that equals 1 when data breach-related keywords appeared in conjunction with the firm name in the two years prior to the CPO announcement	Google News
Firm size (FIRMSIZE)	Logarithm of total assets in the fiscal year prior to the announcement	Thomson Reuters
Firm growth (FIRMGROWTH)	Percentage change of total assets from two years to the one fiscal year prior to the announcement	Thomson Reuters
Risk sentiment (RISKSENT)	The sentiment of the firm is based on the annual report risk section, based on the Loughran-McDonald financial dictionary. We use the R package *SentimentAnalysis* to calculate the sentiment score	EDGAR SEC filings
Time (TIME)	Number of days between the first announcement in our sample to the focal announcement	
Industry sector (SECTOR)	Dummy variables for manufacturing or service sectors are based on the two-digit SIC codes	Compustat

Table 3 Descriptive statistics

Variable	Min	Max	Mean	SD
Data risk	0.00	260.00	49.26	54.69
Media attention	0.00	512.00	28.79	93.18
Firm size (log)	3.02	14.70	8.68	2.54
Firm growth	−0.38	1.47	0.10	0.26
Risk sentiment	−0.10	0.10	−0.04	0.03

of CPOs, we created a dummy variable, indicating whether the firm has made headlines related to data breaches in the two years.

Prior studies indicate that larger firms are more likely to have more members in their top management teams (Arora et al., 2019; Damaraju & Makhija, 2018). We thus control firm size, measured as the logarithm of total assets. Data privacy is a concern for all firms; however, fast-growing firms are usually younger and often more data intensive. These characteristics might lead to a systematically higher appointment rate of CPOs. We thus control firm growth. To offset a potential bias in the number of data-related keywords in a firm's risk statement, we also control for the general sentiment of the risk section. We use the Loughran-McDonald financial dictionary to classify the sentiment, as implemented in the R package *SentimentAnalysis* (Feuerriegel et al., 2018). We also control time and sector effects. Table 3 presents descriptive statistics for our variables.

4.3 Data Analysis

4.3.1 Logit model

To test whether the factors proposed in Hypothesis 1–3 increase the likelihood of CPO appointments, we utilize logistic regression (Hair et al., 2006). Logistic regression estimates the effects of a series of binary or continuous independent variables on a binary dependent variable. The dependent variable indicates the occurrence of an event, such as if the firm has appointed a CPO. Our dependent variable is '1' if the firm has appointed a CPO and '0' if the firm did not appoint a CPO in the observation period of our study. Specifically, we estimate the following logit model:

$$\begin{aligned}\Pr(\text{CPO} = 1|x) = &\Lambda(\gamma_1 + \gamma_2\ \text{DATARISK} + \gamma_3\ \text{MEDIA} \\ &+\gamma_4\ \text{BREACH} + \gamma_5\ \text{FIRMSIZE} + \gamma_6\ \text{FIRMGRWOTH} \\ &+\gamma_7\ \text{RISKSENT} + \gamma_8\ \text{TIME} + \gamma_9\ \text{SECTOR}),\end{aligned}$$

where y indicates the coefficients for our set of variables, x.

4.3.2 Event Study

In Hypothesis 4, we argue that appointing a CPO creates immediate financial benefits for the appointing firm. Similar settings have often been analyzed using an event study, with stock market performance as a measure of (expected) firm financial performance (Arora et al., 2019; Brown & Warner, 1985). The core idea behind an event study is that firm events can create abnormal stock market returns, both positive and negative. The abnormal stock market performance of a firm i on day t (AR_{it}) is understood as the difference between actual stock market returns (R_{it}) and expected stock market returns ($E(R_{it})$).

$$AR_{it} = R_{it} - E(R_{it})$$

To calculate the expected returns, we use the Fama–French four-factor model, as described by Modi et al. (2015):

$$E(R_{\mathrm{it}}) = \alpha_i + \beta_i R_{\mathrm{mt}} + s_i \mathrm{SMB}_t + h_{\mathrm{i}} \mathrm{HML}_t + u_{\mathrm{i}} \mathrm{UMD}_t$$

The model suggests that a firm's expected stock market returns are influenced by the general S&P 500 market performance on day t (R_{mt}), the small-minus-big-size returns (SMB_t), the high-minus-low-book-to-market returns (HML_t), and the momentum factor (UMD_t). For more details on the model, see Fama and French (1993) and Carhart (1997).

Cumulative abnormal returns (CARs) are calculated as the sum of abnormal returns over a specific period. We test mean ARs using event-study-specific t-tests, median ARs using Wilcoxon signed rank tests, and the percentage of negative ARs using generalized sign tests. For a more in-depth description of the event study method, see Kothari and Warner (2007) and applied in, for example, Schmidt et al. (2020).

4.4 Results

First, we examine the drivers of CPO appointments. Table 4 presents the results of our analysis. Column 1 of Table 4 shows the control model, and Column 2 is the main model. Our results indicate that the firm's inherent data risk increases the likelihood of firms appointing a CPO ($\gamma = 0.320$, $p < 0.05$), supporting our Hypothesis 1. The media attention a firm

receives also drives CPO adoption ($\gamma = 0.416$, $p < 0.05$), supporting Hypothesis 2. Surprisingly, having experienced prior data breaches does not drive the appointment of a CPO, as a potential countermeasure. In fact, we find marginal support for an opposite effect ($\gamma = -0.865$, $p < 0.10$). Thus, Hypothesis 3 is not supported by our results.

Next, we explore the financial impact of a CPO announcement using an event study. Table 5 presents the full results. Columns 1 to 5 suggest no significant stock market reaction on the day of the appointment (Day 0) and two days prior and after (Day −2 to Day 2). Subsequently, we see no significant effect during various cumulative windows across our sample (Columns 6–8).

Table 4 Logit regression results

	Predicted	Controls	Main
	sign	(1)	(2)
Hypothesized variables			
Data risks (DATARISK$)	+		0.320*
			(0.158)
Media attention (MEDIA)	+		0.416*
			(0.189)
Prior data breach (BREACH)	+		−0.865+
			(0.512)
Control variables			
Firm size (FIRMSIZE)		0.069	0.019
		(0.082)	(0.094)
Firm growth (FIRMGROWTH)		1.212+	0.532
		(0.686)	(0.720)
Risk sentiment (RISKSENT)		0.732	0.336
		(0.478)	(0.241)
Time (TIME)		−0.014	−0.103
		(0.180)	(0.191)
Sector (SECTOR)		−0.015	−0.103
		(0.524)	(0.191)
AIC		146.07	140.49
R^2		0.03	0.04
N		100	100

+$p < 0.10$, *$p < 0.05$, **$p < 0.01$, *** $p < 0.001$; robust standard errors in parentheses

Table 5 Short-term event study results

	Day −2	Day −1	Day 0	Day 1	Day 2	Days [0, 1]	Days [−1, 0]	Days [−1, 1]
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mean abnormal return	0.01%	0.20%	−0.11%	0.16%	−0.13%	0.05%	0.09%	0.25%
t-test (*t*-statistic)	(-0.49)	(0.34)	(−0.32)	(0.06)	(−0.72)	(−0.19)	(0.02)	(0.05)
Median abnormal return	−0.31%	−0.06%	−0.03%	−0.08%	−0.21%	−0.32%	0.13%	−0.28%
Wilcoxon signed rank test (*Z*-statistic)	(−1.46)	(−0.12)	(−0.56)	(−0.15)	(−1.25)	(−0.98)	(−0.08)	(−0.45)
Percent positive abnormal returns	38.00%	50.00%	44.00%	46.00%	46.00%	40.00%	52.00%	44.00%
Generalized sign test (*Z*-statistic)	(−1.62)	(0.08)	(−0.77)	(−0.49)	(−0.49)	(−1.34)	(0.36)	(−0.77)

$^{+}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$; all tests are one-tailed; $n = 50$

5 Discussion and Future Outlook

The study aims to explore the effects of appointing a CPO. Specifically, we show that firms that inherently are more reliant on new technologies and large amounts of data, and thus potentially experience higher data privacy-related risks, are more likely to appoint a CPO than firms with less data intensity. Media exposure also significantly increases the likelihood of a firm appointing a CPO to implement a proactive data privacy management strategy and react to increasing stakeholder pressure exerted by customers and regulatory entities. We advise managers of data-intensive firms and those under media scrutiny to take data privacy seriously and appoint a CPO. CPO can also help shape a firm's public image, and studies show that positive media perception can have long-term benefits (Beattie et al., 2021; Graf-Vlachy et al., 2020).

Results also indicate that firms, which previously experienced a data breach, are less likely to create a top management position dedicated to data privacy. This is an interesting finding, as it contradicts our initial hypotheses development. We argued that after a negative data privacy-related incident, firms were under more scrutiny to overcome potential shortcomings and prevent future incidents. However, we find a marginally significant opposite effect. One possible explanation is that firms have taken different measures to deal with the incident, and there is no direct need for a dedicated top management position. This might be especially true for firms with a business model less reliant on data, where this type of incident might inherently be lower. We advise future research to explore the relationship between data privacy-related incidents, such as data breaches, and the CPO position in terms of negative impact mitigation and future incident frequency (see Table 6 for an overview of future research questions). Despite our findings, practitioners should consider including a dedicated data privacy position in their response strategy.

We also find no direct effect of CPO announcements on firm value. In our hypothesis development, we argued for a positive effect. However, the literature suggests that top management team changes do not always result in immediate positive stock market effects. For example, Arora et al. (2019) also find no significant stock market reaction following the appointment of sustainability officers. Similarly, Hendricks et al. (2015) examine the financial consequences of chief supply chain management officers, finding that firms do not experience an immediate improvement in stock price. They find, however, that the appointment stops negative operating performance trends. We assume that there could be similar effects for CPO announcements, for example, following a data breach. While appointing a CPO does not directly

Table 6 Opportunities for future research

	Potential research questions
Firm value	• What are potential contingencies under which a CPO appointment can positively impact the firm value? • Are there any long-term effects?
Financial performance	• How can CPOs deliver value? • What are the effects on the firm's financial performance (other than stock market performance)? • Is there an effect on operational performance?
Data breaches	• How does the CPO appointment affect (moderate) the negative impact of data privacy-related events? • Is there a decrease in the frequency of adverse events, such as breaches? • Can CPOs shape the public reaction to data privacy-related events?
Top management teams	• What are the potential interactions of a CPO with other top management functions, such as Chief Information Officers? • What are the appointment and selection criteria of a CPO?

create value in the stock market, we assume that there are indirect benefits that firms should carefully evaluate.

We suggest that future research tackles these questions (see Table 6). In addition, we assume that the financial benefits could be indirect, contingent on a set of firm characteristics. This needs to be explored in future studies, creating opportunities for future work. For example, a long-term event study could provide more insights into the potential financial benefits of CPOs. To facilitate future research on data privacy, we have compiled a set of research areas, including potential research questions.

Our findings contribute to several literature streams and hold new insights for operations and supply chains and data privacy practitioners. This study is among the first to explore antecedents and financial consequences of CPO appointments. The decision to appoint a CPO is also closely linked to general management issues regarding top management teams, team interactions, and general questions regarding board compositions (e.g., Arora et al., 2019; Roh et al., 2016). Our findings also extend the novel stream regarding new technologies and big data in operations and supply chain management (e.g., Araz et al., 2020). Finally, we specifically add to the literature related to data management (e.g., Choi et al., 2018) and the emerging stream on data confidentiality and privacy (e.g., Massimino et al., 2018).

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9

Social Implications of Autonomous Vehicle (AV) Adoption in Supply Chains

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1 Introduction

Autonomous vehicles (AVs) have served the logistics sector in the form of automated guided vehicles (AGVs) for decades. With the advent of Industry 4.0 in 2011, significant advances have been witnessed (Schwab, 2016). The rapid development of droids and drones indicates wider adoption across the industry (Tang & Veelenturf, 2019). Logistics giants such as Alibaba and JD.com in China, and DHL and Amazon in Europe and the USA are applying or testing autonomous vehicles for use in distribution and storage processes (Mohamed et al., 2020). Further, Zipline is a successful drone delivery service providing medical supplies in African countries (Scott & Scott, 2017). However, compared with the rapid progress of technology, academic research into technology usage and the impacts of the applications on society is lagging behind (Monios & Bergqvist, 2020). Previous studies have discussed, simulated, and compared AVs with conventional vehicles with regard to carbon emission levels, energy consumption, as well as various operational costs (Tsolakis et al., 2018). The variables that measure economic and environmental sustainability in supply chains (SCs) are developed and can be used to demonstrate positive performance of AV adoption in these two

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dimensions (Tang & Veelenturf, 2019), while the social sustainability of AV adoption in SCs received much less focus.

There are many important social issues inherent in the adoption of AVs. The adoption of AVs potentially improves traffic safety, as over 90% of fatal road crashes are due to human errors, e.g., fatigued driving in long-haul freight transport (Wang & Li, 2019). They can also reduce traffic congestions on highways by increasing road capacity and traffic flow (Zhao & Lee, 2023). However, the impacts of the adoption of AVs on employment remain controversial—new job creation and labour shortage mitigation in logistics (Fawcett & Waller, 2014) versus unemployment and job replacement (Heard et al., 2018). Risks such as privacy and data security (Tatham et al., 2017) and vehicle crashes (Wang & Li, 2019) also undermine public acceptance. These social concerns lead to strict regulations that restrict the further deployment of AVs: both unmanned ground vehicles (UGVs) and unmanned aerial vehicles (UAVs) are currently not allowed to operate logistics tasks without human monitoring or intervention in public areas (Van Meldert & De Boeck, 2016). Hence, when considering the adoption of AVs in SC processes, these social issues should be highlighted. The social demands raised by stakeholders and the potential social impacts on them must be considered, to allow organisations to create value for them (Hörisch et al., 2014) and improve their well-being (Serrano-Hernandez et al., 2021). Stakeholders are defined as the entities that are involved in organisational activities (i.e., being affected or affecting) (Freeman, 1984). Gaining social approval among stakeholders is essential for wider AV adoption in logistics (Kurniawan et al., 2021), which highlights the importance of social impacts and social sustainability of AV adoption in SCs.

In this context, AVs include both UGVs and UAVs (Rai et al., 2022). UGVs include AGVs or self-driving vehicles that can operate without direct human–driver intervention (Flämig, 2016). Self-driving vehicles follow a widely accepted standard of automated driving level published by the Society of Automotive Engineers (SAE) (SAE, 2021) that categorises autonomous vehicles into level 3 (conditional automation), level 4 (high automation), and level 5 (full automation). These three levels represent the automated driving vehicle types with limited or no human-driver intervention. AGVs, AMRs (Autonomous mobile robots—advanced AGVs with smart sensors and self-navigation systems), and self-driving trucks all fit in this definition (Rai et al., 2022). UAVs on the other hand, have been widely applied in military operations and have more recently been adopted for civil use (DHL, 2014). There are three main categories of UAVs in logistics and SC operations: first and last-mile delivery in urban and rural areas (outdoor), property surveillance

and inspection (indoor), and intralogistics operations (indoor) (DHL, 2014). UAVs supplement and augment conventional logistics operations with their advantages of speed, safety, and cost-efficiency (DHL, 2020).

The conceptual foundation of this research, SCSS is rooted in the concept of corporate social responsibility (CSR), which calls for a corporation to be ethically and socially accountable to a series of stakeholders on the environmental and social outcomes of operations, rather than solely economic performance (Maloni & Brown, 2006). When the requirements extend from a focal company to the SC, Spence and Bourlakis (2009) emphasised a chain-wide consideration. Ethical and social benefits are to be gained within the whole SC, in collaborating with SC partners. Hence, the whole SC behaves ethically and responsibly to accomplish social benefits as a goal. Unlike economic and environmental sustainability, which can be assessed by general quantitative measures, social sustainability is difficult to quantify and often requires qualitative performance indicators instead (Ahi & Searcy, 2015a). Indicators such as health and safety, philanthropy, employment, transparency, equality, labour, human rights, and welfare all define and qualitatively measure SCSS from a general perspective (Ahi & Searcy, 2015a, 2015b; Carter & Rogers, 2008; Yawar & Seuring, 2017). For further conceptualising SCSS, it is necessary to investigate contextual indicators and not all of the items mentioned above apply to a specific context (Gugler & Shi, 2009). Therefore, this study seeks to identify the relevant social sustainability indicators in the AV adoption context to conceptualise SCSS.

Currently, there is a lack of research that comprehensively discusses the social impacts of AV adoption in SCs. The existing relevant studies mainly focus on a specific stakeholder (e.g., end-customers), or a single type of technology. Hence, this study seeks to explore the social impacts of both UGVs and UAVs in both humanitarian and commercial SCs, to understand the SCSS implications of AV adoption. By systematically reviewing the existing literature, the social sustainability implications in this context can be understood from a more encompassing and holistic viewpoint while increasing transparency and minimising biases (Durach et al., 2017; Tranfield et al., 2003). Based on these arguments, the following objectives are to be realised:

1. To synthesise the social impacts of AV adoption in SCs.
2. To identify and map the SCSS indicators that are relevant to the social impacts in the AV adoption context.

The remainder of this chapter is organised as follows. Section 2 details the positive and negative social impacts as well as barriers of AV adoption.

Section 3 establishes connections between the identified impacts and SCSS indicators and discusses research avenues from literature. The final section concludes the study with an overview of contributions and limitations, as well as future research suggestions.

2 Social Impacts of AV adoption in SCs

In total, 58 papers were identified through keyword search (supply chain management, social sustainability, and autonomous vehicles) in three databases (EBSCO, ABI, and SCOPUS) and screened full text. The positive and negative social impacts in both commercial and humanitarian logistics contexts are identified. In addition, this chapter outlines several social barriers to AV adoption.

2.1 Positive Social Impacts

This review identified 13 positive social impacts and classified them into six distinct groups based on the relevance of the identified items. An overview of the positive impacts of AV adoption in both commercial and healthcare humanitarian contexts is shown in Table 1.

2.1.1 Faster and More Customised Delivery Services

The development of e-commerce and urbanisation has led to the increasing importance of urban logistics—especially B2C last-mile logistics (Marsden et al., 2018), and more customer-oriented services for last-mile delivery (Kapser & Abdelrahman, 2020). Reduced delivery time and more customised services are demanded by consumers, and these are challenging for existing conventional delivery modes (Bechtsis et al., 2017), while AVs present feasible solutions to meet customers' needs (Merlino & Sproge, 2017). Among the positive impacts, fast delivery is the most mentioned and significant advantage for AVs, especially drones. For UAVs, this is enabled by their capabilities of ignoring ground traffic and geographical blocks, so that they can operate in direct and uninterrupted routes to conduct delivery tasks (Kellermann et al., 2020). Such advantages in delivery speed reduce consumer waiting time, or customer lead-time, which can satisfy the customers' demands for shorter delivery time (Mohamed et al., 2020). On the other hand, UAVs are considered more flexible, so they can provide more customised services—time slot,

Table 1 Positive social impacts of AV adoption in SC

Source	Technology	Positive Social Impacts												
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**
Wang (2019)	UAV	X												
Sellevold et al. (2020)	UAV			X	X									
Kellermann et al. (2020)	UAV	X	X											
Hwang et al. (2021a)	UAV	X								X				
Hwang et al. (2019)	UAV	X					X			X				X
Rao et al. (2016)	UAV	X												
Mohamed et al. (2020)	UAV	X								X			X	
Rejeb et al. (2023)	UAV	X	X							X			X	X
Serrano-Hernandez et al. (2021)	UAV		X					X						
Hwang et al. (2021b)	UAV	X	X											
Mahroof et al. (2021)	UAV			X										X
Scott and Scott (2017)	UAV	X								X			X	
Tatsidou et al. (2019)	UAV	X											X	
Haula and Agbozo (2020)	UAV	X											X	
Dhote and Limbourg (2020)	UAV	X	X							X				
Tatham et al. (2017)	UAV		X										X	
Poljak and Šterbenc (2020)	UAV	X	X							X			X	
Rosser et al. (2018)	UAV	X	X										X	
Ding (2018)	UAV	X	X											
Comtet and Johannessen (2022)	UAV	X	X										X	
Azmat and Kummer (2020)	UAV & UGV								X	X			X	

(continued)

Table 1 (continued)

Source	Technology	Positive Social Impacts											
Tang and Veelenturf (2019)	UAV & UGV	X		X							X		
Mangiaracina et al. (2019)	UAV & UGV		X					X	X				
Perussi et al. (2019)	UAV & UGV	X	X	X	X		X	X	X				
Slabinac (2015)	UAV & UGV	X	X										
Amarfii-Railean (2018)	UAV & UGV		X										
Rai et al. (2022)	UAV & UGV					X				X			
D'Amico et al. (2021)	UAV & UGV			X	X	X		X					X
Kapser and Abdelrahman (2020)	UGV		X										
Klumpp (2018)	UGV	X				X	X						
Pani et al. (2020)	UGV	X	X							X			
Llopis-Albert et al. (2019)	UGV												X
Sohrabi et al. (2020)	UGV					X		X	X				X
Dadzie and Johnston (1991)	UGV					X							
Bechtsis et al. (2018)	UGV		X			X							X
Sen et al. (2020)	UGV					X		X		X			
Monios and Bergqvist (2020)	UGV							X		X			
Gružauskas et al. (2018)	UGV	X	X	X	X	X		X					
Fagnant and Kockelman (2015)	UGV							X	X				
Marsden et al. (2018)	UGV	X	X										
Bechtsis et al. (2017)	UGV					X							X
Engholm et al. (2021)	UGV						X						
Kim et al. (2022)	UGV						X			X	X		

(continued)

Table 1 (continued)

Source	Technology	Positive Social Impacts												
Kurniawan et al. (2021)	UGV				X	X								
Kapser et al. (2021)	UGV							X						
Nikitas et al. (2021)	UGV				X	X								
Tubis and Poturaj (2021)	UGV													X

1. Fast delivery (consumers, recipients)
2. Customised services (consumers, recipients, patients)
3. Security for assets & products (consumers, logistics managers)
4. Real-time information transparency and visibility (logistics managers and decision makers)
5. New job creation (employees)
6. Fill labour shortage (decision makers)
7. Road traffic safety (road traffic users)
8. Smoother road traffic flows (consumers, road traffic users)
9. Reduced usage of ground space (consumers, road traffic users)
10. Customers/recipients health and safety (customers, recipients)
11. Reduce gender inequality (employees)
12. Fast response and life-saving for emergency scenarios (recipients/patients)
13. Workforce health and safety (employees)

and location of reception (Mangiaracina et al., 2019), and therefore avoid inefficiencies such as delivery failure and repeat visits (Slabinac, 2015). UGVs show similar features in providing rapid and customised delivery services (Pani et al., 2020). Although reliant on existing road infrastructure, there are no driving time restrictions for UGVs. 24/7 operations enable UGVs to conduct transport services with higher flexibility and gain a competitive advantage to conventional ground vehicles (Azmat & Kummer, 2020).

Specifications in fast and customised delivery services are crucial in providing medical supplies in humanitarian and healthcare SC contexts (Ding, 2018). Various medical supplies can be delivered rapidly and directly to recipients with specific needs in urban areas or remote places (e.g., automated external defibrillators for heart attacks, or antivenom serum to combat animal poisoning) (Poljak & Šterbenc, 2020).

2.1.2 *Health and Safety*

Safety is a major social impact. In the context of AV adoption, the discussion of health and safety goes beyond the consumers and recipients to include multiple related entities such as employees, traffic users, etc. As

mentioned above, using AVs to deliver emergency medical supplies showcases their advantages in fast response and rapid delivery speed (Azmat & Kummer, 2020) which are crucial advantages in life-saving emergencies (Kellermann et al., 2020). In commercial contexts, AVs provide touchless customer delivery services, which prevent the spread of contagious viruses, especially after the COVID-19 pandemic (Pani et al., 2020), and therefore, improve end-customers' health and safety. In work environments, AVs help to enhance workforce health and safety by replacing tasks in hazardous conditions (Bechtsis et al., 2017, 2018) and reducing workplace accidents (Llopis-Albert et al., 2019). In the food SC, the use of drones in pesticide application reduces workers' exposure in hazardous chemicals in farming fields (Mahroof et al., 2021). In addition, the introduction of AVs, especially long-haul UGVs for road transport, potentially leads to enhanced safety for traffic users, and the load carried, by taking the place of human drivers (Sen et al., 2020). Road freight transport represents an essential role in logistics; hence, road traffic safety is crucial. More than 90% of traffic accidents are caused by human error—mainly due to fatigue and alcohol or drug abuse (Gružauskas et al., 2018). These human-made fatal accidents can be reduced by the adoption of AVs due to the implementation of smart sensors, big data analytics, and the Internet of Things (IoT) that automatically control trucks in the transport process without human interference (Gružauskas et al., 2018). In sum, employees or workers, other traffic users, and the end-customers, can benefit from improved health and safety performance in SCs by deploying AVs.

2.1.3 New Job Creation and Filling Labour Shortages

Labour shortages in logistics, especially in transportation, present a challenge in many countries. On one hand, ageing and retiring labour forces will result in a decrease in available truck drivers by 25% in the next decade (Klumpp & Zijm, 2019). On the other hand, a low birth rate in Generations Y and Z has led to an insufficient young labour force in SCs (Hwang et al., 2019). Under such circumstances, deploying AVs in the SC, especially in transportation, can reduce labour shortages and ensure stable productivity. The adoption of AVs reduces operational employee requirements for manual and repetitive tasks such as driving fork-lift trucks in warehouses and heavy goods vehicles on motorways (Klumpp, 2018). AVs can create new job opportunities for AGV operators and programmers (Sohrabi et al., 2020) as well as AV and infrastructure maintenance jobs (Sen et al., 2020) while the logistics tasks themselves may benefit from higher speed and greater accuracy

(Tang & Veelenturf, 2019). In addition, the deployment of AVs leads to job transformation in SCs—truck drivers no longer take the steering wheel, but monitor the trucks from screens in control centres or conduct administration tasks on the driving seat (Klumpp, 2018). A workforce that is trained in new technology-related tasks will be increasingly competitive and in demand (Sohrabi et al., 2020).

2.1.4 Reduced Traffic Impacts

Drones use low-altitude air space instead of ground infrastructure and ignore geographic limitations (Mangiaracina et al., 2019), utilising the fastest and most direct routes for delivery (Mohamed et al., 2020). Hence, their use for parcel delivery instead of conventional ground vehicles decreases the number of deployed delivery vehicles, reducing road traffic congestion (Perussi et al., 2019). Smart technologies such as sensors and cloud computing equipment on the UGVs, instead of human operators, help to optimise the distances from other vehicles, which smoothes road traffic flows (Sohrabi et al., 2020) and potentially reduces road congestion.

2.1.5 Product and Asset Security

AVs help to realise the real-time visibility of assets and product flows in SCs efficiently, which remains challenging in remote and locations with poor infrastructure (Sellevold et al., 2020). Equipped with smart sensors, drones are capable of asset and inventory surveillance to guarantee the real-time safety of products in warehouses and pipelines (Perussi et al., 2019), benefiting logistics managers and decision makers in monitoring product flows and intervening to adjust operations in a timely manner (Sellevold et al., 2020). Using built-in sensors, AVs automatically provide real-time status visibility and traceability of products during freight transport (D'Amico et al., 2021), whereas conventional vehicles require extra installations of such sensors to achieve similar benefits.

2.1.6 Improved Gender Equality

There is a serious imbalance of gender distribution in the logistics sector. In freight transport for instance, less than 6% of truck drivers are female in developed economies, and truck driving remains a male dominated profession (Kim et al., 2022). The adoption of AVs holds potential to diminish

this imbalance by creating more gender-neutral job opportunities, without existing labelling such as "male dominant jobs". For example, autonomous trucks take long-haul driving tasks, while human pilots conduct monitoring tasks (Klumpp, 2018). An increased percentage of females are expected to take on such roles leading to greater gender diversity (Kim et al., 2022).

2.2 Negative Social Impacts

Similar to the positive impacts, this study identifies seven negative impacts in five groups as shown in Table 2.

2.2.1 Safety

Despite the positive impacts on health and safety of AV adoption, the negative impacts cannot be ignored. Incidents due to the immaturity of UAV technology such as collisions, loss of signal, and inability to cope with extreme weather, can cause threats to the safety of the public, as well as the security of payloads (Mohamed et al., 2020). In addition, the use of delivery drones to conduct terrorist attacks is mentioned by Aydin (2019), and the operation of drones in commercial airline space as pointed out by Rao et al. (2016) also threaten public safety. Similar concerns exist for UGVs. Further, the reliability is not guaranteed so these vehicles may potentially cause fatal accidents (Fagnant & Kockelman, 2015).

2.2.2 *Privacy and Cybersecurity*

Tightly related to safety problems, weak privacy and cyber security are other negative impacts. The vulnerability to hackers targeting AVs cannot be ignored (Nikitas et al., 2021). Cyberattacks on AV software systems can lead to delivery of parcels to the wrong address, while leaking customers' personal information is a threat to privacy (Aydin, 2019). Further, individuals do not trust organisations and governments to use UAVs responsibly and there is a possibility that their data is being abused by others (Nelson et al., 2019). Hence, privacy and cyber security issues remain valid.

Table 2 Negative social impacts of AV adoption in SC

Source	Technology	Negative Social Impacts						
		1	**2**	**3**	**4**	**5**	**6**	**7**
Wang (2019)	UAV	X						
Kellermann et al. (2020)	UAV	X	X		X			
Aydin (2019)	UAV	X	X		X			
Rao et al. (2016)	UAV	X			X			
Nelson et al. (2019)	UAV				X			
Mohamed et al. (2020)	UAV	X			X			
Cheema et al. (2022)	UAV				X			
Rejeb et al. (2023)	UAV	X			X			
Tatham et al. (2017)	UAV							X
Ding (2018)	UAV				X			
Tatsidou et al. (2019)	UAV	X			X			
Wang et al. (2021)	UAV				X			
Haula and Agbozo (2020)	UAV	X			X			
Dhote and Limbourg (2020)	UAV	X			X			
Poljak and Šterbenc (2020)	UAV	X			X			
Azmat and Kummer (2020)	UAV & UGV		X		X			
Tang and Veelenturf (2019)	UAV & UGV		X		X	X		
Mangiaracina et al. (2019)	UAV & UGV		X					
Perussi et al. (2019)	UAV & UGV	X						
Slabinac (2015)	UAV & UGV	X						
Rai et al. (2022)	UAV & UGV		X		X			
D'Amico et al. (2021)	UAV & UGV	X			X		X	
Klumpp (2018)	UGV		X					
Soni and Soni (2019)	UGV		X				X	
Sohrab et al. (2020)	UGV		X					
Dadzie and Johnston (1991)	UGV		X					
Sen et al. (2020)	UGV		X					
Gružauskas et al. (2018)	UGV		X					
Fagnant and Kockelman (2015)	UGV	X	X		X			
Merlino and Sproǵe (2017)	UGV		X					
Marsden et al. (2018)	UGV	X	X					
Io and Lee (2019)	UGV	X	X		X			
Engholm et al. (2021)	UGV		X					

(continued)

Table 2 (continued)

Source	Technology	Negative Social Impacts					
Kim et al. (2022)	UGV		X		X		
Kurniawan et al. (2021)	UGV		X				
Nikitas et al. (2021)	UGV		X	X	X		
Garus et al. (2022)	UGV	X				X	
Vlachos et al. (2023)	UGV		X				

1. Safety incidents due to crashes, failures, and attacks (consumers, civilians)
2. Unemployment in logistics sectors (employees)
3. Unemployment in supportive industries (employees)
4. Privacy and cybersecurity (consumers, patients, local communities)
5. Social inequality, not accessible (potential consumers)
6. Reduced interactions among employees (employees)
7. Reduced interactions between patients and service providers (patients, victims)

2.2.3 Unemployment

Unemployment is identified as a negative impact in the literature since AVs are likely to replace numerous positions in the logistics sector and result in job losses. Truck drivers who conduct long-haul freight transport, as well as vehicle-related service providers, could potentially lose their jobs (Sohrabi et al., 2020). Similar concerns of potential unemployment are raised for last-mile deliveries, with AVs replacing delivery personnel (Io & Lee, 2019; Marsden et al., 2018). Employees in industries that support conventional vehicles where the requirements differ from AVs (e.g., conventional vehicle maintenance, finance, petrol stations, etc.), will face short-term employment disruptions (Nikitas et al., 2021). Although job transformations are likely to occur (Klumpp, 2018), older and less educated employees with limited transferable skills are more likely to be laid off (Nikitas et al., 2021). Examples of previous industrial revolutions suggest that although new jobs will be generated after disruptive technology deployment, it is inevitable that a large number of lower skilled workers will suffer unemployment, at least in the short-term (Ding, 2018).

2.2.4 *Social Inequality*

Deploying AVs may impact social inequality by enlarging the gap between the rich and poor (Tang & Veelenturf, 2019). Those who cannot afford digital technology are excluded (Garus et al., 2022). Familiarity with the technology plays an important role in public acceptance (Kapser et al., 2021) and those who receive less education regarding AV technology and usage in parcel deliveries are less likely to accept them. Overall, there may be restrictions for certain social classes to access the benefits of AV delivery services.

2.2.5 *Human Interaction*

AVs in SCs reduce human interactions in workplaces such as warehouses or manufacturing plants and in last-mile delivery. As more transportation tasks are handed by AVs, workers may become more isolated (Soni & Soni, 2019). Employees need meaningful interactions when they are working to fulfil their human needs (D'Amico et al., 2021). Tatham et al. (2017) emphasise the importance of building a relationship between healthcare providers and receivers. An AV delivery service potentially reduces human care and weakens the treatments for patients who face cold machines, rather than empathetic human beings they can interact with.

2.3 Social Barriers to AV Adoption

In addition to the social impacts, a series of social barriers are recognised as hindering AVs from wider adoption. They may be potential outcomes of social impacts rather than direct impacts. In this context, social acceptance and legal restrictions are the main barriers mentioned in literature.

2.3.1 *Social Acceptance*

According to Nelson et al. (2019), it is crucial not to hinder the advantages of AVs, by understanding individuals' perceptions and acceptance of the emerging technologies. Public acceptance is relevant to the social impacts mentioned above, especially the negative ones. Aydin (2019) shows that safety and security concerns (for people and products) result in low acceptance, accompanied by privacy concerns (Nelson et al., 2019), unemployment, and working environment safety (Dadzie & Johnston, 1991).

Attitudes towards AVs are rooted in personal experience, or familiarity with the technologies (Eißfeldt et al., 2020). People who see UAVs flying overhead could make unpleasant assumptions instantly (Tatsidou et al., 2019) while consumers who show curiosity in AVs are more willing to be served by AVs (Hwang et al., 2021a). Those who are more familiar with the technology will express a higher willingness to pay—hence a higher acceptance of the technology (Aydin, 2019). When UAVs are conducting humanitarian or healthcare supplies, the public shows a higher acceptance than in commercial contexts (Eißfeldt et al., 2020).

2.3.2 *Regulation Restrictions*

A lack of accepted regulations limits wider AV adoption (Rao et al., 2016) and there are considerable social concerns regarding the negative impacts such as illegal use of AVs for terrorism, cyberattacks, and unemployment. Those concerns drive legislative authorities to formulate restrictions that delay the practical implementation of AVs (Fagnant & Kockelman, 2015). Given the perceived negative impacts and technology immaturity, regulations are still limiting the operations of AVs outdoors (i.e., strict permissions for Beyond-Visual-Line-Of-Sight (BVLOS) flying for UAVs restricting the wider deployment in small parcel delivery services) (DHL, 2020). Therefore, the literature calls for improved regulations and official guidance on AV adoption to be applied, especially for humanitarian and healthcare purposes. When a disaster occurs in a region where drones are not permitted, it hinders relief organisations from planning to deploy UAVs for efficient humanitarian medical supplies (Tatsidou et al., 2019). Therefore, clear licencing and regulations are necessary for AV applications in SCs (Mohamed et al., 2020).

3 Linking Impacts and Indicators

The previous section extracted the positive and negative social impacts of AV adoption from the literature. This section relates the social impacts to relevant SCSS indicators that help to investigate SCSS implications in the AV adoption context. Additionally, a range of suggested research directions are presented.

3.1 Contextual SCSS Indicators

3.1.1 *SCSS Aspects from the Existing Seminal Literature*

An indicator is considered to be a sign of showing the condition or existence of something (Ahi & Searcy, 2015a). SCSS indicators present social performance in various aspects, including welfare, community development, human safety, labour conditions, etc. (Ahi & Searcy, 2015b). After reviewing seminal papers on SC sustainability and the literature included in this review, a summary of SCSS aspects accompanied by the definitions is presented in Table 3.

3.1.2 *Social Impacts and SCSS Indicators*

The collection of positive and negative impacts of AV adoption in SCs serves to fulfil the first objective. Meanwhile, the identified impacts provide the base for the second objective, which relates to social sustainability indicators. On the basis of the listed SCSS indicators and their definitions, the 20 identified social impacts are allocated to the related indicators. It is possible that one impact relates to multiple SCSS indicators, while in this study, one impact is linked with only one most directly and primarily relevant SCSS indicator. For example, the impact of "workforce health and safety" is relevant to both health and safety, and employment. As it concerns the health and safety impacts on the employees, this impact is placed under the "health and safety" indicator.

In this context, there are six indicators found relevant to the identified social impacts. Figure 1 presents the proposed links between the social impacts and SCSS indicators in this context.

3.1.2.1 Health and Safety

This indicator covers broadly the physical and mental health of individuals and the public (Carter & Rogers, 2008; To & Lee, 2017). Positive impacts including life-saving, road traffic safety, workforce safety, and customers' health and safety are included. The safety incidents from the negative impacts are also categorised under this SCSS aspect.

Table 3 Definitions of SCSS indicators

Social indicators	Source*	Description
Community	Community (3)	The interactions between SC organisations and local communities, such as supporting community projects, community complaints, community connection and networks
Culture	Culture (4)	A shared, organisation-wide vision and core value, which also generate internal drive and passion
Customer satisfaction	Customer Satisfaction (3)	Customer perception in terms of the products or services provided by the SC organisations
Diversity and Equality	Gender (2) Disabled/ Marginalised People Inclusion (2) Minority Development (2) Diversity (4)	Equal treatment of different genders, catering to their needs, and assigning equal rights at the work place. Include the disabled/ marginalised people in the workplace, and also treating people equally regardless of their religion, race, and ethnicity
Education	Education (1)	Education includes elements such as the percentage of employees trained, supporting educational institutions
Employment	Employment (1) (3)	Labour inputs and outputs, such as the unemployment issue, labour productivity, and employee satisfaction
Health and Safety	Health (1) Public Safety (1) Safety (4) Health and Safety (2) (3)	Physical and mental health which are directly related to safety and hygiene at work. It also describes hazardous working conditions which could leave long-term effects on the personal health of the worker. Work by children under the age of 18 is hazardous to the physical or mental health of the child
Housing	Housing (1)	Housing indicator includes the public's living conditions, quality of living and the neighbourhood
Human rights	Human Rights (2)	The rights are inherent for all human beings, irrespective of demographic status. Without discrimination is the core of human rights

(continued)

Table 3 (continued)

Social indicators	Source*	Description
	Human Rights (4)	
Labour conditions	Labour conditions (2) Income (1) Child Labour (2)	Working conditions of the employees include low wages, extended hours of working, right to form unions, contract labour and exploitation of the employees
Philanthropy	Philanthropy (4)	The voluntary or discretionary decision of donating money or other corporate resources to social causes
Quality	Quality (3)	The quality and quality improvement of the product or services provided by the SC
Regulation	Regulation (3) (4)	Binding government policies and documents that are influenced by the environmental and social concerns
Reputation	Reputation (4)	The image of an organisation which represents public sentiment in terms of the actions of the organisation
Risk Management	Risk Management (4)	The ability to understand and manage the risks, for more resilient and agile SCs
Transparency	Transparency (4)	Collaborate with stakeholders for sharing of monitoring information and enhancing remediation to improve SC
Welfare	Welfare (3)	Well-being, happiness, and quality of life

*Source** 1. To and Lee (2017); 2. Yawar and Seuring (2017); 3. Ahi and Searcy (2015a, 2015b); 4. Carter and Rogers (2008)

3.1.2.2 Customer Satisfaction

Customer satisfaction represents the customers' thoughts, preferences, and expectations in terms of the products or services offered by SC organisations (Ahi & Searcy, 2015a, 2015b). Following the definition, the impacts regarding faster and more customised delivery services are categorised beneath this SCSS indicator. These two impacts represent the consumers' (end-customers') expectations of the improved delivery services provided, which match the customer satisfaction concept.

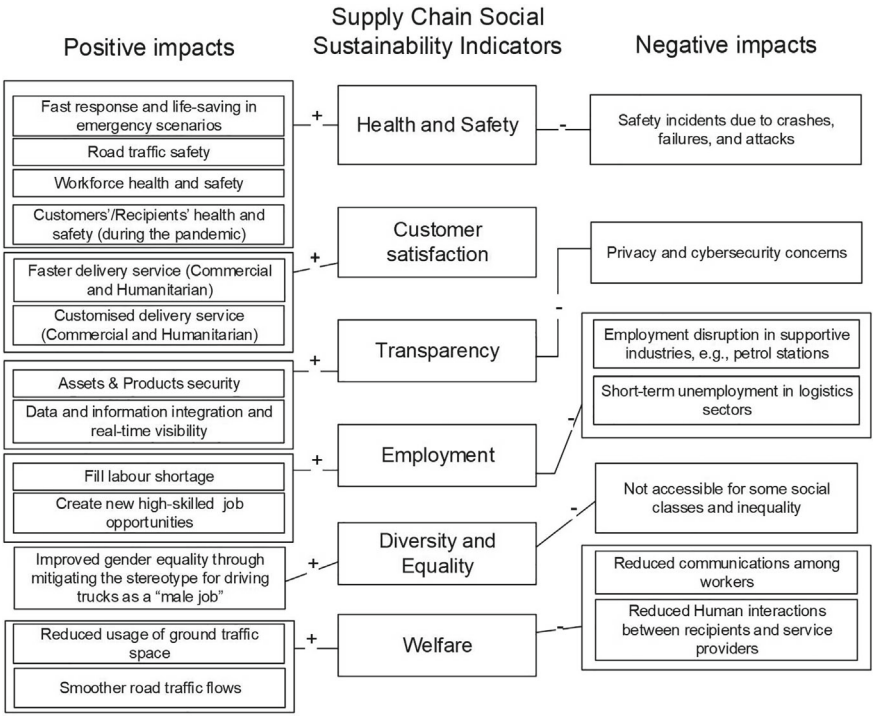

Fig. 1 Links between social impacts and SCSS indicators in AV adoption (*Source* Authors)

3.1.2.3 Transparency

Transparency calls for collaboration with stakeholders for information sharing and monitoring to improve SC visibility (Carter & Rogers, 2008). This means that impacts relating to information visibility and security are appropriate to be classified under this aspect. With this consideration, assets and products security, information integration and visibility, as well as privacy and cybersecurity concerns are categorised under the transparency indicator.

3.1.2.4 Employment

Employment addresses issues including unemployment, labour inputs and outputs, and productivity (Ahi & Searcy, 2015a). Since this study exclusively discusses social implications, elements such as salaries, productivity, and efficiency that are shared within the economic dimension are excluded.

Hence, the impacts including filling labour shortages, new job creation, and unemployment in logistics sectors and supporting industries are relevant.

3.1.2.5 Diversity and Equality

This indicator covers issues including gender equality, diversity, inclusion of disadvantaged groups, and the development of minorities (Carter & Rogers, 2008; Yawar & Seuring, 2017). In this study, the gender equality and stereotype mitigation form the positive list and social class inequality forms the negative list of the social impacts included in this category.

3.1.2.6 Welfare

Based on the description of welfare, "well-being, happiness, and life quality" (Ahi & Searcy, 2015a), two main aspects are included in this indicator: human interaction, and traffic congestion. The positive impacts regarding the reduced usage of ground traffic space as well as smoother road traffic flows are classified under traffic congestion, while reduced communications and human interactions among various entities are involved as the negative aspects.

3.2 Potential Research Directions

As part of this review exercise, research suggestions gathered from the literature pool are mapped into key directions for future research regarding the SCSS of AV adoption. There are 12 key research directions suggested by the literature (presented in Table 4). These directions include four main dimensions: richer data and methods (i.e., more geographical contexts, more data to be collected, more data collection methods), wider SCSS aspects (more social aspects to cover, more stakeholders to focus on, more variables for quantitative studies to form), managerial advice (wider SC processes, new business models for AV adoption, use of more AV types and functions), and technological aspects (human–robot interaction; interoperability with other technologies; transferrable concepts into other smart scenarios).

Table 4 Research directions suggested in the literature

Author	Paraphrased suggestions for future research	Research Directions											
		1	2	3	4	5	6	7	8	9	10	11	12
Haula and Agbozo (2020)	1. More databases to be involved, 2. More geographical regions involved, 3. Security and privacy aspects	X	X				X						
Kapser and Abdelrahman (2020)	More than half of participants are not familiar with the ADV delivery service. Future research should focus on the participants more familiar with this kind of delivery		X	X									
Sellevold et al. (2020)	UAS-logistics system determines the societal, health, safety, labour and social acceptance impacts			X			X						
Klumpp (2018)	Human-artificial interaction and acceptance			X	X		X						
Dhote and Limbourg (2020)	Case of other regions, other attributes of transport—service, accessibility, reliability, safety, security, flexibility	X	X				X						
Kellermann et al. (2020)	Calls for social sciences engaging with emerging transportation technologies in general as well as for disciplines of transport and urban planning or civil society, public acts as a central stakeholder in this innovation process—and avoid potential societal conflicts in the future		X	X		X	X						
Tatham et al. (2017)	Field trials are needed		X										
Sohrabi et al. (2020)	1. More to focus on rural and SAE level 3 & 4 vehicles, 2. Research the uncertainties shown in the framework. More quantitative researches as well		X			X		X			X		

Author	Paraphrased suggestions for future research	Research Directions											
		1	2	3	4	5	6	7	8	9	10	11	12
Betchsis et al. (2018)	The adoption of IAV in last-mile delivery and pharmaceutical SC								X		X		
Hwang et al. (2021a)	1. Other delivery services. 2. meaningful to assess potential barriers in better predicting the formation of consumers' acceptance behaviour towards drone food delivery services			X						X	X		
Hwang et al. (2019, 2021b)	MCI in other contexts—different types of data collection methods	X				X				X			
Sen et al. (2020)	Any aspect that improves the public's standards of living, starting from human health should be prioritised			X			X			X			
Monios and Bergqvist (2020)	1. A new modified model of freight transportation system is needed. 2. New business model							X					
Gružauskas et al. (2018)	The proposed framework to be adopted in other industries, and more sustainable SC strategies to be raised							X					
Aydin (2019)	Developing effective risk mitigation and response strategies after public acceptance to be achieved in the near future			X				X					
Eißfeldt et al. (2020)	1. Further research should also consider different drone types and their functions, as this might have an impact on public acceptance. 2. More investigations focus on the improvement of social acceptance for drone applications that are currently not very high			X			X			X	X		

(continued)

Table 4 (continued)

Author	Paraphrased suggestions for future research	Research Directions											
		1	2	3	4	5	6	7	8	9	10	11	12
Ding (2018)	More empirical data in the future, how smart logistics based on industry 4.0 can provide a sustainable and agile delivery service can be investigated		X				X						
Betchsis et al. (2017)	Currently only on warehouse management and distribution and on the manufacturing division, a lack of exploitation across the entire spectrum of SC operations		X						X				
Io and Lee (2019)	Future studies may focus on analysing comments about the use of drones for delivery		X	X		X							
Mangiaracina et al. (2019)	Targeting under investigated solutions such as mapping the customer behaviour or addressing the implementation of dynamic pricing to sectors different from e-grocery			X			X		X	X			
Cheema et al. (2022)	A drone trajectory optimisation and path planning to be added to enhance the efficiency of the delivery process to be more autonomous. Also, time optimisation for drones to be pursued. Privacy preserving mechanism is required to avoid revealing information			X		X	X						

Author	Paraphrased suggestions for future research	Research Directions											
		1	2	3	4	5	6	7	8	9	10	11	12
Rejeb et al. (2023)	Interoperability of drones with other technologies											X	
Engholm et al. (2021)	1. More environmental and economic impacts of DL-trucks on future transport system and society; 2. network modelling, 3. social impacts resulting from the DL-trucks					X	X	X					
Serrano-Hernandez et al. (2021)	1. More information to be obtained from the population to better understand the significant approval for the aerial distribution—survey; 2. more groups other than resident evaluations—other stakeholders; 3. future work for regulators; 4. more geographic contexts	X	X	X		X							
Kapser et al. (2021)	1. Further research should explicitly target actual users to identify the differences in users' behavioural intentions but also to investigate actual use behaviour; 2. further studies should take a longitudinal approach and should collect data to compare findings before and after the virus outbreak; 3. more geographical regions to be included; 4. other AVs to be investigated	X	X	X							X		

(continued)

Table 4 (continued)

Author	Paraphrased suggestions for future research	Research Directions											
		1	2	3	4	5	6	7	8	9	10	11	12
Vlachos et al. (2023)	1. Future research may rely on multi-case studies that can test and validate the action plan in wider contexts, 2. researchers can use other research methods to collect and test key data; 3. an action plan is needed. 4. future studies can test how the key concepts from Socio-technical theory can be applied in other smart scenarios		X			X		X					X
Comtet and Johannessen (2022)	Proof of concepts: The usage of drones in the organisation processes—to what extent in-depth; whether drones in combination with other technologies have power to transform the healthcare system?								X		X	X	

1. More geographical contexts;
2. More datasets and databases;
3. Focus on more stakeholders
4. Human–robot interactions;
5. More data collection methods;
6. More social aspects to be covered;
7. New business models for AV adoption;
8. Wider SC processes to cover;
9. More variables in quantitative studies;
10. More and detailed AV types and deployment;
11. Interoperability with other technologies;
12. Transferrable concepts into other smart scenarios

4 Conclusion

This study gathers the social impacts of AV adoption in SCs. Positive and negative impacts brought by UAVs and UGVs adopted in the commercial and humanitarian SCs are synthesised from the existing literature. These are then connected to commonly accepted SCSS indicators and research suggestions are outlined.

4.1 Contributions

This review defines the types of autonomous vehicles adopted in SCs and classifies positive and negative social impacts of both UGVs and UAVs in commercial and humanitarian logistics contexts. Thirteen positive and seven negative social impacts are identified. These impacts are connected with six relevant social sustainability indicators: Health and Safety, Customer Satisfaction, Employment, Transparency, Diversity and Equality, and Welfare.

A mapping that links the impacts and sustainability indicators is built upon the outcomes. This finding illustrates how the various SCSS aspects are affected by certain positive and negative social impacts in the context of AV adoption. Hence, this study contributes to SCSS knowledge by exploring the specific context-based social sustainability indicators (Gugler & Shi, 2009).

For practical implications, this review firstly presents a holistic overview of the key social impacts and SCSS aspects to focus on. These outcomes serve as a foundation for organisations to establish a tailored overview of key performance indices regarding social issues. This review also provides a guide for organisations in SCs to enhance the adoption of AVs in operations when considering social demands from stakeholders. Finally, this research raises recommendations to legislative authorities to reflect the existing regulations and enact more suitable laws for broader adoption of AVs in SCs.

4.2 Research Limitations and Future Research Suggestions

This study has a few limitations. First, it focuses on AV adoption in SCs and the social dimension of sustainability, an area that is still in its infancy and hence an emerging field of research. AVs have not been broadly adopted in SCs yet so the impacts and relevant social sustainability indicators should be verified as AVs are being adopted. Second, this study identified several stakeholders that are potentially related to the social impacts and social

sustainability, while much of the existing literature does not explicitly emphasise them. Third, although this study followed a systematic review approach, some relevant literature may have been missed and contributions published in languages other than English are not captured. Finally, the literature pool refers mainly to data gathered from few locations such as Germany, America, and mainland China.

In addition to the 12 classified research directions from the literature, this study provides several research paths based on the outcomes and limitations. First, the relevant social impacts and SCSS indicators in this context can be further refined through empirical observations. Specific stakeholder groups should also be identified in order to strengthen SCSS conceptualisation, since "social sustainability" discusses the "human side of sustainability issues" (Huq et al., 2014). Existing literature provides in-depth investigations among several stakeholder groups, but further exploratory studies could be conducted to identify potential additional stakeholders. Since research into social sustainability is still lagging behind comparing to the other two sustainability dimensions, more studies should focus on identifying contextual SCSS indicators in order to further develop SCSS knowledge. Specifically, the relevance of stakeholders, social impacts, and sustainability indicators should be clarified and studied in wider geographical contexts in order to strengthen the understanding of SCSS.

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10

The Impact of Digitalization on Firm-Level Innovation Performance During the Pandemic

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1 Introduction

The fourth industrial revolution, or Industry 4.0, is one of the biggest trends in the world economy. This phrase, which describes the disruptive changes in business processes, supply chains, and business models, is closely linked to radical innovations and is used to describe the digital revolution in production (Schmidt et al., 2015). According to Schumpeter (1934, 1942), major changes in the world are shaped by radical innovations, and these changes result in creative destruction.

One of the main features of Industry 4.0 is digitalization. Supply chain (SC) digitalization has become a crucial strategy for firms in various industries as it allows businesses to harness the power of digital technologies to optimize their supply chain operations, improve efficiency, and create new opportunities for innovation (Ngo et al., 2023). This is achieved through the adoption of technologies such as data analytics, the Internet of Things,

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artificial intelligence, and virtualization (Cui et al., 2023). Furthermore, digitalization in the logistics and supply chain management industry has the potential to transform traditional paradigms, business models, and industry boundaries (Spieske & Birkel, 2021).

The coronavirus (COVID-19) pandemic has further emphasized the importance of supply chain digitalization, as it has disrupted global supply chains, resulting in significant challenges for businesses in terms of logistics, inventory management, workforce constraints, and demand forecasting. As a response to these challenges, many companies have accelerated their efforts to digitize their supply chains in order to build resilience, enhance their agility and innovation capabilities, and in general adapt to the new normal of the business environment (Hald & Coslugeanu, 2022; Kastelli et al., 2024; Zhao et al., 2023).

Past research provides insights into the impact of digitalization on supply chains; however, to the best of our knowledge, this is the first study to assess the impact of firms' digitalization on their innovation performance during the pandemic. To this end, we use data from World Enterprise Surveys (ES) COVID modules for 2020–2022, which cover forty-five economies. This study aims to contribute to the literature on the role of supply chain digitalization on firm-level innovation performance in two ways.

First, we use cross-country comparable survey data in forty-five economies in Africa, East Asia Pacific, Europe and Central Asia, Latin America & and the Caribbean, and the Middle East and North Africa. Most previous studies on the impact of COVID-19 or other crises are restricted to rich countries, a particular region, or a few developing countries. Among these, the closest study to ours is Heredia et al. (2022). They discuss the role of digital capabilities on firm performance (measured by the change in demand for products and services and the change in sales) for 27 countries. Unlike this work, we use innovation performance as our dependent variable since accelerated digital transformation can bring radical innovations and technological breakthroughs.

Second, we use firm-level data gathered during the pandemic to examine the impact of government assistance to the private sector during the COVID-19 outbreak. Given the disruptive nature of digital innovations, comprehensive government policies are necessary, and they need to go beyond the support of the information and communication technologies (ICT) sector (Nepelski, 2019). Our work complements earlier work by Trieu et al. (2023), Karalashvili and Tamkoç (2022), Bennedsen et al. (2020), and Chetty et al. (2020) who used data from Vietnam, El Salvador and Georgia, US, Denmark, respectively. Contrary to these three articles, we consider forty-five

nations in our analysis and estimate the causal link between the government's support for adopting digital technologies and innovation performance.

The remainder of this chapter is structured as follows: Section 2 reviews the literature on supply chain digitalization, supply chain resilience, and innovation during the COVID-19 pandemic. Section 3 describes the dataset and explains the methodology used in this study. Section 4 presents empirical results. Finally, conclusions and some important policy implications are introduced in Sect. 5.

2 Literature Review

2.1 SC Digitalization and Resilience During the COVID-19 Pandemic

The fourth industrial revolution (Industry 4.0), recognized as a technological framework, combines and expands production processes at the intra- and inter-organizational levels, signifying the start of the fourth industrial revolution (Xue & Li, 2023). It can be defined as "*The sum of all disruptive innovations derived and implemented in a value chain to address the trends of digitalization, autonomization, transparency, mobility, modularization, network-collaboration, and socializing of products and processes", and often referred to as "smart manufacturing" or "smart industry*" (Pfohl et al., 2015). Increased radicalness increases the likelihood of technological breakthroughs (Briggs & Buehler, 2018). And Industry 4.0 is highly correlated with radical innovations, as it disrupts business processes, supply chains, and business models (Schmidt et al., 2015).

The following five primary elements are the basis for Industry 4.0: (1) digitization, optimization, and customization of production; (2) automation and adaptation; (3) value-added services and businesses; (4) automated data exchange and communication; and (5) human–machine interaction (HMI) (Horváth & Szabó, 2019). These characteristics allow businesses to boost productivity and have more effective production processes (Lu, 2017).

One of the dimensions of Industry 4.0 is the digitalization of products/services, which corresponds to the creation of "service-enhanced products" that have sensing, computing, and communication capabilities (Camarinha-Matos et al., 2019). Parida et al. (2019) define digitalization as the "*use of digital technologies to innovate a business model and provide new revenue streams and value-producing opportunities in industrial ecosystems*". Emerging disruptive technologies that enable digital supply chain transformation/

digitalization include cloud computing, big data analytics (BDA), internet of things (IoT), social media, user interface, advanced robotics, 3D printing, augmented reality (AR), blockchain technology, virtual reality (VR), artificial intelligence (AI), drones, and machine learning (Agrawal & Narain, 2023; Bigliardi et al., 2022). These technologies have a significant impact on supply chains and contribute to supply chain resilience by providing flexibility, visibility, agility, and other capabilities (Liu et al., 2023).

During global public health crises, such as the coronavirus disease (COVID-19) pandemic, these characteristics become even more important. The pandemic has caused unprecedented challenges to global supply chains and international production networks. It resulted in negative supply shocks through production reductions and business closures, adverse demand shocks on both the domestic as well as export markets, and also increased demand for critical supplies and personal protective equipment, pressuring healthcare supply chain, as well as internet-based services and information and communication technology (ICT) equipment (Obashi & Kimura, 2021).

Previous studies have demonstrated that digitalized business models of entrepreneurial firms increase their resilience to pandemic crises (Bürgel et al., 2023).

To reduce unexpected disruption risks in uncertain environments, firms need to focus both on resilience and efficiency. Supply chain digitalization enables businesses to gather real-time data and insights, leading to more informed decision-making and improved collaboration across the supply chain network. For example, if a business applies blockchain technologies to oil supply chain management, transportation costs can be reduced, which improves supply chain efficiency, and business processes will be more transparent. Similar technologies affect resilience and efficiency through mechanisms such as early warning and planning and improving visibility, response speed, and recovery capability (Xue & Li, 2023).

In an extensive literature survey, Hald and Coslugeanu (2022) discuss the most effective strategies for global supply chains during the pandemic. These can be summarized as risk management, flexibility, visibility in the upstream SC and end-to-end visibility, collaboration among SC members, localization and regionalization, and flexible and proactive redundancies.

2.2 Role of Digitalization in Innovation Performance

Industry 4.0 has brought about rapid technological advancements that are urging businesses and organizations to integrate knowledge-intensive elements and adopt new digital-based systems in order to achieve their target

outcomes for innovation and establish themselves in the rapidly evolving digital landscape (Kastelli et al., 2024). Businesses now have more need than ever to adapt to quickly evolving business environments as a result of rising globalization and worldwide competitiveness. Firms can achieve this through supply chain innovation (SCI), which can be defined as "an integrated change from incremental to radical changes in product, process, marketing, technology, resource and/or organization, which are associated with all related parties, covering all related functions in the supply chain and creating value for all stakeholders" (Gao et al., 2017). Businesses engage in innovative activities to acquire a persistent competitive advantage, eliminate, or outpace their rivals, or both (Karaman Kabadurmus, 2020).

Digitalization of the supply chain can have a positive impact on innovation performance, especially in times of crisis such as the COVID-19 pandemic. By adopting digital technologies and transforming traditional supply chain processes, firms can enhance their agility and responsiveness to changing market conditions. This enables them to quickly identify and seize new opportunities, adapt their products and services to meet evolving customer needs, and generate innovative solutions to overcome challenges.

Many studies show that digitalization, which facilitates collaboration and knowledge sharing among supply chain members, can enhance innovation performance. Chong et al. (2011) find that collaboration through information sharing in the supply chain can have a positive impact on innovation. This suggests that digitalization, which enables efficient information sharing, can enhance innovation performance.

Jean et al. (2013) highlight the importance of supply chain relationships in fostering innovation generation. The authors emphasize the importance of supply chain members' innovations, rather than individual partners', in the success of all supply chain partners.

Holmström and Partanen (2014) discuss how the combination of digital manufacturing and supply chains can simplify and streamline supply chain processes. This simplification can lead to more effective solutions and ultimately improve innovation performance.

Agrawal et al. (2019) emphasize the importance of organizational structure in the digital transformation of the supply chain. They argue that a customer-driven, agile, integrated, innovative, and adaptive organizational structure enables transparent information flow, knowledge sharing, collaboration, integration, and decision-making. These factors contribute to enhanced innovation performance.

Collaboration in the supply chain through partnerships enhances firm performance, commitment, and innovation performance. According to Shin

et al. (2019) innovation performance positively impacts a firm's productivity, supply chain performance, and competitive advantage. This suggests that digitalization, which facilitates collaboration and innovation in the supply chain, can enhance innovation performance.

Li (2020) examines how supply chain relationship quality affects the innovation performance of businesses. They find that a high-quality supply chain relationship positively affects innovation performance. Therefore, digitalization, which can improve communication, collaboration, and relationship quality in the supply chain, can enhance innovation performance.

Karaman Kabadurmus (2022) show that enterprises' network capabilities and innovation ecosystems contribute to the development of radical innovations that support enterprises to obtain competitive advantages.

Overall, supply chain digitalization has the potential to significantly enhance innovation performance by enabling organizations to adapt to disruptions, identify new opportunities, and make data-driven decisions for continuous improvement and innovation. It enables efficient information sharing, simplifies supply chain processes, fosters collaboration and knowledge sharing among supply chain members, improves relationship quality, and enhances overall firm performance.

3 Data and Empirical Specification

3.1 Data Source

We use the 2020–2022 World Bank Enterprise Survey (ES) COVID-19 module that covers more than 44,000 firms in Africa, East Asia Pacific, Europe and Central Asia, Latin America and the Caribbean, and the Middle East and North Africa. The Enterprise Surveys are representative samples of the private sectors in each nation, allowing for the investigation of trends that impact a substantial part of the overall economy.[1]

The forty-five countries in the sample are Chad, Guinea, Mozambique, Niger, Somalia, Togo, Zambia, Zimbabwe, Mongolia, Albania, Armenia, Azerbaijan, Belarus, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czech Republic, Estonia, Georgia, Greece, Hungary, Italy, Kazakhstan, Latvia, Lithuania, Moldova, Montenegro, North Macedonia, Poland, Portugal, Romania, Russian Federation, Serbia, Slovak Republic, Slovenia, El Salvador, Guatemala, Honduras, Nicaragua, Panama, Jordan, Lebanon,

[1] Please see https://www.enterprisesurveys.org/en/covid-19 for more details on coverage and ongoing data collection projects.

Malta, and Morocco. Topics covered in the module are changes in sales, the demand of products or services, and supply of inputs, changes in the workforce, changes in cashflow availability and source, and government measures. The questionnaire covers three sectors: manufacturing, retail, and other services.

Figure 1 demonstrates the prevalence of innovation across countries for different survey years. In some countries, data is available for only the 1st round, while in some others, there is data for 2nd and 3rd rounds. The firms were asked the following question: "Since the outbreak of COVID-19, did this establishment experience any of the following changes (introduced new or improved products or services) in response to the COVID-19 outbreak?". The figure highlights that most firms did not use innovation as a coping mechanism to stay in business during the pandemic. 66% of the firms in Somalia, 55% in Azerbaijan, and %45 of the firms in Guatemala have said "yes" to new product development in response to COVID-19. The lowest rates are for Croatia (4%), Jordan (7%), and Malta (12%).

3.2 Empirical Model

$$Innovation_{isc} = \varphi\{\alpha_0 + \beta_1 onlinebusiness + \beta_2 delivery + \beta_3 remotework + \beta_4 governmentsupport + \beta_5 X_{ics} + \gamma_c + \rho_s + \varepsilon\} \quad (1)$$

where i denotes the firm, s the industry, and c the country. Country and industry dummies include features unique to a given industry as well as national factors that may be connected to a company's desire to innovate.

$Innovation$ is a dummy variable that takes the value 1 if a firm has developed a new product that is completely new to the firm or is significantly different from its closest existing product. While some innovations may not meet the criteria for being labelled disruptive, we believe that those that do will nevertheless have a sufficient degree of product differentiation to qualify as such for the company, the region, or the nation (Krammer & Jiménez, 2020; Karaman Kabadurmus, 2022).

To measure the digital transformation capabilities of supply chains, we use the following question: "Since the outbreak of COVID-19, did this establishment experience any of the following changes (started or increased business activity online, started or increased delivery or carry-out of goods and services, started or increased remote work arrangement for its workforce) in response to the COVID-19 outbreak?". We then create dummy variables for: $onlinebusiness$, $delivery$, and $remotework$„ following (Heredia et al., 2022). These capabilities are drivers of digital transformation. We associate these with companies' ability to digitalize their goods and services, incorporate and apply digital technologies, and implement new digital business models. Examples could be shifting from product-oriented to digitally

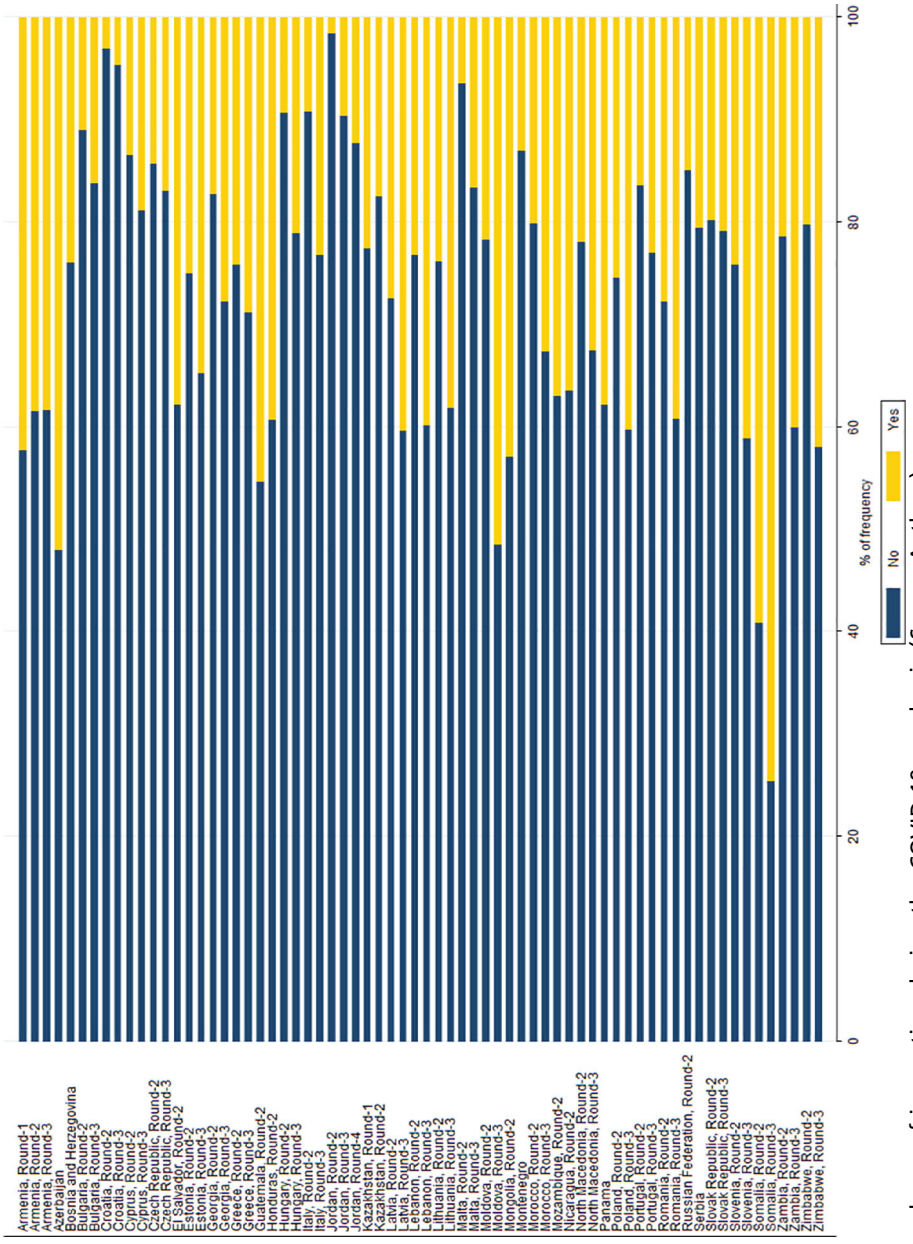

Fig. 1 Prevalence of innovation during the COVID-19 pandemic (*Source* Authors)

serving business operations, rearranging business activities such as supply chain management or innovation and attracting and retaining highly skilled workers (Kastelli et al., 2024).

Enterprise Surveys COVID module and follow-up surveys ask questions about the policies implemented by governments to address problems such as changes in sales, employment, and input purchases, as well as access to finance problems during the pandemic. One of the questions is whether the establishment was given any support—either local or national—from the government in the wake of the crisis, and if so, what kind—technical or financial—for the adoption of digital technologies. We use the responses to this question to construct another key right-hand side variable: $governmentsupport$. Following Karalashvili and Tamkoç (2022), our variable is cumulative, so companies that answered "Yes" in the first, second, or third rounds are taken into account as having received government support. Overall, in the sample, 7433 firms responded to this question, and 6% confirmed to receive digital support from the government.

There could be other firm characteristics that could be tied to digital transformation capabilities and innovation. We capture these control variables in X. These variables include the capacity utilization, i.e. the firm's output produced as a percentage of the maximum output possible if using all the physical capital available ($capacity$) and a dummy coded as 1 if the firm exports directly and/or indirectly ($exporter$). We also control for industry-level and country-level heterogeneities by including industry and country dummies.

Table 1 shows the descriptive statistics of our data; 26% of firms developed a new product or upgraded an existing one after the COVID-19 outbreak. Regarding the supply chain transformation capabilities, 38% of firms started or increased remote work arrangements, 32% started or increased online business activities, and 29% started or increased delivery of goods and services. Only 6% of firms received government support for digital technologies. In addition, 36% of the firms in the sample exported their goods or services, and capacity utilization was around 66%.

4 Results

Table 2 presents our results. We find that supply chain digital transformation capabilities spur innovation.

Capacity utilization (CU) measures how effectively a firm/industry uses its productive capacity (manufacturing facilities, employees, or technology).

Table 1 Descriptive statistics

Variable	Obs	Mean	Std. Dev	Min	Max
Innovation	26,524	0.26	0.44	0	1
Online business	40,080	0.32	0.47	0	1
Delivery	40,759	0.29	0.46	0	1
Remote work	39,252	0.38	0.49	0	1
Government support	7,433	0.06	0.24	0	1
Exporter	44,665	0.36	0.48	0	1
Capacity	19,650	65.77	26.59	0	100

Source *Authors*

Table 2 The impact of supply chain digitalization on new product development

Variables	Innovation	Innovation	Innovation	Innovation
Online business	0.822***	0.743***	0.730***	0.863***
	(0.037)	(0.055)	(0.055)	(0.101)
Delivery	0.790***	0.792***	0.815***	0.648***
	(0.036)	(0.053)	(0.054)	(0.100)
Remote work	0.388***	0.349***	0.303***	0.096
	(0.034)	(0.049)	(0.050)	(0.090)
Capacity		0.007***	0.007***	0.005**
		(0.001)	(0.001)	(0.002)
Exporter			0.319***	0.244**
			(0.052)	(0.095)
Government support				0.784***
				(0.159)
Constant	−1.312***	−1.745***	−1.718***	−1.397*
	(0.137)	(0.250)	(0.251)	(0.714)
Country dummies	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes
Observations	24,065	11,631	11,508	3,429
Log-likelihood	−11,828	−5705	−5619	−1714
chi-square test	3896	1612	1645	453.6

Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ *Source* Authors

A high CU denotes the effective use of production resources, which raises output levels. On the other hand, a low CU indicates that manufacturing capacity is being underutilized, which lowers output levels (Lin & Xie, 2023). Moreover, a high CU indicates the availability of financial resources (Zhang, 2022). Our results indicate that capacity utilization is an important enabler of innovation and firms are more likely to introduce innovations with high-capacity usage.

One of the primary ways that the spread of digital technology influences innovation is through globalization, which lowers technology transfer barriers

and increases productivity and innovation (Skare & Soriano, 2021). Export-oriented firms often face competitive pressures that drive them to invest in research and development to stay ahead in global markets (Lee et al, 2021). We also find that exporters are more likely to innovate. This result confirms earlier studies which argue that exporters can take advantage of technology transfers and are more productive (Cassiman et al., 2010; Saridakis et al., 2019). This focus on innovation can lead to the development of disruptive technologies that give them a competitive edge.

Firms that receive government support for digital technologies are also more likely to innovate. Past research has demonstrated the importance of government's role in supporting especially small and medium-sized enterprises since they face many drawbacks, such as limited access to finance and raw materials and lack of technological capabilities and managerial skills (Chen et al., 2021). Moreover, during the pandemic, firms had to deal with sudden changes and declines in consumer demand, instability in supply chains, a lack of available human resources, and disruptions in their internal operations. Government support for the absorption of shocks and maintaining economic and social orders, such as financial (such as tax reduction, subsidies, loans), operational (government contracts, licenses), and intellectual (such as training and technical assistance programmes), may fuel supply chain resilience (Bodrožić & Adler, 2022). Since digital capabilities have been very critical to remain competitive and survive through the pandemic, government support for digital transformation helped them to gain competitive advantages (Gao et al., 2022).

5 Conclusions and Policy Implications

Businesses and organizations are being urged by Industry 4.0 to incorporate new digital-based methods and adopt knowledge-based systems in order to achieve their desired innovation performance and better position themselves in the rapidly evolving digital environment. Because these new digital technologies provided an essential set of tools to improve business agility as an instrument of adaptation to unprecedented market challenges, their adoption increased dramatically by the unprecedented pressures resulting from the COVID-19 pandemic. This study aimed to analyse the impact of digital transformation capabilities on firm-level innovation performance, using data from World Bank Enterprise Surveys COVID modules. Furthermore, we tested whether the government's support for adopting digital technologies is an enabler of innovation performance.

We used three types of digital capabilities (online business activities, remote work, and delivery) and found positive effects on the probability of innovation, which is consistent with prior research. Moreover, results showed that firm characteristics such as capacity utilization and exporter status are also enablers of innovation.

We also found evidence of the crucial role of governments in driving the digital transformation of businesses and mitigating disruptions during the recovery phase from the pandemic (Huikkola et al., 2022; Mai et al., 2023; Y. Gao et al., 2022; Xiong et al., 2023).

Our results show that digital capabilities and digital transformation of supply chains are effective strategies to remain competitive during unprecedented economic and environmental disruptions. Thus, managers should focus on constructing digital infrastructures and establishing and developing digital technologies. Because most businesses would find it difficult to establish and develop digital capabilities independently, government policies, initiatives, and assistance may be a feasible solution for them.

There were a few limitations encountered during the study. First, due to data limitations, we could not include some firm characteristics such as age, firm size, networking capabilities and the strength of ICT infrastructure, which are important for innovation performance. Second, a high innovation performance can increase capacity utilization in some firms, and this would create a causality problem. Some methods to correct that would be using a proxy variable for capacity utilization or using lagged values of the explanatory variables. However, data limitations prevent us from doing these robustness checks. However, in this study, we also controlled for external government support for digital technologies, and included dummy variables for industries, which reduces the endogeneity caused by omitted variables. An avenue for future research could be providing robustness checks. Another avenue would consider to what extent results generalize to other countries as more surveys are completed for other nations in the world. We think more research in this area is necessary, given the significance of supply chain digitalization in the context of Industry 4.0 and the possibility that these effects will vary among businesses.

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Part III

Transformational Models and Processes in Supply Chains

11

Deep Learning Applications in Inventory Management

Laxmi Pandit Vishwakarma and Rajesh Kumar Singh

1 Introduction

Keeping enough inventory in the warehouse to keep the business moving but not keeping more lists which can lead to draining the limited cash reserves (Hamilton, 2018). This kind of contradiction keeps running in the mind of supply chain managers. Finding a solution to such a situation is very difficult and needs special attention (Guo et al., 2018). Managing inventory revolves around such contradictions. Even a stock out of a single item makes the entire production stand still for hours and sometimes even days. For example, One of Tesla's suppliers failed to deliver the USB cables. As a result, the USD$3 item (USB cable) made the entire Tesla production line standstill till the USB cables were provided for the car's computer systems (Hamilton, 2018).

Inventory management is a crucial part of supply chain management, contributing significantly to the overall cost of supply chain management (Deng & Liu, 2021). Managing inventory is challenging, especially for firms with more than one warehouse or distribution centre. Retailers usually face frequent struggles with inventory management. Common problems such as insufficient stock levels, order mix-ups, and information about the product placed on the wrong shelf disturb the entire inventory management. Identifying customer demands accurately for the products becomes a hot topic for

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inventory and supply chain management (Deng & Liu, 2021). Past studies tried to develop single-point inventory and bulk order strategies to calculate the upcoming demands and achieve reliable inventory management. The single-point stock follows a "centralized processing method" wherein all the orders are placed simultaneously and have a high chance of facing bottleneck issues. The dynamic customer requirements and high uncertainty make traditional methods no longer applicable and highlight the need to adopt new technology. Zhou and Zhou (2019) highlight that safety stock and multi-echelon policies help cope with the changing dynamic environment. To meet the changing requirements, managing inventory is the need of the hour (Praveen et al., 2020).

The rapid advancement of AI technology makes the supply chain intelligent, including inventory management. Intelligent inventory management can meet the changing customer requirements and fulfil customer demands. The subset of AI technology known as deep learning possesses the strength to understand complex inventory management issues and address them quickly. Usuga-Cadavid et al. (2022) highlighted that deep learning algorithms do not require humans to develop the specific programme. Instead, the deep learning algorithms train themselves and perform predictable and incremental actions based on the information provided in the model. Bengio et al. (2021) revealed that deep learning is one of the more recent advanced technologies. Chatterjee et al. (2022) analyse that deep understanding is reshaping the working of manufacturing firms by assisting in both the basic and the complex operations. When combined with other industry 4.0 technologies such as the internet of things, 3D printing, and blockchain technology, the deep learning algorithms help to meet a highly-dynamic object demand (Mashhadi & Monroy, 2020). Deep learning algorithms save more long-term data and extract the information from it. The more the data is stored in the memory, the better the algorithms work and provide highly accurate results (Usuga-Cadavid et al., 2022; Vishwakarma & Singh, 2023; Zhao et al., 2020). The advances brought by the deep learning are anticipated to profit businesses and society in the near future. In addition, deep learning technology proves a strong fit for manufacturing firms, helping them to remain competitive in the local and global environment (Chatterjee et al., 2022). Arunrat et al. (2022) highlight that deep learning algorithms provide accurate and updated information, which helps in better decision-making.

Deng and Liu (2021) consider multiple inventory management factors and develops an optimization model for maximizing profits and minimizing costs. In the developed model, deep learning algorithms address the challenges in

managing inventory. The results of the models show that the average inventory demand prediction increases by 80%. Furthermore, the model could detect the anomaly across inventories more quickly, reducing the inventory cost by 25%. Boute et al. (2021) highlight that deep reinforcement learning is used in the early development of inventory control. The study discusses the deep reinforcement learning algorithms implemented in inventory control. Praveen et al. (2020) discuss demand forecasting as a crucial part of inventory management using machine learning technology. Finally, Stranieri and Stella (2022) explore deep learning and reinforcement learning algorithms to solve supply chain inventory management issues. These issues can be to find the optimal quantity of products to be produced and to ship across different warehouses over a given time. The study develops the model using deep learning algorithms and considers variables such as demand capacities, different topologies, structures, and supply chain costs. The study finds three scenarios with varying demand patterns, product types, and warehouses. The three scenarios are—one product type, one distribution warehouse, one product type, three distribution warehouses, two product types, and two distribution warehouses. The result of the study reveals that even with their different scenarios, deep reinforcement learning algorithms help to achieve optimal solutions.

Tang and Ge (2022) propose an inventory demand forecast model. The model uses material inventory information, sales demand forecast data and material attribute information. The model is a neural network model and works following the error backpropagation algorithm. The proposed model increases the forecast's lead time and helps the supplier reduce the surplus or shortage of supply. It further helped to get longer to prepare the goods for the suppliers. The combination of convolutional neural networks and extended short-term memory models has less prediction error than the convolutional neural network models. But the combination of these models shows optimized forecast results. Sahoo et al. (2023) conducted a bibliometric and systematic literature review to explore deep learning applications in manufacturing operations. The study identifies six themes: forecasting, manufacturing processes, environmental management, intelligent fault diagnosis, intelligent scheduling, and sustainable manufacturing. The study further provides the future research direction for deep learning applications in the fields of Industry 4.0, such as Supply Chain 4.0, Logistics 4.0, Maintenance 4.0, Sustainability 4.0, Manufacturing 4.0, and Quality 4.0.

Wichmann et al. (2020) investigated the buyer–supplier relationship using a deep learning algorithm from natural language text. The study further reveals that deep learning and natural language processing provides multiple

supply chain mapping solutions. These solutions provide additional supplier information, verify existing supply chain maps, and generate new maps. Zhou and Zhou (2019) propose an optimization model using the multi-echelon inventory policy by combining local search metaheuristics and deep learning algorithms. Sahoo et al. (2023) highlighted that despite the increasing academic interest in deep learning technologies. Still, significantly less work is documented in deep learning research. Past literatures also reveal that deep learning technology is a recent development, and its applications for inventory management are still unexplored (Boute et al., 2021; Chatterjee et al., 2022; Deng & Liu, 2021; Sahoo et al., 2023; Tang & Ge, 2022; Usuga-Cadavid et al., 2022; Wichmann et al., 2020). Therefore, we conducted the study to address the applications of deep learning for inventory management.

The sequence of the book chapter is organized as follows. First, Sect. 2 of the book chapter highlights the theoretical background. Section 2.1 discusses deep learning technology and Sect. 2.2 explains deep learning technology and inventory management. Section 3 illustrates the research methodology and is divided into two sections. Next, Sect. 3.1 provides the descriptive analysis, and Sect. 3.2 examines deep learning applications for inventory management. Finally, Sector 4 explains the study's conclusion, limitation, and future scope.

2 Theoretical Background

2.1 Deep Learning

With the emergence of AI in the 1950s, AI technology has drawn the attention of several practitioners and academicians over the years. AI technology is broadly defined as a technology which can "sense, reason, act and adapt" (Jansen Schoonhoven et al., 2018) and can perform human tasks (Ceron, 2019; Marr, 2018; Srivastav, 2020; Usuga-Cadavid et al., 2022; Zhao et al., 2020). AI technology is an umbrella of technology consisting of machine learning, deep learning, artificial neural networks, natural language processing, robotics, visual recognition, speech recognition, etc. (Bengio et al., 2021; Haldorai et al., 2021; Jansen Schoonhoven et al., 2018; Marr, 2018; Vishwakarma & Singh, 2022). The AI algorithms can manage large amounts of data, identify trends, and provide insights from the data, which is extremely hard for a human being to remember (Ceron, 2019). And that is one of the reasons AI technologies are widely adopted by different firms across industries worldwide. Now, when we narrow down the algorithm's capabilities of AI technology, which can "think" on its own

and make decisions or can answer a question, we are discussing the subset of AI technology known as "machine learning" technology (Ceron, 2019; Jansen Schoonhoven et al., 2018). The first use case of machine learning was observed around 1985. Machine learning components include algorithms, datasets, and features. There are four types of machine learning: supervised, unsupervised, semi-supervised, and reinforcement (Bengio et al., 2021; Srivastav, 2020). Machine learning algorithms are good at predictions. The prediction can be anything, for example, predicting "what will be your next order online", "if the loan request is defaulted or accepted", "the multiple products that can be bundled", "the fraudulent behaviour", "the stock market prices", "the trend, can be seasonal or supply–demand related", "a customer will churn a particular firm or not", etc. Machine learning algorithms are made "to fit a model to the data" (Jansen Schoonhoven et al., 2018). Examples of machine learning algorithms include K-mean clustering, decision tree, regression, neural networks, random forest, reinforcement learning, Bayesian networks and deep learning.

As we go further, the algorithm capabilities of AI technology, which can "learn" on its own, are known as "deep learning" technology (Bengio et al., 2021; Ceron, 2019; Raeesi Vanani & Majidian, 2021). "Deep learning" is often called "deep neural learning" or "deep neural network" (Srivastav, 2020). The deep learning algorithms use multi-level hierarchical neural networks. As we drill down, the amount of data needed give rise to the concept of deep learning—the more data, the better the accuracy of the models (Tang & Ge, 2022). The first use case of deep learning was observed after 2010 (Ceron, 2019). The main difference between machine learning and deep learning is that—machine learning comprises an algorithm, whereas deep learning comprises a family of algorithms. Raeesi Vanani and Majidian (2021) provide the main features of deep learning as "big-data training", "unsupervised feature learning", and "strong generalization capability". Deep learning networks can learn from unlabelled and unstructured data without supervision (Marr, 2018; Srivastav, 2020). The deep learning networks are so "deep" that the hardware required for this technology is relatively high. To run the algorithms, specific hardware's needed, such as graphic processing units in addition to a cluster of computer nodes (Bengio et al., 2021; Jansen Schoonhoven et al., 2018). The deep learning algorithms are based on neural networks and therefore require a lot of processing power. The larger the datasets, the better the performance of the deep learning algorithms will be. Deep learning algorithms, when exposed to different situations or patterns of data, the algorithms learn on their own and develop patterns which are never explicitly programmed (Ceron, 2019).

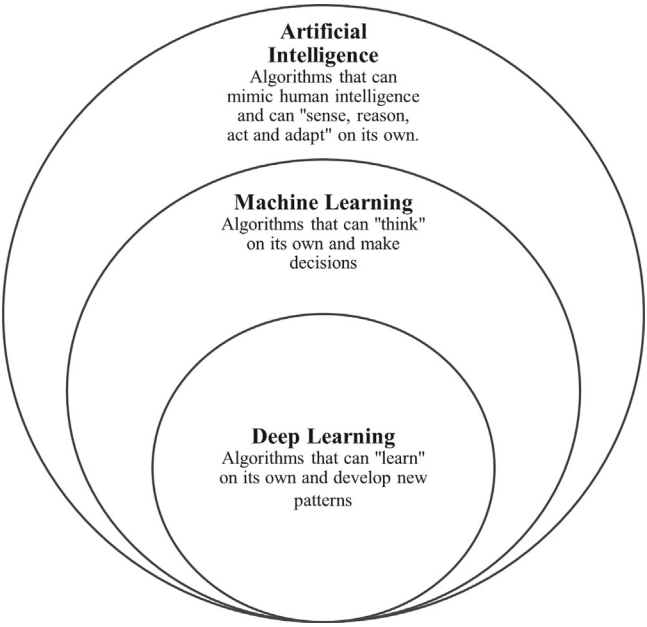

Fig. 1 Concept of deep learning (*Source* Srivastav [2020]; Ceron, [2019] and Jansen Schoonhoven et al. [2018])

In short, AI technology is the superset of deep learning and machine learning. Also, in other terms, machine learning is the subset of AI technology, whereas deep learning is the subset of machine learning (as shown in Fig. 1) (Bengio et al., 2021; Marr, 2018; Usuga-Cadavid et al., 2022; Zhao et al., 2020).

2.2 Deep Learning and Inventory Management

Inventory management plays a critical role in all manufacturing firms. Inventory management involves acquiring, storing, and selling the inventories. These inventories are broadly classified under four different categories known as raw materials, work in progress, finished goods, maintenance, repair, and operations (usually the spare parts). Managing these inventories is a complex task for any organization, especially the one that deals with a large number of product segments, having more than one warehouse and selling hundreds and thousands of products every month. Balancing the demand and supply, ensuring the right amount of inventory is available at the right time and the right location is important for the firm to stay ahead in the business. The traditional warehouse management system can mix up many

of these things and therefore lacks the potential to manage the inventory. Efficient inventory management is also important because excess inventory can lead to high capital, increasing waste products and a rise in storage expenses. While insufficient inventory can lead to production delays, stock-out situations and missing sales opportunities. Hence, the role of AI and deep learning comes into the picture. The integration of deep learning and inventory management can mitigate the above-discussed challenges. Deep learning can help streamline inventory management. The deep learning algorithms help in maintaining optimal inventory levels, reducing uninterrupted production levels, adjusting production volumes, reducing lead time, and fulfilling customer satisfaction (Lim et al., 2021; Mashhadi & Monroy, 2020; Raeesi Vanani & Majidian, 2021; Sahoo et al., 2023; Vishwakarma et al., 2023; Vishwakarma & Singh, 2022). In summary, the application of deep learning can revolutionize inventory management and can help future supply chain managers stay ahead in the highly volatile, dynamic, and competitive environment.

3 Research Methodology

Our study follows the guidelines of Donthu et al. (2021) and follows the research methodology adopted by Cruz-Cárdenas et al. (2021) for conducting the bibliometric analysis and the literature review. Our study starts with developing the research question. After that, we selected Scopus as the database for our study. Then, the keywords used for our research is combined using the "OR" and "AND" string in the title-abstract-keyword section of the search query. Here, we searched using the keywords "deep learning*", "deep reinforcement learning", "deep neural network", "inventory*", and "supply chain*". Initially, we found a total of 976 documents till 29th November 2022.

Further while considering the inclusion and exclusion criteria, we selected the subject areas as the decision sciences, business management, and social sciences. To have a broad understanding of the topic, we considered the articles, book chapters, and conference papers in review and published. We further limited our study to the "English" language. After considering the inclusion and exclusion criteria, we found 233 documents. We additionally conducted the bibliometric analysis and literature review from these documents. The flow chart of the research methodology is shown in Fig. 2.

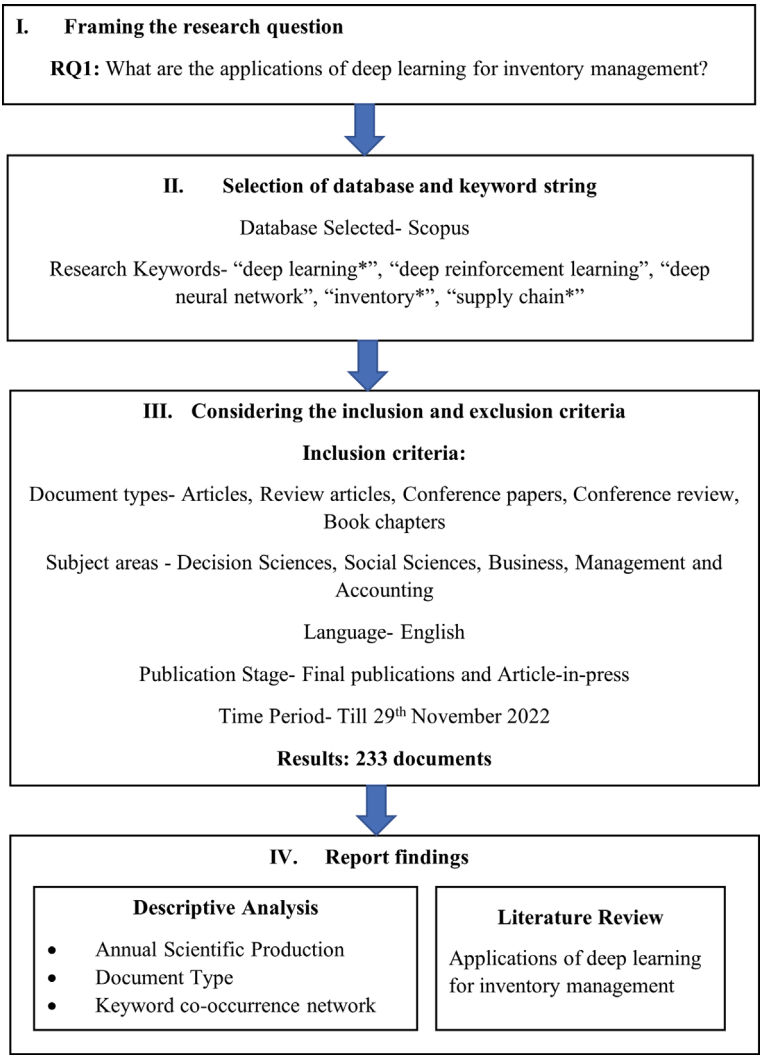

Fig. 2 Flow chart of research methodology (*Source* Donthu et al. [2021] and Cruz-Cárdenas et al. [2021])

4 Research Findings

4.1 Descriptive Analysis

This section provides a descriptive analysis of the study. The study highlights that there were very few publications on deep learning. However, continuous growth is observed after 2013. The majority of the publications were

taken in the recent years after 2019. For example, in 2018, 2019, 2020, and 2021, there were 4, 21, 25, and 54 documents published. From 2022 till 29th November 2022, 88 papers have been published on this topic. The year-wise-publication is shown in Fig. 3.

A large proportion of documents selected for the study are articles 59% and conference papers 27%. As the study also considers the review paper, the review articles constitute 2%, and the conference review contributes 9%. The other 3% contributes to the book chapters considered for the study. Figure 4 shows the document type considered for the study.

The keywords highlight the primary subject matter of the research work. The keyword occurrence is drawn below in Table 1. To illustrate the co-occurrence network of keywords, the minimum number of occurrences of

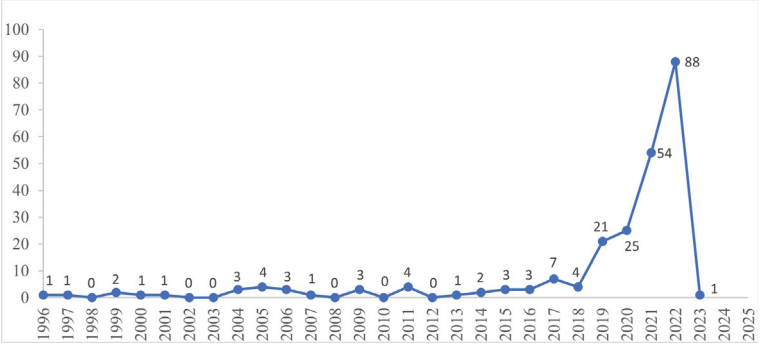

Fig. 3 Year-wise publications (n = 233 till 29th November 2022) (*Source* Authors)

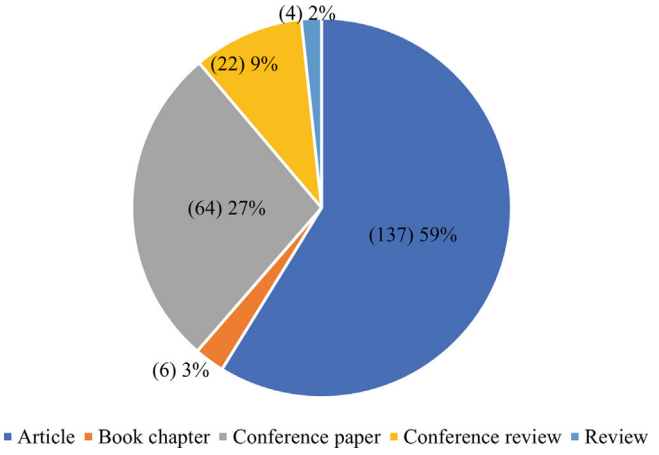

Fig. 4 Document Type (n = 233) (*Source* Authors)

a keyword selected is 5. Out of the 1843 keywords, 79 meet the threshold limit. The keywords with the greatest to the lowest total strength are chosen to draw the co-occurrence network. Figure 5 highlights the overall keyword co-occurrence network visualization. Figure 6 highlights the zoom-in keyword co-occurrence network visualization. Figure 7 highlights the keyword co-occurrence network overlay visualization showing the importance and relevance of deep learning technology over recent years (mainly after 2018).

The number of occurrences of keywords defines the size of the node (Van Eck & Waltman, 2020). The keywords having a high frequency are classified as clusters. The line represents the connection between the keywords. The colour classifies the keywords into different clusters. The keywords which are closer to each other show a stronger relationship compared to the keywords that are farther away (Van Eck & Waltman, 2020). As we look into Fig. 5, we can draw four major clusters discussing deep learning and inventory management. In the red cluster, the most frequent keywords are machine learning, neural networks, decision trees, remote sensing, forestry, object detection, convolution, and computer vision. In this cluster, the keywords are majorly related to the technologies coming under the umbrella of AI (such as decision trees, neural networks, and computer vision). Having these keywords in the output result shows the importance and relevance of these technologies in managing inventory. The strong link between machine learning, neural networks, and deep learning shows the rising research interest of these technologies for inventory management.

In the blue cluster, the most frequent keywords are deep learning, supply chain management, reinforcement learning, and benchmarking. The strong link between deep learning and supply chain management shows the relevance of this study. Reinforcement learning is the science of decision-making. The application of deep learning for decision-making is very important to manage inventory. Timely decisions and right decisions will help in optimizing the inventory which will further help the organization to save cost and stay ahead of the competition. In the green cluster, the most frequent keywords are the learning approach and humans. The strong link between them is related to the importance of human-AI/machine collaboration for managing inventory. The decisions cannot completely rely on the machines and therefore the involvement of humans is a must while making the decisions. A proper human-AI/machine collaboration can be considered a better approach while making decisions. In the yellow cluster, the most frequent keywords are time series, forecasting, backpropagation, demand forecasting,

Table 1 Keywords occurrences and total link strength

Keywords	Occurrences	Total link strength
deep learning	119	501
learning systems	33	202
supply chains	36	197
Forecasting	32	167
machine learning	30	164
deep neural networks	25	119
Human	16	117
inventory control	19	108
neural networks	18	108
Article	15	102
Learning	17	99
supply chain management	20	96
reinforcement learning	18	95
learning algorithms	15	91
Machine learning	14	89
Humans	11	86
big data	15	75
medical education	11	74
Questionnaire	9	65
reinforcement learnings	11	65
inventory management	11	62
Curriculum	8	60
remote sensing	12	60
deep reinforcement learning	16	59
Education	8	59
decision-making	9	58
Sales	11	58
time series	10	58
convolutional neural networks	9	57
long short-term memory	12	57
Teaching	9	57
medical student	8	55
education, medical, undergraduate	6	50
predictive analytics	8	50
object detection	10	49
neural networks	6	48
convolutional	6	46
computer vision	7	44
decision trees	9	44
information management	7	42
Forestry	7	41
student, medical	5	41

(continued)

Table 1 (continued)

Keywords	Occurrences	Total link strength
artificial intelligence	13	40
Female	5	40
learning techniques	5	40
Male	5	40
benchmarking	6	39
geographical information system	5	39
satellite imagery	5	39
artificial neural network	7	38
performance	5	38
convolutional neural network	6	37
demand forecasting	8	37
Skill	5	36
data mining	6	35
industry 4.0	6	34
learning approach	11	33
object recognition	5	32
optimization	6	32
study skills	5	32
recurrent neural networks	6	31
image segmentation	5	30
backpropagation	5	29
statistical tests	5	29
GIS	5	28
supply chain	6	28
Algorithm	6	27
CNN	5	27
image processing	5	27
classification (of information)	5	24
engineering education	8	24
internet of things	5	24
learning approaches	7	24
learning methods	5	23
life cycle	5	23
transfer learning	6	22
deep neural network	5	19
approaches to learning	5	10
higher education	8	9

Source Bibliometric-based analysis using VOSviewer software

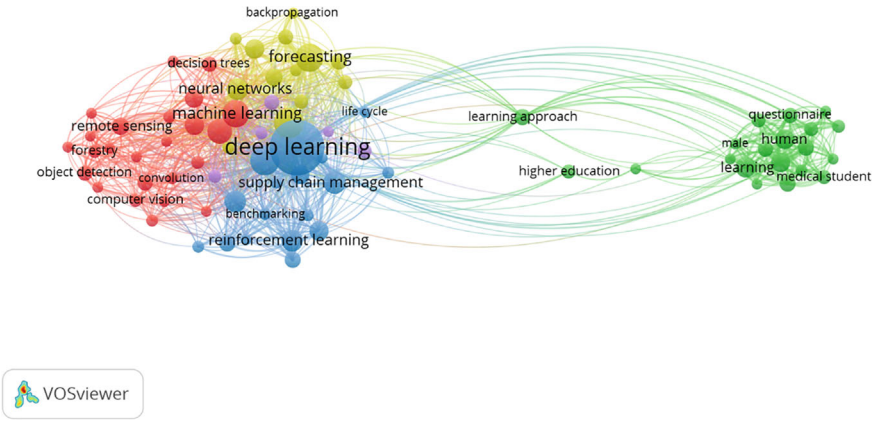

Fig. 5 Overall keyword co-occurrence network visualization (*Source* Bibliometric-based analysis using VOSviewer software)

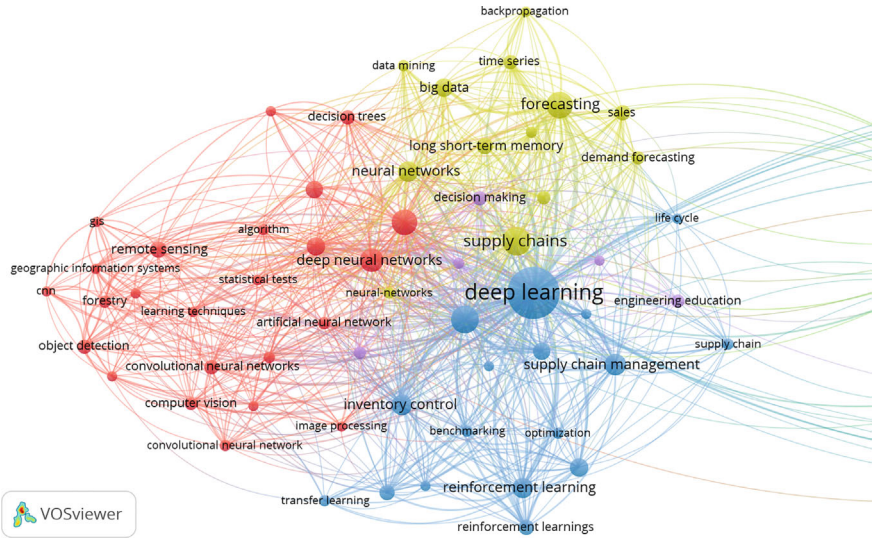

Fig. 6 Zoom in keyword co-occurrence network overlay visualization (*Source* Bibliometric-based analysis using VOSviewer software)

data mining, sales, and big data. This cluster shows the importance of forecasting for managing inventory. Different forecasting tools (such as time series, and backpropagation) are keywords showing high frequency in the cluster. These methods are used for demand forecasting. Demand forecasting is important to reach the optimal inventory levels.

Figure 7 shows the network visualization over the years. The time for the wide spread of these keywords is from 2010 to 2020. The blue cluster, having

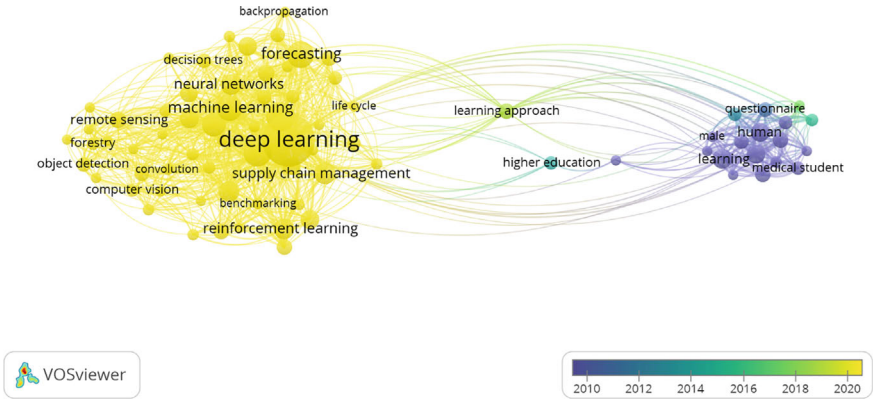

Fig. 7 Keyword co-occurrence network overlay visualization (*Source* Bibliometric-based analysis using VOSviewer software)

a human as a high-frequency keyword, shows the importance of humans since 2010 for managing inventory. The yellow colour shows the emergence of keywords mostly after 2018. This keyword includes deep learning, computer vision, supply chain management, decision tree, reinforcement learning, etc. The results shown in Fig. 8 are also in line with Fig. 3 showing the year-wise publication. Both the figures show the emergence of this topic after 2018. The number of publications has increased after 2018 and the number of frequencies of these keywords has also increased after 2018. With these results, we can also conclude the timely importance and relevance of this topic.

Fig. 8 Applications of deep learning for inventory management (*Source* Authors)

4.2 Applications of Deep Learning for Inventory Management

There are a wide variety of deep learning applications for managing inventory. This section will discuss the applications identified in the current bibliometric and literature review. Figure 8 shows the discussed application of deep learning in inventory management in this chapter.

i. *Automating the inventory inspections*

The deep learning algorithms automate inventory inspection (quality checks) at different stages of supply chain management. The algorithms automate placing inventory orders when the inventory reaches a certain level. The deep learning algorithms make the automated inspection process smoother and more accurate than manual inspections. The deep learning algorithms further improve the supply chain efficiency by automating inventory management (Raeesi Vanani & Majidian, 2021; Sahoo et al., 2023; Vishwakarma & Singh, 2022).

ii. *Reducing the inventory costs*

Mashhadi and Monroy (2020) explore that deep learning algorithms maximize production capabilities and inventory usage. The deep learning algorithms prevent unfair activities, consider raw materials constraints, make production control decisions, and improve production utility, thereby saving inventory costs. The deep learning algorithms, combined with other technologies, find the optimal resource allocation and payments for the buyers.

iii. *Improving decision performances*

Zhou and Zhou (2019) highlighted that the optimization model using the multi-echelon inventory policy by combining local search metaheuristics and deep learning algorithms improves the decision performance. The improved decision performances are in terms of adaptability, profitability, and time efficiency. The reasons behind the decision performances are accurate forecasting, automating inventory management processes, and minimizing inventory risks (Chatterjee et al., 2022; Lim et al., 2021).

iv. *Reducing the bullwhip effect*

Vishwakarma and Singh (2022) highlighted that accurate forecasting at the retailer's end helps manage inventory at the manufacturing and the supplier's end. In addition, deep learning algorithms help capture all the trends and patterns of inventory across different seasons and festivals, which helps the firms manage inventory across the supply chain. Thereby helping to reduce the bullwhip effect (Sahoo et al., 2023).

v. *Minimizing the risk of ineffective inventory management*

The deep learning algorithms reduce the potential risks of ineffective inventory management and help the firm gain a competitive advantage (Chatterjee et al., 2022; Lim et al., 2021; Shi et al., 2020). In addition, as the forecasting is done more accurately using deep learning algorithms, the uncertainties and the risks are predetermined using models. Such steps help to develop or change strategies based on the current situations, thereby minimizing the chances of ineffective inventory management (Boute et al., 2021).

vi. *Extracting inventory features*

Praveen et al. (2020) highlighted that artificial intelligence subset such as deep learning algorithms helps in extracting the inventory features. Extracting the main features provides direction to the managers. For example, Training data contains the product ID and the product name. However, only product ID is required to train the primary model. Therefore, only the main features are extracted and used to prepare the main model. Such extractions improve the accuracy of the models and speed up training the datasets using deep learning algorithms.

vii. *Optimizing inventory management*

Mashhadi and Monroy (2020) highlight that deep learning algorithms help optimize inventory management. The deep learning algorithms customize the constraints according to the business needs. These algorithms are mainly helpful for firms having multiple distribution locations. The deep learning models adjust themselves while considering the independent variables that may affect the stock, product delivery, etc. The deep learning algorithms help to understand how and where products are stored, how employees manage inventory, develop the picking and packaging strategy,

improve the shipping process, and optimize the entire inventory management. Optimum inventory management further helps the organization keep the financial figures in control (Praveen et al., 2020).

viii. *Reducing inaccurate forecasting*

Boute et al. (2021) highlighted that deep reinforcement learning algorithms are used to forecast uncertainty and optimize inventory policies. Many industries heavily rely on forecasting their inventory to understand future requirements. These requirements vary based on seasonality, customer reviews, and many other factors. Over-forecasting or under-forecasting simultaneously leads to wastage of products and loss of opportunity (Vishwakarma & Singh, 2022). Forecasting errors lead to an increase in growing costs for the business. Using deep learning algorithms helps in accurate forecasting by considering the essential factors. These factors can be weather conditions, traffic information, information about the latest news events, historical sales data, competitors' activity, etc. (Boute et al., 2021). These factors are also considered when any new neural network is developed using deep reinforcement learning algorithms. As a result, minor forecasting errors are avoided while finalizing the optimization results. By running multiple optimization models, deep reinforcement learning algorithms are expected to learn by exploring the unobserved actions and states (Boute et al., 2021).

Sahoo et al. (2023) examined that deep learning models are highly used for forecasting. Praveen et al. (2020) highlighted that demand forecasting in the retail section is challenging for firms. Hence, managing inventory plays a vital role in retail supply chain management. The deep learning algorithms help reduce out-of-stock situations and inaccurate inventory forecasting. Tang and Ge (2022) developed a demand forecast model based on the convolution neural network. The model improves demand forecasting accuracy and the firm's work efficiency.

ix. *Developing inventory policies for the dynamic environment*

Boute et al. (2021) highlighted that deep learning algorithms consider all the variables necessary for a dynamic environment. These variables can be market uncertainty, competitor's strategy, profit margins, etc. while developing the algorithms. Therefore, the neural networks help identify the dynamic environment's trends and patterns. For example, they are determining if any recession or market uncertainty is likely to hit in the coming months. Therefore, it helps the supply chain managers and the policymakers

to understand the dynamic scenarios and, depending on them, develop inventory policies. As a result, the deep reinforcement learning algorithms deliver good policies for the inventory models, for which it is challenging to derive well-performing heuristics.

x. *Predicting and managing stock levels*

Understocking and overstocking are severe problems for many firms. The understocking leads to the loss of potential customers and goodwill of the firm. On the other hand, the overstocking of the inventory may lead to loss of the inventory (especially in the case of perishable items like FMCG products, outdated electronic items, and expired food items), engaging the cash flow unnecessarily and finally dissatisfying the customers. Deep learning algorithms deeply understand stock patterns using extensive historical data and predictive analytics. Deep learning and machine learning algorithms analyse past data and predict future sales. The algorithms consider all the variables such as the supplier's location, special occasions such as wedding seasons, festivals such as Diwali or Christmas, time to reach the final destination from the warehouses or the supplier to the manufacturers, the default rate of the machines used for production, etc. (Deng & Liu, 2021; Guo et al., 2018; Hamilton, 2018).

xi. *Providing real-time inventory information*

Mashhadi and Monroy (2020) highlighted that deep learning algorithms provide real-time inventory monitoring. The deep learning algorithms understand how the warehouse, or the distribution centres are mapped for the firms. The algorithms understand the places where the specific inventory is placed and based on it, map the new inventory, helping to get real-time inventory information. In this manner, the deep learning algorithms ease the supply chain manager's work and help them to stay updated.

xii. *Providing vision for autonomous cars, robots, and drones*

Autonomous cars, robots, and drones work on deep learning algorithms. The algorithms understand the pathways and help autonomous vehicles, robots, and drones act like humans and move inventory from one place to another in the warehouse or distribution centres. The deep learning algorithms provide a vision to understand the stop boards and the direction boards to automounts cars, robots, and drones (Marr, 2018). The more these

devices receive the data, the better they can work like human beings. They understand when to pass the inventory to the next stage and are used to handling large amounts of inventory. Such activities improve the efficiency of the supply chain even with a small workforce.

5 Conclusion, Limitation, and Future Scope

The increase in data generation gives rise to the adoption of deep learning algorithms across manufacturing firms. Organizations are using deep learning algorithms in their supply chains to cope with dynamic working environments and reduce uncertainty. The applications of deep learning technology can be observed across the supply chain. However, our study tries to explore the applications of deep learning in inventory management. This chapter discusses twelve deep learning applications for managing inventory for an effective supply chain. These applications automate inventory inspections, reducing the inventory costs, improving decision performances, reducing the bullwhip effect, minimizing the risk of ineffective inventory management, extracting inventory features, optimizing inventory management, reducing inaccurate forecasting, developing inventory policies for the dynamic environment, predicting and managing stock levels, providing real-time inventory information, and providing vision for autonomous cars, robots, and drones. Despite these contributions, our study has some limitations. One of the limitations lies with the research methodology, which is bibliometric analysis, this method sometimes does not capture the motivations and objectives that guide the authors in citing prior articles (Soranzo et al., 2016; Vogel & Güttel, 2013). To overcome it, future researchers can build a research framework or a conceptual model and test it using any qualitative or quantitative method. Further, we hope that our study will help future researchers understand the deep learning concept and its application for managing inventory across the supply chain. This study will help the policymakers and strategy developers to draw strategies and policies for the firms adopting deep learning technology. Future researchers can conduct similar studies using databases such as the Web of Science or EBSCO host.

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12

Smart Farming, Smart Distribution, and Smart Retailing in Agri-Food Supply Chains

Behzad Behdani

1 Introduction

In recent decades, the food supply chain has grown rapidly, with consumers now expecting agri-food products, fresh on their plates, year-round. This has extended the supply chain geographically and across many more parties, making the food chains among the most complex yet important logistics systems. At the same time, the demand for agri-food products has increased significantly (and expected to increase further in the coming decades). A reference estimate is provided by FAO in 2009, estimating that world food production needs to increase by 70% to feed the world population in 2050 (FAO, 2009). Some other studies have presented an even higher figure; for instance, Tilman et al. (2011) present an estimated increase of up to 110% in food demand between 2005 and 2050. This fact together with increased labour shortage in the agri-food sector (Christiaensen et al., 2021) and decreased productivity of resources (Panagos et al., 2021) create a dilemma for policy makers and the main parties in the sector. At the same time, around one-third of edible foods are either lost (in the upstream farming stages) or wasted in retail shops, food services and households (Morone et al., 2019). When agri-food products are lost, all the resources that were used to produce this food—including water, land, energy, labour and capital—go to waste as

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well. To tackle these challenges, several innovation solutions and technologies have been presented in the last two decades.

This chapter provides an overview of technological solutions and opportunities for in agri-food supply chains. Next section discusses the distinctive characteristics of food chains and why digitalization is needed in this sector. After that, the technological and digital solutions for different parts of an agri-food supply chain will be discussed.

2 The Need for Digitalization in Agri-Food Supply Chains

The characteristics of food chains and the trends in the sector have been a driver for using smart digital technologies in managing the processes in agri-food supply chains. Some distinctive characteristics of food chains and key trends in the sectors are discussed in this section.

2.1 Product and Logistics Diversity

A main characteristic of the food sector is the diversity of products and logistics channels. The agri-food sector produces a wide range of products, including non-perishable (processed) food items and perishable temperature-sensitive products. Additionally, based on the temperature requirement for the cargo, we can differentiate between ambient fresh, frozen, and deep-frozen cargo (Behdani et al., 2019, see Fig. 1). Some products may also need specific atmospheric conditions (like the specific level of humidity or CO_2 level). Providing such a tailored atmosphere for perishable products can be costly and may also add to the carbon footprint of agri-food chains (Parashar et al., 2020). Accordingly, different products have different logistics requirements along the chain. Because of this diversity, there is also a need for customized solutions and processes for each agri-food supply chain.

Furthermore, there are many sources of variation in the production and distribution of agri-food products. For instance, there is a high level of intra-field variability in soils, pests, and crop yields at the farming stage (Hamada et al., 2015). Even in a controlled agri-production environment—like a greenhouse—the factors that influence the production yield—like temperature or air circulation—can be very different in different parts of the greenhouse. This can clearly influence the rate of cultivation and maturity of products at the harvesting time. Similar variation can be observed in the

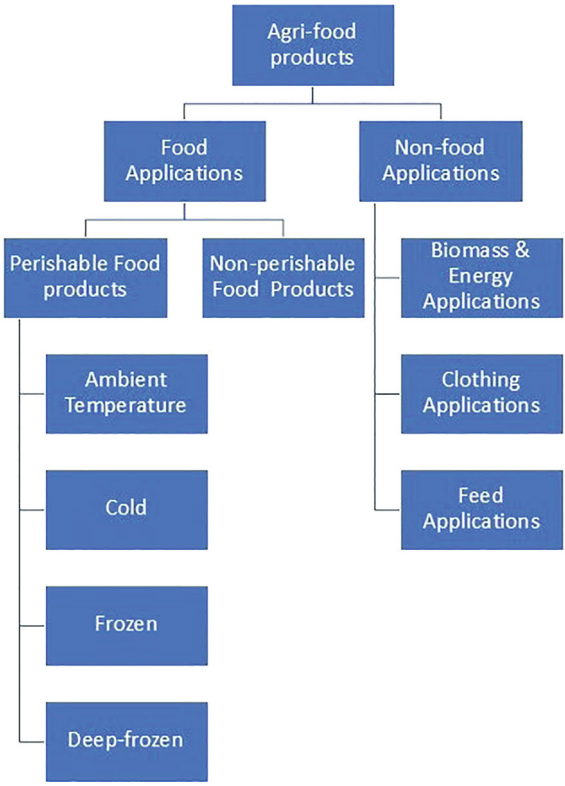

Fig. 1 Product diversity in agri-food sector (*Source* Author)

environment surrounding the products during the storage and transportation of agri-food products (Fan et al., 2021).

At the same time, the distribution and retailing channels for food products are also diverse. The farmer selling directly to consumers (in a local/traditional food chain) is the simplest structure (Fig. 2). In most cases, however, several middle stages can be involved in delivering food from farm to fork. For instance, a wholesaler buys products from several different farmers or manufacturers and delivers those products to retailers or consumers. Clearly, the characteristics of products (like the quality requirements, quality consistency, and price) are different for these different channels as well. Including intermediaries that produce, manufacture, distribute, transport, and sell food products would increase the complexity of food supply chains. Information and Communication technologies can support managing this complexity and better govern the organizational relations in a global chain. Another point often overlooked is the opportunities that come with digitalisation of the food chains for transition towards e-commerce business models

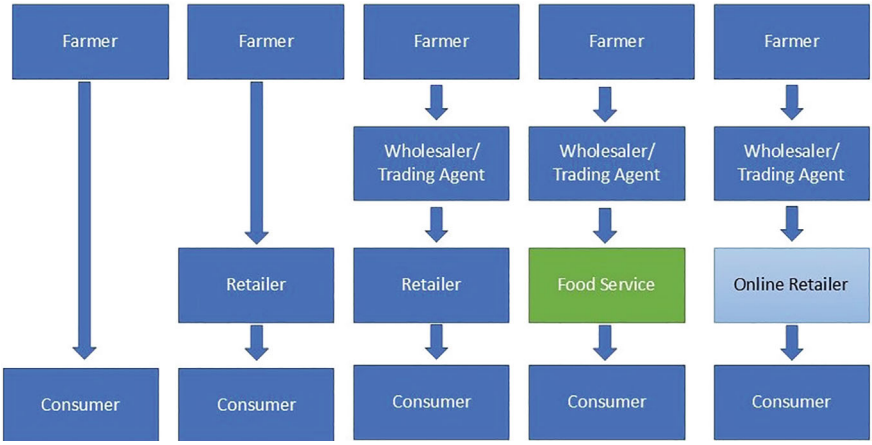

Fig. 2 Different distribution and marketing channels in agri-food sector (*Source* Author)

in the sector (Reardon et al., 2021). This trend has been particularly intensified after the recent COVID-19 pandemic. E-commerce provides substantial opportunities for food trading companies to cut out the middlemen and directly sell to their end consumers.

2.2 Food Quality and Safety

One of the key challenges in managing a food supply chain (and in particular for fresh produce) is keeping the quality of the product. Most agri-food products are living products, and their life processes remain active after harvest. An important life process that must be controlled to maintain the good quality of the fruit is respiration (Martin-Belloso & Fortuny, 2010). Respiration is a complicated induction of chemical reactions that convert starch into sugars and those sugars into energy. The respiration rate also differs between different kinds of fruit, so using the right package for the right fruit is essential for maintaining the quality of the fruit for a long period. Respiration can be influenced by temperature. The higher the temperature of the environment of the fruit, the higher the temperature of the fruit itself. This will result in a faster respiration rate. The faster the respiration rate, the earlier the fruit will spoil (Caleb et al., 2012). To control the respiration rate (and, accordingly the quality of fresh produce), the agri-food supply chains need several stages of cool storage and transportation. Also, there is a need for continuous monitoring of temperature and product environment along the chain (Badia-Melis et al., 2018).

A related issue is food safety. Food safety refers to routines in the preparation, handling, and storage of food meant to prevent foodborne illness and injury (King et al., 2017). Ensuring safety calls for continuous tracking and tracing of the products along the entire food supply chain from farm to fork. This food traceability also helps to quickly track a product back to its source if there is a product recall or unfortunate contamination in the chain (Soon et al., 2020). The product's traceability is, however, a complex matter, as various factors such as microbial impurities, labelling and packaging errors, contamination by metals, or even potentially hazardous substances such as machine oil can happen in different stages of a global chain (King et al., 2017). Smart technologies can capture real-time data for monitoring and tracking food chains.

2.3 Food Regulation and Labelling

Food is a highly regulated sector with international norms and standards—like Codex Alimentarius, established by the Food and Agriculture Organization of the United Nations (FAO) and the World Health Organization (WHO)—and several regional or national regulations (Winickoff & Bushey, 2010). Food regulations are also constantly changing, and automated and transparent processes can better adapt to these changes. Manufacturers are also facing the challenge of encountering the growing demands of well-informed, socially conscious, and environmentally aware customers (Rana & Paul, 2017). They continuously ask for further traceability, safety requirements, and sustainability improvement in all food sectors.

A related factor is the compliance requirements in the agri-food sector. For example, in US, FDA has finalized some major rules to implement Food Safety Modernization Act FSMA, ensuring the safety of the food supply in the global supply chain (for both human and animal food) (Bovay & Sumner, 2018). These rules are defined to guide specific actions that must be taken at each point in a global logistics chain to prevent contamination. With constant contact and follow-up, suppliers will stay in compliance a. Blockchain and IoT sensor technologies can substantially support the implementation of these rules and guarantee compliance with those standards (Sun & Wang, 2019; Zhao et al., 2019).

Additionally, for many (processed) food products, there are mandatory guidelines to mention some detailed information (including the list of ingredients, net quantity, use-by date, nutritional declaration, and allergen information) on the food product labels (Hawley et al., 2013). In most cases, the essential information depends on the product type; for example,

fresh fruit and vegetables are not required to have an ingredient list or an expiry date, but traders and wholesalers must mention the country of origin. Similarly, the sourced country must be mentioned for fresh, pre-packaged, prepared, frozen, or deep-frozen meat (Grunert et al., 2010). Additionally, it is a key trend among food producers (and also requested by some consumers) to produce and deliver organic and authentic food products (Drexler et al., 2018). IT-based solutions can significantly support the reliable labelling process in a global agri-food chain.

2.4 Increased Globalization in Agri-Food Trade

During the last two decades, the world agri-food systems have become increasingly globalized (Fig. 3). Although globalization in food chains has resulted in more diversity and year-round availability of food items for end consumers, it has imposed several challenges for the sector. Like other industries, globalization has reduced the transparency in the agri-food chains (Fan et al., 2020). In many agri-sourcing countries, the farmers and small food producers don't barcode their products -because they do not afford the technology. Additionally, for temperature-sensitive agri-food products, the storage and handling of items must follow specific protocols and cool chain technologies that might be difficult to access and supervise in a global chain. At the same time, different countries may have various regulations and cultures in handling food commodities. These factors, together, make it difficult to track the goods end-to-end through the supply chain, increasing the risk for companies (Rogerson & Parry, 2020). Also, it has become more challenging to manage all safety and compliance needs effectively.

The production of agri-food products is quite labour-intensive. Global food chains are among the sectors that have been criticized by international organizations like ILO for poor working environments and labour rights violations. A large number of workers experience difficult working conditions, particularly in less-developed economies. Migrants, women, and under-aged workers are particularly affected by low wages, long working hours, and a lack of rest and annual leave (Gold et al., 2015; Palacios-Lopez et al., 2017). The situation has been aggravated during the COVID-19 pandemic (Waltenburg et al., 2021). Implementing more automation technologies along the chain—and specifically in the farming sector, as discussed in Sect. 3—can be a potential solution for these challenges. Additionally, IoT applications can provide better visibility for a global food chain and minimize the potential for human rights violations or poor-quality working conditions.

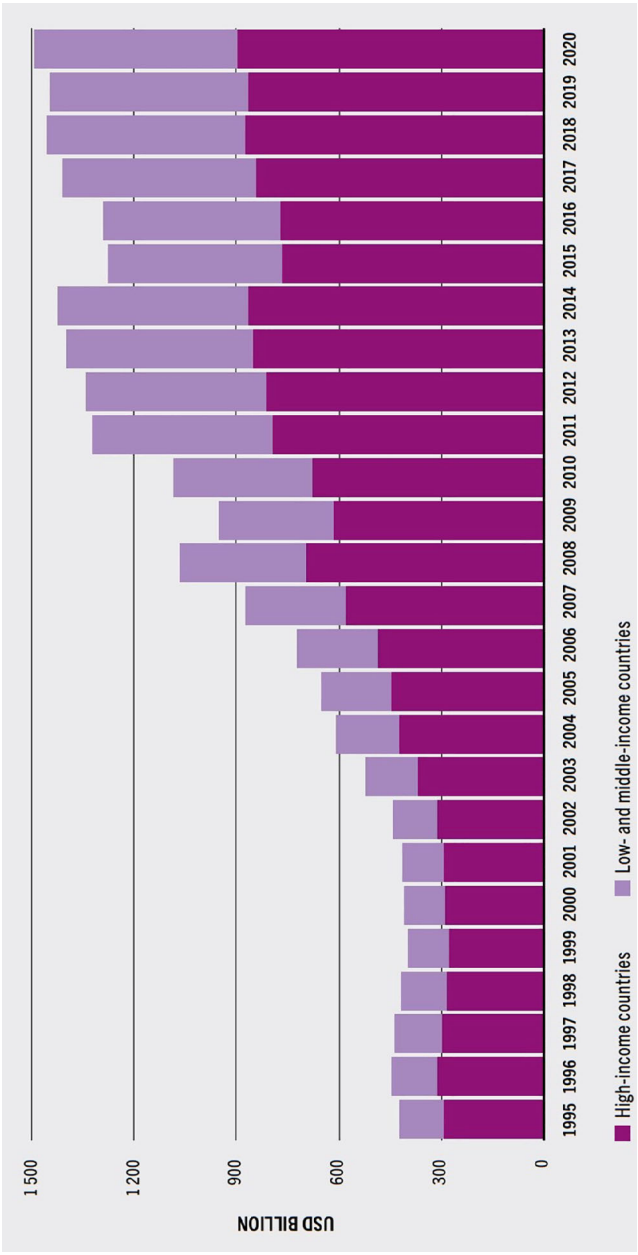

Fig. 3 The evolution of global food and agricultural trade 2020 (*Source* FAO [])

2.5 Climate Change and Resource Scarcity

The agriculture sector, together with forestry and other land uses, contributes nearly a quarter of all anthropogenic greenhouse gas emissions (Clune et al., 2017). At the same time, there has been increasing pressure from climate change, soil erosion, and biodiversity in the agri-food sector. Volatile weather and more extreme events—like floods and droughts—change growing seasons, limit the availability of water, allow weeds, pests, and fungi to thrive, and can reduce crop productivity. Soil erosion is also reducing the amount of land available for agriculture, and declining biodiversity affects the pollination of crops (Tu et al., 2019). Consequently, farmers are under pressure to conserve water and use fewer agricultural inputs. Information and Communication Technologies can provide enormous opportunities for climate-smart practices by preserving the resources, using the resources most efficiently, and developing customized solutions for different lands with different characteristics.

3 Smart Agri-Food Supply Chain

In the last two decades, many firms in the global food chains have started using smart and digital technologies to tackle the challenges and adapt to the trends discussed in Sect. 2. This has resulted in a fundamental transformation of traditional (labour-intensive) food chain processes to digital operations which is termed "Smart agri-food", "Digital Agri-food", or "Agri-Food 4.0" (Miranda et al., 2019). "Agri-Food 4.0" aims to achieve optimum value chain performance through digitized, resilient, innovative systems along with real-time data processing and decision support capabilities (Davis et al., 2021). Some key enabling technologies in this transition are Cloud Computing, Internet of Things, Big Data, Blockchain, Robotics and Artificial Intelligence (Lezoche et al., 2020).

This transition towards digital operations can potentially improve the productivity, enhance the traceability of food products along the chain, improve the quality (Lezoche et al., 2020), leads to better legal compliance (Spanaki et al., 2022) and reduce food waste and losses along the chain (Di Vaio et al., 2020). Additionally, digitalization of agri-food supply chains is a driving force as well as a support for the existing trend towards e-commerce and omnichannel in the food sector.

In this section, we will provide an overview of different digital technologies and practices for different parts of the chain (see Fig. 4).

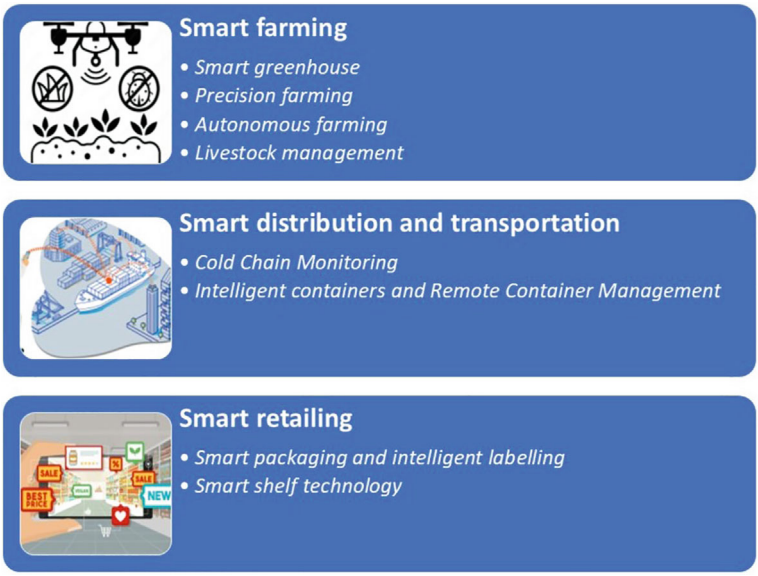

Fig. 4 Key technologies and applications in smart agri-food supply chains (*Source* Author)

3.1 Smart Farming

Growing water scarcity and increasing environmental consciousness among consumers have urged the need to modify agricultural management practices conserving natural resources such as soil, air, and water. Smart farming refers to managing farm operations using modern Information and Communication Technologies to increase the quantity and quality of products (both crops and livestock) while optimizing the required human labour and preserving limited available farming resources (Giua et al., 2022; Walter et al., 2017). Smart farming includes a wide range of technologies and concepts; some important ones are discussed in the following.

3.1.1 *Smart Greenhouse*

Greenhouse farming is referred to as the process of cultivating crops in a greenhouse ecosystem environment. The process is sometimes called controlled environment agriculture or CEA as well (Ramin Shamshiri et al., 2018). Greenhouse farming can increase crop production and income by creating optimal climate conditions for plant growth. Consequently, the crops

grow faster (with more plants per square metre of land used) while retaining as many nutrients as possible.

In a traditional greenhouse, the climate parameters—like humidity, soil moisture, and temperature—are recorded in a manual and—sometimes—inconsistent manner in a pre-defined schedule. The smart greenhouse enables farmers to cultivate crops with minimal human intervention. Climatic conditions are continuously monitored using sensors (including soil moisture sensors, light sensors, humidity sensors, and the temperature sensor), and any variation observed in them triggers an automated action (like regulating irrigation, lighting, or water sprinkle to adjust the temperature) (Kodali et al., 2016).

3.1.2 Precision Farming

Precision farming or Precision agriculture (PA)—also known as site-specific crop management—is an umbrella farming management concept based on (near real-time) observation, measurement, and responses to variability in crops and fields (Finger et al., 2019; Mizik, 2023).

Farmers are continually tasked to determine the correct fertilizer and proportions of ingredients to apply to their fields. This can be a challenge since the soils in every field are inherently variable (Zhang et al., 2010). Accordingly, there is no uniform fertilizer prescription that works for every field or every spot in the field. This is important for the efficiency of farming processes because fertilizer is an expensive agrochemical. Using geospatial technology, farmers can divide their fields into "grids", and programme their spreaders to change the fertilizer mix as they cross different areas of their fields (Pedersen & Lind, 2017). Therefore, precision farming allows decisions per square metre or even per plant/animal rather than for a field.

Precision farming is based on Variable-Rate Application (VRA) technology which means applying materials in such a manner that the application rate is varied based on precise locations (Alameen et al., 2019; Mizik, 2023). For this purpose, VRA is merging different data collection methods. The data can be collected from on-site sensors, like soil moisture and temperature sensors, called sensor-based VRA. It can also come from remote sensing technologies like Global Positioning Systems (GPS) and Geographic Information Systems (GIS) sources—called map-based VRA. Before farmers can take action, the collected data must be analysed to determine field and crop conditions. These analytics use artificial intelligence (AI) algorithms and machine learning to explore the enormous quantities of data generated by imagery and IoT sensors. A common application is Variable Rate Irrigation

systems in which irrigation infrastructure is enabled with automation capabilities and is informed with real-time data from remote sensing (either remote imagery, ground sensors, or weather sensors). This allows an irrigation system to automatically apply more or less water depending upon the moisture levels at specific field areas (Mendes et al., 2019).

3.1.3 Autonomous Farming

Agri-food supply chain—and in particular, the farming sector—is a highly labour-intensive sector. Also, several studies show that farming is not an attractive employment sector for younger generations (White, 2012). Autonomous farming aim at easing labour shortages by minimizing human element intervention in the farming processes.

The main technologies in this area include:

- Unmanned Ground Vehicles (UGV): or autonomous tractors operate using lasers that reflect from transponders located throughout the field (Moysiadis et al., 2021). They allow farmers to take control via an app on their phone or computer. The farmer can use the app to position a tractor, drive the length of a field, turn around, come back, and manoeuvre around obstacles. With sensors, autonomous tractors can typically collect data as they work. This gives farmers the information they need to manage their crops and potentially boost their yield.
- Unmanned Aerial Vehicles (UAV): or drones are also frequently used for crop health assessment, plant counting and yield prediction, plant height measurement, field water pond mapping, crop monitoring, and soil and field analysis (Inoue, 2020). Some recent drone technologies (known as Autonomous drone sprayers) also have the capability of controlled crop spraying (mostly for spraying agrochemicals like insecticides and pesticides). This will significantly increase safety, reduce hazardous components to the environment, and necessitate less manual labour (Ghobadpour, et al., 2022).
- Seed-planting robots: seed sowing requires a substantial amount of human effort and is a time-consuming process. Seed-sowing robots plant seeds in the desired position, taking away the human element usually required (Mahmud et al., 2020). Besides, there have been some recent studies to develop a robot for fruit picking (see, for example, Onishi et al. [2019]). While robotic tools for agriculture have made big progress in recent years, those AI-based tools are mostly used for simple operations like weeding or monitoring soil moisture. However, when it comes to performing physical

work like pruning trees or picking fruit, the existing technologies need further developments before they can be used in the field. For instance, robotic arms and vision systems need further visual perception extension to get clusters of fruit that are hidden behind branches or leaves (Zhou et al., 2022).

3.1.4 Livestock Management

Precision livestock management or Precision livestock farming (PLF) is a group of technologies for real-time monitoring of the health, production, and welfare of livestock (Garcia et al., 2020). It includes several areas like milk harvesting, breeding management, feeding management, and animal comfort management. The aim is to improve the quality of milk and dairy products and contribute to animal welfare.

Many farm owners—particularly, large farm owners—use wireless IoT applications to collect critical data regarding their livestock's location, well-being, and health (Bos et al., 2018). This information helps them identify potentially sick animals and separate them from the herd (subsequently preventing the spread of disease). It also enables better monitoring of the needs of individual animals and adjusting their nutrition accordingly. Additionally, farmers use wireless sensors to gather important information on pregnant animals (Morgan-Davies et al., 2018). The real-time data collected from these sensors will alert farmers in the event a pregnant animal gives birth allowing for improved care and observation.

3.2 Smart Distribution and Transportation

3.2.1 *Cold Chain Monitoring*

Cold chain monitoring refers to using Internet of Things (IoT) technology to continually monitor temperature-sensitive products along the chain.

Smarter tags and sensors accompanied by a cloud-based cold chain visibility platform that can provide a full overview of a food chain in real time (Badia-Melis et al., 2018). A wide range of sensors are used to monitor the temperature-sensitive logistics chains. These sensors help to track and record location, movement history, temperature, and humidity inside the trailer as well as door open/close (using lid sensors). Additionally, vibration and shock monitoring sensors are commonly used to check if the sensitive cargo is transported properly (Rodriguez-Saona et al., 2020). These sensors are also very

common in livestock transport to ensure the animals welfare (Van Engen & Coetzee, 2018).

Without proper monitoring, suboptimal conditions during transport and storage can damage the quality of these products. This can put people's health and safety at risk and potentially damage a company's brand, customer loyalty, and bottom line. Cold chain monitoring can also support the chain operations by:

- Improving fleet management and predictive maintenance based on distance travelled to reduce vehicle and trailer downtime and repair costs.
- Quickly detecting product temperature problems and minimize damage by troubleshooting or dispatching crews to repair malfunctioning equipment.
- Improving compliance and predictive quality, receive alerts when temperatures breach optimal settings and take corrective action and re-routing products in the chain to refrigerated warehouses or other distribution channels based on real-time quality information. The last point is related to a recently developed concept, called quality-controlled logistics (Fan et al., 2020).

3.2.2 Intelligent Containers and Remote Container Management

Another recent technological development for cool chains is the concept of intelligent containers. An intelligent container is a normal reefer equipped with additional monitoring and control units—e.g., temperature logger, humidity, CO_2, O_2, and ethylene sensors (Jedermann et al., 2014). The information provided by the sensors is sent to a freight supervision unit (FSU), which calculates the product quality after receiving information from the sensing network. The telematic enables external communication with intelligent containers, which transmit temperature, humidity, and GPS data to the satellite network. The satellite sends the data to the logistics service provider, who receives warning messages on quality losses and other unexpected events and changes the setting of temperature remotely (Jedermann et al., 2014). If needed, additional sensors can be placed into the ceiling of the container. Using an intelligent container, the atmospheric conditions of perishable products inside a reefer can be continuously monitored and adjusted. The data gathered from an intelligent container can also be used, for example, to analyse the product shelf-life or predict the product quality (Fan et al., 2020). Additionally, by adjusting the environmental conditions

and creating conditions in which the fruits ripen, the container can possibly serve as a ripening chamber for fruits in transit (Fan et al., 2021). This process is called the in-transit ripening process and can be a substitute for post-transit ripening processes. This application of the intelligent container makes the cold chain shorter since the fruits do not need to be ripened in a forced ripening centre in the land (but the ripening can happen during the shipment of fresh produce). An economical and sustainability analysis of using intelligent containers for fresh produce transport is presented by Fan et al. (2021). Their study indicates the economic value of reduced spoiled fruits and improved carbon footprint along a cool chain.

The concept of intelligent containers has been the basis for commercial Remote Container Management (RCM) applications. Current RCM systems used in the market include RCM system developed by Maersk and Hamburg Süd for monitoring the refrigeration units and collecting data measured by the sensors placed on the refrigeration units, such as set-point temperature, supply/return air temperature, ambient temperature, relative humidity, gas level, and power status (Maersk Line, 2020). Furthermore, some sensors developed by Traxens and Orbcomm can also detect unauthorized door openings and theft protection and can improve the security of reefers in the global food chains (Orbcomm, 2020; Traxens, 2020).

3.3 Smart Retailing

3.3.1 *Smart Packaging and Intelligent Labelling*

Packaging has an important role in the protection and safety of food products. Smart packaging is an umbrella term referring to packaging systems that have extended functionalities beyond only protecting the package's contents. Smart packaging (and smart labels) use sensor technologies such as embedded chemical sensors or biosensor technologies to collect and deliver data regarding the product's quality condition, authenticity, and storage (Biji et al., 2015). In general, there are two types of smart packaging (Schaefer & Cheung, 2018):

- Active packaging: this kind of packaging interacts with the product and modifies the immediate surroundings of the product, aiming to improve the product's quality and extend the shelf life. Chemical elements such as moisture controllers, ethylene, and oxygen absorbers, and antimicrobial coatings are the most common agents used in active packaging. A common

example is adding oxygen absorbers to the cap of a plastic beer bottle, which increases the shelf life from three to six months (Biji et al., 2015).

- Intelligent packaging: this type of packaging can do "intelligent" functions such as tracking the product, sensing an attribute of the packaged food or its immediate environment (e.g., pH, temperature), and communicating it to the user (such as the manufacturer, wholesaler, or consumer). Intelligent packaging can also provide data to streamline customer engagement, automation, and marketing. Thus, in this type of packaging, the packaging has evolved from being a mere "container" of food products to a "communicator and diagnostic" tool.

Connected packaging is a subset of intelligent packaging, referring to applications that communicate with other packages, smart appliances, or the internet to provide consumers with unique content or value-added digital experiences (Chen et al., 2023; Kuswandi, 2022).

The most practiced form of smart packaging is time–temperature indicators (TTIs). These devices, initially developed in the early 1950s, are commonly used for cold, frozen, and deep-frozen products to document thermal exposure to excessive temperature along the chain (Wang et al., 2015). They can also estimate the remaining shelf life of perishable products and communicate that to the consumers—mostly by different colouring schemes. Accordingly, consumers can use package colour changes to distinguish between good quality and safe products and those that are spoiled or unsafe.

Time–temperature indicators take several forms (Wang et al., 2015):

- irreversible chemical colour changes resulting from the accumulated effects of time and temperature on a distinctive image.
- enzyme-caused colour changes proportional to cumulative time above an activation temperature.
- thermal expansion thermometers with ongoing data logging.

Besides communicating the product quality, smart packaging can also enhance customer empowerment (Chen et al., 2020). Scanning a QR code on a food package, for instance, can take the customer directly to a resourceful page with all the dietary concerns, ingredient research, and nutritional value. Another example is proprietary technology packaging that may lead customers to "how-to" guides and user manuals accessible via their smartphones (Biji et al., 2015). Additionally, smart, and connected packaging provides food processors with powerful tools for complying with their

product tracing and efficient handling of product recalls with a much lower potential for brand damage (O'Callaghan & Kerry, 2016). Smart packaging technologies can also lead to a transition from fixed and static shelf life for products to dynamic shelf life for agri-food products (Albrecht et al., 2020). The dynamic shelf-life sensor is an active RFID sensor measuring time and temperature to analyse the product´s quality—considering the time that product has been exposed to a certain temperature. Next, an expiring shelf-life date is calculated through an algorithm based on the time and temperature profile. Consumers can use their cell phones to communicate with the sensor and read the dynamic shelf life of the product. This can also substantially reduce food waste at the retail stage (Mezzina et al., 2022). Retailers can also use the information on dynamic shelf life to revise their inventory management and ordering policies (Buisman et al., 2019). Dynamic shelf life can also reduce food waste and potentially increase profitability for retailers, especially for products with short shelf life.

3.3.2 Smart Shelf Technology

Smart shelf technology refers to integrating sensing and monitoring technologies like electronic shelf labels (ESL), digital advertisement signage, or RFID technology (e.g., RFID tags, RFID readers, and antennas) with retail shelves to automatically track inventory in retail stores (Kellermayr-Scheucher et al., 2022).

Using sensors—and in particular RFID sensors—for tracking physical items is not a new concept in the retail sector. In the last couple of decades, RFID technologies have had a noticeable presence in the retail industry. They have been used for tagging products and stock; so staff can automatically track inventory and stock levels while minimizing the risk of loss and theft. However, in the last few years, there has been a trend towards using sensors to track other physical retail assets as well. For instance, a supermarket may equip its trolleys and baskets with long-range RFID tags to avoid losing them and reduce the cost of finding replacements. However, smart shelf technologies could potentially be the most beneficial implementation at the retail stage. The technology includes using RFID tags/readers or weight sensors to automatically track products (amount and location) in retail stores and trigger automatic replenishment at the shops. Additionally, shelves can be equipped with wide-angle HD cameras and computer vision technology to detect customers, products, and potentially suspicious behaviour (Santra & Mukherjee, 2019). This can support retailers in checking inventory levels and knowing when a product is out of stock. Besides, smart shelves can enhance

the shopping experience by changing how customers browse and shop for products in brick-and-mortar stores (Ives et al., 2019). While Amazon Go has already been leading this transition, other conventional retailers, like Tesco, are also using "scan as you shop" concepts to reduce checkout friction at the shops Making the experience more seamless for customers.

Integrating computer vision technology in the smart shelves can also create enormous opportunities for customization at the retail stage. For instance, the shelves can potentially interact with apps installed on customers' smartphones—containing their purchase history information—and provide personalized offers as they walk down an aisle (Spanke, 2020). Also, if customers create a shopping list in the app, smart shelves can guide them to where the products are located.

4 Concluding Remarks and Directions for Future Research

This chapter aims to provide an overview of digitalization landscape and smart solutions in agri-food supply chains. Although there has been a growing tendency towards digitalization in agri-food supply chains (also triggered by the labour shortage and need for enhanced productivity during COVID-19 pandemic situation), most of the existing technological solutions are developed in silos. To handle the challenges in food chains (like traceability or compliance as discussed in Sect. 2), there is a need for further integration and intra-operability in these innovative solutions (see Fig. 5). Data from the sensors and different IoT devices—in different stages of the chain from farm to table—can be shared with the relevant stakeholders either through a local server or the cloud and can be used to improve the decision processes along the chain. One way to make this happen is to employ a supply control tower. Instead of sharing data linearly, from party to party, supply chain network participants send all relevant data to a unified data repository, where relevant partners can access the information. Control towers allow companies to analyse operational data in real-time, allowing for proactive risk management and more detailed control of overall operations. Streamlined access to data is an essential element of a truly smart, sustainable, and resilient agri-food supply chain.

The other relevant area future research is analysing the factors that influence the willingness of different stakeholders (from farmers to food processors and retailers) to adopt digital technologies. Although digital technologies are

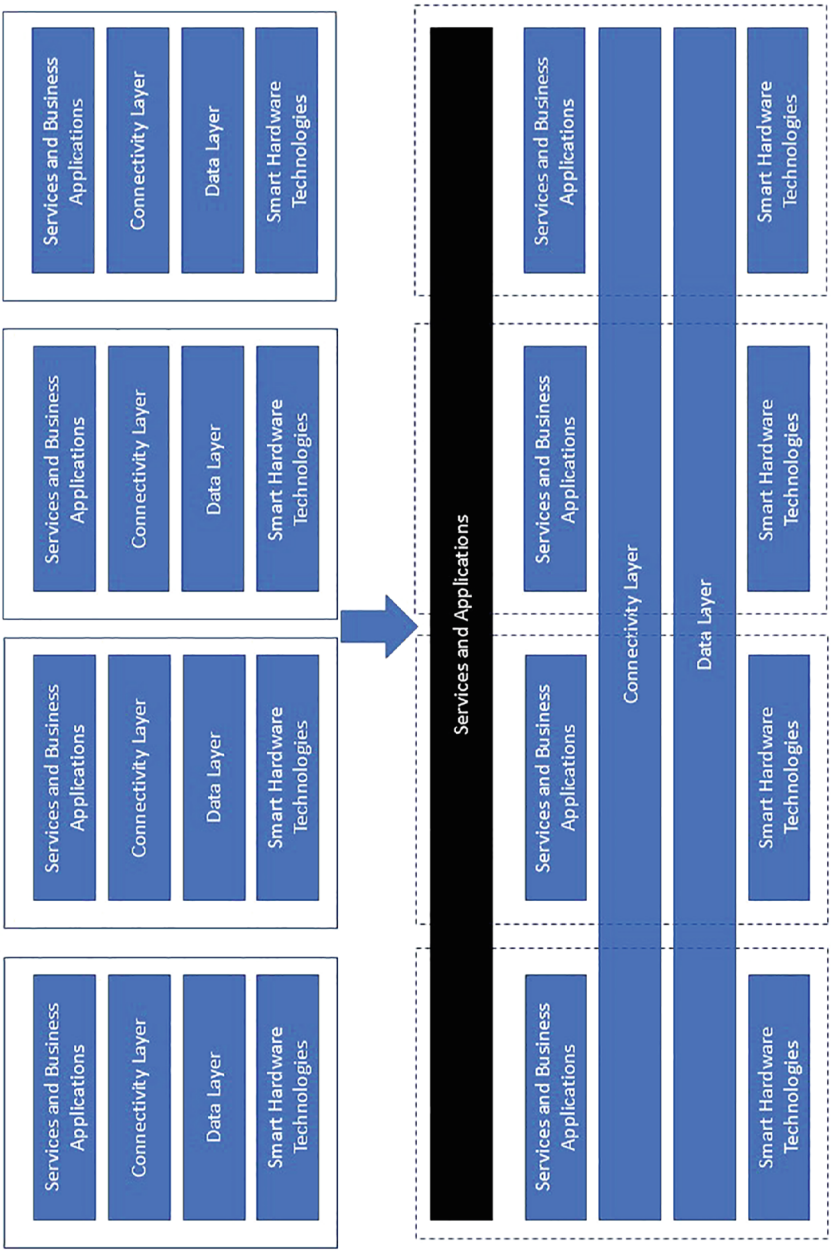

Fig. 5 From silo-based technologies to integrated inter-operable applications (*Source* Author)

vastly developed for the agri-food sector, the technology adoption (and acceptance) has been limited, particularly in the upstream agri-food supply chains. A potential reason is the necessary funds and investment in digitalization, which can challenge small-scale farmers and rural communities with low net profit margins. Developing strategies and financial incentives for engaging farmers and local communities is a potential direction for further research. Additionally, research can contribute to effective digital literacy and training programmes for farmers and agri-food industry workers to maximize the benefits of digital tools.

Finally, developing supply chain management strategies that leverage technologies to improve resilience is a potential area for future research. In particular, climate change significantly threatens the agriculture and food sectors. Researchers can investigate how digital technologies can help agriculture adapt to and mitigate the effects of climate change, including drought, extreme weather events, and shifting climate patterns.

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13

Securing the Supply Chain with Blockchain and IoT

Jims Marchang, Solan Keishing, and Yasanur Kayikci

1 Introduction

A supply chain is a network of parties and stakeholders that deals with item production, processing, distribution, and delivery of goods and services from the origin of the item's sources to the end user delivery for consumption. Supply chain management (SCM) is complex due to the involvement of diverse stakeholders, complex activities, and processes. The system functions based on contracts, legal bindings, and trusts alongside its terms and conditions. In such a system, item tracking and tracing using global positioning system (GPS) may be simple, but tracking the end-to-end condition, tracing counterfeiting or item replacement attempts, knowing item tampering events or attempts, and visibility of every situation of every item movement along the supply chain network is complex. It is also extremely challenging to

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monitor the exact state of the item and the way it has been handled, managed, and shipped including the ambient condition along the supply chain. Thus, it is a daunting task to make the system fraud-free, and tamper-proof, control the supply chain quality, and improve the visibility of the supply chain across every event and process to make the entire system auditable. Even though the supplying, processing, and distributing parties of the supply chain aim to avoid counterfeiting, and product tampering, safeguard data breaches and cyber threats, and meet regulatory compliances and government policies, it is a challenging task to monitor and safeguard every event and activity to deliver what is promised in terms of quality, visibility, and accountability because of system vulnerability, cyberthreats and cyber-attacks. As reported by IBM Security (2022, 2023), manufacturing, finance, business, energy, retail and wholesale, healthcare, transportation, telecom, etc. are among the top 10 most cyber-attack organizations. In the process, it is crucial to maintain a safe and secure way of monitoring, controlling, and accessing the system and preserving data confidentiality and data integrity to maintain trust among the supply chain stakeholders.

Most of the traditional approaches ensure the safety and security of food supply chains using a centralized architecture for collecting sensor data and managing the associated risks (Monakova et al., 2012), which are not the best fit for modern and flexible supply chains of the future. The authors, (Li et al., 2017) designed an Internet of Things (IoT) based tracking and tracing platform using radio frequency identification (RFID), quick response (QR) code, GPS, and other sensors, while Pal and Kant (2018) explores an IoT-based online monitoring system which aims to provide a solution to reducing food waste, increasing transportation efficiency, and tracking food contamination. Different approaches are adopted by different researchers and (Perboli et al., 2018) developed a blockchain framework to help reduce the organization's costs and optimize the operations. Other authors (Mondal et al., 2019) used RFID and designed a blockchain architecture for creating a tamper-proof digital database of the food packages of each instance. Thus, the work of (Azizi et al., 2021) work on harvesting the benefit of IoT applications and blockchain technology to make the supply chain smart and transparent. An analysis-based review of IoT adoption with the integration of blockchain technology shows benefits with a wide range of applications in food supply chain models (Kaur et al., 2022). However, to deeply understand the transitional conditions of the food items of each stage in the entire journey, multiple sensors are required to capture information—how it was stored or transported (temperature, humidity, pressure, light, electrochemical sensors);

how it was handled (accelerometer, gyroscope, proximity); when where and how long (GPS, Time & Date) and so on.

Moreover, when a blockchain system is involved, it is crucial to consider the scalability aspects in terms of storage, real-time validation process, computation, and energy requirements because it is going to be dealing with a huge amount of sensitive sensory data and it is also vital to ensure only reliable data from authenticated and secure sensory nodes are collected. In these blockchain solutions, data protection, and data security are the safeguarding pillars of the application. Supply chains like food and drinks, drugs, and pharmaceutical medicine distribution systems link directly with healthcare, so it is critical to ensure that authentic items and products are delivered to the users as intended and as promised by the suppliers. The diversity of the participation of the stakeholders is many and the geographical coverage is large in these supply chains, so it is crucial to preserve the integrity of the events of the entire system to maintain trust and quality. A framework that may be suitable for any food and drugs supply chain for integration of an IoT system into a blockchain network of a supply chain is shown in Fig. 1. To ensure data and system protection, it is vital to securely collect, transmit, and store the data generated by any sensory device to improve the system's resilience from any cyber-attacks and maintain adoption with a high level of trust and reliability.

The main aim of this chapter is to investigate and explore the key issues and challenges in making the supply chain securely auditable in the process of tracking and tracing items and their condition. The objectives of the

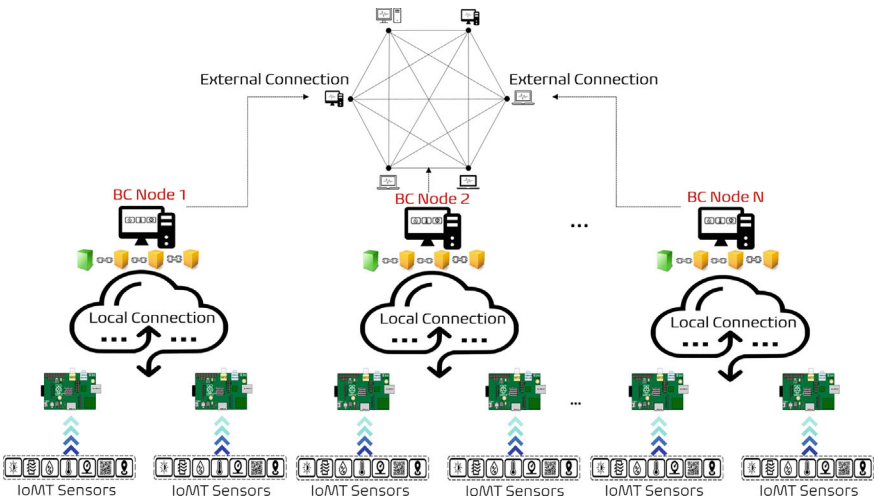

Fig. 1 IoT and Blockchain integration in the supply chain (*Source* Authors)

chapter include exploring and discussing security threats in the supply chain and how secure integration of IoT sensing data into a blockchain system can transform the process of auditing. Other objectives are to explore the entrepreneurial strategies to understand the supply chain stakeholders' needs, use of technology, identity, and competitors within the domain. Lastly, it is to understand the last-mile problem and how it can be addressed using technology and discuss technology's limitations in addressing the issue of the last mile. The chapter conducts a systematic literature review and identifies the research gaps in building a securely auditable supply chain system and develops a theoretical framework solution that uses IoT for sensing and blockchain distributed network for storing the data along the supply chain.

The rest of the chapter will conduct a study on two different types of supply chains (Food and Drugs) and discuss how technology like IoT and blockchain can transform the supply chain. Section 2 provides a background on supply chain security threads and discuss on technology integration to automate and securely manage the supply chain data. Section 3 covers the food supply chain issues and challenges during monitoring and its solution using IoT and blockchain; Sect. 4 addresses the drug supply chain issues and challenges and its solution using IoT and blockchain; Sect. 5 provides entrepreneurial strategies to understand the system's need, while Sect. 6 addresses the last-mile problem. Finally, the chapter ends with a conclusion in Sect. 7.

2 Supply Chain Security Threats and Technology Integration

This section discusses the security threats pertaining to the supply chain and it also discuss about the technological solutions, adoption, and its integration to automate and securely manage the supply chain data of all the items and its participating stockholders to make the entire system security auditable.

2.1 Security Threats in Supply Chain Management Systems

In a supply chain, different kinds of security threats could take place because of the diverse nature of stakeholder participation from different geographical regions which have different governance policies, laws, regulations, data protection rules, and acts. So, the safety and security measures considered by one stakeholder from one region, or a country will not be the same as that of

another country as the goods and services move along the supply chain. The report (Zalud, 2016) highlights vulnerabilities in a supply chain including the issues related to visibility, validation, performance, risk mitigation, and efficiencies related to cost, time, and risk to fight item theft, counterfeiting, etc. Secure collection of item's condition and ambient sensing through IoT is challenging due to the limited resources of these devices and maintaining zero network downtime will be daunting while attempting to record only accurate and reliable monitoring data to the blockchain through smart contracts will be challenging. Since a supply chain generates, transmits, and processes data over a supply chain network to monitor and control quality, the following cyber threats must be considered to ensure the safety of the application, nodes, and data flowing in a supply chain.

- *Data Protection Mechanisms:* Safeguard and authenticate every data source, maintain data confidentiality during transmission, preserve data integrity, and secure data storage from any unauthorized access to protect from data disclosure. Manage automated centralized patch management for every organization of the supply chain stakeholders. Authenticate, Authorize, and Audit every process by controlling any form of data access of every supply chain participant and every access attempt of each organization.
- *Ransomware and Malware:* The system should be safeguarded from any form of ransomware and malware, so that the systems of the supply chain network are not disrupted, operation stalled, and data protected.
- *Insider Threat:* To avoid and address any form of insider threat, every supply chain stakeholder must adopt a zero-trust policy whenever data access is attempted, data sharing is involved, or access rights are permitted.
- *Man-in-Middle Attack:* Adopt an AI-based mechanism such that any form of spoofing attack leading to a man-in-middle attack is detected and prevented. The adoption of intrusion detection and prevention mechanisms must be enforced on all the organizations of every stakeholder of the supply chain.
- *Denial of Service (DoS), Distributed DoS (DDoS):* This attack will affect the availability of the services along the supply chain, so the system should detect and mitigate any form of DoS and DDoS attacks.
- *Data Injection and Interception:* To record and maintain clean synthesized data during the IoT data collection process and to preserve data integrity of the supply chain data, it should be encrypted, and data integrity always preserved before passing it into the blockchain.
- *Data Loss:* Any form of data manipulation, data tampering, and above all data loss should be detected and avoided. It is a daunting task because

along the supply chain, the vulnerable points are many, and maintaining undisturbed network connectivity itself is a challenge. So, a way to preserve the confidentiality, authenticity, and integrity of the data must be considered even if part of the network system is broken and a self-healing network solution must be considered so that network loss does not result in data loss.

- *Standardized Security Practices:* The supply chain partners must have a standard policy on password updates, access control rights and policies, certificate updates, automated patch update mechanisms, etc.
- *Data Recovery and Service Continuity Plan:* It will be challenging to sync across all the parties and stakeholders to enforce a consistent data recovery plan, but it will be critical for service continuity.
- *Complex Legal, Compliance, and Regulatory*: The stakeholders involved in a supply chain are generally from different regions and countries that follow different data regulation policies, privacy management, and laws. So, preserving privacy rights and data security expectations of the end customers and intermediary customers along the supply chain will be a daunting task, especially when it deals with sensitive items and sensitive data, but acceptable standards and solutions must be considered.

2.2 Why IoT and Blockchain?

Data is the key source to understanding and analysing events, and processes, understanding the situation, and context, and even managing quality and gaining insight into any activity. So, collecting and processing high-quality reliable, and trustworthy data is a key to maintaining profitable businesses in a supply chain. In this process, IoT and blockchain can transform the way the data is collected and securely locked in a distributed system. IoT is a way of forming a sensing network to collect data from any objects and environment, making it a perfect way to monitor the items and containers of any supply chain. It means that IoT will allow connection, monitor, and report item's condition and ambient information from anywhere and at any time provided internet connectivity is available. Even when the internet connectivity is broken, an IoT sensor can continue to monitor and collect the data locally and feed into the supply chain network system when the internet goes live e.g. it can collect the temperature, moisture, gases, air quality, lighting, gyroscope, and accelerometer data, etc. to know the condition of the items and the way it is handled along the supply chain. While, the blockchain system is a secure distributed data management and storing system that is

immutable because it is built using a decentralization method, cryptographic hashing mechanism, exchange data using secure public key cryptography methods and digital signature, anonymized for privacy preservation even though transparent, Merkle tree block storing method, resistance to Sybil attacks (fake nodes cannot easily manipulate the network), recording data only after consensus technique is applied, double spending avoided, chaining and linking current data blocks from genesis with a reference to the earlier block, etc., making the entire system resistant to any form of fraud and tampering activity as described by Nakamoto (2008).

Integration of IoT sensing capabilities with a blockchain system in a supply chain will enable the system's visibility of the item's condition (its state, its journey information, how it was handled, how it was stored and managed) in an immutable distributed ledger that does not need the authorization of any trusted third parties (Kayikci et al., 2022a, 2022c). Thus, making the collected data of all the events and activities of the item's movement along the entire supply chain tamper-proof, transparent, and traceable making it auditable with a high level of trust (Kayikci & Subramanian, 2022). These securely collected and stored data will support a quality decision-making process, making concerned stakeholders accountable (what happened to which item when and where), providing a scope for evidence-based insurance, and may enhance product quality, and reduce loss and wastage. Thus, IoT systems provide a better scope of visibility, agility, and adaptability along the supply chain and address the challenges of monitoring the SCM (De Vass et al., 2021).

However, the stored data in a blockchain system cannot be altered, modified, or tampered with, so the IoT system needs to collect only quality and reliable data. Thus, the IoT system needs to operate securely so that the disconnect between the value of real-world data and the virtual data storage system is minimal otherwise the blockchain system will end up storing unreliable data in its un-alterable distributed storage system. Therefore, to avoid any form of collecting unreliable data and avoid disclosure of data to any unauthorized individuals intentionally or unintentionally, it is critical to authenticate and authorize every IoT sensor, control access and maintain data confidentiality, data integrity, and service availability during the data gathering process.

2.3 Blockchain Validation Process for Supply Chain

A centralized blockchain would undermine the technology's purpose because the idea of blockchain is to make a secure system in which the connection and

transactions are conducted peer-to-peer in a decentralized fashion among all (or sub-set of) the participating nodes (Kayikci & Subramanian, 2022). If it is centralized then we are going back to the same principle of relying everything on the so-called trusted parties e.g. central banks, govt., etc. for financial transactions. But the question is: Can we trust the so-called trusted parties? Not really, look at China's policy where they have made all crypto transactions illegal (BBC, 2021). Such governments can manipulate, control, monopolize, etc. and we will live under their mercy to get a loan, for money transfer, to secure a mortgage, etc. A centralized blockchain system will never be fair and transparent, and the system will be open to abuse. In the name of safety and security, the central controller can block or refuse transactions of any user. Moreover, if the blockchain is centralized then accountability and trust will be less e.g. if a bank collapses, what will happen to our money? Autonomy can be directly or indirectly affected e.g. In India, millions of poor people were helped by the Govt. to open a bank account, but a minimum balance is to be maintained otherwise they are penalized (CreditMantri, 2022). This means that even the little money that these poor people have is taken away when they cannot maintain the monthly minimum balance. In addition, charges are applied on transactions that affect poor people the most (Jain, 2019).

The idea of Bitcoin is to be free from a central controller, transparent, and free from unfair and inconsistent regulation that is controlled by the governments. The idea of a distributed ledger will not hold any longer if the blockchain is centralized and it means that the system will be less secure because if the central controller is compromised then everything can be compromised. So, in a centralized system, a single point of failure will exist which is not the case when the ledger is controlled by the distributed nodes in a blockchain system. Lastly, even the idea of tamper-proof cannot be guaranteed because the centralized controller may know and even access the private information of the users as described in Snowden's revelation (UK Parliament, 2015). So, the idea of Bitcoin, which is open, permissionless, and decentralized will not hold if centralized. The idea of blockchain is to empower all the stakeholders of the supply chain so that no individual powerhouse can monopolize the supply chain.

2.4 Scalable Blockchain Solution for Supply Chain

Since a blockchain is decentralized, the distributed ledger is stored in every participating node and the validation is conducted by all i.e. Proof-of-Work (PoW) or a set of nodes i.e. Proof-of-Stake (PoS). In such a network system scalability is a big inherent issue because a blockchain node requires high

storage, high bandwidth capacity, high energy demand, high transaction rates (direct impact on computation power), etc. In a Bitcoin, every node in the blockchain system must verify and validate every transaction and a consensus must be reached to consider the transaction to be valid. The average confirmation time of a Bitcoin network is around 10 minutes (Decentralized Dog, 2021). However, it may vary depending on the network condition, hash rate, network fees, etc.

The first widely adopted blockchain application i.e. Bitcoin adopts the PoW consensus technique which makes the validation process expensive and miners consuming extremely high energy. According to the Cambridge Centre for Alternative Finance (CCAF), Bitcoin currently consumes around 110 Terawatt Hours per year—0.55% of global electricity production. To address the issue of the demand for such high energy, it is reported (Sparkes, 2013), that Ethereum's (second largest crypto fintech) consensus process is upgraded to the PoS approach from the PoW method in Ethereum 2.0, making the upgraded system use less energy for mining and validation processes, and it is reported that the new consensus process has reduced CO_2 emissions by 99.99%. Since all node participates in the mining and validation process of a transaction in PoW, it takes more time and consumes more energy, but such a system is more decentralized and more secure compared to the PoS consensus approaches even though it may reduce the time of the validation process and consume less energy in comparison. However, just changing the consensus technique from PoW to PoS may not translate into a scalable blockchain immediately, but it is one of the transforming factors to consider. In addition, smart contracts empower blockchain automation and make it more efficient, but it does not entirely solve the inherent blockchain challenges to make it scalable. One of the other issues that challenges scalability is the interoperability of the blockchain networks to function as a single system when various blockchain ecosystem exists and it becomes more daunting when it must function alongside the non-blockchain ecosystems, one such study for a supply chain was conducted by Kayikci and Subramanian (2022).

Other ways to make the system more scalable and efficient are to increase the transaction rate and increase the block size to make more transactions per second and to make it scalable, e.g. Bitcoin was hard forked to Bitcoin Cash for the same reason. The block time (time required by miners or validators to validate transactions of a block to produce a new block) of Bitcoin and Ethereum is approximately 10 minutes and 14 seconds, respectively. Therefore, the most efficient way to make blockchain scalable is to consider the PoS consensus mechanism, use smart contracts, increase transaction rate

and block size, and adopt a layer 2 solution in which the transactions are conducted off-chain but linked with the layer 1 blockchain network and harvest all the security features, while the critical time-sensitive transactions are directly linked with the layer 1 blockchain and such technique can also be adopted in exception condition as per the agreement and contract. Thus, to guarantee the scalability aspect of the blockchain application while dealing with paramount real-time enormous IoT data generated within the supply chain, it is crucial to consider the scalability aspect during the design and validation process of the blockchain system.

3 Food Supply Chain Issues and Challenges and Solution Direction

It is important to securely capture the entire journey information of the items including the processing stages to trace the conditions of how food items were stored, transported, processed, packaged, handled, assembled, etc. in such a way that secure intelligent fog computing is incorporated during data collection to filter only meaningful information before it is passed on to the blockchain servers to securely validate and store using consensus smart contracts. In a supply chain, generally, the participating parties trust each other based on their contracts, compliances, and commitment; however, no one knows the exact condition of the item's journey from the harvest to the time when it is delivered to the customer. An example of a food supply chain is given in Fig. 2.

The secure edge IoT system can capture the data about the item's journey easily and has been explored extensively by many researchers, however, building a scalable, reliable, trustable data, and transparent secure system

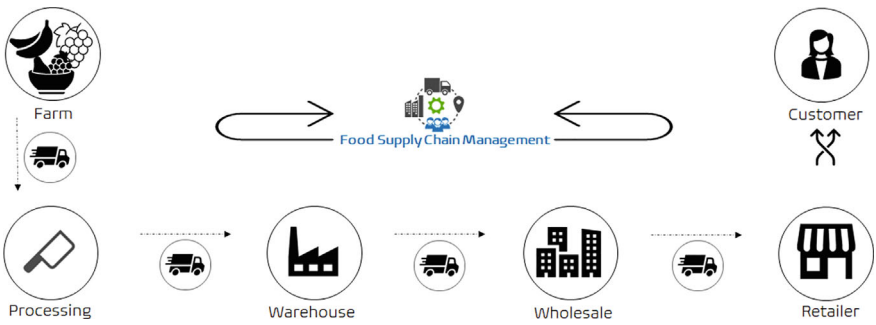

Fig. 2 Food supply chain management (*Source* Authors)

that is acceptable to all supply chain parties is still a challenge. When the supply chain system is opaque to the participating members of the stakeholders and the system fails to maintain data transparency, then there is a potential lack of trust and accountability. Moreover, if the gathered data lacks quality, then it will be challenging to rely on what happened to the items, so it is vital to ensure that the collected data is of high quality and reliable and not simply collect what the sensor records. Unless reliable quality data is recorded securely, it will be challenging to identify the reason for the abnormality, disturbances, and deviation from normal expected sensory information leading to failure to identify and detect loopholes and flaws leading to potential tampering, fraud, and counterfeiting activities. To protect self-interest stakeholders may tamper and manipulate sensory data unless the whole system is made secure and resilient from such fraudulent attempts. Moreover, because of the lack of control, reliability, and visibility of the food journey data to every stakeholder of the supply chain, transparency, and trust will be affected (e.g. How the food boxes were handled by the transport partners and the drivers are not known or visible to the farmers or the producers or the customer in that matter? Food spoiled or damaged caused by the transport team is invisible to the processing or manufacturing units and so on—so, whom to blame for the damage?). The challenge becomes more daunting when data and user privacy are required to be preserved while SCM parties share information to optimize their services. To address these issues and challenges, it is vital to ensure end-to-end data integrity and data confidentiality during the data monitoring and collection process so that all the necessary data including who is handling what item when where, and how is monitored on top of monitoring the item's physical integrity and its ambient securely (Kayikci et al., 2022c). It is also necessary to monitor detailed personal activity securely e.g. securely, and privately, monitor the GPS data of lorries and delivery staff including the driver's identity so that it is easy to track and audit who was responsible for what (if something goes wrong or use the data to even reward and monitor service quality). It will help the system to learn and track the driver's way of management of goods to improve service delivery. However, since it is linked with personal data, it is vital to preserve user privacy, otherwise, it will go against the data protection regulation of General Data Protection Regulation (GDPR) (e.g. of the UK and Europe) and the right to privacy of the user.

On the other hand, blockchain is a potential solution to this paramount challenge of improving management quality, and in the fight against food wastage, and food loss, and maximizing profit by building a trustable and transparent SCM system and making the entire system decentralized,

immutable, secure, transparent, and traceable to make the system auditable securely (Kayikci et al., 2022b, 2022c). It empowers every stakeholder to be accountable and take part in validating any transaction through a consensus process. Moreover, it allows every stakeholder to securely see through the entire journey of every item to maintain trust, accountability, and transparency across the network while preserving privacy by anonymizing and controlling access (How can that be achieved and what could be the trade-off between privacy and transparency is another domain to explore?). It is also a research challenge to note that the overhead of such technology-based solutions should not be too complex to use and too expensive to adopt, but it should be scalable. The IoT-based devices are limited in computation power, storage, and battery life; on the other hand, blockchain technology-based solutions are generally power-hungry, computation-intensive, and memory or storage-demanding in nature as studied by Marchang et al. (2019), so smart ways of building a balanced trade-off to harvest the blockchain benefits are required.

3.1 Discussion and Analysis of Securely Auditable Food Supply Chain

In a food supply chain, some of the critical questions listed below can be addressed with the help of blockchain and the Internet of Things (IoT):

- Does the end customer (consumer) have any idea about the food journey (At what condition the food took the journey to arrive at the table, how the food was stored, shipped, and packed, the environmental condition, etc.)?
- Is the food consumed by the customers the same food that was grown by the farmers (authenticity and origin of the food—no food fraud, no adulteration, etc.)?
- Is fair trade fair?
- Can live health and condition monitoring of any item from anywhere a possibility?
- Does each stakeholder hold accountability based on the facts and reality or simply based on legal bindings and regulations?
- Can the power control and monopoly by a powerful stakeholder of a supply chain be abolished?
- Can every stakeholder take part in every decision-making process?

- Above all, can cyber threats, data breaches, data regulations, and data privacy be protected while maintaining data transparency to make the system auditable and trustable?

Thus, the food and drink journey information from the origin to the customer can be captured at every stage along the supply chain with the help of IoT and stored in a blockchain to make it transparent, traceable, auditable, and tamper-proof. These days, food fraud and food adulteration are becoming more common than ever, in the UK food fraud costs firms a staggering £12 Billion annually (Ridler, 2017). Trusting only the stakeholders and regulatory bodies across all the organizations and governments along with the parties involved in the supply chain is not sufficient. To avoid or reduce fraud it has become mandatory to trace every activity on how the food items were stored, transported, processed, packaged, handled, assembled, and delivered. IoT sensors can collect all the critical and significant information before it is passed on to the blockchain network to securely validate the data and store it using proof-of-stake (PoS) consensus and smart contracts to trace and track every event and activity so that all the stakeholders can transparently take decisions which are beneficial for all the parties involved. This way the system will be more trustworthy, resilient, and reliable. Moreover, in this system, no stakeholder can manipulate, and what went wrong can be easily identified and be visible throughout the food journey, if something goes wrong, it will become clear who holds the responsibility e.g. If the container is opened, the freezer not working, stored in unhealthy ambiance and environment during transportation or storage, etc. This way it will become incredibly difficult to monopolize and control the supply chain and see who is making the most profit, who is being exploited, who should take responsibility, and when. It will become helpful even for the insurance company in deciding on the premiums depending on the conditions and situations. Holistically, it will provide a fair business model for every party involved and the idea of real fair trade can be achieved. Trust should be derived not only from the regulatory bodies but should also rely on technology to make the trusted system more resilient in addressing the food fraud issues ensure fair business and fair sharing of profits and realize benefits among all the stakeholders of the supply chain.

Such a system will also make the supply chain accountable and help detect and report loopholes and flaws at any stage or of any individual party, improving supply chain security and management and reducing food wastage by knowing what happens where, when, why, and how. It will eliminate the opportunity of falsifying data to protect self-interest. Thus, such

blockchain with an IoT system is a potential solution to this paramount challenge in fighting food wastage, food loss, counterfeiting, improving quality, maximizing profit, and building a trustable and transparent supply chain.

In this system, the main customers are the end users who buy and consume the items. So, these consumers have the right to know what they are consuming and how their food is being treated on its way to their dining table. Generally, in the process of making a profitable business, the corporates care less about the customers. Truths can be hidden to avoid loss and maximize profit e.g. replacing organic food and fruits with non-organic foods amounts to lying and cheating the customers (and even affects their health and well-being). So, this supply chain framework with IoT and blockchain will uniquely deliver services with a promise to the customers to make every move of the food transparent to the users while the stakeholders can maximize the profit in all fairness and minimize waste. This model will generate a more honest business model across the food and drinks supply chain and it can be extended to any other business model that involves a supply chain so that fair trade is achieved at the highest possible level across any business model.

4 Drug Supply Chain Issues and Challenges and Solution Direction

Medicine and drug counterfeiting is not a new thing. It prevails throughout the world, and it is on the rise as per the Pharmaceutical Security Institute. As highlighted in Fleck (2023) from the UN Office on Drugs and Crimes, nearly half a million people are killed every year by fake or counterfeit medicines and drugs. In the UK, 15,500 falsified medicine packs were discovered from an authorized medicines supply chain. In addition, illegal medicines and medical devices are in circulation and 3 million medicines and medical devices were sold illegally in the UK (Burns, 2022). So, we need a solution to this problem not only for the UK but for the entire world.

It will be extremely challenging to detect, identify, track, and trace every illegal activity and counterfeiting of drugs and medicine in a healthcare medicine distribution system. It will be very difficult whom to blame for what. It will be very challenging to detect when, where and what medicine is counterfeited or replaced within the supply chain. How do we guarantee that the medicine manufactured by company A is what is received by the distributors or in that matter any stakeholders along the supply chain until it is delivered to the patients? blockchain alone cannot solve this problem, we

need IoT and Artificial Intelligence (AI) along with a trusted intermediary to record medicines and drugs in the system to be able to detect and sense by using IoT and optimize the data collection or analysis using AI. Figure 3 highlights the supply chain model of a drug distribution supply chain system.

The biggest problem in this Industry is that the medicine or drug along the supply chain can be replaced with a fake one using the same bar code or same unique identification (ID) of the medicine or even add a random ID or barcode and introduced or injected along the supply chain and there is no way to detect that it is the same medicine or drug manufactured at the source or it is a counterfeited one when it is delivered to the patient.

- How do we know the source, originality, and learn it is from the trusted supplier using the right ingredients, and components of the drug?
- Is there a way to detect who did it and when or where it happened? The answer is negative. So, how do we even know which stockholder is responsible for what in real terms and identify any component and parties that violate the agreement?
- Who should be held responsible for counterfeiting along the supply chain when something goes wrong? Is it the fault of the manufacturer, the distributor, the pharmacy, the transport, etc.?
- How will the company know what unlikely or abnormal events occur, when, and where?
- How will the system ensure that every stakeholder can see the entire supply chain component transactions and activities?

Generally, it is impossible to track and trace all the questions raised above, but everything is operational because the system relies on trust and contracts and there is no technology in place to address these issues. Moreover, there

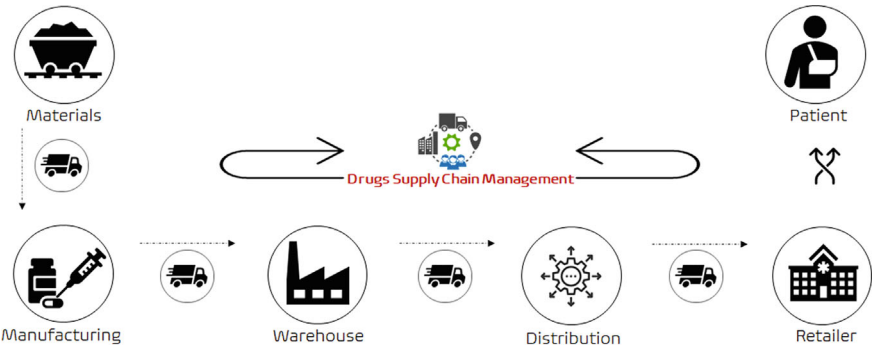

Fig. 3 Drug supply chain management system (*Source* Authors)

is no way to detect it, and the non-transparency of the system makes it even more difficult, what items are replaced, or which are fake ones. The current isolated stakeholders who do not know what happens cannot agree on who should hold responsibility when something goes wrong. Using blockchain and smart contracts to secure, and ensure transparency, traceability, verifiability, immutability, integrity, and efficiency of the supply chain, it is possible to track and trace counterfeited medicine with the support of a smart IoT sensing monitoring system by measuring, collecting, and recording every activity information (end-to-end) along the supply chain to detect any intrusion, alteration or tampering in the process while AI can optimize the detection process (Kayikci & Subramanian, 2022; Kayikci et al., 2022b). The aim is to ensure that the medicine or drug manufactured by authorized manufacturers finally gets delivered at the pharmacy or hospital for the patient and the effectiveness of the medicine can also be recorded in the blockchain (to make the supply chain circular through feedback to control and improve quality). There are multiple works done in the domain of blockchain applications in pharmaceutical industries to address counterfeiting (Haq & Esuka, 2018) tampering issues (Humayun et al., 2022), and fighting drug counterfeiting and trust-based consensus transacting systems (Zoughalian et al., 2022).

Introducing blockchain in this Industry means that the entire process of the way the information is recorded, collected, and stored will be changed, so it will be expensive in terms of the initial infrastructural cost (but the long-term benefit will outweigh the cost). In such a system, it will be challenging to know how the stakeholders will share the financial burden when it is introduced and how they will share the responsibilities. So, new agreements and contracts need to be signed among all the stockholders to have a shared responsibility, bear penalties, bear gains and losses, etc.

4.1 Discussion and Analysis of Securely Auditable Drug Supply Chain

To address the issue of drug counterfeiting (false labelling or fake medicine) and illegal circulation of medicine, a new blockchain-based anti-counterfeiting medicine distribution system should be in place. This system will link the authorized manufacturers, distributors, transportation, wholesalers, pharmacies, supermarkets, and hospitals, and to the extent of who prescribes (doctors) the medicine to whom (patients) to be able to provide feedback on the effectiveness of the medicine to the manufacturers. Other stakeholders could include insurance companies and regulators. The system's

products and services aim to provide authentic and approved medicine to the patients, making stakeholders accountable, voting for selection and promotion on who can remain in the supply chain and who can be penalized, and smart identification of any form of misconduct of stakeholders. So, a trusted medicine distribution system is in place for any national healthcare service in every country or globally in that respect.

A blockchain solution integrated with a smart IoT sensing system will be in place to securely store the data in a distributed system and monitor it by tracking and tracing every item along the supply chain. The scanned ID of an item is securely stored in a blockchain after it is manufactured and then its package movement, environment, and condition are tracked and traced along the supply chain and its data is recorded in the blockchain for every stakeholder to view. The monitoring activities go to an extent when a package is moved, shipped, picked, dropped, opened, etc. to ensure that any event of both normal and abnormal behaviours is securely recorded for analysis and future auditing to maintain product and service quality. The recording distributed blockchain system will have the following features in addition to all the general blockchain features described earlier:

- It will be based on consortium blockchain property (private and public) to provide information that may be related to the public interest and store information in the blockchain that can be accessed only by concerned stakeholders to preserve privacy based on the need and sensitivity of the information. To improve the validation process and increase the level of security, additional features like trust, history, geographical location, etc., of the validators and miners can be used in this system. So, unreliable validators do not participate in the validation process.
- The control of the information who accesses what will be incorporated using a smart contract and based on the role and type of the information needed by the stakeholders or public.
- To improve and address transaction speed and scaling difficulties, the system will adopt a layer 2 blockchain approach where delay-sensitive information (e.g. GPS location information, the opening of a package midway, etc.) is updated in the layer 1 blockchain directly live while the other delay tolerant data (e.g. temperature, pressure, speed of the movement, etc.) are stored in off-chain and update when the network transaction rate requirement is low.
- Every medicine and drug will be recorded in the blockchain system via the authorized and authenticated secure smart IoT recording system to ensure

that only the genuine medicine is recorded via authorized stakeholders along the supply chain until it is delivered to the patients.
- Secure and trusted IoT sensors are used for data collection along the supply chain movement and AI can be adopted for data analysis, data interpretation, and data optimization.

Thus, the drug industries will be able to harvest the benefit of blockchain technology (immutable, transparent, traceable, accountable, and secure) and it will be able to record every medicine and drug from the source i.e. the manufacturer, and will help in delivering that same medicine and drug to the patient, make the system auditable securely and fight against counterfeiting or illegal selling using smart IoT sensing and monitoring devices. However, it will be impossible for the blockchain to guarantee that the recorded data is genuine unless a trusted party or intermediary or sensory information of the IoT and intelligent AI solutions are accurate and reliable.

5 The Four Choices for Entrepreneurial Strategy

In any form of supply chain, understanding the customers, technology application, its unique identity of the technology, and exploring the competitors is important. The following sub-section explores such a strategy.

5.1 Customers

In any supply chain system, the end customers are the key (for food, it is an end consumer, and for the drugs, it is the patients). However, if the supply chain is dissected, then even the distributors, wholesalers, retailers, etc. are all intermediary customers because along the chain they are buying from one to another e.g. the distributors are buying from the manufacturer while the retailers are buying from the wholesalers, etc. and so on. So, the aim is to satisfy every customer and every intermediary participant along the supply chain to optimize the chain's network performance.

5.2 Technology

In an IoT integrating with a blockchain network solution, a consortium permission-based blockchain will be most appropriate considering the diverse

participation of the stakeholders and the trust that needs to be maintained across the supply chain. It is vital to ensure a secure integration of the sensory IoT data with the blockchain network otherwise it will be a disaster during any cyber-attacks and data breaches incident. The system should be developed depending on the need and control of the access to the data to preserve the privacy of the stockholders. Selected or elected stakeholders (if not all) should be involved in the consensus mechanisms during the validation process to maintain trust and guarantee the participation and engagement of a pool of stakeholders based on selection or election. Smart contracts should be used to sign any contracts and agreements between any stakeholders, including the insurance company. The IoT sensors, used to monitor the food or medicine or drug to detect conditions and any form of intrusion or injection of counterfeit food or medicine along the supply chain could be analysed using AI to optimize the detection and improve the efficiency of the process. The recording of the food or medical data will not be done only along the supply chain but what food or medicine is delivered to which customers should also be recorded to study the impact of the food or medicine and the drugs using AI to feed into the supply chain to control quality.

It may even be a possibility, that introducing a new fintech coin to run such a supply chain network will be beneficial and is another dimension for the stakeholders to explore so that the US dollar ($) does not monopolize the market. Such a system will make the entire ecosystem non-dependent on a particular country's currency but provide all stakeholders of the system with an independent currency that does not rely on any country's currency or any trusted third party. It will make the system transparent, more accessible, and cost-efficient. Such a crypto coin for a supply chain blockchain network will run without the need for control by any superpower economy, the entire system will run seamlessly, and it will make auditing, reporting, and traceability easy, but indeed regulation is a must so that no party exploits the system. In such a system, there will be no more issues of information imbalance among the stakeholders since the whole process will be transparent, no stakeholders can monopolize, and no stakeholders can tamper with any information since the system is immutable. All contracts and agreements can be executed using smart contact automatically within the blockchain and it means that there will be no need for stockholder's physical presence and unnecessary meetings and discussions to agree on project targets.

5.3 Identity

The IoT system and the blockchain system should not only be using permission-based blockchain and smart contracts to secure and ensure transparency, traceability, verifiability and guarantee immutability, and data integrity, but also improve the efficiency of the SCM and detect any form of counterfeiting and fraud activities by analysing, tracking, and tracing using efficient AI techniques. Moreover, the IoT system should be a smart system that can collect more reliable and trustworthy data to make the detection mechanisms more efficient. The data collection, monitoring, transmission, and storage in the distributed system should be "secure by design" and "privacy by design", so that the security methods are not an added feature after the system development is completed but be a part of the system's design and development, to make the system more resilient from cyber threats and cyber-attacks. In addition, to generate reliable data, every sensory node should be authenticated, data encrypted, access controlled, and authorized. The system should also be able to study the effectiveness of the supplies, so a customer feedback system should be integrated with the supply chain to mitigate and improve any form of quality improvement, fault, and service satisfaction.

5.4 Competition

There are some leading competitors in food and drinks or any pharmaceutical supply chain industries that use IoT and/or blockchain as a solution to secure and harvest the benefits of its technology, and some of the leading industries are listed below:

- *Ambrosus* is a blockchain-based SCM that manages pharmaceutical industries along with different kinds of supply chains and it addresses the issue of counterfeiting and improves product quality and transparency.
- *IBM Blockchain* also addresses supply chains including pharmaceutical industries to maintain security, transparency, tracking, and tracing of products including medicine and drugs.
- *SAP Pharma Blockchain* enables end-to-end visibility, and traceability and improves efficiency and regulatory compliance.
- *MediLedger* is another blockchain solution that enhances security, traceability, and product verification and improves supply chain efficiency in the pharmaceutical industry.

6 The Last Mile Problem in IoT and Blockchain-Based Supply Chain

There is always a disconnect between the reality of the sensing environment, the data recording platform, and the storage of the data on the online blockchain network during the process of collecting data about the events or activities or the state or journey information of the food or drinks or medicine or drug at any stage of the supply chain e.g. manufacturing process, packaging, during transportation, wholesale distribution, retailing, etc. So, the participation of a trusted intermediary (could be a smart reliable IoT system, AI robots, in conjunction with human participation) to pack the food or record the medicine data into the blockchain is required, otherwise, the blockchain will simply have to trust whatever data it is inputted. This is required, to address both insider threats and external threats. It would be unwise to rely on and leave everything to the technology. In the supply chain, the IoT sensors should collect the condition (item package opened, exposure to a different environment, etc.), health, and movement information of the food or medicine as it moves along the supply chain, and record securely into the blockchain and that needs reliable IoT sensors and AI solutions to find any form of counterfeiting activity e.g. replacing the item, opening package on transit, etc. This checking and sensing happen along the journey of the food or medicine until the final delivery point, i.e. to the customer. However, what happens between packing, unpacking, distribution, etc.? It requires trusted human participation e.g. the scanning of an item at the manufacturing hub before packing and sealing should require trusted human participation (directly or indirectly) because at this point fraud could happen, and insider threats will risk the authenticity of the item if a counterfeited product is swapped and packed. So, a trusted intermediary like human involvement is a must from point to point, otherwise, the immutable blockchain will end up recording fake or tampered data which needs to be avoided at all costs. Thus, relying only on technology like IoT, blockchain or AI will not be able to maintain the quality and reliability of the data. So, reliable, and trustworthy authorized human intermediary participation is a must to collect quality data and in making the system resilient.

7 Conclusion and Future Research Direction

The discussion and analysis of this chapter conclude that the integration of IoT sensing solutions with the blockchain should be secure by design so that data breaches and cyber-attacks can be defended while the privacy of the stakeholders is preserved and protected even when visibility and transparency are maintained by controlling access depending on the needs of each stakeholder including the customers. It is also found that it will not be possible to detect counterfeiting or fraud activity only by using smart IoT and blockchain or AI integration. A trusted intermediary participation like humans is required in addressing the last-mile problem. Moreover, it will be impossible to make fraud detection seamless, accurate, and automated without the support of IoT, blockchain, and AI-based data analytics. To maintain system integrity, any people and devices used in detecting and recording data will not only be authorized at the start but also be authenticated before they participate in recording data into the blockchain, otherwise immutable blockchain system will store tampered or wrong data that cannot be altered once it is recorded. If trusted intermediaries are not involved, then the blockchain system will simply accept every input and store it in the blockchain permanently, so every entry needs to be authenticated and verified before it is consensually agreed upon by the supply chain network making the entire system securely auditable in the process and making every entry traceable and trackable.

The existing solutions do not take a holistic view of the supply chain, because some focus on security, some focus on transparency while others look for ease of management, quality, verifiability, traceability, validation, etc. In the future, the theoretical framework solutions developed in this chapter need to be implemented, tested, and validated. Future work will also include exploring interdisciplinary research to integrate the entire supply chain system by considering other elements like cybersecurity to preserve trust, reliability, accuracy, and integrity, and use AI for detecting any form of fake or counterfeited food or medicine while holding accountability on any concerned stakeholders and making the system securely auditable, and in the process aim to minimize waste, maximize profit, restrict monopoly, increase quality, transparency and finally improve global population health.

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14

Industry 5.0 and the Future of Supply Chains

Behzad Behdani

1 Introduction

Over the last few decades, global supply chains have experienced massive changes. From introducing new handling procedures and equipment to relying on ocean-going vessels within global supply chains and from containerization to computerization, the supply chains have become more complex and—at the same time—more efficient. The rise of e-commerce has also had a transformative impact. It is estimated that the Covid pandemic accelerated the shift to e-commerce by five years, prompting retailers to pivot to omnichannel fulfilment capabilities and increasing the need for home and urban delivery services (Sharma, 2020). This has also led to new fulfilment services, such as same-day delivery, to ensure that customers receive their orders as quickly and conveniently as possible. As more and more consumers have started shopping online, companies have had to adapt their supply chain operations to meet the demands of this new channel.

As supply chains evolve, they move from a purely functional, physical system to a global and interconnected network of assets, data, and organizations. Despite all these changes, the methods used to measure the output and performance of the supply chains have remained the same over the decades. Managing the supply chain was primarily targeted at cost efficiency and

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responsiveness (Chopra & Meindl, 2016). The goal of an efficient supply chain is to save money and maximize profits by optimizing the processes and stages in the supply chain. A responsive supply chain, however, aims at delivering a high service level to the customers by providing a wide range of products or responding quickly to changes in demand. The assumption was that the final consumers were either concerned with the price or the quality of goods and services they received. However, over the last two decades, a new age of conscious consumerism began as many consumer choices expanded to include sustainable products, considering an item's social, economic, and environmental impact before buying it (Hong & Guo, 2019). The Industry 5.0 initiative, recently proposed by the European Union (EU), has also explicitly encouraged industrial parties towards a production model centred not only on technological innovation and economic growth but also on the commitment to environmentally responsible practices.

Although the concept of Industry 5.0 has gained traction over the last few years, the research on the implications for supply chain design and operations is limited. This chapter aims to analyse how Industry 5.0 may impact the supply chain design and operations. This analysis is the basis for developing a new and evolving concept of Supply Chain 5.0. Therefore, the guiding research question in this chapter is: "How can supply chain discipline align itself with the requirements and expectations set by the Industry 5.0 initiative/directives?" and "What is the role of disruptive technologies in achieving this alignment?" To answer these questions, this chapter presents a framework describing the essential dimensions of Supply Chain 5.0. It also discusses how supply chains can align with the Industry 5.0 goals and plans and which capabilities must be focused on. This chapter will further discuss the role of technologies in developing those capabilities and the potential challenges towards implementing Supply Chain 5.0.

2 The Concept of Supply Chain 5.0: An Overview of Key Dimensions

At its core, the concept of Supply Chain 5.0 is the response of the supply chain domain to the industry 5.0 paradigm. Therefore, the basic constructs of this concept are rooted in three primary dimensions of Industry 5.0, i.e. the transition to a human-centric, sustainable, and resilient supply chain (Xu et al., 2021, see Fig. 1). These three dimensions, however, have some specific implications for the supply chain domain and call for investing in some new capabilities, as discussed further below.

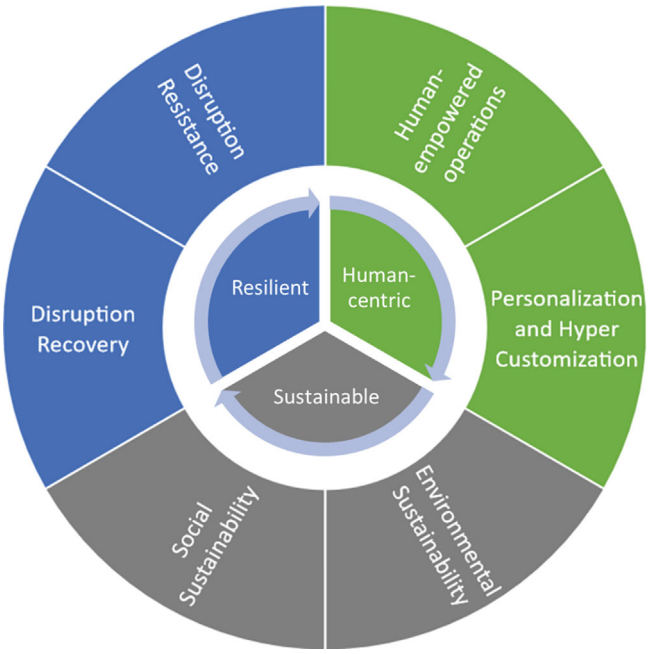

Fig. 1 Key constructs of Supply Chain 5.0 (*Source* Authors)

2.1 Human-Centric

Possibly, the most critical distinction of Industry 5.0 from Industry 4.0 is reintroducing the human element into the manufacturing and supply chain processes. In the context of supply chain operations, the human-centric focus of Industry 5.0 is formalized in two key aspects: "Human Empowered Operations" and "Personalization and Hyper-Customization".

2.1.1 Human-Empowered Operations

A key promise of Supply Chain 5.0 is viewing human needs together with the requirements of production processes. Human capital is a strategic asset, especially, to technology-centred and innovative industries. In addition, with the younger generation of employees (especially in the high-tech sectors), businesses must accommodate the changing needs of new generations. For younger workers, the commitment to environmental and social factors becomes increasingly important in their choice of employer (Bartle & Chatterjee, 2019). This may include a commitment to community initiatives,

flexible working arrangements, and cultivating historically underrepresented minorities in leadership roles.

The goal of Supply Chain 5.0 is designing and implementing human-enabling operations. This close human–machine collaboration has been traditionally discussed as human–machine symbiosis, supply chain ergonomics, and socio-technical systems (Behdani, 2012, see Fig. 2).

Human–machine symbiosis, primarily discussed by MIT scholars in the early sixties, aimed at a vision of interdependent collaborations between computers and humans, each providing complementary abilities towards a common goal. For this purpose, humans and machines are not considered individually but rather as a unit in the form of a system (Gill, 2012). Cooperation is a key focus area in the system design in which humans and machines/robots engage in a reciprocal relationship, i.e. seamless, and interoperable human/machine collaboration (Haesevoets et al., 2021). This is expected to improve the capabilities of both to implement a dynamic division of tasks and create new possibilities and approaches for solving problems (Lu et al., 2021). The human–machine symbiosis is more critical for proximate interaction with mobile robots (instead of remote human–robot interactions). This may be a robot assistant or even include a physical interaction. This is also studied as proximate human–robot teaming (pxHRT) in the recent literature as well (see, for example, Matsumoto and Riek [2019] for more details).

As another study area, Supply Chain Ergonomics is a traditional discipline that analyses the interactions between human activity and the elements of their activity (including task specifications, tools, methods, and work environment). The aim is to develop production and logistics systems that provide

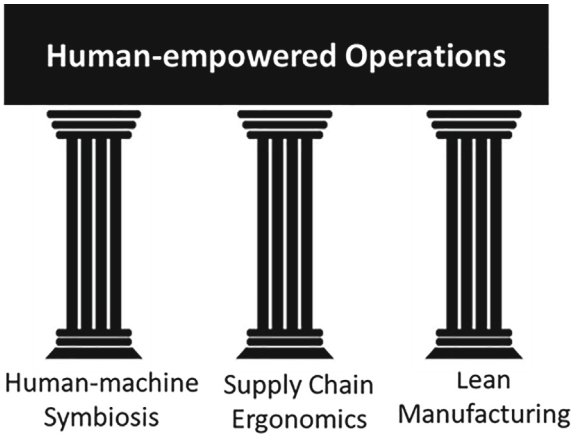

Fig. 2 Traditional supply chain fields supporting human-empowered operations (*Source* Authors)

optimal workplace efficiency, safety, and comfort (Zink & Fischer, 2018). As a result, the effectiveness of human labour will increase (which also enhances the productivity of processes), and the negative impacts on human health will be diminished. Supply chain ergonomics also works with suppliers to provide specifications to minimize injuries, lost work time, human system inefficiencies, and materials management specifications (Thatcher & Yeow, 2016).

As a related field of study, the Lean Manufacturing Philosophy has extensively discussed the importance of human capital (Koemtzi et al., 2022). A fundamental principle in lean is enabling people to lead and contribute to their fullest potential. Employees are considered a fundamental factor in achieving Kaizen or continuous improvement in the lean processes (Gaiardelli et al., 2019). With Kaizen, all employees are accountable for (and authorized to) identify gaps and inefficiencies in the logistics and supply chain processes. Hence, every employee is entrusted to suggest improvements at every level in the organization. In the lean philosophy, the ergonomic aspects are also attached to the "Waste of Motion". A solution for this in the design of workplaces is cellular manufacturing and investing in multi-task capabilities for workers (Zhou, 2016). Cellular manufacturing design allows a person or a few people to operate in an area and perform a multitude of jobs, which is very efficient.

2.1.2 Personalization and Hyper-Customization

The second element of a human-centric supply chain is "hyper-customization" or "mass personalization" in product design and production. To achieve this, we need a supply chain design that combines the efficiency benefits of industrial production (mass production) with individualization (customization). The rationale is that the heterogeneity of customers should be viewed as an opportunity (while it has traditionally been perceived as a risk to the efficiency and productivity of supply chain processes). Also, the assumption is that the existing digital production technologies—like Artificial Intelligence (AI) and additive manufacturing, as described in Sect. 3—will lead to a reasonable reduction of individualized/customized manufacturing (Maddikunta et al., 2022). The validity of these assumptions is, of course, highly industry- and product-dependent. Indeed, products with modular architecture have more potential to realize a mass customization production mode (Zhang et al., 2019).

Achieving hyper-customization implies active engagement of customers in the design process to personalize product designs (in addition to the customized logistics or packaging processes). This distinction, in some literature, is conceived as the contrast between "soft" and "hard" customization methods (Bogner et al., 2017). In the case of soft customization, the customer is not directly involved in the manufacturing (or design) steps—and the personalization takes place outside of the production stage, e.g. in the packaging, labelling, or logistics processes. In hard customization, however, customer preferences have a direct influence on the production of the product.

The intensity of customization in the supply chain literature is mostly formalized in the Customer Order Decoupling Point (CODP) concept (Guo et al., 2021). The Customer Order Decoupling Point describes the process or node (i.e. the inventory point) in the supply chain network where the material flow is tied to a specific customer order. The processes upstream of CODP are driven by forecast-based planning to produce "standard" items pushed downstream. Actual customer requirements drive the downstream processes, and the "customized" items are pulled by the customers or the next step in the chain.

By changing the location of CODP, we have different supply chain designs, as shown in Fig. 3. As more steps in the production and supply chain are demand-driven (pull) instead of supply-driven (push), we can go from soft to hard and hyper-customization.

Achieving hard or hyper-customization needs substantial investment in flexible production technologies—like additive manufacturing, as further explained in Sect. 3. It also asks redefining the product development processes—by creating ultimate modularity in the product design. Additionally, traditional market research methods—designed to find similarities in the customer group and put them in a segment—are becoming obsolete for hyper-customized supply chains (Zhang et al., 2019). Instead, successful

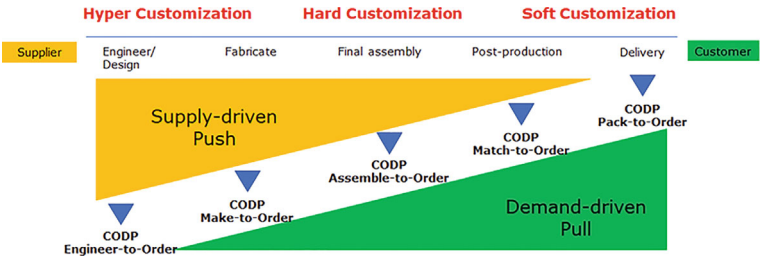

Fig. 3 The impact of CODP on soft, hard and hyper-customization (*Source* Authors)

customization seeks to discover where customers differ—using AI methods, for example—and developing manufacturing capabilities to cater to that individuality.

2.2 Supply Chain Resilience

Supply chain resilience is the capability of a supply chain to mitigate negative unplanned risk factors or shocks that endanger the normal operation of the chain (Tukamuhabwa et al., 2015). Risk management has always been a challenge in global supply chains. However, it has become more challenging due to increased global sourcing and outsourcing of non-core activities. Before globalization, some risk factors, such as exchange rate fluctuations, social instability, and even natural disasters or public health issues (for which the Covid pandemic is a good example), were unimportant or mostly perceived as local or regional challenges (Behdani et al., 2019). However, with the increase in global trade, a disaster in one specific place can influence companies working far from the originating regions and countries. Some recent examples include the COVID-19 pandemic, the war in Ukraine, and the global chip shortage that emerged in 2020.

Globalization has also caused additional challenges in managing supply chains due to communication difficulties and cultural differences. For example, cultural differences and attitudes towards food hygiene in different countries are cited as one of the potential causes for the increasing rate of product recalls in recent years (Roth et al., 2008) or possibly as a critical factor in initiating the COVID-19 pandemic crisis.

In addition to emerging new types of risks, most supply chains have little buffers to bear the disruption's impact (due to emphasis on Just in Time Manufacturing or JIT and supply-base reduction). For this reason, the adverse effects of an initiating disruption spread quickly throughout a supply chain (Tukamuhabwa et al., 2015).

To handle these challenging circumstances, companies need to acknowledge the risks imposed by cost-efficiency trends to the supply chain operation and, simultaneously, seek to systematically improve the resilience of supply chains against threats.

A resilient supply chain aims to develop capabilities to prepare for potential risks and respond to and recover from unexpected events as they occur (Behdani et al., 2019). These capabilities in the literature are classified as:

(1) *Risk Prevention/Resistance:* activities and measures to avoid disruption or minimize exposure to potential shocks and risky events. This is also sometimes referred to as "robustness" and indicates how prepared a supply chain is to predict and lessen the impact of any disruption (Monostori, 2018).
(2) *Disruption Response/Recovery:* the capabilities in a supply chain to detect a disruption and, subsequently, react to that to return the chain to its normal operation and performance (Behdani et al., 2019).

A comprehensive supply chain resilience model combines both resistance and recovery components and a group of second-layer capabilities, as shown in Fig. 4.

- *Redundancy:* One way to manage potential disruptions—particularly to achieve "risk resistance"—is by creating redundancies across the chain (Simchi-Levi et al., 2018). Contracting with a backup supplier to supply the required material (or a part of it) when the primary supplier is disrupted, carrying extra inventory—like storing finished goods to handle

Fig. 4 Supply chain resilience dimensions and capabilities (*Source* Authors)

fluctuations in market demand—and investing in production or warehouse capacity buffers are examples of redundancy in the supply network (Behdani et al., 2019).

- *Flexibility:* Flexibility is the ability to take different positions in response to irregular situations (Simchi-Levi et al., 2018). A necessary condition for flexibility in the supply chain is having multiple interchangeable resources. Flexibility can achieve in different stages of a supply chain, for example, by a flexible supply base (i.e. multiple-sourcing or flexible contracts with suppliers), flexible manufacturing process (through multi-purpose/product plants or using multi-purpose technologies like additive manufacturing), flexible transportation (through multimodal and synchromodal transport), and flexible product configuration (through modulization and manufacturing postponement).
- *Collaboration:* Collaboration is essential as global supply chains are complex systems, and no one actor has all the necessary information for identifying and mitigating the possible risks in the system (Butner, 2010). Additionally, the potential risk remedies might be too expensive to be implemented by a single partner. In these cases, collaboration would help companies to pool resources and share the expenses of disruption response.
- *Visibility and information access:* Supply chain visibility is the ability to track the status of different stages (e.g. the healthiness of different parts, the material in transit in the supply network, etc.) from suppliers to end customers (Somapa et al., 2018). Investing in (joint) performance monitoring and early warning systems is a key ingredient to improve the visibility in the supply chain—and, accordingly, faster detection of disruption. In addition to information availability, developing tools for information analysis can substantially improve visibility (Behdani et al., 2019). IT-driven tools—as discussed in Sect. 3—can support decision-makers by providing insight into the disruption profile, how far disruptions propagate through a supply better chain, and which parts of the system may be affected more by the event.
- *Vigilance:* Vigilance is the organizational capability to quickly identify emerging threats (as well as trends and opportunities) and orchestrate a quick and effective response—in close collaboration with supply network partners. It enables firms to navigate change more efficiently—to anticipate, prepare for, and adapt to abnormal situations—and effectively take advantage of new business opportunities when they arise. Organizational flexibility, employee empowerment, and distributed decision processes are key drivers of higher vigilance in global supply chains (Fayezi et al., 2017). Additionally, external, and internal supply chain integration directly impacts vigilance, positively affecting supply chain performance.

2.3 Supply Chain Sustainability

Sustainable business operation is defined as "adopting business strategies and activities that meet the needs of the enterprise and its stakeholders today while protecting, sustaining, and enhancing the human and natural resources that will be needed in the future" (International Institute for Sustainable Development, 1994: 4). This implies extending the aim of a supply chain beyond the interests of its core shareholders and taking into account the impact on the surrounding of business and broader stakeholders. Sustainability in the supply chain generally focuses on two main aspects: the effect of a supply chain on the "environment" and its impact on "people and society".

2.3.1 Environmental Sustainability

According to the United Nations (UN) World Commission on Environment and Development, environmental sustainability is the commitment to conserve natural resources and protect global ecosystems (Emas, 2015). Because the ecological consequences of business decisions are not felt immediately, a key emphasis of environmental sustainability is its forward-looking nature, simultaneously supporting the health and well-being of current and future generations. The environmental sustainability in a supply chain is primarily discussed from two main perspectives:

- *Supply chain footprint reduction*: The supply chain must be designed and continuously improved to create the minimum footprint, from greenhouse gas emissions to wastewater (Nishant et al., 2020). To classify businesses' impact on the environment, gas emissions are categorised into three groups (or "Scopes") by the Greenhouse Gas Protocol (GHGP). Scope 1 in the GHGP framework comprises direct emissions from resources owned (and controlled) by a company, such as its transport fleet and the production and distribution facilities (Wiedmann et al., 2021). Scope 2 encompasses indirect emissions from purchased heat and electricity. Scope 3 emissions are all those that occur in a company's value chain, such as goods and services purchased or extraction and transportation of raw materials by suppliers. Considering the complexity of a global supply chain, measuring, and reporting the chain footprint demands a high level of transparency and trust. IT systems—particularly blockchain-driven and machine-learning methods—can help to store reliable data and identify inconsistent patterns (Goralski & Tan, 2020).

- *Circular production and resource efficiency:* Another key aspect of environmental sustainability is changing the mindset from "waste" to "resources" and improving efficiency across the supply chain (Farooque et al., 2019). This has been formalized in the Closed-Loop Supply Chain (CLSC) concept. CLSC stresses a cradle-to-cradle design process (i.e. the design and production of products so that, at the end of their life, they can be genuinely recycled or even upcycled). The process starts with Design for Recycling (DFR), i.e. the product design method that considers the ability to disassemble a used product to recover the recyclable parts (Ceschin & Gaziulusoy, 2019). Furthermore, CLSC emphasizes integrating reverse flows with the forward streams in the design and operations of the supply chain processes and, simultaneously, reducing the dependency on the raw materials extracted from the earth—by changing the production technologies or improving the efficiency of existing technologies.

2.3.2 Social Sustainability

Social sustainability often gets less attention than environmental sustainability, even though it is just as important. Social sustainability focuses on "people" related aspects of sustainability, i.e. the supply chain's stakeholders (Klumpp & Zijm, 2019). Stakeholders are groups of people, communities, or organizations who affect or are affected by the activities performed by a firm. These stakeholders fall into three main categories:

- *"Internal to the Firm"* that are stakeholders inside a company, including the shareholders and employees. Shareholders are owners of a company and are entitled to reap the benefits of a business's success. A company has to consider the shareholders' interests in short-term and long-term decisions, ensure the safety of capital investment, and safeguard the business's assets (that, in principle, belong to the company's shareholders). It must also provide the shareholders with regular and accurate information about its working as well as schemes for future growth. The commitment towards shareholders is sometimes categorized as "economic responsibility" in Corporate Social Responsibility (CSR) models, for example, in Carroll's framework of CSR (Carroll, 2016). Employees are the second internal group of stakeholders for a company. The companies are responsible for providing employees with a safe and decent working environment. Ensuring job security, providing equal opportunity for training, promotion, and career development, creating a good relationship between

management and workers, and participating employees in decision-making and possibly profit-sharing programs are among other responsibilities of a company towards its workforce (Mani et al., 2016). This aspect of social sustainability may overlap with the human-centric dimension of Supply Chain 5.0.

- *"External to the Firm but Internal to the Supply Chain Network"* that includes the customers and suppliers for a firm. Consumers are the drivers of the market and the primary source of revenue for businesses. Companies must comply with their commitments and supply quality commodities at the right time and at a reasonable price (Ghamisi et al., 2021). Providing proper after-sale services and avoiding unfair trade practices like black marketing, greenwashing, and false advertising are some of the basic social responsibilities of a company towards its customers. Suppliers' services are also crucial for any business because they supply raw materials, machinery, labour, and other materials. Adopting equitable and competitive trading by paying fair prices for goods in a reasonable time, providing technical advice, and informing suppliers of future developments are some of the responsibilities of a company towards its suppliers. Additionally, the company must ensure that its upstream suppliers follow proper environmental and social practices respecting human and workers' rights—by providing healthy and safe working conditions and avoiding child labour and human rights violations (Klumpp & Zijm, 2019). These principles are particularly important in a global supply chain and some labour-intensive sectors like the agricultural and apparel industries.
- *"External to the Network,"* which includes the stakeholders around a business. A key stakeholder in this group is the governmental and regulatory bodies. Following laws, regulations, and compliances, supporting the government to improve the quality of life, paying taxes and duties honestly and regularly, and avoiding unfair trade practices are some of the primary responsibilities of companies towards the government (Carroll, 2016). These responsibilities are classified as "legal responsibility" in Carroll's Pyramid of Corporate Social Responsibility. Businesses are also responsible for protecting the interests of society and local communities. The community represents the group of people in the immediate surroundings of a business (Mani et al., 2016). Supporting local charities and NGOs, providing financial assistance to local institutions like educational bodies or hospitals, and helping to protect the environment from pollution are some of these responsibilities.

3 Key Enabling Technologies for Supply Chain 5.0

As it has matured over the last decade, Industry 4.0 has provoked a fast development of technologies, allowing a transition towards smart manufacturing and the creation of intelligent factories. Technologies such as the Internet of Things (IoT), Digital Twin (DT), AI, and machine learning are already deeply woven into the manufacturing and supply chain processes. Many of these technologies can be helpful for Industry 5.0 and, accordingly, Supply Chain 5.0 as well. Of course, some technologies like collaborative robots (cobots), Smart Materials and Wearable technologies, and (partially) Additive Manufacturing may have particular relevance for Supply Chain 5.0. As a key technology for Industry 5.0, collaborative robots (or cobots) are a form of robotic automation built to work safely beside human workforce in a shared, collaborative workspace. Robots are assigned the heaviest and repetitive operations, leaving human operators free to carry out more specialist and less fatiguing tasks (Nitsche, 2021). To guarantee a safe operation, the cobots and the workplace must incorporate enhanced sensor (and IoT devices) and vision technologies to enable humans and machines to work side-by-side. Wearable technologies (like smart watches, smart caps, and AR/VR headsets) are also an essential element for Industry 5.0. Besides personal health, improved planning, and productivity, introducing wearables into our workplaces can facilitate communication/collaboration among machines and humans (Maltseva, 2020). Using wearable tech can also promote worker safety by including, for example, CO2 or gas emission sensors or explicit sensors showing the temperatures workers are being exposed to.

An overview of other supporting technologies for Supply Chain 5.0 is presented in Table 1. Some of these technologies—like AI and ML as well as Simulation and DT—can play a more crucial role, partly because they support more dimensions of Supply Chain 5.0. Additionally, Simulation and DT (supported by Blockchain technology) can work as bridging technologies to link and complement different technologies within one system.

4 Challenges and Opportunities for Future Research

Although the existing technologies can provide a solid backbone for the transition towards Supply Chain 5.0, several key challenges remain to be explored, and additional research studies are called for.

Table 1 Role of digital technologies in Supply Chain 5.0

	Artificial Intelligence and Machine Learning	Simulation and Digital Twin	Cobot & Automation	Internet of Things (IoT)	Additive Manufacturing and 3D Printing	Blockchain Technology	Smart Materials and Wearable Technologies
Disruption Resistance	• Switching from periodic to real time risk assessment (Baryannis et al., 2019) • Supporting collaborative risk management by creating secure platforms for information exchange, enabling faster and more efficient decision-making across the supply chain network (Belhadi et al., 2021)	• Using Digital Twin and simulation for root cause and scenario analysis for better-informed risk planning (Burgos & Ivanov, 2021; Ivanov et al., 2019) • Analysing the risk mitigating strategies in the Digital Twin and simulation environment before implementing in practice (Ivanov & Dolgui, 2019)	• Increased resiliency/ agility by complementarity of man and human and possibility of system backup (Ralston & Blackhurst, 2020)		• Reducing the risks (especially the transportation and product warehousing risks) in a supply chain (Thomas, 2016) • Increased supply chain robustness because of decentralized production (Gupta et al., 2022)	• Providing a secure backbone for decentralized decision-making in a supply chain that increases supply chain robustness and agility (Pournader et al., 2020)	

	Artificial Intelligence and Machine Learning	Simulation and Digital Twin	Cobot & Automation	Internet of Things (IoT)	Additive Manufacturing and 3D Printing	Blockchain Technology	Smart Materials and Wearable Technologies
Disruption Recovery	• Automating disruption response and real-time alerts (Belhadi et al., 2022) • Mining patterns in supply chain data that may be related to identify potential/ emerging risks (Baryannis et al., 2019)	• Improved holistic visibility and transparency of the supply chain (assets and operations) during a disruption (Ivanov & Dolgui, 2019) • Avoiding data and organizational silos to understand how well the supply chain operations are performing (Marmolejo-Saucedo, 2020)	• Faster reaction to supply chain abnormalities by decreased throughput time (Manhart et al., 2020)	• Improved visibility—by real time tracking- and asset availability in detecting and reacting to supply chain disruptions (Birkel & Hartmann, 2020) • Developing flexible contingency plans during disruptions by real time asset tracking (Tsang et al., 2018)	• Reduced throughput time for critical items during a disruption (Muir et al., 2018)	• Providing real-time secure visibility and tracking of goods and assets throughout the entire supply chain during a disruption (Manupati et al., 2022) • Increase trust across the supply chain network and the possibility of information/ resource sharing during a disruption (Pournader et al., 2020)	• Creating rapid real-time visibility and alertness in case of machine malfunctions and interruptions (Ambrogio et al., 2022)

(continued)

Table 1 (continued)

	Artificial Intelligence and Machine Learning	Simulation and Digital Twin	Cobot & Automation	Internet of Things (IoT)	Additive Manufacturing and 3D Printing	Blockchain Technology	Smart Materials and Wearable Technologies
Social Sustainability	• Using AI to promote Ethical Labor Practices and detect shady patterns and potential human right violations (Goralski & Tan, 2020)	• Using Digital Twin and simulation to evaluate the impact of supply chains on the local communities (Klumpp & Zijm, 2019)				• Proving reliability, traceability, and authenticity of the information about the source of materials to guarantee the compliance with social and environmental norms—like human right norms or responsible sourcing (Kshetri, 2022)	

	Artificial Intelligence and Machine Learning	Simulation and Digital Twin	Cobot & Automation	Internet of Things (IoT)	Additive Manufacturing and 3D Printing	Blockchain Technology	Smart Materials and Wearable Technologies
Environmental Sustainability	• Using AI to reduce waste in supply chains (e.g. by avoiding excess packaging, unused materials, and obsolete inventory) (Nishant et al., 2020) • Using AI to monitor supply chain operations in real time, identifying areas where waste is being generated and taking action to address the problem (Goralski & Tan, 2020) • Using AI to promote Sustainable Sourcing (e.g. by tracking the raw material sources) (Ghamisi et al., 2021)	• Using Digital Twin and simulation to evaluate the environmental impacts of supply chains (Kamble et al., 2022) • Using Digital Twin and simulation for root cause and scenario analysis to minimize the environmental impacts (Kamble et al., 2022)		• Increasing the resource efficiencies—and minimising idle movements- by providing real time location/ condition data (Yadav et al., 2021) • Reduce energy consumption, optimizing space and minimizing the environmental impact of buildings/ facilities (Manavalan & Jayakrishna, 2019) • Supporting recycling initiatives and reverse logistics by providing the location/ condition data (Thürer et al., 2019)	• Waste reduction by on-demand production and spoilage avoidance (Kunovjanek et al., 2022) • Reducing the need for energy- and resource-intensive manufacturing processes (Ferreira et al., 2023) • Reducing the amount of material required in the product manufacturing process (Kunovjanek et al., 2022)	• Providing a reliable data setup to evaluate the real environmental impacts of products and process over their life cycle (Saberi et al., 2019)	

(continued)

Table 1 (continued)

	Artificial Intelligence and Machine Learning	Simulation and Digital Twin	Cobot & Automation	Internet of Things (IoT)	Additive Manufacturing and 3D Printing	Blockchain Technology	Smart Materials and Wearable Technologies
Personalization and Hyper-Customization		• Using Digital Twin and simulation to analyse the process improvement options (Aheleroff et al., 2020) • Using Digital Twin and simulation to analyse the product improvement and redesign options (Schleich et al., 2017)	• Combing the automation (for repeated tasks) with the human creativity to customize the products (Shi et al., 2020)		• Giving customers the ability to participate in the creation and design process (Oettmeier & Hofmann, 2017) • Speed of first prototype and from prototype to production (Attaran, 2017)	• Providing a secure backbone for digital inventories used in additive manufacturing processes (Perez et al., 2022)	
	• Using AI for customer research, needs assessment and portfolio design (Hermann, 2022) • Using AI to predict the demand patterns and expected irregularity (Kim et al., 2019)						

	Artificial Intelligence and Machine Learning	Simulation and Digital Twin	Cobot & Automation	Internet of Things (IoT)	Additive Manufacturing and 3D Printing	Blockchain Technology	Smart Materials and Wearable Technologies
Human-powered operations	• Using Generative AI (like ChatGPT and GitHub Copilot) to engage with employees and support joint operation of workforce and machines (Wang et al., 2023)		• Improved productivity and enhanced human abilities via human–machine collaboration in a safe way (Nitsche, 2021)	• Improving the safety and convenience of workplace using sensor and monitoring technologies (Molaei et al., 2020)			• Using smart wearables (like Glove Scanners) to improve the speed and reliability of workflows (like scanning parts for assembly work, picking items in a warehouse, or moving and sorting items) (Patel et al., 2022) • Smart PPE (Personal Protective Equipment) to improve the safety of shop floors by alerting the proximity of machines to humans (Adjiski et al., 2019) • Use of wearable devices to monitor the workplace and streamline the resources ((Patel et al., 2022)

4.1 Operational Redesign Challenges

One of the key challenges that Supply Chain 5.0 raises is the fundamental rethinking of supply chain operations. To achieve Supply Chain 5.0, many aspects of supply chain operation, from the supplier and production process to the customer involvement in the product design and chain operations, need rethinking and redesigning. Additionally, ensuring that the processes scale efficiently and are agile enough to adapt to changes that this transition demands is crucial. For example, setting up and implementing hyper-customization can alter a core level of business operations. For every customized product, there might be a need for a different logistics solution (depending on the customized design, shape, and size). For a business, it is not feasible to predict all the different combinations of designs and offerings, enabling various challenges for the operation. Additionally, demand prediction—which has been a driving process in the operations of a supply chain—is complex, making stocking inventory a considerable challenge for Supply Chain 5.0. Indeed, this calls for inventing technologies that create ultimate flexibility—like additive manufacturing and 3D printing technologies—which can potentially lead to other operational challenges, for instance, the time and investment in making the digital inventory, customizing the designs, and converting files into printable formats.

Furthermore, although aiming to provide a high level of personalization, a key component of hyper-customization in Supply Chain 5.0 is automating the bulk of (repetitive steps of) the manufacturing processes. This could be a possible source of confusion. There is a need for theoretical and empirical research to develop frameworks guiding businesses in balancing mass production and customization capabilities in Supply Chain 5.0.

Another challenge associated with mass customization is determining what to do with orders that are cancelled or simply returned by customers. While product return is a common aspect of retailing and sales operations, it becomes a critical challenge when the product is hyper-customized for each specific customer.

4.2 Need for Upskilling Employees

Most of the front-line workers in asset-intensive supply chains are non-tech-savvy. They are used to a well-designed, intuitive workplace that is designed based on productivity and engagement while minimizing training time. Supply Chain 5.0 will likely require new skills and training emphasizing the interaction between machines and operators. Working with robots

sounds fascinating, but workers may need to develop new skill sets, including collaborating with a smart machine. This could also increase work polarization; middle-skill employment is decreasing, and the workforce is split into two communities of highly trained/qualified and low-paid and unqualified employees. Conducting studies to analyse the impact of Industry 5.0 on workforce welfare, job equality, and fairness is needed to understand and minimize the side effects.

4.3 Implementation and Investment Cost

Transforming to a Supply Chain 5.0 demands considerable capital investment. Additionally, training the workforce for new jobs and workplaces can pose additional costs to the companies. In particular, small and medium-sized companies cannot afford the high investment and would be left behind. Additionally, the upstream suppliers in a chain generally have a lower profit margin, reducing incentives for transforming their processes. Developing economic models to support business decisions on the efficient allocation of resources for supply chain transition can be a potential direction for future research. Designing new contracts and joint supplier development initiatives can also support implementing industry 5.0 technologies at the supply chain level.

4.4 Fundamental Operational Mind Shift

Most companies have become very comfortable with their legacy systems and operating processes in a traditional supply chain setting. For Supply Chain 5.0, they need to leave this comfort zone, revamp the existing business strategy, and modify their business models to be customized to meet the requirements of Supply Chain 5.0.

Researching the organizational and cultural aspects and their role in the transition towards Supply Chain 5.0 can guide companies to adapt themselves to this transition better.

5 Concluding Remarks

This chapter aims to elaborate on the concept of Supply Chain 5.0, presenting a framework that defines the key dimension of this concept and discussing key technologies driving this transformation. Although Supply

Chain 5.0 is a new and evolving term, it connects some aspects traditionally perceived as non-related and potentially conflicting in the supply chain literature.

The frameworks presented in this study can support business professionals in understanding the concept better and making well-informed decisions for the transition towards Supply Chain 5.0. Additionally, the research directions discussed in this chapter represent those most pressing and promising from the author's perspective, hopefully encouraging further investigations and continued research in this area. Finally, this study may assist the public authorities and policymakers in devising appropriate policies and developing a regulatory and legal framework supporting companies in their journey towards Supply Chain 5.0.

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Part IV

Digital Design Capabilities for Emerging Technologies in Supply Chains

15

Blockchain Capabilities for Food Supply Chain Traceability

Yasanur Kayikci and Nazlican Gozacan

1 Introduction

Food supply chain traceability (FSC) from farm to fork has become critical for every food system in an increasingly fragmented marketplace due to food safety and quality protection, product counterfeiting, compliance with regulations, and customer concerns about product provenance and authentication (Kayikci et al., 2022b). Before reaching the end consumer, food products frequently pass through a large network of farmers, retailers, distributors, transporters, storage facilities, processors, and suppliers, and must travel through processes such as post-production, harvesting, processing (warehouses, packaging), transportation, distribution, and sales. This long food journey often causes operational complexities and inevitable inefficiencies in both developing and developed countries. In addition to this, today's global challenges such as climate change, natural resource scarcity, population growth etc. point to the need to develop integrated, transparent, and traceable food systems.

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FSC traceability is the backbone of the sustainable production system. Traceability not only responds to *economic claims* arising from miscoordination and mismanagement of operations (resources and processes) such as food insecurity, fraud/tampering and false advertising, spoilage, mass recalls of contaminated items, food loss and waste, food shortages/surplus (Kayikci et al., 2022a) but also responds to *social claims* such as preventing child labour, forced labour and human trafficking in global food operations and supporting wealth generation for local communities (e.g., growers, farmers) (Bhatia et al., 2023; Kayikci et al., 2022a) as well as *environmental claims* such as verifying certified organic, fairtrade, kosher, halal, carbon neutrality (Kayikci et al., 2022a), supporting reforestation (e.g., cocoa plantations) and conservation efforts (Sigles Robert, 2022).

However, traceability alone does not verify the above-mentioned claims. In this sense, it is critical for companies to purposely determine their sustainability goals, to incorporate appropriate data capture and validation methods into their approaches, as well as to choose the right technologies that can meet these needs. Most notably, the implementation of new technologies and innovative solutions plays an important role in transforming the entire FSC system. As a promising tool, blockchain technology with its digital capabilities is well suited to enable traceability of the FSC system, overcome challenges and increase sustainability and resilience in the entire food system. The blockchain can trace the entire FSC from primary manufacture through logistics into manufacturing (Zheng et al., 2023) and downstream operations including wholesale and retail, therefore if contamination takes place, it can easily be detected in real-time. Food monitors have been enabled by blockchain implications in pre-manufacturing and the utilization of substances to modify food authenticity (Pakseresht et al., 2023; Wang & Paliwal, 2007). By utilizing smart contracts, the integration of blockchain in manufacturing could significantly reduce partnership threats (Green, 2018; İndap & Tanyaş, 2023). Moreover, blockchain in logistics procedures enables to maintain of a secured and constant flow of information across the supply chain, minimizing potential hiccups. Retailers and wholesalers can confirm manufacturers' accuracy whenever making transactions by gathering thorough data on factors like batch size and delivery length (İndap & Tanyaş, 2023).

The contribution of this chapter is to close the research gap, to the best of our knowledge, by defining FSC traceability requirements and then associating them with blockchain capabilities. With this motive, the objective of this chapter is to establish traceability requirements throughout the FSC and map the digital capabilities of blockchain technology to meet the traceability

requirements of the FSC to establish farm-to-fork traceability. The following research questions (RQs) are generated to achieve this objective. The RQs are as follows:

RQ1. What are the key traceability requirements for the FSC?
RQ2. How can blockchain capabilities support traceability in the food system?

This study uses a Systematic Literature Review (SLR) with thematic analysis to answer the determined RQs by identifying traceability requirements in the FSC and demonstrating how blockchain capabilities support the FSC to achieve farm-to-fork traceability. Hence, an SLR was conducted to search the relevant literature to determine the traceability requirements in FSC and blockchain capabilities. Then, thematic analysis is done to extract the data. In this method, data obtained and refined in the SLR is analysed to identify, analyse, and report themes (Braun & Clarke, 2006). The thematic analysis focuses on summarizing studies by identifying themes in the literature that are appropriate for the research. Since this study focuses on the traceability requirements of FSC and the blockchain capabilities to support these requirements, the themes identified should be aligned with the purpose of the study.

This chapter is structured as follows. Section 1 outlines the introduction. Following, Sect. 2 defines the research background including Sect. 2.1 explaining FSC traceability and Sect. 2.2 focuses on blockchain-based traceability in FSC. Next, Sect. 3 presents the methodology that consists of Sect. 3.1 describing the systematic literature review, and Sect. 3.2 details the thematic analysis for the systematic literature review. Furthermore, the findings are explained in Sect. 4. Moreover, Sect. 5 discusses the findings including the implications in Sect. 5.1. Finally, Sect. 6 concludes the chapter.

2 Research Background

2.1 Food Supply Chain Traceability

Traceability has different definitions in the literature. The United Nations Global Compact and BSR have suggested a widely accepted definition of traceability according to the International Organization for Standardization (ISO): "The ability to identify and trace the history, distribution, location and application of products, parts, and materials, to ensure the reliability

of sustainability claims, in the areas of human rights, labour (including health and safety), the environment and anti-corruption" (UN-BSR, 2014, p. 6). The emphasis is on providing a sustainable supply chain to maximize product profitability, both by minimizing environmental impacts throughout the entire supply chain and by respecting the social welfare of each supplier's company involved.

The importance of traceability in the FSC lies in its capacity to monitor the movement of food—whether by batch or individual item—both upstream and downstream across all stages of the chain. It enables the assessment of food quality, the identification of safety issues, and, when necessary, the targeted withholding, or recall of hazardous products. Neverthless, global supply chains frequently exhibit insufficient traceability, thereby limiting consumers' ability to ascertain the origin of products and the processes involved in their production (Kayikci et al., 2022a). Thus, it is essential to trace and authenticate the FSC to comprehend its origin in order to pinpoint and manage the causes of contamination in the global FSC (Galvez et al., 2018). Therefore, food product authenticity, which entails adhering to packaging descriptors such as geographical place of origin, date of expiration, conditions of cultivation, cultivating/processing technique and technology, substance, and transparency, is closely tied to these factors (Kör et al., 2021) and a crucial resource for everyone worrying about food quality and safety. Food fraud is an additional prevalent issue that could endanger the customers and could be prevented by effective food traceability technologies (Dos Santos et al., 2019). Food fraud is an increasingly prevalent issue in the modern world, particularly whenever it comes to higher-end or expensive food items like saffron, lobster, and white truffles, etc. In addition to the loss of money, food fraud brings about potential hazards to the well-being of people, for example, if prohibited components are poisonous or infected with pathogenic organisms, or if undeclared alternative solutions along with manufacturing methods result in problems with health such as reactions to allergens (Galvez et al., 2018). Therefore, utilizing distinctive digital identification numbers is extremely important for supplying additional information that makes it possible for food products to be traced throughout the supply chain (Kayikci et al., 2022a; Menon & Jain, 2021).

In line with the previously mentioned problems, traceability requirements that must be met in FSC have been mentioned in the literature. It is important to meet these requirements to ensure the safety of the product and to complete these supply chain processes without risking financial loss as well as customer health. Food origin and ownership, such as where it was cultivated, farmed, or captured, are described by the term "food provenance"

(Hynard, 2020). Additionally, it enables us to comprehend the processes involved in the production and distribution of food. In addition, sustainable operational technologies are that are needed in the process of enhancing traceability in the sustainable supply chain from the perspective of environmental sustainability (Kayikci et al., 2021). The process of structuring, carrying out, and taking care of the movement of a food item from its place of production to its destination is known as food logistics. Transportation, storage, and distribution—including wholesale and retail—are its three primary focal points. These processes involve gathering, temperature, preliminary packaging, preliminary reception on land, shipment, as well as collecting. Since foods are more perishable than other types of goods, logistics requirements additionally take into account the accessibility of conditions of temperature, relative humidity, and moisture, along with additional factors (Kamath, 2018). Moreover, food loss takes place as a result of issues occurring during the steps of the production process, place of storage, the process, and distribution prior to the item of food arriving at its intended recipient (Kayikci et al., 2021). When food has been approved for consumption however is purposefully wasted during the buying process or consumption phases, it is referred to as food waste (Harvard, 2023). Moreover, retailers and businesses are under pressure to find environmentally friendly packaging solutions in order not to generate food waste (Ada et al., 2023) as well as since the comprehensiveness of a food product's characteristics and qualities that the potential customer considers appropriate is considered as food quality, which is another criterion. Dimensions, packaging, form, colour, appearance, and uniformity are a number of the aforementioned food quality characteristics. Due to its direct connection to the well-being of individuals, food quality is a critically important concern in a supply chain (Vangipuram et al., 2022). Furthermore, in the supply chain, the three considerations of infection, deterioration, and compromise contribute to food safety a highly important issue for society (Joo & Han, 2021). Food safety is a developing widely known healthcare problem and implementing the new infrastructure could facilitate a number of corporate processes, such as enhanced operational monitoring for handling food recalls and/or food quality (Casino et al., 2021). Every stage of the food chain manufacturing requires an in-depth comprehension of the safety of food. The role-specific risks related to food safety that have an impact on those who process foods must be properly comprehended, and they must be all set to take action when difficulties come to light (Prashar et al., 2020). The practical definition of food security in public planning consists of three critically important components: food stability (a person's

capacity to receive food throughout times), food access (the cost and transportation of food), and food availability (the accessibility to food generally) (Lin, 2019). The duty to guarantee food security is recognized as being exchanged by all parties involved, particularly consumers, associated administrations, as well as producers of food and merchants (Duan et al., 2020). From the standpoint of environmental sustainability, carbon tracking through FSC is essential to investigate the transforming capability of food traceability for encouraging a food system that is climate-friendly (Liu et al., 2022) taking approaches to farming and application of fertilizers into account (El Hathat et al., 2023). Businesses may identify and monitor the amount of carbon they produce using carbon bookkeeping, which also enables them to decide on decarbonization initiatives with knowledge. Companies may identify the locations where they are emitting the most carbon dioxide by using carbon accounting techniques. Moreover, the term "coordinates" refers to the geographic locations of information regarding location (Galvez et al., 2018), and to confirm that there hasn't been any deforestation (Bager et al., 2022), they are frequently stated in latitude and longitude form and monitor the implementation of sustainable farming techniques. The demand for security in finances to confirm payment information is an element of the economic sustainability perspective (Mangla et al., 2021; Salah et al., 2019; Saranya & Maheswari, 2022) in order to validate allegations of ethical labour carried out by farmers, manufacturers, and employees (Collart & Canales, 2022), liveable wages and living earnings (Friedman & Ormiston, 2022). Lastly, workplace safety is a prerequisite for social sustainability (Luzzani et al., 2021) and environmental safety to guarantee that there are no child workers (Bager et al., 2022; Bai et al., 2022), no involuntary or slave labour exists in FSC (Bai et al., 2022; Mangla et al., 2022).

2.2 Blockchain-Based Traceability in Food Supply Chain

The integration of blockchain technology into farm-to-fork FSC stages is demonstrated in Fig. 1. Pre-harvest and post-harvest are included in the primary production stage. Agriculture, animal husbandry, and fishing are considered pre-harvest activities. Cleaning, sorting, handling, and storing the collected food are considered post-harvest activities (Vangala et al., 2020). Food monitors are currently in a position to enhance food authenticity across fields such as assessment of quality, post-harvest treatment, and such keeping track of poisonous components in livestock feed and food because of blockchain improvements in pre-processing, modelling approaches, and

the use of chemicals (Paksereshtet al., 2023; Wang & Paliwal, 2007). Throughout pre-harvest and post-harvest activities, understanding and information technology provides real-time critically important knowledge, analysis, and innovative innovations that assist with the food supply chain around the world (Chandan et al., 2023). Pre-harvest stages must be monitored, taking into account circumstances about the various types of chemicals and pesticides utilized, the farm's location and additional information, the farmer's specifics, the amount of product provided, the quality and diversity of the food, and the timing of planting, harvesting, and loading (Gupta & Shankar, 2023). The performance of farm machinery and vehicles is subsequently decided by the smart contract depending on its specified circumstances, weather forecasts, water, soil, and other manufacturing components (Galvez et al., 2018). Furthermore, 14% of the world's food production is lost at the post-harvest stage (Vangala et al., 2020). Thus, blockchain functions as a sensor-enabled infrastructure that integrates with real-time stored information to further develop the manufacturing process, maximize shelf life, implement price adjustments based on expiration dates, and ultimately prevent food loss and waste—a crucial aspect of post-harvest handling and product storage (Paksereshtet al., 2023) in FSC as shown in Fig. 1.

After primary production, the following stage in the FSC is process and manufacturing. By using smart contracts to lessen partner jeopardy, the application of blockchain technology potentially assists with the manufacturing industry (Green, 2018; İndap & Tanyaş, 2023). Avoiding the intermediary's issue is another reason why the smart contract is advised (Surasak et al., 2019). With the implementation of smart contracts, agreements and transfers of assets may be automated without necessitating a third party to be accepted. This indicates that the rationale of smart contracts implemented when it

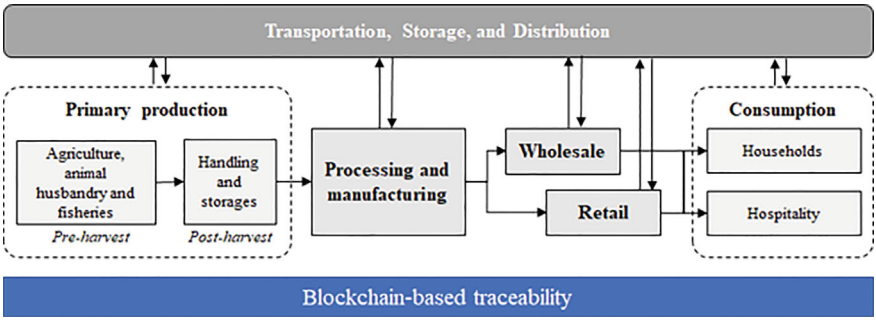

Fig. 1 Integration of blockchain technology into farm-to-fork FSC stages (*Source* Authors)

obtains digital assets is transparent to all parties who are interested. Moreover, monitoring and reliably managing digital and physical assets is possible with the assistance of blockchain asset management, a valuable equipment. A smart contract is a form of computer software that operates on all of the FSC's consensus processes and is stored on the blockchain (Zhao et al., 2019). In a decentralized information system, a consensus mechanism is any strategy that establishes protection, reliability, and acceptance (Yadav et al., 2023). One of the main characteristics of blockchain is decentralization, which makes it possible to create structured frameworks for safe data sharing between departments and business entities. Environmentally friendly preservation of water, processing of food, accessibility, and security of information are all facilitated by the FSC blockchain technology (Yadav et al., 2023; Zhao et al., 2019). Every participant on a blockchain may see every transaction that has occurred. Transactions between two participants, such as a producer and distributor, could not be transparent to a third party in typical supply chains. Every transaction in the supply chain is recorded as an unchangeable, tamper-proof block utilizing blockchain technology. Moreover, the instantaneous, trustworthy tracking mechanism of blockchain can identify dishonest suppliers, unethical hiring practices, and counterfeit merchandise (Luzzani et al., 2021; Yogarajan et al., 2023).

If contamination occurs, it may be quickly identified instantaneously due to the blockchain-based traceability technology, which can follow every stage of the FSC processes from primary production through processing and shipment to system operation (Zheng et al., 2023). By emphasizing visibility and the flow of product tracking throughout the whole supply chain using blockchain, businesses can easily set themselves apart from competitors. Transparency throughout FSCs might be greatly enhanced by blockchain technology. Blockchain technology exhibits more homogeneity in comparison to the conventional electronic traceability method which facilitates data exchange and interoperability more than conventional traceability methods (Iftekhar et al., 2020). Furthermore, promptly determining the source of any contaminated elements in food can assist in minimizing the negative impact of consumer complaints and protect a company's image (Galvez et al., 2018). As a consequence, blockchain enables the FSC increasingly visible in order to encourage food safety (Zheng et al., 2023). Temperature and humidity are substantially more important factors in cold chain distribution, if one of them crosses the threshold set out in the smart contract, an alert will be forwarded to the management to modify the distribution requirement (Galvez et al., 2018).

Following the manufacturing phase, blockchain technology possesses the opportunity to strengthen the ability to connect specific food carriers towards outbreaks of illness caused by food, resulting in fewer events and fewer people affected throughout the food logistics process (İndap & Tanyaş, 2023). Thus, by enabling the secure and uninterrupted flow of information across the supply chain, blockchain technology can assist logistics operations in minimizing any potential disruptions. In the FSC, blockchain is beneficial for maintaining supply–demand equilibrium while maintaining visibility. Every user on a blockchain may see every activity that has occurred (Feng et al., 2020). Every transaction in the supply chain is recorded as an unchangeable, unbreakable block by employing blockchain technology (Lin et al., 2019). Retailers and wholesalers can confirm manufacturers' precise capacity when making transactions by acquiring thorough knowledge about variables such as batch amount and shipping time (İndap & Tanyaş, 2023). In the FSC, downstream merchants are more accountable for food transparency than upstream merchants since buyers frequently accuse merchants of being dishonest if theft of food occurs (Cao et al., 2021; Zheng et al., 2023). Therefore, Smart contracts and blockchain technology can provide assistance in addressing these kinds of obstacles by developing decentralized environments of integrated suppliers, producers, wholesalers, retailers, as well as buyers (Singh & Raza, 2023). Blockchain is an effective instrument to prevent food fraud by, for example, confirming the origins geographically and biologically and using chromatographic fingerprinting together with chemometric methods, which are an effective means of identifying food fraud (Galvez et al., 2018). Blockchain-based platforms that incorporate FSCs collect data on a range of food items to establish food safety requirements for countries, establish auditing authority, and prohibit the distribution of illicit goods (Kayikci et al., 2022a).

3 Methodology

3.1 Systematic Literature Review

The relevant literature was searched as part of a systematic literature review (SLR) to determine the blockchain's capabilities and the traceability requirements for FSC. The SLR corresponds to the following list of Tranfield et al. (2003) recommended phases:

(i) Identifying academic databases: Web of Science is identified as the database.

It is chosen to perform the research using just the high-quality publications that are included in the aforementioned databases.

(ii) Identifying search string: The following search string is utilized in the title, abstract, and keywords of the databases to explore data:

Search string. TITLE-ABS-KEY {"blockchain" AND ("trace*" OR "track*") AND ("food" OR "agricultur*" OR "agri-food" OR "agro-food" OR "milk" OR "dairy" OR "meat" OR "poultry" OR "egg" OR "grain" OR "seafood" OR "fish" OR "aquaculture" OR "beverage" OR "oil") AND ("supply chain" OR "value chain" OR "food system" OR "food chain")}. Limit to: Doctype (article and review); Language (English); Year (2015–2023).

This search string can cover the blockchain and traceability-related keywords for "blockchain; blockchain technology; traceability" and food supply chain-related keywords for "food supply chain; food system; food chain; agriculture; animal; husbandry; fisheries; milk and dairy; fresh fruits and vegetables; grain (wheat, oats, rice, corn, barley, sorghum, rye); meat and meat products; fresh fish and seafood; poultry (chickens, turkeys, geese, ducks) and eggs; beverages (wine, spirit, beer, soft drink, water, juice, ready-to-drink tea, and coffee); oil (olive, coconut, palm, rapeseed, soybean, sunflower); frozen/cooled/chilled food" from the literature.

304 preliminary studies were located from selected databases employing the search strings that were implemented.

(iii) Screening of the studies: The duplicates were eliminated. Then, the inclusion and exclusion criteria that were listed below were used.

Inclusion criteria: Title, abstract, and keywords shall demonstrate traceability, FSC and blockchain as the clear focus/object of the research. The exploration has not been restricted to particular publications in order to encompass all conceivably pertinent research. Since English is the primary language used in supply chain management and logistics research, articles must be published in that language. Articles must appear in journals that are subject to peer review. To provide quality control, only articles and review papers from peer-reviewed publications were accepted.

Exclusion criteria: Research that concentrates on unrelated topics, such as electronics, medicines, or pharmaceuticals, are not included. Chapters and conference papers are not included. The traceability from

an FSC and blockchain technology standpoint is the main topic of this assessment.

(iv) Selection of primary studies: In accordance with the introductions and conclusions, the authors evaluated the research studies. The remaining papers' whole texts were then reviewed by them.

The final paper list did not include the research studies that are outside of its focus. As a result, the primary research list was established, and A total of 209 papers were selected for a thematic analysis.

3.2 Thematic Analysis for SLR

Thematic analysis is employed to extract the data from the publications on the final list. This approach analyses data that have been acquired and improved in the SLR to discover, analyse, and report themes (Braun & Clarke, 2006). The thematic analysis focuses on summarizing studies by identifying themes in the literature that are appropriate for the research. Since this study focuses on the traceability requirements of FSC and the blockchain capabilities to support these requirements, the themes identified should be aligned with the purpose of the study. The steps of the thematic analysis method are as follows (Braun & Clarke, 2006):

(i) Familiarize the data: The authors get familiarized with the data by repeated reading of the data. This will also help us understand the traceability requirements of FSC and the blockchain capabilities.

(ii) Code the data: Initial codes are generated accordingly to answer the research questions. For this, this circle is applied: Identify the requirement, identify where this requirement is needed and determine how the blockchain capability(s) support it. Categorize the study of how the requirement is identified as empirical or non-empirical. Repeat the steps, until all traceability requirements and blockchain capabilities identified in the literature are mapped.

(iii) Translating the codes into themes: All codes that are identified in the previous step are studied, narrowed down, and categorized.

(iv) Reporting the themes: All codes are mapped. Nine themes were created from the analysed data, and the answers were evaluated to ensure the research gap was answered correctly. The results are presented in the findings section.

4 Findings

In this section, the results of thematic analysis are presented. The refined data are analysed, and the traceability requirements are identified. Then, the findings are categorized into three categories according to sustainability pillars. This research demonstrates how blockchain capabilities assist the FSC accomplish farm-to-fork traceability and employs SLR with thematic analysis to identify traceability requirements in the FSC. The SLR procedure proceeded is illustrated in Fig. 2.

Technological developments and innovations are key to improving FSC traceability system. Especially, blockchain capabilities have a great potential to support traceability requirements in FSC. Thematic analysis is performed to analyse the final collection of decided-on publications. We used the following blockchain digital capabilities identified by Kayikci et al. (2022a) in this research: (i) asset management, (ii) auditability, (iii) authentication/provenance, (iv) consensus mechanism, (v) data standardization/security/sharing, (vi) disintermediation/decentralization, (vii) immutability and transparency, (viii) integration/interoperability, (ix) smart contracts, (x) tokenization, (xi) traceability/visibility. Figures 3 and 4 show the themes of food traceability requirements and blockchain capabilities in the notion of the emphasis of this paper.

The mentioned percentages of blockchain capabilities are displayed in Fig. 5. The most mentioned three blockchain capabilities are visibility, smart contracts, and integration. Visibility is the most often listed capability (29% in 168 out of 209 publications). With 17%, smart contracts are the next capability. Integration is the third capability given.

Figure 6 demonstrates the FSC traceability requirements mentioning percentages. The most mentioned first three FSC traceability requirements are logistics, provenance, and quality. The greatest frequently mentioned requirement is food logistics (22% in 144 out of 209 publications). Food provenance appears afterwards with 19% of the requirements. The third requirement is the quality of the food.

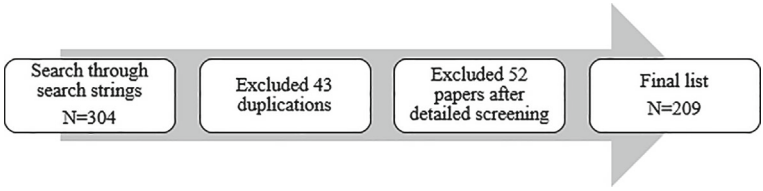

Fig. 2 SLR process (*Source* Authors)

Fig. 3 Themes of blockchain capabilities (*Source* Authors)

The interactive mention of blockchain capability and FSC traceability requirements is visualized in portions in the alluvial diagram in Fig. 7. When the interactions mentioned in the literature were analysed, the interaction of visibility and food logistics is the most mentioned interaction with 44%, and the second interaction is between visibility and provenance with 36%. Blockchain's visibility capability, which receives the most mention, is most associated with food logistics. The second most mentioned blockchain capability, smart contracts, is associated with financial security. The integration capability of blockchain is most associated with food logistics. Immutability, a blockchain capability, is also most frequently paired with food provenance. Additionally, food logistics are most frequently combined with blockchain capability called disintermediation. The blockchain feature of consensus is commonly linked to food provenance. Moreover, financial security and tokenization, a blockchain capability, are frequently combined.

11 blockchain capability themes (Fig. 3), 9 food traceability requirements themes (Fig. 4), and the relationship between these themes, which emerged as a result of examining 209 scientific articles, were also examined within the

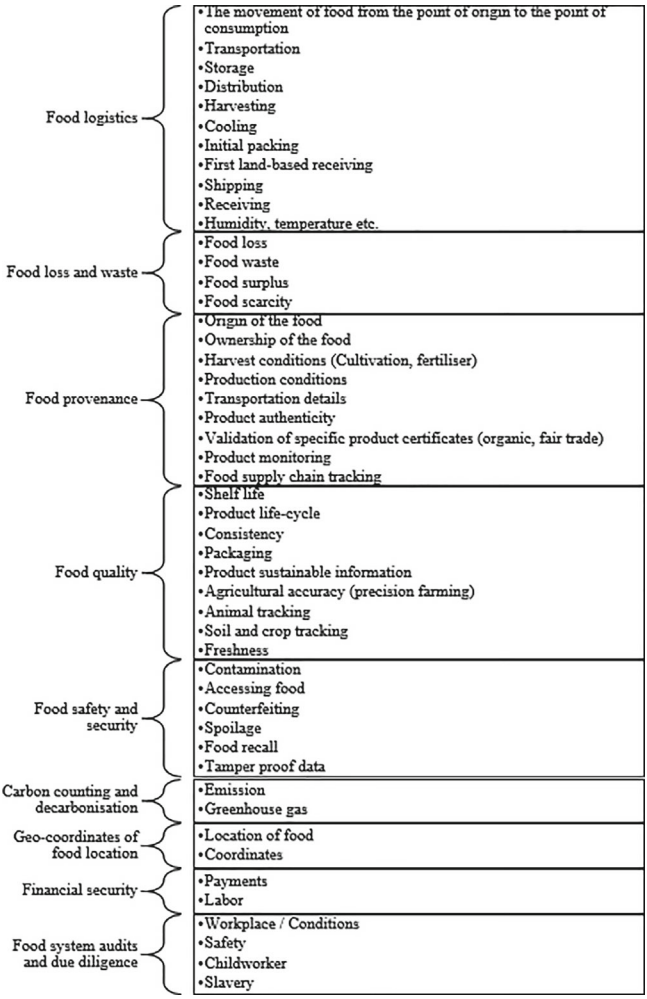

Fig. 4 Themes of food traceability requirements (*Source* Authors)

scope of sustainability's triple bottom line. The following sections explain this relationship and the details of the findings are as follows:

4.1 Traceability Requirements for Economic Claims

Transparency, estimation of demand, operational surveillance, as well as information management are all supported by traceability, which facilitates effective information transfers and flow among numerous FSC members (Agrawal et al., 2021; Sumarliah et al., 2023). Based on the findings, the

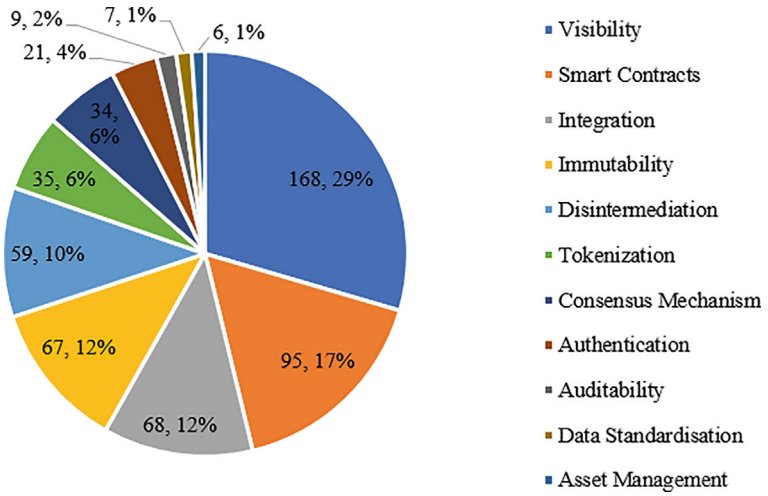

Fig. 5 Blockchain capability mentioning (*Source* Authors)

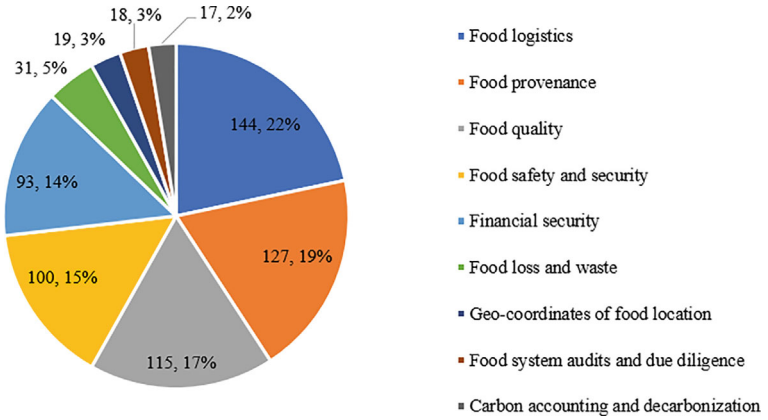

Fig. 6 FSC traceability requirements (*Source* Authors)

most mentioned first three FSC traceability requirements are food logistics, food provenance, and food quality. When the interactions mentioned in the literature were analysed, the interaction of visibility and food logistics is the most mentioned interaction with 44% and the second interaction is between visibility and provenance with 36% as it is mentioned before. The FSC must enhance information visibility and authentication as an aspect of the sustainability management strategy in order to detect and eliminate sources of contamination (Feng et al., 2020; Galvez et al., 2018; Saha et al., 2022). As seen in Fig. 7, considering the most mentioned blockchain capability which is visibility, a blockchain-based food commodity traceability

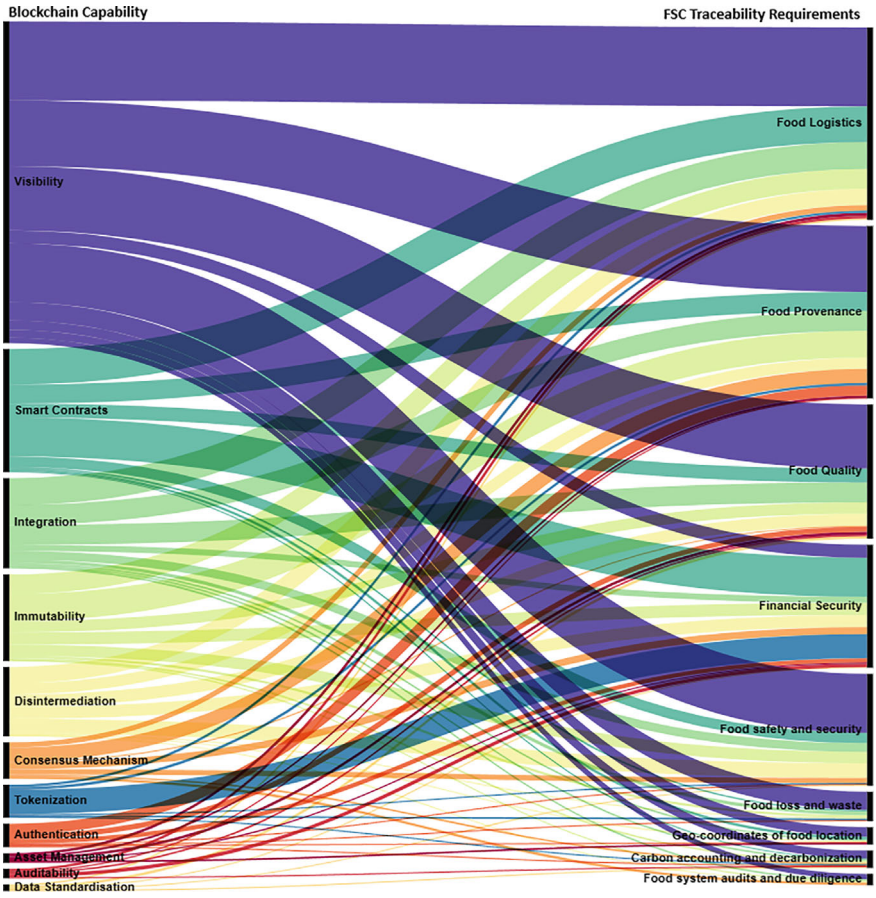

Fig. 7 Alluvial diagram of the interaction (*Source* Authors)

mechanism can keep track of every phase of the product's manufacturing process, from the first stages of manufacturing to the framework's processing and logistics, and in the event of a contamination, it can be detected early enough to be visible in the FSC and ensure food quality, safety and security (Sunny et al., 2020; Zheng et al., 2023). Consequently, to ensure the safety and quality of food products, real-time surveillance and documentation of humidity, temperature, and handling, through farm-to-fork is essential as well as to enhance profitability, efficiency, and sustainability when it is achievable by utilizing sensors and blockchain technology (Casino et al., 2021; Patel et al., 2023). Blockchain has the capability to improve the food system's traceability and visibility whenever implemented appropriately. Thus, sustainability issues may be addressed via minimized food loss and improved FSC networks (İndap & Tanyaş, 2023).

The blockchain makes it possible to trace the authenticity of food items, store, and authenticate electronic addresses using legally encrypted files, and implement smart contracts to minimize transactional costs as well as fraud (Pakseresht et al., 2023). Furthermore, smart contracts are implemented in logistics procedures to eliminate the requirement for a mediator or third party in the contractual process (Choi et al., 2019; Tavana et al., 2023). Additionally, smart contracts on blockchain contribute an immutable recording of interactions, which might enhance the quality and safety of food (Antonucci et al., 2019; Pandey et al., 2022; Xu et al., 2020).

Fundamental components for the social and economic aspects of sustainability include immutability, reliability, traceability, and visibility, as blockchain can detect fraud and abuse, reduce food waste, and recalls, and enforce food safety and rights for individuals (Saha et al., 2022). The immutability of blockchain technology and its capability to provide shared knowledge accessibility across FSC actors can be highly advantageous in attaining FSC knowledge transparency (Rogerson & Parry, 2020; Treiblmaier & Garaus, 2023). Ensuring food safety and quality verification may be achieved by storing farm animal genomic and proteomic information on the blockchain, which can present immutable information on whether the meat and dairy items are generated from either identical or distinct animals (Patel et al., 2023).

Food safety, integrity, security, traceability, quality, and sustainability are just a few of the issues in FSCs that may be resolved with timely and dependable accessibility to pertinent data through a consensus-building process (Ali et al., 2021; Tavana et al., 2023; Yadav et al., 2023). Blockchain-based traceability techniques document details of transactions on blockchains during transportation utilizing a consensus mechanism within FSC organizations, allowing buyers to discover relevant data and confirm the legitimacy of the food items (Pandey et al., 2022). System alterations along with frauds are prohibited by the structure's enhanced reliability and the development of a consensus-based certification mechanism made possible by the decentralised FSC (Mangla et al., 2022).

4.2 Traceability Requirements for Social Claims

Blockchain technology may be incorporated into FSCs to enhance company productivity in social, economic, and environmental domains (Choi et al., 2020; Mangla et al., 2021). Social information quantifies the breadth and depth of social data sharing and visibility in the FSC utilizing blockchain technology. Stakeholders in the FSC may contribute information about

social consequences, human rights, child workforce, and labour regulation (Bai et al., 2022). In order to ensure sustainable labour standards and to validate and track FSC operations, blockchain technology may also enable and establish an environment that supports these objectives (Friedman & Ormiston, 2022) to confirm that there are no forced labour or child labourers in the FSC. Although blockchain technology has the potential to eliminate middlemen, one obstacle to its widespread acceptance is the belief that it would result in reductions in employment (Dutta et al., 2020; Tharatipyakul & Pongnumkul, 2021). Blockchain eliminates child employment, enhances food safety and data sharing, satisfies buyer requirements for more secure food, and brings an end to unethical conduct (Chandan et al., 2023). The fact that smart labelling on blockchain stops illicit fishing is one example of safer food (Tharatipyakul & Pongnumkul, 2021). Blockchain can benefit farmers by facilitating mutual control access and bringing visibility to FSC financing processes (Awan et al., 2021). Blockchain can assist with seamless money transfers along with information collection (Thiruchelvam et al., 2018) by leveraging smart contracts to carry out and validate monetary payments automatically (Galvez et al., 2018) to cultivators, farmers, and employees to substantiate their assertions of sustainable earnings, sustainable wages, and ethical labour. Moreover, the inherent characteristics of blockchain include disintermediation, visibility, and information collecting and data storage, which may further enhance the administration of personnel by offering additional details on human rights and labour laws along the whole FSC as well as safety and health at work, every aspect which can facilitate better management and application of sustainability (Luzzani et al., 2021) and prevent slavery (Howson, 2020).

4.3 Traceability Requirements for Environmental Claims

Environmental data analyse how much and where environmental data are disclosed and visible throughout the FSC using blockchain-based technology (Bai et al., 2022) to confirm the elimination of deforestation and monitor consumption of energy, waste, and emissions, including sustainable farming methods in the FSC. The biological, chemical-based, and ecological characteristics of food are key components of quality monitoring procedures and may be employed to verify the organic farming process, identify the source of food, or detect the inclusion of pesticides in food items (Katsikouli et al., 2021). Given that current FSC practices have resulted in significant deforestation, water shortages, or soil erosion, they cannot achieve sustainable

production of food and agriculture (Awan et al., 2021), Technological framework for a variety of purposes related to environmental sustainability is being provided via blockchain (Howson, 2020). The technology may be utilized to support renewable energy layouts, mitigate global warming, produce food, track e-waste, register land, maintain biodiversity, and create sustainable FSCs (Howson, 2019). For instance, Puma provides data available regarding the reduced consumption of waste, water-based, land, along with carbon pollution available (Bai et al., 2022; Gröschl et al., 2019). To fulfil the growing need for food, increasing land for farming may lead to deforestation, which raises greenhouse gas emissions and diminishes ecosystems.

Farmers possess little knowledge of the fluctuating availability of high-quality insecticides and fertilizers (Kamath, 2018). Blockchain technology may cause such actions to address concerns (Kamath, 2018). Food manufacturing and distribution phases' environmental impacts may be tracked using a blockchain-based approach (Chandan et al., 2023) that encompasses the fertility of the soil, fertilizers applied, the food's biological profile, the climate during the time of production, etc. (George et al, 2019). Moreover, blockchain technology may also be leveraged to investigate how food traceability might revolutionize the FSC and render it less harmful to the environment. Information communicated across the FSC may be confirmed and kept immutably by using automated consensus methods of blockchain, which minimizes the chance of data tampering to almost nothing. Blockchain is being promoted more and more for a variety of services and sectors, such as environmentally friendly fishery management, lowering the use of plastic, decreasing slavery, and visible procurement (Howson, 2020) to maintain food traceability to create a more climate-friendly food system.

5 Discussion

To define traceability requirements across the FSC and demonstrate how blockchain capabilities empower the FSC to achieve farm-to-fork traceability, this research employs a comprehensive literature review and theme analysis. Traceability supports operational monitoring, demand estimates, transparency, and information management by facilitating efficient information flow and transfers across several FSC participants (Agrawal et al., 2021; Sumarliah et al., 2023). For the FSC traceability requirements, there are nine themes. Blockchain technology may be applied to FSCs to improve business efficiency in the social, economic, and environmental aspects while satisfying these requirements (Choi et al., 2020; Mangla et al., 2021). An

FSC traceability system built on blockchain technology can monitor each stage of the product's production, from the initial phases of production to the handling and transportation of the system. In the case of contamination, food quality, safety, and security can be guaranteed since they can be discovered early enough to be seen in the FSC (Sunny et al., 2020; Zheng et al., 2023). Furthermore, the most mentioned three blockchain capabilities are visibility, smart contracts, and integration meanwhile the most mentioned first three FSC traceability requirements are logistics, provenance, and quality. When the interactions mentioned in the literature were analysed, the interaction of visibility and food logistics is the most mentioned interaction with 44%, and the second interaction is between visibility and provenance with 36%. To identify and remove the origins of contamination, the FSC must improve the visibility of data and authentication as part of the sustainability management process (Feng et al., 2020; Galvez et al., 2018; Saha et al., 2022). Therefore, real-time monitoring and recording of temperature, humidity, and handling from farm to fork is crucial for ensuring the safety and quality of food products. It is also necessary to improve economic viability, performance, and sustainability, when possible, by leveraging sensors and blockchain technology (Casino et al., 2021; Patel et al., 2023). Additionally, the blockchain enables the use of legally encrypted data to store and validate virtual addresses, track the provenance of food products, and use smart contracts to reduce fraud and transaction costs (Pakseresht et al., 2023). Furthermore, blockchain-based smart contracts provide an immutable log of transactions that might improve food safety and quality (Antonucci et al., 2019; Pandey et al., 2022; Xu et al., 2020). Using blockchain technology, social data measures the volume and scope of social data exchange and visibility in the FSC. Furthermore, blockchain technology may make it possible to create an atmosphere that promotes these goals (Friedman & Ormiston, 2022) to verify that the FSC does not employ child labour or forced labour. Thus, blockchain may be used to conserve nature, monitor electronic waste, generate food, boost renewable energy style, reduce global warming, and establish sustainable FSCs (Howson, 2019). However, as seen in the results, the least mentioning of FSC traceability requirements are food system audits and due diligence, and carbon accounting and decarbonization.

5.1 Implications

This research contributes literature by defining FSC traceability requirements and then associating them with blockchain capabilities with a map of the

digital capabilities of blockchain technology to meet the traceability requirements of the FSC to establish farm-to-fork traceability. There are a few things to keep in mind while analysing the results in terms of managerial implications. Themes extracted as a result of SLR, and thematic analysis enable management to notice the details required in the FSC tracing plan. It also touches upon which blockchain capabilities can be applied at which stage when including blockchain in this plan. By associating blockchain capabilities with FSC traceability requirements, points that may be unmentioned interactions and research gaps in the literature come to light. From this perspective, food system audits and due diligence, as well as carbon accounting and decarbonization, are the requirements for FSC traceability that are mentioned the least in the results. Additionally, there are a few issues that can be seen as research gaps on this subject: smart contracts and carbon accounting and decarbonization; data standardization and carbon accounting and decarbonization; consensus and quality; tokenization and food safety.

6 Conclusion

Today, every food establishment needs to be able to trace FSC from farm to fork. Sustainable production processes are built based on traceability in FSCs. However, traceability in and of itself does not support the sustainability conditions. To meet these requirements, businesses must consciously determine their sustainability objectives, include appropriate data collection and validation techniques in their processes, and use the right technology. With its digital capabilities, blockchain technology is a viable way to enable FSC traceability, overcome obstacles, and boost robustness and sustainability across FSC. To define traceability requirements across FSC and demonstrate how blockchain capabilities empower FSC to achieve farm-to-fork traceability, this research employs a systematic literature review and thematic analysis. To achieve the FSC's traceability requirements and develop farm-to-fork traceability, this research extends the body of literature by establishing the requirements for FSC traceability and then connecting them with the digital capabilities of blockchain technology through a map. Nine topics regarding the requirements for food traceability are identified in light of the findings. Furthermore, the most mentioned three blockchain capabilities are visibility, smart contracts, and integration. The most mentioned first three FSC traceability requirements are logistics, provenance, and quality. When the interactions mentioned in the literature were analysed, the interaction of visibility and food logistics is the most mentioned interaction with 44%,

and the second interaction is between visibility and provenance with 36%. The utilization of a single database as the source during the SLR procedure is the research's main limitation. However, when assessing the findings in terms of organizational impacts, there are a few points to consider. The specifics needed in the FSC tracing plan are made visible to executives with thematic analysis and themes that are taken from SLR. It also discusses the specific blockchain features that may be used when integrating blockchain into this strategy and when to do so. The connection between FSC traceability standards and blockchain capabilities highlights research needs and potentially overlooked linkages in existing literature. From this perspective, food system audits and due diligence, as well as carbon accounting and decarbonization, are the FSC traceability requirements that are less addressed in the findings. Furthermore, a few topics might be recommended for further research: smart contracts and carbon accounting and decarbonization; data standardization and. carbon accounting and decarbonization; consensus and quality; tokenization and food safety.

Data Availability Statement Data from the reviewed 209 articles that supported the SLR of this study are available on request from the corresponding author.

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Trust Design Framework in Blockchain-Enabled Supply Chains

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1 Introduction

The increased digitalization of supply chains creates vital opportunities for global and sustainable economic growth. Technological innovations such as real-time tagging and tracing of physical items, mobile business-to-business applications, and foundational technologies such as cloud data storage permit extensive data access and analysis. These data offer profound insights into current performance and predictions of both physical and digital processes within supply chains. The growing sophistication of digital technologies such as neural-network-based machine learning, blockchain, and the Internet of Things (IoT) amplifies efficiency gains through heightened data availability and potential transparency in modern supply chains (Patterson et al., 2003).

Blockchain technology, introduced in 2008, plays a pivotal role in shaping transparent processes and holds the promise to foster greater trust within supply chains (Blossey et al., 2019; Wang, Han et al., 2019; Wang, Singgih et al., 2019). It functions as a distributed database, continually expanding

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with a confirmed list of data records maintained by nodes in a peer-to-peer network. In this structure, no single party or entity maintains direct control over the data records, ensuring the accuracy and integrity of the available information (Yli-Huumo et al., 2016). While offering significant business advantages and cost-saving opportunities, the utilization of blockchain technology in supply chains may also raise challenges. This is because of an increased vulnerability of organizations when disclosing and sharing transaction-related information with a completely open or pre-defined network of organizations (Wen et al., 2019).

Finding a balance between the business opportunities and vulnerabilities introduced by the utilization of blockchain technology is increasingly becoming a central focus of new research. This shift is evident as both research and proposed practical applications of the technology have moved beyond the initial hype that surrounded it and its potential use cases. Recent studies are honing in on identifying specific scenarios where blockchain technology can genuinely enhance supply chains, taking into account existing IT applications, the imperative for interoperability, managerial considerations, and the challenges tied to scaling blockchain-based supply chain solutions (Korpela et al., 2017; van Hoek, 2019a, 2019b).

In this evolving landscape, trust assumes a pivotal role in enhancing performance and streamlining logistics, particularly when collaboration is paramount. Few researchers have highlighted that while blockchain has the capability to instil trust, such as technically through time stamps and historically through shared data, the aspect of social trust between parties (e.g., assurance of non-opportunistic behaviour) within blockchain technology is rarely discussed in the blockchain space (Wan et al., 2022). Consequently, trust is often narrowly perceived as a purely technical facet in the context of the blockchain, focusing among others on the aspect of traceability and data integrity (Malik et al., 2019). This perspective overlooks the essential connections between technical, data, and social trust relationships in an exchange network (Wan et al., 2022).

Trust reflects beliefs that the other party has suitable attributed for performing as expected in a specific situation, regardless of the ability to monitor or control that other party (Mayer et al., 1995). When placing trust in a person or organisation, belief in the other party's ability, integrity, and benevolence (human-like trust attributes) determines the level of trust in a specific situation (Lankton et al., 2015). When trusting a technology, scholars have shown that individuals seek system-like trust attributes such as functionality, reliability, and helpfulness in performing a task (Lankton et al., 2014).

In this paper, we understand trust, when utilizing blockchain in supply chains, as emerging from system-like trust attributes. The technical features (e.g., time stamps, consensus mechanism) of the technology lay the basis for trust emergence in supply chains. Concurrently, design aspects (e.g., the extent of automation through smart contracts) alongside technical features are prerequisites for the development of trust-intensive business opportunities and subsequent situations where human-like trust attributes take precedence. In this paper, we use an affordance lens to understand this interplay. In particular, we ask: *What are the trust-related technical, design, and business affordances of blockchain technology in supply chains?*

We address our research question through a structured literature review (Webster & Watson, 2002) aimed at capturing and analysing action potentials relating to trust in existing literature. Trust-related action potentials are then organized according to the three-layered affordances framework proposed by Ostern et al. (2020), categorizing them into technical, design, and business affordances. The outcome of this analysis is a distinctive set of technical, design, and business affordances that interconnect and shed light on trust-related opportunities within supply chains facilitated through implementing blockchain technology.

The remainder of this chapter is structured as follows: The next section discusses related work, mainly, the role of trust in supply chains and blockchain technology. We proceed with a description of the research approach and method, including the data collection and analysis. We present the findings, i.e., technical, design, and business affordances leading to trust in supply chains in the following. The findings and limitations of the research approach are discussed in the following section. Eventually, a conclusion and outlook are provided.

2 Related Work

The following section provides a summary of related work, emphasizing blockchain technology as an emerging force with the potential to impact supply chain operations. Furthermore, we narrow our focus to trust and its pivotal role in expediting the digitalization of supply chains.

2.1 Blockchain Technology

Blockchain is a distributed ledger technology where data are stored in blocks, secured using cryptographic principles, and validated through nodes in a

peer-to-peer system. Consequently, blockchain data are difficult to manipulate or alter, rendering supply chain transactions once referenced on a blockchain, nearly immutable (Chang et al., 2019). The verification and immutability of data, without the requirement for a trusted third party, are what makes a blockchain solution intriguing to both academics and practitioners, sparking discussions about its applications in various contexts, including supply chain support (Chang et al., 2019; Wen et al., 2019; Xia & Yongjun, 2017).

Although blockchain technology gained widespread popularity in 2009 through the e-payment scheme Bitcoin, its components already existed and were well-known to the computer science and information systems community. Therefore, blockchain technology is often described as an amalgamation of IT artefacts that constitute the technology (Du et al., 2019). Drawing upon Du et al. (2019), Table 1 summarizes the main components of blockchain technology.

Given those components, several use cases for blockchain technology in supply chains have been proposed. For instance, the immutable data log in combination with the IoT (e.g., by using sensors or RFID tags) allows the creation of a traceable and integer system where real-time food tracking for all supply chain parties as well as customers could be achieved. This is expected to increase food safety and the reliability of the information on the origin and supply routes of products (Chowdhury et al., 2020; Malik et al., 2019). Similar approaches have been proposed for drug supply chains where blockchain features could be exploited to avoid drug counterfeits (Sahoo et al., 2020). Others have proposed using blockchain technology to increase the efficiency and optimize throughput times of supply chain processes by replacing operational intermediaries with smart contracts that automatically enforce business rules (Chang et al., 2019). Eventually, blockchains have been hypothesized to increase efficiency and competitive advantages through full transparency of supply chain transactions for organizations that are part of the peer-to-peer network (Biggs et al., 2017; J. Lee et al., 2018).

Although these opportunities are frequently discussed, existing research has primarily focused on developing architectures for individual use cases or on conceptually justifying the benefits of blockchain-based solutions in supply chains. This paper, therefore, addresses a research gap in identifying case-independent action potentials arising from blockchain technology in the supply chain industry. This includes the opportunity to discuss action potentials at a level that is not dependent on the use case, which bridges the technical-to-conceptual gap in identifying operationalizable trust-related action potential (i.e., affordances) of blockchain that connect technical

Table 1 Main components of blockchain technology

Component	Description	Reference
Distributed ledger	A database maintained at nodes part of the peer-to-peer system instead of at a central location. The database is identical at each node and is updated in real time	(Beck et al., 2016; Pereira et al., 2019)
Consensus mechanisms	An algorithm facilitating secure updating of information on the blockchain based on a majority vote of all nodes in the peer-to-peer network	(Brenig et al., 2016; Gramoli & Staples, 2018; Viriyasitavat & Hoonsopon, 2019)
Encryption mechanism	A public and private key pair is used to encrypt the data; the private key is used to authenticate the node	(Nakamoto, 2008; Underwood, 2016)
Smart contract	Digitally signed, computable, and self-executing agreements among participants in the peer-to-peer network	(Hans et al., 2017; Magazzeni et al., 2017)
Immutable data log	Nodes can access, read, and add to a ledger (upon validation), thereby creating an audit trail that cannot be altered or entries deleted without majority consensus	(Hofmann et al., 2018; Kshetri, 2018)

Source Authors

features with organizational benefits emerging for organizations that are part of a blockchain network.

2.2 Trust in Technology and the Role of Trust in Supply Chains

The concept of trust refers to a secure willingness to depend on a trustee because of that trustee's perceived characteristics. In today's world, these trustees and associated perceived characteristics are increasingly linked to information technologies (Maida et al., 2012; Söllner et al., 2018). In such

cases, the trustor places trust in technology, either directly or as a mediator between individuals or organizations. This implies a secure willingness to rely on or to be vulnerable to technology. What changes in the latter definition is the trust object, which is now often a specific technology or technologies, in contrast to a particular person or organization (McKnight, 2005).

While the definition of trust in a person or organization compared to trust in technology is slightly different, how people build trusting beliefs differs dramatically. Instead of building trust through the perception of a person's or organization's integrity, ability, and benevolence, humans build trust in technology through system-like trusting attributes such as reliability, functionality, and helpfulness (Lankton et al., 2014; Mcknight et al., 2011). Table 2 lists these attributes and resulting beliefs.

Although trust in technology is frequently investigated from an individual user's standpoint, it also plays a significant role in supply chain management and the perception of trust by diverse stakeholders, organizations, and alliances that collaborate along a supply chain (Sahay, 2003). In particular, with the increasing use of information systems and emerging technologies for supply chain operations, trust hinges on the appropriate use of these technologies and the realization of potential. This, in turn, relies on organizations having trust in the technologies as well as in each other (Lee & Gao, 2005). Upon attainment, trust profoundly influences the overall benefits of partnerships, extending beyond mere commercial value. It fosters strong partner commitment and resistance to short-term gains, promoting the preservation of partnerships and collaboration.

Moreover, the appropriate use of information technologies and trust in them has been shown to enable the reduction or even alleviation of information asymmetries between organizations and stakeholders of a supply chain, this reduction of information asymmetries curtails opportunities behaviour among agents, leading to a decrease contracting and monitoring costs (Singh & Teng, 2016). Emerging technologies furthermore facilitate

Table 2 System-like trust attributes

System-like trust attributes	Trusting beliefs
Functionality	The belief that a specific technology has the capability, functions, or features to perform a specific task
Reliability	The belief that a specific technology will consistently operate properly
Helpfulness	The belief that a specific technology provides adequate and responsive help for users

Source Authors

enhanced information integration, a factor shown to correlate positively with process efficiency, streamlining, improved performance, and data usage (Agrawal & Narain, 2021; Obwegeser & Bauer, 2016; Saeed et al., 2005).

Given these positive impacts, technologies that support supply chain activities and trust therein have been the focus of research for quite some time. Blockchain technology, however, has regained increased interest as it provides a novel infrastructure that potentially allows massive efficiency gains due to complete transparency and traceability of transactions, without the need to trust a single third party.

3 Method

This section elucidates the theoretical underpinnings of this study, rooted in the affordance literature. The choice of affordance theory stems from demonstrated findings regarding its influence on the perception of trust in prior studies (Tuncer, 2021). Following an introduction to the fundamentals of affordances, we expound upon our approach to data collection and analysis. We adopt a structured literature review methodology to gather data. Subsequently, we detail the data analysis process, drawing upon the principles of critical realism to extract the affordances of blockchain technology that foster trust within supply chains.

3.1 Affordances Theory

Affordance theory provides the theoretical backbone for our study. In particular, we chose an affordance lens to identify the action potentials stemming from the usage of blockchain technology in supply chain operations and to analyse their impact on the formation of trust. We chose affordance theory as previous studies have shown that affordances of technologies have a significant impact on the perception of trust (Tuncer, 2021). Particularly, affordances help users to understand the functioning and potentials of a specific technology through which they form trusting beliefs. Thus, affordances provide cues about a technology's functionality, reliability, and helpfulness and thus serve as an antecedent of trust (Lankton et al., 2015).

The concept of affordance stems from the field of ecological psychology, where it has been defined as an action potential provided by an object to a perceiver or user thereof (Gibson, 1977, 1986). In the information systems research and broader management literature, the concept has been defined as a capacity for behaviour (Bygstad, 2010) resulting from the non-deterministic

relationship between an actor or actors resulting in an immediate outcome (Volkoff & Strong, 2013). Given the non-deterministic nature of affordances, this study, therefore, does not aim to identify general attributes of blockchain that predict outcomes, i.e., trust in supply chains. Rather, we aim at understanding the underlying mechanisms that trigger the emergence of particular action potentials, thereby explaining how trust emerges in supply chains using blockchain technology as infrastructure (Bygstad et al., 2015).

In doing so, we focus on so-called perceived affordances. Perceived affordances are affordances that are perceived by an actor, i.e., information is available for perceiving the usability stemming from a specific technology in a given context. The affordance, however, is not yet actualized, i.e., an actor hitherto did not act upon that affordance (Bernhard et al., 2013). Focusing on perceived affordances, thus, allows us to extract affordances from the literature, even though these affordances are not actualized so far (e.g., prototypes, use cases) and trust did not emerge in practice.

3.2 Data Collection and Analysis

To identify mechanisms, we perform a two-phased research approach depicted in Fig. 1. Phase 1 consists of a structured literature review (Webster & Watson, 2002) for which we screened the Web of Science database using the search string ((ALL = (Supply chain* or logist*)) AND TI = (trust*)) AND TI = (blockchain or distribut*). The initial set of articles consisted of 69 papers, from which 7 papers were removed after an abstract screening. Abstract screening ensured the thematic fit of papers, i.e., papers were excluded from the set of articles if they were not concerned with blockchain and supply chains, were abstract-only articles, or were not available to the authors.

Phase 2, the identification of trust-related affordances, builds upon the final set of 62 articles identified through the structured literature review. In identifying affordances, we not only draw upon critical realism, which provides the philosophical underpinning of affordance theory. The basic assumption of critical realism is the existence of a real-world independent of our knowledge of it and typically unobservable (Bygstad et al., 2015). For instance, while we may observe buyers and sellers agreeing on prices and volumes (i.e., the outcome), the underlying mechanisms (i.e., the affordance) is unobservable. Consequently, studies based on critical realism strive to provide theoretical descriptions of mechanisms that could explain the observed results (Bygstad & Munkvold, 2011; Bygstad et al., 2015).

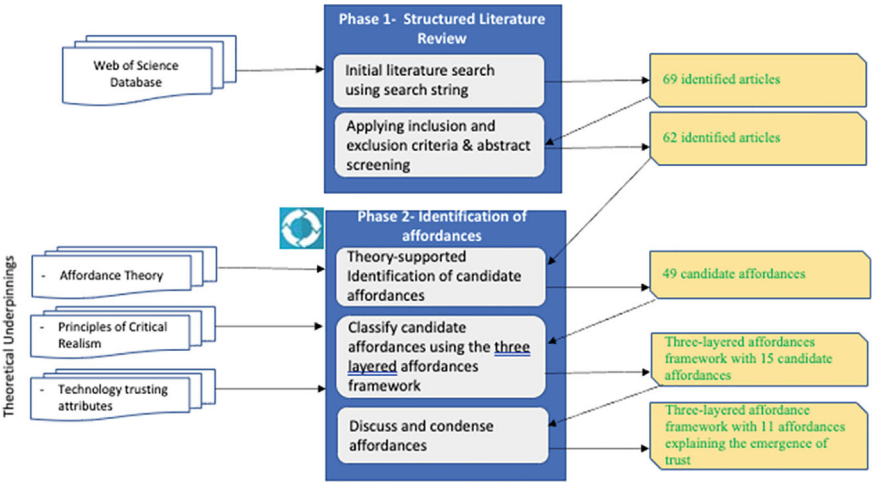

Fig. 1 Research approach (*Source* Authors)

To provide such descriptions, scholars emphasize the need for conducting in-depth studies making use of retroduction, which is the empirical observation and hypothecation of mechanisms that might explain a particular outcome (Danermark et al., 2002; Sayer, 1992). Retroduction, therefore, ensures that outcomes and mechanisms are treated separately. Given the socio-technical nature of digital technologies, additionally, mechanisms should be viewed as a combination of social and technical elements (Bygstad et al., 2015; Easton, 2010). We capture the complexity of affordances by using a three-layered affordances framework as a blueprint for clustering affordances. In particular, the three-layered affordance framework consisting of technical, design, and business affordances informs the clustering processes described below (Ostern et al., 2020). Note that for the sake of convenience, we will explain, technical, design, and business affordance in detail in the next section along with the findings of this study.

To identify affordances, we started coding the 62 articles using open, axial, and selective coding. First, open coding was performed by each author individually. For the first stage open coding, system-like trust beliefs were used as guiding concepts to identify affordances related to the functionality, reliability and helpfulness of blockchain technology in supply chains. This approach yields a set of 49 candidate affordances. In several rounds of discussions, affordances were clustered and summarized using a l code, resulting in 15 candidate affordances. Selective coding eventually yielded 11 affordances based on a further condensation of candidate affordances that represent the final set of affordances that have explanatory power for the observed

outcomes in the studies, i.e., blockchain leading to greater trust in supply chain operations. Eventually, we used the three-layered affordances framework to structure the final set of affordances, through which we consider the socio-technical nature of blockchain technology.

Note that while the open coding and initial identification of affordance were performed by all three authors separately, the emergence, clustering, and condensation of candidate affordances up to a final set of affordances achieved through retroduction were discussed iteratively between by all authors.

4 Findings

Figure 2 depicts the trust-related technical, design, and business affordances that create a trusted ecosystem for supply chain operations as an immediate outcome. Affordances thereby build upon the technical features and their implementation in a blockchain-supported supply chain, such as the applied consensus mechanism, the size of the peer-to-peer network as well as the definition of smart contracts. The differentiation between technical features and affordances is in line with Leidner et al. (2018), emphasizing the fundamental difference between looking at the action potential and merely looking at the technology use of feature use. Notably, the latter describes direct feature use and its observable outcomes, whereas affordances describe a relationship between people and the materiality of the things with which they come in contact, which enables them to use selected features or a combination of features in a meaningful way (Faraj & Azad, 2012; Leidner et al., 2018).

4.1 Technical Affordance

Technical affordances that emerge from blockchain technology in supply chain operations are accessibility, immutability, anonymity, and consistency. Technical affordances are described as a relationship between blockchain features and organizations that identify which action may be opened up by the application of the technology, given the organizations' purpose capabilities, and goals when engaging in supply chain operations (Ostern et al., 2020).

Accessibility is an action potential resulting from blockchain providing a trusted and transparent infrastructure enabling everyone to gain access to information on the blockchain, and thus, be equally informed. For instance, Qian et al. (2020) propose to exploit the affordances accessibility for international food supply chains. In particular, Qian et al. (2020) hypothesize that

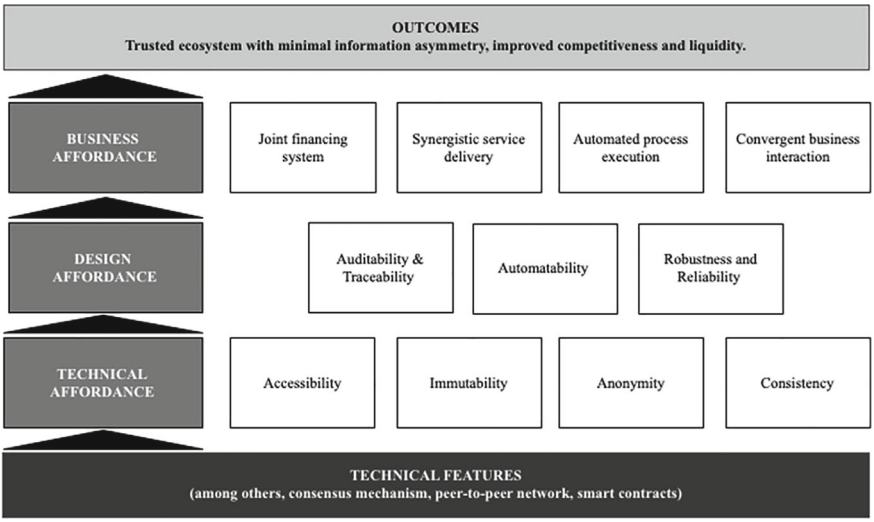

Fig. 2 Trust-related, three-layered affordance framework for blockchain technology in supply chain operations (*Source* Authors)

by documenting food supply chains on the blockchain customers are enabled to access information by scanning the barcode of a product through which they then can trace the origin and transportation routes. This would allow customers to reduce information asymmetries and make informed decisions (Qian et al., 2020). Li et al. (2018) propose blockchain as an infrastructure to support industry clusters for small and medium-sized enterprises where the mutual accessibility of data fosters enterprise alliances and collaborative service components.

Immutability is defined as the action potential to safely store information on the blockchain that cannot be altered ex-post thereby enabling the creation of a trusted and integer data log. Immutability is especially important if the information is prone to intentional or unintended alteration. For instance, Xu et al. (2019) argue that in the food industry mislabelling or the wrong entry of information is not an exception (e.g., eggs that are labelled as cage-free at one stage of the supply chain when they are not.). Blockchain would enable to capture and validate information at every stage in the food supply chain. Once there is consensus it is permanently and immutably recorded. In this case, customers can be sure about the authenticity of food information, and food safety issues can be better managed by retailers (Xu et al., 2019). Mazzei et al. (2020) discuss how immutability as an action potential enables using blockchain technology as a technological foundation for creating immutable digital twins for industrial IoT applications. In particular, blockchains could

support creating immutable and shared records of every transaction associated with a physical asset which can enforce a new level of trust in the Industrial IoT sector (Mazzei et al., 2020).

Anonymity is an action potential of blockchain technology in supply chains that enables parties to collaborate without the need to create interpersonal and organizational trust between collaborators directly. This enables trusted collaboration with a reduced need for monitoring and contractual agreements as requirements are hard-coded into the blockchain. For instance, Jaiswal et al. (2019) propose to use affordance anonymity to conduct auctions in peer-to-peer food trading, where all transactions are performed through participants' IDs and no one learns about suppliers and buyers. This has the benefit that guidelines can be implemented into the blockchain which protects farmers in developing countries from low returns or non-payment (Jaiswal et al., 2019). Others proposed anonymity to be valuable as it enables moderate interaction between agents with conflicting interests and motivations, such as in the car market (Notheisen et al., 2017).

Eventually, *consistency* is defined as the potential for creating a transaction log without friction enabling a complete and trusted database for allowing the digitalization of operations on the supply chain. Consistency allows multiple parties along a supply chain, such as suppliers, manufacturers, carriers, retailers, and customers, to share information in real time. In particular, only if parties have faith in the correctness and consistency of the information provided, and if data processed are not leaked, then there will be a willingness to collaborate and share key information. Thus, consistency of information ensures reduced information costs and enables blockchain to provide collaboration solutions (Wen et al., 2019). Others have proposed a blockchain-based solution to ensure consistency between raw data and sustainability reports, where a consistency token indicates rewards reporting agencies for keeping in accordance with raw data offered by an organization. This would reduce the risks of fraud, e.g., through agencies deliberately overviewing some information or beautifying indices on purpose, and the necessity of close monitoring (Liu et al., 2021).

4.2 Design Affordances

Design affordances allow for the intentional design of action potentials. However, these are not deterministic given the relational relationship between IT artefacts, i.e., the blockchain, and its users along the supply chain, but allow for the development of action potentials along a pre-defined scope (Ostern et al., 2020; Pols, 2012).

Auditability and Traceability allow organizations to create supply chains that enable full transparency and auditability of transactions conducted via a blockchain, leading to the accountability of all stakeholders and parties involved in a supply chain operation. To avoid the inappropriate or misuse of donations in the humanitarian sector, Dubey et al. (2020) propose blockchain technology as an underlying infrastructure that makes actions and donation use through humanitarian actors visible and traceable. Thereby, not only the collaboration between actors is expected to improve, but especially the auditability of actions and, thus accountability for actions and donation use (Dubey et al., 2020). Khadke et al. (2021) propose a system to manage and optimize organizations' and peoples' way of dealing with plastic waste by linking QR codes associated with plastic to a blockchain that makes the origin of waste traceable.

The affordance *automatability* enables organizations to design systems where transactions and processes are automatically triggered and enforced based on the occurrence of an event (e.g., data, time). Thereby, the outcome of a process or supply chain operations gets predictable and variations, such as delays of payments, are reduced. For instance, Schaad et al. (2019) present a use case where business logic is hard-coded into a blockchain, allowing organizations to automatically trigger payment processes, which in turn automatically kickoffs physical actions that are part of the process. Generally, not only payments but the transfer of various types of assets can be enforced, such as assets, property, and value. Therefore, automation serves as a tool for the enforcement of legal contracts that represent stakeholders' understanding and intention (Magazzeni et al., 2017). In this sense, automation enables process design for business operations and could lead to more visible, predictable, and less-intermediated working schemes (Chang et al., 2019).

Robustness and Reliability allow organizations and stakeholders to create an information log that is reliable and trusted by all, given the fact that it is robust to intentional alterations through organizations and stakeholders that are part of the supply chains or external attacks. Scholars have frequently proposed to exploit this affordance for certificates, especially educational certificates that decrease information asymmetries between employers, future employees, and academic institutions (Caldarelli & Ellul, 2021). For that, academic institutions would upload a transcript of a student's performance on specific courses with skills acquired. Notably, such a system would require trusting humans behind this process, i.e., employers of academic institutions. However, once referenced on the blockchain, transcripts and certificates are reliable given their proofed authenticity and immutability (Caldarelli & Ellul, 2021). Others have proposed using blockchain for the robust proof

of provenance along a food supply chain, enabled through non-fragmented, immutable information that reduces the likelihood of deviations and issues regarding the origin and security aspects for food (Garrard & Fielke, 2020).

4.3 Business Affordances

Business affordances allow creating novel business potentials that can be used and exploited by multiple organizations and stakeholders along the supply chain, thereby creating value added to all parties involved in supply chain operation, including customers (Ostern et al., 2020).

The affordance *of convergent business processes* enables organizations and stakeholders along the supply chain to collaborate on a shared and reliable infrastructure that holds everyone accountable for their own actions, which in turn, allows for delivering services and conducting processes in a joint manner. For instance, Xia and Yongjun (2017) propose a joint credibility system based on a blockchain that stores information such as transaction satisfaction, product ability, risk probability, and information concealment from all organizations and parties relevant to a particular supply chain operation. This allows not only organizations to demonstrate their ability to offer fast and reliable services but also reassures customers of the well-functioning of supply chain operations. Seebacher and Schüritz (2017) discuss joint processes on the blockchain to oversee and initiate the execution of processes that involve parties even at the edges of a service system or of a supply chain. Despite improved collaboration and the ability to enforce the execution of actions, convergent business processes also improve resource sharing (e.g., logistics firms can have an increased level of visibility of their freight vehicle capacity, and supply chain partners can make use of scarce resources by sharing them) and joint knowledge creation by sharing market knowledge and understanding of the competitive market (Rejeb et al., 2021). Thereby, the overall quality of supply chain operations and efficiency can be improved for participating organizations, which in turn benefit and leads to increased value on the customers' side.

Another business affordance of blockchain in supply chains is called *joint financing system,* i.e., blockchains' ability to create a trusted environment where increased and closed collaboration enables novel forms of financing operations and solutions along the supply chain. Notably, financing can relate to operations directly linked to the supply chain operation or for incentivizing behaviour external to the supply chain but with a great impact on its functioning. For instance, incentives for improving sustainability approaches for certain goods and products can be supported, which are relevant continued

existence of a sector. In this sense, scholars have proposed the establishment of joint incentivization systems for marine conversation, which is of interest to diverse stakeholders as well as customers of a food supply chain. Therefore, the costs of incentivization can be borne proportionally by those having an interest in it (Howson, 2020). In contrast, Dubey et al. (2020) propose to use a publicly visible infrastructure in humanitarian aid to manage donations and joint financing efforts in charitable causes. Overall, joint financing systems thus enabled finding crowdsourced solutions and liquidity for solving existing issues that are only overcome through close collaboration and trust in operations and stakeholders of a supply chain.

Eventually, the described technical, design, and business affordances have a common outcome, i.e., the creation of a supply chain ecosystem with reduced information asymmetries and increased liquidity, which fosters collaboration and thus competitive advantages in the respective industry. Hence, although affordances identified in the literature stem from diverse contexts (e.g., humanitarian aid, supply chain financing, logistics, marine conservation), we see this outcome as a common denominator of all the affordances mentioned, which altogether led to a trustworthy ecosystem and supply chains that operate within them.

5 Discussion

This article delineates trust-related affordances of blockchain technology within supply chain operations. It comprehensively identifies various types of affordances, ranging from technical to design, and business affordances. Technical affordances emanate directly from the functionalities intrinsic to blockchain technologies. Design affordances, on the other hand, are recognized as action potentials intentionally designed for a supply chain. Although affordances remain non-determining for the perceiver, design affordances can constrain or render the behaviour of supply chain parties and customers more predictable and accountable. Lastly, business affordances represent collaborative endeavours aimed at defining the potential for an ecosystem and multiple parties within a supply chain. This effort focuses on enhancing operational efficiency and rendering financing models more advantageous.

Theoretical contribution: This study provides a summary and condensation of affordances of blockchain in supply chain operations based on a structured literature review. Thereby, we are to the knowledge of the authors among the first that summarize affordances for various use cases of the technology and for various industries where supply chain processes are relevant (e.g., food

supply chains, charitable purposes, etc.). Instead of delving into case-specific feature utilization, the affordances lens permits a more abstract exploration of potential actions, allowing for a broader understanding beyond specific use cases or industries typically explored in the literature. We summarize the identified affordances using the three-level affordances framework (Ostern et al., 2020), which supports theorizing on high-level affordances, i.e., at a design and business level, that abstract from the directly observable technical affordances often referenced in the context of blockchain. Other than offering a comprehensive view of the affordances of blockchain technology in supply chain operations, we hypothesize that the provided framework builds the foundation for future discussion on the affordance-driven design of blockchain-based business models (and architectures) that support supply chain operations and processes, eventually leading to trust.

Practical implications: Besides the theoretical contribution, the insights of this article also provide practical implication. The identified affordances and their relationship displayed in the three-layered affordance framework can support organizations and parties of supply chain operations to systematically identify novel and joint action potential when using or planning to use a blockchain solution. For instance, organizations can start with thinking about design affordances as a way to design for specific and desirable behaviour using a blockchain, by deliberately opening up or limiting the range of actions allowed on a blockchain. Business affordances can be investigated to evaluate the joint business models that are enabled when blockchain is implemented and supports ecosystems with heterogeneous parties that require comprehensive and collaborative solutions.

The insights and implications presented, however, need to be seen in light of this study's limitations. First, the affordances based on a literature review, meaning that no empirical insight is gathered in identifying affordances or reassuring their correctness. While this is a drawback that needs to be addressed in previous studies, the existing literature mostly reports on conceptual blockchain solutions or prototypes used in test environments, such that real empirical insight is hardly available. Second, the literature spans diverse use cases and industries. In identifying affordances, we did analyse the influence of those contextual factors on the emergence of affordances. The affordance approach, however, allows us to abstract from contextual factors and describe high-level affordance that emerges for various industries and use cases. Overall, future work might not only have a closer look at the specifics of affordances for certain industries and use cases but also be intrigued to gather empirical insights into the existence and perception of identified affordances from supply chain parties' perspectives.

6 Conclusion

This study investigates trust-related affordances of blockchain technology for supply chain operations. In doing so, a structured literature review is conducted to extract and condense affordances that are theorized to lead to trust. Affordances are presented in a three-layered affordances framework that displays a gradual build-up of affordances beginning with technical affordances that allow for design and consequently business affordances (also referred to as high-level affordances). The framework supports researchers and practitioners in identifying how affordances on a design and business level evolve from technical affordances and, ultimately, the technical features of blockchain. The identified affordance supports scholars and parties of supply chains to think about optimal designs of blockchain architectures and ecosystems and resulting affordances that support desired behaviour from the multiple, heterogeneous organizations and customers that are part of supply chain operations.

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Blockchain and IoT Integration for Vaccine Supply Chain Trust and Traceability

Ulpan Tokkozhina, Ana Lucia Martins, and Joao C. Ferreira

1 Introduction

Today, the world is moving towards digitalization of processes and operations, where disruptive technologies open new frontiers and possibilities. Common supply chain management (SCM) problems include sensitive information flow, long and complex logistics, unreliable data about capital flow—all of such factors lead to higher costs because trust among supply chain (SC) players is still lacking (Wu & Zhang, 2022). Therefore, for successful operations of SCs, there is a need to determine and create trustful relationships among partnering entities.

One of the technologies that is being vigorously discussed in bringing solutions to SCs is the blockchain technology (BCT). It can be defined as decentralized, distributed ledger, where transactions are logged and added

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in chronological order with the goal of creating permanent and tamper-proof records (Treiblmaier, 2018). BCT has the potential to bring novelties to various fields. It is claimed that where trust in product or service is lacking, BCT opens opportunities for potential application (Durach et al., 2021). BCT brings transparency, enabling actions like record-keeping and tracking; thus, facilitating trust among stakeholders (Kayikci et al., 2022a). Even though BCT-enabled platforms support trust building among entities, it does not fully replace need in trustful relationships among B2B (Brookbanks & Parry, 2022), however, the traceability feature can feasibly improve trust in product with downstream players in terms of B2C relationships (Tokkozhina et al., 2023). Embedded features of BCT create a favourable environment for its application in SC networks that deal with sensitive and critical products and services—e.g., food distribution (Kayikci et al., 2022b), public sector (Tan & Sidhu, 2022) and healthcare (Musamih et al., 2021). When combined with other disruptive technologies, such as Internet of Things (IoT) (Grecuccio et al., 2020), high levels of sensitivity and assurance of product traceability and authenticity could be reached.

IoT allows the communication between a growing number of interconnected smart devices and sensors that are able to connect and match physical and virtual assets with individually identified unique IDs (identities), thus, allowing transparency and simplified tracking of physical status of goods (dos Santos &Canedo, 2019). Usage of IoT technologies in SCM area has a wide spectrum of applications across different stakeholders—it can be used in manufacturing as much as it can also be applied to warehousing or retailing, due to its inherited features, automation of goods identification, real-time data interchange and efficient communication are possible (Tan & Sidhu, 2022).

The purpose of this study is to investigate the ability of a BCT and IoT-based solution in bringing trust to SC players, especially those located downstream in SC. The goal is, therefore, to explore the potential traceability enhancement for SC product flow enabled by a combination of BCT and IoT throughout the SC. Such solution can be applied to various products' distribution, specially making difference for sensitive and dangerous products' flow, such as chemicals, fuels, and other inflammable or hazard goods. In our study, we use vaccine distribution as a demonstrative example and introduce the prototype of Proof of Concept (PoC) for a BCT-based platform powered by DAML smart contract language, which visually indicates the traceability of product flow for trust-enablement across SC players.

The remaining part of this study is organized as follows. Section 2 is going to review literature focusing on BCT, smart contracts, and current applications of BCT-enabled SCs; Sect. 3 will shed light into the methodology used for this study. Section 4 will propose the trust application for the SC distribution process control on an example of a trusted vaccine distribution. Section 5 will approach future research opportunities, gaps, and the limitations of the study.

2 Literature Review

Blockchain as a decentralized shared ledger system and a computational paradigm (Leng et al., 2018) allows peer-to-peer transactions that do not require a third-party intermediary verification (Kamble et al., 2019). Keeping in mind this embedded nature—it is becoming more and more popular for various industries and application cases. From voting systems (Pawlak et al., 2018), modern military solutions (Saadiah, 2021) and certification/diploma issuing (Gresch et al., 2019) to the clothing industry (Agrawal et al., 2021; Benstead et al., 2022) and diamond authentication (Choi, 2019).

SCs today face complexities in terms of multi-tier suppliers and customers, and its overall impacts on the competitive advantage of the focal company (Kappel et al., 2020). By its nature, end-to-end multi-tier SCs are challenged by the need of communicating information and transactions across a wide group of entities, where an instant visibility of information would potentially bring a considerable improvement in terms of SC efficiency (Tokkozhina et al., 2023). Technological capabilities of BCT open a way to improve information transparency across the whole array of SC tiers, from sub-suppliers that are located upstream, all the way to downstream to final customers (Benstead et al., 2022). Blockchain is showing its potential to add value in SCs that are dealing with products that could be sensitive for the health and well-being of final consumers (Dasaklis et al., 2022).

One of the most popular application areas of BCT in SCM context, frequently combined with IoT sensors for temperature/humidity detection, is food industry (Balamurugan et al., 2022; Mirabelli & Solina, 2020; Tsang et al., 2019). Rogerson and Parry (2020) concluded that most feasible BCT applications are those, where consumer is willing to pay extra for the visibility of product origins (e.g., baby food), however, some challenges such as human-generated error, as well as general awareness and trust towards BCT itself, remain open. Capabilities of BCT, such as traceability and decentralization,

were found to provide a more flexible, responsive, and efficient collaboration to stakeholders of food SCs (Kayikci et al., 2022a). The traceability of products that are being kept on BCT ledgers provides assurance to all players of SCs, especially to final consumers, that such products are safe to consume (Keogh et al., 2023). Quickly perishable products—dairy (Casino et al., 2021; Khanna et al., 2022), fish (Ferreira et al., 2021; Tokkozhina et al., 2021a), meat (Cao et al., 2021), as well as alcohol-containing products such as wine (Tokkozhina et al., 2021b) were driving interests of academic research and could be used as good examples of BCT suitability in SC operations. Such perishable products need to be kept and transported under specific conditions in terms of temperature, package etc., and here in a combination with IoT sensors, such information can be kept on BCT ledger without a threat of being compromised. Availability and transparency of records enabled by BCT are not only able to provide data crucial for health safety assurance but also an "ethical" background and origins of food products, that might be of high importance to final consumers—e.g., halal meat products (Rejeb et al., 2022), agricultural products (Antonucci et al., 2019; Balzarova et al., 2022) and general corporate social responsibility compliance of food-providing companies (Rainero & Modarelli, 2021).

Such phenomenon as "cold chain" is applicable not only to the food industry but also in the context of temperature-sensitive products—e.g., vaccine distribution. The cold chain includes such concepts as pre-cooling facilities, cooling warehouses, refrigerated cabinets and transportation trucks, and other (Zhao et al., 2018). An improved visibility is needed to provide product traceability, which can prove authenticity and storage conditions—an urgent aspect for pharmaceutical SCs (Wang et al., 2019). BCT allows to build a common record of product locomotion by connecting various systems and organizations in a secure manner; as well as enhancing end-consumer's visibility for safety measures of product consumption decision-making and allowing relevant SC parties to perform product investigations and recalls in a quicker manner (Tokkozhina et al., 2021a). Having a possibility to trace the storage conditions throughout vaccine journey is a key aim because temperature fluctuations can play a crucial role in the safety of the vaccine itself (Musamih et al., 2021). In the context of sensitive product distribution, BCT combined with IoT temperature tracking sensors are able to provide secure communication with guaranteed immutability of records (Juma et al., 2019), which can ensure safe vaccine supply and exclude possibilities of record fraud and vaccine expiration.

2.1 Blockchain Technology and Smart Contracts Importance in Supply Chains

In the business context, one of the most disruptive areas of blockchain application is SCM—it brings enhancement in visibility among stakeholders, promotes transparency and thus, improves potential trust issues (Kamble et al., 2019). Decentralized environment in BCT-based SCs is built through the implementation of smart contracts (Juma et al., 2019). Smart contracts are pieces of code that contain a predetermined set of business rules agreed between participants, which can automatically trigger the transaction if contract terms were met or to catch if they were violated (Chang et al., 2020; Gurtu & Johny, 2019). Hence, smart contracts ensure the "automatic trading of data between different parties in the supply chain network" (Li et al., 2020). The unique features of smart contracts, combined with BCT nature are able to improve the automation of operations, increasing the level of process synchronization, leading to a more transparent working process (Chang et al., 2019; Tsolakis et al., 2022).

One of the constraints in smart contract adoption is poor coding, since it is also a human factor and smart contract language proficiency that plays a huge role here—if there is a mistake in the predefined contract, then the immutability feature of BCT may bring extra difficulties as it cannot be modified (Cole et al., 2019; Tokkozhina et al., 2022). Another potential difficulty is a dynamic control flow—for instance, when a smart contract interacts with other contracts, if not designed carefully it can cause an increased interconnected number of contracts over time, making it difficult to foresee the behaviour of a contract (Zheng et al., 2020). Hence, the implementation of carefully revised and faultless smart contracts is the priority task for BCT-based SCs.

2.2 Status Quo of Healthcare Blockchain-Enabled Supply Chain Management

In the healthcare field, SCM needs to be precisely effective due to the sensitivity, risk, and complexity of the context; thus, BCT can become a possible solution; taking into consideration it provides features like product provenance and data security (Clauson et al., 2018). The use of smart contracts in healthcare industry is also beneficial, as there is no need for a third-party that will be verifying operations, so administration costs can be decreased (Khatoon, 2020). BCT can detect unauthorized input, error and information-lacking entries and increase data accuracy (Pahontu et al., 2020),

actions like data certification would eliminate counterfeit levels as data is tamper-proof (Chen et al., 2022).

When talking about managing healthcare supply chain (SC) one of the main challenges are security and drug traceability (Uddin, 2021), together with issues like poor leadership and communication, conflict of interests, high costs, and lack of trust between different stakeholders (Omar et al., 2021). Challenges in terms of BCT adoption to healthcare industry also pose scepticism in terms of data storage and transfer and provenance of transactions (Uddin, 2021).

The majority of prior studies with BCT applications to healthcare and drug SCs were focused on anticounterfeit solutions. Various propositions were built to show prototypes of cases: systems based on Hyperledger fabric architecture were simulating different entities in a decentralized way (Pandey & Litoriya, 2021) for a drug counterfeit reduction. Chaincodes were deployed in another counterfeit elimination Medledger application, presenting algorithms and pseudocodes to demonstrate various functions (Uddin, 2021). Electronic product codes were suggested to be used for traceability issues of vaccines in a double-chained BCT structure (Qiu & Zhu, 2021) and claimed to enhance the effectiveness of vaccine anticounterfeiting. In prior studies, pharmaceutical SCs were simulated with temperature alarms, including such classifiers as preconditioning, invalid storage conditions, interior and ambient temperature differences, physical handling, etc., showing that the performance level of the SC decrease when a new temperature range is introduced (Konovalenko & Ludwig, 2021).

Scholars highlight that further system performance improvement and exploration of BCT application in cold-chain logistics industry is needed (Wang et al., 2023). One of the most significant potentials of BCT in drug and medical industry is the decentralized storage of information and transactions, which still requires further examination (Sarkar, 2023). BCT-enabled applications as well as the technical infrastructure and security features design open another potential path that needs to be addressed in literature (Schlatt et al., 2023). According to Völter et al. (2023), there is still a lack of exploration regarding users' trust in scalable BCT adoption cases, moreover research is needed in the direction of BCT and IoT applications for effective trust management (Liu et al., 2023).

Nowadays, majority of pharmaceutical SC looks as follows: central regulatory entity must review and guarantee throughout the whole lifetime and distribution process factors such as safety, confidentiality, security, reliability in product quality, and related transactions (Chiacchio et al., 2020). The

central entity verifies safety and reliability measures of drugs, all the transactions involved throughout the distribution process, creating a need to store some of these data for several years (Bamakan et al., 2021). In this light, BCT is able to eliminate the central authority by providing a decentralized ledger, where each participant is an equal holder of all the information; it can further improve transparency levels and strengthen trust between stakeholders, reducing current human errors (Öztürk & Yildizbaşi, 2020) and documentation flow (Gonczol et al., 2020).

3 Methodology

This study has a purpose of investigating the ability of blockchain-based SC combined with IoT sensors to bring transparency and visibility of the product locomotion, and as a result trust for final consumer—in the context of current study is vaccine recipient. To address the purpose of the study, the obstacles of the trust issue from a vaccine recipient side were investigated in the prior literature. We assume that assuring a quality cold chain for the vaccine distribution, at the same time making the locomotion of it transparent to any vaccine recipient, can potentially increase trust to the end-consumer.

In order to fulfil the goal of this study, we developed a prototype towards a PoC using a BCT platform powered by DAML. This PoC addresses the issue of trust in SCM using IoT devices, persons and SCM stakeholders. This issue of trust appears nowadays with the problem of a vaccine distribution with severe temperature requirements. In a multidisciplinary team of experts, we developed a trust application to control the distribution process and stop SCM distribution if an IoT device detects the noncompliance process. DAML is a powerful smart contract language system, that enables secure record keeping that connects across business processes, it ensures data integrity and fortifies strict authorization within the system, providing actions available for the contract, combined with their parameters, statements, and accurate consequences (DAML 2021).

Through the detection of current SC challenges in literature, this research follows the "hands-on" approach through the introduction of a PoC architecture for vaccine distribution industry. In this PoC, smart contract is created from blueprint "templates" that are a part of the DAML code. Those "templates" include full contract data, such as date, description, network participants involved and their roles (e.g., signature performers). Every contract is a "template" instance stored as a row on the digital ledger.

4 Proof-of-Concept and the Discussion

As mentioned earlier, various systems were proposed in prior studies, mostly focusing on BCT-based anticounterfeit solutions for pharmaceutical SCs. In order to show the possible prototype of a smart contract architecture, we used DAML smart contract language; as product information needs to be measured and validated, smart contracts will be responsible for this stage. Considering BCT's immutable nature, once contract is created on the ledger, it cannot be altered. In order to change it, a new contract with the desired data needs to be created.

Figure 1 shows a simplified example of how blocks are chained to form BCT architecture; each block, in its turn, contains a header and several transactions. Those transactions in a block are hashed to generate a fixed-length hash output which is added to the block header. After the creation of the first block, every subsequent valid block must contain the hash output of the previous block header. Each new block is generated among a pre-defined smart contract that checks pre-defined requirement using inputs from IoT and persons.

As can be seen from Fig. 1, due to the immutability of transactions, BCT is expected to bring enhanced transparency and traceability, and with

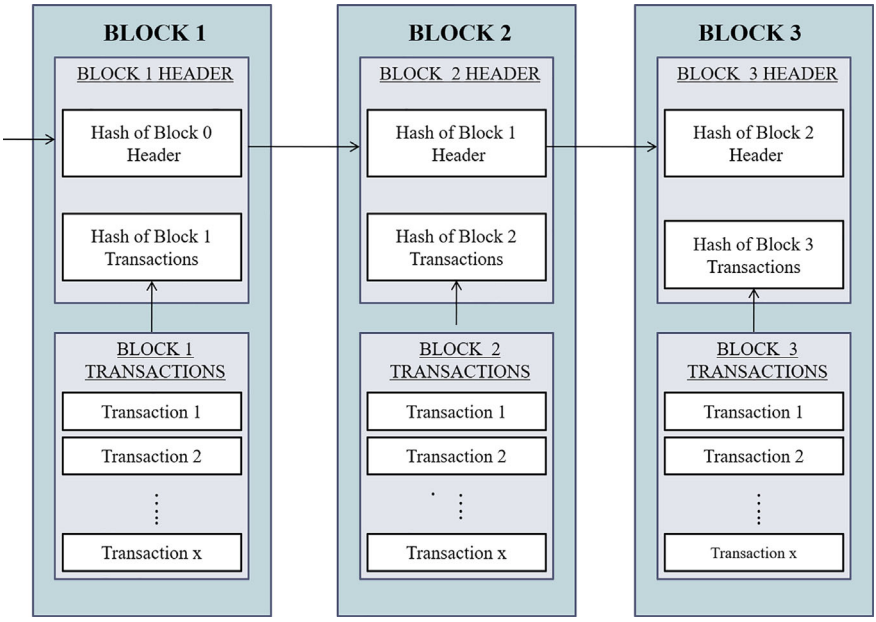

Fig. 1 A simplified example of block sequence in a blockchain (*Source*: Authors)

the use of temperature and IoT devices, it can ensure safety measures of distributed vaccines. Smart contracts can hold and transfer digital assets managed by BCT networks and can invoke other smart contracts stored on the blockchain; codes of smart contract are deterministic and immutable once deployed (Xu et al., 2019). Therefore, network participants who want to implement a digital ledger solution first need to agree on a design that will define the terms of the smart contract and how it will trigger the transactions.

In our case, we used only four main parties involved in the SC processes as an illustration of the PoC, where in a real-life application, it would be a bigger number of stakeholders, e.g., ingredient manufacturers, packaging and vial producer's freezer suppliers, transportation providers, etc. To keep the illustration comprehensive, we chose four main parties, which are:

(1) Manufacturer: is the party that receives all the raw materials, produces the vaccine, divides, and provides packaging of it in doses.
(2) Warehousing: responsible for the collection of ready vaccines and its transportation to the next entity.
(3) Hospital/vaccination centre: these are the ones receiving vaccines, as this is a specific of such product—vaccines are not authorized to be sold in pharmacies—they are usually pre-ordered by a government and are being distributed straight to medical institutions.
(4) Vaccine recipient: an end-consumer, individual that receives the injection. When the vaccine is provided to the recipient, there will be a possibility to track the entire locomotion of the vaccine, like this gaining more confidence in the vaccine authenticity and safety assurance.

During transition phase from one stakeholder to the next, as can be seen in Fig. 2, a new transaction block is generated in BCT based on smart contract defined using a novel approach that combines things and persons inputs. Smart contract will enable encryption of all related transactions, including packaging and serialization process, which will provide a simplified traceability process throughout the whole distribution network (Bamakan et al., 2021). The last party involved in this application is the vaccine recipient, and it is important to note, that this party is not going to have a possibility to create any transaction and is only going to have a full visibility to trace the distribution backwards by scanning a QR code.

It is important to highlight the role of the National Health Authorities in a health-sensitive product distribution and launching. In this light, Ihlen et al. (2022) showed the acknowledged existence of scepticism among general public concerning uncertainty and promised transparency coming

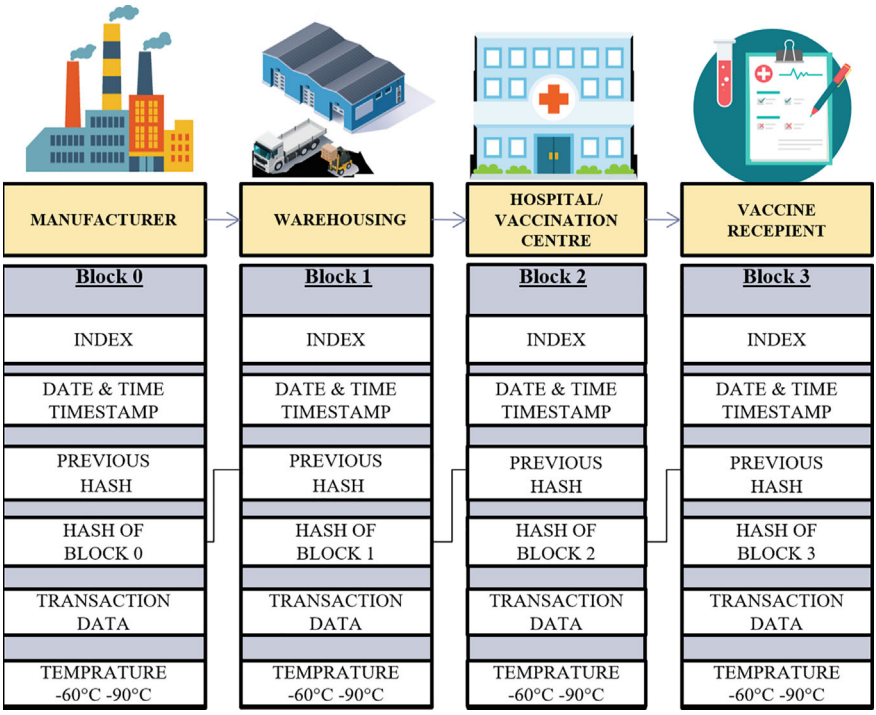

Fig. 2 The architecture of trusted vaccine proof of concept (*Source* Authors)

from Health Authorities in critical events, such as COVID-19 outbreak. Moreover, according to research conducted by Bucchi et al. (2022), citizens' trust in local and National Health Authorities shows an increased acceptance of vaccination process. National Health Authorities are also claimed to mediate between the interests of citizens and the State, therefore, building trustful relationship and gaining positive reputation among general public (Burau et al., 2022).

One of the main established parameters of a smart contract in this application is temperature control. Since vaccines are highly sensitive products, they need to be kept strictly under a certain temperature, so that shelf life and the effectiveness can be guaranteed. Temperature control is going to be realized through IoT devices that will send data to smart contracts, and based on the requirements set, smart contracts will automatically trigger or deny the next transaction. Every vaccine has its own temperature storage requirements; thus, it is important to set smart contract depending on a SC of each type of vaccine. As an illustration, we took an average temperature requirement for this PoC with a range between—60 °C and—90 °C.

In our PoC, every party has two types of contracts:

- Type 1: Product contract (PC)
- Type 2: Contract transfer proposal (CTP)

And then, we propose two different types of roles for this application, due to a health and safety-related nature of a vaccine distribution process.

(1) Watchers—this role is prepared for National Health Authorities (depending on country of application), allowing official government authorities to keep track of the vaccine distribution process and check if everything goes according to the decree and procedures.
(2) Owners—stakeholders of a SC and depending on the stage of the vaccine distribution the ownership changes accordingly, where owner is responsible for the vaccine.

Table 1 represents the responsibilities of the two roles in the PoC, where watchers are playing an important role. This should not be seen as a central authority because watchers *are not able to accept CTP,* but this role is essential for health safety concerns of the populations since it is usually governments that are ordering vaccines; thus, they need to assure the possibility to track the process as well. Watchers are also providing a guarantee of unbiased observers: in case of any suspicious activity coming when ownership is changing between two stakeholders, watchers will have the possibility to reject CTP. Otherwise, if the process goes smoothly and all the conditions of the smart contract are met, it will always be an automatic process, where transactions will be triggered instantly.

Figure 3 demonstrates a BCT transaction workflow between the manufacturing and the warehousing party, with the IoT devices of each party updating

Table 1 Watchers and owners' responsibilities in the vaccine distribution proof of concept (*Source* Authors)

Watchers' responsibilities	Owners' responsibilities
■ Have full visibility of the distribution process ■ Are able to *reject CTP* if it does not meet criteria ■ When rejecting CTP, the PC will remain intact	■ Have the full ownership of vaccines at their own stage—have both PC and CTP ■ Are the only parties that can create and *accept CTP*, also can *reject CTP* if does not meet criteria ■ When accepting the CTP, the old contract (PC) will be archived, and a new PC will take place with the new owner accordingly

the state. The status property present on every contract allows more transparent and trustworthy contracts without the need for each party to have their own validation rules. If the temperature does not meet the established parameters specified on the choice, the properties of the smart contracts are set to false, blocking any further transaction. As shown in Fig. 4, the status property is only checked when the contract is set to be transferred. The contract is not archived in case if temperature requirements fail to be met, it simply makes it impossible to be transferred; thus, protecting SCs from unsafe product distribution.

Like this, the vaccine is changing ownership and is being tracked by a sensor for the temperature conditions to be met, otherwise the transaction will be blocked and will not move further. In this PoC, the last stakeholder is a vaccine recipient, who will be able to trace backwards all the previous transactions to check stored conditions. Therefore, the components of the vaccine can be assured to be effective. Also, granting a role of a watcher to a National Health Authority assures that process is flowing unbiased among stakeholders when the vaccine is changing ownership, also providing a double layer of trust for a vaccine recipient and certainty that the vaccine is not a counterfeit.

PharmacyChainSmartContract: ManufacturerContractOwner

id	status	watchers	owner	iot	date	status	product. quantity	Manufacturer	Manufactureriot	Warehousing
~~#0:0~~	~~archived~~	~~['Warehousing']~~	~~'Manufacturer'~~	~~'Manufactureriot'~~	~~1970-01-01T00:00:00Z~~	~~true~~	~~20~~	X	X	X
~~#1:1~~	~~archived~~	~~['Warehousing']~~	~~'Manufacturer'~~	~~'Manufactureriot'~~	~~1970-01-01T00:00:00Z~~	~~true~~	~~20~~	X	X	X
#5:0	active	['Warehousing']	'Manufacturer'	'Manufactureriot'	1970-01-01T00:00:00Z	true	20	X	X	X

PharmacyChainSmartContract: ManufacturerTransferProposal

id	status	pharmacyCid	newOwner	newWatchers	newIot	Current owner	Manufacturer	Warehousing
~~#2:1~~	~~archived~~	~~#1:1~~	~~'Warehousing'~~	~~['Hospital']~~	~~'Warehousingiot'~~	~~'Manufacturer'~~	X	X

PharmacyChainSmartContract: WarehousingContractOwner

id	status	watchers	owner	iot	date	status	product. quantity	Hospital	Manufacturer	Warehousing	Warehousingiot
~~#0:0~~	~~archived~~	~~['Hospital']~~	~~'Warehousing'~~	~~'Warehousingiot'~~	~~1970-01-01T00:00:00Z~~	~~true~~	~~20~~	X	X	X	X
#1:1	active	['Hospital']	'Warehousing'	'Warehousingiot'	1970-01-01T00:00:00Z	true	20	X	-	X	X

Fig. 3 Scenario with execution fail and output screen from proof of concept (*Source* Authors)

PharmacyChainSmartContract: WarehousingContractOwner

id	status	watchers	owner	iot	date	status	product. quantity	Hospital	Manufacturer	Warehousing	Warehousingiot
#4:1	active	['Hospital']	'Warehousing'	'Warehousingiot'	1970-01-01T00:00:00Z	false	20	X	-	X	X

PharmacyChainSmartContract: WarehousingTransferProposal

id	status	pharmacyCid	newOwner	newWatchers	newIot	Current owner	Hospital	Warehousing
#5:1	active	#4:1	'Hospital'	['Patient']	'Hospitaliot'	'Warehousing'	X	X

Fig. 4 Scenario execution fail as temperature requirements are not met (*Source* Authors)

5 Conclusion

Our study was focused on investigating the ability of BCT-based solution in bringing trust towards the product the health-sensitive SCs by exploring potential visibility and traceability improvement throughout the SC. This PoC is aimed to build a prototype of BCT and IoT-based solution to control the execution of the SC process that is involved in vaccine distribution, using smart contracts. BCT is expected to bring enhanced transparency and traceability with the two main novelties for this application:

- Temperature conditions can be measured by IoT devices and established as predefined parameters for the smart contract. That will ensure safety measures for distributed vaccines, and the maximum effect from the vaccine injection to a recipient.
- BCT architecture for this case proposes a novelty by developing a role of watcher that can be granted for National Health Authorities. This will keep the process unbiased among several stakeholders, since a watcher can deny a transaction if there is any suspicious activity or a violation of a certain procedure, reducing need for further investigations.

Considering these two factors, the vaccine recipient can get an improved trust towards the vaccine due to several reasons as (a) vaccine recipient can personally trace backwards the process by scanning a QR code and see the whole vaccine distribution process before getting an injection; (b) vaccine recipient can check the temperature conditions of the vaccine distributed and how they were kept, which will assure the component effectiveness; (c) National Health Authorities involved in the distribution process provides an extra layer of

confidence, so that the overall process controlled not only by the stakeholders, but by official state representatives.

This was an illustrative PoC application development, which in future can be used as a template for a specific case study. We highlight that for the specific case study, network participants first need to agree on a design for the collaborative process that will define the terms of the smart contract. A more detailed technical description of smart contract coding can be developed in future.

We also suggest paying attention to several topics that remain open: in this study, we considered the record of vial movements throughout the SC and different stakeholders, as vaccines are changing the ownership. However, it would be interesting to create a solution for movement tracking within the same entity. Future studies can focus on tracing the vial within the vaccination centre/hospital from the moment it is pulled out from a freezer until vaccine recipient is going to receive an injection.

Lastly, we indicated a possibility to store the vaccination certificate within the BCT application. For instance, as the vaccine recipient gets the injection, it can further create another block with all personal data of the recipient, which will make vaccination certificate issuing more transparent and trustworthy as it contains all the information of the vaccine vial distribution there is a guarantee of a certificate to be veridical.

Overall, vaccine was used just as an example of a product, however, such BCT and IoT-based application has the potential to bring value to any SC involved in a sensitive product distribution, as it can bring an enhancement in the transparency of the whole distribution process, which in its turn creates a more robust trust from the side of a final consumer or recipient of a product.

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18

Trust Dimensions in Blockchain and Process Industries

Fernando Santiago-Cajaraville, Emel Aktas, and Stuart Bashford

1 Introduction

The flour milling industry plays a vital role in feeding people as flour is an ingredient in the majority of the foods sold in supermarkets. As with any other industry, the milling industry also adopts new technologies, starting from production lines. For example, new smart mills are at the forefront of technology adoption, improving internal operations, increasing the production capacity, and overall increasing the "yield" of the milling operations. However, these technology solutions have historically focused on the material flow from wheat to flour. Information about products, processes, and testing parameters (e.g., water content, ash content, wet gluten) is usually recorded in forms which are maintained in document editing software like MS Word and communicated across the relevant players in the supply chain via emails. Contrary to the advances in the production technology used in the industry, the supply chain information flow between stakeholders is, in many ways, still manual, with prominent use of email, paper-based movement of data sets and data silos in every supply chain step. Moreover, the lack of visibility renders current supply chain information systems inadequate to cope with the new global market requirements (Santiago-Cajaraville, 2021).

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While the ISO 22005:2007 traceability in the feed and food chain outlines the general principles and basic requirements for system design and implementation, it does not prescribe how these principles will be adhered to and what implementation technologies could be used to meet the standard's requirements. To ensure the quality of end-products made from flour, several operations related to the transformation process, such as farming, milling, and baking, should be coordinated with supply chain operations such as drying, blending, storage, and transportation (Nadimi et al., 2023).

This research aims to investigate whether innovative technologies such as blockchain has the potential to improve the data flows in the milling industry, specifically in the flour supply chain. In line with this research aim, the research question we answer in this paper is the following: "*How innovative technologies like blockchain can improve data flows in the flour milling industry?*" Our objectives are to understand if adopting recent technologies can improve supply chain visibility, reduce operation costs, and increase stakeholder collaboration and trust.

Food supply chain visibility, especially food traceability, is a critical aspect of food supply chain management to ensure the food that reaches the consumer is safe to consume, and we can identify the source of contamination if a food product is found to be non-compliant with the standards and specifications expected. Traceability is challenging as a variety of raw materials are mixed, different batches are combined, and safety information is crossed when food is processed. Novel technologies such as artificial intelligence to optimise the mixing of batches, big data to forecast the quality of products and blockchain to ensure credibility in tracing are becoming more prominent in improving the traceability performance of food supply chains, specifically those for processed food products (Qian et al., 2022).

The technology we investigate in this paper is blockchain, which provides a secured platform for the players in the flour supply chain to initially share product quality data records in a digital form and then use the unique version of the data to improve the service and operations downstream and upstream in the supply chain. Blockchain is a distributed, immutable ledger technology that ensures the integrity of data without the need for a third, trusted party (Cocco et al., 2021). The application of a blockchain enables the secure transfer of data between supply chain actors, providing transparency around the exact process parameters being used in the milling of their product.

The research presented in this paper strongly links with three United Nations Sustainable Development Goals: SDG9 Industry, Innovation, and Infrastructure, which aims to build resilient infrastructure, promote sustainable industrialisation, and foster innovation, SDG12 Ensure sustainable

consumption and production patterns, and SDG 3 Good Health and Well-being. The introduction of blockchain technology into the flour milling supply chain provides increased resilience owing to increased visibility of operations (Sharma et al., 2021). In terms of the technology's contribution to sustainable industrialisation and innovation, blockchain reduces the time to identify the source and the transport path of fresh produce from 6.5 days to a few seconds (Qian et al., 2022), enabling further analysis which could inform decisions related to transport optimisation and sourcing. The connection with good health and well-being is underpinned by the information that could be communicated to the consumers, hence facilitating healthy diets and informed food choices (Lopez-Barreiro et al., 2023).

2 Literature Review

The flour supply chain consists of three main facilities: (1) those involved in storing the wheat (i.e., cooperatives/silos), (2) those involved in milling, and (3) those involved in manufacturing products from flour (i.e., bakeries). Not only products, but also information and finances must flow through this supply chain. Over time, with changing supplier capacities, processing costs, and demand locations, it becomes necessary to reoptimize these supply chains (Yousefi-Babadi et al., 2023). However, information visibility is necessary to initiate such optimization studies. Blockchain is a technology that allows the immutable recording of information, but implementing blockchain faces some challenges.

Blockchain implementation in the supply chain necessitates a discussion of trust since supply chain management is built on trust and commitment (Lee et al., 1997). Trust and commitment among the supply chain members promote and increase efficiency, productivity, and effectiveness (Morgan & Hunt, 1994). Trust is expecting positive or non-negative outcomes from the actions of another in an interactive relationship characterised by uncertainty (Sahay, 2003). It is perceived as the cornerstone of strategic partnerships (Spekman, 1988). Trust is the belief that a party's word or promise is reliable, and that party will fulfil their obligations in an exchange relationship (Dwyer et al., 1987).

There is a virtuous cycle between trust and data sharing: the more supply chain actors work together, the more data they will share and the more data they share, the more they will trust one another (Akkermans et al., 2004). A higher degree of interdependence between companies forces them to collaborate and share information, which requires a high level of trust

(Handfield & Bechtel, 2002). Trust is at the centre of theories related to collaboration among supply chain partners. It reduces the uncertainty in the relationship between supply chain actors (Chandra & Kumar, 2001). This uncertainty arises because parties in a supply chain are vulnerable to one another's behaviour (Blois, 1999). Yet the globalisation and decentralisation of supply chains increased the complexity of building trust and sharing data.

Trust within a supply chain has been analysed from different perspectives. It is expressed as the sharing of information and assets (Handfield & Bechtel, 2002). It is categorised into three as (1) Contractual Trust, (2) Competence Trust, and (3) Goodwill Trust (Sako & Helper, 1998). Contractual trust is satisfied when the other party meets the contractual agreement. Competence Trust is satisfied when the other party is capable of doing what they agreed to do. To build goodwill trust, parties make an open-ended commitment to take initiatives for mutual benefit while refraining from unfair advantage-taking, in other words, opportunism. It is the strongest form of trust as it does not depend on a contract or a skill.

Four main reasons push the supply chain actors to look for more trust. (1) *Economic and Financial:* Organisations and individuals calculate the cost or reward of another party cheating or staying in a relationship (Williamson, 1988). The trusting party will assume that it is in the best interest of the other party not to cheat, as the benefits outweigh its drawbacks. (2) *Better Prediction:* Goodwill trust refers to the ability to forecast another party's behaviour. Behaviour uncertainty and potential opportunism negatively impact trust (Kwon & Suh, 2004) among the supply chain members. Past information is pivotal to assessing the credibility of the other party. Delivering on promises increases trust among parties (Swan et al., 1985). (3) *Information Sharing:* Several studies put information sharing as a critical agent for trust-building in supply chain management (Sahay, 2003). Information sharing has a direct impact on reducing behaviour uncertainty, consequently improving the level of trust (Kwon & Suh, 2004). (4) *Meeting Obligations:* Competence trust is about determining another party's ability to meet its obligations (Sako & Helper, 1998). Underpinning this idea of competence, assets dedicated to the relationship between firms will increase the level of trust as it will directly impact the confidence level in fulfilling obligations (Kwon & Suh, 2004).

Trust is the basis of long-term and collaborative relationships (Sahay, 2003). It can lower transaction costs via stable relationships (Heide & John, 1990). It facilitates dispute resolution (Ring & van de Ven, 1994) and reduces legal costs, reducing the extent of formal contracts. Creating a competitive advantage based on trust facilitates long-term relationships among suppliers and makes the supply chain actors more interested in final customer needs

(Zineldin & Jonsson, 2000), facilitating improvements and mutual planning. All these factors enable parties to focus more on long-term benefits than opportunistic behaviours.

Traceability is critical to food supply chains. It is designed to ensure food safety, quality, and provenance and facilitate food recalls. In addition, it requires the supply chain partners to monitor the composition of the final product, including each ingredient along its history. Blockchain's immutable timestamping of transactional data makes it a promising technology for addressing food safety challenges (Qian et al., 2020). Similar to the flour supply chain, the rice supply chain suffers from poor reliability and traceability, which has led to trust issues between supply chain stakeholders (Farooq et al., 2023). The transparency needed to ensure trust in the system can be achieved by implementing a blockchain system which can be used to record readings from sensors and other equipment used for data capture.

The Internet of Things (IoT) technologies are used in the recent implementation of a blockchain for the supply chain management of a particular Italian bread (Cocco et al., 2021). The system guarantees a transparent and auditable traceability of the Carasau bread where each actor in the supply chain can verify the quality of the products and the conformity to the norms about the hygienic-sanitary conditions along the chain. The use of digital technologies reduces data entry errors and the risk of tampering. Furthermore, the system allows end consumers to have a transparent view of the whole journey from raw material to the final product and for supervisory authority to perform online inspections on the products' quality and good working practices. While the blockchain reported in this paper is still experimental, it is one of the projects that are close to real-life implementation.

Blockchain implementation's most frequently highlighted benefits for the agrifood sector are its tamper-proofness and traceability. In a traceability system based on blockchain technology for storage and query of product information in the supply chain of agricultural products, safe sharing of private information and reputation-based smart contracts were implemented (Yang et al., 2021). Furthermore, smart contracts were used to incentivise the nodes in the network to upload traceability data. A performance analysis of the practical application shows that the proposed system improves query efficiency and security of private information, guarantees the authenticity and reliability of data, and meets application requirements.

Gasol flour is made from rice which still has its epidermis (broken skin). Therefore, it has high nutrition, and the production process requires each supplier to know how much raw material or product is sent to a buyer, the price of raw material from the initial supplier, the date and time of the receipt

of raw material or product, the origin of the product, and the quality of the raw material as well as the final product. In the Gasol flour supply chain, a smart contract system based on blockchain technology is implemented to provide a fair and transparent contract for each stakeholder (Kurnia et al., 2020). Contract rules determined by using a tree algorithm were executed in smart contracts to ensure fair distribution of supply chain profits across the supply chain.

As can be seen from these examples, blockchain not only increases the trust of each stakeholder of the supply chain in the system but also helps ensure fair distribution of profits across the supply chain. The potential for implementing the technology is virtually limitless. Still, the initial phase of implementation projects takes time to generate the starting trust from companies already operating without the blockchain in the proposed technology and information-sharing approach.

3 Design/Methodology/Approach

We followed interactive research, a collaborative approach characterised by recurrent interactions and joint learning activities between researchers and practitioners in commonly agreed-upon efforts to study change and innovation in organisations (Sandberg et al., 2022). This entailed participant observation (Vlachos & Malindretos, 2021) in the technology implementation process to document the development and deployment of a data-sharing platform backed by blockchain technology in the milling industry. The process has implied mapping the information flow between a miller and one of its buyers to replicate it in a digital solution.

We show how blockchain ledger records critical product information in a trustworthy manner (Garrard & Fielke, 2020). Furthermore, blockchain provides a unique version of the quality data and the events in the process, ensuring the full traceability of products through a specific part of the supply chain. We present the blockchain implementation as a case study in which we conduct a series of semi-structured interviews and process observations with members of two tiers in the UK milling industry. We have chosen this use-case study on blockchain technology's relatively advanced development and implementation, pioneering its use in the milling industry. We show the methodology followed in five steps: (1) Problem Identification and Sample Groups, (2) Data Collection, (3) Design and Development, (4) Testing and (5) Evaluation. These steps are shown in Fig. 1.

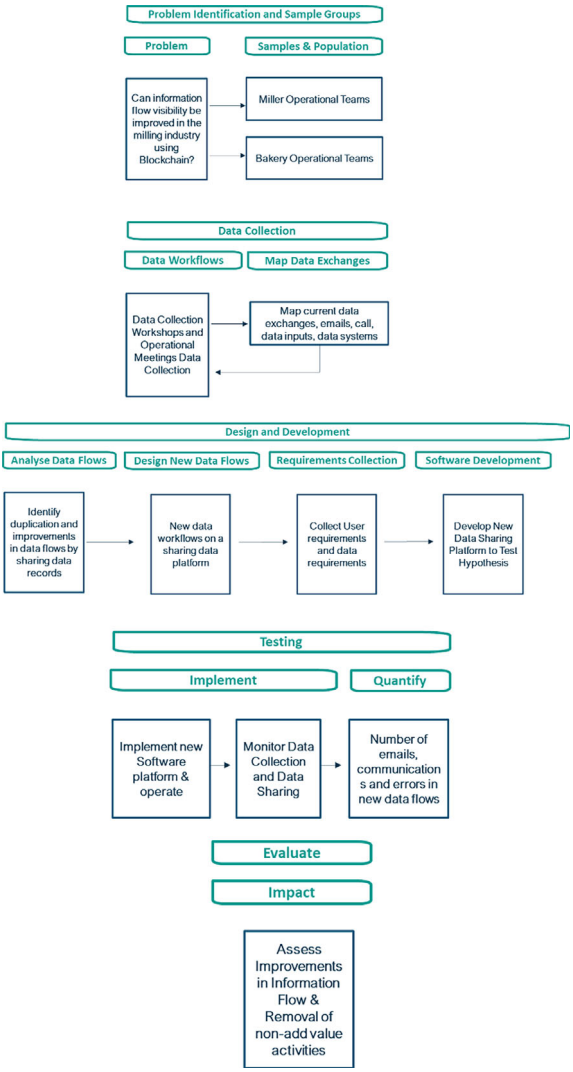

Fig. 1 Research methodology (*Source* Authors)

In the first step, the problem to address was visibility. The project team identified the research problem as improving the visibility of information in the milling industry using the blockchain technology. To understand the problem and perspectives of key stakeholders, the project team interviewed miller operational teams and bakery operational teams. Each team expressed their expectations from the project and what improved visibility of information flow meant for them.

In the second step, the project team run several workshops to collect data about the current data exchanges, how long each information flow took, which medium was used (e.g., emails, calls), where the data was stored and how the data was stored.

In the third step, the design and development of the technology took place. This involved identifying duplication of efforts, multiple data entries, what steps could be removed without negatively affecting the process. A new data sharing platform was devised and new data workflows for sharing data were introduced. User and data requirements were a significant input to software development of the new data sharing platform. Once the platform was developed, it could be used to test hypotheses.

The fourth step involved testing of the platform. Once it was operational, it was used to collect more data in a shared environment. The reduction in email communications was reported against the base case to show the impact of the new tool.

Finally, the fifth step was to evaluate the test results and assess the impact of the solution. This step finalised the project, suggesting to remove non-value added activities from the workflows.

Observations were carried out during the last year of the project implementation. Around 24 meetings were attended and observed, with different stakeholders, technology providers, miller, miller's buyer, developers, and blockchain providers were part of the data collection process. The topics covered in the meetings ranged from operations and information flows mapping, information flow redesign, and automation to quality requirements.

Scrum methodology was used to project manage the development, from requirement gathering to delivering the prototypes and final production. In addition, weekly meetings were attended with development, technology, and blockchain teams.

The roles involved in the solution design and interviewed for this research were technical, quality, and milling directors on the miller side and quality and technical directors on the buyer side. In addition, users were interviewed to know the solution's impact on their daily tasks and operations and to refine the different versions of the solution. Operators range from customer service, transport, and laboratory.

From the conception of the idea to the final solution, it took approximately 36 months to deploy the project on-site and embed the operations of the mill and its buyer on the platform.

4 Findings

The solution combines IoT devices collecting quality data through mill operations and a platform based on blockchain technology to share those quality data. The entire stack has allowed the digitalisation of the information flow among different supply chain members. Furthermore, using a blockchain platform allowed the stakeholders to share unique datasets brought in real-time from the mill operations through IoT devices. The platform provides immutability to the data records and events data, removing redundant reconciliation, double-checking operations, and catalysing the removal of data duplications and data silos.

Consequently, sharing a single version of the information flow improved the transparency supply chain from a translucent supply chain to a transparent one (Lamming et al., 2001). In addition, operation on both sides was improved by removing redundant processes and quality checks. The new information flow is expected to remove the need for contractual and competence trust and increase goodwill trust among business partners (Sako & Helper, 1998).

4.1 Certificate of Analysis

The flour quality specifications are embedded in the commercial agreement among parties. A certificate of analysis accompanies every flour delivery from the mill to a buyer (see Table 1 for an example). The certificate of analysis determines whether a batch is approved or rejected. In addition, it documents the results of all quality tests performed on the final product, along with other delivery information. Without the certificate of analysis, the buyer cannot accept the batch of flour being delivered, as critical data about the product would be missing.

Nowadays, the certificate of analysis has become a data standard in the milling industry. Therefore, it contains the critical data to be shared among suppliers and buyers in the industry. The certificate of analysis is shared on a paper-based form as part of the physical documents that a delivery driver needs to submit to unload a tank on the buyer's sites, along with the delivery note. Therefore, both supply chain tiers need a series of manual and redundant processes to issue a certificate of analysis.

Table 1 An example of a certificate of analysis with main data entry fields

Flour Description		
Customer		Batch
Customer Order Supplying Site		Delivery Date Delivery Number
Test	Results	Specification
Moisture NIR Protein NIR Water Absorption NIR Starch Damage NIR Colour NIR Hagberg NIR Bran % Bran Specs Signed Position Date		

Source Authors

4.2 Release Process

A standard practice in the UK milling industry is assessing batch quality by taking samples. Samples taken from a specific batch are analysed on both sides of the supply chain (supplier and buyer). Until the results are satisfactory for both parties, a certificate of analysis cannot be issued, and consequently, the flour cannot be delivered. Preparing a certificate of analysis implies several manuals and redundant steps in the information flow (see Fig. 2).

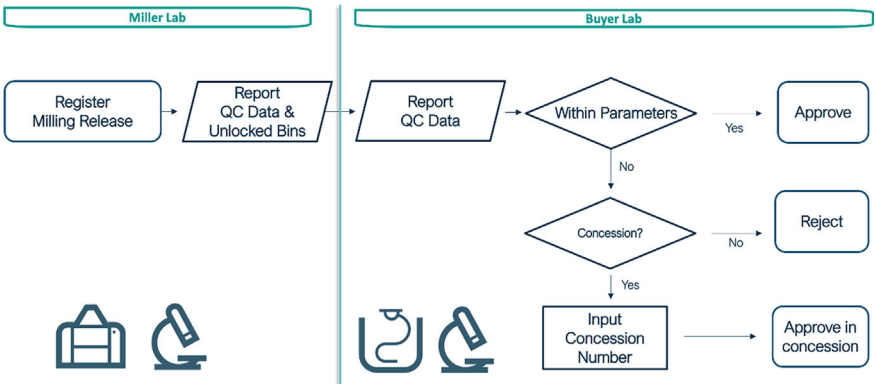

Fig. 2 Supplier release workflow (*Source* Authors)

The sample analysis is carried out on both sides of the supply chain. The current process takes several emails from both companies. Different internal departments and companies need to agree before issuing a certificate of analysis. The technological advances in the milling industry have allowed the development of a new generation of smart mills that can take 15,000 data points in a single milling process. As a result, several IoT devices installed throughout the mill can thoroughly monitor milling operations, mill performance, and quality parameters in real-time.

For quality control, IoT devices such as the NIR Multi Online Analyzer MYRG allows real-time analysis of raw material and end-products at various measuring points. The MYRG systems are equipped with digital cameras capable of detecting colour differences. This image processing allows specks and the flour colour to be automatically measured and quantified. One spectrometer evaluates up to six different measuring points and colours. These data can replace, to a large extent, the current laborious processes of taking a sample and analysing it in the lab.

The data availability was, until now, centralised on the supplier side. The lack of trust in centralised data systems had deterred the acceptance of those data from clients, creating redundant reconciliation processes. Through the Bühler Dash Network, the industry is now in a position to share data directly from MYRG sensors in an immutable and reliable way. Furthermore, sharing data will allow both parties of the process to use those data as the single version of the truth.

The elements of MYRG technology coupled with blockchain technology will allow the removal of redundant processes in Fig. 2. In addition, sharing unique, reliable, and immutable records from Bühler sensors has improved the operations removing manual processes to reconcile and agree on the quality records of a batch. This is part of the current information flow digitalisation in the milling industry. The new information flow is presented in Fig. 3.

Figure 3 shows that with the Bühler DaSh network, no emails, no paper-based exchanges, and no spreadsheet inputs will be needed to share data between the supplier and the buyer. Instead, a single and immutable record is shared among the stakeholders through the Bühler DaSh Network. Non-value-adding processes and times have been removed from the information flow. For example, real-time approval of flour batches has been removed from non-value-adding processes as waiting times to get the approval from both stakeholders' labs, usually a three-days' process.

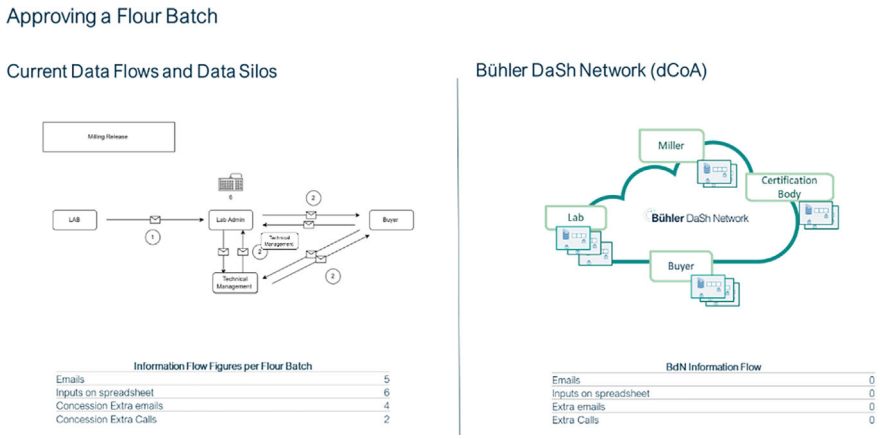

Fig. 3 Mapping of information flow (*Source* Authors)

4.3 Delivery Process

Once the buyer and the miller have agreed upon the milling releases' quality parameters, the operations to deliver the batch in question are initiated. As part of the standards of the flour industry, every single delivery (material flow) should be supported with the right documents (information flow). Therefore, the standard data sharing along with the flour comprises a certificate of analysis for each delivery (quality parameters) and a delivery note.

Before the project started, the information exchange was purely paper-based. For example, a truck driver should carry a delivery note and the subsequent certificate of analysis. To generate these documents, a series of communication is needed among the different departments of the supplier to reconcile delivery numbers, flour quality parameters, and batches. Internal data silos were in place between supplier's departments (Customer Service, Production, Lab, Transport, and Drivers). Several emails are crossed between departments and information systems to build up the data sets to share with the supply chain downstream.

Historically, information systems have been built around functions within the companies; over time, has bequeathed several data silos within the organisations. Turning to spreadsheets and email to link those systems has become standard practice. As a result, the efficiency of information flow has fallen behind the current material flows.

The solution to fill this gap is to create a system to share data among the stakeholders. A blockchain approach allows all the data under one unique and distributed system, removing several non-value-adding processes from the information systems as data gathering and reconciliation tasks. A shared

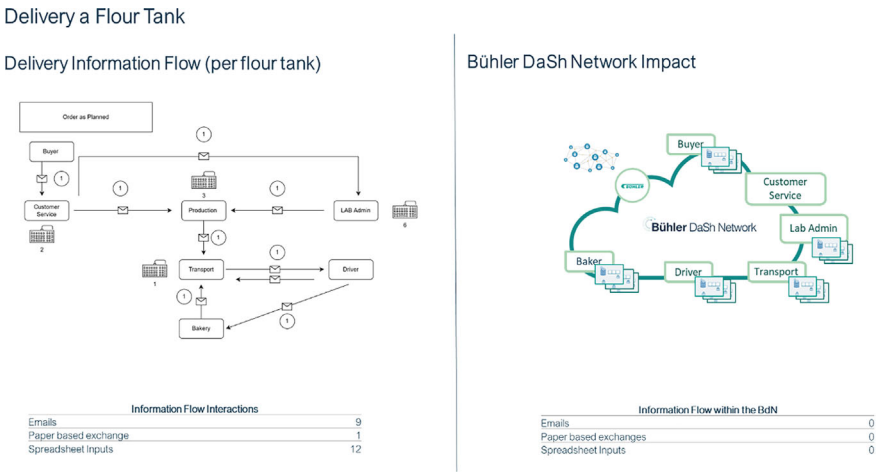

Fig. 4 Customer order workflow (*Source* Authors)

platform has removed "hundreds" of emails among stakeholders, several spreadsheets, and data silos between companies and departments. Instead, stakeholders can use a single version of the data (a single version of the truth) to record events along the flour supply chains. Figure 4 shows the customer order information flow under a unique platform.

Sharing unique records in a common platform, the mill improved its internal processes by removing data silos and communications, and paper-based documents were removed from the system by digitalising the certificate of analysis. The buyer accepts the deliveries digitally and has visibility of the order processing (Fig. 5).

Elimination of redundant and manual processes and communications, reduced lead times, higher data accuracy and granularity of the quality records are some of the recorded impacts of using a platform based on blockchain technology. For example, communications (emails) and data inputs in spreadsheets needed to issue a certificate of analysis are reduced from 223 to zero.

Overall, the solution has the potential to provide visibility, automation, and reliability to the current manual, paper-based, and time-consuming information flows. In addition, this initial work of the supply chain events mapping and digitalisation of the quality data information flow would have the potential to support other data sets relevant to the milling industry motivating a potential extension of the use of technology to capture other data, such as stock levels, vehicle movements, or CO2 emission in the future.

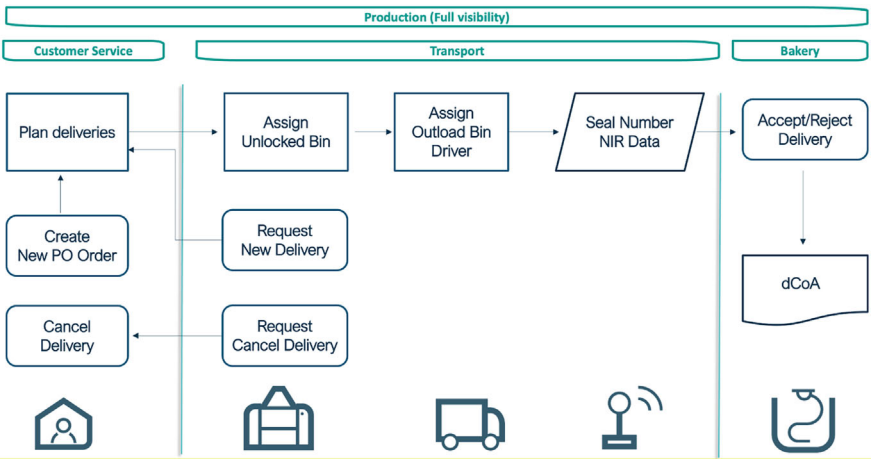

Fig. 5 Miller—buyer workflow under Bühler DaSh network (*Source* Authors)

5 Discussion and Conclusions

This paper set out to investigate how innovative technologies can improve the data flows in the supply chain with a case study from the flour supply chain by reporting blockchain implementation between a flour mill and its buyer. We elaborated on three dimensions of trust: contractual, performance, and goodwill. Blockchain will increase contractual and performance trust but will not impact goodwill trust. Blockchain is an enabling technology that will catalyse trust among stakeholders, but it will not bring trust for itself.

Trust and information sharing go hand-in-hand, perpetuating each other. The more information is shared, the more trust increases, which increases data sharing. In this case study, we take the data-sharing standards already in place and make them digital. It digitalises existing paper records, making immutable records directly from sensors. The same data sets are shared to remove redundant operations, such as reconciliation and double-checking. The blockchain platform is designed to store quality data in the first instance, but it is open to extension by including other operational and logistics data.

The desired outcome of this project is to enable the buyer and supplier supply chain synchronisation, which will allow them to take joint decisions and, in the future, create smart rules to trigger them automatically. Blockchain, as a technology enabler, provides a single version of the truth that is immutable. Therefore, smart contracts running on top of the blockchain platform can trigger automated supply chain decisions embedded in the algorithms that consume immutable data stored in the blockchain (events, delivery times, or orders placed, etc.) to eliminate non-value-added processes

between the supplier and the buyer. Once supply chain actors experience working on a blockchain platform, we expect the level of trust needed to continue the relationship will be lower.

Many blockchain advocates argue that you will not need trust (Morrison, 2016), but this can be misleading. In the short term, you need much more trust between the parties to embark on a blockchain implementation project. Trust will always be needed. While both parties trusted each other to share quality data on paper-based documents, moving these records to digital removes the reconciliation processes needed from a trusted business relationship.

We conclude from our observations and the literature on trust that parties needed to increase their initial levels of trust to get involved in a blockchain environment. Without initial levels of trust between all parties, it would be challenging to realise blockchain implementation.

Long-term commitments between supply chain actors help reduce the uncertainty in the relationship, increasing contractual trust. Commitments to building common systems to be shared over the next decades help increase competence and goodwill trust. The blockchain implementation that started with sharing quality data serves the purpose of building and increasing the level of trust in the network by increasing the visibility and transparency of operations.

In Cocco et al. (2021)'s case study, the IoT is used in the packaging system with Non-Fungible Tokens and Radio Frequency Identification to control the logistics and provenance of the distribution. In our use-case, we use IoT directly in the production line to measure quality specifications of the product as it moves through machines. While Yang et al. (2021) show the use of smart contracts to incentivise the nodes' participation in the network to upload traceability data, in our case, the use of smart contracts is limited to the automatic approval of quality data shared when quality requirements are met.

Once the proof of concept is complete, all parties in the network will be able to see what blockchain implementation can offer to them. It will be easier for all to decide which other data to put on the blockchain, for example, logistics solutions to improve the use of the current transportation resources. By setting up the blockchain channel of communication and information sharing, each actor can explore other collaboration opportunities.

5.1 Contribution to Knowledge

The research findings reported in this chapter have theoretical and managerial implications. Theoretically, we build on the diffusion of innovation theory (Agi & Jha, 2022), and technology organisation, environment framework (Kouhizadeh et al., 2021) to elaborate on how the theory manifests in a real-world setting with a case study from the flour supply chain.

Trust is the centre of any collaboration in the supply chain. We build theoretical knowledge on how blockchain, as an enabling technology, can provide a fairground to share data and consequently build trust. We provide a framework to build trust and data sharing in an encompassed manner.

Managerial implications of the research include guidelines for companies operating in comparable supply chains, specifically those in the food supply chain in adopting blockchain primarily and other Industry 4.0 technologies subsequently to benefit from not only digitising the supply chain information in an immutable manner but also building analytical models on the digitised information for forecasting and optimisation purposes.

5.2 Implications for Practice

At present, material flow operates without human intervention, yet the information flow remains largely manual. In most instances, reports are generated from the system, but the resulting analyses are manually produced and disseminated via email.

The establishment of a unique single register for product quality records harmonises material and information flow. This eliminates quality information silos both upstream and downstream, thereby overcoming the limitations of proprietary quality data.

The adoption of data sharing systems within supply chains will directly influence operational practices for managing information flow and, consequently, material flow. Initially, it will have implications for managing the quality data. However, data sharing systems allow for the synchronisation of not only quality data but also inventories, demand, and production plans. This paves the way for fully autonomous systems in both material and information flow.

Activities that do not add value, such as data entry, sending emails, and other communication tasks can be reduced to a single-entry point in daily operations. Consequently, resources allocated to these tasks can be minimised.

5.3 Limitations and Future Work

This research paves the way toward an autonomous mill, establishing the foundation for supply chain synchronisation. The blockchain application described in this paper and the supporting Bühler DaSh Network will eliminate spreadsheet-based data analysis and stop double data entry, allowing a single version of the truth in one system. It also builds customer intimacy and facilitates further analytics applications on accurate and real-time data.

Limitations include the work being reported on a single case study from the flour supply chain with a miller and one of its customers. While the process explained in this paper is expected to be transferable to other comparable buyer–supplier relationships, there may be company-specific variations in levels of automatisation and digitalisation that necessitate modifications to the explained system. In addition, the social aspects of deploying a new technology to increase visibility are reported only as perceptions of the observers and project participants. The next iteration could devise a questionnaire survey to gather reflections from the system users before and after implementation to measure the impact of deploying this new technology.

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Social Media Engagement in Container Logistics

Sonali Shankar and P. Vigneswara Ilavarasan

1 Introduction

Social media is observed to have reshaped the ways by which businesses are conducted today. The use of social media for product marketing, buying, and obtaining feedback from customers has evolved at a rapid pace over a decade. The use of social media for product marketing, buying, and obtaining feedback from customers has evolved at a rapid pace over a decade. Social media has given rise to innovative business models and adds value to businesses with negligible infrastructure and manpower costs. However, despite the positive impact of social media for promoting businesses, improving return on investment (ROI), increasing customer base, enabling brand engagement, and building brand image, it is not very popularly used in the area of supply chain management. This study is an attempt to bridge this gap and provide social media possibilities for container supply chain and logistics.

The outburst of the COVID-19 pandemic from the city of Wuhan caused multidimensional damages to the world population and economy. On January 31, 2020, the World Health Organization (WHO) declared

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the corona virus as an international public health emergency and by March 11, 2020, it was declared a pandemic. Consequently, the pandemic of such magnitude has gravely affected international relations with an adverse effect on global economic and socio-psychological factors (Pan et al., 2020). It is reported that 94% of Fortune 1000 companies have faced COVID-19 driven supply chain disruptions (Fortune, 2020). The main reason being the presence of tier-one and tier-two suppliers of 938 companies among Fortune 1000 in Wuhan, China (Ivanov, 2020). The measure to combat this virus has placed people across the globe in the largest quasi-quarantine. At this time of global health crisis and unfavourable business complexities, social media provides a significant avenue for collective effort when physical world is at standstill.

This study aims to understand social media usage of container logistics firms and their emerging roles during pandemic. It also explores the nature of the information disseminated by container logistics companies on social media during global lockdown. The plethora of information was available on social media during COVID-19, high level of user participation was observed on microblogging platforms such as Twitter. With the advent of pandemic, this participation was transformed into active participation and subsequently into collective engagement or effort. This study is motivated by the observation that though container supply chain and logistics firms were not very keen on utilizing social media usage until COVID-19 but during pandemic, they were observed to disseminate information on social media platform like Twitter actively and enabled engagement. Consequently, this study analyses the trajectory of how social media usage by container logistics firms on the subject of mutual interest, COVID-19 pandemic. This brings us to the following research question:

> How does social media usage by container supply chain and logistics firms affords collective engagement?

In doing so, the theory of *Affordances* from the information systems scholarships is employed.

The rest of the study is divided into five sections with the theory of affordances explained in Sect. 2. Methodology is presented in Sect. 3 followed by the results and analysis in Sect. 4. The implications of the study for practitioners and academicians are discussed in Sect. 5. In the end, the limitation of the study and the future research endeavours are presented in Sect. 6 along with the conclusion.

2 Theory of Affordances

The affordance theory enhances the understanding of the relationship between an object (technology) and an actor (human being) to foster business information technology (IT) value (Bygstad et al., 2016; Leonardi, 2011; Tim et al., 2018). With the affordance theory, we not only consider the social aspects, but the technical specifics are also given equal importance. The *afford* verb can be found in the dictionary, but its noun *affordance* is a term provided by Gibson, which means 'to offer.' According to Gibson, "*An affordance is an invariant combination of variables, and one might guess it is easier to perceive such an invariant unit than it is to perceive all the variables separately*" (Gibson, 1977, p. 58). For example, consider Fig. 1, in Fig. 1(a), a simple line does not provide us with much information, whereas if all the lines in Fig. 1(b) are considered together in their familiarity and referential totality (Osmundsen et al., 2022), then it can be perceived as a triangle in Fig. 1(c).

The theory has been implemented in various non-indigenous research domains. For instance, affordance is used to establish a relationship between the urban environment and quality of life (QoL) of the residents (Olsen et al., 2019). Similarly, the sustainability of the urban socio-ecological system is studied to determine the association between the environment and people's everyday experiences (Samuelsson et al., 2018). The important reason to use the affordance theory in this study is the ability of the theory to distinguish the actions potential of the technology from its outcome (Du et al., 2019).

The relational concept of affordance theory is elucidated by the relationship between goal-oriented actors and features of technology. The affordances are dependent on the specific intentions of actors because of their relational property (Orlikowski & Scott, 2008; Sun et al., 2019). Consequently, the same feature of technology may give rise to different affordances for different actors (Leonardi, 2013; Vaast et al., 2017). The different affordances render

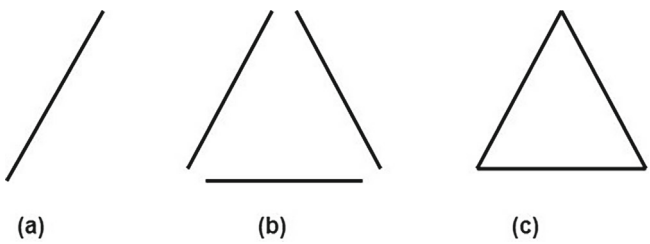

Fig. 1 Perceiving all the variables together (*Source* Authors)

the users using similar features of social media with varying roles of enactment. These affordances are broadly classified as individualized, shared, and collective affordances (Leonardi, 2013).

The present study considers only the shared and collective affordances. This is because of the latent potential of the pandemic to call for a collective engagement via connective action among actors belonging to different groups. The similar needs and uses of technology by the actors give rise to shared affordances (Leonardi, 2013), whereas collective affordances correspond to the affordances actualised by the actors who use technology for their own requirement but as a whole produce collective action (Oborn et al., 2011). The literature has examined the social media affordances for individuals and groups (Fayard & Weeks, 2007; Vaast & Kaganer, 2013) as well as for departments and organizations (Strong et al., 2014; Volkoff & Strong, 2013). However, the technological affordances of social media use for a specific industry are not explored even though the image of a firm on the social media platform largely influences its customer base. This accounts for another major research gap which we will address in this study. Moreover, understanding of technological affordances helps decision-makers to draft their social media policies better.

3 Methodology

In this section, we elaborate upon the theoretical underpinnings of this work, primarily focused on the social media use, affordances, and the emerging roles of the actors in a connective action during a pandemic.

Following are the steps followed in this methodology:

Steps:

Step 1: Collect tweets of Fortune-500 and Alphaliner-100 container logistics companies.

Step 2: Text mining of collected tweets to determine the most popular themes during this period.

Step3: Qualitatively identify the three connective action episodes (CAEs).

Step 4: Collection of as many threads of tweets on each aforementioned connective episode.

*Step 5: A*nalysis of each episode utilizing the text mining and thematic analysis.

Step 6: Clustering of tweets on the basis of feature similarity.

Step 7: Investigating similarities, differences, and interdependence among different clusters in each episode.
Step 8: Determining the enactment roles of actors.
Step 9: Characterization of information disseminated by actors in each action episode.
Step 10: Determining the collective engagement of the actors.

3.1 Data Collection

Only the container logistics companies that have recognition worldwide are selected for the analysis. In order to do so, Fortune-500 and Alphaliner-100 lists containing top companies around the world are considered. The top container logistics companies are selected which are either present in the Fortune-500 or Alphaliner-100 or both lists. The Alphaliner-100 is a worldwide reference in container/liner shipping representing top 100 global liner operators. It is a regularly updated ranking. Consequently, a total of unique 143 container logistics and shipping companies were obtained. Next phase of company's selection dealt with determining their social media presence. Out of 143 companies 48 container logistics companies had active presence (twitting at-least once in a week) on Twitter and hence were selected for the analysis.

We collected tweets of selected companies since January 10, 2020 (first advisory issued by the World Health Organization on novel coronavirus[1]) until April 30, 2020, for more than three-month period from the Twitter.com website. This search resulted in more than 14,000 tweets. The tweets collected earlier were first converted to lowercase text and then the tweets preprocessing was carried out involving removal of the common words, punctuations, chain words, special characters, web and hyperlinks and @usernames. Finally, preprocesses data was used for analysis. In addition, syntactical errors (stop words, punctuations, case corrections, white spaces) correction was being done using R scripts which utilized the existing libraries of R. Semantic correction was more of a slow and manual operation, where each tweet was looked for relevancy and non-English words were eliminated.

Since not all the tweets during the collected period correspond to the COVID-19 pandemic, the inclusion of such tweets would have led the erroneous results. Therefore, refined data is considered for further analysis that

[1] WHO advice for international travel and trade in relation to the outbreak of pneumonia caused by a new coronavirus in China, https://www.who.int/news-room/articles-detail/who-advice-for-international-travel-and-trade-in-relation-to-the-outbreak-of-pneumonia-caused-by-a-new-coronavirus-in-china.

has COVID-19 as prevalent topic of discussion. In order to better understand the twitter contents by container logistics firms, we identified the connective action episodes (CAEs) that talk about COVID-19. The concept of connective action is proposed by (Bennett & Segerberg, 2012a). Connective action represents new form of collective engagement (CE) where multiple actors come together on social media for a common cause (Bennett & Segerberg, 2012b; Vaast et al., 2017).

3.2 Social Media Affordances

The tweets were searched for the hashtag given under *omnipresent* column in Table 1. This resulted in more than 9000 tweets from a total of 14,107 tweets. The corresponding tweets were extracted for further CAEs analysis. The CAEs are identified based on the following selection criteria (Bennett & Segerberg, 2012a; Stohl, 2014; Vaast et al., 2017):

- The multiple actors are involved in social media use.
- The involvement of multiple actors is characterized by a common cause.
- The CAEs are unfolded over time.
- The CAEs involved virtual or on-ground action.

Table 1 List of similar themes emerged as CAEs

Omnipresent	FightCorona	StaySafe	Thankyou
Covid19	FightCorona	StaySafe	Thankyou
Covid19	Fightcoronavirus	StayHome	ThanksForDelivering
COVID19: COVID-19	IndiaFightsCorona	Stayhome	ThankATrucker
Coronavirus	21daylockdown	BeSafe	ToThoseWhoKeepUsMoving
COVID19	Coronavirusindia	Staysafe	EverydayHeroes
Coronavirus	Weareinthistogether	SocialDistancing	HeroesAtSeaShoutOut
Covid_19	WeAreInThisTogether	Socialdistancing	Thankarailroader
Coronavirus	Strongertogether	Safetyfirst	HeroesAtSeaShoutout
CoronaVirus	Togetherness	StayHomeStaySafe	Hlheroes
COVID 19's	AsONEWeCan	Stayathome	EverydayHeroes
COVD 19		Stayhealthy	Thankatrucker
COVID 2019		Stayinformed	ToThoseWhoKeepUsMoving
		HealthandSafety	TRUEHEROES
		HealthAndSafety	WarehouseWarriors
			ToThoseWhoKeepUsMoving:

Source Authors

Various focal CAEs were discussed during the exploratory readings of the Twitter data. It was observed that the majority of tweets were talking about common theme encapsulated in different terminologies. The tweets pertaining to each list were considered as one CAE as shown in Table 1. The logistics firms were observed to exhibit three affordances, namely: *Combat, Resolute,* and *Appreciation* explained under "FightCorona", "StayHome", and "ThankYou" CAEs respectively as discussed in Sect. 4 in detail.

3.3 K-Means Clustering

The fundamental reason to use the clustering technique is to scrutinize if there exist certain shared patterns among users. Clustering is carried out based on similarities in Twitter's feature use. The non-hierarchal K-means is used as a clustering technique for its faster computational ability and potential to handle large dataset (Shankar et al., 2021a, 2021b). It is capable of computing nonlinear clusters of uneven sizes.

The clusters are formed using the similarity in feature use, as listed in Table 2. The tweets that are similar in Twitter feature use of microblogging (tweet), re-tweet, mentions ('@': Usually use to initiate a conversation with a person to whom the tweet is directed to), hashtags, media links, and media contents are grouped. Once the clusters are identified for the entire corpus of tweets, we then examine the nature of these clusters in each episode. For instance, whether the members of the same cluster participated similarly in all clusters or exhibited different patterns in different clusters. This quantitative analysis is combined with qualitative analysis to draw insights on the similar and indifferent nature of the participants evolved over time.

4 Result and Analysis

As discussed earlier, affordance is the 'ability' of the technology (social media here) to address to the needs of the goal-oriented actors (container logistics and supply chain companies here). Consequently, we argue that the social media usage by container logistics companies had facilitated the goal of the companies to fight this pandemic and thus, giving rise to the affordance of *combat.* Similarly, the social media usage gave rise to the affordances of *resolute* and *appreciation* where companies were determined to survive COVID-19 and aimed to showcase their respect towards the ground workers.

Table 2 Key Twitter features used for clustering

Feature	Description
Tweet	It is a way of disseminating content, microblog posts of limited characters either with or without media and links by a user on a Twitter platform
Re-tweet	It is the feature to repost or circulate the already existing tweet in the network
"@"	Mentions, '@,' are used to direct a published content to a specific user regardless of whether they follow you or not. It is a way to draw someone's attention to a corresponding piece of content
"#"	Hashtag symbol is placed in front of words to provide context or allow easy search for it. It facilitates social tagging categorization
URLs	Twitter allows sharing of links of other web-based content like videos, blogs, websites, news, etc. in a microblog post
Media Content	Along with a short message, users can embed media content like photographs and videos in their tweets

Source Authors

The three identified CAEs, namely "Fight Corona", "Stay Safe", and "Thank you" have unfolded over time. The summary of all three episodes is provided in Table 3.

The "Fight Corona" episode calls to fight the pandemic without losing hope. Some of these calls provide information to their network about what initiatives they are taking to fight coronavirus in their capacity to encourage people to work for the welfare of society.

In the episode "Stay Safe", the participants urge users to stay safe and stay home to protect themselves from contagious infection. They aim to instil the sense of responsibility among users to protect themselves since the sea-warriors are working day-in and day-out for them while putting their health risk aside.

During COVID-19, nations came together to thank the healthcare workers, police, and home guards. Meanwhile, the contributions of the container logistics people were not noticed much. The "Thank You" episode corresponds not only to the healthcare workers but a majority of it aims to show gratefulness to the sea-side workers for facilitating the availability of daily essentials.

The episodes showed how actors came together on a social media platform for a common cause. Their collective engagement for the welfare of the society shed down bars of competition among highly competitive industry like container and shipping logistics.

To quote an example:

Table 3 Summary of the three episodes

Episodes	FightCorona	StaySafe	ThankYou
Purpose of connective action	Sharing initiatives to boost morale and disseminate positivity to fight pandemic	Encouraging people to stay safe to fight coronavirus	Appreciating the frontline workers to keep working during the pandemic for the welfare of the society
Period covered	Mostly March–May 2020	Throughout the investigation period	Mostly April–May 2020
Total number of tweeters	380	310	440
Example	APM Terminals Nigeria donates towards the @UN Nigeria Basket Fund to support the government's healthcare response towards COVID-19: https://t.co/s5KutKvks8 #Covid19 #APMTerminals #Maersk #workingtogether	Feel free to share a picture of yourselves (and your families) with the Flag and the hashtags #homebutnotalonesg and #StaySafe For more information, please visit https://t.co/oNFBmMfRNz	The first Maersk Bridge flights landed this morning with Personal Protective Equipment (PPE) for health care workers. #Maerskbridge #ThankYouHealthCareWorkers https://t.co/sCMORepPx0

Source Authors

"We stay on board for you. Stay home for us". Our #hlheroes are sacrificing time with family and friends to make sure you have access to everything you need. So, #stayhome and help us beat this together. @cmacgm @MSCCargo @DHLsupplychain help us share the message! #wearehl".

4.1 Cluster Analysis

Cluster analysis is carried out to identify the enactment roles of the actors. This understanding helps to determine the different social media strategies adopted by the container logistics companies during pandemic.

Cluster 1 is the smallest of all three clusters. It has a maximum amount of "reply". They are dealing with replying to the counterpart companies mentioning counterpart companies' official Twitter handle or to the individual CEOs of the firms associated with them. They also re-tweet the already available content on social media without creating any niche content of their own. Since they are low in original content, they have the least average favourites. However, they show the highest average use of mentions ("@"). The rationale explanation is since they are mostly replying to the users across their network; therefore, to direct their reply to a specific user, they use "@". As a result, cluster 1 shows the highest use of Twitter feature "*Mentions*".

Cluster 3 is the largest that evolved during the analysis. According to (Surucu-Balci et al., 2020), the tweets that have images, videos, or gifs in their post invoke a higher level of engagement and vividness. On comparing cluster 3 with cluster 2, cluster 3 has 3% higher re-tweet posts, 37% more photos, 24% more videos, 83% more hashtags and, 31% more tweets/tweeters than cluster 2. This accounts for the better realization of Twitter features in cluster 3 than cluster 2 with an exception in the presence of *URLs* and *Mentions* in the posts.

The enactment roles of that of the three clusters present analogy with the Miles and Snow's business strategy typology (Miles et al., 1978), namely, ***Defenders***,[2] ***Analysers***,[3] and ***Prospectors***.[4] There is no *Reactor* in this study of three clusters. Though the roles are not conspicuous when clusters are considered in isolation, but they exhibit the near similar characteristics of Miles and Snow model when viewed in its referential totality (Lanamäki et al.,

[2] Defenders produce only a limited set of products directed at a narrow segment of the total potential market (Miles et al., 1978).

[3] Analyser is a unique combination of the Prospector and Defender types and represent a viable alternative to these other strategies (Miles et al., 1978).

[4] In many ways, Prospectors respond to their chosen environments in a manner that is almost opposite of the Defender (Miles et al., 1978).

2015). With this, the role of three clusters becomes clear in relation to the Miles and Snow strategy.

4.2 Role of Actors in Disseminating Information in CAEs

The information circulated by the defenders, analysers, and prospectors in CAEs is situational information. Therefore, it is crucial to develop a proper understanding of the situational information for designing appropriate information publishing strategies for the global crisis. Li et al. (2020) segregated the propagation of situational information on the Chinese equivalent of Twitter named Sina Weibo during the COVID-19 crisis. Based on this characterization, the situational information is divided into seven types namely, a) caution and advice; b) notifications and measures been taken; c) donations of money, goods, or services; d) emotional support; e) help-seeking; f) doubt casting and criticizing; g) counter-rumour. The information propagated by the actors in three CAEs is characterized manually after a critical examination of each tweet in the episodes. The type of information circulated by actors in CAEs is summarized in Table 4. As per the analysis, no tweet in the database corresponds to the *doubt casting and criticizing* or *counter-rumour* category. This is because of the presence of highly professional content in business-to-business (B2B) firms like container logistics and shipping.

The defenders are focused on replying to the already existing content with their replies directed to the individuals or the groups. This makes them rich in notifying or defending their firm's stand during the pandemic and making their network aware of the actions taken by them to combat coronavirus. Besides, they also propagate the information related to the advice and cautions.

The analysers have got the greatest number of favourites among all the actors. They create moderate original content and are rich in re-tweeting. The information propagated by the analysers mostly appeals to the user's emotions by praising their employees, appreciating competitors' social work, and showing gratefulness to the healthcare workers. Apart from providing emotional support, they also share content regarding their selfless contribution to society to combat the global health crisis. This accounts for their popularity with the maximum number of favourites among all three clusters.

As discussed earlier, the analysers and prospectors are observed to be similar in their feature use and are now observed to be similar in their content-type also. The prospectors are also emotionally appealing, with most of its content

Table 4 Information dissemination of actors in three connective action episodes

Episodes	Defenders	Analysers	Prospectors
Roles in connective action	Addresses individuals or firms by personalized expression. Defends their stand	Rekindle interest in connective action using additional information from others	High in original content creation. Increases the visibility of the content via media use
Information type in ***Fight Corona*** episode	Notifications or measures to be taken	Donations of money, goods or services	Providing emotional support
Information type in **Stay Safe** episode	Caution and advice	Providing emotional support	Notifications or measures to be taken
Information type in ***Thank you*** episode	Notifications or measures to be taken	Providing emotional support	Providing emotional support
Most used feature	Mentions	URLs/Re-tweets	Hashtag/Tweets
Example	RT @AAR_ FreightRail: RT @Rodolphe-Saade: Keeping our 110,000 employees safe is our overriding priority. We have taken appropriate actions in line with recommen-dations from medical experts. Having anticipated this quarantine, we are able to ensure the health of our teams as well as business continuity #COVID19 #Staysafe	The CMA CGM Group donates 100,000 face masks to the hospitals of #Marseille and to the Provence-Alpes-Côte d'Azur Regional Health Authority to #fightcorona	Our worldwide managers have a message to share with you! On all five continents and the world seas, from home or onboard, our team members accomplish a remarkable work for our customers #Thankyou #TakeCare

Source Authors

dealing with providing emotional support. Unlike analysers, the prospectors are also a rich source of notifications or measures to be taken related information.

4.3 Emergence of Affordances of Collective Engagement

The facilitating condition, that helps in the realization of the affordances, here are COVID-19 pandemic and the social media presence of container logistics companies. Had there been no pandemic or social media presence of companies, they had not shown zeal to combat together or resolute or be grateful to someone collectively. These facilitating conditions help in the realization of the affordances. As shown in Fig. 2, the clustering of affordances leads to the emergence of affordance of collective engagement. The interesting readers can read about the clustering of affordances in (Thapa & Sein, 2018).

The collective engagement is a phenomenon of interaction among users for the exchange of ideas, information, or emotions, etc. over a mutual interest. The collective engagement affordance has emerged as an output of the clustering of primary affordances of combat, resolute and appreciation. As shown in Fig. 2, the actualization of primary affordances leads to the content propagation. Whereas, actualization of collective engagement (CE) leads to likes, replies, share, etc. which is used to propose a measure of collective engagement. The affordances are actualised with the help of the facilitating condition.

Social media engagement is often considered as the firm's interaction with the users in its network. The state of interaction and responding to the users

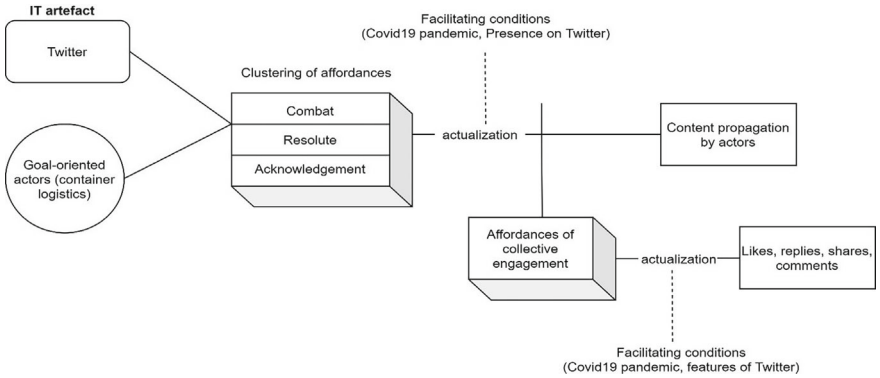

Fig. 2 Emergence of affordance of collective engagement (*Source* Authors)

through official social media account promote co-creation, shared decision making, and mutual understanding (Taylor & Kent, 2014). The involvement of the users in the dialogic loop is observed to enhance social media engagement (Chen et al., 2020; Taylor & Kent, 2014). The similar is supported by the theory of dialogic communication. However, empirical investigation into the responsiveness and social media engagement are lacking in the existing literature (Chen et al., 2020).

Most of the studies have quantified user engagement for a social media platform, for instance, (Bonsón & Ratkai, 2013) quantified the stakeholder engagement as a measure of three dimensions of social media use. According to them, social media engagement is a sum of popularity, commitment, and virality measured by likes, comments, followers, posts, and shares. This social media engagement index has been used widely in the literature (Agostino & Arnaboldi, 2016; Bonsón et al., 2019; Denktaş-Şakar & Sürücü, 2018). Few studies have also used comments, shares, and likes directly as a measure of engagement (Brubaker & Wilson, 2018; Kim & Yang, 2017).

We argue that the social media affordances of CE for a connective action can be quantified as a measure *of popularity, virality, responsiveness, and vividness* of the content propagated by the actors as shown in Table 5. Unlike stakeholder or user engagement matrices, the involvement of the actor (content creator) is as important as a network user's involvement in CE. Therefore, dimensions are considered in a way to take into account the involvement of both actors and users. The popularity and virality are network-centric (obtained by the activities of users/followers in a network) whereas, responsiveness and vividness are actor centric attributes (employed by the participant or the content creator). The proposed matrix for CE can be used to empirically test the engagement of the social media actors.

5 Discussion and Implications

The social media usage by B2B market like container logistics can bring great deal of business opportunities. The main reason for very limited use of social media by B2B companies, especially logistics sector, is their limited awareness to understand social media usefulness.[5] This study is a step towards harnessing the power of social media by logistics companies, not only during the challenging times of COVID-19 but also to see it as an additive business advantage and strategise their business policies accordingly. The

[5] Logistical advantages to harnessing social media, https://supplychaindigital.com/logistics-1/logistical-advantages-harnessing-social-media.

Table 5 Measurement of collective engagement

Representing	Name	Formula	Measure
Users	Popularity	Total favourites/total number of tweets	The average number of favourites per tweet
	Virality	Total retweets/total number of tweets	The average number of retweets per tweet
Actors	Responsiveness	Total replies/total number of tweets	The average replies per tweet
	Vividness	Total photo-video-gif/ total number of tweets	The average embedded media per tweet
CE = Popularity + Virality + Responsiveness + Vividness			

Source Authors

container logistics is a billion dollar industry therefore, two–three percentage of improvement will render huge cost benefits (Shankar et al.,). Social media provide opportunity to gain this benefit at a minimalistic cost.

It is observed in the literature that when the users from diverse professional backgrounds and geographical locations came together on social media because of crisis, they were inter-dependent on each other for their content generation and dissemination. For instance, in Vaast and Kaganer (2013), the obtained three clusters were inter-dependent on each other for their content and its distribution. Whereas when we examined the social media activity of the participants belonging to the same industry, they exhibit independent behaviour. The rationale reasoning for such behaviour is their competitive nature to create the content of their own and display their contribution towards society during this global crisis.

This study provides insights that will help the users to create momentum in connective action as a level of collective engagement with the use of social media. This will enable users to develop a proper understanding of such collective endeavours and can strategize accordingly to put impactful use of social media features forth. This, in turn, will generate a great opportunity to encourage and engage users in an online initiative where offline mobilizations are not possible.

6 Conclusion and Future Directions for Research

This study is probably first of its kind to propose the integration of the theory of the strategic operation (Miles and Snow) into information systems theory (Affordance theory) for social media strategies of B2B firms like container logistics during the pandemic. It surely requires further investigation to establish this research with other firms and examine if the framework still holds good. We observed that the actors of the same industry use the same social media platform differently. This is in agreement with the affordance literature, where similar IT artifact affords different potentials for different goal-oriented actors (Orlikowski & Scott, 2008, 2014).

How social media use by container logistics firms during pandemic leads to the emergence of new forms of affordances and how does it facilitate the collective engagement of the users has been discussed in the chapter. The three social media affordances of *combat, resolute, and acknowledgement* were identified during a pandemic. The enactment roles of the firms were determined by their social media strategies that corresponded to the Miles and Snow typology: defenders, analysers, and prospectors. From the pieces of evidence in the literature, the measure of CE on social media was proposed for COVID-19. The limitation of the study is the use of only one social media platform, Twitter, However, future research can consider more than one social media platform for analysis.

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Part V

Digital Manufacturing Capabilities for Emerging Technologies in Supply Chains

20

Digitalization in Bioproduction—AI & Digital Twins for Process Optimization

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1 Introduction

Lactic acid bacteria have been used in foods for centuries for their beneficial properties. As science has advanced, these bacteria have been identified and many of these bacteria and their products have been granted GRAS (Generally Recognized as Safe) status by the FDA (Food and Drug Administration). Lactic acid bacteria prolong the shelf life of foods by inhibiting the growth of bacteria that cause food spoilage and combating disease-causing bacteria growth in food. To achieve this, one of the strategies they use is to lower the pH of the environment and thus inhibit the growth of harmful bacteria,

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and another is to specifically stop the growth of these pathogens with various molecules they produced called antimicrobial peptides. The most well-known group of antimicrobial agents are bacteriocins. Nisin, one of the bacteriocins, is produced by certain strains of *Lactococcus lactis*, a lactic acid bacterium, and has been used in many foods, especially cheese, since 1951 in many countries. The fact that the toxic effect of nisin for humans is equivalent to that of table salt, makes it a very attractive option compared to the harmful chemical food preservatives that can be substituted.

The bottleneck to the more widespread use of nisin as a food preservative is that nisin is not chemically synthesized, but only produced by bacteria and then purified from the cellular environment. Of course, scientific developments have produced many options for solving this problem. The process started with optimizing the growth conditions of the producer cell, *Lactococcus lactis*, and now, with genetic modifications both in the producer strain and in other bacteria that produce nisin heterologously, it is possible to produce nisin at higher yields. Whatever bacteria are used in nisin production, and whatever genetic modifications are made to these bacteria, it is essential to optimize the growth conditions for high yields of nisin production under fermentation conditions. These conditions include many variables such as temperature, pH, dissolved oxygen concentration, stirring speed, etc., and considering that only one value of each of these variables needs to be changed in each experiment, this process will be both very costly and time consuming. At this point, taking advantage of digitalization is the most rational solution.

2 Role of the Lactic Acid Bacteria in Food Safety

Lactic acid bacteria (LAB) constitute a broad bacterial group mainly characterized as non-motile, Gram-positive, catalase-negative, nonsporulating, and facultative anaerobic cocci or rods with low GC content genomes. LAB is often found in contact with food and plant material, and their habitats could be animal and human. According to their metabolic traits, this group of bacteria is classified as homofermentative, facultative heterofermentative, and heterofermentative (Gänzle, 2015). Homofermentative metabolism causes the production of lactic acid, while different products such as formate, acetate, and ethanol, in addition to lactic acid, are yielded at the end of heterofermentative metabolism (Zaunmüller et al., 2006).These microorganisms were announced to be at GRAS status by the FDA (George et al., 2018), and they have been used extensively in numerous applications over the years

for the benefit of humans. For example, some members of LAB are used as a starter culture for dairy products and some beverages to give them a unique taste and texture. Due to their probiotic nature, they can be used directly as a food additive to improve gut health (Perdigón et al., 2002). In recent years, the potential use of these microorganisms in biotechnological fields has been increasing, such as their use as live vaccines (Azizpour et al., 2017), as a source of various pharmaceutical agents, e.g., organic acids (Bangar et al., 2022) and B-group vitamins (Zhu et al., 2009), and their role in controlling serum cholesterol levels (Gilliland, 1990) and some types of cancer (Goh et al., 2022). Moreover, LAB and its products are believed to have a high potential for protecting against and treating viral infections, including COVID-19 disease (Umair et al., 2022). LAB prevents spoilage of foodstuffs and can provide safe foods with prolonged shelf life by combating pathogens in different ways. Lactic acid production inhibits pathogens by lowering the pH to almost below 4.0. Release of the antimicrobial peptides (AMPs) is another defence strategy of these bacteria (Eijsink et al., 2002; Kleerebezem & Quadri, 2001). AMPs are mainly produced as "guns" to compete with other bacteria in an ecological niche, thus providing a competitive advantage to the producers. Among the AMPs, bacteriocins, ribosomally synthesized peptides, have attracted increasing attention thanks to their broad antimicrobial activity spectrum and safety for human consumption (Verma et al., 2022). Because of these properties, bacteriocins are often used as natural food preservatives.

According to their biochemical and genetic features, bacteriocins are separated into four groups (Bhattacharya et al., 2022). Class I and Class II bacteriocins are both heat-stable peptides; the ones subjected to post-translational modifications belong to Class I (also called lantibiotics), while the others without any modification are classified as Class II (Maurya et al., 2021). Class III are heat-labile proteins (Preciado et al., 2016), and Class IV consists of complex proteins containing lipid and carbohydrate moieties (Sabo et al., 2014). Nisin is the best-known member of Class I bacteriocins (lantibiotics) with a molecular weight of 3354 Daltons and consists of 34 amino acids with many post-translational modifications (Cheigh & Pyun, 2005). Nisin A and Nisin Z are the most abundant variants of nisin in nature. Nisin is produced by certain strains of *L. lactis,* which are naturally found in cheese and raw milk (Gijtenbeek et al., 2021). It shows its antimicrobial activity via lipid II in the bacterial cell wall by two distinct mechanisms: highly specific pore formation and interruption of bacterial cell wall synthesis (Wu et al., 2023). It is especially effective on Gram-positive bacterial cells. On the other hand, it can also be active on Gram-negative cells in the presence

of a chelating agent such as EDTA (Delves-Broughton, 1993; Stevens et al., 1991).

Nisin biosynthesis is carried out by a gene cluster of 11 genes (Fig. 1). Among these genes, the *nisA*(*Z*) gene encodes the NisA(Z) precursor peptide consisting of 57 amino acid residues. The N-terminal leader peptide of 23 amino acids modifies the nisin precursor. Nisin prepeptide is then converted into its active form by various post-translational modifications and, subsequently, the cleavage of the N-terminal leader peptide. NisB and NisC proteins are responsible for the post-translational modification of the nisin prepeptide. Modified nisin molecules are then transported out of the cell by the ABC transport protein NisT. The proteolytic cleavage of the leader peptide is accomplished by the serine protease NisP. NisI and NisFEG proteins work together in the immunity of nisin-producing cells against nisin. Nisin triggers its own biosynthesis at the early growth stage. The quorum sensing system regulates nisin production by the NisR and NisK regulatory proteins (Cheigh & Pyun, 2005; Kleerebezem & Quadri, 2001).

Nisin has enormous commercial potential. The global market volume of nisin is growing steadily and is expected to exceed US$ 500 million by 2025 (Weixler et al., 2022). Among the bacteriocin AMPs, only nisin and pediocin PA-1 are commercially available as food additives. Nisin (sold as Nisaplin) is used as a preservative in many foodstuffs in more than 50 countries around the world, while pediocin PA-1 (sold as Alto®2341) is authorized for use in meat products only (Sidhu & Nehra, 2021; Silva et al., 2018). Although they are promising candidates for the food and medical industry, new bacteriocins fail to enter the market due to the high cost of production and demanding purification processes of these substances. For example, natural nisin-producer strains of *L. lactis* grown in batch fermentation using a complex medium resulted in commercial preparations with only 2.5%

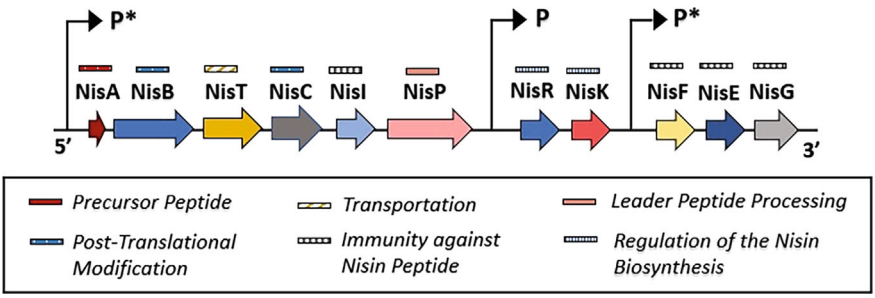

Fig. 1 The gene cluster responsible for nisin biosynthesis (*Source* Authors)

nisin (Chikindas et al., 2018), indicating an urgent need to improve these processes.

3 Genetic Modification Strategies and Bioreactor Systems for the Efficient Production of Antimicrobial Peptides

Since their production is expensive and requires complex downstream processes, the development of different approaches is crucial for more convenient and high-level production of bacteriocins. For this, one way is to optimize cultural conditions to improve the yield of AMPs by the natural producers. The other way is to generate recombinant cells by genetic manipulations. Tables 1 and 2 summarize the studies involving individual and combination fermentation strategies and manipulations of different *L. lactis* producer strains by genetic engineering to overcome the low rate of nisin production, respectively.

4 Digitalization of Bioproduction Systems

As explained in the previous sections, bacteriocins are highly important as natural preservatives in food safety. The most significant limitation in their high-level manufacturing is the high cost due to the challenges in the purification step and the requirement of a complex nutritional condition to cultivate lactic acid bacteria. Therefore, fermentation technology should be improved to increase the yield by reducing the manufacturing costs (Jawan et al., 2021). To this aim, digital transformation in the food industry has been accelerated, with a focus on Industry 4.0, which is driven by technological advances including artificial intelligence, machine learning, and sensor systems. The computational strategies applied to improve bacteriocin production are discussed in the following sections. Furthermore, digitalization concept is comprehensively covered from the perspective of food industry.

4.1 Computational Optimization Studies for Enhanced Bacteriocin Production

In the conventional medium optimization methods (e.g., one-factor-at-a-time experiments), one factor is often changed over a specific range while

Table 1 Examples of individual and combination fermentation strategies to increase nisin yield

Lactococcus lactis Strain	Fermentation System	Growth parameters: Growth Medium (Modified Content)	Temperature (°C)	pH	Agitation (rpm)	Aeration (Dissolved O_2)	Cell Dry Weight (CDW, g/L)	Nisin Activity (IU/ml)	Reference
NIZO 22186	Batch fermentation	CM medium (10 g/L Sucrose)	30	6.8	50	NA	2.34	1.793	Vuyst and Vandamme (1992)
NIZO 22186	Batch fermentation	CM medium (4% Sucrose, 5% K_2HPO_4)	30	6.8	50	NA	1.65	3.500	Vuyst and Vandamme (1993)
ATCC 11454	Batch fermentation	CM medium (40g/L Sucrose, 20g/L Nitrogen)	30	6.8	100	NA	4.2	2.371	Lv et al. (2004)
	Fed-batch fermentation	CM medium (2g/L Sucrose—constant feeding, 20g/L Nitrogen)	30	6.8	100	NA	3.8	3.887	

Lactococcus lactis Strain	Fermentation System	Growth parameters: Growth Medium (Modified Content)	Temperature (°C)	pH	Agitation (rpm)	Aeration (Dissolved O_2)	Cell Dry Weight (CDW, g/L)	Nisin Activity (IU/ml)	Reference
	Fed-batch fermentation	CM medium (40g/L Sucrose, 2g/L Nitrogen—constant feeding)	30	6.8	100	NA	4.2	4.131	
	Fed-batch fermentation	CM medium (2g/L Sucrose, 4g/L Nitrogen—slow feeding)	30	6.8	100	NA	4	3.844	
	Fed-batch fermentation	CM medium (2g/L Sucrose, 4g/L Nitrogen—fast feeding)	30	6.8	100	NA	4.3	3.116	
ATCC 11454	Batch fermentation	CM medium (30 g/L Sucrose)	30	6.8	100	NA	NA	2.658	Lv et al. (2005)
	Fed-batch fermentation	CM medium (2 g/L Sucrose)	30	6.8	100	NA	0.85*	4.961	
	Fed-batch fermentation	CM medium (16 g/L Sucrose)	30	6.8	100	NA	3.30*	3.370	
	Fed-batch fermentation	CM medium (20 g/L Sucrose)	30	6.8	100	NA	4.35*	3.498	
ATCC 11454	Batch fermentation	MCD medium (25 g/L Glucose)	30	6.8	100	DOT 60%	NA	3.122	Papagianni et al. (2007)
	Batch fermentation	MCD medium (35 g/L Glucose)	30	6.8	100	DOT 60%	3.25	3.000	

(continued)

Table 1 (continued)

Lactococcus lactis Strain	Fermentation System	Growth parameters: Growth Medium (Modified Content)	Temperature (°C)	pH	Agitation (rpm)	Aeration (Dissolved O_2)	Cell Dry Weight (CDW, g/L)	Nisin Activity (IU/ml)	Reference
	Fed-batch fermentation	MCD medium (10 g/L Glucose)	30	6.8	100	DOT 60%	2.9	6.100	
CGMCC NO. 3050	Batch fermentation	Optimized CM medium	37	NA	150	NA	NA	22.216	Guo, Zhang and et al., (2010a, 2010b)
LD2	Batch fermentation	Optimized medium (23 g/L Glucose—aerobic)	37	6.8	NA	DOT 30%	1	11.932	Jiang et al. (2015)
	Batch fermentation	Optimized medium (23 g/L Glucose—anaerobic)	37	6.8	NA	NA	6.5	10.700	
	Fed-batch fermentation	Optimized medium (6 g/L Glucose—aerobic)	37	6.8	NA	DOT 30%	6.5	15.367	

Lactococcus lactis Strain	Fermentation System	Growth parameters: Growth Medium (Modified Content)	Temperature (°C)	pH	Agitation (rpm)	Aeration (Dissolved O_2)	Cell Dry Weight (CDW, g/L)	Nisin Activity (IU/ml)	Reference
	Fed-batch fermentation	Optimized medium (6 g/L Glucose—anaerobic)	37	6.8	NA	NA	0.9	12.000	
ATCC 11454	Batch fermentation	SI medium	37	7.6	150	NA	NA	4.657	Zheng et al. (2015)
E15	Fed-batch fermentation	Optimized medium (5–15 g/L Sucrose)	30	6.5	100	NA	NA	5.370	Zhu (2017)
	Batch fermentation	Optimized medium	30	6.8	100	NA	NA	3.037	
NCIM 2114	Batch fermentation	MRS medium	30	NA	25 rpm	DOT 0.2 vvm	5.71	50.400	Singh et al. (2016)
UL719	Batch fermentation	SWP (0.1% Tween 80 and 1% YE)	30	6	NA	DOT 60%	NA	20.500	Amiali et al. (1998)
	Batch fermentation	SWP (0.1% Tween 80 and 1% YE)	30	6	NA	DOT 90%	NA	16.400	

* The values are calculated and given approximately based on Fig. 3 in the article (Lv et al., 2005). rpm: revolutions per minute, NA: Not Applicable, DOT: Dissolved Oxygen Tension, vvm: Aeration air-flow rate

Table 2 Examples of manipulations of producer strains by genetic engineering studies to increase nisin yield

Lactococcus lactis Strain	Mutation type	Phenotype	Nisin Activity (IU/mL)	Reference
ATCC11454	Cloning of *aox*1 gene	Increased oxidative stress tolerance	7.900	Papagianni and Avramidis (2012a)
ATCC11454	Cloning of *pfk*13, *pka*C, and *aox*1 genes	Induced glycolytic activity and resistance to oxidative stress	14.000	Papagianni and Avramidis (2012b)
A164	Cloning of multicopy genes of *nis*Z, *nis*RK, *nis*FEG	Increased nisin Z production	13.0208	Cheigh et al. (2005)
N8	Prophage-related fragment (PRF) was deleted	Nisin immunity of producer strain increased	6.500	Qiao et al. (2020)
F44	Genome shuffling mutation was applied to the producer strain	Nisin Z production and glucose tolerance increased	4.023	Zhang et al. (2014)
F44	*ldh*(Z), *murG*, and *hde*AB genes were cloned and co-expressed	Enhanced the acidic tolerance and nisin production	5.563	Zhang et al. (2016)
F44A	*asn*H gene was cloned and overexpressed	Enhanced the acidic tolerance and nisin production	5.346	Hao et al. (2017)
F44 XY	*yvh*B/*xyn*D genes were cloned and co-overexpressed	Enhanced the acidic tolerance and nisin production	5.571	Cao et al. (2018)
ATCC11454 3087 ATCC11454 3823	Mutations were introduced via the adaptive mutation process	The mutant strain showed increased nisin production compared to the parent strains	2.160 2.140	Zheng et al. (2016)
F44	*gad*B and *gad*C genes were cloned	Enhanced the acidic tolerance and nisin production	3.758	Liu et al. (2020)

(continued)

Table 2 (continued)

Lactococcus lactis Strain	Mutation type	Phenotype	Nisin Activity (IU/mL)	Reference
PTCC 1336	The mutant strain was obtained by the adaptive laboratory evolution method	Enhanced oxidative and acid resistance of producer strain	6.000	Papiran and Hamedi (2021)
PLAC7	Cloning of chitin-binding domain (ChBD)	Nisin was produced in a continuous fermentation system in the presence of chitin	10.500	Şimşek, (2014)

the remaining factors are kept constant. Thus, the impact of each factor on the production of the desired metabolite is investigated one by one. Computational techniques, on the other hand, are promising to design economically feasible bioproduction processes by reducing laborious and time-consuming experimental methods. This has prompted researchers to use different computational approaches to meet the market demand in bioproduction processes by dramatically decreasing the number of the required experimental setups (Jawan et al., 2021). To date, several computational approaches such as response surface methodology (RSM) and machine learning algorithms (e.g., artificial neural network (ANN), genetic algorithm (GA), and decision trees) have been used to optimize the production of bacteriocins and bacteriocin-like inhibitory compounds.

RSM is a widely used statistical method in designing experimental conditions by identifying the relationship between operational parameters (Prakash Maran et al., 2013). Although the RSM-based models are remarkable alternatives to conventional models for quadratic approximations, they provide accurate predictions for only a limited number of parameters (Guo, Zhang et al., 2010a, 2010b). To address this issue, machine learning algorithms have been commonly used. Of these approaches, ANNs, which have been inspired by the learning system in the human brain, are convenient to learn and solve complicated nonlinear systems without a need for understanding the mathematical background of the problems (Lahiri et al., 2021). Thus, this data-driven modelling approach predicts the optimum parameters with a low number of prior assumptions (Youssefi et al., 2009). The main drawback of the ANNs is their black box modelling structure that does not provide any information about the individual effect or interaction effect among the

factors on the response values. This method can improve the prediction accuracy coupled with GA (ANN-GA). In this hybrid approach, GA is beneficial to optimize ANN parameters according to the principles of biological evolution like natural selection (Guo, Zhang et al., 2010a, 2010b). ANN-GA can be preferred to solve optimization problems with discontinuous, non-differentiable, or nonlinear objective functions, which cannot be solved via standard optimization algorithms (Singh et al., 2017). Decision trees are the other non-parametric machine learning models relying on a series of decision principles and their possible consequences (Kabari & Nwachukwu, 2012). Importantly, all these computational approaches hold great promise for developing cost-effective bioproduction technologies by optimizing the crucial factors associated with microbial growth and metabolism.

Modelling the bacteriocin-producing food microorganisms has increasingly attracted attention due to their potential roles in the food industry. In a recent study, using RSM and ANN approaches, bacteriocin production by *Rhizobium leguminosarum* DM20 was effectively optimized for temperature, pH, and incubation time (Lahiri et al., 2021). The production of another bacteriocin, macedocin, by *Streptococcus macedonicus* ACA-DC 198 was investigated at different temperatures, pH, and salt concentrations. ANNs were demonstrated to represent superior performance than the gamma concept-based conventional model according to the growth behaviour and macedocin production (Poirazi et al., 2007). More recently, Upendra and colleagues used RSM and ANN-GA to support the enhanced yield of bacteriocins by *Pediococcus pentosaceus* Sanna 14 via the optimization of culture conditions (Upendra et al., 2021).

Even if diverse microorganisms can produce bacteriocins, lactic acid bacteria are commonly screened for their bacteriocin-producing capacity. As highlighted before, the prominent antimicrobial peptide produced by *L. lactis*, nisin, has considerable commercial value. Though many environmental factors associated with nisin production have been reported (Table 1), there is a limited attempt to promote its bioproduction at a large scale via computational approaches. Guo and colleagues applied a computation pipeline with techniques, including the ANN model and GA, to reveal optimal culture conditions for nisin production by *L. lactis* (Guo, Yang et al., 2010a, 2010b). Similarly, the fermentation medium was optimized using a combinatory strategy based on the RSM combined with ANN-GA. Considering the individual effects and mutual interaction effects of the tested variables, optimum parameters were determined with increased predictive accuracy for the highest nisin titer (Guo, Zhang et al., 2010a, 2010b). Using the RSM, Kördikanlioğlu and colleagues determined optimum glucose, hemin,

and oxygen concentrations to enhance nisin production in a fed-batch fermentation system by promoting the biomass formation of *L. lactis* N8 (Kördikanlıoğlu et al., 2015). Akben and colleagues preferred to use the decision tree method to determine the optimum values of the selected variables and to achieve the highest nisin levels in a fed-batch fermentation system. Compared to the RSM, they highlighted this approach's superiority in predicting variable ranges (Akben, 2017).

Bioproduction is a crucial type of manufacturing to achieve sustainability goals, and digitalization is a remarkable concept supporting the efficiency and productivity of bioproduction systems. A suitable digitalization strategy can allow us to reach the Industry 5.0 goals announced by the European Commission (Xu et al., 2021). In the context of digitalization, the technologies applied to industrial bioproduction systems are described in the following sections.

4.2 Towards Digitalization in Industrial Bioproduction Systems

Optimization of the bioproduction processes needs the monitoring and control of the manufacturing systems effectively, and it can only be achieved using sensor technology. Sensors can detect metabolic and environmental alterations through the real-time monitoring of the bioprocesses. Thus, they can automatically configure the growth-associated parameters to design and maintain the optimal conditions (Reyes et al., 2022).

Thermal sensors, pH meters, gas sensors, cameras, current sensors, vibration meters, gyroscopes, accelerometers, and flow meters are widely used in manufacturing systems. Of these sensors, vibration meter, gyroscope, and accelerometer are used to measure the amount of vibration for precise automation or maintenance. Flow meters are useful to detect the amount of fluid flow. A pH sensor can be divided into two main categories: probes and contactless sensors. Probes contain two electrodes and measure the pH regarding the voltage between electrodes. Contactless pH sensors consist of a spot and optical device to measure the change of colour in the spot. The spot contacts the fluid whose pH will be measured. However, there is no contact between the measuring device and the fluid.

There are various ways to measure temperature. Three typical thermal measurement applications are possible in industry (Kim & Oh, 2020; Mahal et al., 2022; Zhu & Chen, 2023). In most cases, fibre optic cables are used to measure fluid temperature. Other thermal sensors, like pH sensors, can be

a probe or a contactless sensor. Fibre sensors collect the light of thermal radiation with a fibre optic cable and transmit it to a contactless sensor. To make more stable measurements, fibre sensors are considered as a better option than contactless sensors, especially for blended fluid (Zhu & Chen, 2023).

Various gases (e.g., O_2, CO, CO_2, and CH_4) are consumed or produced in biological reactions. The changed gas levels in a reactor system can allow us to calculate the reaction speed, which is highly related to productivity. Analysis of the gases can be done in two ways (Nagai et al., 2013; Ramanathan et al., 2023). One way is heating or burning the gas with a laser, and the regarding thermal characteristics determines the amount and type of the gas. The other way is the use of metals, which can react with the corresponding gas. After the metal-gas reaction, the amount of gas can be calculated based on the varying voltage along the metal.

The camera systems consist of a photodetector array converting the light into electrical signals. Photodetector array is a rectangle containing some equally sized detectors called pixels. When taking cell images with an appropriate pixel-sized camera, *phase contrast microscopy, quantitative phase contrast microscopy, fluorescent microscopy, and holotomography* are used to measure and visualize the specific parts of the cells. The phase contrast microscopy uses an optical mechanism to increase the contrast of the image while the quantitative phase contrast microscopy facilitates to take the images of cells at different focal planes from the same exposure. Fluorescence microscopes enable to take images of cells by staining them with a fluorescence dye. The holotomography is based on reconstructing multiple 2D holographic images into 3D (Barsanti et al., 2023). These techniques allow to measure or detect the cells or specific compounds before or after a reaction.

4.3 Digital Twin Concept

Liu et al. (2023) proposed the digital twin (DT) as a digital copy of a real system. In this context, the most important contribution of DT technology is that the DT of a system or material can be tested or operated in a virtual environment under diverse environmental conditions without the need for real environmental test data. Thus, using the DT concept, the result of a real event can be predicted in a virtual environment without incurring extra costs or the loss of time and resources. So, the optimization of the product design in the R&D phase and the production process can be realized with less trial and error and less cost (Liu et al., 2023).

The most crucial point to be considered in the transition from the currently used simulations in production systems to DT technology is that

the DT concept focuses on modelling the working mechanism of the system. Therefore, the simulations focus on the behaviour of the system. For instance, the failure distributions of bioreactors are among the basic parameters in a production system simulation. Failure occurs randomly in the simulation model regarding the distribution. DT calculates the changes in the operating characteristics. In this way, the failure time is not generated randomly as in the simulation (Meraghni et al., 2021).

Mathematical and physical models or data-driven prediction models with artificial intelligence can be used to build the DT models of bioreactors. It is possible to use the estimations or calculations of the DT models in a wide range of areas like sustainability (e.g., energy consumption and the rate of waste), scheduling, and predictive maintenance studies. Generation of the data-driven DT models requires data pre-processing, model creation, and model validation processes.

4.4 Data-driven Digital Twins in Bioproduction Process

In a bioreactor, it is possible to make all biological productions under the controlled conditions provided by the reactor. Therefore, bioreactors are manufacturing machines with a comprehensive production capacity. In this regard, a dynamic control mechanism is required to monitor and optimize operation parameters. This requirement increased the prominence of the data-driven DT proposed by Kapteyn et al. (2022), which is an effective control model for bioreactors. The first step of the data-driven DT generation is real-time data communication between the DT model and the production system. In this step, the real-time data collected from the shopfloor is transmitted to DT, and the outputs of the DT model are transmitted to the automation system. The second step is data pre-processing to create a proper dataset for DT generation or execution. In this step, noise removal, data smoothing, and data normalization are applied to generate a proper dataset. After the dataset preparation step, a machine learning or artificial intelligence model is built, and it establishes a relationship between the sensor data and the outputs of the bioproduction process as a third step. The developed model must be validated by comparing model outputs and process outputs via F-score, error metrics, or statistical tests (e.g., t-test and ANOVA). When the developed model is proven to generate similar results to the outputs of the real system, the execution step of the DT model, which is the 4^{th} step, is available to start.

4.5 Artificial Neural Networks in Bioproduction

As previously highlighted, ANNs have been developed with inspiration from the working principle of neurons. Thus, they are multi-node intelligent systems with activation functions, weights, bias, and specified layers. Common layers of the ANN are input, output, and hidden layers for multi-layered networks. The typical behaviour of artificial neural networks is to detect correlations between inputs and outputs by changing the weight of each node in the network and a single bias value in every layer (except for the input layer). Investigation of the optimal weight and bias is called the training process of the ANN.

The working systematic of a single neuron (node) in the network is to calculate the output (o) by multiplying inputs (i) with weight (w) of network, adding the bias (b) and by applying an activation function (f), as defined in Eq. 1.

$$o = f(i*w + b) \tag{1}$$

4.6 Activation Functions

The role of the activation function is to fit the output of the node into a specific range by adjusting the value calculated with respect to the set of inputs transmitted to the node. Activation functions strengthen some output values while weakening others to scale the effect on the result. The commonly used activation functions are given below (Eq. 2–9).

$$softmax(x_i) = \frac{e^{x_i}}{\sum_{j=1}^{n} x_j} \tag{2}$$

$$logistic(x) = \frac{1}{1 + e^{-x}} \tag{3}$$

$$softplus(x) = \ln\left(1 + e^{x}\right) \tag{4}$$

$$sigmoid(x) = \frac{e^{x}}{e^{x} + 1} = 1 - sigmoid(-x) \tag{5}$$

$$swish(x) = xsigmoid(x) \tag{6}$$

$$\tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}} \tag{7}$$

$$relu(x) = \max(0, x) \tag{8}$$

$$gelu(x) = \frac{x \tan\left(\sqrt{\frac{2}{\pi\left(x + 0.44715x^{3}\right)}}\right)}{2} \tag{9}$$

In the training step, optimal weights and biases are iteratively searched with a gradient descent algorithm. The amount of the change in weights and biases is determined by the *learning rate* parameter of the neural network, which is multiplied by the derivatives of the activation functions called *gradients*.

4.7 Underflow—Overflow

During the calculation of the output (Eq. 1), high-value weights and denominators of the activation function may cause the result to round to zero. When the output equals zero, the weight and bias cannot be updated during training, as there is no change in the output during training iterations. This situation, where calculations get stuck at zero, is referred to as underflow. In contrast, overflow occurs when the output rounds to positive or negative infinity due to significant weight and bias values. For instance, a small x value (nearly zero) causes the overflow for the *softmax* function (Eq. 2), because the function will be calculated nearly $\frac{1}{0}$ in that case. Similarly, a big negative x value causes underflow for the *softplus* function (Eq. 4), because the result of the function will approximately be $\ln(1 + 0)$. Underflow or overflow is also called *poor conditioning* of the ANN.

Poor conditioning causes underfitting or overfitting in the ANNs. However, underfitting and overfitting can be seen when there is no underflow or overflow. To detect the underflow or overflow, the change of two-loss function values is investigated based on the training dataset and test dataset. An ongoing positive correlation between the two features means the existence of an underfitting situation, and training must continue in this situation. When the correlation started to decrease or turned into a negative correlation, it shows the initiation of overfitting. Figure 2 presents the underfitting zone, overfitting zone, and optimal learning point regarding the loss function.

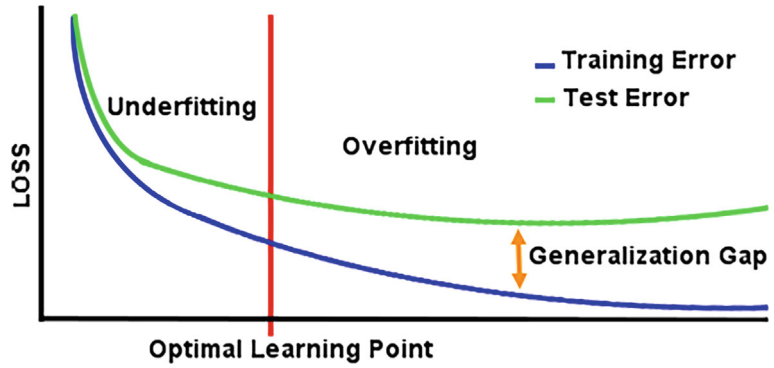

Fig. 2 Underfitting, optimal learning point, overfitting (*Source* Authors)

a

40	35	32	89	3
35	3	16	15	1
45	12	16	71	9
68	86	11	0	2
18	80	60	66	12

b

1	0	0
0	1	0
0	0	1

c

59	122	56
58	19	89
191	89	28

Fig. 3. **a** Inputs **b** Convolution kernel **c** Result of convolution (*Source* Authors)

4.8 Convolution—Pooling

The convolution process allows to emphasize various dataset features for especially images. Input data in Fig. 3a is altered by applying a kernel filter in a convolution layer. The aim of applying the filter is to increase the difference between common inputs and different inputs being minority. Convolution kernels are matrixes with a specific value, as given in Fig. 3b. A zone in inputs having equal size to the convolution kernel is multiplied with the kernel by exact positions. The sum of the multiplications is the value of the left top cell, as given in Fig. 3c. When the matrix shown in Fig. 3a is applied convolution with 3×3 kernel in Fig. 3b, the first output is "$40x_1 + 35x_0 + 32x_0 + 35x_0 + 3x_1 + 16x_0 + 45x_0 + 12x_0 + 16x_1 = 59$" (Fig. 3c).

Pooling is an abstraction process to eliminate common inputs. Features are emphasized when pooling is applied after convolution, and common inputs are eliminated. Elimination is performed with aggregation functions for a matrix with a particular dimension. Max pooling, min pooling, and average

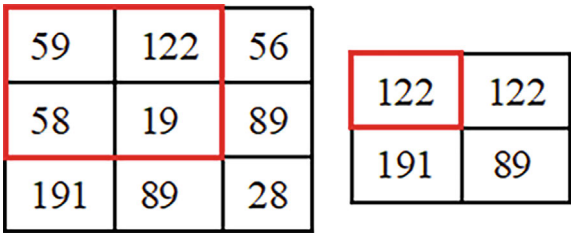

Fig. 4 An example of the pooling step (*Source* Authors)

pooling are commonly applied in the pooling layer. For instance, when the max pooling with 2 × 2 filter is applied to the output matrix given in Fig. 3, the maximum value (122) of the 2 × 2 matrix (59, 122, 58, 19) truncated from the output matrix is written to the first row and first column of the output matrix (Fig. 4).

4.9 Dropout

Dropout is a layer preventing the ANN from overfitting. The dropout mechanism eliminates the nodes of the ANN whose output causes overflow. Therefore, this layer forces ANN to use more robust features, which increases the time complexity of ANN. As a result of increasing time complexity, execution time also increases. However, the dropout layer reduces training time. In some cases, the dropout layer can be used to use convolution and pooling layers in sequence to achieve a similar performance with a faster ANN.

5 Case Study

Here, we presented a case study on the optimization of nisin bioproduction. This process includes three significant steps: 1) the growth of *L. lactis* cells under different culture conditions by setting several parameters (e.g., carbon source, temperature, and pH) and the estimation of nisin levels for each condition, 2) ANN-based modelling of the nisin production via this dataset to predict the optimum parameters, and 3) the evaluation of in silico predictions through the additional experimental setups (Fig. 5). Accordingly, bacterial cell cultures were grown under different conditions (e.g., 1, 2, and 3). The amount of nisin (IU/ml) was estimated via agar well diffusion assay for an indicator strain (*M. luteus)* based on a standard curve of the nisin. Nisin production was modelled using this dataset via ANN approach to elucidate the optimum environmental factors. Lastly, in silico predictions were

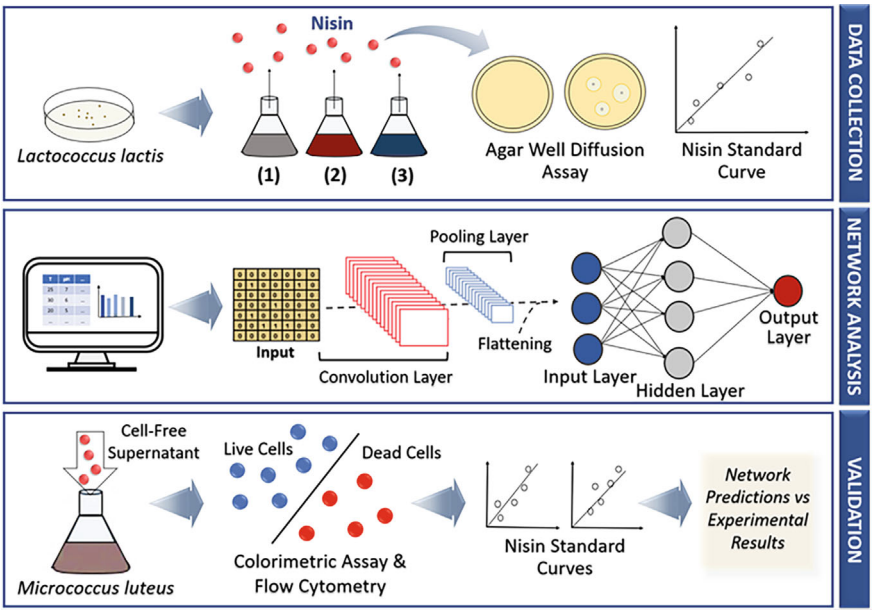

Fig. 5 Identification of the optimum growth parameters for nisin production by *L. lactis* (*Source* Authors)

confirmed through in vitro methods including colorimetric assay and flow cytometry with a high accuracy and precision of the nisin quantification.

5.1 Bacterial Strains and Datasets

L. lactis subsp. *lactis* (ATCC 11454) was used as the nisin producer, and *M. luteus* NCIMB 8166 (ATCC 10240) was selected as the indicator microorganism to measure nisin activity. For nisin production, the starter culture of *L. lactis* was grown in M17 broth at different temperatures and pH values for 18–20 hours. The impact of each dietary content on nisin production was evaluated for several carbon sources (Girgin Ersoy, 2013). The medium without any supplements was used as the control. For each growth condition, nisin activity was determined with an easy and inexpensive agar well diffusion assay (Wu et al., 2009). In this method, nisin activity was investigated considering the size of nisin-caused inhibition zones of *M. luteus*. The final nisin concentrations were calculated using the standard curve obtained with the standard nisin solutions (Girgin Ersoy, 2013), prepared according to (Cheigh et al., 2002).

5.2 Artificial Neural Network Model

An ANN model was developed to reveal the correlation between different factors (i.e., pH, temperature, carbon source, and the amount of nisin production). To use the tangent hyperbolic activation function, the dataset was scaled between −1 and 1 using the formula in Eq. 7. X_{norm} is the result of normalization, X_{max} is the maximum value of the selected feature in the dataset, and X_{min} is the minimum value of the selected feature in the dataset.

$$X_{norm} = 2 \text{ x } \frac{X - \frac{X_{max} - X_{min}}{2}}{X_{max} - X_{min}} \tag{10}$$

Only one hidden layer consisting of 40 nodes was enough to predict the amount of nisin production. The networks were trained for 8000 epochs. The training process of the network is completed with R^2 0.9998 and approximately 2.5% mean absolute percentage error. We used the ANN model for numerical analysis to calculate the amount of nisin production. The range of the numerical analysis includes the temperatures 25–28 °C and pH 5–7 for 13 carbon sources. According to the numerical analysis results, the highest nisin production was reached with galactose and lactose at pH 7.0 and 25 °C growth temperature.

5.3 Validation of the Computational Predictions

Flow cytometry and colorimetric assay were used to confirm the optimum predictions obtained by the ANN models. In the nisin production process, *L. lactis* cells were inoculated into flasks containing M17 broth (pH 7.0) supplemented with glucose, galactose, or a combination of lactose and galactose. Bacterial cells were cultivated at 25 °C and 30 °C, respectively. For the flow cytometry measurements, 100 µL of the cell-free *L. lactis* culture supernatant containing nisin was added to 900 µl of the *M. luteus* culture (~10^7 CFU/ml) grown in S1 medium. After the incubation at 37 °C with agitation for 1.5 h, cell viability was assessed (Budde & Rasch, 2001). In this step, 100 µl of the culture was mixed with the staining buffer at the ratio of 1:10 in addition to 5 µl of the propidium iodide, which binds to the free nucleic acids of the dead cells, and incubated in the dark for 5 min. Then, the cell viability was analysed by measuring fluorescence excitation at the wavelength of 480 nm using a flow cytometer. Nisin concentrations (IU/ml) were estimated through a standard nisin curve ($R^2 = 0.991$).

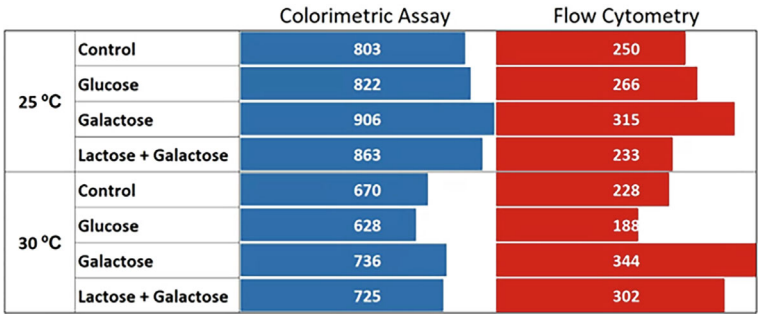

Fig. 6 Nisin concentrations (IU/mL) produced by *L. lactis* cells that were cultivated in M17 broth (pH 7.0) supplied with different carbon sources at 25 °C and 30 °C. Blue and red bars represent the nisin levels measured by colorimetric assay and flow cytometry, respectively (*Source* Authors)

Nisin susceptibility of the *M. luteus* was also evaluated via a colorimetric assay (Wang et al., 2007). 90 µl of the bacterial cell culture (~10^5 CFU/ml) was mixed with 10 µl of the cell-free *L. lactis* supernatants in a 96-well plate. For the control samples, 10 µl of GM17 broth (M17 containing 0.5% glucose) was added instead of the supernatants, while GM17 and S1 broths were used as the blank. After the incubation for 4 h (at 37 °C, 180 rpm), the cells were directly treated with 10 µl of MTS reagent at 37 °C with agitation (130 rpm) in the dark. Then, the viable cells were quantified by an ELISA reader at 490 nm, and the nisin concentrations (IU/ml) were estimated based on the standard curve ($R^2 = 0.963$) of the nisin. The results of both methods demonstrated that galactose supplementation considerably enhanced nisin production compared to glucose (Fig. 6). Lactose and galactose were previously reported to induce nisin biosynthesis in *L. lactis* from *nisA* promoter even in the absence of nisin in the growth environment (Chandrapati & Sullivan, 1999).

6 Conclusions

We aimed to propose a data-driven intelligent model to represent the nisin production process. To ensure a valid model, we established a couple of experiments and collected data on nisin production. ANN was utilized to build the model of nisin production process, and successfully predicted the nisin yield based on pH, temperature, and carbon sources. By integrating our model with a backtracking search algorithm, we found optimal nisin production conditions. The regulatory relationship between these carbon sources and

nisin biosynthesis was supported by both ANNs and in vitro results. Interestingly, the colorimetric assay exhibited more consistent results with ANN predictions regarding the optimum temperature. To ensure the optimum temperature value, several temperature values in the 25 °C and 30 °C range should be experimentally tested in future studies. In addition, a more sensitive method may be preferred to distinguish small differences in the input data even if the agar well diffusion assay is a widely used practical and inexpensive approach for quantifying the antimicrobial activity of microbial extracts. Though further research is required to confirm the network predictions, this case study can provide valuable insights into the comprehensive analysis and optimization of the bioproduction processes. This strategy has significant potential to design suitable experimental setups for the cost-effective production of food preservatives. In the future, digital twins are expected to become a common planning tool for the industry. The food industry and healthcare systems utilizing bioreactors for the production of beneficial microorganisms and microbial byproducts as well as studying on pathogenic organisms highly need precise automation. Thus, our approach is promising to increase digital twin technology-based bioreactor utilization.

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21

Predictive Maintenance: Digital Twins in CNC Machine Failure Detection

Elif Cesur, Muhammet Raşit Cesur, and Şeyma Duymaz

1 Introduction

Continuous and efficient operation of production systems is one of the most important factors that ensure the maximization of system performance. Continuity and efficiency reduce unit costs, deliver orders without delay, and improve sustainability with less resource consumption. To ensure system continuity, production system components (machines, tools, workbenches) must be constantly available for production. The planning function that ensures that the production system is available (suitable) at the desired level is called maintenance planning. Maintenance planning aims to find a way to ensure the production system meets the targeted level of availability with a certain budget or at the minimum cost (Tsang, 2002). At this point, the goal of this study was to employ Digital Twin (DT) technology to identify faults in Computer Numerical Control (CNC) machines by analysing the present behaviour of the stepper motor, a component that is frequently seen in these

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kinds of machines. The novelty of this paper lies in addressing the core challenge of comprehending and effectively implementing DT technology for CNC machines, crucial components in modern production systems.

The maintenance planning production system is carried out with different approaches according to the characteristics of the components, component costs, and supply times of the components. To implement the most appropriate maintenance policy, it is necessary to understand the failure characteristics of the components of the production system. Since the DT can be used to model machines and tools in the production system, it is one of the most important tools that can be used in creating DT maintenance plans. It is very important to understand when previous errors have started in machines, to predict and prevent when a failure will occur, to minimize disruptions in production, and to reduce critical losses due to failures. The primary challenge addressed in this study pertains to the limited comprehension and practical implementation of DT creation for CNC machines, which play an essential role in production systems (Luo et al., 2020). The principal contribution of this study lies in the enhancement of maintenance planning through the integration of DT and machine learning algorithms to proactively predict and mitigate equipment failures.

The general structure of the paper is as follows: First, a literature review of the existing studies is presented. Then, to understand the research purpose, the theoretical background of DT and maintenance planning is discussed. In Sect. 4 we explain the Artificial Neural Networks (ANN), Support Vector Machines (SVM), and Statistical Quality Control (SQC) used in the methodology. After a detailed explanation of the case study, the paper concludes with the discussion and conclusion sections.

2 Literature Review

In the literature review section, firstly, studies related to DT applications, especially those conducted in the production domain, are introduced. Subsequently, DT technologies are discussed. Finally, DT studies, particularly within the framework of predictive maintenance, are presented. At the end of a comprehensive literature review, two research gaps have been identified.

Studies have been conducted that focus on taking a holistic approach to production systems and aim to create a DT of a factory. Um et al. proposed a data model with the DT to improve the plug-and-play feature in multiple assembly lines (Um et al., 2017). Cimino et al. developed a DT model for the assembly line of a mobile phone company (Cimino et al., 2019).

Kocabay used the DT to determine, the time when a product that enters the system unplanned will leave the production line (Kocabay, 2019). At the same time, in this study, the model can determine which process will create a queue and the queue lengths. Zhuang et al. (2018) proposed a framework for DT-based intelligent manufacturing management and a control approach for complex product assembly shops (Zhuang et al., 2018). There are four basic techniques included in the framework. These are (1) acquiring, organizing, and managing physical assembly line data in real-time, (2) generating assembly shop DT, (3) DT and bigdata-based prediction for the assembly shop, and (4) DT-based assembly shop production management and control service. Detailed implementation of the proposed DT-based smart production management and control approach in a satellite assembly workshop scenario is carried out to detail how the proposed approach will be applied to reality. Roy et al. developed a DT model for a welding technique called FSW (Roy et al., 2020). For the DT application, data were taken from the FSW machine with various sensors (force sensor, power sensor, tool rotation speed, linear speed sensors, turbidity sensor, flow rate sensor, etc.). Maintenance measures have been taken as a result of the DT. Zheng et al. created a DT of a welding production line in their paper (Zheng et al., 2019). On the other hand, starting from the product design stage by using DT technologies, the monitoring studies of the process were also discussed, and studies were carried out in this context. Zhang et al. proposed a three-layer framework for an intelligent manufacturing logistics system that can implement resource modelling and self-organizing configuration (Zhang et al., 2018). Qi et al. predicted product lifecycle with DT to control product status (Qi & Tao, 2018). As a result, it has made operations and maintenance more efficient. Ma et al. presented a DT model to develop a cleaner production strategy (Ma et al., 2019).

When the literature is reviewed in terms of technologies used, it is seen that the studies focused on cloud and fog computing, artificial intelligence and autonomous robots, virtualization, and data transfer. Coronado et al. proposed a new production system supported by a cloud computing system that collects data from operators and integrates it with production data (Urbina Coronado et al., 2018). Rosen et al. aimed to keep the data received in various ways (for example, produced in the sensor or system) by the DT (Rosen et al., 2015). The DT, which collects the data, will be able to provide information about the future process with forward-looking simulation studies. He and Bai (2021) and Tao et al. (2019) suggested that DT can receive real environment data from sensors, model equipment, and subsystem

components, and control the flow between product design data and production environment data since the DT is intended to be a digital copy of a real system (He & Bai, 2021). Cheng et al. (2018) determined the main framework that would enable this flow to be managed in layers and drew attention to the importance of the correct management of data (Cheng et al., 2018).

In terms of case studies, we can say that DT applications mostly focus on predictive maintenance and fault detection. Ribeiro et al. discussed the application of a new DT model to diagnose mechanical faults in electric motors through a non-contact approach and vibration analysis based on a digital signal processing (DSP) device in an optical computer mouse (Ribeiro et al., 2019). An experimental bench was set up to test the bearing damage detection. Based on the concept of predictive maintenance, the proposed method has the potential to be an efficient and low-cost tool to predict failures. Wang et al. used the Gray Relation method to analyse the failure risk in their study of predictive maintenance and stated that applying the deep learning method will also produce efficient results (K. J. Wang et al., 2021). Joung et al. implemented a cloud-based engine condition monitoring system to detect the abnormal operation of engines (Joung et al., 2020). In their study, three-axis vibration sensors were used to collect data. Liu et al. with the help of sensors attached to the CNC head, the amount of movement that the machine should make, and the amount of movement realized by the Sliding Window method, and thus the quality control of the produced product was made (Liu & Xu, 2017). Luo et al. developed a hybrid predictive maintenance algorithm by proposing a system with a DT to perform reliable predictive maintenance on CNC machines (Luo et al., 2020). Wang et al. predicted 90% of equipment failures (C. P. Wang et al., 2020). Zheng et al. used DT to detect defects, reflect on the real situation, and predict possible faults (Zheng et al., 2019).

This study aims to address several major gaps in the scientific literature in the domain:

(1) More research is needed on the development of a DT model for production environments, particularly for the benefit of practitioners.
(2) There is a shortage of information regarding the comprehension of DT applications and the integration of machine learning algorithms for predictive maintenance prediction.

3 Theoretical Background

3.1 Digital Twin

When the modelling studies carried out in the scientific literature are considered, it is seen that there are studies that can be called DT in the past. However, the concept of DT was first introduced by Grieves in 2003 (Grieves, 2005). DT, which is a digital copy of real systems that allows the calculation of their behaviour in real environments in a virtual environment, has become a popular subject with the digitalization trend that started in the industry with Industry 4.0. The contribution of DT, which is one of the important building blocks of digitalization, to the digitalization of production systems is apparent. In production systems, DT contributes to the performance of critical functions of the system such as production engineering, production planning, and maintenance planning. Product design in production engineering and determination of optimal production parameters; scheduling in production planning; In maintenance planning, DT technology is used in predictive maintenance applications.

Most of the studies related to DT are in the field of production. Especially in areas such as the modelling of production systems, fault detection, and maintenance planning, DT applications are frequently used. The studies aimed to create a cyber-physical system that integrates digital resources and physical and real objects into a coherent environment. Semenkov et al. aimed to analyse the properties of the models and compare them with the modelling of the digital and physical components of a cyber-physical system (Semenkov et al., 2020). Qi et al. talked about how production and DT can be combined in their study (Qi & Tao, 2018). They discussed how to use various DT models. Terkaj and Urgo (2014) presented a DT model to support the development and evaluation of production systems in their study (Terkaj & Urgo, 2014). Kádár et al. used a DT for the product lifecycle in a virtual factory (Kádár et al., 2013). Qi et al. describe in their article how the production services and the DT are brought together and how the various components of the DT are used by manufacturers as services (Qi et al., 2018). Tao et al. (2019) aimed to create a new product design and production process with the DT. With the real-time data and historical data obtained in this system, the product manufacturer can understand the working status of the product in real-time and can use the extra virtual product data (Tao et al., 2019).

During the operation of the machine, the measurement data obtained from the sensors changes depending on the working parameters of the machine. To understand whether the operating conditions are normal or not,

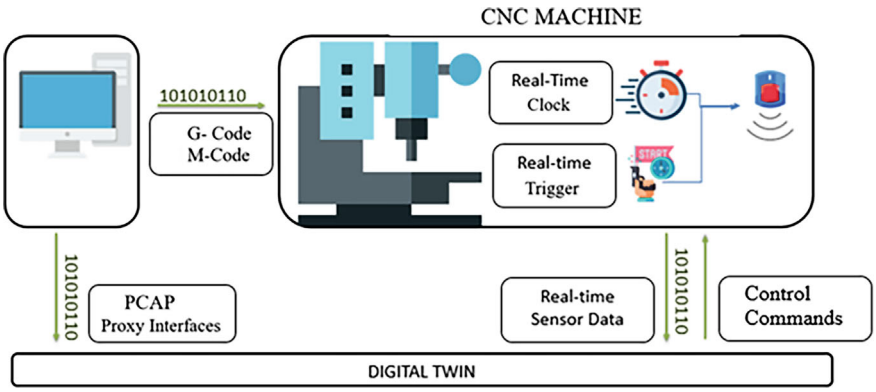

Fig. 1 Structure of Digital Twin (*Source* Authors)

it is necessary to establish a correlation between the operating parameters and the sensor data. The DT can analyse the malfunction status of the machine by using the operating parameters of the machine and the sensor information generated during operation. When the DT detects any anomaly or a situation that will cause an anomaly in the machine, it will intervene in the operating parameters and ensure that the machine finishes the process without any fault. The main framework of the DT is presented as follows in Fig. 1.

3.2 Maintenance Planning

Maintenance policy is divided into two categories: planned and unplanned maintenance. Unplanned maintenance is a policy based on performing maintenance only when breakdowns occur. In cases of breakdown, either the faulty component or the entire unit is replaced. It is applied to equipment without bottlenecks, easily repairable, or those with available spare parts. It is a reactive approach. Planned maintenance includes (1) Periodic Maintenance, (2) Preventive Maintenance, and (3) Predictive Maintenance. Within the scope of the study, the focus will be on predictive maintenance.

Predictive maintenance is based on understanding whether workbenches and machines need maintenance as a result of various measurements. Predictive maintenance focuses on determining the risk and time of failure of the device. In this context, predictive maintenance models are being developed with three different approaches: survival models, similarity models, and degradation models. Survival models provide the correlation of measurement levels with failure rates by looking at the failure data. For example, when the vibration level in the machine reaches a certain point, the risk of failure is

calculated with these models. The maintenance unit performs maintenance when the measurement data reaches the determined levels according to the availability target. Similarity models try to identify previously recorded fault conditions based on measurements. If faults are observed in the machine, maintenance is performed before failure occurs. Degradation models, similar to survival models, are used to predict when a certain threshold is reached. Threshold values are at levels known to fail and are usually determined by machine manufacturers.

Duration of maintenance depends on the type of failure, tooling of maintenance, and proficiency of the maintenance team. Types of common failures such as (1) Wear, corrosion, and erosions, (2) Cracks, breaks, cuts, and ruptures, (3) Deformations, (4) Change of material density, (5) Aging, or (6) Loosening of attachment points.

Failures can occur depending on various reasons regarding some factors like environmental conditions, usage, the technical design of the machine, or maintenance. Common reasons for failures such as (1) Excessive vibration, (2) Uncalibrated controller devices, (3) Overload operation over the limits, (4) Structural causes such as wear or poor quality in parts, (5) Irregularity and dirty working environment, (6) Inappropriate environmental conditions, (7) Failure to fix minor faults, (8) Improper maintenance or lack of maintenance.

Reasons for failures are important to be monitored to prevent the system from failure. Minor faults can cause a change in the working characteristics of machines. Thus, monitoring true parameters provides a preventing system from failure with minor maintenance operations which take less time and cause less cost. The failure symptoms could be (1) Cracks, (2) Increase in vibration and loudness, (3) Overheating, (4) Rot and stench, (5) Improper working, (6) Leaks, (7) Fluctuations in energy consumption.

4 Methodology

In terms of method, machine learning algorithms were mostly used in DT applications. Süve used a DT for various investigations of machine learning methods, setting simulation parameters, interacting with the system, and visualizing data, and the DT of the enterprise was created (Süve, 2021). Magargle et al. run a DT with simulation-based modelling in an integrated brake system that can be used for tracking, prediction, and detection (Magargle et al., 2017). et al. worked on the creation of a simulation-based virtual factory (Yildiz et al., 2021). Roy et al. classified the availability of equipment with SVM in the DT model they developed (Roy et al., 2020).

The choice of preferred machine learning algorithms often depends on the characteristics of the problem and the dataset. Algorithms such as Logistic Regression, SVM, Gradient Boosting, Naive Bayes, ANN, and K-Nearest Neighbours (KNN) are commonly favoured for binary classification employed in this study. These algorithms tend to work faster and produce accurate results compared to other algorithms in binary classification. The selection of the appropriate algorithm depends on factors such as the dataset's attributes, size, linearity, risk of overfitting, and performance requirements. SVM can effectively work with both linear and non-linear datasets. ANN utilizing deep learning techniques can yield successful outcomes in large and complex datasets. Lastly, SQC methods are applied in addition to machine learning algorithms for fault detection. This chapter is divided into three parts under the headings: (1) ANN, (2) SVM, and (3) SQC.

4.1 Artificial Neural Network (ANN)

ANNs implement algorithms that strive to achieve performance akin to neurological connections, such as learning from experiences, making generalizations from similar situations, and judging situations where poor outcomes were obtained in the past. The history of ANNs dates back to the early 1940s; however, it was only in the mid-1980s that these algorithms became scientifically robust and feasible. Since the late 1980s, ANNs have been applied in numerous industrial applications. Today, ANNs are applied to a wide range of real-world industrial problems, from functional prediction to system modelling (especially in cases where the understanding of physical processes is rather complex). These applications span from pattern recognition engines to robust classifiers, enabling decisions about uncertain input data and recognizing situations of the same type as encountered during the learning of the ANN (Kubat, 1999).

ANNs are theoretical outcomes of applying biological principles to scientific research (Meireles et al., 2003). In other words, they are artificial intelligence methods that draw ideas from certain behavioural and structural characteristics of biological individuals or the biological world. ANNs are known by various names such as field, connectivity, parallel distributed processing, neurocomputation, natural intelligent systems, and machine learning algorithms. ANNs learn from examples, much like humans. This attempt involves simulating multi-layered neuron layers with specialized hardware or advanced software. Each neuron is connected to a few of its neighbours, and these neighbours represent the strong aspects of these

connections. Neural networks can find non-linear definitions or dependencies among various variables, moving from input space to output space. This way, they can recognize cases of the same type as those during learning (Gong et al., 2021).

A neural network consists of simple processing units called neurons. Generally, a neural network has three layers: input, output, and hidden layers. Neurons in the input layer receive input from the external environment. This layer doesn't perform any computations. The hidden layer, which receives input from the input layer, performs computations and forwards outputs to the output layer. The output layer consists of neurons that convey the system's output to the user or the external environment (Korteling et al., 2021).

The process of learning from experience is natural for the human brain. ANNs are sometimes referred to as machine learning algorithms because changing the connection weights leads the network to learn how to solve problems. The strength of connections between neurons is stored as a weight value for a specific connection. The learning capability of ANNs is determined by their architecture and the algorithmic method chosen for training (Abiodun et al., 2019). If there is enough information about the problem, the weights can be pre-calculated. If an ANN can provide accurate solutions when presented with test data different from the training data, it is said to have been generalized. A system like an ANN-based operator support system can minimize operator errors and assist in adverse situations (Santosh et al., 2007).

The learning process in ANNs is usually applied with examples, also known as "training". This is because the learning process is accomplished by iteratively adjusting the connection weights. The learning process in ANNs can be categorized into three types: supervised, unsupervised, and reinforcement learning. Supervised learning is based on a direct comparison between the actual output and the expected output. Optimization algorithms can be used to iteratively adjust connection weights and thus minimize errors. Reinforcement learning is a special case of supervised learning; it depends solely on whether the actual output is correct. Unsupervised learning is based solely on the correlation of input data. The essence of a learning algorithm is the learning rule that determines weight updates. Several popular learning rules include the delta rule, Hebbian rule, and competitive learning rule (Ding et al., 2013).

4.2 Support Vector Machine (SVM)

Artificial intelligence is the autonomous execution of behaviours by machines that are typically carried out by humans. Its purpose is to mimic human intelligence through computer-based processes and impart learning capabilities to computers. Machine learning is an application of artificial intelligence. Machine learning is the general term for computer algorithms that model a problem based on the data specific to that problem (Pirim, 2006).

Classification tasks with artificial intelligence are widely used today. One of the significant machine learning algorithms developed to solve the classification problem, which is one of the tasks of data mining, is SVMs. It is commonly employed in supervised learning methods for classification processes. SVMs have been witnessed to deliver successful results on numerous occasions. Their generalization performance is high and effective.

Unlike other machine learning algorithms, which often require a large amount of training data, SVMs tackle issues like low convergence rates and getting trapped in local minima. They work based on the principle of structural risk minimization. One of the classification applications is SVM classification. SVMs perform remarkably well in scenarios involving high-dimensional yet limited datasets.

SVM constructs an n-dimensional hyperplane that best separates the data into two categories. It's closely related to ANN. SVM utilizing a sigmoid kernel function possesses a two-layer feedforward ANN model (Kubat, 1999).

4.3 Statistical Quality Control (SQC)

Statistics is the organization of numerical data to derive meaningful conclusions, while control involves monitoring activities within various standards. When processes deviate from control, variability becomes a factor (Aydin & Arikan Kargi, 2018). Quality control aims to standardize production in a way that encompasses the entire process to meet customer requirements (Flores et al., 2021). SQC involves the application of statistical techniques at every stage of the process. In the production process, various factors can lead to product variations. SQC is applied not only in manufactured products but also in all industrial and service processes. There are two different situations regarding the acceptability of product differences: acceptable and unacceptable. If it is acceptable, it is under control; otherwise, it is considered out of control. Variability is a significant concern in manufactured products. Techniques used to detect changes and find errors are evaluated as part

of SQC studies. SQC not only assesses the state of control but also analyses the reasons behind out-of-control situations. SQC aims to prevent the production of defective products. Before implementing SQC techniques, data related to the work area must be collected, which may include data collection techniques such as observation, surveys, sampling, and others.

Control charts, one of the basic levels of SQC techniques, are the method used to reveal the out-of-control situation in a short time due to identifiable reasons. The data are compared with the specified limits. It shows the problems but does not give us the reason for it. Data are divided into measurable and unmeasurable. If it has measurable values, control charts are drawn blindly to the sample size. It is divided into two quantitative and qualitative. Quantitative control charts are used where measurable values are available. It examines the distribution of sample means. Standard deviation or ranges of variation are used as a measure of dispersion. Qualitative control charts are a type of control that cannot be calculated numerically. Products are classified as defective and flawless.

5 Case Study

When a lost step fault occurs in the stepper motors used in CNC machines, the workbench cannot complete its movement. In this case, measurement errors or surface quality problems occur. Current and vibration data are used to detect this failure. The amount of current drawn is recorded as data. Cesur and Cesur (2021) utilized this current data in their fault detection study for lost steps as shown in Fig. 2 (Cesur & Cesur, 2021). In the 3rd, 5th, and 7th cycles shown in Fig. 2, the synchronization lost step fault occurred.

When the instantaneous change amounts in the current data in Fig. 2 are calculated, the graph given in Fig. 3 is obtained. The sections marked on the graph are the moments when the lost step fault occurs. The behaviour of the graph for the correct detection of a lost step fault is expressed by the 8 classes: (1) Idle, (2) Beginning of the motion, (3) Motion, (4) End of the motion, (5) Beginning motion with lost step fault, (6) Lost step fault occurrence during motion, (7) Motion after lost step fault, (8) End of motion after lost step fault (Cesur & Cesur, 2021).

According to Fig. 4, while there is a fluctuation in the change graph when the lost step fault occurs, there is current consumption above a certain level at these moments. It is understood that a successful classification can be made by using the data created from the two graphs given. Binary classification models were produced using ANNs and SVMs for the classification process. In the

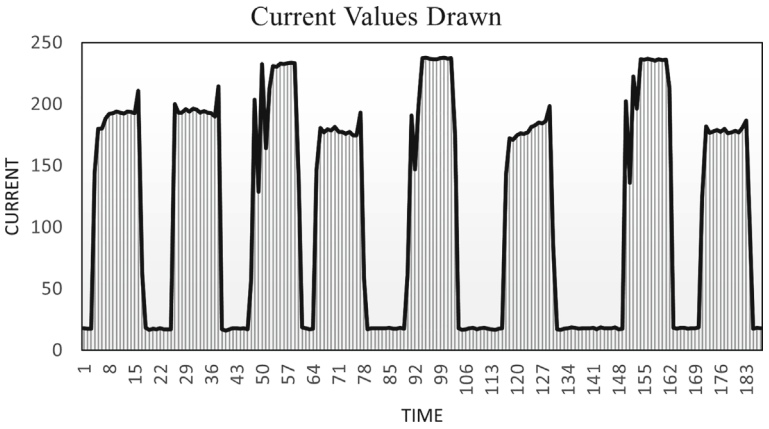

Fig. 2 The behaviour of current values (*Source* Cesur and Cesur [2021])

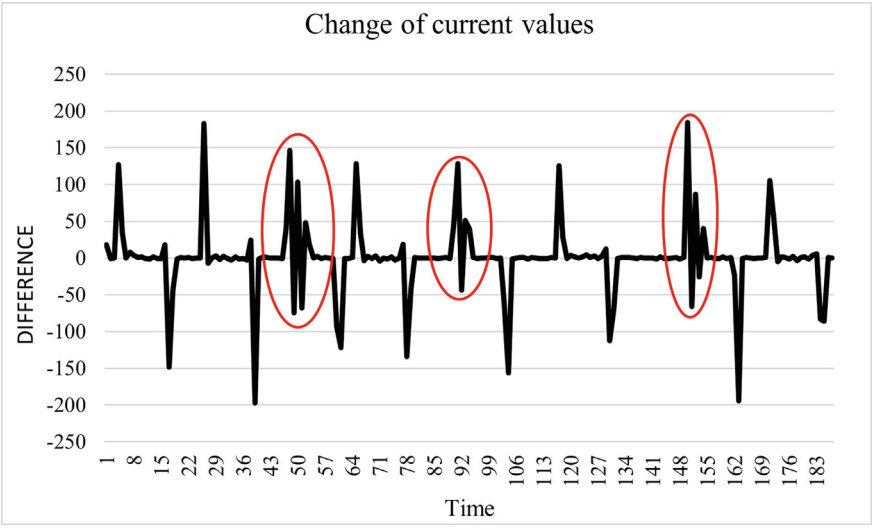

Fig. 3 Changes in current values (*Source* Authors)

binary classification model, current and current changes are used as inputs, while the number of classes is the output. As a result of the classification process, the value of only one output is 1 and the others are 0. By looking at the output with a value of 1, the class determined by the model is understood.

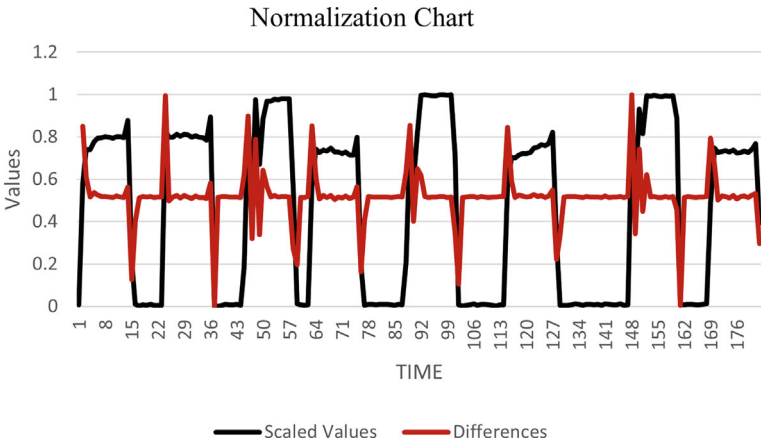

Fig. 4 Behaviour of current values after normalization (*Source* Authors)

5.1 Result of ANN

ANN is one of the effective methods used for binary classification. Cesur and Cesur (2021) have successfully identified 8 situations that they have determined using deep learning in their binary classification study (Cesur & Cesur, 2021). Hyper-parameters of the deep neural network model with four hidden layers are given in Table 1.

The source code developed in Python is given below. The model is compiled via Tensorflow and Keras libraries. To create a Tensorflow session "*session=tf.Session()*" command line is applied. After creating the session, "*K.set_session(session)*" command is applied to enable Keras. The model is defined with "*ke.Sequential()*" command. All layers are added to the model by the "*add*" function. Finally, the model is created with "*model.compile(loss='mse', optimizer='adam', metrics=['mse'])*" command. After creating the model, training is started with "*fit*" function.

Table 1 ANN topology architecture

Nodes in layers	300, 400, 200, 50, 8
Activation functions in each layer	softplus, softplus, softplus, softplus, softplus
Learning rate	0.01
Optimizer	Adam
Epocs	7500
Loss function	Mean Squared Error

Source Authors

```
from keras.layers import Dropout
from keras import backend as K
from keras import layers as L
import keras as ke
import numpy as np
import tensorflow.compat.v1 as tf

#train_X, train_Y, test_X, test_Y must be imported from dataset
np.random.seed(1)
tf.compat.v1.disable_eager_execution()
tf.set_random_seed(1)

sesssion = tf.Session()
K.set_session(session)
model = ke.Sequential()
model.add(L.Dense(300, input_dim=4, activation='softplus'))
model.add(L.Dense(400, activation='softplus'))
model.add(L.Dense(200, activation='softplus'))
model.add(L.Dense(50, activation='softplus'))
model.add(L.Dense(8, activation='softplus'))
model.compile(loss='mse', optimizer='adam', metrics=['mse'])
history = model.fit(train_X, train_Y, epochs=7500, validation_data=(test_X,
test_Y))

X_val=np.concatenate((train_X, test_X), axis=0)
Y_val=np.concatenate((train_Y, test_Y), axis=0)
predictions = np.array(model.predict(X_val).flatten()).reshape(-1,8)
results=np.round(predictions)
mean_abs_percentage_error=np.average(np.abs((results-Y_val)/ Y_val))
```

5.2 Result of SVM

Binary classification with the deep neural network is also performed using SVM for the same 8 classes. The reason for using SVM is that the training time is much shorter than that of a deep neural network. Especially when hyperparameter optimization is desired or when the same model is desired to be adapted to different CNC machines, it is an important advantage that the training time is short. Since there is a possibility to use a non-linear kernel in SVM, it is possible to obtain outputs close to the deep neural network. Therefore, the SVM has been evaluated as an alternative method. The source code of the SVM model developed using R is given below. The SVM library is loaded with the command "library(e1071)". With

the command "library(caret)", the "caret" library, which contains the section "confusionMatrix" function, is loaded. This library will be used to calculate the F-Score. With the "as.factor" command, the outputs are converted to a class. Then the data set is divided into a training set and a test set. The model was created with the "svm" function. With the "predict" function, the model detects the fault status based on the current data. Finally, the F-Score was calculated with the function "caret::confusionMatrix".

```
library(e1071)
library(caret)

data$output<-as.factor(data$output)
set.seed(nrow(data))
trainIndex<-sample(1:nrow(data), size=0.7*nrow(data))
trainSet<-data[trainIndex,]
testSet<-data[trainIndex,]

linearModel<-svm(output~., data=trainSet, kernel= "linear")
radialModel<-svm(output~., data=trainSet, kernel= "radial")
polnomialModel<-svm(output~., data=trainSet, kernel= "polynomial")

predLinear<-predict(linearModel,testSet)
predradial<-predict(radialModel,testSet)
polinomial<-predict(polnomialModel,testSet)

caret::confusionMatrix(predLinear, testSet$output)
caret::confusionMatrix(polinomial, testSet$output)
caret::confusionMatrix(predradial, testSet$output)
```

In the data set consisting of 49 rows of data, 35 rows contain normal operation, and 14 rows contain losing step status. 30% of the data set was allocated as a test set and the performance of the linear model was calculated as 86%. The performance of the radial and polynomial models was calculated as 93%.

5.3 Result of SQC

SQC methods can be used to determine the occurrence of the losing step error. The classification process in artificial intelligence and machine learning cannot be done with these methods. However, it can be understood by these methods whether an abnormality occurs during the operation of the engine. Therefore, the change in current values was examined by SQC methods and

it was tried to decide whether the change in values was abnormal or not by using SQC diagrams.

To create the R chart given in Fig. 5 using the change data, a data set consisting of two consecutive measurements and a test set containing incorrect and error-free situations were created. Since we determine the range in the R chart, a range has been determined to distinguish all faulty situations.

When creating the X-Bar chart given in Fig. 6, two consecutive measurements were used for sampling, as in the X-Bar chart. When the performances of the two charts given in Table 2 are compared, it is seen that the X-Bar chart gives better results.

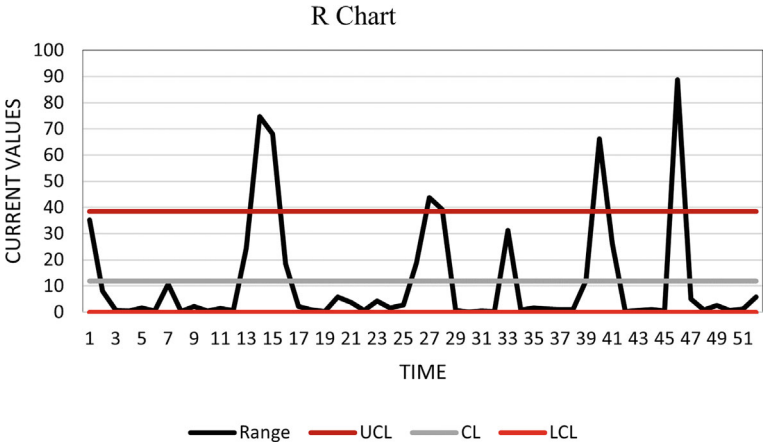

Fig. 5 R chart according to the change in current values (*Source* Authors)

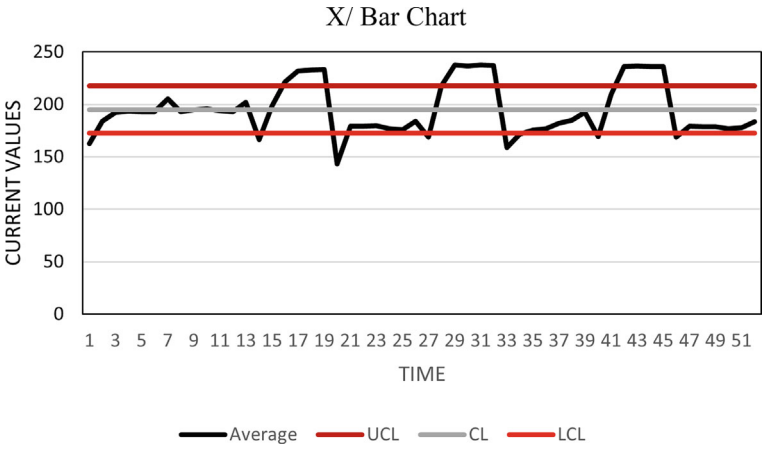

Fig. 6 X/Bar chart according to the change of current values (*Source* Authors)

Table 2 Comparison table of methods

z	R Bar	X Bar	MV	SVM linear	SVM radial
Accuracy	0.72	0.88	0.72	0.87	0.93
Precision	0.92	0.76	0.79	1.00	1.00
Recall	0.28	1.00	0.43	0.82	0.91
FI	0.42	0.87	0.56	0.90	0.95

5.4 Comparison of Methods

Losing step error is a synchronization problem that causes measurement defects, surface defects, and position errors in CNC systems or robotic systems using stepper motors. In the case study, a DT model was developed that would detect the losing step error and transmit the occurrence of this error to the CNC control software. The DT model works by using the data of instantaneous changes in the amount of current consumed by the stepper motors. The data-driven DT model has been proposed so that the model can adapt to different CNC machines and robots. In the development of the model, deep learning, machine learning, and SQC methods were used. The detection performances of the methods used are given in Table 2. The method with the highest F-Score value is deep learning. The best method after deep learning is "SVM with non-linear kernel". The fact that the linear kernel and non-linear kernel and precision values are 1, indicates that the losing step errors detected in both kernels are detected correctly, but some losing step errors cannot be detected because the recall and F-Score values are less than 1. The rate of misdetection is lower in the non-linear kernel. Because of this, its performance has been better. In the X-Bar chart, it is understood that there is a false alarm state because the recall value is 1 and the precision value is less than 1. In the R-Bar chart, there is both a false alarm situation and the inability to detect all losing step errors.

Although DNN is better than other methods in terms of performance, it has a long training period. Therefore, the creation of the network is difficult. Of the SVM models that can be created in less time than DNN, the non-linear one has better performance. Since the developed "SVM with non-linear kernel" model does not produce false alarms, it may be preferred in cases where it will take a long time to create the DNN model. Statistical control diagrams were evaluated to determine the losing step status without wasting time for training. It was concluded that the X-bar chart and R-bar chart produced false alarms. We also assess that it is more appropriate not to use them because these control diagrams are the methods with the lowest detection performance.

The research presented in this paper offers valuable insights for manufacturing companies, enabling them to enhance their equipment monitoring and predictive maintenance capabilities using DT. By doing so, organizations can significantly boost equipment availability and prevent unforeseen downtime, which ultimately translates into improved operational efficiency and cost savings.

6 Conclusion

The use of DT technology has proven to be an effective means of anticipating and preventing equipment breakdowns, which can lead to significant financial losses for manufacturing organizations. DT provides a comprehensive, real-time, and data-driven approach to asset management, significantly influencing maintenance strategies. Several benefits and advantages of incorporating DT into maintenance plans include real-time monitoring, reduced maintenance costs, decreased downtime, lower energy consumption, and a positive environmental impact. In this study, the objective was to employ DT technology to identify defects in CNC machines by analysing the current behaviour of the stepper motor, a commonly used component in these machines. Machine learning methods and SQC techniques were applied to the data collected from the current sensor to detect anomalies and characterize malfunctions. Data preparation processes were executed, and eight current behaviour patterns were established. Both ANNs and SVMs were employed to analyse the data for anomalies. Additionally, control charts were created to flag malfunctions as events falling outside the lower and upper boundaries.

For future research, it is recommended to base the establishment of these lower and upper boundaries on reliability when formulating effective maintenance plans. To implement reliability-based predictive maintenance, a risk function will be developed for determining these boundaries. Consequently, the system's behaviour, as assessed by sensors, will be correlated with malfunctions within the context of reliability.

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Implementing Smart Production Through Lean Digitalization

Ralph Riedel and Susanne Franke

1 Introduction

While the digital transformation of production companies has already led to large amounts of data being acquired and stored, much of this data still remains unused. One reason why *Smart Manufacturing* has still not reached its full potential becomes apparent when looking at the prevalent situation in small and medium-sized enterprises (SMEs): The digitalization of the production is realized as a gradual process and leads to many isolated solutions, which usually are implemented without following an overall concept and without having a holistic view on the production process as well as the company's business processes in general (Mittal et al., 2019). Additionally, a large amount of manual inputs instead of automated data acquisition, limited knowledge regarding the wide range of possibilities to use the data and financial restrictions are typical concerns and drawbacks for SMEs. As a result, the introduction of digitization solutions can quickly lead to missing the

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mark and setting up inefficient, counterproductive structures and processes (Garzoni et al., 2020).

According to Alieva and Haartman (2020), Smart Manufacturing shares similar priorities with Lean Production like for instance efficiency gain and waste reduction. Being a "lean company" requires specific lean principles to be implemented as well as followed and is achieved by realizing different measures depending on the company's characteristics (for SMEs, this means, e.g., the available machinery or financial restrictions) and targets (strategy, product program design, customer requirements (Schuh et al., 2008)).

The basis for Smart Manufacturing and data-driven process optimization is set with proper data acquisition. Hereby, we define *Lean Data Acquisition* as the economical and efficient use of equipment and human resources to acquire data having been identified as relevant for production and optimization processes. Matt et al. (2015) address strategies for digital transformation from a manager's point of view. However, considering the realization of digital transformation itself, research and application papers reveal the lack of standardized approaches (Teichert, 2019). Simultaneously, there is a high demand of companies to implement efficient procedures, starting with data acquisition. One possibility to improve the efficiency of data acquisition is the application of lean principles. The question arises how production companies can identify and prioritize lean ways for data acquisition as a basis for, e.g., the implementation of Smart Manufacturing processes. Therefore, we investigate how proven approaches from Lean Production can be adapted in order to apply to data acquisition processes. Lean production is achieved by adhering to key lean principles such as the value-adding principle. In the context of Lean Data Acquisition, this corresponds to producing as little digital waste as possible and minimizing non-value-adding activities.

Typical approaches for measuring the ability of an organization for continuous improvement in a particular discipline are maturity models (as, e.g., proposed by Ifenthaler and Egloffstein [2020] for digital transformation) or the identification, weighing and evaluation of the factors describing this maturity, which allows the combination of different factors as well as a comparison of different solution approaches. In order to choose measures leading to an efficient and effective data acquisition, we create an *assessment logic* for making the targets measurable and for evaluating implementation possibilities, thereby following defined steps representing lean principles: We provide methods for the operationalisation of operational performance indicators that are relevant for the company (e.g., KPIs to be evaluated) and measure their fulfilment. Subsequently, we evaluate the expedient choice and realization of *data acquisition-specific lean practices* to achieve the identified

goals. For a holistic consideration of the company's demands, we combine the view of different customers, where the customers belong to the company, thereby creating synergies in the realization.

We explicitly focus on data acquisition and not data analysis since this would be beyond the scope of this contribution; however, the usage of data to be acquired is clarified roughly in advance (in accordance with the company's goals) in order to choose methods and procedures as well as devices for data collection in the required quality, frequency, etc. and means for data storage which allow an expedient data utilization. This is achieved by following and mapping the data value stream.

Regarding the question of the proper order of implementing Lean or Smart Manufacturing and the mutual support of both, we refer the interested reader, e.g., to Powell et al. (2021) and Davies et al. (2017). We want to mention that the proposed methodology supports the expedient application of the lean approach while realizing Smart Manufacturing (regardless of the current lean state of the company) since our approach provides a method for lean digitization under consideration of the current state of both Lean and Smart Manufacturing.

This contribution is structured as follows. After a thorough investigation of the relevant theoretical background in Sects. 2 and 3 comprises the development of the methodology. In Sect. 4, a practical application of this methodology is provided. The contribution closes with a conclusion.

2 Theoretical Background

Lean Production has already been widely investigated. There exist approaches for its implementation as well as studies on dimensions to be considered during this implementation process (Kenett & Shmueli, 2016; Shah & Ward, 2003). In connection with Lean Production, Meudt et al. (2017) and Hartmann et al. (2018) provide holistic methods for a value stream mapping with the primary goal of continuous material flow for shortest throughput times. Fan et al. (2011) propose a method to convert acquired data into lean data using a data management platform with an open architecture. However, they define lean data as data gained by the application of algorithms on the pre-processed acquired data.

Whether the automation of lean principles leads to digital waste (as production waste) with negative impact on the added value has recently been investigated by Alieva and Powell (2021). There, digital waste was identified

both in connection with suppliers and customers in product- and service-oriented technology. Additionally, different definitions of digital waste are compiled. We follow the authors and define digital waste as "uncollected, unprocessed, or misinterpreted data in the production process". Following the classical lean concept, digital waste does not contribute to the customers' benefit.

Smart Manufacturing first requires the data to be provided. This can be achieved by the steps acquisition, transfer, storage, transfer, and utilization (Franke et al., 2020). Note that data acquisition comprises both digitization, which addresses the installation of devices to provide the data, and digitalization, which refers to processing and making use of the data by implementing digital technologies.

Recent developments show that there is a trend to combine Lean and Smart Manufacturing. However, there only exist few sources, as shown, e.g., in the literature overview by Ejsmont et al. (2020), and the focus hereby is usually set on value-added processes. Florescu and Barabas (2022) for example present a horizontal and vertical integration model for Smart Manufacturing with the application of Lean principles. A fundamental gap exists in a Lean way to acquire data, which is addressed in this contribution.

Regarding Lean Production and factors for the evaluation of the performance of a system, there already exists a thorough research basis. Table 1 gives a short overview on prior research in the fields of Lean Production and performance indicators that has influenced our work.

In the following, the content of the respective sources is shortly summarized with respect to the application in this contribution. The first category we investigate more deeply is operational performance indicators. Buer et al. (2021) identify five key dimensions of operational performance (speed, quality, flexibility, dependability, cost) which are operationalized into specific indicators (e.g., speed into production lead time). This builds the basis for considering all possibly relevant target dimensions from the customer's point of view, this is included in our provision of information to the customer in Step 1 of the upcoming methodology. It also allows an evaluation which dimensions are usually important for the customers. Operational performance is additionally important in connection with continuous improvement processes—it is useful to check which of these dimensions haven't been addressed yet as well as why and if they should be included in the improvement process in the future.

Tan et al. (2007) propose operational capability as an important impact factor for value creation in production companies. Its elements are new

Table 1 Relevant contributions in the literature

Topic	Sources	Application
Operational performance indicators	Buer et al. (2021) Tan et al. (2007)	Customers' targets and continuous improvement process Measuring influencing factors; different approach: operational capability
Lean principles and dimensions	Womack and Jones (1996) Shah and Ward (2003, 2007) Romero et al. (2018) Reis (2018)	5 lean principles as basis of our methodology Difference between lean production and lean data acquisition Use 7 kinds of digital waste as basis for waste reduction within methodology (Step 2b) Use parts of the assessment logic dimensions, but adapt them to lean data acquisition

product design and development, Just in Time (JIT) and Quality Management. The proper realization of these elements is measured by certain influencing factors (e.g., factors for the element new product design and development are concurrent engineering, standardization of parts). This approach is reflected in our methodology as measuring the influence of certain factors to achieve highest possible outcome with respect to the target. However, our targets are set differently: the focus is set on the customers' goals, they are evaluated focusing on data acquisition.

Considering the second category, we focus on Lean Production. A fundamental source is given by Womack and Jones (1996), who identify and describe five lean principles to reduce waste, produce efficiently and provide the customers the desired products. This builds the basis for our methodology: we follow the five lean principles and formulate the necessary content and work with respect to Lean Data Acquisition. These five principles form the basis for the methodology we propose.

Shah and Ward (2003) focus on Lean Production and identify 22 lean practices as well as 4 bundles (JIT, Total Quality Management, Total Productive Maintenance, Human Resource Management) of these practices. Accordingly, in Shah and Ward (2007), dimensions of Lean Production are formulated, e.g., supplier feedback and customer involvement. They roughly

follow Womack and Jones (1996) but add supplier view and production-oriented aspects like set up time reduction and maintenance. A direct transmission of these practices or dimensions to Lean Data Acquisition is not possible (cf. the remark at the beginning of Sect. 3), but parts of these practices find application within the methodology in an adapted way (e.g., aspects to be considered while defining the value and taking the customers' needs into account).

One fundamental idea of Lean is the reduction of waste, or, in the context of Lean Data Acquisition, digital waste. A thorough investigation of ways to decrease or prevent the seven types of digital waste is given by Romero et al. (2018). Based on the seven kinds of waste by Ohno (1988), the authors distinguish active and passive digital waste and formulate digital/Smart Manufacturing technologies as well as their expedient application to reduce digital waste, also indicate possible risks in connection that may arise when using these technologies. The descriptions find application in the realization of Step 2b of the upcoming methodology, which is concerned with waste reduction while implementing Lean Data Acquisition technologies.

Reis (2018) provides a methodology to measure the quality of information generated in data-centric activities regarding the chemical processing industry. He investigates eight assessment dimensions for determining the value of information which stem originally from Kenett and Shmueli (2016), in the context of data acquisition and expedient usage. Regarding this special application, the focus is set on evaluating, analysing, and improving the quality of information generated in the variety of data-driven activities and empirical studies. We have investigated these dimensions with respect to our research topic, filter the relevant ones and describe the respective content, use this as basis for identifying other important dimensions or factors.

3 Development of the Methodology

In this section, we describe the methodology to identify suitable means for Lean Data Acquisition. As mentioned in Sect. 2, we follow Womack and Jones (1996) and use the five lean principles *Define value*, *Map the value stream*, *Create flow*, *Establish pull* and *Pursue perfection* as the basis for our methodology. This is depicted in Fig. 1. *Define value* focuses on the customer's needs. *Map the value stream* includes defining every necessary process step that transforms the input into the output (which is then provided to the customer). *Create flow* focuses on identifying and deleting

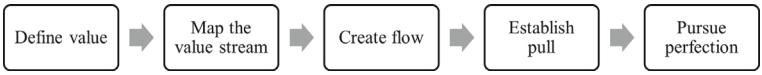

Fig. 1 Five lean principles (*Source* Womack and Jones [1996])

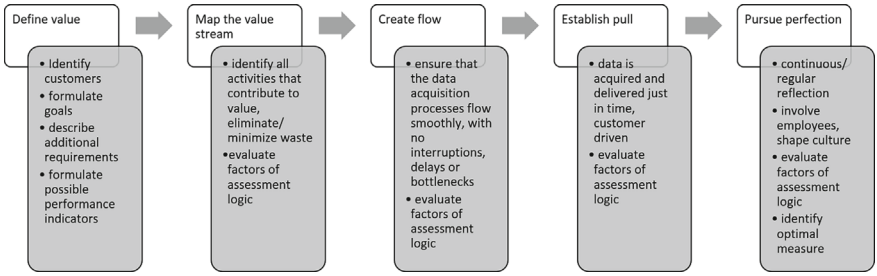

Fig. 2 Application of assessment logic to implement Lean Data Acquisition (*Source* Authors)

interruptions, bottlenecks, and delays slowing down the process. With *Establish pull*, the created flow is considered from the customer's point of view who pulls the value to avoid high inventory. *Pursue perfection* demands continuous improvement for all the above-mentioned steps.

Lean Data Acquisition is realized by the implementation of certain measures. To evaluate and prioritize different measures, we provide an assessment logic. The different factors which are part of the assessment logic are evaluated within these steps. As the result, the most suitable measure is identified. This procedure is illustrated in Fig. 2.

3.1 Assessment Logic

The assessment logic addresses two main parts: the operationalization of the customer's targets as well as the designation and evaluation of relevant assessment factors for Lean Data Acquisition.

As described in Sect. 1, the goals formulated by the company have to be operationalized in order to evaluate their degree of realization.

According to the derived performance indicators, means and ways for the acquisition of data have to be identified and evaluated with respect to specific factors. The factors resolution, integration, and operationalisation were identified by Reis (2018). In connection with Lean Data Acquisition, we identified additional factors affecting the lean character: transferability, necessity, and costs as influencing factors. In the following, the factors are described shortly.

Resolution/Precision: Depending on the type of analysis the data will be used for; both the frequency and granularity of data acquisition have to be specified. While a high resolution provides a higher precision, it negatively affects the amount of storage and the calculation time of the data analysis. Additionally, devices providing a higher resolution/precision are usually costly. Hence, a balance between expense (in the sense of time and cost) and benefit has to be achieved.

Costs: This is of high interest especially for small companies, which usually cannot finance holistic solutions, but iteratively extend their existing infrastructure.

Operationalisation: This factor assesses the complexity of implementing the solution approach and achieving the desired results. It encourages the user to detect possible challenges like combining retrofit approaches with available data before the actual implementation to solutions which are not realizable. It includes the temporal expenditure regarding installation, usage, and maintenance.

Transferability: This factor describes the potential of solving similar tasks with the available means. It leads to a higher utilization of the devices.

Impact: There are usually gradations in the influence of certain implementations have on the outcome. The factor impact reflects the satisfaction of the operational performance indicators.

Integration: The data acquisition devices cannot be regarded separately, as some devices might provide data for different tasks or might provide full data only in combination with another device. This can also be process-related (e.g., data from both input and output are needed for the calculation of efficiency key performance indicators).

In practical application, certain limitations like costs restrict the solution space and function as initial hurdle before a deeper investigation of the remaining factors follows. Hence, the factors are divided into the two categories *initial requirements* and *realization*, which are considered in this order. The *realization* itself is partitioned into *evaluation of objective function* and *feasible combinations*. The former category comprises evaluation factors which contribute to the calculation of the objective function value, i.e., the evaluation of the value of the respective solution possibility. The latter category allows the combination of different solution possibilities, which are realized together in order to fulfil the identified goals. This can be considered if for example the goals are very versatile and address different areas.

Table 2 provides an overview of the relevant assessment factors, their categorization into initial requirements to form the feasible set and realization for the determination of the degree of fulfilment. Note that factors whose initial

fulfilment is required nevertheless do appear in the second category. Here, they are used for comparison operations in case there exist different solution possibilities. For each factor, the evaluation style and, if necessary, the scale for a rating in connection with the fulfilment of operational performance indicators are formulated.

Note that if initial requirements are irrelevant for a certain use case, then the upper/lower limits can be set sufficiently high/low or as the maximum/minimum value of the solution possibilities.

Assuming there exist m different solution possibilities, the respective performance value has to be calculated. Therefore, every influencing factor is evaluated. For every factor $f_i, i = 1, \ldots, n$ (with n being the total number of factors to be evaluated), an assessment value a_{ij} is calculated to represent the value of this solution j $(j = 1, \ldots, m)$. Hereby, the higher the value, the better the solution is ranked. Furthermore, a weight factor w can be assigned. (If weights are not to be considered, then $w_i = 1$ for

Table 2 Relevant factors for data acquisition

Factor f_i	Evaluation	Scale (feasible assessment values)
Initial requirements: feasible set		
Resolution/precision	Lower limit	–
Costs	Upper limit	–
Realization: evaluation of objective function		
Resolution/precision	Ranking based on the resolution/precision	[1, …, |solution possibilities|]
Costs	Ranking based on the total cost	[1, …, |solution possibilities|]
Operationalisation	Ranking based on the description of the complexity of the solution	[1, …, |solution possibilities|]
Transferability	Ranking based on the number of possibilities to expediently extend Lean Data Acquisition	[1, …, |solution possibilities|]
Impact	Ranking based on the contribution to the calculation of the operational performance indicators	[1, …, |solution possibilities|]
Realization: feasible combinations		
Integration	Formulation of combinations of solution possibilities to acquire all necessary data	–

Source Authors

$i = 1, \ldots, n$.) After assigning each solution possibility the assessment values for all factors, it is the goal to find the solution $j \in \{1, \ldots, m\}$ such that $\sum_{i=1}^{n} w_i a_{ij}$ achieves the maximum value or a feasible combination of solutions $\{1, \ldots, k\} \in \{1, \ldots, m\}$ such that $\sum_{j=1}^{k} \sum_{i=1}^{n} w_i a_{ij}$ achieves the maximum value (with $k \leq m$).

The question arises how the valid solution possibilities as well as the respective assessment values can be identified properly. For a precise assessment, it is necessary to evaluate the solution possibilities with respect to their characteristics. This is achieved by considering the relevant steps for Lean Data Acquisition as they are formulated in the upcoming part: In the following section, we assign the evaluation of the different factors to the steps of the methodology.

3.2 Methodology

We start this section with a few remarks on the proposed methodology.

Difference between value stream and data value stream: An established lean method to analyse the current state of a system and design the desired target state is value stream analysis and mapping. Hereby, typical performance indicators to evaluate the fitness of a manufacturing system as the basis to formulate and rate the target state are lead time, flow time, and the availability of certain resources. This scheme has to be adapted to Lean Data Acquisition as it focuses on the information flow and is evaluated using different performance indicators. Furthermore, Shah and Ward (2003) describe practices to establish lean production (e.g., lot size reductions, maintenance optimization). Those can as well not be transferred directly and completely to Lean Data Acquisition, as it is illustrated in the following using two examples.

First, the basis for the evaluation and optimization of current processes is provided by customer-related data: One important performance characteristic is (customer) tact time, which corresponds to taking the customer value into consideration. In the sense of data acquisition however, we transfer this approach and define internal customers who describe their goals as foundation for the formulation of the target state. Since the data acquisition we consider here (as basis for production process optimization) regards internal processes, the respective knowledge carriers and/or users of the acquired data are regarded as internal customers. This can for example be managers, production managers, or employees of quality control. Each internal customer has his or her own specific view depending on the respective work level. For example, managers aim at increasing productivity, whereas production managers focus, e.g., on high machine availability and

regular reports on machine failures. Hence, for deriving effective performance indicators, it is required to meet and combine all these demands.

Second, material and information flow are crucial factors in connection with value stream mapping. For Lean Data Acquisition, the mapping of the production process itself is important as it leads to a transparent overview which data can be provided at which places. The material flow itself however is not as important as the value-adding focus is not set on production. Nevertheless, the material flow has to be modelled carefully since certain goals to be achieved may refer to it, like the minimization of layover in a certain production line.

Three Ms: It is the goal of Lean Production to minimize overload, imbalance and waste, also known as Muri, Mura and Muda, the three Ms. They are interconnected and cannot be addressed separately: Strong fluctuation for example leads to (temporary) overload, which itself causes waste. Starting from the non-value adding waste (Muda), the origin lies in Mura and Muri (Pieńkowski, 2014). We illustrate this with two Lean Data Acquisition examples regarding Muda as described in Romero et al. (2018).

First, digital motion waste can occur in a manufacturing company when an employee has to make inputs on order data into an excel file at a local workstation several defined times a day, which forces him to interrupt his work process. The solution is the usage of wearable human machine interface (HMI) for the employee to make inputs. However, it is important that the employee understands the working principle of HMI, which, e.g., requires additional training.

Second, according to Hasan and Burns (2011), digital defects waste can occur in the form of unintentional or unwanted data. This kind of waste occurs for example when instead of just storing changes in the current production, a complete image of the current production plan is stored because the triggering action is set wrongly. This slows down data storing processes and occupies storage capacity.

Application of the assessment logic: Note that the upcoming methodology including the application of the assessment logic is not only applied to evaluate the expedience of new digital solutions, but also to evaluate the quality of existing ones. This allows the evaluation of their contribution to Lean Data Acquisition and possibly existing optimization potential which influences the choice of new solutions.

Based on these preliminary considerations, we can formulate the methodology.

For every lean principle, the data acquisition-specific lean practices which to be evaluated are assigned within the upcoming steps as displayed. Thereby,

the focus is set on the customers as the recipients of the goods produced in the value-added process. As mentioned at the beginning of this section, for the realization of Lean Data Acquisition, the customers are members of the company who make use of the data to be acquired, e.g., manager, production planner, employees of quality control.

3.2.1 Step 1: Define Value

First, the relevant customers are identified in order to acquire their goals. Based on the customers' demands, the data to be acquired and its utilization (i.e., which use the customers want to make of this data) are derived. These goals usually differ among the customers. For example, the company's manager wants certain key performance indicators (KPIs) to map the achievement of the company's goals, whereas the production planner needs data, e.g., to get automated feedback on the progress of the production orders. Slack et al. (2007) define five key dimensions of operational performance: speed (e.g., production lead time), quality (e.g., product quality), flexibility (e.g., process flexibility), dependability (e.g., process uptime) and cost (e.g., production cost per unit). Depending on the customer, they can function as reference points for the formulation of the performance indicators.

Furthermore, it is necessary to identify unimportant or even interfering/disturbing characteristics. For example, acquiring data in a very high frequency and automatically distributing reports on machine status instead of push messages when a problem is detected leads to an overload of unnecessary information and distracts the recipients from work. With this knowledge, the requirements of the solution are derived. Such requirements are, for example, the frequency in which the KPIs are evaluated or the necessary information to calculate the KPIs.

This step is performed iteratively for each customer by identifying the respective goal, realization, required data and expected result. Subsequently, the findings are combined. Note that it might be necessary to re-approach customers when new information of other customers is available to match the different needs. Techniques to acquire the information of the customers (goals, demands, and additional requirements) are for example surveys or interviews.

Assessment logic: This step functions as a preparation step for the application of the assessment logic. The customers provide their goals, which are operationalized into performance indicators. Other initial requirements are acquired, and the data necessary for the evaluation are defined.

3.2.2 Step 2: Map the Value Stream

This step aims at using the customers' values as reference point to identify all activities that contribute to this value. Any other activities not contributing to this value are waste and have to be minimized.

We divide this step into two sub steps and describe them separately.

Step 2a. Define target and actual state Before deriving the necessary new techniques for data acquisition, both the current and the desired state are identified. The current production process is mapped with the focus being set on which data is currently acquired at which positions (e.g., machines) having which characteristics (e.g., frequency, format, quality) and is transmitted in which way. Then it is deduced which data has to be acquired at which positions having which characteristics to achieve the goals formulated under Step 1, hence the target state is derived from customers' requirements. Hereby, both material flow (e.g., with respect to production process chains) and information flow (e.g., centralized/decentralized data storage, digital/analogue data acquisition) are of interest. Hartmann et al. (2018) describe possibilities for optimizing the value stream. For example, the material flow can be improved using automatic machine parametrization, and the information flow can be enhanced with active traceability and the reduction of media breaks.

Assessment logic: The formulation of the target and the actual state is the starting point to identify means and ways for achieving a high operational performance regarding the goals defined in Step 1. Already existing solutions as well as suitable new solution possibilities are identified and described properly to continue with evaluation and prioritization. For the new solution possibilities, both *resolution/precision* and *costs* are identified and serve as initial requirements to be fulfilled. The factor *integration* is evaluated to identify feasible combinations of solutions possibilities which increase benefit or might contradict each other.

Step 2b. Waste reduction Lean thinking focuses on reducing non-value adding activities since these activities increase costs and lower efficiency. This is achieved in this step. Using the actual state overview, it is derived which data is currently acquired but not used and also not used in the future because it is not needed within the target state. Based on this information, it is checked whether stopping this data acquisition or changing its characteristics is beneficial for reducing waste. In this context, it has to be evaluated whether the currently chosen ways of data acquisition and storage are expedient. For the realization of this step, the seven kinds of waste can be distinguished and addressed separately. Romero et al. (2018) describe the seven kinds of digital waste.

Note however that there exist necessary and unnecessary waste. Necessary waste helps improving the processes but does not directly add value. It typically occurs in planning and testing processes. If this kind of waste is detected, careful consideration if and how it can be minimized is required.

Assessment logic: In connection with the identified solution possibilities, the waste they produce is evaluated. Especially the factors operationalization and impact map the waste reduction, which can, e.g., be achieved by automating cumbersome manual data entries (this can happen for example when products are moved between interim storage units).

3.2.3 Step 3: Create Flow

After the waste has been eliminated, it is ensured that the remaining (i.e., non-waste-producing) processes flow smoothly and with no interruptions, delays or bottlenecks. Proper means for regular, smooth data acquisition, direct transmission with minimal transfer points and secure storage have to be evaluated under consideration of the relevant places and authorities that will make use of the data. For example, storage concepts include centralized (Cloud Computing) and decentralized approaches (Fog and Edge Computing) as well as purely local solutions.

Assessment logic: This step refers to the evaluation of, e.g., retrofitting possibilities to acquire data in comparison to the investment in new technology. Typical strategies are breaking down the processes (top-down approach) to properly assign the data acquisition task to the relevant sub steps as well as analysing the interdependencies between different departments. Additionally, the necessary employee training to implement and properly use the data acquisition tools is considered. Hence, especially the factor *integration* and *operationalisation* are evaluated.

3.2.4 Step 4: Establish Pull

The flow system creates ideal conditions for just-in-time: For Lean Data Acquisition, this corresponds to providing the relevant information every customer demands at the correct place and time and in the desired quality, following the principle “product steers process”. Implementation strategies are based on the value stream map created in Step 2: The value stream is followed backwards through the production system from the data recipient to its origin to identify, e.g., the ideal places for provision.

Assessment logic: To avoid instabilities and fluctuations, the temporal expenditure for the usage of the different solution possibilities is of high interest. This is included in the factor *operationalisation*. A unification allows an easier usage and maintenance for the employees.

3.2.5 Step 5: Pursue Perfection

The developed and implemented solution is not static but subject to continuous improvement: As the environmental conditions change, several factors have to be checked regularly: Are the KPIs mapped properly, is it necessary to include new KPIs? Is the data acquired, stored, and used as planned? Are new technologies on the market that would improve the system's performance significantly? In this context, it is essential to make lean thinking and process improvement part of the corporate culture and actively involve all employees.

Assessment logic: This step addresses the regular check of the timeliness of the implemented solutions and requires regular meetings with the responsible customers to identify changed demands. Concerning the continuous improvement, the *transferability* of the existing solution is evaluated. Furthermore, a regular check of the solution possibilities is necessary to identify optimization potential arising from technological innovations. The results of the aforementioned steps have to be evaluated regularly.

4 Prototypical Application

For a prototypical application, we followed a case study approach of Yin (2018) and considered a SME in the textile industry that chemically treats, dyes, and prints on textiles on a specific customer demand before those textiles are further processed into garments. The company processes a wide variety of materials, namely polyester, cotton, silk, and viscose. Accordingly, the processes are also very diverse. Roughly speaking, they can be divided into the steps of incoming goods inspection, pre-treatment (bleaching, washing), dyeing or printing, post-treatment (steaming, washing, drying), each with integrated quality assurance, and finally packaging. The process chain is visualized in Fig. 3.

The company follows a Make To Order (MTO) strategy. The orders have different quantities. Challenges lie in the short throughput times demanded by customers and in balancing order quantities with sensible production lot sizes; for the latter, fixed sizes such as fabric bales or fabric rolls play a major role.

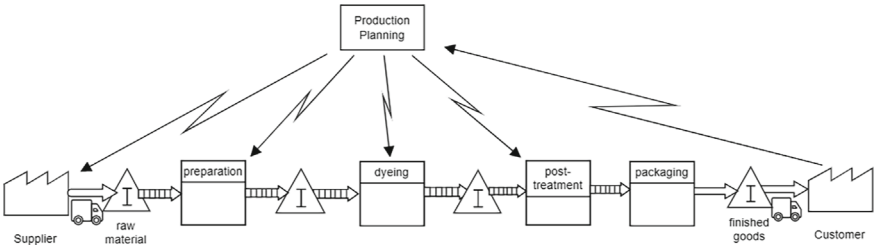

Fig. 3 Process chain of the textile SME (*Source* Authors)

Pre-treatment in particular aims to achieve a clean surface of the textiles, free of fluff. For this purpose, thermal (gas flame) and chemical (oxidative) processes are combined. Depending on the material and the desired results, special chemical mixtures as well as individual process parameters such as throughput speed, temperature, material tension, etc. are relevant—these are set in advance according to a recipe and must be controlled in the process. The company identified various optimization potentials regarding process, quality, and logistics, which were to be unlocked with the help of digitization. The existing machinery is relatively old, and there was no automated data collection and processing in the process. The technologies are comparatively stable, investments in new machines are not necessary. The company decided to do a retrofit to shorten throughput times, especially in pre-treatment, to increase the value-added time at the machines, to better track the progress of orders and thus to be able to better predict delivery times as well as delivery dates. These optimizations were to be achieved by collecting order, process/machine, environment, and position data. At the same time, the risk of overengineering had to be considered since cost was a major factor. Therefore, lean practices for data acquisition had to be implemented in the described context-oriented way. The prototypical and exemplary application of the proposed assessment logic is presented in Fig. 4 and explained in detail in the following paragraphs.

4.1 Step 1: Define Value

The production planner and the supervisor of pre-treatment were identified as internal customers. The relevant targets were identified based on their needs (see Table 3).

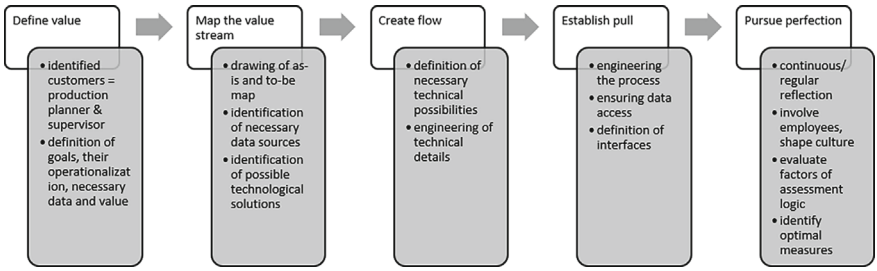

Fig. 4 Methodology and assessment logic in the case study (*Source* Authors)

4.2 Step 2: Map the Value Stream

The pre-treatment process was analysed in detail, see Fig. 5, for an overview. Technologically determined dwell times are necessary between each of the up to four work steps. Quality control is carried out after each process step. The total lead time of an order can be up to 40 hours.

Production planning works with cycle times for the entire pre-treatment process. The supervisor of the pre-treatment department does a manual detailed planning based on known or estimated times for the individual processing steps, machine availability, and available personnel capacity. During processing, the detailed planning is continually adjusted based on current progress and material/personnel availability. If necessary, there is feedback to production planning, especially with regard to delivery dates.

On the basis of the targets and identified waste, a future state was drawn up which focused on continuous and timely order tracking, material and order localization, a reduction in lead times/setup times, and an increase in process stability and thus also availability. The necessary data sources for these purposes were defined. In this application, we focus on order tracking as well as material and order localization as the first step before optimization processes for, e.g., lead time reduction can be realized. Therefore, real-time locating systems (RTLS) were identified as possible solutions. Three different versions were deeper investigated and evaluated by the introduced logic. The company decided to not include preliminary limits on costs, whereas the precision of the tracking system was set to be 0.5 m. Table 4 comprises the evaluation values resulting from the following steps.

4.3 Step 3: Create Flow

In this step, the necessary technical possibilities were investigated and defined in order to create a lean, smooth process of data acquisition, transmission,

Table 3 Step 1: define value

Customer	Production planner	Supervisor pre-treatment
Goal	• Transparency of order progress • Reliable prediction of lead times and delivery dates • Evaluation of new orders, especially rush orders	• Reduction of lead times, especially setup times • Avoidance of search times for material • Increase of quality
Realization	• Booking of the beginning of the order for each intermediate step • Localization of the orders between the machining processes • Recording/ evaluation of (available) machine capacity	• Improvement of process control by means of a (self-learning) model that considers, among other things, environmental conditions • More accurate prediction of machining times and material availability in order to better plan setup activities
Required data	• Order booking (order, machine, time) • Processing, setup, dwell times • Location per container/batch/ order	• Location per container/batch/order • Process parameters and chemical recipe directly at the order/material • Process and material parameters • Environmental data (temperature, air pressure, humidity) • Residual material per fabric roll
Result	• Order progress in real-time • Current and remaining processing time, probable completion date • Available machine capacity	• Real-time progress of processes • Optimal process/machine parameters • Material localisation • Optimal setup times

Source Authors

and use. The already partially practiced manual recording and collection of environmental and process data were replaced by sensor-based recording and direct forwarding to the place of further processing (machine control). Likewise, material data from the quality laboratory were made available for further processing for process optimization. Certain process parameters, for example material tension, speeds, etc., do not have to be recorded separately

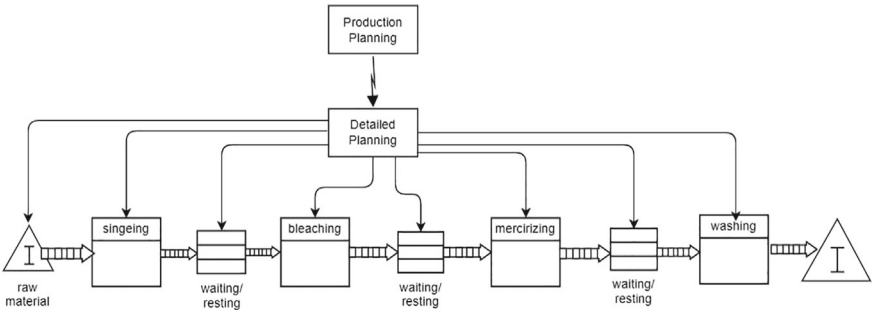

Fig. 5 Description of pre-treatment (*Source* Authors)

Table 4 Evaluation of the solution possibilities

Factor	Weight of the factor	RTLS Solution 1 (UWB)	RTLS Solution 2 (Wi-Fi)	RTLS Solution 3 (Bluetooth)
Initial requirements				
Resolution/precision	–	0.1 m	0.5 m	0.15 m
Realization				
Resolution/precision	0.15	3	1	2
Costs	0.05	1	3	2
Operationalisation	0.35	2	3	2
Transferability	0.25	3	1	1
Impact	0.2	3	1	1
Result		2.1	1.65	1.25

Source Authors

by sensors but can be captured from the machine control system. The locations of containers/orders will be recorded via the RTLS solution, and an order booking will be made at the start of processing on a machine. This data will be available to both the supervisor and the production planner.

4.4 Step 4: Establish Pull

In the future, the pre-treatment supervisor will be able to access the processing progress of individual orders and also order/material locations and use this information when planning transport, material provision, and setup activities. The data from process planning (for example, machine speeds, remaining quantity of material) will be used to calculate the expected end of processing or the time for required setup operations (new fabric roll). Machine parameters and recipes will be provided with the order/material in the future and will be read in by the machines before processing. When

scheduling new orders, the production planner can access the processing progress of the orders and the current and already planned machine assignment. Rush orders in particular can thus be evaluated in terms of their feasibility.

4.5 Step 5: Pursue Perfection

After the implementation, regular evaluations are carried out to determine whether the goals pursued with the retrofit/digitization have been achieved. It may be necessary to readjust recording locations, times, frequencies, degree of resolution, etc. The data quality must also be checked. In terms of waste to be avoided, it should also be questioned whether the collected data are actually fully necessary to achieve the goals and whether there are more efficient ways of collection (for example, through integrated sensor technology), storage, processing. The effects of changes in machines, processes, environment, and infrastructure must also be evaluated, for example, when a new enterprise resource planning (ERP) system is introduced.

In summary, the ultra-wideband (UWB) solution was identified as the most suitable one for the company. Regarding the operationalisation, besides the installation especially the signal strength and range were a critical criterion, whereas precision and minimal interference with surrounding technologies also played a role in the evaluation and decision-making process.

From a methodological point of view, we can conclude that our proposed assessment logic was applicable for that specific case, that it supported structuring the problem-solving process and that it led to a lean and purposeful solution which meets the company's needs.

5 Conclusion

In this contribution, lean thinking is extended to data acquisition as the basis for Lean Smart Manufacturing. With a proper strategy for lean data acquisition, companies are able to acquire data in a target-oriented way and do not create digital waste, which would slow down their processes. This builds the basis for, e.g., production monitoring, smart maintenance and automated quality control.

Therefore, we formulated an assessment logic which allows the evaluation of different solution possibilities for data acquisition. In order to address all relevant aspects of the lean concept, the evaluation is performed

within the lean methodology by Womack and Jones (1996). The methodology and assessment logic are applied in a retrofit use case. In this context, the customers are defined as company-internal since they make use of the provided data. However, regarding digital business models, the transfer to external customers can be considered as well.

Regarding the assessment logic, the current factors considered for the evaluation are typical for digital transformation. However, it is of course possible to include company-specific additional factors and define the respective evaluation and scale. For example, not only the transferability of the solution possibilities, but also the generalisability might be of interest.

The methodology provides companies an easy approach to evaluate existing solutions as well as to identify new means for Lean Data Acquisition. The company-specific goals are mapped into the assessment logic, which provides a clear way to identify optimal strategies for data acquisition as the basis for Smart Manufacturing.

However, regarding complex digital transformation projects with many stakeholders, the evaluation might lack in clarity (for example when different stakeholders influence the evaluation or many). Then, the assessment logic can, e.g., be applied in different steps (for example first for each stakeholder separately before these results are combined). Note that the methodology is limited to data acquisition and does not cover the actual usage of this data. However, simultaneously, the company is required to already know roughly which use they want to make of the data as the evaluation of certain factors like, e.g., resolution/precision depends on the actual application (for example, energy consumption can be monitored daily but also in a much higher frequency when, e.g., peak loads are relevant). Future research will address additional practical applications to SMEs of different branches to extent the case study. Additionally, we will investigate which enablers for digital transformation (Lokuge & Duan, 2023) have impact on Lean Data Acquisition capability.

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Additive Manufacturing and Industry 4.0: A Path to Digital Transformation

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1 Introduction

The term "Industry 4.0" (I4.0) pertains to a manufacturing paradigm centred around technology, wherein the cyber and physical worlds are interconnected through heightened automation (Adnan et al., 2021; Choi et al., 2022). Following the emergence of I4.0, manufacturing businesses have increasingly adopted various digital technologies such as Big Data, the cloud computing, the Internet of Things (IoT), Augmented Reality (AR) and Machine Learning (ML) (de Sousa Jabbour et al., 2018; Lasi et al., 2014). According to (Dalenogare et al., 2018) and (Fettermann et al., 2018), the incorporation of I4.0 technology is anticipated to result in notable enhancements in goods, processes, services, and overall organisational performance. Additive Manufacturing (AM) refers to a set of technological processes that enable the

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fusion of materials to fabricate products based on Three-Dimensional (3D) model data. This method often involves the sequential layering of materials, in contrast to the subtractive manufacturing approaches. In the given context, AM, previously known as 3D printing, assumes a pivotal role by actualising the conceptualisation. Various AM procedures encompass a wide spectrum of techniques, including polymerisation, jetting, extrusion, and deposition (Mercado Rivera & Rojas Arciniegas, 2020). The physical aspect of smart factories is constrained by the capabilities of the current production technologies. The inclusion of AM establishes its significance as a crucial element within the framework of I4.0. In order to meet the demands of mass customisation in the context of I4.0, it is imperative to create unconventional production techniques. Therefore, AM has the potential to emerge as a significant technology in the production of personalised goods, owing to its capacity to produce intricate items with enhanced characteristics.

Numerous studies have examined the advantages and assimilation of I4.0 technologies in conjunction with AM. Automation and streamlining of AM processes through the use of I4.0 technologies can increase productivity and efficiency (Ceruti et al., 2019; Majeed et al., 2021; Phua et al., 2022; Urhal et al., 2019). By monitoring the production process and detecting potential issues at an early stage, I4.0 technologies can enhance the quality of AM products. By increasing the efficiency of resource utilization and process optimisation, I4.0 technologies may aid in the cost reduction of AM (Guo et al., 2022). AM can be rendered more adaptable and flexible by I4.0 technologies, allowing businesses to react more rapidly to fluctuations in demand and consumer requirements (Ahuja et al., 2015). I4.0 technologies can enable businesses to develop new AM products and services, including on-demand spare parts production and mass customisation. The capacity of AM to generate intricate geometries and one-of-a-kind designs is in ideal harmony with the focus of I4.0 on widespread customisation. IoT-connected AM systems can receive customer specifications directly, enabling on-demand production of personalized products. Big Data analytics and AI algorithms can analyze design data and simulate product performance, leading to optimised designs for strength, weight, and manufacturability. AM's rapid prototyping capabilities enable quick iterations and testing of new designs, accelerating product development cycles and reducing time to market (Xiong et al., 2023). AM can produce custom tooling and manufacturing aids, such as jigs, fixtures, and moulds, tailored to specific production needs, enhancing manufacturing efficiency and accuracy. On-demand production of spare parts using AM reduces inventory costs and downtime, improving maintenance efficiency and supply chain agility.

These arguments give rise to new research questions such as: the manner in which a specific I4.0 technology enhances the AM and its operation; how can I4.0 technologies be incorporated into an AM system, and what are the obstacles to doing so? The principal aim of the study is to illustrate the potential of AM and I4.0 to enhance productivity, thereby influencing the trajectory of contemporary manufacturing. Taking into account the distinctive characteristics of these two technologies, the purpose of this research is to determine how I4.0 facilitates AM and what effect AM has on I4.0. I4.0 is built upon robotics, Artificial Intelligence (AI), and IoT; therefore, the primary focus of the review and discussion will be on the effects of these I4.0 technologies on AM systems. The main objective of this study is to examine the potential benefits and obstacles associated with the integration of I4.0 technologies in AM. These technologies encompass Big Data analytics, robotics, AI, IoT, cybersecurity, simulation techniques, blockchain, cloud computing, and system integration.

2 Impact of I4.0 Technologies on AM

2.1 Big Data and AM

Big Data is the collection, analysis, and use of large datasets. Big Data analytics is the process of using advanced analytical techniques to extract insights from Big Data datasets. Big Data analytics can be used to improve decision-making, identify new opportunities, and optimise business processes. AM systems generate a lot of data, including sensor data, process data, and part data. This data can be used to monitor and control the printing process, optimise process parameters, and develop new materials and processes. The data generated by AM systems is very varied, including structured data (e.g., temperature, pressure, build time), unstructured data (e.g., images of the printing process), and semi-structured data (e.g., CAD files). This variety of data can be used to gain a more complete understanding of the AM process and identify areas for improvement. AM systems generate data rapidly, especially when printing large or complex parts. This data needs to be processed and analyzed in real time in order to monitor and control the printing process effectively. The integration of Big Data with AM is a growing trend that has the potential to revolutionise the way products are designed and manufactured. Big Data is used to monitor and analyze AM processes in real time, identify potential problems, and make adjustments to ensure that parts are produced to the desired quality and specifications (Majeed et al.,

2021). Big Data can be used to optimise AM processes, reduce waste, and improve efficiency. Big Data can be used to identify new materials that are suitable for AM and to develop new AM processes that are faster, more efficient, and more cost-effective. Big Data can be used to create customised AM products that meet the specific needs of individual customers. The integration of Big Data with AM is still in its early stages, but it has the potential to transform the manufacturing industry (Gupta et al., 2022). As Big Data analytics tools and technologies continue to develop, manufacturers will be able to gain even more insights from their AM data and use these insights to improve their products, processes, and efficiency.

2.2 Artificial Intelligence and Machine Learning with AM

Artificial Intelligence (AI) deals with the creation of intelligent agents, which are systems that can reason, learn, and act autonomously. AI is having a major impact on the AM industry. An AI system that can generalise from its training data can be used to print these new parts without the need for additional training. Adaptation ability is important for AM because the printing process can be affected by factors such as temperature, humidity, and vibration. An AI system capable of adapting to these changes can contribute to the successful completion of the printing process. An AI system that can learn from data can adapt to printing process variations and produce high-quality parts consistently. AI can be used to generate new and innovative AM designs that would be difficult or impossible to create using traditional design methods. AI can further be used to optimise existing designs for AM (Jin et al., 2020). AI can be used to optimise AM process parameters, such as print speed, layer thickness, and support structure design. This can lead to improved part quality, reduced production time, and lower costs. With respect to quality control, AI can be used to inspect AM parts for defects in real time. The can help to ensure that parts are produced to the highest quality standards (Siyaev et al., 2023). AI can be used to develop ML models that can predict the outcome of AM processes. This information can be used to improve process optimisation, quality control, and other aspects of AM production.

Machine Learning (ML) is a subset of AI that allows computers to learn without being explicitly programmed. ML is being used to improve AM processes in a number of ways. ML can be used to identify and optimise AM process parameters, such as laser power, scan speed, and layer thickness. The result can lead to improved part quality, reduced production time, and lower costs (Knofius et al., 2021). ML can be used to monitor

AM processes in real time and identify defects. The result can help to ensure that parts are produced to the highest quality standards (Razvi et al., 2019). ML can be used to predict when machines need maintenance, which help avoid unplanned downtime. Product design: ML can be used to design products that are specifically optimised for AM. The result can lead to lighter, stronger, and more efficient parts.

2.3 Cloud Computing and AM

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction. Cloud-based AM design and simulation software allows users to design and optimise AM parts without having to invest in expensive hardware. This makes AM more accessible to a wider range of users, including small businesses and individuals. Cloud-based AM manufacturing services allow users to print parts on demand without having to own their AM machines. Cloud-based AM data management and analytics tools can help users to collect, store, and analyze data from their AM processes. Cloud computing allows businesses to access AM resources on demand, without having to invest in and maintain their AM machines. The service can be a cost-effective solution for businesses that need to produce AM parts on a limited basis (Wang et al., 2019). Cloud computing is scalable, meaning that businesses can scale up or down their AM resources based on their needs. Such an arrangement can be helpful for businesses that experience seasonal fluctuations in demand. Cloud computing can facilitate collaboration between different stakeholders in the AM process, such as designers, engineers, and manufacturers. Such an arrangement can help to improve the efficiency and effectiveness of the AM process. Cloud computing can be used to collect and analyze data from AM processes (Baumann & Roller, 2017). Such an arrangement data can be used to improve process optimisation, quality control, and other aspects of AM production.

2.4 Cyber Security and AM

Cyber security is the practice of protecting systems, networks, and data from unauthorised access, use, disclosure, disruption, modification, or destruction. AM systems and data are increasingly vulnerable to cyberattacks. This fact is because AM systems are often connected to the internet and contain

sensitive data, such as Computer Aided Design (CAD) files and process parameters. Attackers can steal CAD files, disrupt AM production, and exfiltrate AM data. The integration of cybersecurity with AM is essential to protect AM processes and products from cyberattacks. AM is a complex process that involves several of digital technologies, including design software, slicing software, and AM machines (Brandman et al., 2020). This makes AM vulnerable to a variety of cyberattacks, such as data theft. Attackers can steal AM data, such as design files, slicing profiles, and process parameters, and this data can be used to counterfeit AM products or to sabotage AM processes. The possibility of malicious code injection cannot be ruled out. Attackers can inject malicious code into AM software or into the firmware of AM machines. This code can be used to alter AM designs, disrupt AM processes, or even cause AM machines to fail. That apart Denial-of-Service (DoS) attacks can be possible wherein attackers can launch DoS attacks against AM systems to make them unavailable or to degrade their performance. Such actions can disrupt AM production and lead to financial losses (Bridges et al., 2015).

2.5 Blockchain and AM

Blockchain utilises distributed ledger technology to securely and transparently record transactions and track assets. Blockchain has the potential to revolutionise the AM industry by improving the traceability, security, and efficiency of AM supply chains (Lupi et al., 2023). Blockchain can be used to track and trace AM parts throughout their lifecycle, from design to manufacturing to delivery. Such an approach can help to ensure the authenticity and quality of AM parts, and it can also help to identify and prevent counterfeiting. Blockchain can be used to secure AM data, such as design files, process parameters, and material data. This can help to protect AM data from unauthorised access and tampering. Blockchain can be used to automate AM payments (Eskindarov et al., 2019). These benefits can help to reduce the costs and risks associated with AM transactions. Blockchain can be used to create AM marketplaces where buyers and sellers can connect and transact securely. These developments can help to make AM more accessible and affordable for businesses of all sizes. Blockchain can have a significant impact on the AM supply chain by improving transparency, traceability, and security. Blockchain can provide transparency into the AM supply chain by tracking the movement of parts and materials from raw materials to finished products. Blockchain can also improve traceability in the AM supply chain by tracking the provenance of parts. Blockchain can also improve security

in the AM supply chain by making it more difficult to tamper with data or products.

2.6 Digital Twin and AM

A digital twin is a virtual representation of a physical object or system. It is a digital replica that can be used to monitor, simulate, and optimise the physical object or system. Digital twins are used in a variety of industries, including AM, to improve product design, process optimisation, and quality control. This can help to identify and eliminate design flaws and to optimise the design for performance and manufacturability. Digital twins serve various purposes in the realm of AM. They facilitate the prediction of the performance of AM parts during service, monitor the real-time AM manufacturing process, optimise production by pinpointing inefficiencies, and simulate AM techniques like laser powder bed fusion and directed energy deposition (Mandolla et al., 2019). These features can help to ensure the quality of AM parts and to reduce the need for trial-and-error experimentation, hence cost and response time. Digital twins can be used to monitor AM production in real time. Such monitoring can help to identify and address problems early on before they lead to defective parts or downtime.

2.7 Robotics and AM

Robotics and cobotics are playing an increasingly important role in AM. Robotics is the use of robots to perform tasks that are difficult, dangerous, or repetitive. Cobots, or collaborative robots, are a type of robot that is designed to work safely alongside humans. Cobots are typically smaller and more agile than traditional robots, and they can be equipped with sensors that allow them to interact with humans without causing harm. Cobots can be used to automate a variety of AM tasks that are currently performed by humans, such as: assembling AM parts, packing and shipping AM parts, providing quality control and inspection. The integration of robotics with AM is a rapidly developing field with the potential to revolutionise the way AM products are designed, manufactured, and inspected. Robots can be used to load and unload AM machines, which can free up human workers to focus on other tasks. These devices can also improve the efficiency and productivity of AM production. Robots can be used to transport AM parts between different AM machines and other workstations (Urhal et al., 2019). This process can help to reduce the risk of damage to AM parts and to improve the efficiency of

AM production. Robots can be used to perform post-processing operations on AM parts, such as support structure removal and surface finishing (Sheydaeian & Toyserkani, 2017). These actions can help to improve the quality and appearance of AM parts. Robots can be equipped with sensors to inspect AM parts for defects. These features can help to ensure the quality of AM parts and to reduce the need for manual inspection.

2.8 Augmented Reality and AM

AR is a technology that superimposes a computer-generated image on a user's view of the real world, thus providing a composite view. The integration of AR with AM is a rapidly developing field with the potential to revolutionise the way AM products are designed, manufactured, and inspected. AR is a technology that superimposes a computer-generated image on a user's view of the real world, thus providing a composite view (Etonam et al., 2019). AR can be used to represent 3D AM processes and designs. The method can assist designers in identifying prospective issues and gaining a deeper understanding of their designs. During AM production, AR can also be used to train AM operators and provide real-time assistance (Xiong et al., 2023). AR is capable of detecting flaws in AM components, including porosity, cracking, and deformation. AR can also be used to verify that AM components adhere to specifications and to measure their dimensions. AR can assist in the assembly process of AM elements by providing detailed instructions to employees and helping detect and correct assembly errors. AR has the potential to enhance AM product development in numerous ways. By utilizing AR to visualize AM components in their actual environments, designers are able to determine the parts' functionality and fit. AR has the potential to offer direction and support throughout the AM manufacturing procedure. AR can be used, for instance, to superimpose instructions onto the AM machine or deliver instantaneous feedback regarding the manufacturing procedure. AR is capable of detecting defects in AM components. AR, for instance, can detect discrepancies by comparing the actual component to the CAD model. AR has the potential to serve as a guiding and supporting tool throughout the AM manufacturing process, thereby enhancing efficiency and minimizing defects. These improvements may result in increased productivity and decreased discard rates, both of which contribute to decreased manufacturing expenses.

2.9 Simulation and AM

Simulation is the process of creating a virtual model of a real-world system or process and then running experiments on that model to predict how the system or process will behave. Simulation has some unique features for AM like predictive power, scalability, and flexibility. Simulation can be used to identify and eliminate potential problems in a product or process before it is built or implemented. This can improve the quality of the product or process and reduce the risk of failures. Simulation can optimise product designs and processes to reduce costs. For example, simulation can be used to identify the most cost-effective way to manufacture a product or to reduce the energy consumption of a process. Simulation can identify and eliminate bottlenecks in a process, which help improve throughput and reduce response time. For example, simulation can be used to optimise the layout of a factory or to design a new production line. The integration of simulation with AM is a powerful tool that can be used to improve the design, development, and production of AM parts. Simulation can be used to predict the performance of AM processes, such as laser-based powder bed fusion and directed energy deposition (Rodgers et al., 2017). This can help to identify and optimise process parameters, such as laser power, scan speed, and layer thickness. Simulation can be used to identify potential AM defects, such as porosity, cracking, and warping. These findings can help to develop strategies to prevent these defects from occurring. Simulation can be used to improve the design of AM parts for manufacturability and performance. For example, simulation can be used to identify areas of a part that are likely to experience high stress or deformation, and to modify the design accordingly (Singh et al., 2023).

2.10 Internet of Things and AM

IoT devices are physical objects that are embedded with sensors and software that allow them to collect and transmit data. IoT devices are increasingly equipping AM machines, enabling remote monitoring and control (Wang et al., 2019). It is possible to remotely monitor AM machines with IoT devices, which can aid in the early detection and resolution of issues. The outcome may result in decreased downtime and increased output. By using IoT data to forecast the maintenance requirements of AM machines, one can prevent unforeseen idleness and increase the operational longevity of such equipment (Haghnegahdar et al., 2022). We can optimise AM processes like laser power, scan speed, and layer thickness by utilising IoT data. Such

optimization may result in cost savings and enhanced quality. In real time, IoT devices can be used to inspect AM components for defects. The results may aid in enhancing the quality of AM components and decreasing the requirement for revision. Data collection from products and processes can be facilitated by IoT devices at every stage of the product development lifecycle. An analysis of this information can subsequently detect trends, patterns, and anomalies. We can use this data to improve the design of products and processes, reduce errors, and improve quality. Additionally, IoT devices can be used to gather longitudinal data regarding the effectiveness of processes and products. Subsequently, this information can be used to construct potential maintenance models capable of foreseeing potential issues. These actions may result in decreased maintenance and outage expenses. The use of IoT devices enables effective collaboration and communication among diverse product development teams. For instance, IoT devices can facilitate data sharing among the design, engineering, and manufacturing departments. This has the potential to enhance the efficacy and efficiency of the product development procedure.

To summarise the above, I4.0 components, including digital data transfer, simulations, IoT, Big Data, cloud computing, and cyber security, can improve the AM process considerably. AR is used to discover defects in a product by comparing it to its true equivalent during manufacture. To evaluate product performance, AM can use the Computer-Aided Design (CAD) file to run numerical simulations. The file might enable the creation of a digital replica for product construction or attribute analysis. ML and AI algorithms can increase digital twin performance and produce more realistic results. To reduce cyberattack threats, data must be protected during digital transmission over the Internet. Protective measures are necessary to avoid product damage and AM system damage. Large amounts of data must be processed and sent to suitable channels. Big Data analytics, AI, and ML can improve AM quality. Digital twins can forecast issues accurately using complex simulations. Thus, the seamless integration of all I4.0 enabling technologies for AM specialised applications is a field of progress. The application of various I4.0 technologies in the AM process covered in the literature is listed in Table 1.

2.11 Integration of I4.0 Technologies with AM

The integration of I4.0 technologies with AM is a rapidly developing field with the potential to revolutionise the manufacturing industry. The proposed integration model in Fig. 1 provides a framework connecting the

Table 1 Application of I4.0 in AM process

#	AM process parameters	Application of I4.0 technologies
1	Process Optimisation	Big Data, AI, ML, IoT, Digital twin
2	Quality control	Big Data, AI, ML, IoT, AR, Robotics
3	Machine maintenance	ML, IoT
4	Product design	ML, AR
5	Generative design	AI, ML
6	Data analytics	Cloud computing
7	Scalability	Cloud computing
8	Collaboration	Cloud computing
9	Access to AM resources	Cloud computing
10	Intellectual property	Cyber security
11	Disruption in production	Cyber security
12	Security risk	Cyber security
13	Track and trace AM parts	Blockchain
14	Secure AM data	Blockchain
15	Automate AM payments	Blockchain
16	Create an AM marketplace	Blockchain
17	Simulate AM process	Digital twin
18	Predict AM defects	Digital twin, Simulation
19	Monitor AM production	Digital twin
20	Transporting AM machines, parts	Robotics
21	Post-processing AM parts	Robotics
22	Visualisation of AM design	AR
23	Assembly of AM parts	AR
24	Predict AM process performance	Simulation
25	Improve AM part design	Simulation
26	Remote monitoring	IoT
27	Predictive maintenance	IoT

Source Authors

different technologies and processes involved in these two fields in a way that maximises the benefits and minimises the challenges. The integration model for I4.0 and AM is to divide it into four layers: process layer, data layer, analytics layer, system layer (Cañas et al., 2021). The process layer involves integrating the different stages of the AM process, such as design, printing, post-processing, and quality control. This can help improve the efficiency and effectiveness of the AM process. The data layer involves connecting the various data sources in the AM process, such as sensors, machines, and design software. This data can then be used to monitor the AM process, identify problems early on, and make predictions about future performance. The analytics layer is responsible for analyzing the data collected in the data layer to identify patterns and trends, and to make predictions about the AM process. The system layer involves integrating the AM process with other

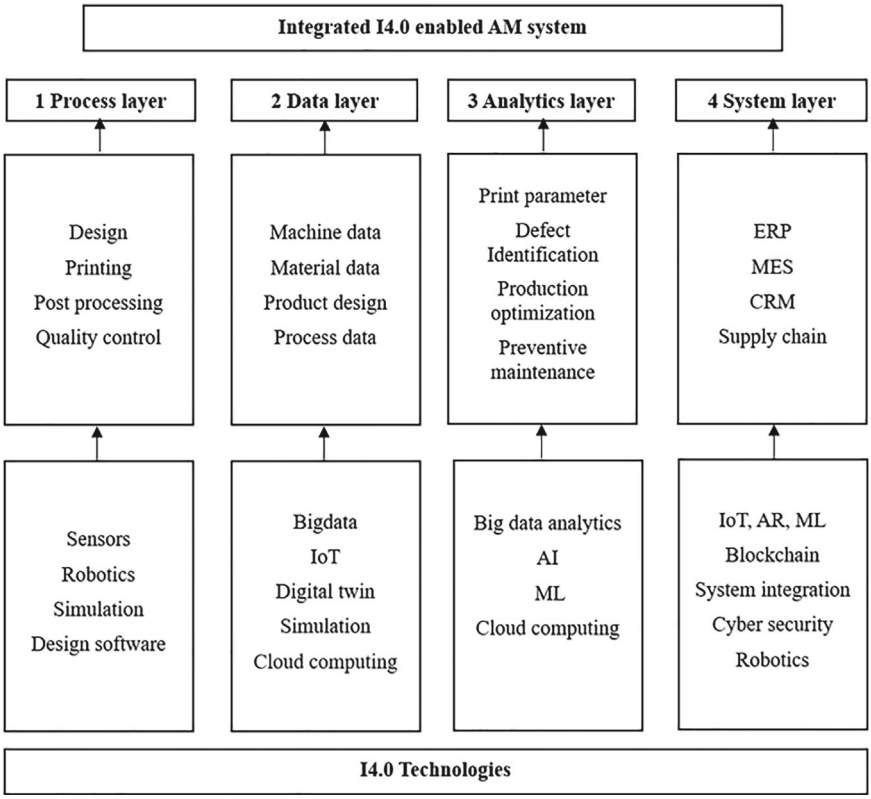

Fig. 1 Proposed integration model for I4.0 and AM (*Source* Authors)

systems in the manufacturing enterprise, such as ERP, MES, and CRM systems. These systems can help to improve the visibility and traceability of AM products.

3 Challenges of Integrating I4.0 Technologies with AM

Though there are many benefits while integrating I4.0 technologies with AM systems, there are also some challenges to integrating I4.0 technologies with AM, including;

- *Cost:* I4.0 technologies can be expensive to implement. This can be a barrier for small and medium-sized enterprises.

- *Security:* I4.0 technologies can be vulnerable to cyberattacks. Manufacturers need to take steps to protect their data and systems from cyberattacks.
- *Skills shortage:* There is a shortage of skilled workers with the necessary skills to operate and maintain I4.0 technologies. This can make it difficult for manufacturers to implement and integrate these technologies.

There are a number of challenges that need to be addressed before Big Data can be fully integrated with AM. One challenge is the lack of standardisation in AM data. Different AM machines generate different types of data, and there is no standard format for storing and sharing this data. This makes it difficult to develop Big Data applications for AM. Another challenge is the cost and complexity of Big Data analytics. Big Data analytics requires powerful computers and specialised software. This can be a significant investment for small and medium-sized businesses (Ahuja et al., 2015; Majeed et al., 2021).

We must address these challenges before fully integrating AI with AM. One challenge lies in the scarcity of training data. AI models need to be trained on large datasets of data to be effective. There is currently a shortage of training data for AM, which is limiting the development of new AI applications. Another challenge is the complexity of AM processes. AM processes are complex and involve several of variables. Such complexity makes it difficult for AI models to accurately predict the outcome of AM processes (Liu et al., 2022).

One of the biggest challenges to integrating ML with AM is the lack of standardised data. Different AM machines generate different types of data, and there is no standard format for storing and sharing this data. This problem makes it difficult to develop ML models that can be applied to a wide range of AM machines. Another challenge is the cost and complexity of ML software. ML software can be expensive to purchase and maintain, and it can be difficult to use for people who are not familiar with ML (Qi et al., 2019).

One of the primary challenges to integrating cloud computing with AM is the security of AM data. AM data is often confidential and sensitive, and businesses need to be confident that their data is secure when it is stored and processed in the cloud. Another challenge is the reliability of cloud computing services. Cloud computing services can be unreliable at times, such problems this can lead to disruptions in AM production (Haghnegahdar et al., 2022).

The integration of cybersecurity with AM is a complex and challenging task. There is currently no standard format for AM data, which can make it difficult to develop cybersecurity solutions that are compatible with a wide range of AM systems. AM systems are complex and involve several of different components, including software, hardware, and materials. This complexity can make it difficult to identify and mitigate all security vulnerabilities. Connecting AM systems to the internet and other networks often heightens the risk of cyberattacks. Many AM users are not aware of the cybersecurity risks associated with AM. This lack of awareness can make it difficult to implement effective cybersecurity measures (Corallo et al., 2023).

There is currently no standard format for AM data, which can make it difficult to develop blockchain-based solutions that are compatible with a wide range of AM systems. AM systems are complex and involve several of different components, including software, hardware, and materials. This complexity can make it difficult to integrate blockchain technology with AM systems. Blockchain technology can be expensive to implement and maintain. Blockchain technology is still in its early stages of development, and it is not yet scalable to meet the demands of large-scale AM production (Lupi et al., 2023).

AM systems are complex and involve several of different components, including software, hardware, and materials. This complexity can make it difficult to create accurate digital twins for AM systems. Digital twins require large amounts of data to be effective. This data can be difficult and expensive to collect. There is currently no standard format for AM data. These limitations can make it difficult to develop digital twins that are compatible with a wide range of AM systems (Fuller et al., 2020).

This complexity of AM systems can make it difficult to integrate robots with AM systems. Robots can be expensive to purchase and maintain. There is a shortage of skilled workers who can program and operate robots (Dörfler et al., 2022).

Despite the benefits, there are also some challenges associated with the integration of AR with AM. AR devices and software can be expensive. AR systems can be complex to set up and use. There is a shortage of skilled workers who can develop and maintain AR systems (Ceruti et al., 2019b).

AM processes are complex and involve several of different factors, such as laser power, scan speed, layer thickness, and material properties. This complexity can make it difficult to develop accurate simulation models of AM processes. Simulation models require large amounts of data to be effective. This data can be difficult and expensive to collect. There is currently no standard format for AM data. These limitations can make it difficult to

develop simulation models that are compatible with a wide range of AM machines and materials (Michopoulos et al., 2018).

AM systems and IoT devices are complex and integrating them can be a challenging task. IoT devices can be vulnerable to cyberattacks, and it is important to implement security measures to protect AM systems and data from cyberattacks. IoT devices and the infrastructure needed to support them can be expensive. There is currently no standard format for AM data, which can make it difficult to integrate IoT devices with AM systems (Haghnegahdar et al., 2022). Challenges for integrating I4.0 with AM are summarised in Table 2.

Table 2 Challenges for integration of I4.0 with AM

#	I4.0 technologies	Challenges for integration with AM
1	Big Data	Standardisation in AM data, cost, and complexity of data analytics
2	AI	Shortage of training data for AM, complexity of AM process
3	ML	Standardisation in AM data, cost and complexity of ML software
4	IoT	Security and privacy, connectivity, data management, compatibility, and standardisation
5	AR	The challenges include a lack of skilled workers, high costs, and the complexity of additive manufacturing systems
6	Robotics	The challenges include a lack of skilled workers, high costs, and the complexity of additive manufacturing systems
7	Cloud computing	Security of AM data, reliability of cloud service
8	Cyber security	Lack of standardisation, complexity to identify and mitigate all security vulnerabilities, internet connectivity, limited awareness
9	Blockchain	AM system complexity can make it difficult to integrate blockchain technology, lack of standardisation, cost, scalability for large-scale production
10	Digital twin	AM system complexity makes it difficult to create accurate digital twins of AM systems, difficulty and expensive data collection, and standards for compatibility
11	Simulation	Complexity of AM processes to develop an accurate simulation model, requirement of a large amount of data, compatibility with a large no of AM machines and materials

Source Authors

4 Research Gaps and Future Scopes

Though there is lots of positivity associated with AM, there are problems as well in terms of the cost, quality, and ecosystem. AM needs people with the right skills to run these tools. Another problem is that these machines are costly. I4.0 will help make industry more technologically advanced in the future and set the stage for the new industrial revolution. This technology fosters a customer-centric market that swiftly and effortlessly adjust to their requirements. With the help of AM, I4.0 is a new level of production efficiency and the improvement of the whole manufacturing system around the world. The integration of I4.0 technologies with AM is a rapidly developing field with many opportunities for future research. There are a few specific areas where further research is needed:

Research should focus on the development of new I4.0 technologies specifically designed for AM. For example, new sensors could be developed to monitor the AM process in more detail, or new AI algorithms could be developed to optimise the printing process (Jan et al., 2023). I4.0 technologies should be integrated with the entire AM supply chain. This procedure includes integrating I4.0 technologies with design, materials, post-processing, and quality control (Lupi et al., 2023).

I4.0 enables the development of new business models for AM. For example, companies could use I4.0 technologies to offer on-demand custom manufacturing or to provide predictive maintenance services for AM machines (Battaglia et al., 2023; Bootz et al., 2022). The study aims to evaluate the social and economic impact of integrating I4.0 technologies with AM. This includes research on the impact of these technologies on employment, productivity, and innovation (Agrawal et al., 2022).

Here are some specific research questions that could be explored in AM areas:

1. Researchers should focus on developing new I4.0 technologies specifically for AM
   - What are the most important process parameters to monitor in AM?
   - How can sensors be used to monitor these process parameters more effectively?
   - How can AI be used to optimise the AM process?
2. Integration of I4.0 technologies with the entire AM supply chain
   - How can I4.0 technologies be used to integrate design, materials, post-processing, and quality control with AM?

- How can I4.0 technologies be used to create a more efficient and sustainable AM supply chain?

3. The development of new business models for AM is enabled by I4.0
   - What are the most promising new business models for AM enabled by I4.0?
   - How can companies use I4.0 technologies to offer on-demand custom manufacturing or predictive maintenance services for AM machines?
4. Evaluation of the social and economic impact of the integration of I4.0 technologies with AM
   - What is the impact of the integration of I4.0 technologies with AM on employment, productivity, and innovation?
   - How can the negative impacts of this integration be mitigated?

The research in these areas will help to accelerate the adoption of I4.0 technologies in AM and to unlock the full potential of this technology.

5 Conclusion

There are many benefits to integrating I4.0 technologies with AM. I4.0 technologies can help to automate tasks such as machine setup and process planning. These capabilities can free up workers to focus on more complex tasks and can help to improve the overall efficiency of the production process. I4.0 technologies can be used to monitor and control AM processes in real time. Such monitoring can help to identify and correct problems early on, which can lead to improved product quality. I4.0 technologies can help to reduce costs by optimizing the production process and reducing waste. For example, AI can identify the most efficient way to print a particular part, which can help to reduce the amount of material used. I4.0 technologies can produce custom products on demand. This can help companies to meet the needs of their customers' more effectively and reduce inventory costs. I4.0 technologies can make AM more sustainable by reducing waste and energy consumption.

Adding I4.0 technologies to AM is changing the way things are made and opening the door to a new age of customised, efficient, and environmentally friendly production. This convergence is transforming the boundaries of product design and manufacturing. It's making way for a future where creativity and flexibility are the most important parts of industrial production. Researchers, schools, and businesses all need I4.0 to work. AM is a key

technology for this fourth revolution, and it is a key reason for the new development's success. AM is a new technology that changes many uses and solves many problems related to I4.0. Global production companies can now use AM technologies, which offer new hybrid methods and lower costs. Companies are now contemplating about how to use these technologies to make I4.0 a reality, and AM is in a favourable situation to lead the way. For I4.0 to become a reality in an exact and adaptable way, AM is a key part. Using design tools, 3D printers, and processes together makes them more efficient, which is important for implementing I4.0. I4.0 is the new way of doing things for smart workplaces of the future, and AM helps it reach its full potential. I4.0 and AM are coming together to change the future of production in many ways. AM enables production to take place near the product's intended use. This cuts down on shipping costs and lets people customise the product in their own area. AM's capacity to create intricate shapes with minimal waste and its utilisation of recycled materials contribute to sustainable production practices. When you combine the freedom of design that AM gives you with the data-driven insights of I4.0, you create new goods that work better and have more features. A lot of data analysis and AI can help companies ensure that the designs of AM goods are strong, light, and easy to make.

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Part VI

Digital Delivery and Use Capabilities for Emerging Technologies in Supply Chains

24

The Role of UAV Automation in Inbound Logistics and Warehousing

Cisem Lafci, Yalcin Berberoglu, and Yasanur Kayikci

1 Introduction

Unmanned aerial vehicles (UAVs), commonly known as drones, are powered unmanned vehicles that may be flown remotely or autonomously via planned software-controlled routes of flight that operate in conjunction with GPS devices and sensors fitted aboard (Renduchintala et al., 2019). UAVs, which were originally utilized for military objectives, have grown in popularity, with technical breakthroughs, and increasing industrial and research awareness, broadening their applications (Alladi et al., 2020; Alwateer et al., 2019). The major goal in designing the UAVs was to create a low cost, basic, and flexible first model that may be readily adaptable to numerous purposes and circumstances (Fernández-Caramés et al., 2019). Despite its original usage, the integration of UAVs in industrial practices has garnered an increasing amount of interest in different domains, such as the logistics and retailing industries and so on (Yurek & Ozmutlu, 2018). UAVs or drones, have lately

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acquired popularity when many large online retailers, including Amazon, DHL, Walmart, and Google incorporated drones into their last-mile delivery processes (Macrina et al., 2020; Yoo et al., 2018). Due to enhanced licenses provided by national authorities such as Australia, Singapore, Iceland, and Switzerland, the logistics business is leading in UAV use-cases, rather than the Urban Air Mobility sector (Hader & Baur, 2020). Hence, many companies have increasingly adopted UAVs in their processes for package delivery in recent years, allowing them to carry products directly from distribution facilities to consumer locations (Moshref-Javadi et al., 2021).

The reason behind the adoption and integration of UAVs into the industrial sectors is that these UAVs are becoming increasingly affordable, simple to use, and simple to control (Jordan, 2019) in terms of maintaining various smart operations efficiently. The usage of all these aerial delivery services is beneficial for logistics company workers who are trained in business-to-business (B2B) or business-to-consumer (B2C) operations (Kornatowski et al., 2018). Since 2013, after the usage of Amazon package delivery, logistics has been regarded as one of the most inventive use-cases of UAVs (Cristiani et al., 2020). In the logistics sector, UAVs are commonly used for last-mile delivery of comparatively small products, order picking in automated facilities, and semi-automated physical inventory (Ferraro et al., 2023; Remondino & Zanin, 2022). Due to inbound and outbound procedures to and from warehouses, as well as internal and external materials handling and transport operations, which are all part of logistics (Zijm et al., 2019), UAV deployments in warehouses have lately increased dramatically. Drones are being utilized in warehouses to do automated inventory checks, addressing problems such as worker safety, mistakes, operating expenses, and time (Chauke et al., 2023). They may scan inventory data manually or automatically, allowing for planned cycle audits (Chauke et al., 2023). Automated or semi-automated systems that employ drone-scanning technology can improve warehouse operations in many ways such as inventory inspections and stock taking, as well as storage procedures, minimize human injury hazards, and eliminate downtime (Đurić et al., 2018). Furthermore, data collecting by drone in conjunction with a sophisticated information system enables improved data management and distribution while maintaining dependability and correctness (Rhiat et al., 2022).

Therefore, UAVs are beginning to draw traction in warehouse and inventory management due to their automation of operations like as inventory tracking, stock counting, and data collecting, which improves operational efficiency while eliminating human involvement and mistakes. Especially with robotics advancements, businesses started to create new mechanisms

towards order planning, management of inventory, the delivery process, and so on (Euchi, 2021). The usage of UAVs in logistics such as delivery/transportation services, distribution, and storing and warehousing operations has only begun, implying that we can anticipate widespread adoption and implementation of autonomous systems in supply chain activities in the future (Škrinjar et al., 2019). For instance, inventory inspection utilizing UAVs at warehouses provides various benefits over manual inspection by human employees, including the ability to reduce time and money as compared to employing high-reach forklifts, as well as minimize the potential harm possible to workers performing manual barcode scanning on a rapidly moving high-reach forklift (Kwon et al., 2020). It may be inferred that, in addition to optimization benefits, the employment of UAVs has an impact on worker safety, job quality, error reduction, and other factors (Škrinjar et al., 2019).

The major purpose of this book chapter is to give an in-depth and insightful examination of the function of UAVs or drones in inbound logistics operations as warehouse management. Therefore, this chapter tries to highlight the multiple advantages of incorporating UAV automation, ranging from cost savings and increased efficiency to improvements in inventory tracking accuracy and worker safety in warehouse management. Furthermore, the study will dive into UAV automation capabilities and their potential to transform warehouse management by decreasing human interaction and mistakes. This research will not only give a detailed analysis of the existing scene but will also analyze future implications and trends connected to UAVs in the logistics industry, paving the way for a more efficient and technology-driven approach to supply chain operations.

The following is the rest of this book chapter: The second section largely explored and offered a variety of research for subjects relating to inbound logistic operations, most notably warehouse activities. The third segment looked at the possible advantages and downsides of using UAVs or drones in warehouse inventory management. The fourth section mostly highlighted the best practices of companies that used UAVs in their supply chain operations. Finally, the last section ended with a summary of the study and a presentation of the essential topics to focus on, limits, and future research suggestions.

2 Literature Review of UAV Applications in Inbound Logistics Operations

A few publications have been published that investigate the viability of UAV systems in warehouses for inventory management. For instance, Ong et al. (2007) used a preliminary simulation for feasibility visualization to explore the design and application hurdles of a Radio-frequency identification (RFID)-reader-equipped multi-UAV system for warehouse automation. Similarly, Bae et al. (2016) worked on the feasibility of UAV systems and proposed a novel approach to monitoring inventory in open storage yards using the technologies of UAV and RFID which reduces costs and mismatching between real and digital inventory and has been tested with a prototype system to demonstrate feasibility. Derpich et al. (2018) also examined the feasibility of using drones in a manufacturing factory for multiple transports, analyzing their low autonomy and load capacity while minimizing energy consumption.

Guérin et al. (2016) present a novel warehouse inventory scheme by adopting an experimental study related to Unmanned Ground Vehicle (UGV) and a UAV cooperatively. Also, an effective drone-assisted inventory management technique is put forth by Cho et al. (2018). It uses an infrared camera to automatically identify target barcodes, which lowers power consumption and allows for effective drone path planning. In addition to that, the use of micro aerial vehicles (MAVs) is boosting indoor applications. In this context, a high-performance autonomous inventory MAV is developed by Beul et al. (2018) for warehouse operations, detecting stock, and avoiding obstacles. Xu et al. (2018) created algorithms for autonomously extracting barcodes from video data in warehouse logistics utilizing three techniques (Hough transform-based algorithm and Harris corner detector, quick histogram strategy, and connectivity and geometry features). Using a novel model in a known indoor environment, Li et al. (2018) studied universal path planning for indoor drone navigation, presenting safe shortest, and safe least-cost pathways for optimal drone navigation. Tubis et al. (2021) found a risk assessment technique for intangible services, notably UAV systems in warehouse operations, that employs fuzzy logic to detect and prioritize bad occurrences.

Another type of issue with UAV adoption in warehouse management is incomplete information exchange, cargo statistics disorder, and manual inventory, Chen et al. (2020) introduced a warehouse management system utilizing UAV depending upon digital twin and 5G technology. The study of Malang et al. (2023) also reviewed UAV adoption in warehouse management

and revealed hardware and software as the most influential factors. Their findings provided insights for decision-makers and practitioners on effective UAV-based warehouse management, enhancing efficiency and effectiveness. The study by Shen et al. (2021) examined drone-to-warehouse task, allocation rules, and handling of batteries in a multi-warehouse drone delivery system. Olivares and Córdova (2015) aimed to simulate and evaluate the operation of a drone fleet in manufacturing plants, focusing on the impact of inventory reduction and changes in productive resources capacity. In addition, Sorbelli et al. (2019) explored the adoption and usage of drones in an automated warehouse picking system to compare the efficiency of the drone system with the traditional methods. According to this system, the drone collects items from each customer order, locating the cart at a location that minimizes the drone's distance. Bartoli et al. (2019) proposed a drone delivery solution for small cargo last-mile logistics using mixed EM-grids, which stands for Euclidean and Manhattan distance metrics. The system minimizes distances between destinations and warehouses.

Companik et al. (2018) proposed a framework for why supply chain executives must invest in automated machines such as robotics, UAV technology, etc. Because Industry 4.0 is revolutionizing manufacturing by utilizing emerging technologies like UAVs for automation and traceability, Fernández-Caramés et al. (2019) demonstrated a UAV-embedded system for storing and validating inventory data utilizing blockchain technology and distributed ledgers. The system's performance in an industrial warehouse demonstrates its ability to obtain fast inventory data and estimate item position. Otto et al. (2018) also reviewed the existing literature on optimizing UAVs in terms of civil applications, highlighting their potential cost savings, potential disaster management benefits, and potential modelling approaches, offering insights for future research.

A growing number of different studies are looking into the effectiveness of delivery methods that use drones in warehouse and inventory management by adopting UAVs and related technologies. For instance, Kwon et al. (2020) established an autonomous inventory inspection system for warehouse settings by merging visual SLAM algorithms and outlier elimination into a low-cost multi-sensor fusion framework. Radácsi et al. (2022) also presented a model for controlling inventory in a warehouse logistics using a drone for imaging and transmission. Without external orientation tools, the model involves pre-processing, real-time drone movement, and post-processing for quick response (QR) code identification and inventory control. Yoon et al. (2021) generated a position estimate approach for drones and

items depending upon a QR code segmentation model, allowing for effective warehouse management in storage yards by calculating drone location and three-dimensional (3D) coordinates using ground QR codes. Similarly, Vanhie-Van Gerwen et al. (2021) employed the flexible sensor fusion platform OASE to examine the influence of many sensors on 3D indoor position accuracy in order to present a feasible solution for three indoor use scenarios. Furthermore, Manjrekar et al. (2021) studied an affordable drone-based inventory management approach that allows for accurate tracking of warehouse items using Wi-Fi and camera modules, as well as efficient coverage via marker-based navigation. Liu et al. (2021) offered a task-planning model for tobacco sector organizations that use drones equipped with RFID readers to manage inventories. They suggested a hybrid approach based on lion swarm optimization to overcome location information loss in traditional differential evolution algorithms, which they tested with data from the tobacco industry warehouse.

To present, UAV applications have been the primary focus of large outdoor warehouses for inventory management, infrastructure development, localization, and indoor navigation, among other things. Therefore, inventory management is seen as critical in supply chains particularly in big outdoor warehouses (Benes et al., 2022), thus, they offered an RSSI-based approach for identifying products at least 2 metres away, overcoming obstacles in detecting the precise location of commodities. Another study claimed that large warehouses could employ autonomous indoor drones, but doing so would require precise localization techniques. Thus, Macoir et al. (2019) made a battery-powered infrastructure anchor node solution for Ultra-Wideband (UWB) accessible as open-source software by updating the rate of up to 2892 Hz and an average current consumption of 27 mA. A related study by Kalinov et al. (2020) also presented a UAV-based robotic system that uses convolutional neural networks for real-time barcode detection and scanning to lower scanning errors and stocktaking time to improve localization by optimizing flight paths according to barcode locations in warehouses with low light levels. A generic architecture for UAV-embedded inventory control in large warehouses is put forth by Cristiani et al. (2020), who also addressed technical concerns such as restricted flight autonomy, package identification, and indoor navigation. Moreover, Fontaine et al. (2021) suggested deploying self-driving drones in Industry 4.0 automation to identify warehouse assets using RFID tags and machine learning for efficient, precise tracking that is adaptive to different settings and cost-effective RFID localization. Similarly, Li et al. (2021) presented a passive RFID localization scheme using

UAVs for warehouse inventory management. To find tagged items laterally, this approach uses phase, received signal strength indicator (RSSI), and readability.

3 Potential Benefits and Disadvantages of UAVs or Drones in Warehouse Management

UAVs are predicted to have various benefits over traditional vehicles, including continuous and rapid speed of travel (Thiels et al., 2015), minimum physical road infrastructure requirements (Foth, 2017), direct transport, and no interaction with congested roads and traffic (Moshref-Javadi et al., 2021; Thomalla, 2017). They appear to have the ability to reduce time and money over traditional modes of transportation, as well as make relief products supplied to cut-off locations conceivable in the first place (Rabta et al., 2018; Tiwari & Dixit, 2015). Because they are not impacted by road traffic, they may move at a continuously high speed, resulting in shorter delivery times to clients (Joerss et al., 2016; Moshref-Javadi et al., 2021). The usage of UAVs is not only confined to road infrastructure and may be employed in locations with a weak road network (Ackerman & Strickland, 2018). UAVs have also shown to be incredibly useful in a wide range of industries, involving remote sensing, tracking in real-time, disaster management, border, and crowd surveillance, army usage, product delivery, precision agriculture, infrastructure examination, and entertainment and media (Fernández-Caramés et al., 2019; Shakhatreh et al., 2019; Hassanalian & Abdelkefi, 2017). UAVs provide considerable benefits to companies by being economically inexpensive, readily controlled and deployed, precisely and correctly manoeuvrable, and efficient surveillants (Malang et al., 2023). The benefits of using UGVs in city or industrial contexts consist of, such as increased security and safety (Azmat & Kummer, 2020; Azmat et al., 2019), reduced congestion (Gkartzonikas & Gkritza, 2019; Wintersberger et al., 2019), higher fuel economy (Azmat et al., 2018), fewer emissions (Mahmassani, 2016; Milakis et al., 2017), more affordable ownership costs (Kyriakidis et al., 2015; Litman, 2023), eliminating driving time constraints (Bierstedt et al., 2014) and so on.

UAVs are seen as critical technologies for smart factories, warehouses, and operations since they allow for the fulfilment of repetitive and dangerous tasks with minimal or no human involvement or supervision (Fernández-Caramés et al., 2019). In addition to that they serve a variety of functions in

warehouse operations literature, including real-time inventory tracking, stock retrieval and restocking, security and surveillance, maintenance and inspection, environmental monitoring, workflow optimization, order picking and sorting, emergency response, and transportation, all of which contribute to increased efficiency, safety, and overall operational effectiveness within warehouse environments. For instance, in terms of the particular advantages of UAVs, they have been extensively proposed and used for different inventory and traceability activities; (Fernández-Caramés et al., 2019; Wawrla et al., 2019) as inventory inspection, cycle counting, stock taking, regular surveillance, and so on (Malang et al., 2023). In Fig. 1, the warehouse activities, or operations that UAV technologies can contribute to have been represented.

Also, recent research in related fields suggests integrating drone adoption and implementation in warehouse management, encompassing areas such as barcode identification, decoding, and automated warehouse operations using computer vision and machine learning (Cho et al., 2018; Kalinov et al., 2020). Considering this information, below presented table tries to examine the ten basic warehouse management areas that can be advanced by the UAVs or drone implementations. In Table 1, the basic processes for warehouse management and UAVs' contribution to these operations are explained.

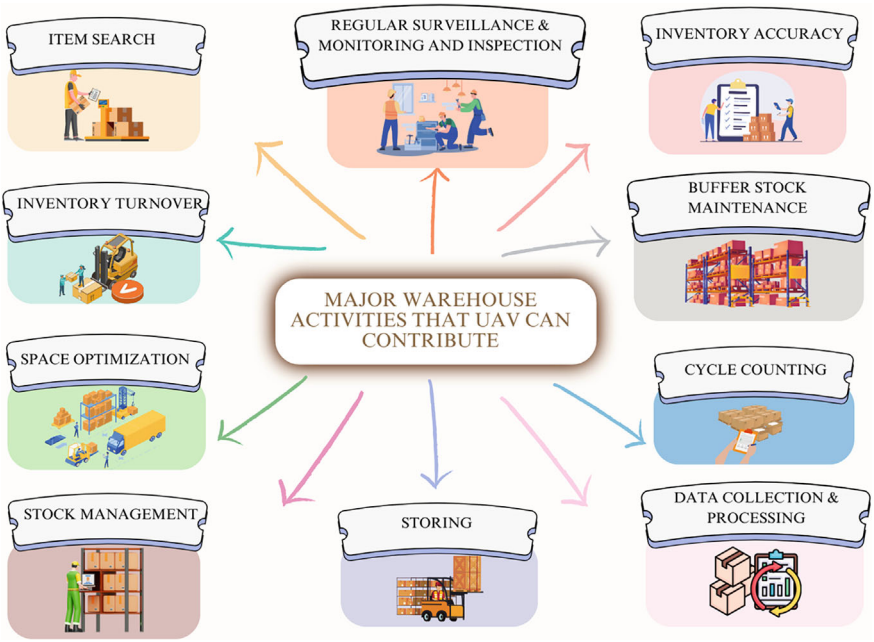

Fig. 1 Major warehouse activities that UAV technologies can contribute (*Source* Authors)

Table 1 Major warehouse activities through UAVs

No	Major warehouse activities	Description	UAV contribution
1	Assuring Inventory Accuracy	Inventory accuracy is defined as a frequent comparison between the amount of stock computed or counted physically in the warehouse and ensuring that the quantity of goods in stock matches the electronically recorded quantity so that stockouts or surplus stock do not occur	Efficient Surveillance, Real-time Data of Inventory, Enhanced Inventory Accuracy, Time and Cost Savings, Enhanced Inventory Safety; Data Analysis of Stocks, Automation, Inventory Tracking, Quick Response to Discrepancies, Improved Record-Keeping, and Stock Accuracy Audits
2	Buffer Stock Maintenance	Buffer stock maintenance refers to the monitoring and upkeep of the buffer or safety stock, which is the additional kept or reserved inventory on hand, to reduce the risk of shortages and fulfil any unexpected future supply or demand shifts in the event of a crisis, emergency, supply chain failure, transportation delays, and so on	Inventory Monitoring, Stock Replenishment, Stock Accuracy Audits, Security and Surveillance, Location Tracking, Shipment Verification, Task Automation

(continued)

Table 1 (continued)

No	Major warehouse activities	Description	UAV contribution
3	Cycle Counting	Cycle counting is a method for regular and periodic inventory management and auditing in which entire physical inventory items are verified, tracked, and assured to be accurate in a regularly repeated procedure	Inventory Sampling, Inventory Data Accuracy, Time Efficiency, Eliminate Human Error, Access to Hard-to-Reach Areas, Real-time Data, Cost Savings, Enhanced Safety, Historical Data Tracking, and Improved Inventory Management
4	Data Collection & Processing	Data collection and processing for warehouse management entail extracting, transferring, loading, processing, organizing, and reacting to inventory data, movements, operations, and conditions to aid decision-making or efficient inventory control using tools and technologies such as UAVs, sensors, barcode scanners, and so on	Data Capture, Barcode Scanning, Integration with Inventory Systems, Real-time Updates, Inventory Tracking, Data Analysis, Cost Saving, Inventory Accuracy, Better Traceability, Less Time Correcting Errors, Immediate Reporting of Movements and Changes in Inventory

(continued)

Table 1 (continued)

No	Major warehouse activities	Description	UAV contribution
5	Inventory Turnover	Inventory turnover is a financial measurement metric or unit that measures the rate at which a warehouse's inventory is sold, consumed, and replaced over a certain period, indicating the warehouse's efficiency in handling and restocking stock	Cycle Counting, Real-time Monitoring, Stock Replenishment, Data Accuracy, Performance Analytics, Auditing and Compliance
6	Item Search	Item search is crucial for warehouse operations like order fulfilment, restocking, quality assurance, and inventory management. It requires significant effort and time for employees to locate and retrieve items from storage spaces	Visual Scanning, Barcode Scanning, Real-time Navigation, Inventory Mapping, Emergency Retrieval, Time and Cost Savings, Real-time Item Tracking
7	Regular Surveillance & Monitoring and Inspection	Regular surveillance, monitoring, and inspection are crucial for warehouses, ensuring efficiency, safety, and quality of operations. These activities help identify and address issues in a dynamic environment	Security Surveillance, Environmental Monitoring, Equipment & Infrastructure Inspection, Quality Control Inspections, Emergency Response, Shipment Verification, Inventory Audits, Emergency Item Retrieval, Reduced Inspection Time, Enhanced Record-Keeping

(continued)

Table 1 (continued)

No	Major warehouse activities	Description	UAV contribution
8	Space Optimization	Space optimization, which aims to maximize the effective use of limited storage space while minimizing wasted space, is an essential component of warehouse management	Inventory Mapping, Storage Rack Assessment, Inventory Placement, Stock Density Analysis, Traffic Flow Optimization, Reconfiguration, Guidance, Inventory Consolidation
9	Stock Management	Stock management is a crucial aspect of warehouse operations that includes inventory planning, organization, control, tracking, restocking, and accurate record keeping ensuring efficient operations	Inventory Monitoring, Stock Replenishment, Cycle Counting, Inventory Audits, Real-time Updates, Data Accuracy, Stock Location Tracking, Performance Analytics
10	Storing	Storing is an essential warehouse function that guarantees the safe and orderly storage of goods for processing or distribution, maintains inventory accuracy, and guarantees the timely fulfilment of customer orders	Inventory Placement, Cargo Inspection, Racking System Assessment, Inventory Mapping, Space Optimization, Localization, Inventory Safety, Data Collection and Decision-Making

Source Authors

UAVs in warehouse inventory management confront various challenges, including getting industry acceptability, not being fully automated, and having difficulty scanning tiny products (Tella, n.d.). The disadvantages of employing UAVs in warehouses can also involve the costly start-up costs in drone technology and equipment, possible security and privacy problems associated with data collecting, and the requirement for competent operators to ensure secure and efficient drone deployment. Furthermore, drones may be limited in their ability to handle highly large or delicate inventory items, and

poor weather conditions may impair their functioning. Hence, existing warehouse management systems may not be suited for drone integration, making ensuring optimal performance and accuracy difficult (Tella, 2023).

4 Best Practices for UAV Technologies for Both Inbound and Outbound Logistics

As the demand for e-commerce items grows, leading to larger and more sophisticated warehouses, the warehouse industry is increasingly turning to innovative technological solutions like robotics, RFID tags, and automated picking systems to boost efficiency and maintain a competitive edge (Newcastle Systems, 2023). Many businesses are keen to implement these technologies, particularly in warehouse systems, in order to reap the benefits, they provide. Walmart, Amazon, and UPS have all invested in UAV technology areas to remain competitive in delivery automation (Dukowitz, 2021). The deployment of UAVs in warehouses is still in the works and is lagging when compared to last-mile deliveries. However, UAV technologies have enormous potential for use in warehouse operations. Companies noticed the potential for implementing these technologies in logistics and warehouses based on the implementations of the technologies mentioned below.

For instance, Amazon has registered patents for a flying drone, underwater, and beehive-like warehouse, and moveable ground-based fulfilment hubs. Their patent underscores Amazon's dedication to fulfilment and logistics innovation by permitting drone deliveries along high-demand routes (CB Insights, 2018). Walmart has also submitted to the Federal Aviation Administration (FAA) for authorization to delivery drone testing in parking lots and houses, following Amazon and other smaller corporations in requesting exemptions from federal drone laws. Aerial technology will also be used to assess the retail giant's buildings, warehouses, and delivery hubs (Malcolm & Weise, 2015). In addition to that, Walmart also plans to enhance its drone delivery network to 37 locations by the end of the year, allowing batteries and Hamburger Helper to be delivered to 4 million households in Arizona, Arkansas, Florida, Texas, Utah, and Virginia. This move intends to accelerate e-commerce growth and outperform Amazon (Repko, 2022). Despite cost inefficiencies owing to delivery distances and aircraft weight constraints, UPS is incorporating drone technology, or UAVs into its delivery procedures intending to increase efficiency by combining drones with human drivers for parallel activities (Stewart, 2017). Mercedes-Benz and drone maker Matternet developed a trial project in 2017 for on-demand e-commerce

delivery utilizing a driverless vehicle named the "Vision Van". Robots at a warehouse would autonomously load the truck, and autonomous drones would transport things to their destination (Macrina et al., 2020).

5 Conclusion

This chapter delves into the usage of UAVs or drones in warehouse management and logistics. With its rising popularity, potential advantages to warehouse operations, and technological improvements, it underscores the reasons why the adoption of these technologies is a requirement rather than a luxury. UAVs can improve productivity and eliminate human mistakes by tracking inventory, counting supplies, and collecting data. They are also becoming more popular because of their low cost, ease of use, and appropriateness for smart operations. Inventory accuracy, buffer stock maintenance, cycle counting, data collecting, and inventory turnover are key areas where UAVs may have a substantial influence on warehouse management. This chapter also addresses the different domains or operations that these technologies might integrate and benefit from in warehouses, as well as the constraints of deploying UAVs in warehouses, such as industry acceptance, operational complexity, and data privacy issues. In addition to that, UAV technologies or drones may considerably speed up labour-intensive procedures such as order picking and put away by acting as substitutes to conveyor belts or forklifts, lowering costs and increasing efficiency. Drone integration can result in workforce transformation, allowing warehouse personnel to become expert drone operators. Furthermore, drones contribute to more precise and effective inventory management, as well as increased security and monitoring, which deters possible intrusions and protects the workplace.

Despite significant interest and investment in UAV technologies for last-mile delivery, one noticeable issue mentioned in this chapter is the warehouse sector's relative lack of attention. Due to the limited implementation, shortage of literature, quick technology progress, context-specific nature of warehouse settings, regulatory frameworks, and industry variances, the study on drones in warehouse operations has limitations. While there is a rising emphasis on employing UAVs for delivery automation, the enormous potential of UAVs within warehouses has not been completely explored. Therefore, this study may not completely investigate these difficulties, and data pertaining to individual drone installations may be secret and difficult to get. Drones' operational and technological maturity in warehousing may be in its early phases, limiting the depth and breadth of the results. These

constraints demonstrate a wasted opportunity to deploy UAV technology to improve warehouse productivity, safety, and accuracy. Warehouse management provides a unique set of issues and possibilities that might be greatly aided by UAV integration. As a result, there is a need to devote attention and resources to fully realize the potential of UAVs in the warehouse industry. Further research on UAVs in warehouse management can concentrate on optimizing path planning algorithms through the integration of IoT devices, particularly drones or UAVs in warehouse operations, for real-time monitoring, investigating cooperative multi-drone systems for challenging tasks, addressing quality and safety concerns in hazardous material handling, and so on. Addressing social and ethical consequences is critical for responsible UAV deployment; future issues may include privacy, job displacement, and psychological effects on workers.

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The Physical Internet and Sustainable Urban Logistics

Rod Franklin, Gero Niemann, and Malte Spanuth

1 Introduction

Cities are complex entities composed of multiple interacting networks of actors, systems, technologies, and regulations (Batty, 2013). As a structure built from the interaction of multiple networks, a city is not a planned construct, but an emergent concept reflecting the multitude of interactions between its underlying components (Fromm, 2004). Historically, the approach cities have taken to managing this complexity has been to organize functionally and, within function, to focus on either technical or geographic components of the function (Minnett, 1975). Unfortunately, such a structure has encouraged the development of "silo" thinking, which ignores the interconnections of the various networks that operate within a city. As Christopher Alexander has noted, a city is not a hierarchical construct that can be neatly laid out and managed as if what happens along branch "x" has no impact

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on things going on along branch "y" (Alexander, 1965). The complex interactions that people, organizations, services, etc., have within a city preclude this tidy tree like thinking (Jacobs, 1961).

Overseeing the various services that are accessed by the entities that deliver services and benefits to the city's citizens are the city's public works department, the transportation department, the police and fire departments, the tax department, a building inspection department, a business license department, an education department, a social services department, a parks and recreation department, and numerous other city departments. In addition, water companies, electric companies, Internet service providers, gas companies and telephone companies all are also managing a set of services that allow the city to operate and for its citizens to live their lives with some level of comfort. And, of course, there is that logistics service provider who is being asked to replenish the stock of the retailer, deliver a package to the consumer, or deliver office supplies to the various city departments that enable the retailer to sell to the citizen who wishes to park their car in front of the retailer's shop and shop at the retailer.

As cities have grown, the services that they are expected to deliver to their citizenry have also grown. These services, provided by numerous city functional departments and external commercial entities intersect, facilitate, and constrain the various recipients of the services in an emergent manner that no individual department or company can foresee or control. Given the city's inability to systemically control or understand what is happening at the operational level means that interventions are as likely to harm as they are to help in achieving higher-level goals being set by citizens or regulatory authorities. Another approach to managing the operations of the city is required.

2 The Physical Internet (PI)

One approach to potentially addressing the problems of increasing complexity in city operations is to recognize that the city's physical infrastructure, as well as its citizens, are all attached to sensors of various types. These sensors allow for the interaction of various remote agencies (telephone companies, electric utilities, Internet Service Providers, water companies, etc.) with citizens, businesses, and the city in a completely "online" manner. In addition, as cities have implemented sensors for control of their infrastructures (the "smart city" urban model), cities have the ability to monitor and manage their physical infrastructure in an online manner as well. With all of these physical entities attached to sensors, it is not an outlandish idea to

think of the physical world becoming very similar to the digital world. Using this analogy, we might be able to manage all of the complexity faced by a city by conceiving of the city as a "Physical Internet".

In the Internet, data is transmitted from a source host through different networks to a destination host (Fig. 1). There are various devices and protocols (rules) that enable the smooth flow of data over these networks and between the hosts. Routers decide over which link (pathway) the data is sent. If the physical medium must be changed, for example from Ethernet cable to Wireless Local Area Network (WLAN), modems are used. To enable the flow of data over the networks, two protocols are particularly important: the Transmission Control Protocol (TCP) and the Internet Protocol (IP). TCP defines how data and/or messages can be broken down into transmittable packets, manages the flow of these packets across the Internet, and ensures that the data that was transmitted reaches its destination. IP provides routing and addressing rules for data being transmitted so that routers can send the data to the appropriate destination host.

The Internet is not one single network, but a network-of-networks. Through the use of gateway protocols, the Internet ensures that each independent network (called Autonomous Systems in Internet terminology) can send messages across the many networks making up the Internet. The PI, in an analogous manner, enables the interconnection of the numerous city silos providing them with a means of managing cross silo interactions via standardized protocols while allowing each silo to manage its internal operations in a manner appropriate to its functional and operational objectives. This physical network-of-networks is envisioned as enabling the efficient flow of

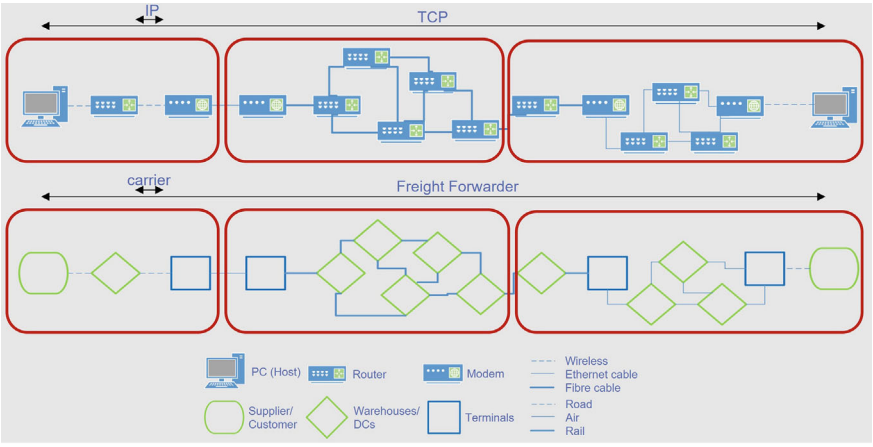

Fig. 1 Comparison internet vs. physical internet (*Source* Authors)

goods, people, resources, and information between originators and consumers through distribution, consolidation and multi-modal physical centres/hubs (routers in the Internet) in a manner similar to that employed in the movement of data over the Internet (Dong & Franklin, 2021). Figure 1 illustrates some analogies between the Internet and the PI.

Figure 1 shows a simplified comparison of how a digital packet is sent over the Internet (top diagram) and how a physical package would flow over the PI (note that similar analogies can be developed concerning how people, other resources, or information could flow over a Physical Internet). A translation between the steps shown in the two models (Digital Internet-DI vs. Physical Internet-PI) can be found in Table 1. Using the movement of parcels in an urban environment as our example the shipment process starts with a supplier wanting to send parcels to an end-customer. The parcels are sent through various logistics networks until they reach the end-customer. Warehouses and distribution centres take on the role of routers, deconsolidating and consolidating parcels and deciding which link to use for sending the parcel to the next "router". Terminals are involved when the mode of transport needs to be changed, for example from air to road. In the PI, terminals are the equivalent of modems in the Internet. Freight forwarders, like the TCP protocol in the Internet, ensure that the packets reach the end-customer at the right time. Carriers take over the parcel movement task of the IP and ensure secure transport of the parcels between the nodes.

3 Components of a Physical Internet

To fully understand the relationship between sensor enabled physical assets (including people) and Internet packets requires some abstract thinking. However, if one is willing to invest the time to do such thinking, a productive analogy can be developed that provides some interesting insights into how a city might manage its operations more effectively and deliver the services that its citizens demand in a more economically, timely, and environmentally beneficial way. To help the reader envision how the concept of a Physical Internet might work in an urban environment, the following sections use the example of a package being sent from some retailer to a consumer in the city. While this is a restricted application of the idea of the Physical Internet, which applies generally to any network of sensor enabled objects, it is a close analogy to how digital packets are managed over the Internet and should provide a good starting point for envisioning how a Physical Internet

Table 1 Components internet vs. physical internet

DI	Explanation	PI	Explanation
PC/Host	Sends/receives data	*Supplier/ End-customer*	Sends/receives parcels
Router	Divides data packets if they are too big for subsequent network Decides which link is used next to transmit data packets	*Warehouses/ Distribution centres*	Deconsolidates, consolidates, and redistributes parcels
Modem	Are used when the physical media is changed	*Terminals*	Are used when the mode of transport is changed
Wireless *Ethernet Cable* *Fiber Cable*	Infrastructure to transmit data Links PCs, Routers, and modems	*Road* *Air* *Rail*	Infrastructure for carrier to transport parcels Links supplier, warehouses, terminals, and end-customer
Transmission Control Protocol (TCP)	Ensures that the information reaches its destination in the correct order	*Freight Forwarder*	Ensures that the parcels reach their destination at the correct time
Internet Protocol (IP)	Ensures the transport of the data packets on the links between the routers	*Carrier*	Ensures the transport of the parcels on the links between warehouses
TCP/IP (together)	Orchestrate the transmission of data packets in a reliable, ordered manner between applications running on hosts communicating via an IP network	*Freight Forwarders & Carriers (in collaboration)*	Organises the entire freight shipping process (including transport, documentation, warehousing etc.) from origin to destination in a logistics network

Source Authors

might work. It also serves as a good introduction to later sections that directly address the challenges of urban logistics and last mile delivery of goods.

3.1 PI Nodes

3.1.1 What Are PI Nodes?

To develop a comprehensive overview of where PI nodes sit in the overall framework of the PI, the delivery process of a package is briefly described. To move goods from a starting point A to a destination B several stakeholders, physical vehicles and logistics facilities are involved. The following overview focuses on the distribution of shipments within a single country and only includes a road-based transport discussion. International or non-road modes of transport are not discussed in this simple model.

As soon as a customer has placed an order, employees at fulfilment centres start to pick and pack the order, which is subsequently consolidated with other parcels and loaded into a line haul trailer to transport it to a sorting centre. At the sorting centre, loads are organized and consolidated into trucks for further middle haul transport. Once the goods end the middle haul at a delivery station, employees sort parcels for the last mile according to their delivery route into vans and the customer is notified that their parcels are "out for delivery", see in Fig. 2.

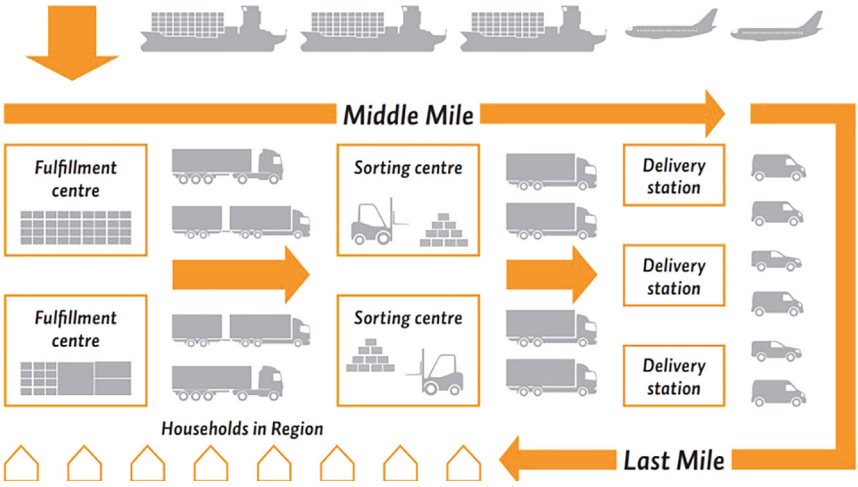

Fig. 2 Delivery process (*Source* Amazon [2021])

As noted, the delivery process incorporates several stations where parcels are deconsolidated, sorted, consolidated, and loaded again. These stations are defined as nodes in the PI framework. Parallels can be made to the Internet where nodes such as routers also play a fundamental role in the deconsolidation, sorting, consolidation, and forwarding of data packets. The close resemblance between the flow of data packets and physical parcels makes it useful to understand the role of nodes in the Internet to be able to draw analogies for node operations in the PI.

Various nodal technologies are employed for moving data packets over the Internet. These technologies are termed modems, switches, hubs, and routers. Modems are needed when the mode of transmission is changed (e.g., WLAN to ethernet cable). The task of a switch is to connect devices within a local network and to transfer data packets between them. Hubs distribute information to all devices within the local area network. Today's modern routers incorporate the functions of switches and hubs, and this term (router) will be used in the remainder of this document to cover the functions of these other technologies. A router receives data packets that flow into its receiving queues. The router reads the header information on the incoming packet and determines where its final destination is. In determining where the packet is going, the router may determine that the packet's destination is outside the network to which the router belongs. If this determination is made, the router repackages for transmission across the gateway between its home network and the external network. Once the destination has been determined and any repackaging performed, the router moves the packet to an outbound queue for retransmission to the next stop in its journey (Dong & Franklin, 2021).

PI nodes are analogous to the various nodal technologies used in transmitting data packets over the Internet. However, PI nodes are physical locations such as terminals, depots, warehouses, distribution centres, parcel lockers, pick up and drop off (PUDO) operations, etc. PI terminals, such as seaport, airport, inland waterway, and rail terminals, facilitate mode switch analogous to modems for Internet transmission mode switching. They also reroute shipments, so their operations, while a bit more complex than a standard router like node, can still be envisioned under the concept of a router. PI nodes, keeping in mind the prior discussion of terminals, act as routers handling deconsolidation, reconsolidation, and outbound lane determination for parcels that they handle. The workflows within the PI nodes are the responsibility of the node operators, just as the routing algorithms in each Internet router are determined by the router manufacturer. However, the inbound and outbound protocols, as well as the cross-PI shipment management protocol all need to adhere to the PI's equivalent TCP/IP protocols to

allow for the seamless transfer and flow of goods over the PI (note that these protocols are, at this point in time, still being discussed and defined).

Several different node hierarchies can be developed based on the perspective that one wishes to take concerning the flow of goods over the PI. One might be interested in how goods flow through a geography where nodes are clustered depending on their geographical location. Another viewer might be interested in the function of the node and therefor cluster nodes based on the function each nodes serves. Nothing prevents someone from combining both of these views and this combined view, as described in Montreuil et al. (2018), is employed in the discussion of urban flows that follows.

The highest-level node in the urban model acts as a gateway. It connects different cities with one another, receiving goods from outside its home city for distribution to locations within the city or, for shipments simply passing through the gateway, retransmission onwards. The gateway also receives goods from within its home city for distribution to other cities or for retransmission to other peer (large cities may have several gateway nodes) or lower-level nodes within the city. The gateway node is responsible for receiving shipments, deconsolidation and reconsolidation of shipments, determination of best lane for their onward journey, and embarking them on that journey. The same functions are performed by the lower-level local consolidation centres and microhubs, which are the next two hierarchy levels in the nodal hierarchy. The main difference between hierarchical level is that the volume of goods handled is reduced at each level due to the reduction in geographic coverage of the node (Montreuil et al., 2018).

To ensure better comprehensibility the hierarchy of the nodes with a short description of their main function is displayed in Fig. 3 and Table 2.

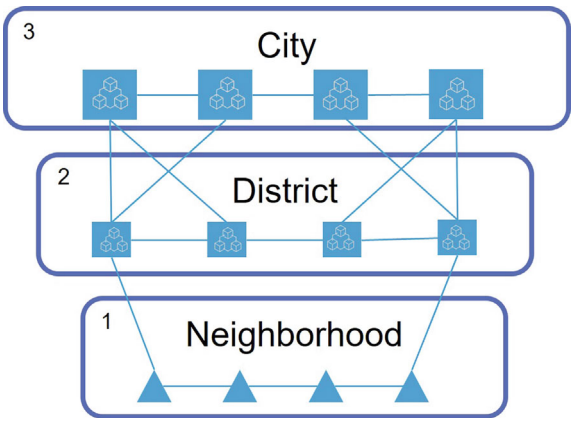

Fig. 3 Node hierarchy

Table 2 Node hierarchy

National or regional distribution centre • Distributes/consolidates the parcels to/from the UCCs • Links areas (cities)	Primary autonomous system
Urban gateway or city consolidation centre • Distributes/consolidates the parcels to/district consolidation centres or microhubs • Links areas (urban districts)	Secondary autonomous system
Urban consolidation centre (UCC) • Distributes/consolidates the parcels to/from microhubs or delivers directly to customer • Links local cells within a district	Campus/local area network
Microhub • Distributes/consolidates the parcels to/from the customer • Links Unit zones (neighbourhoods)	Local area networks

Source Montreuil et al. (2018)

3.1.2 The Role of PI Nodes

PI nodes are entities that enable input and output flow in the logistics network. In the last mile delivery setting, defining the roles and responsibilities of a node is of importance because it provides visibility and accessibility to the logistics operations for involved stakeholders when services and operations are shared in an integrated network. When multiple logistics service providers (LSPs) operate nodes independently, there is limited communication and visibility between node levels for stakeholders external to the LSP. Unlike the Internet where nodes share information on loads, congestion, and connection status, LSPs operating their independent networks generally ignore other LSPs unless they have a contracted relationship with them. Even when under contract, only a small amount of information is shared between the LSPs. This leads to inefficiencies in the operation of the system of urban logistics (not necessarily in each LSP's operation of their nodes). PI nodes exhibit two critical differences. First, they provide an open physical infrastructure that enables logistics services to be operated collaboratively, such as open regional distribution centres, Urban consolidation centres (UCCs), and microhubs. Secondly, these nodes are the main enablers of standard and accessible service operations. As open and collaborative transfer points, PI nodes are similar to the Internet nodes that perform inbound flow management, outbound lane selection, packet aggregation by lane, and outbound flow management without regard to whose packets they are handling.

3.2 PI Networks

Current logistics networks consist of interconnected transportation and storage nodes, the links between them and the modes that are operated on the links, as well as services that are provided within the network boundaries. The primary purpose of these networks is to enable the movement of goods from one point to another subject to specific objectives. These objectives may be cost efficiency, fastest delivery time or lowest environmental impact. PI networks propose a solution to optimize logistics performance under equal provision of several objectives.

3.2.1 What Is a PI Network?

Current logistics networks belong to LSPs such as DHL, UPS, or FedEx that manage their networks in a highly optimized manner. In these networks control is through a single company with limited visibility beyond the boundary of the network. These private networks include nodes that operate in a manner like routers on the Internet. DHL for example operates numerous warehouses, distribution centres, and consolidation hubs that represent nodes on the journey of a package from the sender to the customer. PI networks extend these private networks (Physical Intranets) by connecting them through gateways (Ballot et al., 2022). In the Internet these inter-network-connections are established through routers running Boundary Gateway Protocols. In the PI any node can act as an inter-network connection as long as it implements a standard set of PI interconnection protocols that enable goods being controlled by one LSP to be seamlessly handed over to another LSP for onward movement and control in the second LSP's network (Ballot et al., 2022).

3.2.2 PI Network Operations

PI networks connect various existing networks through gateway nodes into a seamless interconnected logistics and transportation system. PI connection nodes may include warehouses, distribution centres, ports, transportation hubs, and other logistics facilities. The operations within the PI network provide seamless door-to-door service where users state their transport demand form origin to destination and leave the execution to the network and its actors. Effective and efficient collaboration and consolidation of loads within this network-of-networks maximizes the network performance and

ensures service levels are met. Routing within the network is dynamically adjustable to absorb disturbances or demand fluctuations. Inventory can be stored within the shared nodes based on expected end-customer demand to reduce transport requirements and increase response times.

To ensure that all the operations in the PI are feasible, certain prerequisites on the physical, operational and digital levels have to be established (Crainic & Montreuil, 2016).

- **Physical Level:** The PI network provides shippers with Infrastructure as a Service (IaaS) that is shared between different stakeholders. This infrastructure can be provided by cities (roads, curbs, parking spaces, sidewalks, open space, etc.) or private organizations (DHL, UPS, FedEx, et al.) that provide operational assets such as warehouses, distribution centres, or microhubs. Infrastructure and assets are, therefore, a service provided by the PI network (Franklin et al., 2023).
- **Operational Level:** On the operational level a smooth flow of goods across the network must be ensured. PI protocols and operating processes are employed to ensure goods reach their destination on time and in the most efficient and effective way. Protocols ensure the dynamic consolidation and coordination of shipments based on real time routing, scheduling, pricing, and inventory deployment rules. Goods can also be stored in nodes to enable decentralized inventory management to increase service levels and reduce safety stocks.
- **Digital Level:** To enable the operation of the PI network, information must be exchanged between LSPs and infrastructure systems. Smart logistics technologies, such as the Internet of Things (IoT), Artificial Intelligence (AI), and Big Data Analytics (BDA), enable monitoring and real time tracking of goods allowing for upstream and downstream visibility across the supply chain (Rizopoulos et al., 2023). Blockchain and Smart Contracts technologies can help establish a secure and transparent system for information exchange and payments, by automating and enforcing agreements and facilitating efficient coordination and collaboration between all parties involved in the network (Khan et al., 2021).

3.2.3 PI Network Management

Managing a network, whether it be the PI or the Internet, requires that participants adhere to standard protocols. In the Internet, different Internet Service Providers (ISPs) provide different Internet services (Dong & Franklin, 2021).

Table 3 Five-layer Internet protocol model applied to PI

	Internet	PI
Application Layer	Data (HTTP, POP3, SMTP) Communicates application data	Physical parcels equipped with all the relevant information that is needed to flow over the PI
Transport Layer	Segment (TCP, UDP) Provide error recovery service to application layer & end-to-end delivery of data	Shipment management by shipper or freight forwarder to ensure end-to-end delivery (sender to end-customer, mailing address (name, street, city, country)), according to Service Level Agreement (Flow & Error Control); Consolidation at the last node
Network Layer	Packet (IP, ICMP) addressing and routing	Address and handling requirements for shipment to ensure that shipment can be moved from origin to destination
Data Link Layer	Frame (Ethernet) Sending data over physical link governed by rules and conventions	Carrier selection process and carrier movement of goods per handling requirements and lane/region/country legal requirements
Physical layer	Frame (Cable) Transmits bits over physical link	Physical infrastructure (road, rail, air, sea, river, etc.) used to transport shipment along a path between origin and destination

Source Authors

These ISPs are very similar to the LSPs (or different city departments) that ensure the smooth flow of physical goods in the PI. To manage the flow of data packets, the Internet applies standards that are incorporated in protocols and are organized in layers. Such a protocol stack is also required for the PI to manage the operation of the PI network. Table 3 provides a mapping between the five layers in the Internet protocol stack used to manage the operations of the Internet to a similar protocol stack for the management of the PI network.

3.3 PI System

The PI is a network-of-networks. This interconnected set of networks is necessary for the development of the PI, but it is not sufficient. In addition to the interconnected networks, interconnection of information and financial flows, regulatory requirements, and operating protocols are required to ensure that the PI operates as the envisioned open management system.

To establish such a PI system for last mile delivery operations requires the connection of several existing entities. Crainic and Montreuil (2016) identify eight interconnections that must exist for the urban PI system to operate.

1. Interconnection of Cities as Nodes: Cities represent a node in larger regional, national, and international transport networks as they handle inbound and outbound flows that are processed by the logistics operations within the city. Connection between cities and regions is fundamental for the logistics flow and reachability within the city.
2. Interconnection of Standardized Systems: Encapsulation protocols and interfaces are fundamental for the operations of the PI. As most LSPs operate in many different cities, regions, and countries, they would have difficulty performing their services efficiently if every geographic entity established separate protocols for operating within their borders.
3. Interconnection of City Logistics Activities: The PI goes beyond the simple movement of goods to create an all-encompassing logistics system including the way goods are moved, stored, realized, supplied, and used. Pooling, sharing, and collaboration between stakeholders provides for a more efficient and effective use of both the assets used for goods distribution within the city and the city's own infrastructure.
4. Interconnection of City Networks into an Urban Web Architecture: Through interconnecting the current city logistics architecture a shift to a distributed, multi-zone, multimodal, web-like approach enforces the PI's open logistics model. Dynamic, multi-modal logistics flows and routing guided by policies and business decisions are the result of this integrated web architecture.
5. Interconnection of Multiple Urban Logistics Centres: The PI sets the focus on exploiting the capabilities of a multiplicity of existing urban logistic centres, rather than emphasizing one or a few new city distribution centres. These existing logistic facilities, spaces, public and private distribution centres, platforms, warehouses, parking spaces and facilities connected in the urban web provide the capacity for a highly connected and efficient urban PI.
6. Interconnection of Stakeholders into an Open System: The PI incorporates the engagement of multiple stakeholders that collaborate to achieve overall system performance. This collaboration requires supportive agreements on at least three levels: (1) On the operation level, the city logistics web requires multiple multi-modal LSPs to collaborate, consolidate and synchronize. (2) On the business level, collaboration seeks for clear contractual arrangements between providers but also users to regulate

pricing, service level, liability, insurance, risk, cost, revenue, and profit that leads to interconnected business models. (3) On the public-service level, city authorities are asked to engage with LSPs to set up a proper environment with legislation, regulations and policies ensuring efficient and effective operations.

7. Interconnection of People Mobility and Freight: Existing infrastructure dedicated to people mobility should be interconnected to freight logistics for the sake of exploiting synergies through movements that are taking place anyway, e.g., a tramway that is circulating the city can be used to transport goods between nodes.
8. Interconnection of (urban) Infrastructure Planning and Logistics: Urban logistics are strongly dependent on other developments and activities within a city. For example, urban layout and land-use policies have a large impact on city logistics performance. Therefore, comprehensive, and systematic urban planning through an interplay of freight and people transport providers as well as city authorities should be established.

3.3.1 Managing the Interaction of Networks Within the PI

The interaction of networks within the PI requires security, privacy, and trust. As these prerequisites are established, interoperability and accessibility of resources further evolves towards the PI model. Secure protocols and services are enablers for the PI model. They also ensure operational efficiency of freight movement to increase the efficiency and effectiveness of the transport system regardless of freight characteristics, nodal operations or mode (Ballot et al., 2022). A further requirement is the extensibility of the system. New innovative protocols addressing new business models, services or further use cases may be interactively integrated into the system in an agile plug-and-play manner.

3.3.2 Information Exchange Between Networks, Nodes, etc.

Besides the operational aspects of the PI, it is also important to establish protocols and services for the flow of information and payments. Operations require visibility in terms of information across the PI system. Information concerning not only visibility to the supply chain is required, but also visibility to infrastructure availability and other factors that might impact the flow of goods is required. Payments must also be handled, requiring that the

PI system manage who does what and ensuring that when contracted activities are completed the contracting entity is paid for the services performed.

3.4 PI Access

Easy and non-bureaucratic access by users to the PI is a crucial prerequisite for the effective and efficient implementation of the PI. Small LSPs, in a manner similar to their aggregation into groupage or pallet networks, can gain in reach and efficiency by being able to seamlessly access the PI. By integrating loads from a larger group of shippers and LSPs, the PI increases load and operational efficiencies, reducing overall costs and environmental and social impacts (Briand et al., 2022). The PI will not establish itself overnight. As with the Internet, small "experimental" operations are needed to demonstrate the benefits of the PI. Access to a system always requires an initial induction point that connects to the larger system. Considering the Internet, the modern router represents this access point. The Internet is open and accessible to every individual wishing to connect. The PI is also envisioned as an open network-of-networks accessible to any user wishing to connect and link other entities within its own network or across networks.

3.4.1 Who Can Use the PI?

The PI is open to everyone that complies with its protocols. Every user that agrees to the procedures of the PI and every network that meets the requirements for a particular use case should be able to participate, collaborate, share assets, and pool resources in a fair and open manner. In the logistics context, any LSP, whether a major international player such as a DHL, Kuehne + Nagle, Maersk, or UPS, or a small organization or individual taking a single parcel over the "last mile" should be able to use the PI if they comply with its protocols. All of these LSPs provide a logistics service enabled by the PI that moves their loads from an origination point to a destination point.

3.4.2 The Role of Shippers, Forwarders, Carriers, Hubs (Nodes), End-Customers in the PI

For the use of the PI for shipping parcels, there are several parties that participate in the shipment of goods over the PI. Shippers and their customers are two crucial entities that initialize the PI process. The customer must create demand to start the PI process. As soon as there is demand, the shipper inserts

the shipment into the system. The shipper and their customer can be the same entity but generally are different actors. The roles of these entities can also change depending on the circumstance (e.g., a customer can become a shipper if they are returning an item). As with the Internet when it is used to send an email message, shippers (senders of the parcel) and customers (receivers of the parcel) need to trust that the PI will deliver their shipment as specified even though they do not know who will do the transportation or which route the parcel will take to get to its final destination.

Freight forwarders, carriers, hubs (nodes), and shippers that utilize their own internal "control tower" services are critical entities managing flows over the PI. Freight forwarders and shipper control towers take over the management role of planning and monitoring the journey of the shipment. Freight forwarders and shipper control towers rely on the operations of carriers and hubs to execute the movement, transfer, and consolidation of shipments per service levels established for the shipments. Through their oversite, the freight forwarders and shipper control towers act as the PI's TCP service ensuring that shipments arrive at their destinations on time and in full.

Hubs are the nodes in the network that deconsolidate and consolidate loads and move them out along connecting lanes to nodes closer to their final destination. In performing these services hubs act like routers in the Internet. Carriers are the entities that physically move the loads. Carriers can be large asset heavy companies such as Hapag Lloyd, Maersk, Lufthansa Cargo, Atlas Air, Deutsche Bahn, Burlington Northern, DB Schenker, or Schneider Trucking. They can also be small parcel carriers, last mile specialists, barge operators, city transit systems, etc. Their role is to move the freight physically from an origin point to a destination point over whichever infrastructure (road, air, sea, river, rail, etc.) they provide services over. In this function, these carriers combine part of the Internet's IP protocol covering the construction of messages and the physics of transporting electronic packets over wire, fibre, air, etc.

3.4.3 Management of a Shipment in the PI

The management of the PI from a general perspective is facilitated through protocols, the rules, guidelines, and conventions that define how goods, resources, people and information should be handled as entities traverse the PI. These protocols, as with the protocols originally developed for the Internet, are required primarily for the smooth transfer of physical entities between connected networks. These separate networks interconnect through the gateway protocols of the PI ensuring that information

concerning the contracted requirements for the entity flows are seamlessly transferred downstream and information concerning the flow's status is passed upstream. Adherence to this minimum set of gateway protocol standards allows users that operate their own networks (Autonomous Systems as defined in the Internet) to continue to operate them as they have. However, their handling of flows that they have received from external networks must be per the service requirements that they received through the gateway protocol connecting them to the external network and, should they transfer the flow outside of their internal system to another network, they will need to transfer that flow along with the information required by the gateway protocol (Gontara et al., 2018).

3.5 PI Governance

Governance of the PI should be analogous to how the Internet is governed, which assures that no government or commercial entities are allowed to decide who can or cannot use the network. The Internet has grown based on its decentralized structure and freedom from governmental management (van Eeten & Mueller, 2013). However, there are organizations, such as the Internet Engineering Task Force (IETF), that develop standards that are used by a majority of users. These standards ensure, among other things, the secure transfer of data between sender and receiver.

The goal of governance in the PI is to make the universal interconnection between user networks technically feasible and economically profitable, which creates acceptance by society and industry. PI governance incorporates the rules, trust building processes and mechanisms that are defined by the stakeholders to evolve the PI. A bottom-up approach is favoured where nodes, networks, systems, and their stakeholders collaboratively develop mechanisms that converge from PI islands to a global PI. Different layers for governance processes must be defined, e.g., systems and operations. Clear guidelines for integration and access of external stakeholders in vertically integrated networks, asset sharing, and competition have to be defined to provide an integrated governance approach.

3.5.1 Who "Owns" the PI?—Coordination Mechanisms

Whether an end state governance system for the PI can be designed before there is a true PI is an open question. Today's Internet has organically grown, and its governance structure has changed over time. In its earliest years, the

Internet was an academic and research project of the Defence Advanced Research Agency (DARPA) in the United States. DARPA's oversight was loose and one more focused on testing ideas, approaches, and theories of networking. This loose oversight allowed the academics and commercial organizations involved in developing the Internet to guide its development along the model of academic freedom assuming that open discussions and collaboration with colleagues would create a workable system benefiting their needs. During this informal and developmental period the voluntary governance mechanisms that are characteristic of today's Internet were established (Hoffman & Harris, 2006).

The Internet began when only a few small networks existed. Connecting them through the original Interface Message Processor (IMP), the precursor to the boundary gateways used today to connect Autonomous Systems, was a relatively straight forward and welcome process as these isolated networks could now communicate nationally and soon, internationally. The PI, unfortunately, proposes to connect many already existing networks that span not only local areas, but regions, nations, and the globe. The operators of these networks have had many years to develop their particular approaches to managing asset flows and competing with other networks to generate revenue. Asking these independent and highly competitive networks to voluntarily collaborate and connect their networks is a daunting task.

The large conceptual barrier that collaboration brings to these independent network operators may not be as high as they and most others think. Today, these large network operators already cooperate on a limited basis. Shipping companies, for example, carry containers managed by their competitors. Freight forwarders utilize the services of some of their competitors for operations in areas where their networks do not exist, or for the provision of services that they do not provide. In addition, they regularly have their loads consolidated with those of their competitors when employing third parties to move portions of their end-customer shipments. As an example, most freight forwarders do not own cargo ships or aircraft. They employ, therefore, the services of operators who do have these assets and who also provide similar services to their competitors, consolidating the loads provided and moving them in a controlled and successful manner.

However high the barriers might be, there is still a need for developing an open technical committee to help in developing the protocols to be used in the PI and to begin discussions between the various stakeholders that will utilize the PI. A task force similar to the Internet's IETF that is open to all interested stakeholders is needed (Fig. 4). A group such as the non-profit European Alliance for Logistics Innovation through Collaboration in Europe

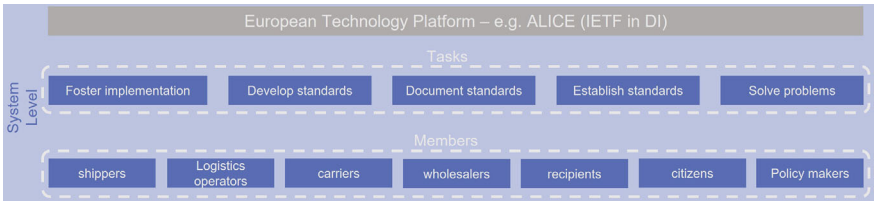

Fig. 4 Governance task force on system level (*Source* Authors)

(ALICE) could be considered for this function. This group could act in a role very similar to the IETF soliciting input on standards, establishing forums for discussion of new services and standards, and encouraging users to work together to create an international system that operates more efficiently and effectively.

3.5.2 Standardization—Using Protocols to Manage Flows

The ICONET project,[1] funded by the European Union's Horizon 2020 programme, developed protocols for the PI that build heavily on the seven-layer OSI model adapted to the PI by Montreuil et al. (2012) as the OLI model (Open Logistics Interconnection model). The ICONET model utilizes a central orchestrator that oversees and manages the operations in the PI by applying the protocols that the project developed. While this approach is one conceptual approach to managing the PI but however the governance model develops, it should be developed by the actors using and operating the PI. It should not be imposed as this will most likely lead to rejection and a failure of many actors to utilize the PI.

3.5.3 Open Standards

Development of operational protocols are only the beginning of realizing a functional PI. Users of the PI must be able to monitor whether service levels are being adhered to and regulatory requirements are being met. Examining how the PI should be overseen from a nation's perspective, there are three levels of oversight that arise (Fig. 5). If the PI is viewed at the national level, it is overseen by the national government. National governmental agencies

[1] ICONET Project. New ICT infrastructure and reference architecture to support Operations in future PI Logistics NETworks. Grant agreement ID: 769,119. Available online: https://cordis.europa.eu/project/id/769119.

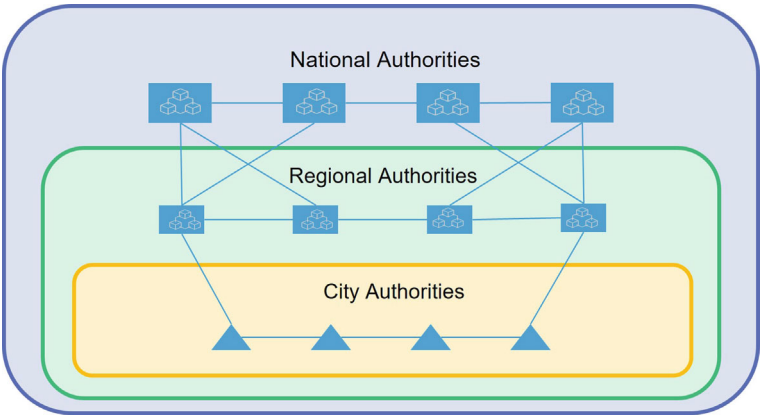

Fig. 5 Organizations involved in operations governance (*Source* Authors)

agree to employ the PI's protocols and ensure, among other things, that operators within the PI adhere to governmental regulations, laws, and national competitive requirements. Dropping down to the regional level, the regional authorities manage the PI and ensure compliance with regional regulations and laws. In doing so, the regulations of the higher level must be complied with. Moving down one level further to urban operations, the PI is governed by the city authorities. Here too, the regulations of the higher-level structures must of course be considered.

Governance also considers the following aspects:

- Regulation and policy, ensuring social and environmental responsibility.
- Intellectual property considerations
- Liabilities
- Privacy and security
- Sustainability.

3.6 PI Adoption

The roles of the various stakeholders in the use of the PI have already been discussed in the sections above. In addition, it has been elaborated how access is enabled, how operations are performed, and how a governance structure permits creation of the PI. In order to achieve a high level of adoption by stakeholders, which in turn enables the smooth flow of parcels through the PI, underlying process enablers such as the data exchange requirements and payment processes must be examined in more detail. It is important to

note that the adoption and widespread implementation of the PI will not be achieved in one dramatic switch of behaviours. Roles, access, governance, and operations will evolve over time. How this evolution will sort itself out remains to be seen. However, a little clarity on data exchange and payment processes may help allay certain adoption fears.

3.6.1 Data Sharing—Data Exchange Requirements

It is important to point out that data that is required to tranship parcels through the PI is already available (such availability may not exist when considering other flows that might use the PI). Figure 6 illustrates an example of a shipping label that is currently used for shipping parcels. The information that is printed on it are weight, sender and customer address, size, shipment-id, and barcodes containing this information. It is also known which service the customer booked, and thus within what timeframe the parcels must be delivered. One further piece of information required concerns the nature of the goods, whether they are hazardous or not. Based on this information the parcels can be routed through existing networks, nodes can deconsolidate and consolidate the parcels as part of larger shipments, and the parcels can be assigned to an appropriate delivery mode.

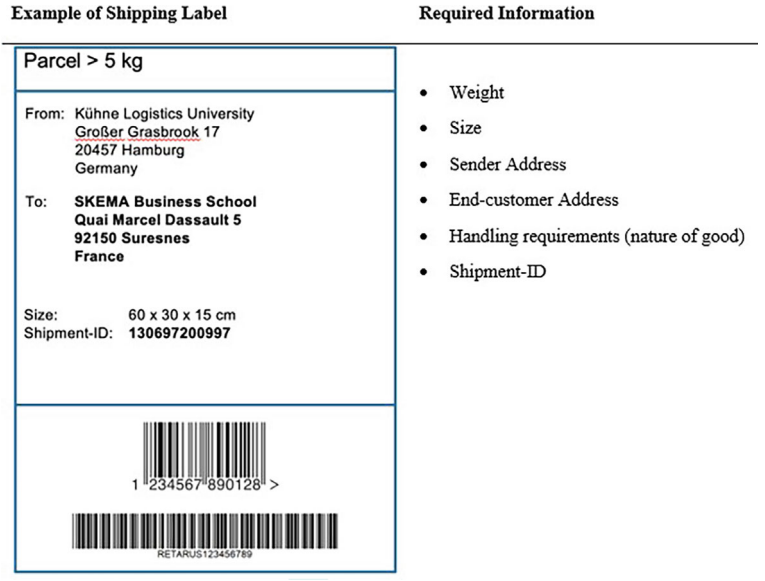

Fig. 6 Example parcel label (*Source* Authors)

This minimum information set is all that is needed to ship a parcel today, and it is all that is needed to ship a parcel across the PI. However, this information is insufficient to allow the various entities that handle the parcel to be paid, or for the entities that are managing the shipment to properly understand its status and to ensure that it reaches its destination on time, in full, and without damage.

In order to exploit the full potential of the PI, the sharing of additional data is essential. This includes information about the traffic situation on a lane, handling times at nodes, capacity constraints at nodes, or available space at the curb side. One could store this information in a multitude of different ways and aggregate it "on demand" so that decisions concerning deliveries and routes could be made and payments processed. However, because the PI is essentially a black box to the shipper and end-customer, should a dispute arise, simple database storage is insufficient. What is required is a formal non-repudiation system. One such system being employed today is the blockchain system. This approach to managing information about shipments requires that contractual information be instantiated in executable code and that the progress of the shipment be monitored so that contracted events can be logged and stored on the blockchain. Disputes can now be addressed as the logging of events to the blockchain means that they cannot be denied (repudiated) and, therefore, are permissible as legal evidence of something having occurred.

Apart from the minimum data requirements for sending a parcel as can be found on a parcel label, the PI requires some further data exchange. This data exchange can be termed as "communication" between nodes. It is not competition-critical but it is crucial for smooth operations (Briand et al., 2022). "Operation" refers in this context to operations across the supply chain but not in the node. Operations in the node, e.g., loading, unloading, or storing, are the responsibility of the node operator.

From a high-level-perspective, an urban node must communicate with its adjacent nodes. Adjacent nodes on the last mile are the customers receiving the parcel (downstream node) and a node that is further upstream the supply chain, such as UCCs. Figure 7 illustrates the communication between adjacent nodes. The communicated data is crucial for the node to make accurate decisions. As previously mentioned, data to be shared are the shipment information (address, timing of delivery and dimensions) and the handling requirements (refrigeration, fragile, glass). This information is communicated by the upstream node to enable the next node to properly process the parcel for delivery. When a node is becoming overloaded, this information must be communicated to avoid receiving further shipments that would be delayed

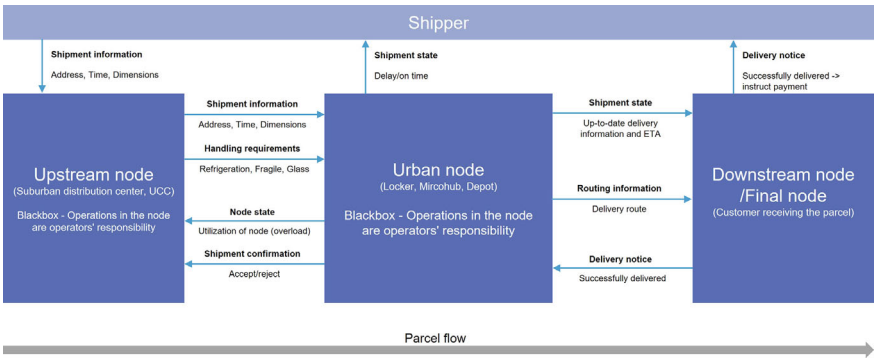

Fig. 7 Data exchange diagram (*Source* Authors)

by the node's overloaded capacity. This information can be referred to as node state. A node can always accept or reject an inbound shipment that is announced to it. Rejections may occur if the node is facing congestion of parcel flows though the node or cannot adhere to the handling requirements of the parcel (e.g., no freezer capacity for frozen items).

Shippers may want to stay up-to-date about the state of their shipment. This means that the PI must continually update its estimates on delivery times based on the current state of the PI. Shippers and end-customers can subscribe to this information for their shipments/orders to determine whether the parcels that have begun to flow through the PI will be delivered on time or if they will be late. End-customers want to know the up-to-date time window of delivery and receive notice if a delay is going to occur. LSPs need to know when a parcel will be handed over to them so that they can plan their deliveries to comply with service levels. To end the delivery cycle, a notice about a successful delivery must be communicated to the urban node and the shipper to trigger the closure of delivery and payment of all participants in the delivery cycle.

3.6.2 Tracking Activities, Events, and Ensuring Delivery

Today, LSPs offer detailed tracking information to their customers. Starting with the notification that the parcel's delivery data has been transmitted by the shipper to the delivery LSP, through the arrival of the parcel in the customer's region, to the successful delivery. This service must also be available to the customer on the PI. But here the information is not only relevant for the customer, but also the events that occur as a result of the transmission of the shipment information. Each time the customer receives an

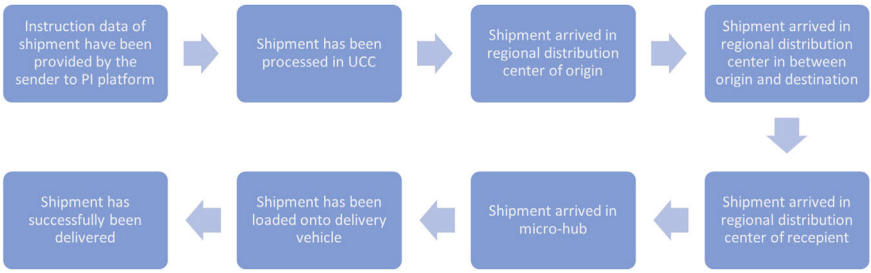

Fig. 8 Tracking information provided to the end-customer (*Source* Authors)

update on the progress of the package, a stakeholder completes its service. For example, once the shipment has been processed in the UCC, the UCC operator has provided a service and must be compensated for it. For clarity, Fig. 8 provides an overview of each step (for an indicative scenario), including the information provided to the customer.

3.6.3 Payments

The previous section pointed out the different stakeholders participating in a delivery process. Each of these stakeholders requires payment for their services. A manually operated payment process would require significant resources and coordination when dealing with multiple stakeholders. In addition, manual processes are often prone to errors. This showcases the importance/need for an automated payment system for stakeholders to be compensated once they have completed their service according to their service agreements. One approach to addressing this requirement would be through the use of smart contracts. A smart contract is a contract translated into computer code that is automatically triggered and executed when a previously defined event occurs. Events can be generated by IoT devices, for example, when a package is detected by a Radio-Frequency Identification (RFID) reader. Another possibility is to scan a barcode in the incoming goods department and thus confirm that the package arrived and been received at a UCC. A major benefit is that the events related to the smart contracts are stored on a blockchain and thus profit from transparency and immutability. For ease of understanding, Fig. 9 shows a simplified payment process for a stakeholder.

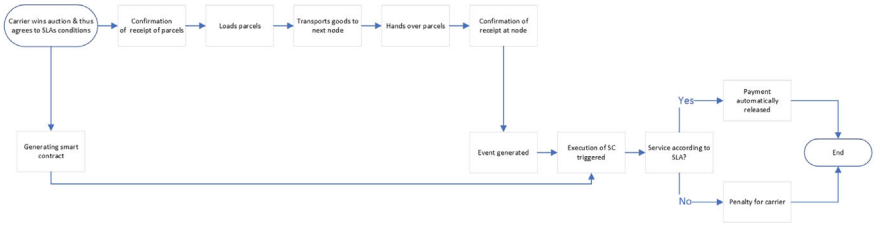

Fig. 9 Payment process example (*Source* Authors)

4 The PI and Last Mile Delivery

Looking at the supply chain of a parcel from a bird's eye view (Fig. 10), there is a first mile, where the package is picked up and inserted into the system, a last mile, where package is delivered to its destination and the long haul in between. Protocols cannot simply adapt from one stage in the supply chain to another stage. The first and the last mile are different from what is in between.

In the urban environment, where parcel volumes are aggregated and density of flows get larger and larger, a solid mechanism for flow management is required. Today parcel flows are predominantly handled by large carriers delivering to customers by van. This is effective but not very efficient. Zooming in to the urban last mile, which can also be considered the first mile when considering return shipments, the parcel flow incorporates two stages (Fig. 11), (1) from the UCC to urban microhubs and (2) from there to the destination considering different modes of transport as well as pick-up and drop-off options. Starting with the first stage at a UCC, which is located outside of the inner city, vehicles are equipped with parcels that need to be moved into city neighbourhoods. The UCC acts as a gate to the urban last mile as the transformation from long haul transport to urban city logistics takes place. Depending on regulations and conventions, e.g., zero emission zones, the right mode of transport has to be selected. The second stage starts with the microhub that serve as the local allocation and distribution facility. This microhub receives inflows that are subsequently deconsolidated and reconsolidated to several smaller outflows allocated to a mode that fits predefined criteria. The assigned last mile LSP delivers the parcel via one of

Fig. 10 Parcel logistic chain model (*Source* Authors)

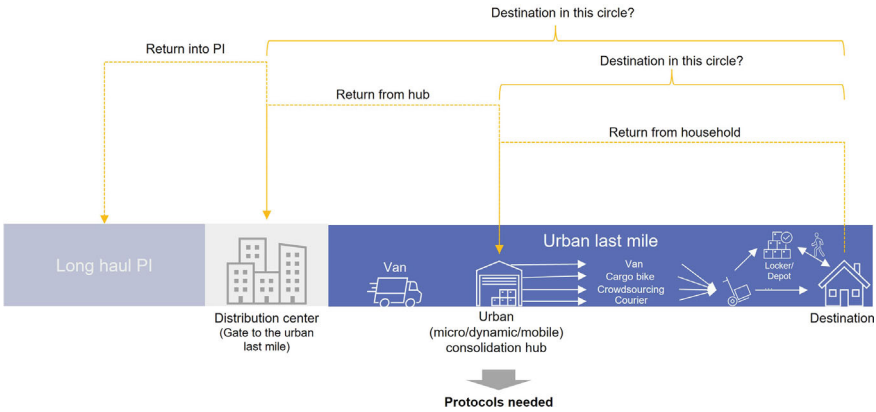

Fig. 11 Stylized last mile delivery (*Source* Authors)

the predefined modes, e.g., cargo bike, either directly to its destination or to a nearby pick-up point, e.g., parcel locker. On its round trip, the LSP also picks up return shipments that are returned to the UCC for movement up the supply chain to an appropriate returns centre.

4.1 Application of PI Protocols—A Hierarchy of Flow Management

As logistics flows cross the boundaries of a city, these streams become condensed and need proper management on the last mile to reach their final destinations. The flows can be carried by several different modes each having advantages and disadvantages. What modes should be selected depends heavily on the city's regulations. For example, low emission zones with narrow streets can exploit cargo bikes because of lower emissions, agility, and less space utilization, while larger parcel volumes or large parcels can exploit the higher capacity of delivery vans. It is certain that relying on one mode will not solve the problems arising in the urban environment. There is a trade-off between the new innovative delivery solutions, e.g., cargo bikes, crowdsourcing, etc., and the conventional van. What is required is to balance the trade-offs between the different modes for effective and efficient delivery within the particular city neighbourhood. Flow management protocols can tackle the problems that arise.

The funnel shown in Fig. 12 is a stylized example of one model of a flow management protocol. It should be noted that the hierarchical model depicted is one of several conceptual models that might be utilized to determine how to allocate loads to different transport technologies. In the "real

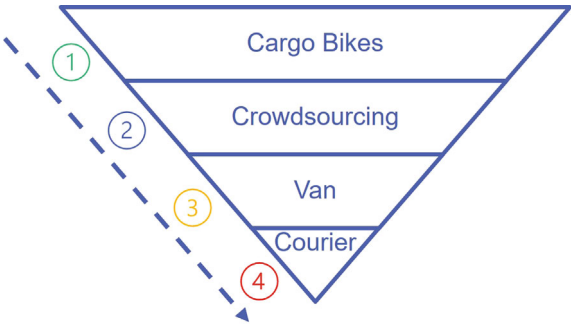

Fig. 12 Hierarchy of modes (*Source* Authors)

world" the process would most likely be messier and would sort itself out based on what was "best" for the city and distribution operations employed.

This example proposes the use of cargo bikes as a preferred mode of last mile delivery, followed by crowdsourcing. Certainly, crowdsourcing should be tied to restrictions as it is not the goal to artificially create traffic through individuals carrying parcels as their "mini-job". Crowdsourcing is meant to be an option for individual trips that will be performed anyway. As a third option, a van is considered as an option for bulky parcels, e.g., a flatscreen television. If none of previous options are available or workable, a van operator (courier in the figure) can be contracted to take over the delivery. The courier is a kind of back-up solution where the mode, used by the courier, is difficult to control.

Figure 13 presents an example of an allocation procedure based on the hierarchy discussed previously. This flow chart represents the visualization of a protocol as a structured step-by-step recipe for flow allocation. Things that should be considered are as follows:

- Is the parcel eligible to be carried by cargo bike or is a van required (dimensions, weights)? A flatscreen, for example, may not be eligible for a cargo bike.
- Which parcels can a van additionally drop off along its route while delivering the bulky parcels?
- Does the selected mode have available capacity?
- Is the required reliability given by the selected mode?
- Has the selected carrier the right certification?
- Are further vans required to handle the high volume of parcels?
- Can a crowd shipper walk to the delivery point?

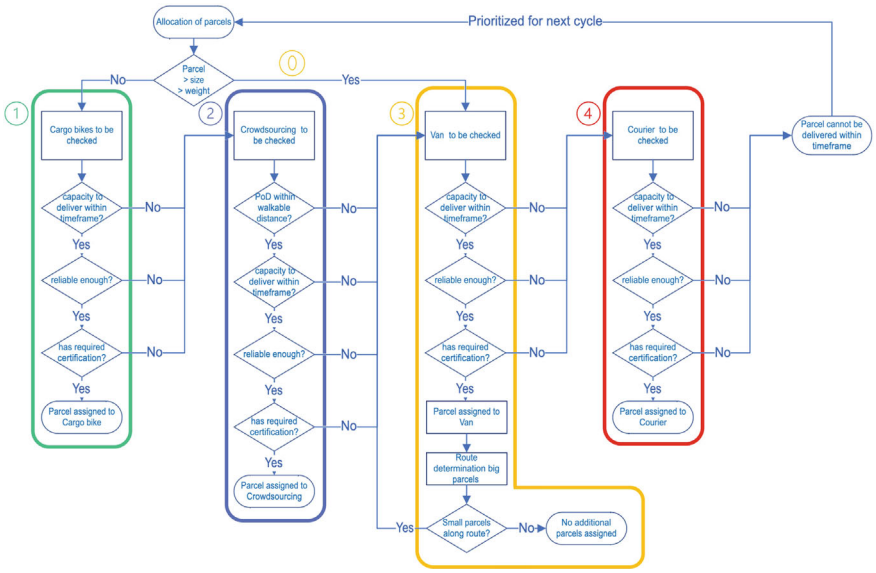

Fig. 13 Allocation procedure (*Source* Authors)

Apart from the outbound distribution flows, reverse flows need to be managed. Customers need a reliable process for their return shipments. They can either drop the parcel at a nearby parcel locker, post office, microhub or preregister a pickup at home. For pickup at home, a preregistration is required as last mile LSPs need to know about the reverse flows and plan their routes and capacity accordingly.

5 Barriers to PI Adoption

The PI is an interesting concept and has captured the imagination of academics and industry as a potential solution to logistics related problems. Unfortunately, the uptake of the concept has been slowed by a number of actual and perceived barriers. It is not the intention of this document to list all barriers that have been discussed concerning the PI. Interested readers are encouraged to make use of the ALICE website (www.etp-logistics.eu) to explore this topic and many more concerning the PI. To give the reader a short understanding of issues that have arisen that do act as barriers to the adoption of this model for logistics operations, the following list should suffice.

- *Lack of Standardization*—people, processes, and systems within and between LSPs, their end-customers, regulatory bodies, modes of transport, and many other interacting components in the logistics world have all developed independently from one another over the centuries. This has created a true "Tower of Babel" in which communications between parties in supply chain operations is extremely difficult. Achieving the "plug and play" connections that have become common for Internet access is still a long way off for the logistics industry.
- *Infrastructure Constraints*—modes of transport have evolved for centuries independently from one another. While the PI focuses on the best and most efficient use of assets to transport goods, the inability to synchronize and quickly transfer goods from one mode to another creates blockages for realizing environmental, asset utilization, and efficiency potentials postulated in PI models.
- *Cultural Barriers/Lack of Trust and Collaboration*—competitors in a market economy are extremely hesitant to collaborate with one another. Worries about revealing pricing, volumes, end-customer data, etc., based on real past experiences make collaboration are hard sell to these individuals.
- *Regulatory/Legal Barriers*—further delaying collaboration between competitors in the logistics space is the relatively recent set of competition penalties that were levied on the major actors by the EU for price fixing. The cease-and-desist requirements placed on these actors and the current interpretation of competition laws make most LSPs, and industry competitors in general, extremely hesitant to talk about anything with their competitors.
- *Data Sharing and Privacy Concerns*—layered on top of all the other legal issues, the recent General Data Protection Regulation (GDPR) rules developed to protect individual privacy act as another barrier to cooperation. Worries about what might happen if shared data is somehow stolen makes sharing data that much more difficult.
- *Implementation Costs*—nothing is free in the world of business and when your margins are typically very low the issue of what something will cost takes a very high priority. Logistics service companies are not academic institutions where research is a key element of their reason for existence. Spending money on "research experiments" is not something that resonates in the executive suites of these companies.
- *Complexity*—as is noted throughout this report, the operation of supply chains is an extremely complex undertaking. Trying to sort out where a new model of operation fits in and where it might need to be modified to

accommodate the infinite variety of ways that things are moved is not a simple task.

6 Guidance for Adoption of PI

To help in implementing sustainable urban logistics and last mile delivery processes in line with the PI model, certain actions from the adopters are required. A short list of some of these actions appears below.

1. Define objectives for improving city congestion, emissions, and social impacts based on improvements in the operation of urban logistics and last mile delivery (LMD) activities.
2. Define the nodes and links in each use case: Identify the physical locations and the routes connecting the nodes.
3. Define potential PI-inspired innovations (e.g., PI protocols) to be included in the nodes of the Living Labs.
4. Define the scope and boundaries of the logistics model, considering the specific geographic area, the types of goods being transported, and the modes of transportation being used.
5. Identify the actors and define the roles as per the PI Framework: Determine the various stakeholders involved in the network, including shippers, LSPs, and customers.
6. Meet with the actors to discuss options, determine expectations, and refine plans for implementing the urban logistics and PI model.
7. Get engaged with city authorities to jointly discuss and define regulations that provide value socially, ecologically, and economically.
8. Define infrastructure (physical and digital) for operating the network according to PI principles.
9. Identify the services that are offered such as transportation, warehousing, inventory management, order processing, and delivery. Define the requirements, relationships and interactions between the service providers and those requiring the use of the services (e.g., retailers, other LSPs, end-customers).
10. Develop an ongoing dialogue with LSPs, retailers, customers and city authorities concerning governance and commercial issues: Define how the elements within the urban network will operate, who will manage which of the elements, and how the operations will become self-supporting.

11. Map flows for the movement of goods through the network. Define type of products, indicative volumes of goods being transported, the capacity of the nodes, travel times associated with different modes: Develop processes and procedures to capture this data from ongoing operations to ensure that objectives are met, and improvements continue.
12. Incorporate technical assets in city transport management and planning: Consider how technology can be used to optimize the performance of urban logistics operations and its potential for building true PI functionality, including the use of sensors, digital twins, data analytics, blockchain, etc., to plan and manage in real time urban logistics operations.
13. Build on common performance metrics to evaluate the effectiveness of urban logistics operations and PI services, including e.g., delivery times, inventory accuracy, and order processing times.
14. Validate/test models and operations: Models of operations and actual operations should be validated and tested using real-world data to ensure that they accurately represent the performance of the urban network and PI implementation.
15. Self-assess PI maturity at regular intervals and
16. Use models to make predictions/take decisions about network design, operations, and management.

The tasks outlined above map to the requirements for developing a strategic urban logistics plan (SULP) per the guidelines developed by the European Commission (see CIVITAS.eu website—https://civitas.eu/ for more information on SULP requirements). Employing the principles of the PI to improve urban logistics operations and last mile delivery activities will help in achieving the improvement objectives established for cities and help deliver the net zero objectives that have been established for the European Union.

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26

Blockchain for Hydrogen Supply Chain Optimization

Muhammad Huzaifa Najmi, Sharfuddin Ahmed Khan, and Yasanur Kayikci

1 Introduction

This study unravels the synergy between Blockchain Technology (BCT) and the Hydrogen Supply Chain (HSC). In light of the increasing climate concerns, the global energy landscape is undergoing a profound metamorphosis. Amid this transformative era, hydrogen as an energy resource paves way for a cleaner, more sustainable future. It has enormous potential to mitigate carbon emissions and power a spectrum of sectors, from transportation to industry. However, the journey towards a hydrogen-based economy is riddled with complexities, with one of the primary concerns being the efficient management of its supply chain (Qazi, 2022).

Apart from its roots in cryptocurrency, blockchain's decentralised and transparent nature offers a robust solution to the intricacies of the HSC. It envisions a system in which every step from the production of eco-friendly 'green' hydrogen to the meticulous logistics of distribution is visible and verifiable in real-time. This is the potential that blockchain offers to the hydrogen industry (Namasudra & Akkaya, 2023).

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BCT is at its core a digital ledger system in decentralised and distributed format, allowing multiple parties to record and verify transactions securely and transparently. It was originally designed for use with cryptocurrencies, but it has subsequently found applications in fields other than banking (Saraji, 2023). A blockchain is essentially a chain of blocks, each containing a collection of transactions. These blocks are connected chronologically to form an immutable chain. This improves safety and security by dispersing control and eliminating a single point of failure (Umarovich & Bakhtiyorovich, 2021).

The extensive network of interrelated processes, organisations, resources, and activities involved in the development, manufacture, distribution, and delivery of products and services from their raw material origins to the end users is referred to as a supply chain (Kayikci et al., 2022a, 2022b). It consists of several steps, including raw material sourcing, manufacture, distribution, retailing, and eventually client consumption. The goal of supply chains is to optimise the movement of products, information, and funds while responding quickly and effectively to changes in customer demand and market dynamics (Khan et al., 2022).

Suppliers, manufacturers, distributors, retailers, and customers, as well as various supporting functions such as logistics, inventory management, and data analytics, work together to ensure that products are moved in a timely and cost-effective manner while maintaining quality and meeting customer expectations (Kayikci et al., 2022a). Supply chains have grown increasingly complicated in today's globalised economy, frequently spanning international borders. Consequently, their adequate management is critical for organisations seeking to gain a competitive edge, improve customer satisfaction, and react to fluctuating market conditions.

A notable development in this field is the Hydrogen Economy, which provides a conceptual framework and vision in which hydrogen gas (H2) as a clean and adaptable energy carrier plays the key role in combating numerous energy and environmental concerns. Using this method, hydrogen is produced from renewable or low-carbon sources, stored, delivered, and used in a variety of industries to replace fossil fuels and reduce greenhouse gas emissions (Amirthan & Perera, 2022). Electrolysis (using electricity to split water into hydrogen and oxygen), steam methane reforming (converting natural gas into hydrogen), and gasification of biomass are ways to produce hydrogen. If renewable energy sources such as wind, solar or hydroelectric power are used in the production process, the resulting "green hydrogen" may be nearly emission-free (Kabir et al., 2023).

The hydrogen economy, as a crucial component of the energy transition, is inherently linked to the principles of the green supply chain, integrating sustainable practices across its production, distribution, and utilisation processes. The green supply chain approach is essential for ensuring the sustainability and environmental integrity of the entire HSC, from production to consumption. The interrelated processes involved in creating, storing, transporting, and using hydrogen as an energy carrier are referred to as the HSC.

By delving into existing literature, real-world applications, and potential future developments, we seek to highlight the transformative power of blockchain in this pivotal sector. Examining successful implementations and understanding the hurdles previously faced, this research aims to pave the way towards a transparent, sustainable, and efficient hydrogen economy.

The research questions (RQs) of this chapter are:

RQ1. What are the application potentials of blockchain technology in the hydrogen supply chain?
RQ2. What are the future directions of blockchain technology in hydrogen supply chain?

Our research methodology employs a systematic literature review and analysis of case studies, adhering to the PRISMA protocol. We commenced by searching for existing publications using relevant keywords and then integrated these resources to construct a comprehensive body of literature. We conducted an exploratory analysis to evaluate the applicability of blockchain technologies within the HSC, leading to the proposition of future research directions that prioritise efficiency, sustainability, and regulatory compliance. We acknowledged the limitations of our current study and highlighted potential areas for further investigation.

The remainder of this chapter is structured as follows: Sect. 2 presents an in-depth literature review encompassing established knowledge on blockchain, hydrogen economy, supply chain, green supply chain, HSC and blockchain capability in HSC, as well as the research gap. Section 3 presents the research design and methodology, divulging the research process and data collection techniques used. The findings are demonstrated in Sect. 4. Section 5 discusses some implications. Finally, the last section concludes the chapter.

2 Literature Review

2.1 Blockchain Technology and Its Applications in Supply Chain

The openness and security of blockchain have found applications far beyond cryptocurrency. It has been used in supply chain management to monitor the origin and travel of things, in healthcare for secure patient data exchange, in voting systems for tamper-proof elections, and in digital authentication, to name a few applications (Kayikci et al., 2022b, 2022c). Blockchain has also enabled smart contracts, which are self-executing contracts with built-in rules, automating complex procedures and removing the need for intermediaries (Raja Santhi & Muthuswamy, 2022). However, BCT has some disadvantages. Dependence on consensus techniques can cause scaling issues, resulting in slower transaction times compared to traditional databases. Energy usage is another issue because many blockchains use resource-intensive mining operations. As technology advances, efforts are being undertaken to solve these issues and unlock its full potential in various sectors (Merrad et al., 2022).

Now on the other side of the spectrum, Table 1 presents a holistic look at BCT which is rapidly being used to improve transparency, traceability, and efficiency in supply chain management. Several important breakthroughs have occurred at the convergence of BCT and supply chain.

These examples demonstrate how BCT is being used to revolutionise supply chain management across several sectors, improving transparency, trust, and sustainability throughout global supply networks.

2.2 Hydrogen Economy

Hydrogen economy is a conceptual framework and vision in which hydrogen gas (H2) is important in addressing numerous energy and environmental concerns as a clean and adaptable energy carrier. In this approach, hydrogen is produced from renewable or low-carbon sources, stored, delivered, and used in a variety of industries to replace fossil fuels and reduce greenhouse gas emissions (Amirthan & Perera, 2022). Electrolysis (using electricity to split water into hydrogen and oxygen), steam methane reforming (converting natural gas into hydrogen), and gasification of biomass are ways to produce hydrogen. If renewable energy sources such as wind, solar or hydroelectric power are used in the production process, the resulting "green hydrogen" might be just about emission-free (Kabir et al., 2023). Once hydrogen is produced, it may be utilised in a variety of applications. It can be used in fuel

Table 1 Blockchain inclusion in supply chain in different industries

Area	Explanation	Reference
Food Traceability	The IBM Food Trust platform was developed in collaboration with Walmart, IBM, and other partners. This blockchain approach allows food goods to be tracked throughout the supply chain. For example, Walmart utilises it to track the origin of mangoes, cutting the time it takes from days to seconds and improving food safety	Damoska Sekuloska and Erceg (2022); Kayikci et al. (2022c)
Diamond Tracing	De Beers, the diamond corporation, has unveiled Tracr, a BCT for tracking diamonds from mine to retail. Tracr assures ethical sourcing, lowers the danger of conflict diamonds, and boosts customer confidence	Kshetri (2023)
Coffee Supply Chain	Bext360 has created a BCT that records the coffee supply chain, allowing for openness in sourcing and price. It enables customers to discover more about the origins of their coffee beans as well as the economic impact on producers	Köhler et al. (2022)
Traceability of Seafood	Thai Union, one of the world's largest seafood enterprises, collaborated with startup Viant to develop a blockchain-based system for tracing the supply chain of seafood items. This resolve worries about illegal fishing and sustainability by tracing seafood from the ocean to the plate	Tolentino-Zondervan et al. (2022)

(continued)

Table 1 (continued)

Area	Explanation	Reference
Automotive Parts Authentication	Renault, a French automaker, collaborated with Microsoft to develop a blockchain-based system for authenticating spare auto parts. This keeps counterfeit components out of the supply chain and ensures safety and quality	Hao et al. (2022)
Pharmaceutical Authentication	MediLedger is a coalition of pharmaceutical businesses that uses blockchain to verify the legitimacy of prescription medications. This improves patient safety by lowering the possibility of counterfeit pharmaceuticals entering the supply chain	Shashi and Gossett (2022)
Wine origin	In the wine business, blockchain has been utilised to verify the authenticity and origin of premium wines. Scanning QR codes on wine bottles provides consumers with information about the wine's provenance, verifying its validity	Brookbanks and Parry (2022)
Cobalt Supply Chain	Companies which include IBM and Ford are utilising blockchain to track the cobalt supply chain, which is a vital component in lithium-ion batteries. They hope to address issues about child labour and environmental effect in cobalt mining by assuring ethical and responsible sourcing	Bernards et al. (2022)

cells to generate energy, with the sole exhaust being water vapour. This makes it a viable alternative to internal combustion engines as well as a solution for powering numerous modes of transportation (Terlouw et al., 2022).

The hydrogen economy is more than just transportation. It may be incorporated into industrial processes such as steel and cement manufacture, where hydrogen can be used as a clean substitute for fossil fuels. Furthermore, hydrogen can assist to balance intermittent renewable energy sources by storing extra energy during peak production and releasing it when consumption becomes excessive. However, the hydrogen economy has its challenges. In particular, the production cost of green hydrogen is currently higher than traditional methods. The storage and distribution of hydrogen requires specialised infrastructure due to its low energy density and potential leakage difficulties. Additionally, breakthroughs in fuel cell technology as well as regulatory support are needed to accelerate its deployment (Ferahtia et al., 2023). While the hydrogen economy holds promise in terms of decarbonization and energy source diversification, its actualization will necessitate a collaborative effort between governments, industries, and researchers to develop cost-effective technologies, create the necessary infrastructure, and develop policies that encourage its adoption as a key component of a sustainable energy future.

2.3 Green Supply Chain

Green supply chain, also known as a sustainable supply chain, is a strategic method to managing a product's or service's whole lifespan with a heavy emphasis on minimising environmental effect. It entails incorporating environmentally friendly practises into every level of the supply chain, from raw material procurement through manufacturing, distribution, and even end-of-life disposal. A green supply chain's major goal is to avoid negative environmental consequences, save resources, and promote social responsibility while satisfying consumer expectations and guaranteeing economic sustainability (Li et al., 2022). A green supply chain includes practises such as sustainable procurement, where businesses select suppliers based on their commitment to eco-friendly ideals. This also entails improving energy efficiency in industrial processes, lowering waste generation, introducing recycling activities, using eco-friendly packaging, and implementing sustainable transportation routes to reduce carbon emissions. Collaboration with partners, stakeholders, and regulatory authorities is essential to align aims and achieve common sustainability targets (Alghababsheh et al., 2022).

2.4 Hydrogen Supply Chain

Hydrogen is regarded as a clean energy source since its combustion or conversion in fuel cells emits only water vapour, resulting in no carbon emissions. Electrolysis is one way of its production, as is reformation (Scovell & Walton, 2023). Depending on the desired purpose, hydrogen can be kept in various forms after generation, such as compressed gas or liquid. It can be carried by pipelines, as with natural gas, or via specialised carriers like tube trailers. The hydrogen is then stored and used in fuel cells to create energy for a variety of industries, including transportation, industrial activities, and power generation (Zhang et al., 2022).

Creating a viable HSC necessitates the establishment of a network of manufacturing facilities, storage infrastructure, transportation systems, and end-user applications. This all-encompassing strategy intends to make hydrogen a flexible and long-term option for decarbonizing energy-intensive sectors and creating a greener energy future (Riera et al., 2023).

2.5 Blockchain Capability in Hydrogen Supply Chain

In the context of modernising energy systems and tackling sustainability concerns, BCT and the HSC are both critical. The decentralised and tamper-proof nature of blockchain, transforms traditional data management methods, making it critical for a variety of sectors, including energy. In Table 2, we go more in depth towards the aspects of blockchain which will strengthen HSC.

Table 3 shed some light upon as to how crucial hydrogen is in our foreseeable future. The HSC is critical to reaching sustainability goals and switching to greener energy sources.

Essentially, the synergy between BCT and the HSC has the potential to change the energy landscape. While the HSC contributes to sustainable energy solutions, environmental stewardship, and economic advancement, BCT provides openness, security, and efficiency in handling the intricacies of the HSC.

To close these gaps, industry players, legislators, and technological specialists must work together. By effectively integrating BCT, these problems will be overcome, paving the way for a more efficient, transparent, and sustainable HSC.

Table 2 Benefits of blockchain's inclusion in hydrogen supply chain

Area	Explanation	Reference
Transparency & Traceability	In the HSC, every stage of hydrogen synthesis, transit, and utilisation can be identified and validated, promoting accountability and quality assurance through documentation in an immutable ledger	Mangla et al. (2022)
Security & Trust	BCT's cryptographic algorithms protect data and decrease the possibility of unauthorised access or manipulation. This is crucial for ensuring the integrity of sensitive data across the complicated HSC	Namasudra and Akkaya (2023)
Efficiency & Collaboration	BCT improves communication and cooperation across HSC stakeholders by offering a common, real-time digital platform. This simplifies operations, lowers paperwork, and decreases delays	Raja Santhi and Muthuswamy (2022)
Implementation of Smart Contracts	The use of smart contracts has expedited operations in the HSC, automating functions like payments and compliance checks. This has resulted in increased efficiency and fewer human mistakes	Wang et al. (2022)
Carbon Emissions Monitoring	BCT has enabled accurate carbon emissions monitoring throughout the hydrogen lifetime, assisting in emission reduction and environmental stewardship	Lu et al. (2022)

Table 3 Hydrogen's significance primarily from sustainability's standpoint

Area	Explanation	Reference
Clean Energy Transition	Hydrogen is a versatile clean energy carrier, providing an option for decarbonizing numerous industries such as transportation and industrial. Its implementation has the potential to drastically cut greenhouse gas emissions	Jiang et al. (2023)
Energy Storage	Hydrogen may be used to store excess renewable energy, hence improving grid stability, and allowing for greater utilisation of intermittent energy sources such as solar and wind	Jiang et al. (2023)
Industrial Applications	Hydrogen is employed in a variety of industrial operations, including chemical manufacture and refining, which reduces emissions and dependency on fossil fuels	Qazi (2022)
Energy Security	By minimising reliance on a single energy resource, hydrogen diversification can improve energy security	Noussan et al. (2020)
Innovation & Job Creation	The HSC promotes innovation, research, and investment in hydrogen production, storage, and utilisation technologies. As a result, new employment possibilities and economic growth are created	Mneimneh et al. (2023)

2.6 Sustainability Factor

Like anything worth our time, effort & dedication; adoption of the HSC as a sustainable energy solution comes with difficulties, notably in terms of carbon dioxide (CO_2) emissions and their influence on global warming. To guarantee that the HSC effectively helps to minimising these concerns, many critical obstacles must be addressed as mentioned in Table 4.

To address these issues, governments, companies, and research institutes must work together. To guarantee that the HSC efficiently contributes to decreasing CO_2 emissions and combating global warming, a holistic approach that prioritises low-carbon manufacturing methods, efficient transportation, and supporting regulations will be required.

Table 4 Challenges in the path of hydrogen supply chain

Area	Explanation	Reference
Carbon Intensity of Hydrogen Generation	The carbon intensity of hydrogen generation is a key problem. While hydrogen is a clean energy carrier when used, its generation processes can have varying environmental impacts. The production of hydrogen from fossil fuels using technologies such as steam methane reforming can result in large CO_2 emissions. To fully realise the HSC's potential, a transition towards low-carbon and renewable hydrogen generation technologies, such as electrolysis fuelled by renewable energy sources, is required	Incer-Valverde et al. (2023)
Emissions from the Supply Chain	The HSC involves several steps, including manufacturing, transit, and utilisation. Emissions may occur at any point as a result of energy use, leakage, or other circumstances. To reduce emissions in the supply chain, efficient transportation, and distribution techniques, as well as leak detection systems, are essential	Li et al. (2022)
Energy Transition Challenges	Moving to a hydrogen-based economy entails considerable infrastructural expenditures as well as research and development. To prevent increasing carbon emissions, the energy necessary to produce and transport hydrogen must originate from sustainable sources	Wanniarachchi et al. (2023)

(continued)

Table 4 (continued)

Area	Explanation	Reference
Infrastructure Development	Building the infrastructure required for hydrogen production, storage, transit, and utilisation is a daunting task. Retrofitting existing facilities or establishing new ones calls for major expenditures and regulatory changes	Mneimneh et al. (2023)
Economic Viability	Creating a competitive hydrogen market while conforming to environmental regulations is difficult. It is critical for widespread adoption to ensure that the prices of generating and consuming hydrogen are competitive with other energy sources	Kumar et al. (2023)
Scaling Up	As businesses and sectors migrate to greener energy, demand for hydrogen is likely to skyrocket. The logistical and technological constraints of increasing hydrogen production to fulfil this demand while minimising emissions and ensuring sustainability are significant	Odenweller et al. (2022)
Lifecycle Emissions Analysis	A thorough examination of hydrogen's lifecycle emissions is essential. This covers not just accounting for emissions during manufacturing, but also throughout transit, storage, and utilisation	Li et al. (2022)

2.7 Research Gaps

Although significant progress has been achieved; and we have had the basics covered in the integration of BCT with the HSC, Table 5 examines some key gaps and obstacles that need to be solved.

Table 5 Gaps towards the adoption of blockchain in hydrogen supply chain

Area	Explanation
Standardisation	The lack of standardised protocols for BCT applications in the HSC impedes seamless integration and interoperability among various systems and stakeholders
Scalability	As the HSC adopts BCT, scalability issues develop because to the computing intensity of blockchain networks, possibly slowing down transactions and data processing
Data Privacy	Maintaining data privacy in a transparent blockchain system is a difficult task. It is vital to strike a balance between transparency and the protection of sensitive information
Energy Consumption	Blockchain networks, especially those that use Proof of Work consensus techniques, may require a lot of energy. If the energy consumed is not from renewable sources, this undermines the HSC's environmental goals
Regulatory and Legal Frameworks	Many areas lack clear legislation regarding BCT applications in the HSC. It is vital to establish legal frameworks that accommodate the unique elements of BCT
Adoption Challenges	Because of the learning curve involved with BCT, adoption difficulties exist. Industry-wide education and training are required to realise the full potential of BCT in the HSC
Infrastructure Expenditure	Establishing blockchain infrastructure and upgrading legacy systems require considerable expenditure, which may dissuade some stakeholders from adopting new technologies

3 Methodology

In this section, we utilised the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) protocol to conduct a structured literature review, focusing on identifying key dimensions while ensuring relevance in a subject area that can be broadly scoped. The subsequent subsections detail the application of the PRISMA framework (Page et al., 2021).

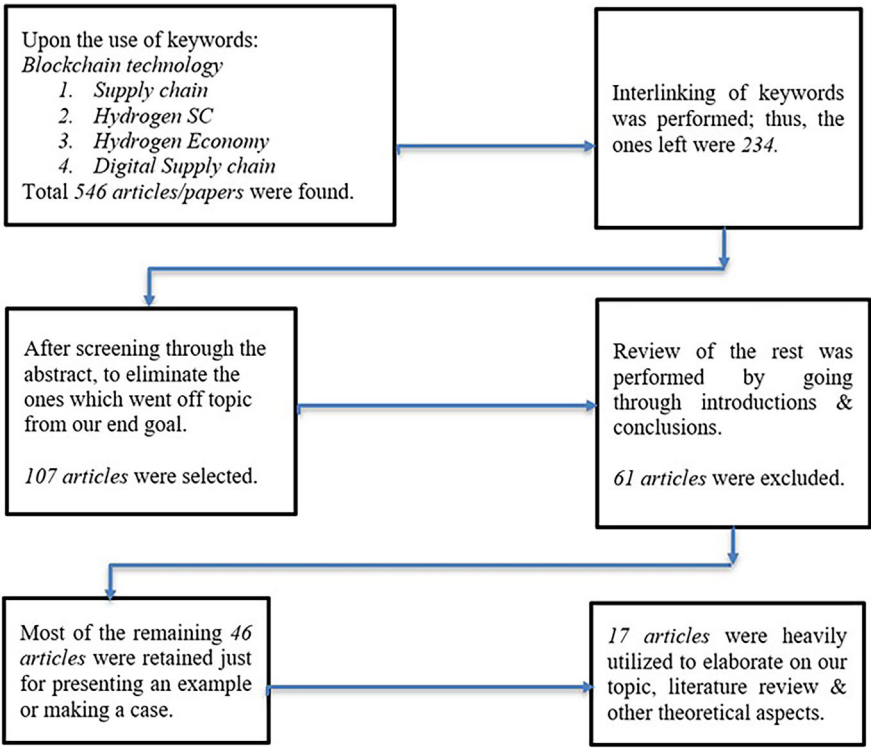

Fig. 1 PRISMA analysis flowchart

3.1 Prisma Template

Figure 1 presents the details how PRISMA methodology was conducted.

3.2 Literature Review

The literature search was conducted on research databases for *Web of Science, Scopus* and *Science Direct* to search recent research papers (not older than 3 years) on the *Hydrogen Supply Chain, Hydrogen Economy, Blockchain Technology*, and their sector-specific applications.

Initial Screening: Started with a broad selection of articles and systematically narrowed down the list based on relevance, ensuring a targeted and current literature review.

3.3 Hydrogen Supply Chain Evaluation

Data collection specific to the HSC was also done to analyse its viability for our future. Selected recent papers meeting the criteria, emphasising blockchain integration and other pertinent aspects.

3.4 Blockchain Technology Evaluation

Separate evaluation of blockchain platforms and consensus algorithms relevant to supply chains was done before focusing on our objective.

Comparative Analysis: Systematically compared technical aspects, security features, and scalability of blockchain technologies from selected papers to evaluate their applicability in the HSC.

3.5 Case Studies and Exploratory Analysis

Many research papers were selected for the sole purpose of citing the existing real-world applications of our study, which presents clear potential of our work.

Case Study Analysis: Analysed methodologies, outcomes, challenges, and lessons from selected case studies, ensuring a comprehensive understanding of practical applications.

This methodology, relying solely on *Web of Science, Scopus* and *Science Direct*, involved a systematic approach to source recent and relevant literature. The process included initial broad searches followed by focused selection, ensuring the research's depth and applicability.

4 Findings

4.1 Application Potentials of Blockchain Technology in Hydrogen Supply Chain

BCT integration into the HSC has potential for improving transparency, security, and efficiency in the administration of hydrogen-related operations. At the junction of HSC and BCT, several efforts and breakthroughs have emerged which are discussed in Table 6.

Table 6 Real-world case studies of blockchain's integration in hydrogen supply chain

Areas	Explanations	Reference
Green Hydrogen Certification	Organisations such as the Energy Web Foundation and Energy Web Chain have created blockchain-based systems to validate the origin and techniques of generation of green hydrogen. These systems enable manufacturers to independently verify the renewable energy sources utilised in hydrogen manufacturing, assuring transparency and legitimacy	Mould et al. (2022)
Carbon Emissions Tracking	Hydrogen Europe, a hydrogen industry group, is investigating the use of blockchain to track and report carbon emissions linked with hydrogen production and delivery. This includes collecting emissions data at various stages and providing a full picture of the environmental effect of the HSC	Gallo et al. (2023)
Supply Chain Traceability	Hydrogen businesses such as Hydrogenious LOHC Technologies are using blockchain to track hydrogen shipments. They use blockchain to ensure that data linked to the loading, shipping, and unloading of hydrogen carriers is safe and verifiable	Ferraro et al. (2023)

(continued)

Table 6 (continued)

Areas	Explanations	Reference
Hydrogen Smart Contracts	Companies such as ENGIE and the French utility company Gazel have been testing blockchain-based smart contracts for hydrogen transactions. When established circumstances are satisfied, these contracts execute payments and transfers of ownership automatically, expediting commercial operations	Wang et al. (2022)
Platforms for Trading Hydrogen Platforms	Some initiatives, such as the Hydrogen in blockchain platform, attempt to develop marketplaces for trading hydrogen certificates and quotas. These platforms enable buyers and sellers to trade hydrogen qualities such as provenance and sustainability in a transparent manner	Panah et al. (2022)
Collaboration on the Hydrogen Supply Chain	Major firms in the energy sector, including Siemens and Tennet, have partnered on the 'Element 3' initiative. The blockchain is being used in this endeavour to optimise the whole value chain of green hydrogen generation, storage, transportation, and use	Riva Sanseverino et al. (2022)

(continued)

Table 6 (continued)

Areas	Explanations	Reference
Emissions Tracking and Offset	Some initiatives, such as Mitsubishi Power and NTT's Hydrogen-Blockchain Initiative, use blockchain to track carbon emissions throughout the hydrogen lifetime. The openness of blockchain allows stakeholders to track emissions and even facilitate offsetting via tokenised systems	Mould et al. (2022)
Decentralised Energy Networks	Blockchain-based decentralised energy networks are being investigated in places such as Australia for controlling hydrogen production and delivery. These networks enable more flexible and localised energy systems that incorporate hydrogen and renewable energy sources	Karumba et al. (2023)

These examples show how blockchain technology is being used to improve several areas of the HSC, such as certification, emissions tracking, transactions, and cooperation. These projects, while still in their early stages, set the groundwork for a more transparent, efficient, and sustainable hydrogen economy.

Now simply put, Hydrogen generation is at the crux of HSC; thus, there are terms "green" and "blue" referring to distinct techniques of hydrogen generation with differing environmental implications and amounts of carbon emissions.

Green hydrogen is created by powering the electrolysis process using renewable energy sources such as solar, wind, or hydroelectric power. Water molecules are divided into hydrogen and oxygen by electrolysis. Because renewable energy sources emit little carbon, hydrogen generated in this manner is considered ecologically beneficial and sustainable. Green hydrogen helps to reduce carbon emissions in the energy industry and has a wide range

of uses, including powering fuel cells in automobiles and industrial operations (Kumar & Lim, 2022).

Blue hydrogen is produced from fossil fuels, often natural gas, via a process known as steam methane reforming (SMR). However, CO_2 created during the SMR process is captured and stored using carbon capture and storage (CCS) technology to reduce carbon emissions. This keeps CO_2 from entering the atmosphere and contributes to global warming. While blue hydrogen cannot produce zero emissions because of its relationship with fossil fuels, the use of CCS technology can significantly lower its carbon footprint when compared to standard fossil fuel techniques (AlHumaidan et al., 2023).

Green and blue hydrogen are both important components of the shift to a low-carbon energy economy. Green hydrogen is seen as the perfect answer owing to its reliance on renewable energy sources, which results in zero carbon emissions during manufacturing. Blue hydrogen, on the other hand, is viewed as a temporary solution since it allows existing natural gas infrastructure to be repurposed while cutting emissions through CCS (Noussan et al., 2020).

Balancing the use of green and blue hydrogen is critical for fulfilling rising energy demands while also lowering carbon emissions in a variety of industries.

4.2 Future Directions of Blockchain Technology in the Hydrogen Supply Chain

In the realm of exploring BCT in the HSC, there are promising avenues as can be seen in Tables 7 and 8, for future investigation that could shape sustainable energy systems.

Exploring the use of BCT in the HSC entails numerous approaches to maximising transparency, efficiency, and sustainability:

Through the exploration of these pathways, integrating BCT into the HSC can unlock efficiency enhancements, ensure transparency, and contribute to the evolution towards a more sustainable energy landscape.

5 Implications

The task of integrating BCT in HSC is full of challenges and hurdles/gaps; to overcome them, industry players, legislators, and technological specialists must work together in the areas mentioned in Table 9, paving the way for a more efficient, transparent, and sustainable HSC.

Table 7 Areas requiring further exploration

Areas	Explanation	Reference
Initial Testing	Commencing pilot initiatives within the HSC can showcase how BCT can trace hydrogen's origin, production techniques, transport pathways, and carbon emissions. Such trials can substantiate the viability and advantages of BCT integration	Ferraro et al. (2023)
Data Fusion	Combining data from different HSC stakeholders on a blockchain platform can establish a transparent and consolidated record. This ensures seamless data exchange while upholding data security and confidentiality	Keller et al. (2023)
Automated Agreements	Creating and implementing smart agreements can mechanise processes in the HSC, including hydrogen dealings, quality certifications, and adherence checks. These agreements ensure accuracy, diminish manual intervention, and heighten operational efficiency	Ferraro et al. (2023)
Carbon Footprint Monitoring	Developing blockchain-driven mechanisms to monitor and counterbalance carbon emissions across the HSC can amplify sustainability. Smart agreements can trigger offset dealings upon achieving predetermined emission thresholds	AlHumaidan et al. (2023)

(continued)

Table 7 (continued)

Areas	Explanation	Reference
Collaborative Networks on Global Scale	Constructing collaborative blockchain platforms involving producers, distributors, regulators, and users can streamline communication and coordination. This optimization can enhance hydrogen production, distribution, and utilisation	Keller et al. (2023)
Validation and Authentication	Implementing blockchain-based validation and authentication systems can verify the legitimacy of green hydrogen and follow its production via renewable sources, thus enhancing market credibility	De Blasio and Hua (2021)
Research and Innovation	Investing in research to tackle issues such as scalability, energy efficiency, and confidentiality can improve the applicability of BCT within the HSC	Merrad et al. (2022)
Public–Private Collaborations	Partnerships between public entities, private corporations, and research institutions can propel the creation and deployment of BCT solutions within the HSC	Merrad et al. (2022)
Advocacy for Policies	Encouraging policymakers to formulate favourable regulations and norms for BCT within the energy sector can establish a conducive environment for its adoption	De Blasio and Hua (2021)
Training and Education	Providing training programs and workshops to stakeholders can augment comprehension of BCT's potentials and encourage inventive applications within the HSC	Raja Santhi and Muthuswamy (2022)

Table 8 Future prospects in the amalgamation of blockchain in hydrogen supply chain

Areas	Explanation	Reference
Interoperability	Research efforts might concentrate on creating standards for interoperable blockchains that would allow various participants in the HSC to easily share information and carry out transactions across various blockchain platforms	Schmid (2023); Kayikci and Subramanian (2022)
Scalability	Addressing scalability issues will be essential as the HSC expands. Future research can look on methods to improve blockchain networks so they can handle the growing amount of data and transactions	Keller et al. (2023)
Regulatory Frameworks	For blockchain applications in the HSC, it will be crucial to build regulatory frameworks and industry standards. To ensure compliance and security, researchers can help to shape these frameworks	Ferahtia et al. (2023)
Integration with IoT and AI	Using blockchain in conjunction with IoT (Internet of Things) and AI (Artificial Intelligence) can result in a HSC that is more reliable and effective. Future studies might look at creative fusions of these technologies for real-time decision-making and monitoring	Dedeoglu et al. (2023)

(continued)

Table 8 (continued)

Areas	Explanation	Reference
Data Privacy and Consent	Future research can investigate ways to deal with data privacy issues and make sure that participants' consent is upheld because blockchain entails the sharing of data across several parties	Karumanchi et al. (2022)
Energy Efficiency	Since the energy consumption of blockchain technology has been an issue in some applications, research might also concentrate on making blockchain networks more energy efficient	De Blasio and Hua (2021)
Long-Term Impact Assessment	It's necessary to assess the long-term effects of integrating BCT into the HSC, ensuring it genuinely contributes to cleaner energy goals	Ferraro et al. (2023)

By embarking on these paths, researchers have the potential to significantly advance the HSC by integrating blockchain technology, leading to increased efficiency, transparency, and sustainability.

Table 9 Combined endeavour towards widescale adoption

Area	Explanation
Standardisation	The lack of standardised protocols for BCT applications in the HSC impedes seamless integration and interoperability among various systems and stakeholders
Scalability	As the HSC adopts BCT, scalability issues develop because to the computing intensity of blockchain networks, possibly slowing down transactions and data processing
Data Privacy	Maintaining data privacy in a transparent blockchain system is a difficult task. It is vital to strike a balance between transparency and the protection of sensitive information
Energy Consumption	Blockchain networks, especially those that use Proof of Work consensus techniques, may require a lot of energy. If the energy consumed is not from renewable sources, this undermines the HSC's environmental goals
Regulatory and Legal Frameworks	Many areas lack clear legislation regarding BCT applications in the HSC. It is vital to establish legal frameworks that accommodate the unique elements of blockchain technology
Adoption challenges	Because of the learning curve involved with blockchain technology, adoption difficulties exist. Industry-wide education and training are required to realise the full potential of BCT in the HSC
Infrastructure Expenditure	Establishing blockchain infrastructure and upgrading legacy systems require considerable expenditure, which may dissuade some stakeholders from adopting new technologies
Save Energy with Blockchain	Finding methods to make blockchain use less energy, especially in situations where it needs a lot of power, could be crucial. This way, the process aligns better with the environment-friendly goals of the HSC
Use Smart Computers	Computers that can learn from data might help improve how the HSC works. They could predict when things need fixing or come up with ways to use resources more wisely

(continued)

Table 9 (continued)

Area	Explanation
Set Rules for Safety	Creating clear guidelines for safe and legal blockchain use within the HSC is important to avoid any unwanted issues
Facilitate Hydrogen Trade	Simplifying hydrogen trading through blockchain could make transactions smoother and more efficient, benefiting various stakeholders

6 Conclusion

This study delves into the integration of BCT within the HSC, aiming to uncover its potential in enhancing transparency, efficiency, and eco-friendliness across hydrogen-related processes. The study showcases achievements encompassing heightened transparency, the automation of smart contracts, diligent monitoring of carbon emissions, and tangible implementations spanning diverse sectors. It also sheds light on hurdles, including the need for scalability solutions, safeguarding data privacy, and managing energy consumption. The research contends that BCT bears the capability to optimise the HSC by allowing instantaneous tracking, secured data exchange, and the automation of key operations. The study emphasises the significance of partnerships among stakeholders, research geared towards energy-efficient blockchain models, and the formulation of regulatory frameworks to ensure secure implementation. Standardised protocols, the amalgamation of artificial intelligence into the ecosystem, and the establishment of tokenised hydrogen assets are cited as means to amplify the effectiveness of BCT within the HSC. Moreover, it calls for educational endeavours to disseminate awareness about BCT's latent capacities and its pivotal role in fostering the sustainable production and dissemination of hydrogen. Ultimately, the study advocates for the continual advancement of research, cross-sectoral collaboration, and the comprehensive evaluation of enduring sustainability ramifications. These actions are poised to harness the transformative potential of BCT, steering the HSC toward a more streamlined and ecologically conscious energy landscape.

In conclusion, the HSC's use of blockchain in the future has enormous potential to improve efficiency, transparency, and sustainability. To fully realise the potential of blockchain in this sector, more study is needed that addresses technological issues, regulatory issues, and interaction with other developing technologies.

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27

Late Last Mile Delivery in a Physical Internet Enabled Context

Konstantinos Zavitsas and Gerasimos Kouloumbis

1 Introduction

Delays in the last mile are typically cumulative in nature reducing delivery capacity and causing second delivery attempts, while increasing cost and environmental impact. Currently, second delivery attempts represent around 20% daily extra cost for operations due to the additional amount of kilometres needed to either come back and try delivery on the same day or to return to base and plan the delivery for the next or a later date. This problem is operational and of dynamic nature due to its real-time aspect. Delivery rounds are typically fully designed daily, prior to initiating their implementation and consider parameters such as the fleet (i.e., the number and capacity of delivery vehicles available) and local accessibility constraints such as Low Emissions Zones (LEZs) or Zero Emissions Zones. LEZs, that are increasingly popular in European cities (Urbanaccessregulations, n.d.).

The last mile delivery round design process involves the pairing of each delivery location with a delivery vehicle as part of a sequence of time-dependent visits, while the time and operational constraints are respected.

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This is typically achieved using a Vehicle Routing Problem (VRP), a popular operations research problem with wide applicability, that has been extensively researched and documented in the literature since the 1960s (Toth & Vigo, 2015). The VRP is a well-known NP-hard problem because it includes the Travelling Salesman Problem (TSP) as a special case (Garey & Johnson, 1979). The largest VRP instances that can be consistently solved by the most effective exact algorithms proposed so far contain about 50 customers, whereas larger instances may be solved to optimality only in particular cases. Heuristics and approximate methods are typically used for solving larger instances to near optimality, as last mile operators may typically deal with 2000 or more deliveries at a daily level.

The last mile of urban parcel deliveries is typically associated to high uncertainty due to the busy city environment. UKs Department for Transport argues that the last mile accounts for around 30% of operators' costs (DfT, 2019). This includes fuel, space rental for consolidation centres, staff employment, vehicle purchase, and other logistics' fixed costs such as insurance and maintenance. Furthermore, for urban parcel delivery rounds, about 50% of the time of a delivery tour, is not spent while driving, but while the vehicle is parked and the handover to the customer takes place (GLA, 2017). Allen et al. (2018) study 25 vehicle rounds in the busy "West End" of Central London and identify that 62% of the total round time was spent with the vehicle parked while the driver sorted, unloaded, and delivered parcels.

A survey of last mile delivery professionals measures the influence of operational issues on the efficiency of the carrier operations in the inner-city area (Aljohani & Thompson, 2020). It finds that parking availability received the highest rating (29%) while regulations of on-street loading zones (OLZ) are ranked as the 3rd most negative issue (13%). Issues related to traffic congestion and physical design of the city centre, were viewed as the second most negative aspects, as traffic congestion received a rating of 24%, whereas street design received a modest rating of 5%. Road traffic conditions worsen in high congestion hours, typically between 6–10 AM and 3–7 PM. During peak traffic periods traffic speed may reduce up to 40%. The average traffic speed (including out-of-peak hours) has been consistently declining in the inner-city areas. This adversely impacts the travel time and the reliability of deliveries. This figure may vary from city to city. For example, vehicle delays are reported to have increased by 31% in central London (Allen et al., 2018).

With respect to parking availability, a study performed in Seattle, USA, revealed that couriers had used the kerbside to park their vehicles during deliveries to about 87% of all buildings in the downtown area (UW SCT & LC, 2018). Similarly, it is reported that couriers had to use the kerbside for

about 95% of deliveries in central London (Allen et al., 2018). The productivity of couriers decreases with longer walking distances, as they become less able to carry multiple parcels in each walking trip and therefore parking their vehicles very close to the retailers and businesses can have a significant efficiency impact.

Urban delivery uncertainty arises from urban road traffic, parking spot availability as well as during the handover process significantly influencing last mile delivery operations. This paper proposes an algorithm to update delivery schedules of vehicles on the go, by sharing the load of late running vehicles with the ones that are ahead of their schedule or have a lesser workload. This paper is structured into six sections. Section two presents past studies on the state-of-the art of last mile delivery focusing on operational aspects and extending to the concepts of collaboration between operators and the Physical Internet (PI). Section three is a methodological section where the parcel reshuffling algorithm is introduced. Section four describes the implementation of the algorithm into a use case in Madrid, Spain, through a tailor-made simulation approach, and the parametrisation of the simulated environment. Section five presents the results of the case study and discusses the findings on the performance of the algorithm and its limitations. The paper concludes with Section six with a summary of results and recommendations for further work.

2 Literature Review

This paper examines the last mile delivery problem from an operational aspect that exploits vehicle and operator collaboration. This approach is highly relevant to the concept and principles of the Physical Internet as it utilizes the benefits associated to dynamic tracking of parcel deliveries and vehicle fleet, the modularized containerization of parcels as well as open collaboration between operators. The literature review is divided into three sub-sections dealing with the solution algorithms for last mile delivery route design, innovations for collaboration in last mile delivery and the concept of the Physical Internet.

Operators currently monitor the implementation of last mile delivery rounds through vehicle GPS sensors, and software that track the delivery progress (typically through parcel barcode scanning). The delivery rounds progress is then visualized on dashboards as the one shown in Fig. 1, where each row represents a delivery round. Its progress is captured through a series of delivery timestamps that represent the planned (or executed) delivery

time of a specific parcel, spanning in this case from 9AM to 9PM. Green vertical lines represent successfully completed deliveries, orange lines represent pending deliveries, while red line represent unsuccessful deliveries. The visualization of the delivery rounds enables the manual tracking of delivery progress, and the identification of severe delays. The red vertical line at 3 pm, captures the current time, and enables progress inspection.

The sequence at which van drivers choose to implement the delivery round does not always align with the delivery planned route, as experienced delivery drivers have tacit knowledge about the complex operational environment in which they serve customers daily. They know which roads are hard to navigate, when traffic is bad, when and where they can easily find parking, which stops can be conveniently served together, and many other things that are hard, if not impossible to formalize in an optimization model. This tacit information is therefore not contained in most route planning tools used in the industry, causing drivers to frequently deviate from originally planned route sequences. Taking into account their tacit knowledge, drivers follow a deviated actual route sequence instead, which is potentially more convenient under real-life operational conditions. For example, in Fig. 1 route C17 (first row) seems to follow the proposed sequence and be roughly on-time, while for round C24 (last row) the driver seems to have adjusted the delivery sequence.

A recent stream of research proposes a two-step approach that first uses machine learning to predict the "real" route taken by a driver, and then integrates this prediction into the Travelling Salesman Problem with Time Windows (TSPTW). This is achieved by adding a constraint which limits the extent of deviation between a suggested tour and the predicted driver tour (Dieter et al., 2023). Introducing Machine Learning in the route determination process requires the availability of rich historical delivery data. As part of 2021 Amazon Last Mile Routing Research Challenge, a real-world dataset that includes rich historical last mile delivery data in the USA has been made available that enables model training (Merchán et al., 2022).

Fig. 1 Delivery round monitoring dashboard (*Source* Authors)

To address the uncertainty in logistics and last mile delivery beyond reliable routing, the more efficient utilization of vehicle and infrastructure resources is examined. Synchromodality and collaboration are two key concepts exploring a sustainable and, at the same time, efficient supply chain that reduces its environmental impact while optimizing resource utilization. Synchromodality is an evolution of multimodal, intermodal, and combined transport and it involves different and strongly integrated transportation modes. Synchromodality builds on the vision of high-level flexible management of corridors proposed by intermodal transport to meet the wider interests of the whole supply chain, not only for a specific shipment or company (Giusti et al., 2019). In the context of last mile delivery multiple modes including delivery vans, regular and electric cargo bikes, push carts, drones and crowd-sharing can contribute towards improving sustainability. The Physical Internet builds on the concept of synchro-modality and aims to implement a network of independent physical distribution networks in which operations for transport, transshipment and storage are performed openly through fair and evident mechanisms based on standardized protocols. This physical network-of-networks is envisioned as enabling the efficient flow of goods between senders and end-customer through distribution, consolidation, and intermodal centres/hubs (Dong & Franklin, 2021).

Operational collaboration in last mile delivery has also been examined separately to the overarching concept of the Physical Internet. Collaboration between operators can be facilitated through platforms to negotiate and allocate orders and capacity among participating carriers (Guo et al., 2021; Los et al., 2020). Operational collaboration is typically examined in the context of commonly established and operated urban consolidation centres (UCC) and the responsibility of arranging fleet capacity for last mile deliveries is of the UCC operator (Handoko et al., 2016; McLeod et al., 2020). The concept of dynamic collaboration during delivery involving the redesign and redistribution of vehicles on the go is rarely considered.

2.1 Dynamic Collaborative Parcel Reshuffling Algorithm

As delivery round delays are being undertaken, the original planning and design of the rounds might require to be updated due to external uncertainty. Operational constraints, such as delivery time windows (including no late-night deliveries) and driver shift hours, should not be violated. As uncertainty arises, a fleet operator tries to identify delivery rounds that might finish early or be ahead of schedule and dispatch them for helping the round

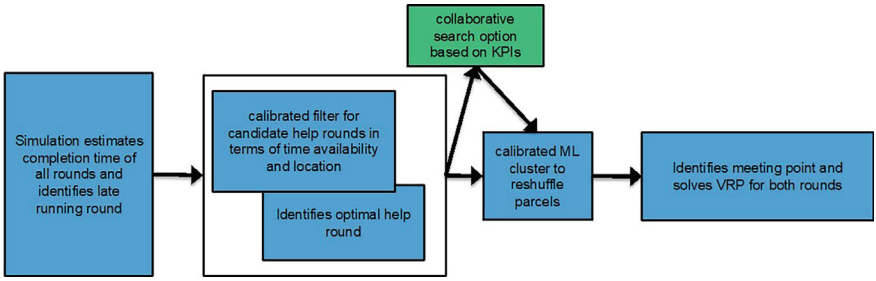

Fig. 2 Parcel reshuffling algorithm with collaborative search option (*Source* Authors)

running late. The process of identifying van availability, van suitability and then redesigning the delivery rounds, that involves identifying which parcels will be moved from the original van to the helping van, and where the two should meet for the parcel exchange will take place is currently undertaken manually. A dynamic algorithm is proposed for automating this process, aiming to alleviate human related constraints and costs and enable the faster identification of considerably more efficient solutions. The algorithm can be extended to account for collaboration between last mile operators, as higher participation is last mile logistics is typically found to yield route reduction and distance savings (Aktas et al., 2021). The reshuffling model (see Fig. 2) is an algorithm that seeks the better utilization of resources during delivery through informed redistribution of parcel delivery jobs assignment. The algorithm can be implemented either within a single company context, when it fosters collaboration between drivers and vehicles, or in an open multi-operator collaborative context. For the multi-operator collaboration to be implementable a platform that collects delivery progress for all vehicles operating within a specific area is required. Such a platform is envisaged as a municipality initiative with the aim of delivering efficient and sustainable operations.

The algorithm is composed of three stages, with the first stage identifying the nearest available help rounds, the second stage addressing the redistribution of parcels, and the third the redesign of the delivery routes.

2.1.1 Step 1: Nearby delivery rounds identification

As illustrated in Fig. 3, the first step of the process involves identifying all the delivery rounds operating in proximity. Following the openness principles of the PI, the proposed algorithm can consider the delivery rounds of one or more operators as candidates for helping the delivery round running late. The

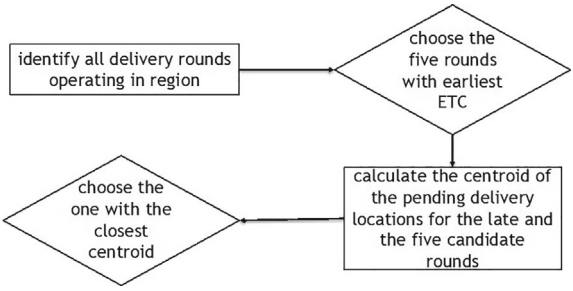

Fig. 3 Algorithm for nearby and available help round identification (*Source* Authors)

process firstly filters the rounds in terms of estimated time of round completion, to identify the ones with higher availability, and then identifies the average centroid for each round considering all pending delivery locations.

2.1.2 Step 2: Parcel delivery job redistribution

Once the optimal help round has been identified, and it is confirmed that it is operating in proximity to the late running round, as well as that it has sufficient spare time for handling additional parcel deliveries, the task of reshuffling the parcels pending delivery for the two vehicles is initiated. The aim of this task, is to identify which of the pending parcels of the two rounds, should be delivered by which vehicle, to alleviate overall delivery delay. The late running round requires to share some of its load with the helping round, however up to this point it is not clear which ones should be transferred.

A Machine Learning K-means clustering algorithm is applied to the dataset with the number of clusters parameter of the K-means algorithm set to two, to reflect each vehicle. The K-means algorithm utilizes a calculation based on the moving centroid of its cluster to determine to which cluster a node will belong. The K-means algorithm compulsorily assigns all nodes to one of the two clusters, leaving no nodes unassigned. The algorithm is applied using real-world travel times as a distance criterion for vector separation of a node to the cluster centroid.

The output of the clustering algorithm is a new tag for each node of the population that corresponds to a unique delivery round. Each tag is then associated to each of the two delivery rounds, by considering the original assignments, which ensures that the correct delivery round tag is set on each cluster. The parcels can then be classified into the ones remaining in the late running round, the ones remaining in the help round, and the ones moving from the late running round to the help round.

2.1.3 Step 3: Meeting point and delivery round redesign

After reshuffling the parcel delivery locations, and establishing the area moving to the help round, it is required to convert that information to instructions for the two van drivers. In effect, this includes the new routes for both vehicles, that incorporate a meeting point, and the information on which parcels require to be transferred from the one van to the other.

The meeting point requires to be determined prior to addressing the vehicle routing decision. The meeting point necessitates proximity of the two vans, as well as limiting the waiting time for transhipment involved in the process. To identify two locations with proximity that are suitable for serving as the meeting point, the locations of the parcels remaining on the late running round, and the location of the parcels remaining on the help round are considered. The locations of the parcels moving from the late running round to the help round are excluded from this process, as prior to the exchange at the meeting point, they are loaded on the incorrect van. The distances between all locations are calculated and the two points with the closest distance are identified. From the pair, the meeting point is set to be a location originally included in the late round to alleviate further delay. This location is then added to the locations the help round requires to visit.

The meeting point represents a proximity location suitable for both vans to visit, however there is no guarantee up to this point that the two vans arrive there simultaneously. To address this, a common time window is set on both vans for reaching the meeting point. The time window may be set to start in 30 minutes from the current time, and last for 30 minutes. Start time and duration of the time window may be adjusted based on the position of the vehicles in comparison to the meeting point and the time available until the 9 pm cut-off. If no routing solution can be found the meeting point time window is relaxed, by either delaying its start time, or expanding it, or both.

A Travelling Salesman Problem with time windows is then solved for each vehicle, including a common time window for reaching the meeting point, while no time window constraints are considered for all other locations.

3 Case Study—Madrid, Spain

There are two datasets provided by CityLogin, one for 16 July 2021 that contains 1842 delivery jobs and one for 29 November 2021 that contains 4513 jobs. To enable the multi-operator analysis, for both datasets the jobs have been evenly split into four operators.

To enable the development of solution strategies and for assessing the effectiveness of the proposed algorithm in addressing last mile delivery rounds delays, a simulation model has been developed that starting at 9AM, records the progress of all delivery rounds at one-minute intervals. The simulation terminates at 9PM when all rounds require to be completed. Furthermore, the simulation applies stochastic delays to the vehicle rounds based on traffic data. Furthermore, a stochastic parking and handover duration is assumed using a normal distribution with mean of 6 minutes and standard deviation of 3 minutes. The simulation does not consider deviations from the round sequence as there is no way of telling what kind of impact such deviation might have in the route design process. In the simulation, if for any delivery round the estimated time of completion is found to be later than 9PM, then an alert is triggered providing the information on the vehicle round that appears to be running late, and its current location.

Figure 4 illustrates all the delivery locations for the late running round (leftmost light blue) and the five-candidate help rounds. The figure confirms the quantitative finding obtained using the delivery round centroid calculation approach that the round depicted in pink is operating in proximity to the late running round (light blue) and is therefore considered as the ideal assistance option.

The next step of the algorithm is to undertake the parcel reshuffling. Figure 5 illustrates how the boundary separating the delivery locations of the late running and helping delivery rounds changes before (dashed line) and after (continuous line) the parcel reshuffling step. The delivery locations border for the help round, which is operating at the bottom right of Fig. 5, moves further north claiming the shaded area that after the parcel reshuffling is served by the help round.

The algorithm then identifies a convenient meeting point and reassesses vehicle routing. Figure 6 reflects the redesign of the two rounds that includes a meeting point. The two vehicles meet simultaneously (at 12:18 which is the start of the time window), to exchange the parcels, and then the help round (shown as blue) delivers the additional parcels. Both vehicles' rounds are scheduled to be completed before the 9 pm cut-off, effectively addressing the delay.

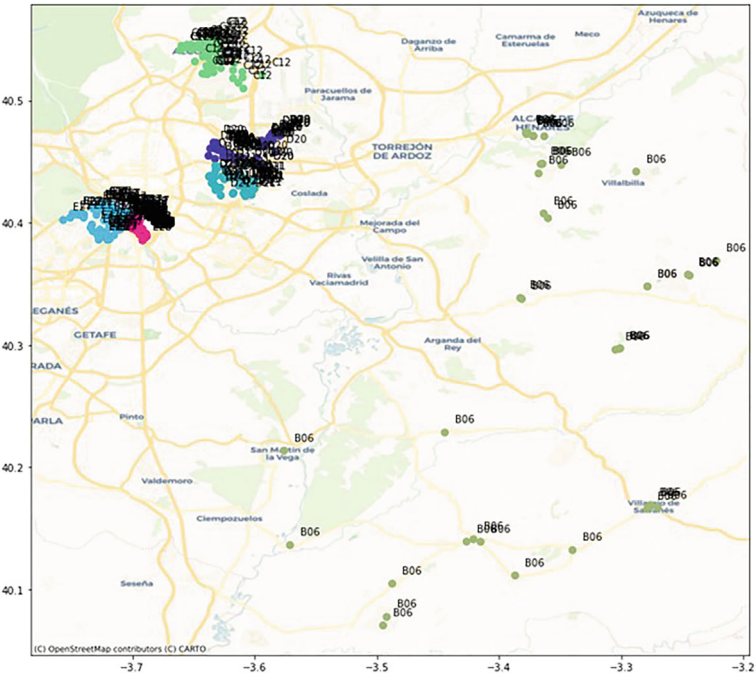

Fig. 4 Pending delivery locations for the late running round and the five-candidate help rounds (*Source* Authors)

Fig. 5 Area served by the help round after parcel reshuffling (*Source* Authors)

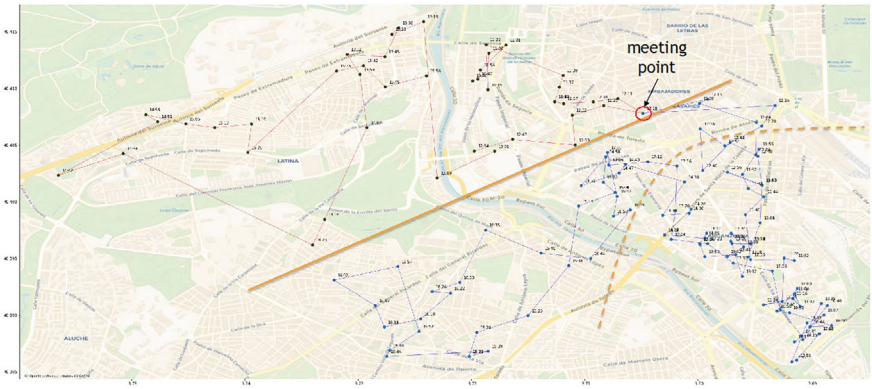

Fig. 6 Vehicle rounds redesign with meeting point using VRP with time windows (*Source* Authors)

4 Results and Discussion

The aim of this case study is to establish a comparative standard in last mile operations when a delay arises. The comparison is between the following four scenarios:

- *Scenario 1 (Sc1): Worst case:* There is no action when a delay is observed. In this case the vehicle will have to return with all undelivered parcels back to the distribution facility and attempt to redeliver them the following day in a dedicated run. This is the baseline and worst-case scenario.
- *Scenario 2 (Sc2): Current practice:* When a delay arises, the operator searches manually for a vehicle that is operating ahead of schedule to assist. The vehicles meet, exchanges parcels and the drivers are responsible for adjusting their routes. When a help vehicle is not available the parcels are returned to the distribution facility and if possible, they are loaded another round the following day, leading primarily to loss of capacity. As illustrated in Fig. 4, a suitable help vehicle can be identified effectively using only visual means. The parcel's reshuffling, the identification of a suitable meeting point and producing updated routing instructions are challenging to be undertaken manually. As it is difficult to capture those inefficiencies in a simulated context, the scenario is not considered further.
- *Scenario 3 (Sc3): Reshuffling service:* The reshuffling service is responsible for identifying the help vehicle from the fleet of the operator, re-assigning the parcels, identifying a meeting point, and redesigning the routes.

- *Scenario 4 (Sc4): Marketplace reshuffling:* A marketplace is made available where vehicles of any operator can assist the delayed vehicle. The reshuffling service is responsible for identifying optimal help from any vehicle operating in proximity regardless of operator, re-assigning the parcels and redesign the routes.

Table 1 summarizes the findings of the analysis and allocates a qualitative scale ranging from, meaningless to no improvement to useful.

The parcel reshuffling algorithm is found to provide meaningful improvements in the cases of low demand when reshuffling within a single operator's fleet, while delivering no improvement if other operators fleet is considered. In the high demand case, no value adding solutions were identified when searching within a single operator fleet, however, reshuffling was particularly

Table 1 Qualitative performance assessment of parcel reshuffling algorithm

	Low demand	High demand
Sc3: Reshuffling	*Useful*—A vehicle with lighter workload was identified and a meaningful collaboration was set up enabling both vehicles to complete all their tasks on-time	*Meaningless*—All candidates who help vehicles were found to be busy and offering assistance to a late running round was causing more disruption (meeting point, parcel exchange) than assistance. Therefore, in this context the added value of the proposed solution is minimal
Sc4: Marketplace	*No improvement*—A vehicle with lighter workload was identified and a meaningful collaboration was set up enabling both vehicles to complete all their tasks on-time. However, the solution did not appear to offer a significant improvement over the solution provided in Sc3. As the model does not consider the loss of volume and data sharing, operators might find that the disadvantages outweigh the advantages	*Useful*—A vehicle with lighter workload was identified and a meaningful collaboration was set up enabling both vehicles to complete all their tasks on-time

Source Authors

meaningful when multiple operator fleets were considered. The results indicate a correlation between the demand and the opportunities that arise in identifying suitable help vehicles. In low demand, a small fleet was sufficient for identifying suitable help, and increasing the fleet did not provide much additional benefit. In high demand, a small fleet was not sufficient for identifying suitable help, however, when multiple operator fleets became available in Sc4 a meaningful solution was identified.

5 Conclusions

The dynamic algorithm developed offers a solution to the last mile delivery rounds delays in the urban context. The tool is useful for last mile operators addressing delivery delays, but it also offers an opportunity for collaborative last mile logistics, as it creates fair criteria for assessing the helping rounds available in a region. The algorithm is end-to-end in determining, and identifying optimal help rounds, as well as dealing with parcel reshuffling and delivery round redesign through a meeting point. As multiple last mile operators run delivery rounds in proximity, integrating the proposed methodology in an urban collaborative logistics platform offers fair criteria for identifying and implementing mutually beneficial collaboration opportunities.

Higher first attempt delivery success is key for last mile delivery operators, as each package returned to the warehouse due to a failed or out of time delivery generates economic, social, and ecological costs. Currently, second delivery attempts represent around 20% daily extra cost for operations due to the additional amount of kilometres needed to either come back and try delivery on the same day or to return to base and plan the delivery for the next or a later date. The solution presented in this paper enables an algorithm to minimize the impact of problems that affect the entire last mile industry and arise due to delivery uncertainty. The algorithm was assessed in a case study based on real-world data in Madrid, Spain, in low and high demand and in single or multiple operator contexts. The results indicate a correlation between the delivery demand and the helping vehicle pool size. Further research will investigate this relationship further and attempt to integrate an intermediate step for identifying appropriate conditions for expanding the helping vehicle pool to other operators' fleet.

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28

Revolutionizing Last-Mile Delivery: Discovering Disruptive Technologies and Innovative Solutions

Ozgur Kabadurmus, Yasanur Kayikci, and Fatma Nur Karaman Kabadurmus

1 Introduction

One of the most critical parts of a customer's order experience is the last-mile delivery (LMD), which handles the last stage of an order's transportation. Customer satisfaction regarding delivery speed, accuracy, and overall service are determined mostly during LMD operations. High quality last-mile operations can increase customer satisfaction (Byrd, 2020; Southin, 2018), and therefore, help increasing customer retention. On the other hand, poor last-mile performance (e.g., missed deadlines, incomplete deliveries, or a lack of transparency) can result in a decrease in customer satisfaction, a deterioration of trust, or even the loss of business to the competition. In order to meet (and

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exceed) customer expectations in the increasingly competitive business environment of today, last-mile service providers (LSPs) and online retailers must improve the effectiveness of their LMD operations. LMD is often inefficient and costly, especially in urban areas, as up to 40% of a product's shipping costs occur in LMD (Kayikci et al., 2023).

There are a number of difficulties with LMD performance brought on by the COVID-19 epidemic (Kim, 2021). LSPs have experienced major delays, bottlenecks, and many supply chain disruptions as a result of the sharp rise in e-commerce activities driven by lockdowns, social distance requirements, and reluctance for in-person buying (Byrd, 2020; ERTICO, 2020; Lund et al., 2021). LMD activities became more complicated during COVID-19 to safeguard the health of customers and delivery personnel (Bombelli, 2020). Additionally, there are now more issues with LMD regarding infrastructure, technology, and personnel as a result of customer demand for contactless and on-time deliveries (Cordon, 2020). Lastly, demand forecasting and resource allocation became much more challenging because of the pandemic's unexpected course. These difficulties raised the possibility of service delays, dissatisfied consumers, and financial strain for businesses using LMD.

To overcome these challenges in the LMD operations, the pandemic facilitated innovative applications, such as, Artificial Intelligence (AI), Machine Learning (ML), Big Data Analytics (BDA), Blockchain, Digital Twin, Internet of Things (IoT), 3D printing, and Augmented Reality (AR)/Virtual Reality (VR). These new disruptive technologies can help companies to improve LMD performances and remain competitive (Ivanov, 2021; Queiroz et al., 2022). In response to the increasing demands of their customers for speed, convenience, and transparency, the majority of LSPs modified their distribution procedures, strategically relocated their facilities, and made investments in new machinery. Some of these LMD businesses also made investments in the newest innovative technologies with the goal of improving their fleet management, inventory control, warehousing, capacity planning, and delivery operations. Our review of the existing literature shows that there is no existing research investigating the use of disruptive technologies and innovative solutions in LMD operations. Therefore, this study contributes to the literature by filling this research gap. The primary objective of this chapter is to examine the advancements in utilizing disruptive technologies and state-of-the-art distribution strategies in LMD operations. We also investigate the possibilities these technologies offer as well as their implications, advantages, and challenges from the perspectives of both LSPs and customers. By revealing the potential of these technologies, researchers and professionals can effectively harness them within LMD systems to not only meet customer

expectations but also surpass them more efficiently. This study seeks to answer two research questions: *What are the disruptive technologies and innovative solutions available in LMD operations? What are the potential and adoption challenges of disruptive technologies in LMD?*

This research is conducted based on compilation and analysis of existing literature and real practices to answer these research questions.

This research is conducted based on the collection, compilation and analysis of existing literature and real practices to answer the above-mentioned research questions. The rest of the chapter is organized as follows. The second section provides an overview of the LMD system and LMD processes. The third section presents current applications of disruptive technologies for LMD operations, where delivery vehicles, delivery operations and distribution strategies, and the potential use of disruptive technologies and innovative solutions in LMD processes are illustrated. The adoption issues of disruptive technologies in LMD operations are discussed in the fourth section. The last section concludes the chapter.

2 Overview Last-Mile Delivery System

In general, LMD consists of five processes (Huang et al., 2009) along the LMD chain which extends from the suburban warehouse through urban consolidation centre and microhub with connection to the physical store or mobile depot and to the delivery point/pickup point in the LMD network: shopping (online and/or in-store), packing, delivery, pickup, and return process as presented in Fig. 1 with the use of some LMD vehicles and solutions. These steps are defined in detail:

(1) *Shopping:* Although in-store purchases can also start the LMD process, online orders have increased significantly and correspond to the largest share of LMD orders. This process can be initiated from all omnichannel options and can result in delivery from the warehouse or picking up from a store and a pickup point.
(2) *Packing process:* The preparation of products for shipment involves several steps. Typically, this process includes boxing up the products, labelling the boxes, and palletizing or containerizing the cargo to be loaded to the vehicle. Also, more packing may be needed to be done in the warehouse or urban consolidation centre, and or at the store.

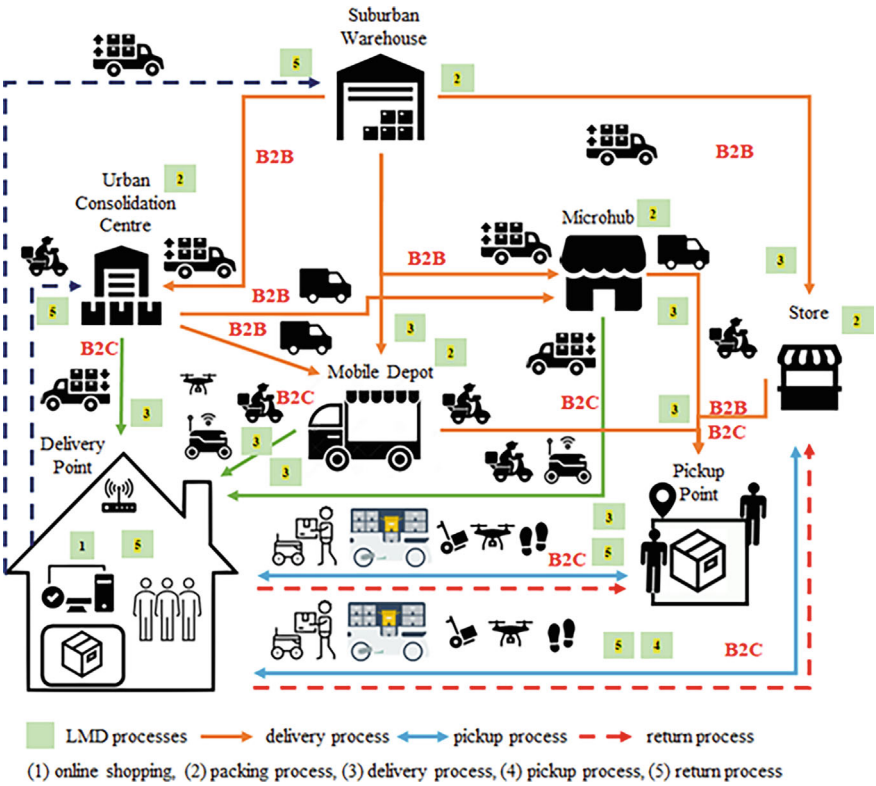

Fig. 1 A typical of last-mile delivery system for omnichannel retail (*Source* Authors)

(3) *Delivery process:* This step involves delivering the product to the customer's home or pickup points using the delivery network of LMD service provider. There are multiple deliveries during this step: delivery from the warehouse to the consolidation centre, from the consolidation centre to the store, from the store to the pickup point or residential home.

(4) *Pickup process:* This process can be done using various pickup points, such as hub/microhub, parcel lockers, collection point, store (in-store pickup), post office, another transfer centre. This process is also termed as Pick-Up and Drop-Off (PuDo).

(5) *Return process:* This process is required in case a return is initiated by the customer. Similar to distribution, the return process is also omnichannel. After the customer starts the return process, the product is either taken from customer's home by an LMD operator or brought back to a store, pickup point, or post office to complete the return. At the end of the return process, the customer can get the original form of payment, exchange for another item (identical or different), or store credit.

3 Applications of Disruptive Technologies for Last-Mile Delivery Operations

The general procedure of traditional LMD operations starts when the parcel arrives at the urban consolidation centre. Upon sorting the parcels according to their destination area (usually by postal code and delivery time window), vehicle routes are determined. The routes can be optimized dynamically according to the current demand or there can be fixed routes that each vehicle covers. The vehicles distribute their loaded parcels according to their routes, but any undelivered cargo returns to the distribution centre to be rescheduled for another delivery or waited for customer pickup from a pickup location.

In addition to this delivery operation, there can be returns or a new shipment from a pickup point to another point within or outside of the distribution area of the vehicle. In the traditional LMD operations, as the routes are fixed before the vehicles started distribution, parcels are taken to the consolidation centre to be scheduled for delivery.

As the next subsections explain in detail, new disruptive technologies have been started to be used in LMD operations to improve the efficiency of this traditional operation.

3.1 Delivery Vehicles

For the last decade, low- or zero-emission vehicles and autonomous delivery technologies have been matured for widespread usage. Thus, new vehicle types and delivery technologies became available for LMD providers. We categorized delivery vehicles into low- or zero-emission vehicles, cargo bikes (cargo cycles), autonomous vehicles, hybrid LMD vehicles, and other vehicles as follows:

- Low- or zero-emission vehicles: Electric vehicles called as electric drive vehicles (EDVs) or battery-electric vehicles (BEVs) are driven solely by one or more electric motors powered by energy stored in batteries.
  - o *Light electric vehicle* (LEV),[1] small vehicle weighing less than 100 kg that run on electricity.
  - o *Electric light commercial vehicle* (e-LCV), used for commercial purposes, they have a gross vehicle weight of no more than 3.5 metric tonnes—defined by the EU as an N1 vehicle.

[1] https://ulaads.eu/low-and-zero-carbon-vehicles-for-urban-last-mile-deliveries/

- *Electric delivery van (E-van)* is one of the electric drive vehicles (EDVs), regular-sized van that runs on electricity instead of petrol and an internal combustion engine.
- *Low-emission vehicle* (CNG and EURO V Diesel Trucks)
- *Plug-in Hybrid Electric Vehicle* (PHEV), a hybrid vehicle with a battery pack that can be recharged both internally and externally by plugging a charging cable into an external electric power source, as well as by its onboard internal combustion engine (ICE)-powered generator.

- Cargo bikes (cargo cycles) (ULaaDS, 2020):
  - *E-assist cargo bike*
    Small e-cargo bike (15 kg, 2 wheels) – e.g., Ebike4delivery
    Medium e-cargo bike (100 kg, 2 wheels) – e.g., Urban arrow L
    Medium e-cargo bike with trailer (360 kg, 4 wheels) – e.g., Larry vs Harry Bullitt + Carla Cargo
    Large e-cargo bike (200 kg, 3 wheels) – e.g., Rytle Movr25
    Large e-cargo bike with trailer (200 kg, 4 or more wheels) – e.g., Velove Armadillo, UPS eQuad, EAV 2cubed
  - *Pedal-only cargo bike* (Narayanan & Antoniou, 2022)
    Postal delivery bikes (50–75 kg, 2 wheels)
    Longtail bike (50–100 kg, 2 wheels)
    Frontloader/Long John bike (150–200 kg, 2–3 wheels)
    Backloader, heavy load (500 kg, 3–4 wheels)
- Autonomous vehicles (ULaaDS, 2020):
  - *Indoor autonomous vehicles*
    Intra-logistical drones
    - Visual inspection
    - In-house transport

    Robot assisted automation system.
    - Cube-storage automation - e.g., AutoStore
    - Smart robot platform – e.g., Ocado

    Ground-based autonomous vehicles
    - Automated guided carts (under ride) and vehicles (AGC and AGV)
    - Autonomous mobile robot (AMR)
    - Automated forklifts
  - *Road-based autonomous vehicles*
    Sidewalk robots

- Follower sidewalk robots – e.g., Starship[2], ALEES[3], LMAD[4] for on-site parcel deliveries
- (Semi-) autonomous sidewalk robots[5] – e.g., Amazon Scout[6], UrbANT[7], TeleRetail, Eliport, BoxBot, Kiwibot, Udelv, Nuro, Robby technologies etc.

Road robots (<3.5 tonnes)

- Low speed vans
- High speed vans

Road robots (>3.5 tonnes)

- Autonomous trucks
- Platooning (Truck Platooning)

o *Aerial autonomous vehicle*

Drones

- Multirotor drones – e.g., Amazon prime air drone[8]
- Hybrid drones

- Hybrid LMD vehicles:

o *Ducktrain*

The trailer duck
The follow me duck
The auto duck

o *Truck-robot vehicle*

- Other vehicles:

o *Small delivery truck*
o *Small delivery van*
o *Motorbike / motorcycle delivery*
o *Scooter delivery*
o *Quads / quad bike (ATV) delivery*
o *Buggy delivery*

[2] Startship's autonomous package delivery robot - https://www.starship.xyz/

[3] ALEES, self-driving logistical electric units for urban environments https://vil.be/en/project/alees/

[4] LMAD https://www.lmad.eu/news/adr-industrial-site-nokia-france/

[5] 10 autonomous robots for last-mile deliveries, https://www.practicalecommerce.com/10-autonomous-robots-for-last-mile-deliveries.

[6] Amazon Scout—fully-electric delivery system – https://www.aboutamazon.com/news/transportation/meet-scout

[7] UrbANT, development of an urban, automated, user-oriented (in German "nutzerorientiert") transport platform - https://urbant.de/en/

[8] Prime Air drone deliveries - https://www.aboutamazon.com/news/transportation/amazon-prime-air-prepares-for-drone-deliveries

Electric vehicles and hybrid electric vehicles are other new technologies that the logistics service providers have been using for LMD. Even before the pandemic, cargo companies started to include electric trucks into their fleets. For example, UPS invested in 10,000 electric trucks made by Arrival just before the start of the COVID-19 pandemic (UPS, 2020). Among many startups for producing electric delivery vans, Rivian Automotive, Arrival and Canoo made significant contracts with logistics providers for electrification of their fleets (Sutherland & Bryant, 2021). Also, other startup companies have been working on creating a variety of electric vehicles, including e-cargo bikes. For example, an Italian company, I-FEVS (I-FEVS, 2023), has been developing interactive, fully electrical vehicles and three-wheeled electric bicycles for last-mile distribution. The transformation of regular diesel fleet to hybrid or electric trucks provides significant savings for LMD companies in terms of total cost of ownership and operating costs (Winston, 2018). However, the use of electric vehicles is limited by the number of charging stations over the distribution network (Jochem et al., 2019).

According to a recent review (Narayanan & Antoniou, 2022), bikes and e-bikes have also become popular for LMD operations. A few examples of these electric and electric assisted LMD vehicles are UPS's *electric cargo tricycle*[9], *UPS eQuad electric Quadricycle*[10], *and EAV'S 2CUBED*[11]. Additionally, New York City (NYC) Department of Transportation (DOT) started the Commercial Cargo Bike Pilot program, which allowed local carriers to use bicycles with pedal-assist in 2019 and the number of cargo bicycle deliveries increased by 109% during the first year of COVID-19 pandemic (from May 2020 to January 2021) (NYC DOT, 2023). NYC DOT also estimates that a delivery truck can be replaced by two pedal-assisted bicycles to reduce CO_2 emissions by 14 tonnes per year. As of August 2023, NYC plans to allow the use of larger pedal-assist cargo bikes (a change from 36 inches wide bikes to 48 inches) for distribution (New York City, 2023). The city also reports that there have been 130,000 cargo bike trips in 2022 only, which corresponds to the delivery of more than 5 million packages, even though 90% of NYC's freight distribution relies on cargo trucks.

The use of autonomous delivery vehicles (ADVs) as alternative LMD vehicles (such as droids and drones) has been attracting attention. Worldwide, companies are increasingly adopting light autonomous vehicles for parcel

[9] UPS's electric cargo tricycle: https://about.ups.com/us/en/our-stories/innovation-driven/lessons-from-the-e-bike-journey-of-ups.html.

[10] UPS's eQuad electric Quadricycle: https://www.nbcnewyork.com/new-york/ups-tries-new-way-to-efficiently-deliver-cargo-in-big-cities-like-nyc-4-wheel-cycles/3734162/ (max 200 kg payload, 2000 litres).

[11] EAV'S 2CUBED: https://eav.solutions/ecargo/ (max 150 kg payload, 2000 litres).

delivery. For example, FedEx Express has collaborated with Neolix to test autonomous deliveries in China[12], while DPD has partnered with Clevon to implement similar autonomous delivery solutions in Estonia[13]. These initiatives reflect a growing trend towards automation in last-mile logistics, aiming to enhance efficiency and reduce operational costs. Ground-based autonomous vehicles (i.e., droids) are of interest because they may replace delivery workers. A few examples of those droids are follower sidewalk robots (e.g., Starship[14], ALEES[15], LMAD[16]) and autonomous/semi-autonomous sidewalk robots[17]—e.g., Amazon Scout[18], UrbANT[19]. There are also road robots that can be designated as light (<3.5 tonnes) and heavy (>3.5 tonnes) autonomous vehicles. Some of these road vehicles have the capability of platooning, which allows them to move as a group in a coordinated way so that they drive safely at high speeds with very small spacing between the vehicles (Martinez-Diaz et al., 2021). Similarly, "ducktrain"[20] is another system similar to platooning, but it operates multiple automated light electric vehicles ("ducks") from a microhub to a delivery location (Schomakers et al., 2022). Different from the platoons, after the ducktrain reaches to the delivery zone, its individual vehicles can separate from the train and complete their deliveries individually before forming the ducktrain and returning to the microhub. Even though this innovative solution attracted much attention from companies, ducktrain declared bankruptcy in June 2023, showing the difficulties of staying in business with new technologies and solution development in LMD operations.

Electric battery drones are mainly being researched for the quick delivery of small packages and essential items like medications because of their limited load capacity. Due to their capability of bypassing traffic congestion and providing direct transportation, autonomous aerial vehicles (i.e., drones) have become increasingly appealing to delivery companies. This capability can be extremely helpful for humanitarian missions or disaster relief operations, as well as vaccine delivery during pandemics. For example, (Enayati et al.,

[12] https://newsroom.fedex.com/newsroom/asia-english/fedex-tests-autonomous-delivery-vehicle-in-china

[13] https://www.parcelandpostaltechnologyinternational.com/news/vehicles-fleet/self-driving-delivery-robot-launches-in-estonia.html

[14] https://www.starship.xyz/

[15] https://vil.be/en/project/alees/

[16] https://www.lmad.eu/news/adr-industrial-site-nokia-france/

[17] https://www.practicalecommerce.com/10-autonomous-robots-for-last-mile-deliveries

[18] https://www.aboutamazon.com/news/transportation/meet-scout

[19] https://urbant.de/en/

[20] https://www.tuvie.com/ducktrain-electric-automated-logistic-vehicle-for-future-urban-last-mile-delivery/

2023) presented a mathematical model and solution methodology for multimodal vaccine delivery operations by drones in Vanuatu. Although Amazon first introduced its plans for drone delivery operations in December 2013 and other companies (such as Alphabet, Walmart, and UPS) started testing drone deliveries, only real drone delivery operation is conducted in Africa by "Zipline Drone Delivery & Logistics"[21] for global health (Campbell, 2022). However, the use of drones in urban areas with dense populations is still in progress due to certain limitations with the technology and its potential application (such as air corridors, suitable drop points, and operational issues due to inclement weather). Even though LMD using drones is not readily available for logistics companies, several efficient use cases of drones—for example, drone assisted truck delivery ("flying sidekicks") models (Murray & Chu, 2015; Murray & Raj, 2020)—are identified by researchers. These sidekick operations can be extended to droids and vans for urban LMD operations (Ostermeier et al., 2022). An example of sidekick droids and vans for LMD is "Robovan", which integrates specially adapted Mercedes-Benz vans with autonomous delivery robots developed by Starship Technologies[22].

3.2 Delivery Operations and Distribution Strategies

Microhubs and parcel lockers are two new distribution strategies that have been practised more widely since the pandemic. Both microhubs and parcel lockers are closer to the delivery point than the distribution centres (Janjevic & Ndiaye, 2014). Customers can retrieve their package using the parcel lockers and self-storage units located near their address, at their convenience without waiting for the delivery. Microhubs (i.e., micro-distribution facilities or micro-depots) are small consolidation centres located closer to the actual distribution zone (Katsela et al., 2022). They are smaller than urban consolidation centres since urban consolidation centres serve a larger zone that can only be covered by many microhubs and also have a smaller distribution radius (Janjevic & Ndiaye, 2014). Due to the limited distribution range of microhubs, last-mile providers are able to use electric vehicles, bicycles (electric or traditional) or self-driving vehicles/drones. Katsela et al. (2022) summarized 17 microhub applications from Northwestern Europe (seven companies from Belgium, France, and the UK), Scandinavia (three companies), the United States (five companies from the states of New York,

[21] https://www.flyzipline.com/

[22] https://www.starship.xyz/press_releases/robovan-by-starship-technologies-and-mercedes-benz-vans-future-proof-local-delivery/

Oregon, and Washington) and Canada (two companies from Quebec and Ontario). In addition to establishing a specific location for a microhub, some companies use on-demand ad-hoc microhubs at various places, such as vacant parking lots and curbside of low-traffic streets, to increase consolidation and distribution efficiency (Katsela et al., 2022).

Another hybrid LMD operation is called cargo hitching (Fig. 2), where the cargo transportation is integrated into existing public transportation modes, such as buses or trams (Meyer et al., 2022).

Various companies (e.g., Amazon and UPS in the U.S., DHL in Germany, Bpost in Belgium, Yamato Transport in Japan, and Flipkart in India) started to use parcel lockers as an alternative delivery model for picking up cargo. The parcel pickup systems provide a secure location (either stationary or mobile) for parcels that are waiting to be picked up by customers at a later convenient time (Schwerdfeger & Boysen, 2022). Parcel lockers are operated mostly unattended and can either be stationary or mobile, and some of them are capable of receiving customer returns as well as pickups. Notable examples of mobile parcel locker delivery using autonomous vehicles include Carrefour's Mobile Drive Service[23], which offers a flexible and efficient parcel collection system. Another example is Renault's Autonomous Driving Platform EZ-GO[24], which is designed to integrate autonomous technology into urban delivery solutions. These innovations demonstrate the evolving role of automation in last-mile logistics.

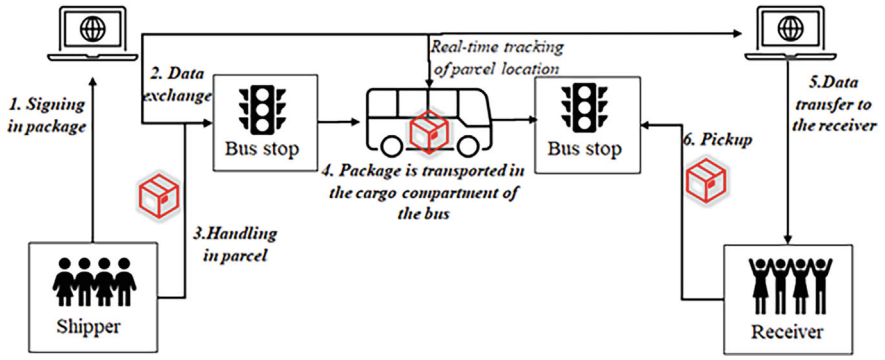

Fig. 2 Example of cargo hitching using bus transportation (*Source* Meyer et al. [2022])

[23] https://trans.info/en/carrefour-autonomous-shuttle-317362

[24] https://itchronicles.com/automation/the-last-mile-to-automation-how-autonomous-vehicles-could-solve-the-last-mile-delivery-problem/

Crowdshipping is another delivery method integrating people's mobility and parcel distribution to complete LMD, which can also employ parcel lockers, public transportation, and electric vehicles (Kızıl & Yıldız, 2023; Rougès & Montreuil, 2014). As summarized by Alnaggar et al. (2021), there are several crowdsourced delivery platforms for e-retailers (e.g., Amazon Flex and Walmart Spark in North America) and courier crowdsourced delivery services (e.g., Postmates, Deliv, DoorDash, Kanga, UberEats, Hitch, Roadie, PiggyBee, Nimber, DHL MyWays, UberFreight, Truxx, and BuddyTruk). Each of these systems has a different delivery window (varies from 0.5 hours to 2–3 days), payment schedule (per hour, delivery, or order), and other business model parameters such as how to determine shipping rate—company or shipper/driver, target market—businesses or individuals, minimum pay guarantees, minimum working hours/shift, short/long haul, and type of delivery items (Alnaggar et al., 2021).

3.3 Disruptive New Technologies in Last-Mile Delivery Steps

Figure 3 demonstrates the potential usage of disruptive new technologies for LMD processes. These technologies can be used in various stages of the LMD, starting from the suburban warehouse to the pickup or delivery point through an urban consolidation centre and a microhub.

AR adds digital and real-time information to the physical environment and, therefore, can improve warehouse and consolidation centres operations (Rejeb et al., 2020; Stoltz et al., 2017). For example, AR-enabled smart glasses can direct workers to the precise location of items on the shelves, which can significantly reduce receiving, storing, picking, and shipping errors and improve overall labour productivity (Stoltz et al., 2017). Inventory management can also be improved by AR because of real-time and accurate inventory tracking with the use of digital information on physical shelves. Similarly, AR technology can assist drivers by offering navigation updates and suggested routes while forklift operators or warehouse workers can benefit from glasses that enhance their ability to visualize and optimize truck space availability.

Similarly, a variety of uses for VR technology can be found in consolidation and storage facilities (Attaran, 2020; Kembro & Norrman, 2022). Using VR to practice jobs in a simulated virtual environment without making mistakes is one of the most popular uses of VR for workers. VR can also help visualize the structure of a warehouse or consolidation centre to maximize space usage. In a similar way, it can be used to optimize the cargo space utilization within a vehicle. Moreover, companies can utilize AR and VR to provide training

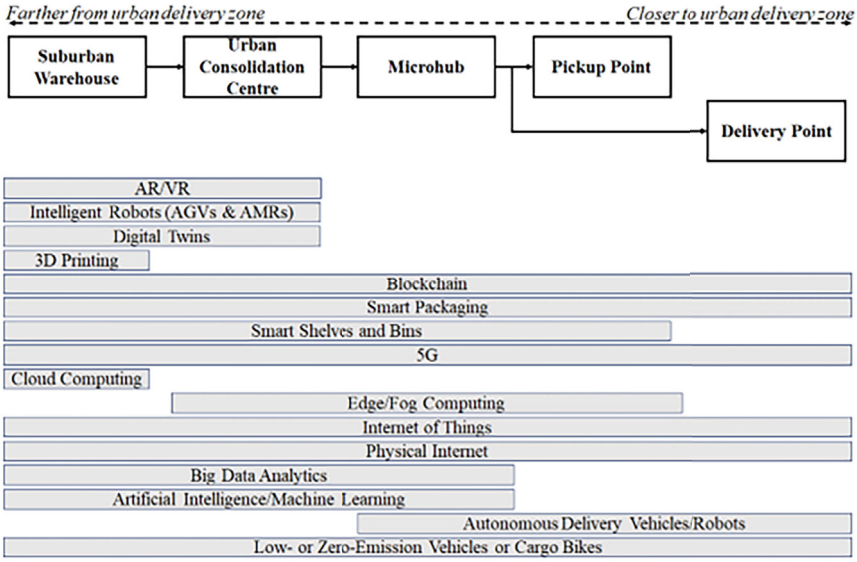

Fig. 3 Conceptual mapping of disruptive new technologies for last-mile delivery processes (*Source* Authors)

for new employees, incorporating interactive and immersive simulations. This approach effectively reduces the time it takes for them to become proficient in their roles (Rejeb et al., 2020). They can also help train personnel for specific safety modules and prepare for operational challenges.

Digital Twins can create digital replicas of facilities, which enables real-time monitoring, analysis, and optimization of the processes (Govindan et al., 2022). This allows Digital Twins to identify issues with inventory, machinery, and personnel to eliminate bottlenecks and increase productivity. It can also be used for redesigning the facility layout. Existing layouts can be optimized and tested for reallocating the workforce and the changing demand structure.

Warehouse and urban consolidation centre efficiency can be improved by integrating intelligent robots. Material handling tasks are carried out by both Automated Mobile Robots (AMRs) and Automated Guided Vehicles (AGVs) (Turhanlar et al., 2022). While AMRs can independently travel in unstructured areas, AGVs follow predetermined routes.

Smart packaging, as well as smart shelves and bins, equipped with Radio-Frequency Identification (RFID) tags and sensors (Schaefer & Cheung, 2018), can help with real-time inventory tracking within warehouses, urban consolidation centres, microhubs, and pickup points. Smart shelves and bins can also help automated replenishment when the inventory levels fall below predetermined reorder points. In addition, smart shelves can help increase

space utilization in the facilities since they increase inventory visibility and transparency. At pickup points, the smart packaging and shelves can automatically generate notifications to customers upon arrival of the package. Smart packaging can also monitor the actual condition of a product through various sensors (e.g., temperature and humidity) and control the safety and integrity of products (Kabadurmus et al., 2023). To ensure compliance with regulations or ensure food safety, it can also enhance the traceability of the product through its delivery process. For this purpose, blockchain technology can also be used to further increase data safety (Issaoui et al., 2019). For example, as a countermeasure for multiple E. Coli outbreaks in the United States caused by Romaine Lettuce produce, Walmart required all its suppliers for leafy green vegetables to upload data to its blockchain tracking system in 2018 (Wang, 2021). RFID-enabled smart packaging can streamline the sorting process in consolidation centres and work with intelligent robots in order picking and transportation operations. To increase security during residential deliveries, smart packaging can allow contactless delivery verification by employing technologies such as geofencing and Quick Response code (QR code) verification.

3D printing can be used to personalize products or even build them on demand. This can lower inventory levels and increase product availability. As a result, order lead times can be shortened (Demir et al., 2021). However, the cost of 3D printing can be expensive and its limited capability for manufacturing products can be challenging to use it for widespread applications (Mckinnon, 2016).

With the help of 5G—the 5th generation mobile network, data exchange among IoT devices can be enabled seamlessly. This real-time connectivity can help inventory visibility, tracking and more responsive LMD operations (Wanganoo & Patil, 2020). They can also provide real-time data for the optimization of delivery vehicle routes (Tsang et al., 2021). At pickup or delivery points, these technologies also allow for contactless transactions. Using IoT and AI technologies, the status of equipment and delivery vehicles (especially the delivery robots) can be monitored effectively, and predictive maintenance can be performed to reduce downtime or increase reliability.

Cloud computing can process vast amounts of real-time data for warehouse operations and provide cloud-based analytics. This can allow data-driven decision-making for inventory management, demand forecasting, and other operational issues. However, due to the increased number of connected devices due to IoT applications and, therefore, massive amount of data created by these devices, network traffic towards data centres has increased significantly (Dolui & Datta, 2017). Also, the long response time for the

generated data to be sent to the cloud for processing and retrieving the result reduces the quality of service of cloud computing (Laroui et al., 2021; Yousefpour et al., 2019). Thus, by getting some of the computing resources closer to the data source by fog and edge computing, the decisions can be made faster with the real-time data provided by IoT devices and sensors (Dolui & Datta, 2017; Laroui et al., 2021). So, edge and fog computing can help providing better insights for urban consolidation centres, microhubs, pickup points and delivery points. According to Yousefpour et al. (2019), edge computing operates at the edge (immediate first hop from IoT devices, e.g., access points or gateways); however, fog computing can be anywhere between the cloud and the IoT devices. These technologies can also help automated delivery vehicles (including drones, droids, and other robots) to work more effectively for delivery and warehouse operations where localized and faster decision is often needed. BDA can utilize the data generated by warehouses, urban consolidation centres, microhubs and delivery vehicles to provide insights for inventory management and order fulfilment. BDA and ML can be used together to make more accurate demand forecasts (Darendeli et al., 2021; Özemre & Kabadurmus, 2020). AI and ML methods can also be used along with BDA to make better decisions related to optimizing inventory levels, identify demand changes, optimizing vehicle routes according to traffic and weather conditions, improving staff scheduling and resource allocation by using real-time data. All these new technologies—AI, IoT, cloud/edge/fog computing, 5G, BDA, and AI/ML—work together to increase LMD performance.

As explained in Sect. 3.1 in detail, low or zero-emission vehicles, cargo bikes (electric or regular), autonomous vehicles, and robots have already been started to be used in LMD operations. In addition to their significant promise for economic benefits, improved efficiency and flexibility, these delivery solutions also help reduce the carbon footprint. The multimodal nature of these transport solutions can also allow designing a more sustainable and more resilient delivery network design (Kabadurmus & Erdogan, 2020) through suburban warehouses, urban consolidation centres, microhubs, pickup points, and delivery points.

Physical Internet (PI) is a global open and shared logistics system in order to enhance an effective, seamless, hyperconnected, and standardized global logistics network by applying the principles of the digital Internet to physical objects and transportation systems (Montreuil et al., 2013). The whole transport network operates under internationally recognized protocols and is open to any user. The goods in PI are transported in standardized boxes or containers, which can allow automated and efficient handling of goods, whereas the bulk freight, pallets, and non-containerised goods are not

transported within PI. The standardized containers are in standard sizes or modules (like 20-and 40-foot ISO containers), environmentally sustainable (made from eco-friendly materials), smart (trackable and routable with integrated RFID and GPS-Global Positioning System), secure, can be interlocked to create larger units, and, in most cases, capable of being collapsed for more efficient empty storage and return. The standardized modular boxes are used in different combinations to fill the size of a standardized container and they can be built up and broken down like building blocks in the game of Tetris (Fig. 4).

These standardized boxes can be used in LMD vehicles (e.g., small delivery vans, sidewalk robots) and solutions (e.g., smart parcel lockers) in urban areas. PI uses global system of standardization initiatives like UNSPSC[25] (United Nations Standard Products and Services Code), GS1[26] (Global System Standards) and EDI (Electronic Data Interchange) in order to improve value chains. GS1 has already defined standards for barcodes, and it enables to search, look up, and verify information about any barcode number: GTIN (Global Trade Item Number), EAN (European Article Number)/UPC (Universal Product Code), location: GLN (Global Location Number) or company. GS1 is a member of the open mail market initiative (OPAL), which defines the rules for the transport sector such as other GS1 key: SSCC (Serial Shipping Container Code). GS1 EDI[27] provides global standards for electronic business messaging allowing automation of business transactions. As an emerging phenomenon, PI is still in its infancy, but the use of disruptive

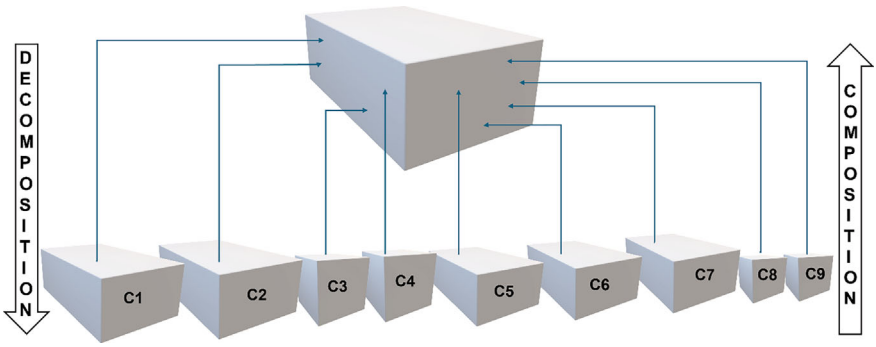

Fig. 4 Different modular box combinations to fit the container size in PI (*Source* Montreuil [2011])

[25] https://www.unspsc.org/

[26] https://www.gs1.org/

[27] https://www.gs1.org/standards/edi

technologies such as blockchain, AR/VR, 5G, and AI/ML has an impact on the growing potential and proliferation of PI's implementation.

4 Adoption Issues of Disruptive Technologies in Last-Mile Delivery Operations

In the context of LMD operations, disruptive technologies are transforming the traditional methods of delivering goods to the end consumers. However, 84% of the companies fail at the digital transformation efforts[28]. Therefore, the adoption of disruptive technologies in LMD operations faces several challenges.

In a comprehensive literature survey on the adoption of green vehicles (such as AVs, EVs, and drones) in logistics, Patella et al. (2021) argue that the key challenge is the cooperation problem between logistics service providers, businesses, the public sector, and the citizens. In addition, Rizet et al. (2016) point out the problem of traffic congestion in mixed fleet scenarios as the e-commerce sector is growing, and the number of green vehicles travelling in city areas is increasing. This extends the range of green vehicles even further, which is not a cost-effective solution for adopters because research indicates that low daily mileage is more cost-efficient when energy prices or consumptions are high (compared to diesel prices or consumptions) or when an extended warranty does not cover the battery (Taefi et al., 2017).

A substantial body of research has been done on the use of drones for last-mile deliveries. The major challenges include optimizing the vehicle routes for real-time traffic information and changing customer demand, short delivery range and operational time due to battery limitations, and the operational cost in the assignment of drones to the customer; choosing the optimal locations of LMD facilities and recharging stations, and finally, determining the optimal size of the fleet size (multiple trucks equipped with multiple drones) to reduce greenhouse gas and energy use (Benarbia & Kyamakya, 2022). Moreover, the increasing demand for e-commerce strains the smooth functioning of supply chains in low-density areas since packages are more widely dispersed, which increases operational costs dramatically (Borghetti et al., 2022). Another difficulty faced by drone usage is the legal restrictions such as no-fly zones (airlines, hospitals, highways, etc.), pilot quality (need of insurance and flier licence), and the requirement for documentation of all flights (Fehling & Saraceni, 2023).

[28] https://www.forbes.com/sites/brucerogers/2016/01/07/why-84-of-companies-fail-at-digital-transformation/?sh=2768891f397b

Adoption of low- or zero-emission vehicles in LMD operations faces technical and operational barriers such as cost and high upfront retail prices, range limitations due to availability of charging infrastructure, reduced payload (due to size and weight of batteries in EVs), and grid concerns (capacity of the grid and the local electrical circuits) (Morganti & Browne, 2018).

Several barriers to adopting cargo bikes have been identified in the literature. For example, Thoma and Gruber (2020) indicate vehicle restrictions, risks, and concerns (about theft and damage, implementation of cargo cycles in their routines), riders' concerns, and infrastructure limitations as significant adoption barriers. One of the most significant issues is technical shortcomings (or technological failures), such as lack of sufficient charging stations, limited autonomy (particularly in hilly areas during rain), and occasional engine and battery malfunctions (Nocerino et al., 2016). Infrastructure constraints such as the requirement to share the road with cars and the absence of overnight storage facilities are also decisive factors. Another issue that discourages the use of cargo bikes is the time required to plan the change from vans and cars to cargo bikes (Narayanan et al., 2022).

Using crowdsourcing technology platforms like Amazon Flex, GrubHub, or Instacart is also challenging because of increased agent independence as drivers become more selective on locations based on how much customers tip them. This creates additional uncertainty and could compromise operational performance, especially in densely populated areas (Castillo et al., 2022).

Several barriers to adopting disruptive technologies in the logistics industry have been identified in the literature. For example, after a thorough literature review, Rathore et al. (2022) identified 12 barriers. Using Fuzzy Delphi Method and Interpretive Structural Modelling, they received opinions from logistics experts who identified ten of those barriers as critical. The two rejected barriers were "lack of communication", and "lack of advanced analytical skills". Rathore et al. (2022) also identified the interrelationships among these barriers with the help of experts' opinions. According to their results, the most important barrier to adopting disruptive technologies in logistics companies is the "lack of top management support". Several other significant barriers were also identified as "legal and regulatory frameworks", "infrastructure", and "resistance to change". The remaining barriers, "Data management", "technical issues", "lack of adequate resources", "lack of trust", "security/privacy", and "lack of reliability" were identified as important but mostly affected by previous barriers. As this study was conducted by collecting the opinions of 15 logistic experts in India with at least ten years of work experience in various areas of logistics, we can also conclude that these

barriers can also be applicable to the adoption of disruptive technologies in LMD systems.

5 Conclusion

This chapter presented a comprehensive review of the usage of new disruptive technologies and innovative solutions including AI/ML, BDA, PI, Blockchain, Digital Twins, IoT, 3D printing, AR/VR, low/zero emission, and autonomous vehicles, microhubs, and mobile parcel lockers in LMD operations. The state-of-the-art applications of new technologies in LMD were reviewed, future directions were outlined, and finally potential barriers for adoptions were discussed. This research was based on compilation and analysis of existing literature and real practices. In the prior research studies, there has been limited research conducted in this field, which has not addressed to comprehensively analyze the usage of disruptive technologies in LMD operations. Therefore, this research is the first to give a comprehensive review to serve the literature. The findings of this chapter provide professionals, decision makers and practitioners in LMD operations with insight into what disruptive technologies and innovative solutions are available in this field and what the key challenges are to implement. The limited number of studies conducted so far also limits the scope of this chapter. In the future, more research needs to be conducted to examine the true potential of these technologies and innovations to improve operational outcomes in LMD. Future research will make a significant contribution to the literature by examining the applications of disruptive technologies in LMD with real industry cases from a sustainability aspect, considering social, economic, and environmental perspectives.

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Part VII

Conclusion

29

Disruptive Technology and Supply Chain: Final Thoughts and Future Research Directions

Nachiappan Subramanian, Yasanur Kayikci, Atanu Chaudhuri, and Michael Bourlakis

1 3D Printing in Reverse Supply Chains

The widespread adoption of 3D printing (3DP) has empowered many industries to innovate at an accelerated pace (Choi et al., 2022). In particular, the application of 3DP in closed-loop supply chains holds significant promise as an enabler of reverse logistics. By disrupting the economics of traditional manufacturing value chains, 3DP offers the potential to facilitate small-scale, local production that converts waste plastics into 3D printing feedstock, creating a more sustainable and economically viable process (Garmulewicz et al., 2018).

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However, several barriers hinder the full-scale adoption of 3DP in reverse supply chains. Among these are the quality of 3D-printed products made from either virgin or recycled materials, as well as the unfavorable value proposition of converting waste plastics into economically viable printed products (Garmulewicz et al., 2018). Unruh (2018) examined the application of biosphere rules—such as material efficiency, value cycles, energy autonomy, and sustainable product platforms—to explore how reverse logistics could be supported by 3DP infrastructure.

Furthermore, Basu et al. (2023) introduced the concept of 3D printing-enabled trucks as a solution to extend the lifespan of discontinued vehicles no longer manufactured by original equipment manufacturers (OEMs). This innovation aims to alleviate the burden of reverse logistics processing, particularly in regions like India, where an estimated 7.6 million vehicles were discontinued between 2010 and 2020. Vehicle owners often face challenges when sourcing replacement spare parts for these vehicles, especially when on the road. To address this, Basu et al. (2023) propose the use of mobile trucks equipped with 3D printing capabilities. These trucks could rapidly produce the required spare parts on-site, regardless of the brand or model, within just a few hours. By triggering the nearest truck through an online order, the mobile units can be dispatched to the location where they can manufacture and deliver the parts directly to service teams. This model significantly enhances spare parts availability, making it more reliable across service centers, and adds critical agility, flexibility, and resilience to the spare parts supply chain.

In line with these advances, Choi et al. (2022) suggest several avenues for future research, including:

- The development of optimal production portfolio planning combining 3DP and mass production.
- The creation of optimal production scheduling models that integrate 3DP.
- Investigating the impacts of 3DP on supply chain agents and consumers.

2 Drones and Intermodal Logistics

Drones have become increasingly pivotal in last-mile delivery, which represents the final leg of the e-commerce supply chain—connecting brands with customers by delivering purchased products. In the wake of the e-commerce boom and the COVID-19 pandemic, the importance of last-mile delivery has surged, as it is key to meeting customer expectations for free and fast shipping while simultaneously bolstering the seller's reputation. This has made

last-mile delivery significantly more expensive for sellers, who now face the reality that it accounts for 53% of total shipping costs. Consequently, sellers are looking toward innovative technologies—such as cargo bikes, drones, autonomous robots, and even Amazon's future airships—to reduce delivery costs without sacrificing customer satisfaction (Boysen et al., 2021).

To put this into perspective, the cost of shipping a package for a retailer average around $27 per item, while Amazon manages to keep its shipping costs to about $5 per package (fareye.com). Retailers expect deliveries to be on time, accurate, efficient, and sustainable (fareye.com, Boysen et al., 2021). However, several challenges significantly affect delivery times, costs, and customer satisfaction. According to fareye.com and Boysen et al. (2021), the ten most prominent challenges are as follows:

- Traffic congestion that delays delivery times.
- Limited visibility while packages are en route.
- Missed deliveries, leading to added costs and customer frustration.
- Complex returns management, requiring significant resources and coordination.
- Customer-preferred delivery times, which introduce additional scheduling constraints.
- Rising costs in urban areas due to traffic and logistical challenges.
- Unpredictable demand patterns, resulting in inefficiencies and cost increases.
- Security risks, such as theft of packages, leading to increased costs, and customer dissatisfaction.
- Environmental impact, including pollution and packaging waste.
- Limited access to rural areas due to inadequate infrastructure.

In their comprehensive review, Boysen et al. (2021) suggest several key research directions related to storage, transport, and handover mechanisms, which could offer future avenues for optimizing last-mile logistics. These include central depots, micro-depots, mobile depots, parcel lockers, parcel shops, and even car trunks for storage, as well as vans, cargo bikes, drones, autonomous bots, crowdshipping, trains, vessels, aircraft, and public transport for transportation, and home delivery (both attended and unattended) and self-service for handovers. Specifically, some promising future research topics include:

- Designing optimal territory partitions for routing truck-drone tandems efficiently—a challenging and high-potential research topic.

- Leveraging real-world data, such as street networks, parking spaces, footways, access restrictions, and precise customer locations, to benchmark both standard and non-standard delivery methods.
- Investigating the ideal mix of delivery methods for different regions, such as combining delivery vans and drones for rural areas (where drones can land in backyards) and using cargo bikes or autonomous bots for urban delivery scenarios.
- Customer segmentation, categorizing customers as priority, non-urgent, or price-sensitive, and assigning them suitable delivery methods to optimize efficiency and customer satisfaction.

3 Autonomous Vehicles and Robotics

Autonomous vehicles (AVs)—also referred to as automated fleets, self-driving fleets, or driverless fleets (Faisal et al., 2019; US DOT, 2023)—are transforming the transportation landscape. In particular, the use of AV taxis offers a combination of traditional car-sharing and taxi services, with major players like Uber and Lyft exploring shared AV fleets (Faisal et al., 2019).

AV Capabilities: The capabilities of AVs extend beyond basic transport functions, encompassing a range of operational features that enhance performance on public roads. These include advanced hardware and software systems designed to facilitate platooning, fuel efficiency, eco-driving, adaptive cruise control, crash avoidance, lane keeping, lane changing, valet parking, traffic sign recognition, and the detection of cyclists and pedestrians. Additionally, AVs are equipped with systems for safe maneuvering that contribute to overall driving safety (Anderson et al., 2014; Faisal et al., 2019).

Planning and Systemic Interventions: The advent of AVs will have profound implications at the systemic and planning authority levels. These interventions will span various sectors, including infrastructure design, transport modeling, capital investment, fleet ownership, land use, employment, energy consumption, traffic safety, and public health (Dixit et al., 2016; Faisal et al., 2019). Countries such as the US, UK, Sweden, and the Netherlands are leading the charge with the development of legislation and policies tailored to AVs. Despite setbacks, such as the high-profile Uber self-driving car crash involving a pedestrian in the US (Faisal et al., 2019; Vellinga, 2017), the ongoing excitement surrounding AV innovation remains palpable.

However, significant challenges remain, particularly in privacy and cybersecurity. The transmission of real-time location data, historical travel patterns, and future journey plans could lead to concerns over illegal tracking and

surveillance if data protection measures are not adequately enforced. There is an urgent need to develop defensive mechanisms within AV systems to safeguard against potential cyberattacks in the future.

4 Intralogistics: Autonomous Guided Vehicles (AGVs) and Autonomous Mobile Robots (AMRs)

In the realm of intralogistics operations, Automated Guided Vehicles (AGVs) are commonly employed, particularly in environments that require repetitive, stable transport. These AGVs are computer-controlled, wheel-based load carriers designed for use within a confined setting, requiring no human drivers. Autonomous Mobile Robots (AMRs), by contrast, are more flexible and adaptable, finding application across sectors not only for material handling and transport but also for tasks like patrolling and collaborating with human operators to boost productivity and quality while reducing worker fatigue (Fragapane et al., 2021).

As of 2017, over 13,000 AGV and AMR systems had been installed globally, with AMRs becoming more affordable over time. These advanced systems surpass traditional AGVs by not only managing scheduling, routing, and dispatching tasks but also by engaging in decentralized decision-making and facilitating communication between machines and other operational systems (Bechtsis et al., 2017; Fragapane et al., 2021).

Several key research areas in intralogistics remain ripe for exploration (Fragapane et al., 2021): While manufacturing and warehouse intralogistics applications have been well-studied, there is significant opportunity for further research in other intralogistics domains.

- The profitability of intelligent mobility—particularly through decentralized control mechanisms—warrants deeper investigation across a broader range of intralogistics applications.
- The analysis of strategic, tactical, and operational-level decisions, incorporating multiple objectives (e.g., determining the number of vehicles, optimizing service point locations, and simultaneously scheduling and planning paths), is crucial for advancing the efficiency of these systems. AI-driven tools can play a vital role in enabling these decisions.

5 Blockchain and SME Supply Chains

It is widely recognized that blockchain technology ensures transparency and traceability among partners in the supply chain. However, outside of high-value product supply chains, blockchain adoption has been limited to only a few cases. As noted by Choi et al. (2022), further research is needed to address several critical questions, such as identifying the bottlenecks in supply chain operations, understanding the potential risks of implementing blockchain in low-value product supply chains, and assessing the costs and risks associated with blockchain adoption.

This research will help to shape new business models for SMEs looking to adopt blockchain, fostering enhanced trust and highlighting the cost implications of implementation. Additionally, blockchain technology enables automated computer-based contracting mechanisms, with smart contracts facilitating deeper collaboration among supply chain partners. As a result, smart contracts are gaining increasing popularity. However, Choi et al. (2022) suggest further empirical studies are necessary to explore the implementation of both complete and incomplete contracts in blockchain systems.

Moreover, numerous gray areas and controversies exist in contract law, particularly within the context of business agreements, which must be further researched to make blockchain feasible at the SME level. Several research avenues warrant further investigation, including:

1. The impacts of smart contracts on social welfare and supply chain systems.
2. The issue of moral hazard and information asymmetry in supply chains using Initial Coin Offerings (ICOs), akin to crowdfunding models.
3. Business case studies for incorporating government regulation into smart contracts for SMEs.

6 Internet of Things (IoT) and Smart Replenishment

While research in the Business-to-Business (B2B) domain has garnered considerable attention, recent studies have shifted focus toward the Business-to-Consumer (B2C) landscape. In the context of the Product-as-a-Service model, manufacturers are now able to retrieve valuable point-of-consumption data—such as usage patterns and inventory levels—from smart household appliances. This data empowers manufacturers to offer consumers a broader

range of augmented functionalities. One prominent example is smart replenishment, a system wherein manufacturers leverage this point-of-consumption information to seamlessly and automatically restock consumables as needed (Weißhuhn & Hoberg, 2021, see Table 1).

Key indicators influencing Fast-Moving Consumer Goods (FMCG) manufacturers to adopt smart replenishment include factors such as sales volume, demand uncertainty, profit margins, cost structures, and the necessity of the product. Other considerations involve the customer base characteristics, the existing online channel infrastructure, and the efficiency of outbound warehouse operations. Additionally, the extent to which IoT technology is integrated into household appliances plays a significant role in determining the feasibility of implementing a smart replenishment system (Weißhuhn & Hoberg, 2021).

Table 1 Smart replenishment using IoT

Consumables used in household appliances		Consumables stored in the refrigerators		Consumables stored in the shelf	
Product as a service	Brand	Product as a service	Brand	Product as a service	Brand
Coffee machines and capsules	Nespresso, Illy, Hamilton Beach	Beer fridges	Budweiser, Drinkshift	Sensor-equipped shelves & RFID-equipped products	Amazon
Washing machines and detergent	Miele	Fridge with camera monitoring of inventories through smart phone	Samsung, LG	Trashcans with barcode scanner	GeniCan
Electronic tooth brush and heads	Oral-B, Philips	Sensor-equipped egg trays	Egg minder	Sensor-equipped jars	Neo

(Adapted from Weißhuhn & Hoberg, 2021)

7 Packaging Logistics: A Growing Discipline

Packaging logistics is an emerging field that is quickly gaining recognition due to its potential to provide strategic advantages for companies (ModusLink, 2023). Valued at US$ 424 billion in 2016, the global packaging market is projected to grow at an annual rate of 11%. Despite this significant growth, packaging has historically been underappreciated by manufacturers, retailers, and leading brands, often seen merely as a functional tool for transporting products or handling returns. This limited view has overlooked packaging's broader potential, particularly in ensuring the safe delivery of goods and preserving the freshness of perishable items during shipping (Deloitte, 2018a; Thirupathi Vasuki et al., 2023).

However, the rise of smart packaging is set to revolutionize the industry. By incorporating advanced technologies, innovative materials, and thoughtful design, smart packaging offers additional functionalities that go beyond traditional packaging solutions. These innovations not only protect products during transit but also enable features such as real-time tracking, temperature control, and even inventory management. A comprehensive overview of various types of smart packaging and their applications is provided in Table 2.

The primary aim of smart packaging is to ensure product authenticity, combat counterfeiting, elevate customer satisfaction, and improve accessibility. In addition to these core benefits, innovations in materials and technologies are paving the way for enhanced quality assurance, increased safety, fortified security against theft, and comprehensive traceability. These advancements also support businesses in meeting their sustainable development goals by enabling more efficient and eco-friendly packaging solutions.

Smart packaging holds immense potential to transform both forward and reverse logistics, offering a more streamlined and intelligent approach to managing supply chains. However, a study by Deloitte (2018a) highlighted several challenges that must be addressed for successful implementation. These include commercial hurdles such as monetization concerns and prohibitive costs, legal complexities like regulatory and privacy issues along with data ownership challenges, technological barriers surrounding the standardization of systems, and organizational factors such as the need for collaboration and fostering an environment conducive to agile innovation.

Table 2 Types of smart packaging and applications

Type of smart packaging	Definition	Example
Design led	Design that improves form and function supports the value perception of the product and enhances user perceptions	PillPack offers the convenience of delivering portioned and easily accessible doses of medication that are tailored to the individual preferences of customers. The packaging is designed to be user-friendly and effortless to open
Active	Packaging that utilizes advanced materials or chemistry to enhance functionality or appearance, such as moisture retention or temperature control	Freshmax pouches incorporate oxygen absorbers to maintain freshness
Connected	Packaging that incorporates technology (sensors/tags) capable of generating data that can be captured, analyzed, or communicated to individuals or machines, enabling meaningful insights in the real world	Ferngrove Vineyards utilizes radio frequency identification (RFID) device chips embedded in wine bottles to allow customers to track and trace the quality of the product and engage with the brand

(Adapted from Deloitte, 2018a)

8 Augmented and Virtual Reality

Craig (2013) defines augmented reality (AR) as a dynamic medium where digital content is overlaid on the physical world, creating an interactive, real-time experience that blends spatial and temporal elements. AR enhances the real-world environment by integrating valuable computer-generated information such as text, graphics, video, sound, haptic feedback, GPS data, and even, in some cases, smell (DHL, 2011). It's essential to recognize the distinction between augmented reality (AR) and virtual reality (VR). While AR combines virtual and real-world information at a specific moment in time (often referred to as mixed reality), VR immerses users in a completely computer-generated, three-dimensional environment, typically viewed through LED screens or stereoscopic displays (Cheriakkara & Alex, 2020; DHL, 2011).

Rejeb et al. (2021) proposed several avenues for future research regarding the use of augmented reality smart glasses in logistics and supply chain management, including:

- Assessing the suitability of smart glasses under challenging conditions such as bright sunlight, low light, moisture exposure, and noisy industrial environments.
- Investigating the efficiency and effectiveness of smart glasses in real-world forklift truck operations, particularly when performing complex tasks.
- Exploring the integration of smart glasses with emerging technologies such as cloud computing, blockchains, Internet of Things (IoT), smart gloves, and wristbands.
- Analyzing the potential of smart glasses in enhancing logistics training programs.
- Examining the privacy and security standards necessary for leveraging smart glasses in logistics.
- Studying the maturity level of smart glasses in developing and emerging economies.
- Investigating how smart glasses can contribute to sustainable logistics practices.

Future research could delve further into user experience and contextual interaction, conduct cross-case analyses to evaluate the value for both retailers and consumers in adapting to AR, and explore longitudinal studies to track the technology's maturation over time, including its impact on usage and customer value.

The virtual reality (VR) market is projected to witness significant growth, with estimates predicting its value to reach $48.5 billion by 2025. Notably, the use of VR headsets in mid- to high-end applications surged, with 52.3 million units sold in 2020. One key advancement in VR technology is the inside-out tracking feature, which uses an internal camera to track the position of objects. However, several barriers hinder VR's widespread adoption, such as limitations in computational power, cost concerns, security issues, and user perceptions.

In the context of supply chain applications, VR can be leveraged in areas like product and process design, virtual collaboration, and experience-based learning. The benefits of using VR in these areas include reduced design time, visualization of complex data, mitigation of operational hazards, improved workforce collaboration, and enhanced training effectiveness (Caboni & Hagberg, 2019; Deloitte, 2018b). Within the digital factory context, VR has

found particular use in manufacturing, including tasks such as factory layout planning, robot path planning, virtual prototyping, and training (Sekearan et al., 2021).

Deloitte (2018b) outlines critical considerations for implementing VR within a supply chain, structured around five key attributes: functional deployment, technology infrastructure, security, regulations, and operating environment. By carefully evaluating these attributes, organizations can make informed decisions to maximize their return on investment in VR technology.

Below are key evaluation and adaptation criteria for each of the five key attributes, as identified by Deloitte (2018b):

Functional Deployment:

- Identify functions with high potential for rapid value capture during the initial rollout.
- Assess the availability of existing software and applications suitable for your use cases, or determine if your organization has the capability to develop them.
- Evaluate whether current workflows need to be re-engineered to incorporate VR.
- Determine the type and quantity of devices required to achieve expected benefits, considering the option to leverage existing corporate mobile devices for VR applications.

Technology Infrastructure:

- Evaluate the necessary changes in the technology stack to accommodate VR integration.
- Assess the digital maturity of your supply chain or network.
- Ensure the VR hardware and software seamlessly integrate with existing systems.

Security:

- Develop mitigation strategies to address potential network disruptions, especially given the increased reliance on device-enabled workflows.
- Ensure the presence of a robust security network to protect operational technologies.
- Identify which hardware and software components can function on private networks to enhance security.

Regulations:

- Understand regulatory differences across countries where you operate, particularly those impacting wearables in the workplace.
- Formulate a plan for managing intellectual property (IP) created globally via VR devices.
- Stay informed on how IP for VR content is managed within your industry and region.

Operating Environments and Perception:

- Anticipate how the workforce will respond to working with VR and ensure the physical environment supports VR implementation.
- Consider challenges around sharing VR devices among employees and plan accordingly.
- Develop comprehensive training programs to familiarize employees with new VR technologies.
- Evaluate the need for insurance policies to cover VR-related sickness or discomfort.

By considering these comprehensive evaluation criteria, organizations can better prepare for and optimize the implementation of VR technology within their supply chain operations, aligning it with broader strategic objectives. Craig (2013) defines augmented reality as a medium in which digital content is superimposed over the physical world, creating a spatial and temporal experience that is interactive in real-time. Augmented reality enriches the real environment by adding valuable computer-generated information, including text, graphics, video, sound, haptic feedback, GPS data, and in some cases, even smell (DHL, 2011). It is important to note that there is a significant difference between augmented reality (AR) and virtual reality (VR). AR combines virtual and real information in a specific point in time, also known as mixed reality. On the other hand, VR creates a completely computer-generated, immersive, and three-dimensional environment displayed on LED screens or stereoscopic displays (Cheriakkara & Alex, 2020; DHL, 2011).

A few research directions proposed by Rejeb et al. (2021) regarding the use of augmented reality smart glasses in logistics and supply chain management are listed below:

- Assess the suitability of smart glasses in harsh conditions, including sunlight, low light, exposure to moisture, and noisy industrial environments.
- Investigate the impact of using smart glasses on forklift trucks in realistic scenarios to understand the interaction's efficiency and effectiveness when performing complex tasks.

- Explore the integration of smart glasses with emerging technologies such as cloud computing, blockchains, the Internet of Things (IoT), smart gloves, and wristbands.
- Analyze the effectiveness of using smart glasses in logistics training.
- Examine privacy and security standards necessary for leveraging smart glasses in logistics.
- Study the level of maturity of smart glasses in developing and emerging economies.
- Investigate how smart glasses can contribute to sustainable logistics developments.

Future research suggestions include additional research on user experience and context interaction, cross-case analysis to compare the value for both retailers and users in adapting to AR, and longitudinal research to understand the technology's maturity over time in terms of usage and value to customers.

Virtual reality (VR) market is projected to experience significant growth, with a predicted value of $48.5 billion by 2025. In particular, the growth of VR headsets in mid and high-end applications reached 52.3 million units in 2020. An innovative feature called "inside-out tracking" technology enables the VR headset to track the position of objects using an internal camera. However, there are several well-known barriers hindering the growth of VR, including computational power limitations, pricing concerns, security issues, and user perception.

Moreover, VR can be leveraged in various supply chains applications, such as product and process design, virtual collaboration, and experience-based learning. The utilization of VR in these areas offers several benefits, including reduced product and process design time, visualization of complex enterprise data, mitigation of operational hazards, improved workforce collaboration, and enhanced employee training effectiveness (Caboni & Hagberg, 2019; Deloitte, 2018b). In the context of the digital factory, VR has predominantly been applied in manufacturing, followed by activities such as factory layout planning, robot path planning, virtual prototyping, and training (Sekearan et al., 2021).

Deloitte (2018b) provides several questions to consider in order to make informed decisions and maximize the return on investment when implementing VR in a company's supply chain. These questions revolve around five key attributes: functional deployment, technology infrastructure, security, regulations, operating environment, and perception. It is crucial to assess these attributes to ensure a successful VR implementation.

Furthermore, it is important to have a clear understanding of the firm's unique operational model and how it aligns with strategic objectives when adopting any technology. This understanding will help ensure that the chosen technology, in this case VR, is a suitable fit for the company's overall goals and objectives.

Below are the vital evaluation and adaptation criteria related to the five key attributes that one must consider before implementing VR, as per Deloitte's (2018b) findings:

Functional Deployment:

- Identify functions that offer the best use case and potential for rapid value capture during the initial rollout.
- Determine if there are existing software and unique applications available for your desired use cases. If not, assess if your organization has the capabilities to build them.
- Evaluate if existing workflows need to be re-engineered to incorporate virtual reality.
- Determine the type and quantity of devices required to achieve expected benefits. Explore the possibility of utilizing existing corporate mobile devices for VR applications.

Technology Infrastructure:

- Assess the necessary changes in the existing technology stack to facilitate VR integration.
- Evaluate the digital maturity level of your supply chain or supply network.
- Determine if your preferred VR hardware and software can seamlessly integrate with your existing systems.

Security:

- Develop mitigation strategies for network disruptions considering the increased use of device-enabled workflows.
- Ensure the presence of a comprehensive security network for operational technology.
- Identify which hardware and software components can operate on private networks.

Regulations:

- Understand and account for regulatory differences across the countries where you operate.

- Determine industry requirements for wearables in your workplace.
- Establish a plan to manage intellectual property (IP) created globally through VR devices.
- Stay informed about how IP for VR content is being addressed in your industry.

Operating Environments and Perception:

- Anticipate and assess the response of the current workforce to working with VR.
- Ensure that the proposed physical environment meets the necessary requirements for VR implementation.
- Consider if employees will be sharing VR devices and address any associated challenges.
- Plan the required training to familiarize employees with the new technology.
- Evaluate the need for insurance policies to accommodate VR-related sickness.

By considering these criteria, organizations can make informed decisions and effectively implement VR technology within their supply chain operations.

9 AI and Machine Learning in Supply Chain

Based on Hendriksen's (2023) insights, several promising research avenues for the integration of AI and machine learning in supply chain management have emerged. These research opportunities are organized into strategic, tactical, and operational perspectives:

Strategic Perspective:

- Develop a differentiated knowledge management strategy that addresses the unique needs of diverse groups within the supply chain, each with differing attitudes toward information sharing.
- Examine the necessity of a formalized and systematic AI strategy for supply chain professionals, viewed through the lens of knowledge management, to streamline AI integration effectively.
- Investigate AI's potential to facilitate knowledge sharing between supply chain partners across the end-to-end value chain, fostering collaboration and data-driven decision-making.

Tactical Perspective:

- Explore tactical practices and approaches tailored to various AI applications in different supply chain scenarios, ensuring relevance and efficacy in each context.
- Investigate the application of Generative AI (Gen AI) tools and identify the best strategies to harness tacit knowledge that provides a competitive edge to organizations.
- Examine the integration of Gen AI tools with emerging technologies, such as IoT and blockchain, to optimize supply chain processes and create synergies across activities.

Operational Perspective:

- Identify effective methods for extracting valuable insights from AI, considering the challenges posed by information overload and the need for targeted, actionable intelligence for supply chain professionals.
- Evaluate the impact of Gen AI tools on customer outcomes, assessing how AI-driven innovations can enhance service levels and drive customer satisfaction.
- Develop novel techniques for analyzing Gen AI outcomes related to delivery, integrating real-time transportation data with contextual elements like images, videos, and live streams to support dynamic decision-making in logistics and scheduling.
- Create innovative Gen AI algorithms that process diverse data types, such as images, videos, and live streams, to assess transportation scenarios and examine their seamless integration into transportation management systems.
- Explore strategies to safeguard customer privacy when analyzing transport-related data, ensuring that sensitive personal information (e.g., home addresses, workplace details) is securely handled and protected in AI-powered supply chain processes.

These research pathways provide rich opportunities to advance the application of AI and machine learning in supply chain management, fostering greater efficiency, innovation, and customer-centric solutions.

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